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Method for smart energy device infrastructure

Abstract

A method for characterizing a state of an end effector of an ultrasonic device is disclosed. The ultrasonic device including an electromechanical ultrasonic system defined by a predetermined resonant frequency. The electromechanical ultrasonic system further including an ultrasonic transducer coupled to an ultrasonic blade. The method including applying, by an energy source, a power level to the ultrasonic transducer, measuring, by a control circuit coupled to a memory, an impedance value of the ultrasonic transducer, comparing, by the control circuit, the impedance value to a reference impedance value stored in the memory; classifying, by the control circuit, the impedance value based on the comparison; characterizing, by the control circuit, the state of the electromechanical ultrasonic system based on the classification of the impedance value; and adjusting, by the control circuit, the power level applied to the ultrasonic transducer based on the characterization of the state of the end effector.

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6849074	12/2004	Chen et al.	N/A	N/A
6852219	12/2004	Hammond	N/A	N/A
6863650	12/2004	Irion	N/A	N/A
6869430	12/2004	Balbierz et al.	N/A	N/A
6869435	12/2004	Blake, III	N/A	N/A
6911033	12/2004	de Guillebon et al.	N/A	N/A
6913471	12/2004	Smith	N/A	N/A
6937892	12/2004	Leyde et al.	N/A	N/A
6945981	12/2004	Donofrio et al.	N/A	N/A
6951559	12/2004	Greep	N/A	N/A
6962587	12/2004	Johnson et al.	N/A	N/A
6978921	12/2004	Shelton, IV et al.	N/A	N/A
6988649	12/2005	Shelton, IV et al.	N/A	N/A
7000818	12/2005	Shelton, IV et al.	N/A	N/A
7009511	12/2005	Mazar et al.	N/A	N/A
7030146	12/2005	Baynes et al.	N/A	N/A
7032798	12/2005	Whitman et al.	N/A	N/A
7041941	12/2005	Faries, Jr. et al.	N/A	N/A

7044352	12/2005	Shelton, IV et al.	N/A	N/A
7044911	12/2005	Drinan et al.	N/A	N/A
7044949	12/2005	Orszulak et al.	N/A	N/A
7048775	12/2005	Jornitz et al.	N/A	N/A
7053752	12/2005	Wang et al.	N/A	N/A
7055730	12/2005	Ehrenfels et al.	N/A	N/A
7073765	12/2005	Newkirk	N/A	N/A
7077853	12/2005	Kramer et al.	N/A	N/A
7077856	12/2005	Whitman	N/A	N/A
7081096	12/2005	Brister et al.	N/A	N/A
7094231	12/2005	Ellman et al.	N/A	N/A
7097640	12/2005	Wang et al.	N/A	N/A
7103688	12/2005	Strong	N/A	N/A
7104949	12/2005	Anderson et al.	N/A	N/A
7118564	12/2005	Ritchie et al.	N/A	N/A
7121460	12/2005	Parsons et al.	N/A	N/A
7137980	12/2005	Buyse et al.	N/A	N/A
7140528	12/2005	Shelton, IV	N/A	N/A
7143923	12/2005	Shelton, IV et al.	N/A	N/A
7143925	12/2005	Shelton, IV et al.	N/A	N/A
7147139	12/2005	Schwemberger et al.	N/A	N/A
7155316	12/2005	Sutherland et al.	N/A	N/A
7164940	12/2006	Hareyama et al.	N/A	N/A
7169145	12/2006	Isaacson et al.	N/A	N/A
7177533	12/2006	McFarlin et al.	N/A	N/A
7182775	12/2006	de Guillebon et al.	N/A	N/A
7207472	12/2006	Wukusick et al.	N/A	N/A
7208005	12/2006	Frecker et al.	N/A	N/A
7217269	12/2006	El-Galley et al.	N/A	N/A
7230529	12/2006	Ketcherside, Jr. et al.	N/A	N/A
7232447	12/2006	Gellman et al.	N/A	N/A
7236817	12/2006	Papas et al.	N/A	N/A
7246734	12/2006	Shelton, IV	N/A	N/A
7252664	12/2006	Nasab et al.	N/A	N/A
7278563	12/2006	Green	N/A	N/A
7294106	12/2006	Birkenbach et al.	N/A	N/A
7294116	12/2006	Ellman et al.	N/A	N/A
7296724	12/2006	Green et al.	N/A	N/A
7317955	12/2007	McGreevy	N/A	N/A
7328828	12/2007	Ortiz et al.	N/A	N/A
7334717	12/2007	Rethy et al.	N/A	N/A
7343565	12/2007	Ying et al.	N/A	N/A
7344532	12/2007	Goble et al.	N/A	N/A
7353068	12/2007	Tanaka et al.	N/A	N/A
7362228	12/2007	Nycz et al.	N/A	N/A
7371227	12/2007	Zeiner	N/A	N/A
7380695	12/2007	Doll et al.	N/A	N/A
7383088	12/2007	Spinelli et al.	N/A	N/A
7391173	12/2007	Schena	N/A	N/A
7407074	12/2007	Ortiz et al.	N/A	N/A
7408439	12/2007	Wang et al.	N/A	N/A
7413541	12/2007	Konishi	N/A	N/A
7422136	12/2007	Marczyk	N/A	N/A
7422139	12/2007	Shelton, IV et al.	N/A	N/A
7422586	12/2007	Morris et al.	N/A	N/A
7423972	12/2007	Shaham et al.	N/A	N/A
D579876	12/2007	Novotney et al.	N/A	N/A
7457804	12/2007	Uber, III et al.	N/A	N/A
D583328	12/2007	Chiang	N/A	N/A
7464847	12/2007	Viola et al.	N/A	N/A
7464849	12/2007	Shelton, IV et al.	N/A	N/A
7496418	12/2008	Kim et al.	N/A	N/A
D589447	12/2008	Sasada et al.	N/A	N/A
7515961	12/2008	Germanson et al.	N/A	N/A
7518502	12/2008	Austin et al.	N/A	N/A
7554343	12/2008	Bromfield	N/A	N/A
7563259	12/2008	Takahashi	N/A	N/A
7568604	12/2008	Ehrenfels et al.	N/A	N/A
7575144	12/2008	Ortiz et al.	N/A	N/A
7597731	12/2008	Palmerton et al.	N/A	N/A
7617137	12/2008	Kreiner et al.	N/A	N/A
7621192	12/2008	Conti et al.	N/A	N/A
7621898	12/2008	Lalomia et al.	N/A	N/A
7631793	12/2008	Rethy et al.	N/A	N/A
7637410	12/2008	Marczyk	N/A	N/A

7637907	12/2008	Blaha	N/A	N/A
7641092	12/2009	Kruszynski et al.	N/A	N/A
7644848	12/2009	Swayze et al.	N/A	N/A
7667592	12/2009	Ohyama et al.	N/A	N/A
7667839	12/2009	Bates	N/A	N/A
7670334	12/2009	Hueil et al.	N/A	N/A
7694865	12/2009	Scirica	N/A	N/A
7699772	12/2009	Pauker et al.	N/A	N/A
7699860	12/2009	Huitema et al.	N/A	N/A
7717312	12/2009	Beetel	N/A	N/A
7720306	12/2009	Gardiner et al.	N/A	N/A
7721934	12/2009	Shelton, IV et al.	N/A	N/A
7721936	12/2009	Shalton, IV et al.	N/A	N/A
7722603	12/2009	McPherson	N/A	N/A
7736357	12/2009	Lee, Jr. et al.	N/A	N/A
7742176	12/2009	Braunecker et al.	N/A	N/A
7743960	12/2009	Whitman et al.	N/A	N/A
7753245	12/2009	Boudreaux et al.	N/A	N/A
7757028	12/2009	Druke et al.	N/A	N/A
7766207	12/2009	Mather et al.	N/A	N/A
7766905	12/2009	Paterson et al.	N/A	N/A
7770773	12/2009	Whitman et al.	N/A	N/A
7771429	12/2009	Ballard et al.	N/A	N/A
7776037	12/2009	Odom	N/A	N/A
7782789	12/2009	Stultz et al.	N/A	N/A
7784663	12/2009	Shelton, IV	N/A	N/A
7803151	12/2009	Whitman	N/A	N/A
7810692	12/2009	Hall et al.	N/A	N/A
7818041	12/2009	Kim et al.	N/A	N/A
7819298	12/2009	Hall et al.	N/A	N/A
7832612	12/2009	Baxter, III et al.	N/A	N/A
7833219	12/2009	Tashiro et al.	N/A	N/A
7836085	12/2009	Petakov et al.	N/A	N/A
7837079	12/2009	Holsten et al.	N/A	N/A
7837680	12/2009	Isaacson et al.	N/A	N/A
7841980	12/2009	Minosawa et al.	N/A	N/A
7845537	12/2009	Shelton, IV et al.	N/A	N/A
7857185	12/2009	Swayze et al.	N/A	N/A
D631252	12/2010	Leslie	N/A	N/A
7862560	12/2010	Marion	N/A	N/A
7862579	12/2010	Ortiz et al.	N/A	N/A
7865236	12/2010	Cory et al.	N/A	N/A
7884735	12/2010	Newkirk	N/A	N/A
7887530	12/2010	Zemlok et al.	N/A	N/A
7892337	12/2010	Palmerton et al.	N/A	N/A
7907166	12/2010	Lamprecht et al.	N/A	N/A
7913891	12/2010	Doll et al.	N/A	N/A
7918230	12/2010	Whitman et al.	N/A	N/A
7918377	12/2010	Measamer et al.	N/A	N/A
7920706	12/2010	Asokan et al.	N/A	N/A
7927014	12/2010	Dehler	N/A	N/A
7932826	12/2010	Fritchie et al.	N/A	N/A
7942300	12/2010	Rethy et al.	N/A	N/A
7945065	12/2010	Menzl et al.	N/A	N/A
7945342	12/2010	Tsai et al.	N/A	N/A
7950560	12/2010	Zemlok et al.	N/A	N/A
7951148	12/2010	McClurken	N/A	N/A
7954682	12/2010	Giordano et al.	N/A	N/A
7954687	12/2010	Zemlok et al.	N/A	N/A
7955322	12/2010	Devengenzo et al.	N/A	N/A
7956620	12/2010	Gilbert	N/A	N/A
7963433	12/2010	Whitman et al.	N/A	N/A
7966269	12/2010	Bauer et al.	N/A	N/A
7967180	12/2010	Scirica	N/A	N/A
7976553	12/2010	Shelton, IV et al.	N/A	N/A
7979157	12/2010	Anvari	N/A	N/A
7980443	12/2010	Scheib et al.	N/A	N/A
7982776	12/2010	Dunki-Jacobs et al.	N/A	N/A
7988028	12/2010	Farascioni et al.	N/A	N/A
7993140	12/2010	Sakezles	N/A	N/A
7993354	12/2010	Brecher et al.	N/A	N/A
7993954	12/2010	Wieting	N/A	N/A
7995045	12/2010	Dunki-Jacobs	N/A	N/A
8005947	12/2010	Morris et al.	N/A	N/A
8007494	12/2010	Taylor et al.	N/A	N/A

8007513	12/2010	Nalagatla et al.	N/A	N/A
8010180	12/2010	Quaid et al.	N/A	N/A
8012170	12/2010	Whitman et al.	N/A	N/A
8015976	12/2010	Shah	N/A	N/A
8016855	12/2010	Whitman et al.	N/A	N/A
8019094	12/2010	Hsieh et al.	N/A	N/A
8025199	12/2010	Whitman et al.	N/A	N/A
8027710	12/2010	Dannan	N/A	N/A
8035685	12/2010	Jensen	N/A	N/A
8038686	12/2010	Huitema et al.	N/A	N/A
8038693	12/2010	Allen	N/A	N/A
8043560	12/2010	Okumoto et al.	N/A	N/A
8054184	12/2010	Cline et al.	N/A	N/A
8054752	12/2010	Druke et al.	N/A	N/A
8062306	12/2010	Nobis et al.	N/A	N/A
8062330	12/2010	Prommersberger et al.	N/A	N/A
8066721	12/2010	Kortenbach et al.	N/A	N/A
8074861	12/2010	Ehrenfels et al.	N/A	N/A
8075571	12/2010	Vitali et al.	N/A	N/A
8095327	12/2011	Tahara et al.	N/A	N/A
8096459	12/2011	Ortiz et al.	N/A	N/A
8116848	12/2011	Shahidi	N/A	N/A
8118206	12/2011	Zand et al.	N/A	N/A
8120301	12/2011	Goldberg et al.	N/A	N/A
8123764	12/2011	Meade et al.	N/A	N/A
D655678	12/2011	Kobayashi et al.	N/A	N/A
8128625	12/2011	Odom	N/A	N/A
8131565	12/2011	Dicks et al.	N/A	N/A
8136712	12/2011	Zingman	N/A	N/A
8146149	12/2011	Steinkogler et al.	N/A	N/A
D657368	12/2011	Magee et al.	N/A	N/A
8147486	12/2011	Honour et al.	N/A	N/A
8155479	12/2011	Hoffman et al.	N/A	N/A
8157145	12/2011	Shelton, IV et al.	N/A	N/A
8157150	12/2011	Viola et al.	N/A	N/A
8157151	12/2011	Ingmanson et al.	N/A	N/A
8160098	12/2011	Yan et al.	N/A	N/A
8160690	12/2011	Wilfley et al.	N/A	N/A
8161977	12/2011	Shelton, IV et al.	N/A	N/A
8170396	12/2011	Kuspa et al.	N/A	N/A
8172836	12/2011	Ward	N/A	N/A
8181839	12/2011	Beetel	N/A	N/A
8185409	12/2011	Putnam et al.	N/A	N/A
8206345	12/2011	Abboud et al.	N/A	N/A
8208707	12/2011	Mendonca et al.	N/A	N/A
8210411	12/2011	Yates et al.	N/A	N/A
8214007	12/2011	Baker et al.	N/A	N/A
8216849	12/2011	Petty	N/A	N/A
8220688	12/2011	Laurent et al.	N/A	N/A
8225643	12/2011	Abboud et al.	N/A	N/A
8225979	12/2011	Farascioni et al.	N/A	N/A
8229549	12/2011	Whitman et al.	N/A	N/A
8231042	12/2011	Hessler et al.	N/A	N/A
8239066	12/2011	Jennings et al.	N/A	N/A
8241322	12/2011	Whitman et al.	N/A	N/A
8255045	12/2011	Gharib et al.	N/A	N/A
D667838	12/2011	Magee et al.	N/A	N/A
8257387	12/2011	Cunningham	N/A	N/A
8260016	12/2011	Maeda et al.	N/A	N/A
8262560	12/2011	Whitman	N/A	N/A
8292639	12/2011	Achammer et al.	N/A	N/A
8292888	12/2011	Whitman	N/A	N/A
8295902	12/2011	Salahieh et al.	N/A	N/A
8308040	12/2011	Huang et al.	N/A	N/A
8321581	12/2011	Katis et al.	N/A	N/A
8322590	12/2011	Patel et al.	N/A	N/A
8328065	12/2011	Shah	N/A	N/A
8335590	12/2011	Costa et al.	N/A	N/A
D675164	12/2012	Kobayashi et al.	N/A	N/A
8343065	12/2012	Bartol et al.	N/A	N/A
8346392	12/2012	Walser et al.	N/A	N/A
8360299	12/2012	Zemlok et al.	N/A	N/A
8364222	12/2012	Cook et al.	N/A	N/A
D676392	12/2012	Gassauer	N/A	N/A
8365975	12/2012	Manoux et al.	N/A	N/A

D678196	12/2012	Miyauchi et al.	N/A	N/A
D678304	12/2012	Yakoub et al.	N/A	N/A
8388652	12/2012	Viola	N/A	N/A
8393514	12/2012	Shelton, IV et al.	N/A	N/A
8397972	12/2012	Kostrzewski	N/A	N/A
8398541	12/2012	DiMaio et al.	N/A	N/A
8403944	12/2012	Pain et al.	N/A	N/A
8403945	12/2012	Whitfield et al.	N/A	N/A
8403946	12/2012	Whitfield et al.	N/A	N/A
8406859	12/2012	Zuzak et al.	N/A	N/A
8411034	12/2012	Boillot et al.	N/A	N/A
8413871	12/2012	Racenet et al.	N/A	N/A
8422035	12/2012	Hinderling et al.	N/A	N/A
8423182	12/2012	Robinson et al.	N/A	N/A
8428722	12/2012	Verhoef et al.	N/A	N/A
8429153	12/2012	Birdwell et al.	N/A	N/A
8439910	12/2012	Greep et al.	N/A	N/A
8444663	12/2012	Houser et al.	N/A	N/A
8452615	12/2012	Abri	N/A	N/A
8453906	12/2012	Huang et al.	N/A	N/A
8454506	12/2012	Rothman et al.	N/A	N/A
8461744	12/2012	Wiener et al.	N/A	N/A
8468030	12/2012	Stroup et al.	N/A	N/A
8469973	12/2012	Meade et al.	N/A	N/A
8472630	12/2012	Konrad et al.	N/A	N/A
8473066	12/2012	Aghassian et al.	N/A	N/A
D687146	12/2012	Juzkiw et al.	N/A	N/A
8476227	12/2012	Kaplan et al.	N/A	N/A
8478418	12/2012	Fahey	N/A	N/A
8489235	12/2012	Moll et al.	N/A	N/A
8499992	12/2012	Whitman et al.	N/A	N/A
8500728	12/2012	Newton et al.	N/A	N/A
8500756	12/2012	Papa et al.	N/A	N/A
8503759	12/2012	Greer et al.	N/A	N/A
8505801	12/2012	Ehrenfels et al.	N/A	N/A
8506478	12/2012	Mizuyoshi	N/A	N/A
8512325	12/2012	Mathonnet	N/A	N/A
8512365	12/2012	Wiener et al.	N/A	N/A
8515520	12/2012	Brunnett et al.	N/A	N/A
8517239	12/2012	Scheib et al.	N/A	N/A
8521331	12/2012	Itkowitz	N/A	N/A
8523043	12/2012	Ullrich et al.	N/A	N/A
8533475	12/2012	Frikart et al.	N/A	N/A
8540709	12/2012	Allen	N/A	N/A
8546996	12/2012	Messerly et al.	N/A	N/A
8554697	12/2012	Claus et al.	N/A	N/A
8560047	12/2012	Haider et al.	N/A	N/A
8561870	12/2012	Baxter, III et al.	N/A	N/A
8562598	12/2012	Falkenstein et al.	N/A	N/A
8566115	12/2012	Moore	N/A	N/A
8567393	12/2012	Hickle et al.	N/A	N/A
8568411	12/2012	Falkenstein et al.	N/A	N/A
8571598	12/2012	Valavi	N/A	N/A
8573459	12/2012	Smith et al.	N/A	N/A
8573465	12/2012	Shelton, IV	N/A	N/A
8574229	12/2012	Eder et al.	N/A	N/A
8585631	12/2012	Dacquay	N/A	N/A
8585694	12/2012	Amoah et al.	N/A	N/A
8590762	12/2012	Hess et al.	N/A	N/A
8591536	12/2012	Robertson	N/A	N/A
8595607	12/2012	Nekoomaram et al.	N/A	N/A
8596513	12/2012	Olson et al.	N/A	N/A
8596515	12/2012	Okoniewski	N/A	N/A
8604709	12/2012	Jalbout et al.	N/A	N/A
8608044	12/2012	Hueil et al.	N/A	N/A
8608045	12/2012	Smith et al.	N/A	N/A
8616431	12/2012	Timm et al.	N/A	N/A
8617155	12/2012	Johnson et al.	N/A	N/A
8620055	12/2012	Barratt et al.	N/A	N/A
8620473	12/2012	Diolaiti et al.	N/A	N/A
8622275	12/2013	Baxter, III et al.	N/A	N/A
8623027	12/2013	Price et al.	N/A	N/A
8627483	12/2013	Rachlin et al.	N/A	N/A
8627993	12/2013	Smith et al.	N/A	N/A
8627995	12/2013	Smith et al.	N/A	N/A

8628518	12/2013	Blumenkranz et al.	N/A	N/A
8628545	12/2013	Cabrera et al.	N/A	N/A
8631987	12/2013	Shelton, IV et al.	N/A	N/A
8632525	12/2013	Kerr et al.	N/A	N/A
8636190	12/2013	Zemlok et al.	N/A	N/A
8636736	12/2013	Yates et al.	N/A	N/A
8641621	12/2013	Razzaque et al.	N/A	N/A
8652086	12/2013	Gerg et al.	N/A	N/A
8652121	12/2013	Quick et al.	N/A	N/A
8652128	12/2013	Ward	N/A	N/A
8657176	12/2013	Shelton, IV et al.	N/A	N/A
8657177	12/2013	Scirica et al.	N/A	N/A
8663220	12/2013	Wiener et al.	N/A	N/A
8663222	12/2013	Anderson et al.	N/A	N/A
8666544	12/2013	Moll et al.	N/A	N/A
8679114	12/2013	Chapman et al.	N/A	N/A
8682049	12/2013	Zhao et al.	N/A	N/A
8682489	12/2013	Itkowitz et al.	N/A	N/A
8685056	12/2013	Evans et al.	N/A	N/A
8688188	12/2013	Heller et al.	N/A	N/A
8690864	12/2013	Hoarau	N/A	N/A
8701962	12/2013	Kostrzewski	N/A	N/A
8708213	12/2013	Shelton, IV et al.	N/A	N/A
D704839	12/2013	Juzkiw et al.	N/A	N/A
8719061	12/2013	Birchall	N/A	N/A
8720766	12/2013	Hess et al.	N/A	N/A
8733613	12/2013	Huitema et al.	N/A	N/A
8740840	12/2013	Foley et al.	N/A	N/A
8740866	12/2013	Reasoner et al.	N/A	N/A
8747238	12/2013	Shelton, IV et al.	N/A	N/A
8752749	12/2013	Moore et al.	N/A	N/A
8757465	12/2013	Woodard, Jr. et al.	N/A	N/A
8761717	12/2013	Buchheit	N/A	N/A
8763879	12/2013	Shelton, IV et al.	N/A	N/A
8768251	12/2013	Claus et al.	N/A	N/A
8771270	12/2013	Burbank	N/A	N/A
8775196	12/2013	Simpson et al.	N/A	N/A
8779648	12/2013	Giordano et al.	N/A	N/A
8790253	12/2013	Sunagawa et al.	N/A	N/A
8794497	12/2013	Zingman	N/A	N/A
8795001	12/2013	Lam et al.	N/A	N/A
8799008	12/2013	Johnson et al.	N/A	N/A
8799009	12/2013	Mellin et al.	N/A	N/A
8800838	12/2013	Shelton, IV	N/A	N/A
8801703	12/2013	Gregg et al.	N/A	N/A
8814996	12/2013	Giurgiutiu et al.	N/A	N/A
8818556	12/2013	Sanchez et al.	N/A	N/A
8819581	12/2013	Nakamura et al.	N/A	N/A
8820603	12/2013	Shelton, IV et al.	N/A	N/A
8820607	12/2013	Marczyk	N/A	N/A
8820608	12/2013	Miyamoto	N/A	N/A
8827134	12/2013	Viola et al.	N/A	N/A
8827136	12/2013	Hessler	N/A	N/A
8840003	12/2013	Morgan et al.	N/A	N/A
D716333	12/2013	Chotin et al.	N/A	N/A
8851354	12/2013	Swensgard et al.	N/A	N/A
8852174	12/2013	Burbank	N/A	N/A
8864747	12/2013	Merchant et al.	N/A	N/A
8875973	12/2013	Whitman	N/A	N/A
8876857	12/2013	Burbank	N/A	N/A
8882662	12/2013	Charles	N/A	N/A
8885032	12/2013	Igarashi et al.	N/A	N/A
8886790	12/2013	Harrang et al.	N/A	N/A
8893946	12/2013	Boudreaux et al.	N/A	N/A
8893949	12/2013	Shelton, IV et al.	N/A	N/A
8899479	12/2013	Cappuzzo et al.	N/A	N/A
8905977	12/2013	Shelton et al.	N/A	N/A
8912746	12/2013	Reid et al.	N/A	N/A
8914098	12/2013	Brennan et al.	N/A	N/A
8917513	12/2013	Hazzard	N/A	N/A
8918207	12/2013	Prisco	N/A	N/A
8920186	12/2013	Shishikura	N/A	N/A
8920414	12/2013	Stone et al.	N/A	N/A
8920433	12/2013	Barrier et al.	N/A	N/A
8930203	12/2014	Kiaie et al.	N/A	N/A

8930214	12/2014	Woolford	N/A	N/A
8931679	12/2014	Kostrzewski	N/A	N/A
8934684	12/2014	Mohamed	N/A	N/A
8936614	12/2014	Allen, IV	N/A	N/A
8945095	12/2014	Blumenkranz et al.	N/A	N/A
8945163	12/2014	Voegelé et al.	N/A	N/A
8955732	12/2014	Zemlok et al.	N/A	N/A
8956581	12/2014	Rosenbaum et al.	N/A	N/A
8960519	12/2014	Whitman et al.	N/A	N/A
8960520	12/2014	McCuen	N/A	N/A
8962062	12/2014	Podhajsky et al.	N/A	N/A
8967443	12/2014	McCuen	N/A	N/A
8967455	12/2014	Zhou	N/A	N/A
8968276	12/2014	Zemlok et al.	N/A	N/A
8968296	12/2014	McPherson	N/A	N/A
8968309	12/2014	Roy et al.	N/A	N/A
8968312	12/2014	Marczyk et al.	N/A	N/A
8968337	12/2014	Whitfield et al.	N/A	N/A
8968358	12/2014	Reschke	N/A	N/A
8974429	12/2014	Gordon et al.	N/A	N/A
8979890	12/2014	Boudreaux	N/A	N/A
8986288	12/2014	Konishi	N/A	N/A
8986302	12/2014	Aldridge et al.	N/A	N/A
8989903	12/2014	Weir et al.	N/A	N/A
8991678	12/2014	Wellman et al.	N/A	N/A
8992565	12/2014	Brisson et al.	N/A	N/A
8998797	12/2014	Omori	N/A	N/A
9002518	12/2014	Manzo et al.	N/A	N/A
9010608	12/2014	Casasanta, Jr. et al.	N/A	N/A
9010611	12/2014	Ross et al.	N/A	N/A
9011366	12/2014	Dean et al.	N/A	N/A
9011427	12/2014	Price et al.	N/A	N/A
9016539	12/2014	Kostrzewski et al.	N/A	N/A
9017326	12/2014	DiNardo et al.	N/A	N/A
9020240	12/2014	Pettersson et al.	N/A	N/A
D729267	12/2014	Yoo et al.	N/A	N/A
9023032	12/2014	Robinson	N/A	N/A
9023071	12/2014	Miller et al.	N/A	N/A
9027431	12/2014	Tang et al.	N/A	N/A
9028494	12/2014	Shelton, IV et al.	N/A	N/A
9035568	12/2014	Ganton et al.	N/A	N/A
9038882	12/2014	Racenet et al.	N/A	N/A
9043027	12/2014	Durant et al.	N/A	N/A
9044227	12/2014	Shelton, IV et al.	N/A	N/A
9044244	12/2014	Ludwin et al.	N/A	N/A
9044261	12/2014	Houser	N/A	N/A
9050063	12/2014	Roe et al.	N/A	N/A
9050083	12/2014	Yates et al.	N/A	N/A
9050120	12/2014	Swarup et al.	N/A	N/A
9052809	12/2014	Vesto	N/A	N/A
9055035	12/2014	Porsch et al.	N/A	N/A
9055870	12/2014	Meador et al.	N/A	N/A
9060770	12/2014	Shelton, IV et al.	N/A	N/A
9060775	12/2014	Wiener et al.	N/A	N/A
9066650	12/2014	Sekiguchi	N/A	N/A
9072523	12/2014	Houser et al.	N/A	N/A
9072535	12/2014	Shelton, IV et al.	N/A	N/A
9072536	12/2014	Shelton, IV et al.	N/A	N/A
9078653	12/2014	Leimbach et al.	N/A	N/A
9078727	12/2014	Miller	N/A	N/A
9084606	12/2014	Greep	N/A	N/A
9089360	12/2014	Messerly et al.	N/A	N/A
9095362	12/2014	Dachs, II et al.	N/A	N/A
9095367	12/2014	Olson et al.	N/A	N/A
9099863	12/2014	Smith et al.	N/A	N/A
9101358	12/2014	Kerr et al.	N/A	N/A
9101359	12/2014	Smith et al.	N/A	N/A
9101374	12/2014	Hoch et al.	N/A	N/A
9106270	12/2014	Puterbaugh et al.	N/A	N/A
9107573	12/2014	Birnkrant	N/A	N/A
9107662	12/2014	Kostrzewski	N/A	N/A
9107684	12/2014	Ma	N/A	N/A
9107688	12/2014	Kimball et al.	N/A	N/A
9107689	12/2014	Robertson et al.	N/A	N/A
9107694	12/2014	Hendriks et al.	N/A	N/A

9111548	12/2014	Nandy et al.	N/A	N/A
9113880	12/2014	Zemlok et al.	N/A	N/A
9114494	12/2014	Mah	N/A	N/A
9116597	12/2014	Gulasky	N/A	N/A
9119617	12/2014	Souls et al.	N/A	N/A
9119655	12/2014	Bowling et al.	N/A	N/A
9119657	12/2014	Shelton, IV et al.	N/A	N/A
9123155	12/2014	Cunningham et al.	N/A	N/A
9125644	12/2014	Lane et al.	N/A	N/A
9129054	12/2014	Nawana et al.	N/A	N/A
9137254	12/2014	Bilbrey et al.	N/A	N/A
9138129	12/2014	Diolaiti	N/A	N/A
9138225	12/2014	Huang et al.	N/A	N/A
9149322	12/2014	Knowlton	N/A	N/A
9155503	12/2014	Cadwell	N/A	N/A
9160853	12/2014	Daddi et al.	N/A	N/A
9161803	12/2014	Yates et al.	N/A	N/A
9168054	12/2014	Turner et al.	N/A	N/A
9168091	12/2014	Janssen et al.	N/A	N/A
9168104	12/2014	Dein	N/A	N/A
9179912	12/2014	Yates et al.	N/A	N/A
9183723	12/2014	Sherman et al.	N/A	N/A
9186143	12/2014	Timm et al.	N/A	N/A
9192375	12/2014	Skinlo et al.	N/A	N/A
9192447	12/2014	Choi et al.	N/A	N/A
9192707	12/2014	Gerber et al.	N/A	N/A
9198711	12/2014	Joseph	N/A	N/A
9202078	12/2014	Abuelsaad et al.	N/A	N/A
9204830	12/2014	Zand et al.	N/A	N/A
9204879	12/2014	Shelton, IV	N/A	N/A
9204995	12/2014	Scheller et al.	N/A	N/A
9211120	12/2014	Scheib et al.	N/A	N/A
9216062	12/2014	Duque et al.	N/A	N/A
9218053	12/2014	Komuro et al.	N/A	N/A
9220502	12/2014	Zemlok et al.	N/A	N/A
9226689	12/2015	Jacobsen et al.	N/A	N/A
9226751	12/2015	Shelton, IV et al.	N/A	N/A
9226766	12/2015	Aldridge et al.	N/A	N/A
9226767	12/2015	Stulen et al.	N/A	N/A
9226791	12/2015	McCarthy et al.	N/A	N/A
9232883	12/2015	Ozawa et al.	N/A	N/A
9237891	12/2015	Shelton, IV	N/A	N/A
9237921	12/2015	Messerly et al.	N/A	N/A
9241728	12/2015	Price et al.	N/A	N/A
9241730	12/2015	Babaev	N/A	N/A
9241731	12/2015	Boudreaux et al.	N/A	N/A
9247996	12/2015	Merana et al.	N/A	N/A
9250172	12/2015	Harris et al.	N/A	N/A
9255907	12/2015	Heanue et al.	N/A	N/A
9259282	12/2015	Azizian et al.	N/A	N/A
9265429	12/2015	St. Pierre et al.	N/A	N/A
9265585	12/2015	Wingardner et al.	N/A	N/A
9265959	12/2015	Drew et al.	N/A	N/A
9272406	12/2015	Aronhalt et al.	N/A	N/A
9277956	12/2015	Zhang	N/A	N/A
9277961	12/2015	Panescu et al.	N/A	N/A
9277969	12/2015	Brannan et al.	N/A	N/A
9280884	12/2015	Schultz et al.	N/A	N/A
9282962	12/2015	Schmid et al.	N/A	N/A
9282974	12/2015	Shelton, IV	N/A	N/A
9283045	12/2015	Rhee et al.	N/A	N/A
9283054	12/2015	Morgan et al.	N/A	N/A
9289211	12/2015	Williams et al.	N/A	N/A
9289212	12/2015	Shelton, IV et al.	N/A	N/A
9295514	12/2015	Shelton, IV et al.	N/A	N/A
9299138	12/2015	Zellner et al.	N/A	N/A
9301691	12/2015	Hufnagel et al.	N/A	N/A
9301753	12/2015	Aldridge et al.	N/A	N/A
9301755	12/2015	Shelton, IV et al.	N/A	N/A
9301759	12/2015	Spivey et al.	N/A	N/A
9301810	12/2015	Amiri et al.	N/A	N/A
9302213	12/2015	Manahan et al.	N/A	N/A
9307894	12/2015	von Grunberg et al.	N/A	N/A
9307914	12/2015	Fahey	N/A	N/A
9307986	12/2015	Hall et al.	N/A	N/A

9314246	12/2015	Shelton, IV et al.	N/A	N/A
9314308	12/2015	Parihar et al.	N/A	N/A
9320563	12/2015	Brustad et al.	N/A	N/A
9325732	12/2015	Stickle et al.	N/A	N/A
9326767	12/2015	Koch et al.	N/A	N/A
9326770	12/2015	Shelton, IV et al.	N/A	N/A
9331422	12/2015	Nazzaro et al.	N/A	N/A
9332987	12/2015	Leimbach et al.	N/A	N/A
9333042	12/2015	Diolaiti et al.	N/A	N/A
9336385	12/2015	Spencer et al.	N/A	N/A
9341704	12/2015	Picard et al.	N/A	N/A
9345481	12/2015	Hall et al.	N/A	N/A
9345490	12/2015	Ippisch	N/A	N/A
9345544	12/2015	Hourtash et al.	N/A	N/A
9345546	12/2015	Toth et al.	N/A	N/A
9345900	12/2015	Wu et al.	N/A	N/A
9351726	12/2015	Leimbach et al.	N/A	N/A
9351727	12/2015	Leimbach et al.	N/A	N/A
9358003	12/2015	Hall et al.	N/A	N/A
9358685	12/2015	Meier et al.	N/A	N/A
9360449	12/2015	Duric	N/A	N/A
9364200	12/2015	Whitman et al.	N/A	N/A
9364230	12/2015	Shelton, IV et al.	N/A	N/A
9364231	12/2015	Wenchell	N/A	N/A
9364249	12/2015	Kimball et al.	N/A	N/A
9364294	12/2015	Razzaque et al.	N/A	N/A
9370400	12/2015	Parihar	N/A	N/A
9375282	12/2015	Nau, Jr. et al.	N/A	N/A
9375539	12/2015	Stearns et al.	N/A	N/A
9381003	12/2015	Todor et al.	N/A	N/A
9381058	12/2015	Houser et al.	N/A	N/A
9386984	12/2015	Aronhalt et al.	N/A	N/A
9386988	12/2015	Baxter, III et al.	N/A	N/A
9387295	12/2015	Mastri et al.	N/A	N/A
9393017	12/2015	Flanagan et al.	N/A	N/A
9393037	12/2015	Olson et al.	N/A	N/A
9398905	12/2015	Martin	N/A	N/A
9398911	12/2015	Auld	N/A	N/A
9402629	12/2015	Ehrenfels et al.	N/A	N/A
9404868	12/2015	Yamanaka et al.	N/A	N/A
9414776	12/2015	Sillay et al.	N/A	N/A
9414940	12/2015	Stein et al.	N/A	N/A
9419018	12/2015	Sasagawa et al.	N/A	N/A
9421014	12/2015	Ingmanson et al.	N/A	N/A
9433470	12/2015	Choi	N/A	N/A
9439622	12/2015	Case et al.	N/A	N/A
9439668	12/2015	Timm et al.	N/A	N/A
9439736	12/2015	Olson	N/A	N/A
9445764	12/2015	Gross et al.	N/A	N/A
9445813	12/2015	Shelton, IV et al.	N/A	N/A
9450701	12/2015	Do et al.	N/A	N/A
9451949	12/2015	Gorek et al.	N/A	N/A
9451958	12/2015	Shelton, IV et al.	N/A	N/A
9463022	12/2015	Swayze et al.	N/A	N/A
9463646	12/2015	Payne et al.	N/A	N/A
9468438	12/2015	Baber et al.	N/A	N/A
9474565	12/2015	Shikhman et al.	N/A	N/A
D772252	12/2015	Myers et al.	N/A	N/A
9480492	12/2015	Aranyi et al.	N/A	N/A
9485475	12/2015	Speier et al.	N/A	N/A
9486271	12/2015	Dunning	N/A	N/A
9492146	12/2015	Kostrzewski et al.	N/A	N/A
9492237	12/2015	Kang et al.	N/A	N/A
9493807	12/2015	Little et al.	N/A	N/A
9498182	12/2015	Case et al.	N/A	N/A
9498215	12/2015	Duque et al.	N/A	N/A
9498219	12/2015	Moore et al.	N/A	N/A
9498231	12/2015	Haider et al.	N/A	N/A
9498291	12/2015	Balaji et al.	N/A	N/A
9509566	12/2015	Chu et al.	N/A	N/A
9516239	12/2015	Blanquart et al.	N/A	N/A
9519753	12/2015	Gerdeman et al.	N/A	N/A
9522003	12/2015	Weir et al.	N/A	N/A
9526407	12/2015	Hoeg et al.	N/A	N/A
9526499	12/2015	Kostrzewski et al.	N/A	N/A

9526580	12/2015	Humayun et al.	N/A	N/A
9526587	12/2015	Zhao et al.	N/A	N/A
9532827	12/2016	Morgan et al.	N/A	N/A
9532845	12/2016	Dossett et al.	N/A	N/A
9539007	12/2016	Dhakad et al.	N/A	N/A
9539020	12/2016	Conlon et al.	N/A	N/A
9542481	12/2016	Halter et al.	N/A	N/A
9546662	12/2016	Shener-Irmakoglu et al.	N/A	N/A
9549781	12/2016	He et al.	N/A	N/A
9554692	12/2016	Levy	N/A	N/A
9554794	12/2016	Baber et al.	N/A	N/A
9554854	12/2016	Yates et al.	N/A	N/A
9561038	12/2016	Shelton, IV et al.	N/A	N/A
9561045	12/2016	Hinman et al.	N/A	N/A
9561082	12/2016	Yen et al.	N/A	N/A
9561982	12/2016	Enicks et al.	N/A	N/A
9566708	12/2016	Kurnianto	N/A	N/A
9572592	12/2016	Price et al.	N/A	N/A
9579099	12/2016	Penna et al.	N/A	N/A
9579503	12/2016	McKinney et al.	N/A	N/A
9585657	12/2016	Shelton, IV et al.	N/A	N/A
9585658	12/2016	Shelton, IV	N/A	N/A
9592095	12/2016	Panescu et al.	N/A	N/A
9597081	12/2016	Swayze et al.	N/A	N/A
9600031	12/2016	Kaneko et al.	N/A	N/A
9600138	12/2016	Thomas et al.	N/A	N/A
9603024	12/2016	Wang et al.	N/A	N/A
9603277	12/2016	Morgan et al.	N/A	N/A
9603609	12/2016	Kawashima et al.	N/A	N/A
D783675	12/2016	Yagisawa et al.	N/A	N/A
D784270	12/2016	Bhattacharya	N/A	N/A
9610114	12/2016	Baxter, III et al.	N/A	N/A
9610412	12/2016	Zemlok et al.	N/A	N/A
9615877	12/2016	Tyrrell et al.	N/A	N/A
9622684	12/2016	Wybo	N/A	N/A
9622808	12/2016	Beller et al.	N/A	N/A
9628501	12/2016	Datta Ray et al.	N/A	N/A
9629560	12/2016	Joseph	N/A	N/A
9629623	12/2016	Lytle, IV et al.	N/A	N/A
9629628	12/2016	Aranyi	N/A	N/A
9629629	12/2016	Leimbach et al.	N/A	N/A
9630318	12/2016	Ibarz Gabardos et al.	N/A	N/A
9636096	12/2016	Heaton et al.	N/A	N/A
9636112	12/2016	Penna et al.	N/A	N/A
9636188	12/2016	Gattani et al.	N/A	N/A
9636239	12/2016	Durand et al.	N/A	N/A
9636825	12/2016	Penn et al.	N/A	N/A
9641596	12/2016	Unagami et al.	N/A	N/A
9641815	12/2016	Richardson et al.	N/A	N/A
9642620	12/2016	Baxter, III et al.	N/A	N/A
9643022	12/2016	Mashiach et al.	N/A	N/A
9649110	12/2016	Parihar et al.	N/A	N/A
9649111	12/2016	Shelton, IV et al.	N/A	N/A
9649126	12/2016	Robertson et al.	N/A	N/A
9649169	12/2016	Cinquin et al.	N/A	N/A
9652655	12/2016	Satish et al.	N/A	N/A
9655614	12/2016	Swensgard et al.	N/A	N/A
9655616	12/2016	Aranyi	N/A	N/A
9656092	12/2016	Golden	N/A	N/A
9662116	12/2016	Smith et al.	N/A	N/A
9662177	12/2016	Weir et al.	N/A	N/A
9668729	12/2016	Williams et al.	N/A	N/A
9668732	12/2016	Patel et al.	N/A	N/A
9668765	12/2016	Grace et al.	N/A	N/A
9671860	12/2016	Ogawa et al.	N/A	N/A
9675264	12/2016	Acquista et al.	N/A	N/A
9675354	12/2016	Weir et al.	N/A	N/A
9681870	12/2016	Baxter, III et al.	N/A	N/A
9686306	12/2016	Chizeck et al.	N/A	N/A
9687230	12/2016	Leimbach et al.	N/A	N/A
9690362	12/2016	Leimbach et al.	N/A	N/A
9700292	12/2016	Nawana et al.	N/A	N/A
9700309	12/2016	Jaworek et al.	N/A	N/A
9700312	12/2016	Kostrzewski et al.	N/A	N/A
9700320	12/2016	Dinardo et al.	N/A	N/A

9706993	12/2016	Hessler et al.	N/A	N/A
9710214	12/2016	Lin et al.	N/A	N/A
9710644	12/2016	Reybok et al.	N/A	N/A
9713424	12/2016	Spaide	N/A	N/A
9713503	12/2016	Goldschmidt	N/A	N/A
9717141	12/2016	Tegg	N/A	N/A
9717498	12/2016	Aranyi et al.	N/A	N/A
9717525	12/2016	Ahluwalia et al.	N/A	N/A
9717548	12/2016	Couture	N/A	N/A
9724094	12/2016	Baber et al.	N/A	N/A
9724100	12/2016	Scheib et al.	N/A	N/A
9724118	12/2016	Schulte et al.	N/A	N/A
9733663	12/2016	Leimbach et al.	N/A	N/A
9737301	12/2016	Baber et al.	N/A	N/A
9737310	12/2016	Whitfield et al.	N/A	N/A
9737335	12/2016	Butler et al.	N/A	N/A
9737355	12/2016	Yates et al.	N/A	N/A
9737371	12/2016	Romo et al.	N/A	N/A
9740826	12/2016	Raghavan et al.	N/A	N/A
9743016	12/2016	Nestares et al.	N/A	N/A
9743929	12/2016	Leimbach et al.	N/A	N/A
9743946	12/2016	Faller et al.	N/A	N/A
9743947	12/2016	Price et al.	N/A	N/A
9750499	12/2016	Leimbach et al.	N/A	N/A
9750500	12/2016	Malkowski	N/A	N/A
9750522	12/2016	Scheib et al.	N/A	N/A
9750523	12/2016	Tsubuku	N/A	N/A
9750560	12/2016	Ballakur et al.	N/A	N/A
9750563	12/2016	Shikhman et al.	N/A	N/A
9753135	12/2016	Bosch	N/A	N/A
9753568	12/2016	McMillen	N/A	N/A
9757126	12/2016	Cappola	N/A	N/A
9757128	12/2016	Baber et al.	N/A	N/A
9757142	12/2016	Shimizu	N/A	N/A
9757152	12/2016	Ogilvie et al.	N/A	N/A
9763741	12/2016	Alvarez et al.	N/A	N/A
9764164	12/2016	Wiener et al.	N/A	N/A
9770541	12/2016	Carr et al.	N/A	N/A
9775611	12/2016	Kostrzewski	N/A	N/A
9777913	12/2016	Talbert et al.	N/A	N/A
9782164	12/2016	Mumaw et al.	N/A	N/A
9782169	12/2016	Kimsey et al.	N/A	N/A
9782212	12/2016	Wham et al.	N/A	N/A
9782214	12/2016	Houser et al.	N/A	N/A
9788835	12/2016	Morgan et al.	N/A	N/A
9788836	12/2016	Overmyer et al.	N/A	N/A
9788851	12/2016	Dannaher et al.	N/A	N/A
9788902	12/2016	Inoue et al.	N/A	N/A
9788907	12/2016	Alvi et al.	N/A	N/A
9795436	12/2016	Yates et al.	N/A	N/A
9797486	12/2016	Zergiebel et al.	N/A	N/A
9801531	12/2016	Morita et al.	N/A	N/A
9801626	12/2016	Parihar et al.	N/A	N/A
9801627	12/2016	Harris et al.	N/A	N/A
9801679	12/2016	Trees et al.	N/A	N/A
9802033	12/2016	Hibner et al.	N/A	N/A
9804618	12/2016	Leimbach et al.	N/A	N/A
9805472	12/2016	Chou et al.	N/A	N/A
9808244	12/2016	Leimbach et al.	N/A	N/A
9808245	12/2016	Richard et al.	N/A	N/A
9808246	12/2016	Shelton, IV et al.	N/A	N/A
9808248	12/2016	Hoffman	N/A	N/A
9808249	12/2016	Shelton, IV	N/A	N/A
9814457	12/2016	Martin et al.	N/A	N/A
9814460	12/2016	Kimsey et al.	N/A	N/A
9814462	12/2016	Woodard, Jr. et al.	N/A	N/A
9814463	12/2016	Williams et al.	N/A	N/A
9820699	12/2016	Bingley et al.	N/A	N/A
9820738	12/2016	Lytle, IV et al.	N/A	N/A
9820741	12/2016	Kostrzewski	N/A	N/A
9820768	12/2016	Gee et al.	N/A	N/A
9826976	12/2016	Parihar et al.	N/A	N/A
9826977	12/2016	Leimbach et al.	N/A	N/A
9827054	12/2016	Richmond et al.	N/A	N/A
9827059	12/2016	Robinson et al.	N/A	N/A

9830424	12/2016	Dixon et al.	N/A	N/A
9833241	12/2016	Huitema et al.	N/A	N/A
9833254	12/2016	Barral et al.	N/A	N/A
9839419	12/2016	Deck et al.	N/A	N/A
9839424	12/2016	Zergiebel et al.	N/A	N/A
9839428	12/2016	Baxter, III et al.	N/A	N/A
9839467	12/2016	Harper et al.	N/A	N/A
9839470	12/2016	Gilbert et al.	N/A	N/A
9839487	12/2016	Dachs, II	N/A	N/A
9844321	12/2016	Ekvall et al.	N/A	N/A
9844368	12/2016	Boudreaux et al.	N/A	N/A
9844369	12/2016	Huitema et al.	N/A	N/A
9844374	12/2016	Lytle, IV et al.	N/A	N/A
9844375	12/2016	Overmyer et al.	N/A	N/A
9844376	12/2016	Baxter, III et al.	N/A	N/A
9844379	12/2016	Shelton, IV et al.	N/A	N/A
9848058	12/2016	Johnson et al.	N/A	N/A
9848877	12/2016	Shelton, IV et al.	N/A	N/A
9861354	12/2017	Saliman et al.	N/A	N/A
9861363	12/2017	Chen et al.	N/A	N/A
9861428	12/2017	Trees et al.	N/A	N/A
9864839	12/2017	Baym et al.	N/A	N/A
9867612	12/2017	Parihar et al.	N/A	N/A
9867651	12/2017	Wham	N/A	N/A
9867670	12/2017	Brannan et al.	N/A	N/A
9867914	12/2017	Bonano et al.	N/A	N/A
9872609	12/2017	Levy	N/A	N/A
9872683	12/2017	Hopkins et al.	N/A	N/A
9877718	12/2017	Weir et al.	N/A	N/A
9877721	12/2017	Schellin et al.	N/A	N/A
9883860	12/2017	Leimbach	N/A	N/A
9888864	12/2017	Rondoni et al.	N/A	N/A
9888914	12/2017	Martin et al.	N/A	N/A
9888919	12/2017	Leimbach et al.	N/A	N/A
9888921	12/2017	Williams et al.	N/A	N/A
9888975	12/2017	Auld	N/A	N/A
9895148	12/2017	Shelton, IV et al.	N/A	N/A
9900787	12/2017	Ou	N/A	N/A
9901342	12/2017	Shelton, IV et al.	N/A	N/A
9901406	12/2017	State et al.	N/A	N/A
9905000	12/2017	Chou et al.	N/A	N/A
9907196	12/2017	Susini et al.	N/A	N/A
9907550	12/2017	Sniffin et al.	N/A	N/A
9913642	12/2017	Leimbach et al.	N/A	N/A
9913645	12/2017	Zerkle et al.	N/A	N/A
9918326	12/2017	Gilson et al.	N/A	N/A
9918730	12/2017	Trees et al.	N/A	N/A
9918778	12/2017	Walberg et al.	N/A	N/A
9918788	12/2017	Paul et al.	N/A	N/A
9922304	12/2017	DeBusk et al.	N/A	N/A
9924941	12/2017	Burbank	N/A	N/A
9924944	12/2017	Shelton, IV et al.	N/A	N/A
9924961	12/2017	Shelton, IV et al.	N/A	N/A
9931040	12/2017	Homyk et al.	N/A	N/A
9931118	12/2017	Shelton, IV et al.	N/A	N/A
9931124	12/2017	Gokharu	N/A	N/A
9936863	12/2017	Tesar	N/A	N/A
9936942	12/2017	Chin et al.	N/A	N/A
9936955	12/2017	Miller et al.	N/A	N/A
9936961	12/2017	Chien et al.	N/A	N/A
9937012	12/2017	Hares et al.	N/A	N/A
9937014	12/2017	Bowling et al.	N/A	N/A
9937626	12/2017	Rockrohr	N/A	N/A
9938972	12/2017	Walley	N/A	N/A
9943230	12/2017	Kaku et al.	N/A	N/A
9943309	12/2017	Shelton, IV et al.	N/A	N/A
9943312	12/2017	Posada et al.	N/A	N/A
9943377	12/2017	Yates et al.	N/A	N/A
9943379	12/2017	Gregg, II et al.	N/A	N/A
9943918	12/2017	Grogan et al.	N/A	N/A
9943964	12/2017	Hares	N/A	N/A
9949785	12/2017	Price et al.	N/A	N/A
9962157	12/2017	Sapre	N/A	N/A
9968355	12/2017	Shelton, IV et al.	N/A	N/A
9974595	12/2017	Anderson et al.	N/A	N/A

9980140	12/2017	Spencer et al.	N/A	N/A
9980769	12/2017	Trees et al.	N/A	N/A
9980778	12/2017	Ohline et al.	N/A	N/A
9987000	12/2017	Shelton, IV et al.	N/A	N/A
9987068	12/2017	Anderson et al.	N/A	N/A
9987072	12/2017	McPherson	N/A	N/A
9990856	12/2017	Kuchenbecker et al.	N/A	N/A
9993248	12/2017	Shelton, IV et al.	N/A	N/A
9993258	12/2017	Shelton, IV et al.	N/A	N/A
9993305	12/2017	Andersson	N/A	N/A
10004491	12/2017	Martin et al.	N/A	N/A
10004497	12/2017	Overmyer et al.	N/A	N/A
10004500	12/2017	Shelton, IV et al.	N/A	N/A
10004501	12/2017	Shelton, IV et al.	N/A	N/A
10004527	12/2017	Gee et al.	N/A	N/A
10004557	12/2017	Gross	N/A	N/A
D822206	12/2017	Shelton, IV et al.	N/A	N/A
10010322	12/2017	Shelton, IV et al.	N/A	N/A
10010324	12/2017	Huitema et al.	N/A	N/A
10013049	12/2017	Leimbach et al.	N/A	N/A
10016199	12/2017	Baber et al.	N/A	N/A
10021318	12/2017	Hugosson et al.	N/A	N/A
10022090	12/2017	Whitman	N/A	N/A
10022120	12/2017	Martin et al.	N/A	N/A
10022391	12/2017	Ruderman Chen et al.	N/A	N/A
10022568	12/2017	Messerly et al.	N/A	N/A
10028402	12/2017	Walker	N/A	N/A
10028744	12/2017	Shelton, IV et al.	N/A	N/A
10028761	12/2017	Leimbach et al.	N/A	N/A
10028788	12/2017	Kang	N/A	N/A
10034704	12/2017	Asher et al.	N/A	N/A
10037641	12/2017	Hyde et al.	N/A	N/A
10037715	12/2017	Toly et al.	N/A	N/A
D826405	12/2017	Shelton, IV et al.	N/A	N/A
10039546	12/2017	Williams et al.	N/A	N/A
10039564	12/2017	Hibner et al.	N/A	N/A
10039565	12/2017	Vezzu	N/A	N/A
10039589	12/2017	Virshek et al.	N/A	N/A
10041822	12/2017	Zemlok	N/A	N/A
10044791	12/2017	Kamen et al.	N/A	N/A
10045704	12/2017	Fagin et al.	N/A	N/A
10045776	12/2017	Shelton, IV et al.	N/A	N/A
10045779	12/2017	Savage et al.	N/A	N/A
10045781	12/2017	Cropper et al.	N/A	N/A
10045782	12/2017	Murthy Aravalli	N/A	N/A
10045813	12/2017	Mueller	N/A	N/A
10048379	12/2017	Markendorf et al.	N/A	N/A
10052044	12/2017	Shelton, IV et al.	N/A	N/A
10052102	12/2017	Baxter, III et al.	N/A	N/A
10052104	12/2017	Shelton, IV et al.	N/A	N/A
10054441	12/2017	Schorr et al.	N/A	N/A
10058393	12/2017	Bonutti et al.	N/A	N/A
10069633	12/2017	Gulati et al.	N/A	N/A
10076326	12/2017	Yates et al.	N/A	N/A
10080618	12/2017	Marshall et al.	N/A	N/A
10084833	12/2017	McDonnell et al.	N/A	N/A
D831209	12/2017	Huitema et al.	N/A	N/A
10085748	12/2017	Morgan et al.	N/A	N/A
10085749	12/2017	Cappola et al.	N/A	N/A
10092355	12/2017	Hannaford et al.	N/A	N/A
10095942	12/2017	Mentese et al.	N/A	N/A
10097578	12/2017	Baldonado et al.	N/A	N/A
10098527	12/2017	Weisenburgh, II et al.	N/A	N/A
10098635	12/2017	Burbank	N/A	N/A
10098642	12/2017	Baxter, III et al.	N/A	N/A
10098705	12/2017	Brisson et al.	N/A	N/A
10102926	12/2017	Leonardi	N/A	N/A
10105140	12/2017	Malinouskas et al.	N/A	N/A
10105142	12/2017	Baxter, III et al.	N/A	N/A
10105470	12/2017	Reasoner et al.	N/A	N/A
10111658	12/2017	Chowaniec et al.	N/A	N/A
10111665	12/2017	Aranyi et al.	N/A	N/A
10111679	12/2017	Baber et al.	N/A	N/A
10111703	12/2017	Cosman, Jr. et al.	N/A	N/A
D834541	12/2017	You et al.	N/A	N/A

10117649	12/2017	Baxter et al.	N/A	N/A
10117651	12/2017	Whitman et al.	N/A	N/A
10117702	12/2017	Danziger et al.	N/A	N/A
10118119	12/2017	Sappok et al.	N/A	N/A
10130359	12/2017	Hess et al.	N/A	N/A
10130360	12/2017	Olson et al.	N/A	N/A
10130361	12/2017	Yates et al.	N/A	N/A
10130367	12/2017	Cappola et al.	N/A	N/A
10130432	12/2017	Auld et al.	N/A	N/A
10133248	12/2017	Fitzsimmons et al.	N/A	N/A
10135242	12/2017	Baber et al.	N/A	N/A
10136246	12/2017	Yamada	N/A	N/A
10136887	12/2017	Shelton, IV et al.	N/A	N/A
10136891	12/2017	Shelton, IV et al.	N/A	N/A
10136949	12/2017	Felder et al.	N/A	N/A
10136954	12/2017	Johnson et al.	N/A	N/A
10137245	12/2017	Melker et al.	N/A	N/A
10143526	12/2017	Walker et al.	N/A	N/A
10143948	12/2017	Bonifas et al.	N/A	N/A
10147148	12/2017	Wu et al.	N/A	N/A
10149680	12/2017	Parihar et al.	N/A	N/A
10152789	12/2017	Carnes et al.	N/A	N/A
10154841	12/2017	Weaner et al.	N/A	N/A
10159044	12/2017	Hrabak	N/A	N/A
10159481	12/2017	Whitman et al.	N/A	N/A
10159483	12/2017	Beckman et al.	N/A	N/A
10164466	12/2017	Calderoni	N/A	N/A
10166025	12/2018	Leimbach et al.	N/A	N/A
10166061	12/2018	Berry et al.	N/A	N/A
10169862	12/2018	Andre et al.	N/A	N/A
10172618	12/2018	Shelton, IV et al.	N/A	N/A
10172687	12/2018	Garbus et al.	N/A	N/A
10175096	12/2018	Dickerson	N/A	N/A
10175127	12/2018	Collins et al.	N/A	N/A
10178992	12/2018	Wise et al.	N/A	N/A
10179413	12/2018	Rockrohr	N/A	N/A
10180463	12/2018	Beckman et al.	N/A	N/A
10182814	12/2018	Okoniewski	N/A	N/A
10182816	12/2018	Shelton, IV et al.	N/A	N/A
10182818	12/2018	Hensel et al.	N/A	N/A
10188385	12/2018	Kerr et al.	N/A	N/A
10189157	12/2018	Schlegel et al.	N/A	N/A
10190888	12/2018	Hryb et al.	N/A	N/A
10194891	12/2018	Jeong et al.	N/A	N/A
10194907	12/2018	Marczyk et al.	N/A	N/A
10194913	12/2018	Nalagatla et al.	N/A	N/A
10194972	12/2018	Yates et al.	N/A	N/A
10197803	12/2018	Badiali et al.	N/A	N/A
10198965	12/2018	Hart	N/A	N/A
10201311	12/2018	Chou et al.	N/A	N/A
10201349	12/2018	Leimbach et al.	N/A	N/A
10201364	12/2018	Leimbach et al.	N/A	N/A
10201365	12/2018	Boudreaux et al.	N/A	N/A
10205708	12/2018	Fletcher et al.	N/A	N/A
10206605	12/2018	Shelton, IV et al.	N/A	N/A
10206752	12/2018	Hares et al.	N/A	N/A
10213201	12/2018	Shelton, IV et al.	N/A	N/A
10213203	12/2018	Swayze et al.	N/A	N/A
10213266	12/2018	Zemlok et al.	N/A	N/A
10213268	12/2018	Dachs, II	N/A	N/A
10219491	12/2018	Stiles, Jr. et al.	N/A	N/A
10220522	12/2018	Rockrohr	N/A	N/A
10222750	12/2018	Bang et al.	N/A	N/A
10226249	12/2018	Jaworek et al.	N/A	N/A
10226250	12/2018	Beckman et al.	N/A	N/A
10226254	12/2018	Cabrera et al.	N/A	N/A
10226302	12/2018	Lacal et al.	N/A	N/A
10231634	12/2018	Zand et al.	N/A	N/A
10231733	12/2018	Ehrenfels et al.	N/A	N/A
10231775	12/2018	Shelton, IV et al.	N/A	N/A
10238413	12/2018	Hibner et al.	N/A	N/A
10245027	12/2018	Shelton, IV et al.	N/A	N/A
10245028	12/2018	Shelton, IV et al.	N/A	N/A
10245029	12/2018	Hunter et al.	N/A	N/A
10245030	12/2018	Hunter et al.	N/A	N/A

10245033	12/2018	Overmyer et al.	N/A	N/A
10245037	12/2018	Conklin et al.	N/A	N/A
10245038	12/2018	Hopkins et al.	N/A	N/A
10245040	12/2018	Milliman	N/A	N/A
10251661	12/2018	Collings et al.	N/A	N/A
10251725	12/2018	Valentine et al.	N/A	N/A
10258331	12/2018	Shelton, IV et al.	N/A	N/A
10258359	12/2018	Kapadia	N/A	N/A
10258362	12/2018	Conlon	N/A	N/A
10258363	12/2018	Worrell et al.	N/A	N/A
10258415	12/2018	Harrah et al.	N/A	N/A
10258418	12/2018	Shelton, IV et al.	N/A	N/A
10258425	12/2018	Mustufa et al.	N/A	N/A
10263171	12/2018	Wiener et al.	N/A	N/A
10265004	12/2018	Yamaguchi et al.	N/A	N/A
10265035	12/2018	Fehre et al.	N/A	N/A
10265066	12/2018	Measamer et al.	N/A	N/A
10265068	12/2018	Harris et al.	N/A	N/A
10265072	12/2018	Shelton, IV et al.	N/A	N/A
10265090	12/2018	Ingmanson et al.	N/A	N/A
10265130	12/2018	Hess et al.	N/A	N/A
10271840	12/2018	Sapre	N/A	N/A
10271844	12/2018	Valentine et al.	N/A	N/A
10271850	12/2018	Williams	N/A	N/A
10271851	12/2018	Shelton, IV et al.	N/A	N/A
D847989	12/2018	Shelton, IV et al.	N/A	N/A
10278698	12/2018	Racenet	N/A	N/A
10278778	12/2018	State et al.	N/A	N/A
10283220	12/2018	Azizian et al.	N/A	N/A
10285694	12/2018	Viola et al.	N/A	N/A
10285698	12/2018	Cappola et al.	N/A	N/A
10285700	12/2018	Scheib	N/A	N/A
10285705	12/2018	Shelton, IV et al.	N/A	N/A
10292610	12/2018	Srivastava	N/A	N/A
10292704	12/2018	Harris et al.	N/A	N/A
10292707	12/2018	Shelton, IV et al.	N/A	N/A
10292758	12/2018	Boudreaux et al.	N/A	N/A
10292769	12/2018	Yu	N/A	N/A
10292771	12/2018	Wood et al.	N/A	N/A
10293129	12/2018	Fox et al.	N/A	N/A
10299792	12/2018	Huitema et al.	N/A	N/A
10299868	12/2018	Tsuboi et al.	N/A	N/A
10299870	12/2018	Connolly et al.	N/A	N/A
10305926	12/2018	Mihan et al.	N/A	N/A
D850617	12/2018	Shelton, IV et al.	N/A	N/A
10307159	12/2018	Harris et al.	N/A	N/A
10307170	12/2018	Parfett et al.	N/A	N/A
10307199	12/2018	Farritor et al.	N/A	N/A
10311036	12/2018	Hussam et al.	N/A	N/A
10313137	12/2018	Aarnio et al.	N/A	N/A
10314577	12/2018	Laurent et al.	N/A	N/A
10314582	12/2018	Shelton, IV et al.	N/A	N/A
10321907	12/2018	Shelton, IV et al.	N/A	N/A
10321964	12/2018	Grover et al.	N/A	N/A
10327764	12/2018	Harris et al.	N/A	N/A
10327779	12/2018	Richard et al.	N/A	N/A
10335042	12/2018	Schoenle et al.	N/A	N/A
10335147	12/2018	Rector et al.	N/A	N/A
10335149	12/2018	Baxter, III et al.	N/A	N/A
10335180	12/2018	Johnson et al.	N/A	N/A
10335227	12/2018	Heard	N/A	N/A
10339496	12/2018	Matson et al.	N/A	N/A
10342543	12/2018	Shelton, IV et al.	N/A	N/A
10342602	12/2018	Strobl et al.	N/A	N/A
10342623	12/2018	Huelman et al.	N/A	N/A
10343102	12/2018	Reasoner et al.	N/A	N/A
10349824	12/2018	Claude et al.	N/A	N/A
10349939	12/2018	Shelton, IV et al.	N/A	N/A
10349941	12/2018	Marczyk et al.	N/A	N/A
10350016	12/2018	Burbank et al.	N/A	N/A
10357184	12/2018	Crawford et al.	N/A	N/A
10357246	12/2018	Shelton, IV et al.	N/A	N/A
10357247	12/2018	Shelton, IV et al.	N/A	N/A
10362179	12/2018	Harris	N/A	N/A
10363032	12/2018	Scheib et al.	N/A	N/A

10363037	12/2018	Aronhalt et al.	N/A	N/A
10368861	12/2018	Baxter, III et al.	N/A	N/A
10368865	12/2018	Harris et al.	N/A	N/A
10368867	12/2018	Harris et al.	N/A	N/A
10368876	12/2018	Bhatnagar et al.	N/A	N/A
10368894	12/2018	Madan et al.	N/A	N/A
10368903	12/2018	Morales et al.	N/A	N/A
10376263	12/2018	Morgan et al.	N/A	N/A
10376305	12/2018	Yates et al.	N/A	N/A
10376337	12/2018	Kilroy et al.	N/A	N/A
10376338	12/2018	Taylor et al.	N/A	N/A
10378893	12/2018	Mankovskii	N/A	N/A
10383518	12/2018	Abu-Tarif et al.	N/A	N/A
10383699	12/2018	Kilroy et al.	N/A	N/A
10384021	12/2018	Koeth et al.	N/A	N/A
10386990	12/2018	Shikhman et al.	N/A	N/A
10390718	12/2018	Chen et al.	N/A	N/A
10390794	12/2018	Kuroiwa et al.	N/A	N/A
10390825	12/2018	Shelton, IV et al.	N/A	N/A
10390831	12/2018	Holsten et al.	N/A	N/A
10390895	12/2018	Henderson et al.	N/A	N/A
10398348	12/2018	Osadchy et al.	N/A	N/A
10398434	12/2018	Shelton, IV et al.	N/A	N/A
10398517	12/2018	Eckert et al.	N/A	N/A
10398521	12/2018	Itkowitz et al.	N/A	N/A
10404521	12/2018	McChord et al.	N/A	N/A
10404801	12/2018	Martch	N/A	N/A
10405857	12/2018	Shelton, IV et al.	N/A	N/A
10405859	12/2018	Harris et al.	N/A	N/A
10405863	12/2018	Wise et al.	N/A	N/A
10413291	12/2018	Worthington et al.	N/A	N/A
10413293	12/2018	Shelton, IV et al.	N/A	N/A
10413297	12/2018	Harris et al.	N/A	N/A
10417446	12/2018	Takeyama	N/A	N/A
10420552	12/2018	Shelton, IV et al.	N/A	N/A
10420558	12/2018	Nalagatla et al.	N/A	N/A
10420559	12/2018	Marczyk et al.	N/A	N/A
10420620	12/2018	Rockrohr	N/A	N/A
10420865	12/2018	Reasoner et al.	N/A	N/A
10422727	12/2018	Pliskin	N/A	N/A
10426466	12/2018	Contini et al.	N/A	N/A
10426467	12/2018	Miller et al.	N/A	N/A
10426468	12/2018	Contini et al.	N/A	N/A
10426471	12/2018	Shelton, IV et al.	N/A	N/A
10426481	12/2018	Aronhalt et al.	N/A	N/A
10433837	12/2018	Worthington et al.	N/A	N/A
10433844	12/2018	Shelton, IV et al.	N/A	N/A
10433849	12/2018	Shelton, IV et al.	N/A	N/A
10433918	12/2018	Shelton, IV et al.	N/A	N/A
10441279	12/2018	Shelton, IV et al.	N/A	N/A
10441344	12/2018	Notz et al.	N/A	N/A
10441345	12/2018	Aldridge et al.	N/A	N/A
10448948	12/2018	Shelton, IV et al.	N/A	N/A
10448950	12/2018	Shelton, IV et al.	N/A	N/A
10456137	12/2018	Vendely et al.	N/A	N/A
10456140	12/2018	Shelton, IV et al.	N/A	N/A
10456193	12/2018	Yates et al.	N/A	N/A
10463365	12/2018	Williams	N/A	N/A
10463367	12/2018	Kostrzewski et al.	N/A	N/A
10463371	12/2018	Kostrzewski	N/A	N/A
10463436	12/2018	Jackson et al.	N/A	N/A
10470684	12/2018	Toth et al.	N/A	N/A
10470762	12/2018	Leimbach et al.	N/A	N/A
10470764	12/2018	Baxter, III et al.	N/A	N/A
10470768	12/2018	Harris et al.	N/A	N/A
10470791	12/2018	Houser	N/A	N/A
10471254	12/2018	Sano et al.	N/A	N/A
10478181	12/2018	Shelton, IV et al.	N/A	N/A
10478182	12/2018	Taylor	N/A	N/A
10478185	12/2018	Nicholas	N/A	N/A
10478189	12/2018	Bear et al.	N/A	N/A
10478190	12/2018	Miller et al.	N/A	N/A
10478544	12/2018	Friederichs et al.	N/A	N/A
10485450	12/2018	Gupta et al.	N/A	N/A
10485542	12/2018	Shelton, IV et al.	N/A	N/A

10485543	12/2018	Shelton, IV et al.	N/A	N/A
10492783	12/2018	Shelton, IV et al.	N/A	N/A
10492784	12/2018	Beardsley et al.	N/A	N/A
10492785	12/2018	Overmyer et al.	N/A	N/A
10496788	12/2018	Amarasingham et al.	N/A	N/A
10498269	12/2018	Zemlok et al.	N/A	N/A
10499847	12/2018	Latimer et al.	N/A	N/A
10499891	12/2018	Chaplin et al.	N/A	N/A
10499914	12/2018	Huang et al.	N/A	N/A
10499915	12/2018	Aranyi	N/A	N/A
10499994	12/2018	Luks et al.	N/A	N/A
10507068	12/2018	Kopp et al.	N/A	N/A
10507278	12/2018	Gao et al.	N/A	N/A
10510267	12/2018	Jarc et al.	N/A	N/A
10512413	12/2018	Schepis et al.	N/A	N/A
10512461	12/2018	Gupta et al.	N/A	N/A
10512499	12/2018	McHenry et al.	N/A	N/A
10512509	12/2018	Bowling et al.	N/A	N/A
10512514	12/2018	Nowlin et al.	N/A	N/A
10517588	12/2018	Gupta et al.	N/A	N/A
10517595	12/2018	Hunter et al.	N/A	N/A
10517596	12/2018	Hunter et al.	N/A	N/A
10517686	12/2018	Vokrot et al.	N/A	N/A
10524789	12/2019	Swayze et al.	N/A	N/A
10531579	12/2019	Hsiao et al.	N/A	N/A
10531874	12/2019	Morgan et al.	N/A	N/A
10531929	12/2019	Widenhouse et al.	N/A	N/A
10532330	12/2019	Diallo et al.	N/A	N/A
10536617	12/2019	Liang et al.	N/A	N/A
10537324	12/2019	Shelton, IV et al.	N/A	N/A
10537325	12/2019	Bakos et al.	N/A	N/A
10537351	12/2019	Shelton, IV et al.	N/A	N/A
10537396	12/2019	Zingaretti et al.	N/A	N/A
10542978	12/2019	Chowaniec et al.	N/A	N/A
10542979	12/2019	Shelton, IV et al.	N/A	N/A
10542982	12/2019	Beckman et al.	N/A	N/A
10542991	12/2019	Shelton, IV et al.	N/A	N/A
D876466	12/2019	Kobayashi et al.	N/A	N/A
10548504	12/2019	Shelton, IV et al.	N/A	N/A
10548612	12/2019	Martinez et al.	N/A	N/A
10548673	12/2019	Harris et al.	N/A	N/A
10552574	12/2019	Sweeney	N/A	N/A
10555675	12/2019	Satish et al.	N/A	N/A
10555748	12/2019	Yates et al.	N/A	N/A
10555750	12/2019	Conlon et al.	N/A	N/A
10555769	12/2019	Worrell et al.	N/A	N/A
10561349	12/2019	Wedekind et al.	N/A	N/A
10561422	12/2019	Schellin et al.	N/A	N/A
10561470	12/2019	Hourtash et al.	N/A	N/A
10561471	12/2019	Nichogi	N/A	N/A
10561560	12/2019	Boutoussov et al.	N/A	N/A
10561753	12/2019	Thompson et al.	N/A	N/A
10568625	12/2019	Harris et al.	N/A	N/A
10568626	12/2019	Shelton, IV et al.	N/A	N/A
10568632	12/2019	Miller et al.	N/A	N/A
10568704	12/2019	Savall et al.	N/A	N/A
10575868	12/2019	Hall et al.	N/A	N/A
10582928	12/2019	Hunter et al.	N/A	N/A
10582931	12/2019	Mujawar	N/A	N/A
10582962	12/2019	Friedrichs et al.	N/A	N/A
10582964	12/2019	Weinberg et al.	N/A	N/A
10586074	12/2019	Rose et al.	N/A	N/A
10588623	12/2019	Schmid et al.	N/A	N/A
10588625	12/2019	Weaner et al.	N/A	N/A
10588629	12/2019	Malinouskas et al.	N/A	N/A
10588630	12/2019	Shelton, IV et al.	N/A	N/A
10588631	12/2019	Shelton, IV et al.	N/A	N/A
10588632	12/2019	Shelton, IV et al.	N/A	N/A
10588711	12/2019	DiCarlo et al.	N/A	N/A
10592067	12/2019	Merdan et al.	N/A	N/A
10595844	12/2019	Nawana et al.	N/A	N/A
10595882	12/2019	Parfett et al.	N/A	N/A
10595887	12/2019	Shelton, IV et al.	N/A	N/A
10595930	12/2019	Scheib et al.	N/A	N/A
10595952	12/2019	Forrest et al.	N/A	N/A

10602007	12/2019	Takano	N/A	N/A
10602848	12/2019	Magana	N/A	N/A
10603036	12/2019	Hunter et al.	N/A	N/A
10603128	12/2019	Zergiebel et al.	N/A	N/A
10610223	12/2019	Wellman et al.	N/A	N/A
10610224	12/2019	Shelton, IV et al.	N/A	N/A
10610286	12/2019	Wiener et al.	N/A	N/A
10610313	12/2019	Bailey et al.	N/A	N/A
10617412	12/2019	Shelton, IV et al.	N/A	N/A
10617414	12/2019	Shelton, IV et al.	N/A	N/A
10617482	12/2019	Houser et al.	N/A	N/A
10617484	12/2019	Kilroy et al.	N/A	N/A
10624635	12/2019	Harris et al.	N/A	N/A
10624667	12/2019	Faller et al.	N/A	N/A
10624691	12/2019	Wiener et al.	N/A	N/A
10631423	12/2019	Collins et al.	N/A	N/A
10631858	12/2019	Burbank	N/A	N/A
10631912	12/2019	McFarlin et al.	N/A	N/A
10631916	12/2019	Horner et al.	N/A	N/A
10631917	12/2019	Ineson	N/A	N/A
10631939	12/2019	Dachs, II et al.	N/A	N/A
10639027	12/2019	Shelton, IV et al.	N/A	N/A
10639034	12/2019	Harris et al.	N/A	N/A
10639035	12/2019	Shelton, IV et al.	N/A	N/A
10639036	12/2019	Yates et al.	N/A	N/A
10639037	12/2019	Shelton, IV et al.	N/A	N/A
10639039	12/2019	Vendely et al.	N/A	N/A
10639098	12/2019	Cosman et al.	N/A	N/A
10639111	12/2019	Kopp	N/A	N/A
10639185	12/2019	Agrawal et al.	N/A	N/A
10653413	12/2019	Worthington et al.	N/A	N/A
10653476	12/2019	Ross	N/A	N/A
10653489	12/2019	Kopp	N/A	N/A
10656720	12/2019	Holz	N/A	N/A
10660705	12/2019	Piron et al.	N/A	N/A
10667809	12/2019	Bakos et al.	N/A	N/A
10667810	12/2019	Shelton, IV et al.	N/A	N/A
10667811	12/2019	Harris et al.	N/A	N/A
10667877	12/2019	Kapadia	N/A	N/A
10674897	12/2019	Levy	N/A	N/A
10675021	12/2019	Harris et al.	N/A	N/A
10675023	12/2019	Cappola	N/A	N/A
10675024	12/2019	Shelton, IV et al.	N/A	N/A
10675025	12/2019	Swayze et al.	N/A	N/A
10675026	12/2019	Harris et al.	N/A	N/A
10675035	12/2019	Zingman	N/A	N/A
10675100	12/2019	Frushour	N/A	N/A
10675104	12/2019	Kapadia	N/A	N/A
10677764	12/2019	Ross et al.	N/A	N/A
10679758	12/2019	Fox et al.	N/A	N/A
10682136	12/2019	Harris et al.	N/A	N/A
10682138	12/2019	Shelton, IV et al.	N/A	N/A
10686805	12/2019	Reybok, Jr. et al.	N/A	N/A
10687806	12/2019	Shelton, IV et al.	N/A	N/A
10687809	12/2019	Shelton, IV et al.	N/A	N/A
10687810	12/2019	Shelton, IV et al.	N/A	N/A
10687884	12/2019	Wiener et al.	N/A	N/A
10687905	12/2019	Kostrzewski	N/A	N/A
10695055	12/2019	Shelton, IV et al.	N/A	N/A
10695081	12/2019	Shelton, IV et al.	N/A	N/A
10695134	12/2019	Barral et al.	N/A	N/A
10702270	12/2019	Shelton, IV et al.	N/A	N/A
10702271	12/2019	Aranyi et al.	N/A	N/A
10709446	12/2019	Harris et al.	N/A	N/A
10716489	12/2019	Kalvoy et al.	N/A	N/A
10716615	12/2019	Shelton, IV et al.	N/A	N/A
10716639	12/2019	Kapadia et al.	N/A	N/A
10717194	12/2019	Griffiths et al.	N/A	N/A
10722222	12/2019	Aranyi	N/A	N/A
10722233	12/2019	Wellman	N/A	N/A
10722292	12/2019	Arya et al.	N/A	N/A
D893717	12/2019	Messerly et al.	N/A	N/A
10729458	12/2019	Stoddard et al.	N/A	N/A
10729509	12/2019	Shelton, IV et al.	N/A	N/A
10733267	12/2019	Pedersen	N/A	N/A

10736219	12/2019	Seow et al.	N/A	N/A
10736498	12/2019	Watanabe et al.	N/A	N/A
10736616	12/2019	Scheib et al.	N/A	N/A
10736628	12/2019	Yates et al.	N/A	N/A
10736629	12/2019	Shelton, IV et al.	N/A	N/A
10736636	12/2019	Baxter, III et al.	N/A	N/A
10736705	12/2019	Scheib et al.	N/A	N/A
10743872	12/2019	Leimbach et al.	N/A	N/A
10748115	12/2019	Laster et al.	N/A	N/A
10751052	12/2019	Stokes et al.	N/A	N/A
10751136	12/2019	Farritor et al.	N/A	N/A
10751768	12/2019	Hersey et al.	N/A	N/A
10755813	12/2019	Shelton, IV et al.	N/A	N/A
D896379	12/2019	Shelton, IV et al.	N/A	N/A
10758229	12/2019	Shelton, IV et al.	N/A	N/A
10758230	12/2019	Shelton, IV et al.	N/A	N/A
10758294	12/2019	Jones	N/A	N/A
10758310	12/2019	Shelton, IV et al.	N/A	N/A
10765376	12/2019	Brown, III et al.	N/A	N/A
10765424	12/2019	Baxter, III et al.	N/A	N/A
10765427	12/2019	Shelton, IV et al.	N/A	N/A
10765470	12/2019	Yates et al.	N/A	N/A
10772630	12/2019	Wixey	N/A	N/A
10772651	12/2019	Shelton, IV et al.	N/A	N/A
10772673	12/2019	Allen, IV et al.	N/A	N/A
10772688	12/2019	Peine et al.	N/A	N/A
10779818	12/2019	Zemlok et al.	N/A	N/A
10779821	12/2019	Harris et al.	N/A	N/A
10779823	12/2019	Shelton, IV et al.	N/A	N/A
10779897	12/2019	Rockrohr	N/A	N/A
10779900	12/2019	Pedros et al.	N/A	N/A
10783634	12/2019	Nye et al.	N/A	N/A
10786298	12/2019	Johnson	N/A	N/A
10786317	12/2019	Zhou et al.	N/A	N/A
10786327	12/2019	Anderson et al.	N/A	N/A
10792038	12/2019	Becerra et al.	N/A	N/A
10792118	12/2019	Prpa et al.	N/A	N/A
10792422	12/2019	Douglas et al.	N/A	N/A
10799304	12/2019	Kapadia et al.	N/A	N/A
10803977	12/2019	Sanmugalingham	N/A	N/A
10806445	12/2019	Penna et al.	N/A	N/A
10806453	12/2019	Chen et al.	N/A	N/A
10806454	12/2019	Kopp	N/A	N/A
10806499	12/2019	Castaneda et al.	N/A	N/A
10806506	12/2019	Gasprede et al.	N/A	N/A
10806532	12/2019	Grubbs et al.	N/A	N/A
10813638	12/2019	Shelton, IV et al.	N/A	N/A
10813703	12/2019	Swayze et al.	N/A	N/A
10818383	12/2019	Sharifi Sedeh et al.	N/A	N/A
10828028	12/2019	Harris et al.	N/A	N/A
10828030	12/2019	Weir et al.	N/A	N/A
10835206	12/2019	Bell et al.	N/A	N/A
10835245	12/2019	Swayze et al.	N/A	N/A
10835246	12/2019	Shelton, IV et al.	N/A	N/A
10835247	12/2019	Shelton, IV et al.	N/A	N/A
10842473	12/2019	Scheib et al.	N/A	N/A
10842490	12/2019	DiNardo et al.	N/A	N/A
10842492	12/2019	Shelton, IV et al.	N/A	N/A
10842522	12/2019	Messerly et al.	N/A	N/A
10842523	12/2019	Shelton, IV et al.	N/A	N/A
10842575	12/2019	Panescu et al.	N/A	N/A
10842897	12/2019	Schwartz et al.	N/A	N/A
D904612	12/2019	Wynn et al.	N/A	N/A
10849697	12/2019	Yates et al.	N/A	N/A
10849700	12/2019	Kopp et al.	N/A	N/A
10856768	12/2019	Osadchy et al.	N/A	N/A
10856867	12/2019	Shelton, IV et al.	N/A	N/A
10856868	12/2019	Shelton, IV et al.	N/A	N/A
10856870	12/2019	Harris et al.	N/A	N/A
10863984	12/2019	Shelton, IV et al.	N/A	N/A
10864037	12/2019	Mun et al.	N/A	N/A
10864050	12/2019	Tabandeh et al.	N/A	N/A
10872684	12/2019	McNutt et al.	N/A	N/A
10881399	12/2020	Shelton, IV et al.	N/A	N/A
10881401	12/2020	Baber et al.	N/A	N/A

10881446	12/2020	Strobl	N/A	N/A
10881464	12/2020	Odermatt et al.	N/A	N/A
10888321	12/2020	Shelton, IV et al.	N/A	N/A
10888322	12/2020	Morgan et al.	N/A	N/A
10892899	12/2020	Shelton, IV et al.	N/A	N/A
10892995	12/2020	Shelton, IV et al.	N/A	N/A
10893863	12/2020	Shelton, IV et al.	N/A	N/A
10893864	12/2020	Harris et al.	N/A	N/A
10893884	12/2020	Stoddard et al.	N/A	N/A
10898183	12/2020	Shelton, IV et al.	N/A	N/A
10898186	12/2020	Bakos et al.	N/A	N/A
10898189	12/2020	McDonald, II	N/A	N/A
10898256	12/2020	Yates et al.	N/A	N/A
10898280	12/2020	Kopp	N/A	N/A
10898622	12/2020	Shelton, IV et al.	N/A	N/A
10902944	12/2020	Casey et al.	N/A	N/A
10903685	12/2020	Yates et al.	N/A	N/A
10905415	12/2020	DiNardo et al.	N/A	N/A
10905418	12/2020	Shelton, IV et al.	N/A	N/A
10905420	12/2020	Jasemian et al.	N/A	N/A
10912559	12/2020	Harris et al.	N/A	N/A
10912567	12/2020	Shelton, IV et al.	N/A	N/A
10912580	12/2020	Green et al.	N/A	N/A
10912619	12/2020	Jarc et al.	N/A	N/A
10916415	12/2020	Karancsi et al.	N/A	N/A
10918385	12/2020	Overmyer et al.	N/A	N/A
10930400	12/2020	Robbins et al.	N/A	N/A
D914878	12/2020	Shelton, IV et al.	N/A	N/A
10932705	12/2020	Muhsin et al.	N/A	N/A
10932772	12/2020	Shelton, IV et al.	N/A	N/A
10932784	12/2020	Mozdzierz et al.	N/A	N/A
10932804	12/2020	Scheib et al.	N/A	N/A
10932806	12/2020	Shelton, IV et al.	N/A	N/A
10932872	12/2020	Shelton, IV et al.	N/A	N/A
10939313	12/2020	Eom et al.	N/A	N/A
10943454	12/2020	Shelton, IV et al.	N/A	N/A
10944728	12/2020	Wiener et al.	N/A	N/A
10945727	12/2020	Shelton, IV et al.	N/A	N/A
10950982	12/2020	Regnier et al.	N/A	N/A
10952708	12/2020	Scheib et al.	N/A	N/A
10952732	12/2020	Binmoeller et al.	N/A	N/A
10954935	12/2020	O'Shea et al.	N/A	N/A
10959727	12/2020	Hunter et al.	N/A	N/A
10959729	12/2020	Ehrenfels et al.	N/A	N/A
10959744	12/2020	Shelton, IV et al.	N/A	N/A
10959788	12/2020	Grover et al.	N/A	N/A
10960150	12/2020	Zergiebel et al.	N/A	N/A
10962449	12/2020	Unuma et al.	N/A	N/A
10966590	12/2020	Takahashi et al.	N/A	N/A
10966791	12/2020	Harris et al.	N/A	N/A
10966798	12/2020	Tesar et al.	N/A	N/A
10973516	12/2020	Shelton, IV et al.	N/A	N/A
10973517	12/2020	Wixey	N/A	N/A
10973520	12/2020	Shelton, IV et al.	N/A	N/A
10973682	12/2020	Vezzu et al.	N/A	N/A
10980536	12/2020	Weaner et al.	N/A	N/A
10980537	12/2020	Shelton, IV et al.	N/A	N/A
10980560	12/2020	Shelton, IV et al.	N/A	N/A
10980595	12/2020	Wham	N/A	N/A
10980610	12/2020	Rosenberg et al.	N/A	N/A
10987102	12/2020	Gonzalez et al.	N/A	N/A
10987178	12/2020	Shelton, IV et al.	N/A	N/A
10992698	12/2020	Patel et al.	N/A	N/A
10993715	12/2020	Shelton, IV et al.	N/A	N/A
10998098	12/2020	Greene et al.	N/A	N/A
11000276	12/2020	Shelton, IV et al.	N/A	N/A
11000278	12/2020	Shelton, IV et al.	N/A	N/A
11007004	12/2020	Shelton, IV et al.	N/A	N/A
11007022	12/2020	Shelton, IV et al.	N/A	N/A
11013563	12/2020	Shelton, IV et al.	N/A	N/A
11026687	12/2020	Shelton, IV et al.	N/A	N/A
11026712	12/2020	Shelton, IV et al.	N/A	N/A
11026713	12/2020	Stokes et al.	N/A	N/A
11026751	12/2020	Shelton, IV et al.	N/A	N/A
11039834	12/2020	Harris et al.	N/A	N/A

11045191	12/2020	Shelton, IV et al.	N/A	N/A
11045192	12/2020	Harris et al.	N/A	N/A
11045197	12/2020	Shelton, IV et al.	N/A	N/A
11045591	12/2020	Shelton, IV et al.	N/A	N/A
11051817	12/2020	Shelton, IV et al.	N/A	N/A
11051836	12/2020	Shelton, IV et al.	N/A	N/A
11051873	12/2020	Wiener et al.	N/A	N/A
11051876	12/2020	Shelton, IV et al.	N/A	N/A
11051902	12/2020	Kruecker et al.	N/A	N/A
11056244	12/2020	Shelton, IV et al.	N/A	N/A
11058423	12/2020	Shelton, IV et al.	N/A	N/A
11058498	12/2020	Shelton, IV et al.	N/A	N/A
11058501	12/2020	Tokarchuk et al.	N/A	N/A
11064997	12/2020	Shelton, IV et al.	N/A	N/A
11069012	12/2020	Shelton, IV et al.	N/A	N/A
11071560	12/2020	Deck et al.	N/A	N/A
11071595	12/2020	Johnson et al.	N/A	N/A
11076921	12/2020	Shelton, IV et al.	N/A	N/A
11083458	12/2020	Harris et al.	N/A	N/A
11090047	12/2020	Shelton, IV et al.	N/A	N/A
11090048	12/2020	Fanelli et al.	N/A	N/A
11090075	12/2020	Hunter et al.	N/A	N/A
11096688	12/2020	Shelton, IV et al.	N/A	N/A
11096693	12/2020	Shelton, IV et al.	N/A	N/A
11100631	12/2020	Yates et al.	N/A	N/A
11103268	12/2020	Shelton, IV et al.	N/A	N/A
11109866	12/2020	Shelton, IV et al.	N/A	N/A
11109878	12/2020	Shelton, IV et al.	N/A	N/A
11114195	12/2020	Shelton, IV et al.	N/A	N/A
11116485	12/2020	Scheib et al.	N/A	N/A
11123070	12/2020	Shelton, IV et al.	N/A	N/A
11129611	12/2020	Shelton, IV et al.	N/A	N/A
11129634	12/2020	Scheib et al.	N/A	N/A
11129636	12/2020	Shelton, IV et al.	N/A	N/A
11129669	12/2020	Stulen et al.	N/A	N/A
11129670	12/2020	Shelton, IV et al.	N/A	N/A
11132462	12/2020	Shelton, IV	N/A	G16H 40/63
11134942	12/2020	Harris et al.	N/A	N/A
11141160	12/2020	Shelton, IV et al.	N/A	N/A
11141213	12/2020	Yates et al.	N/A	N/A
11147607	12/2020	Yates et al.	N/A	N/A
11160551	12/2020	Shelton, IV et al.	N/A	N/A
11160605	12/2020	Shelton, IV et al.	N/A	N/A
11166716	12/2020	Shelton, IV et al.	N/A	N/A
11166772	12/2020	Shelton, IV et al.	N/A	N/A
11179150	12/2020	Yates et al.	N/A	N/A
11179151	12/2020	Shelton, IV et al.	N/A	N/A
11179155	12/2020	Shelton, IV et al.	N/A	N/A
11179175	12/2020	Houser et al.	N/A	N/A
11179204	12/2020	Shelton, IV et al.	N/A	N/A
11179208	12/2020	Yates et al.	N/A	N/A
11185325	12/2020	Shelton, IV et al.	N/A	N/A
11185330	12/2020	Huitema et al.	N/A	N/A
11191539	12/2020	Overmyer et al.	N/A	N/A
11191540	12/2020	Aronhalt et al.	N/A	N/A
11197668	12/2020	Shelton, IV et al.	N/A	N/A
11202570	12/2020	Shelton, IV et al.	N/A	N/A
11207065	12/2020	Harris et al.	N/A	N/A
11207067	12/2020	Shelton, IV et al.	N/A	N/A
11207090	12/2020	Shelton, IV et al.	N/A	N/A
11213293	12/2021	Worthington et al.	N/A	N/A
11213294	12/2021	Shelton, IV et al.	N/A	N/A
11213359	12/2021	Shelton, IV et al.	N/A	N/A
11218822	12/2021	Morgan et al.	N/A	N/A
11219453	12/2021	Shelton, IV et al.	N/A	N/A
11224426	12/2021	Shelton, IV et al.	N/A	N/A
11229436	12/2021	Shelton, IV et al.	N/A	N/A
11229471	12/2021	Shelton, IV et al.	N/A	N/A
11234756	12/2021	Shelton, IV et al.	N/A	N/A
11241230	12/2021	Shelton, IV et al.	N/A	N/A
11253256	12/2021	Harris et al.	N/A	N/A
11253315	12/2021	Yates et al.	N/A	N/A
11257589	12/2021	Shelton, IV et al.	N/A	N/A
11259806	12/2021	Shelton, IV et al.	N/A	N/A
11259807	12/2021	Shelton, IV et al.	N/A	N/A

11259830	12/2021	Nott et al.	N/A	N/A
11266409	12/2021	Huitema et al.	N/A	N/A
11266468	12/2021	Shelton, IV et al.	N/A	N/A
11272931	12/2021	Boudreaux et al.	N/A	N/A
11273001	12/2021	Shelton, IV et al.	N/A	N/A
11273290	12/2021	Kowshik	N/A	N/A
11278280	12/2021	Shelton, IV et al.	N/A	N/A
11278281	12/2021	Shelton, IV et al.	N/A	N/A
11284890	12/2021	Nalagatla et al.	N/A	N/A
11284936	12/2021	Shelton, IV et al.	N/A	N/A
11291440	12/2021	Harris et al.	N/A	N/A
11291441	12/2021	Giordano et al.	N/A	N/A
11291444	12/2021	Boudreaux et al.	N/A	N/A
11291445	12/2021	Shelton, IV et al.	N/A	N/A
11291465	12/2021	Parihar et al.	N/A	N/A
11291495	12/2021	Yates et al.	N/A	N/A
11291510	12/2021	Shelton, IV et al.	N/A	N/A
11298128	12/2021	Messerly et al.	N/A	N/A
11298129	12/2021	Bakos et al.	N/A	N/A
11298130	12/2021	Bakos et al.	N/A	N/A
11298148	12/2021	Jayme et al.	N/A	N/A
11304699	12/2021	Shelton, IV et al.	N/A	N/A
11304720	12/2021	Kimball et al.	N/A	N/A
11304745	12/2021	Shelton, IV et al.	N/A	N/A
11304763	12/2021	Shelton, IV et al.	N/A	N/A
11308075	12/2021	Shelton, IV et al.	N/A	N/A
11311306	12/2021	Shelton, IV et al.	N/A	N/A
11311342	12/2021	Parihar et al.	N/A	N/A
D950728	12/2021	Bakos et al.	N/A	N/A
D952144	12/2021	Boudreaux	N/A	N/A
11317915	12/2021	Boudreaux et al.	N/A	N/A
11317919	12/2021	Shelton, IV et al.	N/A	N/A
11317937	12/2021	Nott et al.	N/A	N/A
11322248	12/2021	Grantcharov et al.	N/A	N/A
11324557	12/2021	Shelton, IV et al.	N/A	N/A
11331100	12/2021	Boudreaux et al.	N/A	N/A
11331101	12/2021	Harris et al.	N/A	N/A
11337746	12/2021	Boudreaux	N/A	N/A
11344326	12/2021	Faller et al.	N/A	N/A
11350932	12/2021	Shelton, IV et al.	N/A	N/A
11350959	12/2021	Messerly et al.	N/A	N/A
11350978	12/2021	Henderson et al.	N/A	N/A
11357503	12/2021	Bakos et al.	N/A	N/A
11364075	12/2021	Yates et al.	N/A	N/A
11369377	12/2021	Boudreaux et al.	N/A	N/A
11373755	12/2021	Shelton, IV et al.	N/A	N/A
11376002	12/2021	Shelton, IV et al.	N/A	N/A
11376098	12/2021	Shelton, IV et al.	N/A	N/A
11382697	12/2021	Shelton, IV et al.	N/A	N/A
11382715	12/2021	Arai et al.	N/A	N/A
11389164	12/2021	Yates et al.	N/A	N/A
11389188	12/2021	Gee et al.	N/A	N/A
11399858	12/2021	Sawhney et al.	N/A	N/A
11406382	12/2021	Shelton, IV et al.	N/A	N/A
11406390	12/2021	Shelton, IV et al.	N/A	N/A
11410259	12/2021	Harris et al.	N/A	N/A
11413042	12/2021	Shelton, IV et al.	N/A	N/A
11419606	12/2021	Overmyer et al.	N/A	N/A
11419630	12/2021	Yates et al.	N/A	N/A
11419667	12/2021	Messerly et al.	N/A	N/A
11423007	12/2021	Shelton, IV et al.	N/A	N/A
11424027	12/2021	Shelton, IV	N/A	N/A
D964564	12/2021	Boudreaux	N/A	N/A
11432885	12/2021	Shelton, IV et al.	N/A	N/A
11446052	12/2021	Shelton, IV et al.	N/A	N/A
11464514	12/2021	Yates et al.	N/A	N/A
11504191	12/2021	Mccloud et al.	N/A	N/A
11571212	12/2022	Yates et al.	N/A	N/A
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2004/0243148	12/2003	Wasielewski	N/A	N/A
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2008/0059658	12/2007	Williams	N/A	N/A
2008/0077158	12/2007	Haider et al.	N/A	N/A
2008/0083414	12/2007	Messerges	N/A	N/A
2008/0091071	12/2007	Kumar et al.	N/A	N/A
2008/0114212	12/2007	Messerges	N/A	N/A
2008/0114350	12/2007	Park et al.	N/A	N/A
2008/0129465	12/2007	Rao	N/A	N/A
2008/0140090	12/2007	Aranyi et al.	N/A	N/A
2008/0164296	12/2007	Shelton et al.	N/A	N/A
2008/0167644	12/2007	Shelton et al.	N/A	N/A
2008/0177258	12/2007	Govari et al.	N/A	N/A
2008/0177362	12/2007	Phillips et al.	N/A	N/A
2008/0200940	12/2007	Eichmann et al.	N/A	N/A
2008/0223904	12/2007	Marczyk	N/A	N/A
2008/0234708	12/2007	Houser et al.	N/A	N/A
2008/0235052	12/2007	Node-Langlois et al.	N/A	N/A
2008/0245841	12/2007	Smith et al.	N/A	N/A
2008/0255413	12/2007	Zemlok et al.	N/A	N/A
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2008/0281678	12/2007	Keuls et al.	N/A	N/A
2008/0296346	12/2007	Shelton, IV et al.	N/A	N/A
2008/0306759	12/2007	Ilkin et al.	N/A	N/A
2008/0312953	12/2007	Claus	N/A	N/A
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2009/0030437	12/2008	Houser et al.	N/A	N/A
2009/0036750	12/2008	Weinstein et al.	N/A	N/A
2009/0036794	12/2008	Stubhaug et al.	N/A	N/A
2009/0043253	12/2008	Podaima	N/A	N/A
2009/0046146	12/2008	Hoyt	N/A	N/A
2009/0048589	12/2008	Takashino et al.	N/A	N/A
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2009/0076409	12/2008	Wu et al.	N/A	N/A
2009/0090763	12/2008	Zemlok et al.	N/A	N/A
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2009/0114699	12/2008	Viola	N/A	N/A
2009/0128084	12/2008	Johnson et al.	N/A	N/A
2009/0157072	12/2008	Wham et al.	N/A	N/A
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2009/0217932	12/2008	Voegelé	N/A	N/A
2009/0234352	12/2008	Behnke et al.	N/A	N/A
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2010/0036405	12/2009	Giordano et al.	N/A	N/A
2010/0038403	12/2009	D'Arcangelo	N/A	N/A
2010/0057106	12/2009	Sorrentino et al.	N/A	N/A
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2010/0069939	12/2009	Konishi	N/A	N/A
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2010/0070417	12/2009	Flynn et al.	N/A	N/A
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2016/0287912	12/2015	Warnking	N/A	N/A
2016/0292456	12/2015	Dubey et al.	N/A	N/A
2016/0296246	12/2015	Schaller	N/A	N/A
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2016/0310055	12/2015	Zand et al.	N/A	N/A
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2016/0314716	12/2015	Grubbs	N/A	N/A
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2016/0321400	12/2015	Durrant et al.	N/A	N/A
2016/0323283	12/2015	Kang et al.	N/A	N/A
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2016/0342753	12/2015	Feazell	N/A	N/A
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2016/0345857	12/2015	Jensrud et al.	N/A	N/A
2016/0350490	12/2015	Martinez et al.	N/A	N/A

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2016/0354162	12/2015	Yen et al.	N/A	N/A
2016/0361070	12/2015	Ardel et al.	N/A	N/A
2016/0367305	12/2015	Hareland	N/A	N/A
2016/0367401	12/2015	Claus	N/A	N/A
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2016/0374723	12/2015	Frankhouser et al.	N/A	N/A
2016/0374762	12/2015	Case et al.	N/A	N/A
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2017/0005911	12/2016	Kasargod et al.	N/A	N/A
2017/0007247	12/2016	Shelton, IV et al.	N/A	N/A
2017/0027603	12/2016	Pandey	N/A	N/A
2017/0042604	12/2016	McFarland et al.	N/A	N/A
2017/0068792	12/2016	Reiner	N/A	N/A
2017/0079530	12/2016	DiMaio et al.	N/A	N/A
2017/0079730	12/2016	Azizian et al.	N/A	N/A
2017/0086829	12/2016	Vendely et al.	N/A	N/A
2017/0086930	12/2016	Thompson et al.	N/A	N/A
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2017/0105787	12/2016	Witt et al.	N/A	N/A
2017/0116873	12/2016	Lendvay et al.	N/A	N/A
2017/0127499	12/2016	Unoson et al.	N/A	N/A
2017/0132374	12/2016	Lee et al.	N/A	N/A
2017/0132385	12/2016	Hunter et al.	N/A	N/A
2017/0132785	12/2016	Wshah et al.	N/A	N/A
2017/0143284	12/2016	Sehnert et al.	N/A	N/A
2017/0164997	12/2016	Johnson et al.	N/A	N/A
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2017/0172565	12/2016	Heneveld	N/A	N/A
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2017/0172674	12/2016	Hanuschik et al.	N/A	N/A
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2017/0178069	12/2016	Pattera et al.	N/A	N/A
2017/0185732	12/2016	Niklewski et al.	N/A	N/A
2017/0196583	12/2016	Sugiyama	N/A	N/A
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2017/0202608	12/2016	Shelton, IV et al.	N/A	N/A
2017/0209145	12/2016	Swayze et al.	N/A	N/A
2017/0215944	12/2016	Keffeler	N/A	N/A
2017/0224332	12/2016	Hunter et al.	N/A	N/A
2017/0224428	12/2016	Kopp	N/A	N/A
2017/0231553	12/2016	Igarashi et al.	N/A	N/A
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2017/0245809	12/2016	Ma et al.	N/A	N/A
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2017/0304020	12/2016	Ng et al.	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 16/209,458, titled METHOD FOR SMART ENERGY DEVICE INFRASTRUCTURE, filed Dec. 4, 2018, now U.S. Patent Application Publication No. 2019/0201047, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/773,778, titled METHOD FOR ADAPTIVE CONTROL SCHEMES FOR SURGICAL NETWORK CONTROL AND INTERACTION, filed Nov. 30, 2018, to U.S. Provisional Patent Application No. 62/773,728, titled METHOD FOR SITUATIONAL AWARENESS FOR SURGICAL NETWORK OR SURGICAL NETWORK CONNECTED DEVICE CAPABLE OF ADJUSTING FUNCTION BASED ON A SENSED SITUATION OR USAGE, filed Nov. 30, 2018, to U.S. Provisional Patent Application No. 62/773,741, titled METHOD FOR FACILITY DATA COLLECTION AND INTERPRETATION, filed Nov. 30, 2018, and to U.S. Provisional Patent Application No. 62/773,742, titled METHOD FOR CIRCULAR STAPLER CONTROL ALGORITHM ADJUSTMENT BASED ON SITUATIONAL AWARENESS, filed Nov. 30, 2018, the disclosure of each of which is herein incorporated by reference in its entirety. (2) U.S. patent application Ser. No. 16/209,458 claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/750,529, titled METHOD FOR OPERATING A POWERED ARTICULATING MULTI-CLIP APPLIER, filed Oct. 25, 2018, to U.S. Provisional Patent Application No. 62/750,539, titled SURGICAL CLIP APPLIER, filed Oct. 25, 2018, and to U.S. Provisional Patent Application No. 62/750,555, titled SURGICAL CLIP APPLIER, filed Oct. 25, 2018, the disclosure of each of which is herein incorporated by reference in its entirety. (3) U.S. patent application Ser. No. 16/209,458 also claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/729,183, titled CONTROL FOR A SURGICAL NETWORK OR SURGICAL NETWORK CONNECTED DEVICE THAT ADJUSTS ITS FUNCTION BASED ON A SENSED SITUATION OR USAGE, filed Sep. 10, 2018, to U.S. Provisional Patent Application No. 62/729,177, titled AUTOMATED DATA SCALING, ALIGNMENT, AND ORGANIZING BASED ON PREDEFINED PARAMETERS WITHIN A SURGICAL NETWORK BEFORE TRANSMISSION, filed Sep. 10, 2018, to U.S. Provisional Patent Application No. 62/729,176, titled INDIRECT COMMAND AND CONTROL OF A FIRST OPERATING ROOM SYSTEM THROUGH THE USE OF A SECOND OPERATING ROOM SYSTEM WITHIN A STERILE FIELD WHERE THE SECOND OPERATING ROOM SYSTEM HAS PRIMARY AND SECONDARY OPERATING MODES, filed Sep. 10, 2018, to U.S. Provisional Patent Application No. 62/729,185, titled POWERED STAPLING DEVICE THAT IS CAPABLE OF ADJUSTING FORCE, ADVANCEMENT SPEED, AND OVERALL STROKE OF CUTTING MEMBER OF THE DEVICE BASED ON SENSED PARAMETER OF FIRING OR CLAMPING, filed Sep. 10, 2018, to U.S. Provisional Patent Application No. 62/729,184, titled POWERED SURGICAL TOOL WITH A PREDEFINED ADJUSTABLE CONTROL ALGORITHM FOR CONTROLLING AT LEAST ONE END EFFECTOR PARAMETER AND A MEANS FOR LIMITING THE ADJUSTMENT, filed Sep. 10, 2018, to U.S. Provisional Patent Application No. 62/729,182, titled SENSING THE PATIENT POSITION AND CONTACT UTILIZING THE MONO-POLAR RETURN PAD ELECTRODE TO PROVIDE SITUATIONAL AWARENESS TO THE HUB, filed Sep. 10, 2018, to U.S. Provisional Patent Application No. 62/729,191, titled SURGICAL NETWORK RECOMMENDATIONS FROM REAL TIME ANALYSIS OF PROCEDURE VARIABLES AGAINST A BASELINE HIGHLIGHTING DIFFERENCES FROM THE OPTIMAL SOLUTION, filed Sep. 10, 2018, to U.S. Provisional Patent Application No. 62/729,195, titled ULTRASONIC ENERGY DEVICE WHICH VARIES PRESSURE APPLIED BY CLAMP ARM TO PROVIDE THRESHOLD CONTROL PRESSURE AT A CUT PROGRESSION LOCATION, filed Sep. 10, 2018, and to U.S. Provisional Patent Application No. 62/729,186, titled WIRELESS PAIRING OF A SURGICAL DEVICE WITH ANOTHER DEVICE WITHIN A STERILE SURGICAL FIELD BASED ON THE USAGE AND SITUATIONAL AWARENESS OF DEVICES, filed Sep. 10, 2018, the disclosure of each of which is herein incorporated by reference in its entirety. (4) U.S. patent application Ser. No. 16/209,458 also claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/721,995, titled CONTROLLING AN ULTRASONIC SURGICAL INSTRUMENT ACCORDING TO TISSUE LOCATION, filed Aug. 23, 2018, to U.S. Provisional Patent Application No. 62/721,998, titled SITUATIONAL AWARENESS OF ELECTROSURGICAL SYSTEMS, filed Aug. 23, 2018, to U.S. Provisional Patent Application No. 62/721,999, titled INTERRUPTION OF ENERGY DUE TO INADVERTENT CAPACITIVE COUPLING, filed Aug. 23, 2018, to U.S. Provisional Patent Application No. 62/721,994, titled BIPOLAR COMBINATION DEVICE THAT AUTOMATICALLY ADJUSTS PRESSURE BASED ON ENERGY MODALITY, filed Aug. 23, 2018, and to U.S. Provisional Patent Application No. 62/721,996, titled RADIO FREQUENCY ENERGY DEVICE FOR DELIVERING COMBINED ELECTRICAL SIGNALS, filed Aug. 23, 2018, the disclosure of each of which is herein incorporated by reference in its entirety. (5) U.S. patent application Ser. No. 16/209,458 also claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/692,747, titled SMART ACTIVATION OF AN ENERGY DEVICE BY ANOTHER DEVICE, filed on Jun. 30, 2018, to U.S. Provisional Patent Application No. 62/692,748, titled SMART ENERGY ARCHITECTURE, filed on Jun. 30, 2018, and to U.S. Provisional Patent Application No. 62/692,768, titled SMART ENERGY DEVICES, filed on Jun. 30, 2018, the disclosure of each of which is herein incorporated by reference in its entirety. (6) U.S. patent application Ser. No. 16/209,458 also claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/691,228, titled METHOD OF USING REINFORCED FLEX CIRCUITS WITH MULTIPLE SENSORS WITH ELECTROSURGICAL DEVICES, filed Jun. 28, 2018, to U.S. Provisional Patent Application No. 62/691,227, titled CONTROLLING A SURGICAL INSTRUMENT ACCORDING TO SENSED CLOSURE PARAMETERS, filed Jun. 28, 2018, to U.S. Provisional Patent Application No. 62/691,230, titled SURGICAL INSTRUMENT HAVING A FLEXIBLE ELECTRODE, filed Jun. 28, 2018, to U.S. Provisional Patent Application No. 62/691,219, titled SURGICAL EVACUATION SENSING AND MOTOR CONTROL, filed Jun. 28, 2018, to U.S. Provisional Patent Application No. 62/691,257, titled COMMUNICATION OF SMOKE EVACUATION SYSTEM PARAMETERS TO HUB OR CLOUD IN SMOKE EVACUATION MODULE FOR INTERACTIVE SURGICAL PLATFORM, filed Jun. 28, 2018, to U.S. Provisional Patent Application No. 62/691,262, titled SURGICAL EVACUATION SYSTEM WITH A COMMUNICATION CIRCUIT FOR COMMUNICATION BETWEEN A FILTER AND A SMOKE EVACUATION DEVICE, filed Jun. 28, 2018, and to U.S. Provisional Patent Application No. 62/691,251, titled DUAL IN-SERIES LARGE AND SMALL DROPLET FILTERS, filed Jun. 28, 2018, the disclosure of each of which is herein incorporated by reference in its entirety. (7) U.S. patent application Ser. No. 16/209,458 claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/665,129, titled SURGICAL SUTURING SYSTEMS, filed May 1, 2018, to U.S. Provisional Patent Application No. 62/665,139, titled SURGICAL INSTRUMENTS COMPRISING CONTROL SYSTEMS, filed May 1, 2018, to U.S. Provisional Patent Application No. 62/665,177, titled SURGICAL INSTRUMENTS COMPRISING HANDLE ARRANGEMENTS, filed May 1, 2018, to U.S. Provisional Patent Application No. 62/665,128, titled MODULAR SURGICAL INSTRUMENTS, filed May 1, 2018, to U.S. Provisional Patent Application No. 62/665,192, titled SURGICAL DISSECTORS, filed May 1, 2018, and to U.S. Provisional Patent Application No. 62/665,134, titled SURGICAL CLIP APPLIER, filed May 1, 2018, the disclosure of each of which is herein incorporated by reference in its entirety. (8) U.S. patent application Ser. No. 16/209,458 also claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/659,900, titled METHOD OF HUB COMMUNICATION, filed on Apr. 19, 2018, the disclosure of which is herein incorporated by reference in its entirety. (9) U.S. patent application Ser. No. 16/209,458 also claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/650,898, filed on Mar. 30, 2018, titled CAPACITIVE COUPLED RETURN PATH PAD WITH SEPARABLE ARRAY ELEMENTS, to U.S. Provisional Patent Application No. 62/650,887, titled SURGICAL SYSTEMS WITH OPTIMIZED SENSING CAPABILITIES, filed Mar. 30, 2018, to U.S. Provisional Patent Application No. 62/650,882, titled SMOKE EVACUATION MODULE FOR INTERACTIVE SURGICAL PLATFORM, filed Mar. 30, 2018, and to U.S. Provisional Patent Application No. 62/650,877, titled SURGICAL SMOKE EVACUATION SENSING AND CONTROLS, filed Mar. 30, 2018, the disclosure of each of which is herein incorporated by reference in its entirety. (10) U.S. patent application Ser. No. 16/209,458 also claims the benefit of priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/649,302, titled INTERACTIVE SURGICAL SYSTEMS WITH ENCRYPTED COMMUNICATION CAPABILITIES, filed Mar. 28, 2018, to U.S. Provisional Patent Application No. 62/649,294, titled DATA STRIPPING METHOD TO INTERROGATE PATIENT RECORDS AND CREATE ANONYMIZED RECORD, filed Mar. 28, 2018, to U.S. Provisional Patent

Application No. 62/649,300, titled SURGICAL HUB SITUATIONAL AWARENESS, filed Mar. 28, 2018, to U.S. Provisional Patent Application No. 62/649,309, titled SURGICAL HUB SPATIAL AWARENESS TO DETERMINE DEVICES IN OPERATING THEATER, filed Mar. 28, 2018, to U.S. Provisional Patent Application No. 62/649,310, titled COMPUTER IMPLEMENTED INTERACTIVE SURGICAL SYSTEMS, filed Mar. 28, 2018, to U.S. Provisional Patent Application No. 62/649,291, titled USE OF LASER LIGHT AND RED-GREEN-BLUE COLORATION TO DETERMINE PROPERTIES OF BACK SCATTERED LIGHT, filed Mar. 28, 2018, to U.S. Provisional Patent Application No. 62/649,296, titled ADAPTIVE CONTROL PROGRAM UPDATES FOR SURGICAL DEVICES, filed Mar. 28, 2018, to U.S. Provisional Patent Application No. 62/649,333, titled CLOUD-BASED MEDICAL ANALYTICS FOR CUSTOMIZATION AND RECOMMENDATIONS TO A USER, filed Mar. 28, 2018, to U.S. Provisional Patent Application No. 62/649,327, titled CLOUD-BASED MEDICAL ANALYTICS FOR SECURITY AND AUTHENTICATION TRENDS AND REACTIVE MEASURES, filed Mar. 28, 2018, to U.S. Provisional Patent Application No. 62/649,315, titled DATA HANDLING AND PRIORITIZATION IN A CLOUD ANALYTICS NETWORK, filed Mar. 28, 2018, to U.S. Provisional Patent Application No. 62/649,313, titled CLOUD INTERFACE FOR COUPLED SURGICAL DEVICES, filed Mar. 28, 2018, to U.S. Provisional Patent Application No. 62/649,320, titled DRIVE ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, filed Mar. 28, 2018, to U.S. Provisional Patent Application No. 62/649,307, titled AUTOMATIC TOOL ADJUSTMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, filed Mar. 28, 2018, and to U.S. Provisional Patent Application No. 62/649,323, titled SENSING ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, filed Mar. 28, 2018, the disclosure of each of which is herein incorporated by reference in its entirety. (11) U.S. patent application Ser. No. 16/209,458 also claims the benefit of priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, to U.S. Provisional Patent Application No. 62/611,340, titled CLOUD-BASED MEDICAL ANALYTICS, filed Dec. 28, 2017, and to U.S. Provisional Patent Application No. 62/611,339, titled ROBOT ASSISTED SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of each of which is herein incorporated by reference in its entirety.

BACKGROUND

(1) In a surgical environment, smart energy devices may be needed in a smart energy architecture environment.

SUMMARY

(2) In one aspect the present disclosure provides a method for characterizing a state of an end effector of an ultrasonic device. The ultrasonic device comprising an electromechanical ultrasonic system defined by a predetermined resonant frequency. The electromechanical ultrasonic system further comprising an ultrasonic transducer coupled to an ultrasonic blade. The method comprising: applying, by an energy source, a power level to the ultrasonic transducer, measuring, by a control circuit coupled to a memory, an impedance value of the ultrasonic transducer, comparing, by the control circuit, the impedance value to a reference impedance value stored in the memory; classifying, by the control circuit, the impedance value based on the comparison; characterizing, by the control circuit, the state of the electromechanical ultrasonic system based on the classification of the impedance value; and adjusting, by the control circuit, the power level applied to the ultrasonic transducer based on the characterization of the state of the end effector.

(3) In another aspect the present disclosure provides a method for characterizing a function of an end effector of an ultrasonic device. The ultrasonic device comprising an electromechanical ultrasonic system defined by a predetermined resonant frequency. The electromechanical ultrasonic system further comprising an ultrasonic transducer coupled to an ultrasonic blade. The method comprising: applying, by an energy source, a power level to the ultrasonic transducer, measuring, by a control circuit coupled to a memory, an impedance value of the ultrasonic transducer, comparing, by the control circuit, the impedance value to a reference impedance value stored in the memory; classifying, by the control circuit, the impedance value based on the comparison; characterizing, by the control circuit, the function of the electromechanical ultrasonic system based on the classification of the impedance value; and adjusting, by the control circuit, the power level applied to the ultrasonic transducer based on the characterization of the function of the end effector.

(4) In another aspect the present disclosure provides a method for characterizing a tissue in contact with an end effector of an ultrasonic device. The ultrasonic device comprising an electromechanical ultrasonic system defined by a predetermined resonant frequency. The electromechanical ultrasonic system further comprising an ultrasonic transducer coupled to an ultrasonic blade. The method comprising: applying, by an energy source, a power level to the ultrasonic transducer, measuring, by a control circuit coupled to a memory, an impedance value of the ultrasonic transducer, comparing, by the control circuit, the impedance value to a reference impedance value stored in the memory; classifying, by the control circuit, the impedance value based on the comparison; characterizing, by the control circuit, the tissue in contact with the end effector based on the classification of the impedance value; and adjusting, by the control circuit, the power level applied to the ultrasonic transducer based on the characterization of the tissue in contact with the end effector.

Description

FIGURES

(1) The features of various aspects are set forth with particularity in the appended claims. The various aspects, however, both as to organization and methods of operation, together with further objects and advantages thereof, may best be understood by reference to the following description, taken in conjunction with the accompanying drawings as follows.

(2) FIG. 1 is a block diagram of a computer-implemented interactive surgical system, in accordance with at least one aspect of the present disclosure.

(3) FIG. 2 is a surgical system being used to perform a surgical procedure in an operating room, in accordance with at least one aspect of the present disclosure.

(4) FIG. 3 is a surgical hub paired with a visualization system, a robotic system, and an intelligent instrument, in accordance with at least one aspect of the present disclosure.

(5) FIG. 4 is a partial perspective view of a surgical hub enclosure, and of a combo generator module slidably receivable in a drawer of the surgical hub enclosure, in accordance with at least one aspect of the present disclosure.

(6) FIG. 5 is a perspective view of a combo generator module with bipolar, ultrasonic, and monopolar contacts and a smoke evacuation component, in accordance with at least one aspect of the present disclosure.

(7) FIG. 6 illustrates individual power bus attachments for a plurality of lateral docking ports of a lateral modular housing configured to receive a plurality of modules, in accordance with at least one aspect of the present disclosure.

(8) FIG. 7 illustrates a vertical modular housing configured to receive a plurality of modules, in accordance with at least one aspect of the present disclosure.

(9) FIG. 8 illustrates a surgical data network comprising a modular communication hub configured to connect modular devices located in one or more operating theaters of a healthcare facility, or any room in a healthcare facility specially equipped for surgical operations, to the cloud, in accordance with at least one aspect of the present disclosure.

(10) FIG. 9 illustrates a computer-implemented interactive surgical system, in accordance with at least one aspect of the present disclosure.

(11) FIG. 10 illustrates a surgical hub comprising a plurality of modules coupled to the modular control tower, in accordance with at least one aspect of the present disclosure.

(12) FIG. 11 illustrates one aspect of a Universal Serial Bus (USB) network hub device, in accordance with at least one aspect of the present disclosure.

(13) FIG. 12 illustrates a logic diagram of a control system of a surgical instrument or tool, in accordance with at least one aspect of the present disclosure.

(14) FIG. 13 illustrates a control circuit configured to control aspects of the surgical instrument or tool, in accordance with at least one aspect of the present disclosure.

(15) FIG. 14 illustrates a combinational logic circuit configured to control aspects of the surgical instrument or tool, in accordance with at least one aspect of the present disclosure.

(16) FIG. 15 illustrates a sequential logic circuit configured to control aspects of the surgical instrument or tool, in accordance with at least one aspect of the present disclosure.

(17) FIG. 16 illustrates a surgical instrument or tool comprising a plurality of motors which can be activated to perform various functions, in accordance with at least one aspect of the present disclosure.

(18) FIG. 17 is a schematic diagram of a robotic surgical instrument configured to operate a surgical tool described herein, in accordance with at least one aspect of the present disclosure.

(19) FIG. 18 illustrates a block diagram of a surgical instrument programmed to control the distal translation of a displacement member, in accordance with at least one aspect of the present disclosure.

(20) FIG. 19 is a schematic diagram of a surgical instrument configured to control various functions, in accordance with at least one aspect of the present disclosure.

(21) FIG. 20 is a system configured to execute adaptive ultrasonic blade control algorithms in a surgical data network comprising a modular communication hub, in accordance with at least one aspect of the present disclosure.

(22) FIG. 21 illustrates an example of a generator, in accordance with at least one aspect of the present disclosure.

(23) FIG. 22 is a surgical system comprising a generator and various surgical instruments usable therewith, in accordance with at least one aspect of the present disclosure.

(24) FIG. 23 is an end effector, in accordance with at least one aspect of the present disclosure.

(25) FIG. 24 is a diagram of the surgical system of FIG. 22, in accordance with at least one aspect of the present disclosure.

(26) FIG. 25 is a model illustrating motional branch current, in accordance with at least one aspect of the present disclosure.

(27) FIG. 26 is a structural view of a generator architecture, in accordance with at least one aspect of the present disclosure.

(28) FIGS. 27A-27C are functional views of a generator architecture, in accordance with at least one aspect of the present disclosure.

(29) FIGS. 28A-28B are structural and functional aspects of a generator, in accordance with at least one aspect of the present disclosure.

(30) FIG. 29 is a schematic diagram of one aspect of an ultrasonic drive circuit.

(31) FIG. 30 is a schematic diagram of the transformer coupled to the ultrasonic drive circuit shown in FIG. 29, in accordance with at least one aspect of the present disclosure.

(32) FIG. 31 is a schematic diagram of the transformer shown in FIG. 30 coupled to a test circuit, in accordance with at least one aspect of the present disclosure.

(33) FIG. 32 is a schematic diagram of a control circuit, in accordance with at least one aspect of the present disclosure.

(34) FIG. 33 shows a simplified block circuit diagram illustrating another electrical circuit contained within a modular ultrasonic surgical instrument, in accordance with at least one aspect of the present disclosure.

(35) FIG. 34 illustrates a generator circuit partitioned into multiple stages, in accordance with at least one aspect of the present disclosure.

(36) FIG. 35 illustrates a generator circuit partitioned into multiple stages where a first stage circuit is common to the second stage circuit, in accordance with at least one aspect of the present disclosure.

(37) FIG. 36 is a schematic diagram of one aspect of a drive circuit configured for driving a high-frequency current (RF), in accordance with at least one aspect of the present disclosure.

(38) FIG. 37 is a schematic diagram of the transformer coupled to the RF drive circuit shown in FIG. 34, in accordance with at least one aspect of the present disclosure.

(39) FIG. 38 is a schematic diagram of a circuit comprising separate power sources for high power energy/drive circuits and low power circuits, according to one aspect of the present disclosure.

(40) FIG. 39 illustrates a control circuit that allows a dual generator system to switch between the RF generator and the ultrasonic generator energy modalities for a surgical instrument.

(41) FIG. 40 illustrates a diagram of one aspect of a surgical instrument comprising a feedback system for use with a surgical instrument, according to one aspect of the present disclosure.

(42) FIG. 41 illustrates one aspect of a fundamental architecture for a digital synthesis circuit such as a direct digital synthesis (DDS) circuit configured to generate a plurality of wave shapes for the electrical signal waveform for use in a surgical instrument, in accordance with at least one aspect of the present disclosure.

(43) FIG. 42 illustrates one aspect of direct digital synthesis (DDS) circuit configured to generate a plurality of wave shapes for the electrical signal waveform for use in surgical instrument, in accordance with at least one aspect of the present disclosure.

(44) FIG. 43 illustrates one cycle of a discrete time digital electrical signal waveform, in accordance with at least one aspect of the present disclosure of an analog waveform (shown superimposed over a discrete time digital electrical signal waveform for comparison purposes), in accordance with at least one aspect of the present disclosure.

(45) FIG. 44 is a diagram of a control system configured to provide progressive closure of a closure member as it advances distally to close the clamp arm to apply a closure force load at a desired rate according to one aspect of this disclosure.

(46) FIG. 45 illustrates a proportional-integral-derivative (PID) controller feedback control system according to one aspect of this disclosure.

(47) FIG. 46 is an elevational exploded view of modular handheld ultrasonic surgical instrument showing the left shell half removed from a handle assembly exposing a device identifier communicatively coupled to the multi-lead handle terminal assembly in accordance with one aspect of the present disclosure.

(48) FIG. 47 is a detail view of a trigger portion and switch of the ultrasonic surgical instrument shown in FIG. 46, in accordance with at least one aspect of the present disclosure.

(49) FIG. 48 is a fragmentary, enlarged perspective view of an end effector from a distal end with a jaw member in an open position, in accordance with at least one aspect of the present disclosure.

(50) FIG. 49 is a system diagram of a segmented circuit comprising a plurality of independently operated circuit segments, in accordance with at least one aspect of the present disclosure.

(51) FIG. 50 is a circuit diagram of various components of a surgical instrument with motor control functions, in accordance with at least one aspect of the present disclosure.

(52) FIG. 51 illustrates one aspect of an end effector comprising RF data sensors located on the jaw member, in accordance with at least one aspect of the present disclosure.

(53) FIG. 52 illustrates one aspect of the flexible circuit shown in FIG. 51 in which the sensors may be mounted to or formed integrally therewith, in accordance with at least one aspect of the present disclosure.

(54) FIG. 53 is an alternative system for controlling the frequency of an ultrasonic electromechanical system and detecting the impedance thereof, in

accordance with at least one aspect of the present disclosure.

(55) FIG. 54 is a spectra of the same ultrasonic device with a variety of different states and conditions of the end effector where phase and magnitude of the impedance of an ultrasonic transducer are plotted as a function of frequency, in accordance with at least one aspect of the present disclosure.

(56) FIG. 55 is a graphical representation of a plot of a set of 3D training data S, where ultrasonic transducer impedance magnitude and phase are plotted as a function of frequency, in accordance with at least one aspect of the present disclosure.

(57) FIG. 56 is a logic flow diagram depicting a control program or a logic configuration to determine jaw conditions based on the complex impedance characteristic pattern (fingerprint), in accordance with at least one aspect of the present disclosure.

(58) FIG. 57 is a circle plot of complex impedance plotted as an imaginary component versus real components of a piezoelectric vibrator, in accordance with at least one aspect of the present disclosure.

(59) FIG. 58 is a circle plot of complex admittance plotted as an imaginary component versus real components of a piezoelectric vibrator, in accordance with at least one aspect of the present disclosure.

(60) FIG. 59 is a circle plot of complex admittance for a 55.5 kHz ultrasonic piezoelectric transducer.

(61) FIG. 60 is a graphical display of an impedance analyzer showing impedance/admittance circle plots for an ultrasonic device with the jaw open and no loading where red depicts admittance and blue depicts impedance, in accordance with at least one aspect of the present disclosure.

(62) FIG. 61 is a graphical display of an impedance analyzer showing impedance/admittance circle plots for an ultrasonic device with the jaw clamped on dry chamois where red depicts admittance and blue depicts impedance, in accordance with at least one aspect of the present disclosure.

(63) FIG. 62 is a graphical display of an impedance analyzer showing impedance/admittance circle plots for an ultrasonic device with the jaw tip clamped on moist chamois where red depicts admittance and blue depicts impedance, in accordance with at least one aspect of the present disclosure.

(64) FIG. 63 is a graphical display of an impedance analyzer showing impedance/admittance circle plots for an ultrasonic device with the jaw fully clamped on moist chamois where red depicts admittance and blue depicts impedance, in accordance with at least one aspect of the present disclosure.

(65) FIG. 64 is a graphical display of an impedance analyzer showing impedance/admittance plots where frequency is swept from 48 kHz to 62 kHz to capture multiple resonances of an ultrasonic device with the jaw open where the gray overlay is to help see the circles, in accordance with at least one aspect of the present disclosure.

(66) FIG. 65 is a logic flow diagram of a process depicting a control program or a logic configuration to determine jaw conditions based on estimates of the radius and offsets of an impedance/admittance circle, in accordance with at least one aspect of the present disclosure.

(67) FIGS. 66A-66B are graphical representations of an ultrasonic transducer current hemostasis algorithm, where

(68) FIG. 66A is a graphical representation of percent of maximum current into an ultrasonic transducer as a function of time and

(69) FIG. 66B is a graphical representation of ultrasonic blade temperature as a function of time and tissue type, in accordance with at least one aspect of the present disclosure.

(70) FIG. 67 is a logic flow diagram of a process depicting a control program or a logic configuration to control the temperature of an ultrasonic blade based on tissue type, in accordance with at least one aspect of the present disclosure.

(71) FIG. 68 is a logic flow diagram of a process depicting a control program or a logic configuration to monitor the impedance of an ultrasonic transducer to profile an ultrasonic blade and deliver power to the ultrasonic blade on the profile according to one aspect of the present disclosure.

(72) FIGS. 69A-69D is a series of graphical representations monitoring the impedance of an ultrasonic transducer to profile an ultrasonic blade and deliver power to the ultrasonic blade on the profile according to one aspect of the present disclosure, where

(73) FIG. 69A is a graphical representation of the initial impedance of the ultrasonic transducer as a function of time,

(74) FIG. 69B is a graphical representation of power delivered to the ultrasonic blade as a function of time based on the initial impedance,

(75) FIG. 69C is a graphical representation of a new impedance of the ultrasonic transducer as a function of time, and

(76) FIG. 69D is a graphical representation of adjusted power delivered to the ultrasonic blade based on the new impedance.

(77) FIG. 70 is a system for adjusting complex impedance of the ultrasonic transducer to compensate for power lost when the ultrasonic blade is articulated, in accordance with at least one aspect of the present disclosure.

(78) FIG. 71 is a logic flow diagram of a process depicting a control program or a logic configuration to compensate output power as a function of articulation angle, in accordance with at least one aspect of the present disclosure.

(79) FIG. 72 is system for measuring complex impedance of an ultrasonic transducer in real time to determine action being performed by an ultrasonic blade, in accordance with at least one aspect of the present disclosure.

(80) FIG. 73 is a logic flow diagram of a process depicting a control program or a logic configuration to determine action being performed by the ultrasonic blade based on the complex impedance pattern, in accordance with at least one aspect of the present disclosure.

(81) FIG. 74 is a logic flow diagram depicting a control program or a logic configuration of an adaptive process for identifying a hemostasis vessel, in accordance with at least one aspect of the present disclosure.

(82) FIG. 75 is a graphical representation of ultrasonic transducer current profiles as a function of time for vein and artery vessel types, in accordance with at least one aspect of the present disclosure.

(83) FIG. 76 is a logic flow diagram depicting a control program or a logic configuration of an adaptive process for identifying a hemostasis vessel, in accordance with at least one aspect of the present disclosure.

(84) FIG. 77 is a graphical representation of ultrasonic transducer frequency profiles as a function of time for vein and artery vessel types, in accordance with at least one aspect of the present disclosure.

(85) FIG. 78 is a logic flow diagram depicting a control program or a logic configuration of a process for identifying a calcified vessel, in accordance with at least one aspect of the present disclosure.

(86) FIG. 79 is a logic flow diagram depicting a control program or a logic configuration of a process for identifying a calcified vessel, in accordance with at least one aspect of the present disclosure.

(87) FIG. 80 is a logic flow diagram depicting a control program or a logic configuration of a process for identifying a calcified vessel, in accordance with at least one aspect of the present disclosure.

(88) FIG. 81 is a diagram of a liver resection with vessels embedded in parenchymal tissue, in accordance with at least one aspect of the present disclosure.

(89) FIG. 82 is a diagram of an ultrasonic blade in parenchyma but not contacting a vessel, in accordance with at least one aspect of the present disclosure.

(90) FIGS. 83A-83B are ultrasonic transducer impedance magnitude/phase plots with curves for parenchyma shown in bold line, in accordance with at least one aspect of the present disclosure.

(91) FIG. 84 is a diagram of an ultrasonic blade in parenchyma and contacting a large vessel.

(92) FIGS. 85A-85B are ultrasonic transducer impedance magnitude/phase plots with curves for a large vessel shown in bold line, in accordance with at least one aspect of the present disclosure.

(93) FIG. 86 is a logic flow diagram depicting a control program or a logic configuration of a process for treating tissue in parenchyma when a vessel is detected, in accordance with at least one aspect of the present disclosure.

(94) FIG. 87 is an ultrasonic device configured to identify the status of the ultrasonic blade and determine the clocked clamp arm status to determine whether a disposable portion of a reusable and disposable ultrasonic device has been installed correctly, in accordance with at least one aspect of the present disclosure.

(95) FIG. **88** is an end effector portion of the ultrasonic device shown in FIG. **52**.
(96) FIG. **89** is an ultrasonic device configured to identify the status of the ultrasonic blade and determine whether the clamp arm is not completely distal to determine whether a disposable portion of a reusable and disposable ultrasonic device has been installed correctly, in accordance with at least one aspect of the present disclosure.
(97) FIG. **90** is a logic flow diagram depicting a control program or a logic configuration to identify the status of components of reusable and disposable devices, in accordance with at least one aspect of the present disclosure.
(98) FIG. **91** is a three-dimensional graphical representation of tissue radio frequency (RF) impedance classification, in accordance with at least one aspect of the present disclosure.
(99) FIG. **92** is a three-dimensional graphical representation of tissue radio frequency (RF) impedance analysis, in accordance with at least one aspect of the present disclosure.
(100) FIG. **93** is a graphical representation of carotid technique sensitivity where the time impedance (Z) derivative is plotted as a function of initial radio frequency (RF) impedance, in accordance with at least one aspect of the present disclosure.
(101) FIG. **94A** is a graphical representation of impedance phase angle as a function of resonant frequency of the same ultrasonic device with a cold (solid line) and hot (broken line) ultrasonic blade; and
(102) FIG. **94B** is a graphical representation of impedance magnitude as a function of resonant frequency of the same ultrasonic device with a cold (solid line) and hot (broken line) ultrasonic blade.
(103) FIG. **95** is a diagram of a Kalman filter to improve temperature estimator and state space model based on impedance across an ultrasonic transducer measured at a variety of frequencies, in accordance with at least one aspect of the present disclosure.
(104) FIG. **96** are three probability distributions employed by a state estimator of the Kalman filter shown in FIG. **95** to maximize estimates, in accordance with at least one aspect of the present disclosure.
(105) FIG. **97A** is a graphical representation of temperature versus time of an ultrasonic device with no temperature control reaching a maximum temperature of 490° C.
(106) FIG. **97B** is a graphical representation of temperature versus time of an ultrasonic device with temperature control reaching a maximum temperature of 320° C., in accordance with at least one aspect of the present disclosure.
(107) FIG. **98** is a graphical representation of the relationship between initial frequency and the change in frequency required to achieve a temperature of approximately 340° C., in accordance with at least one aspect of the present disclosure.
(108) FIG. **99** illustrates a feedback control system comprising an ultrasonic generator to regulate the electrical current (i) set point applied to an ultrasonic transducer of an ultrasonic electromechanical system to prevent the frequency (f) of the ultrasonic transducer from decreasing lower than a predetermined threshold, in accordance with at least one aspect of the present disclosure.
(109) FIG. **100** is a logic flow diagram of a process depicting a control program or a logic configuration of a controlled thermal management process to protect an end effector pad, in accordance with at least one aspect of the present disclosure.
(110) FIG. **101** is a graphical representation of temperature versus time comparing the desired temperature of an ultrasonic blade with a smart ultrasonic blade and a conventional ultrasonic blade, in accordance with at least one aspect of the present disclosure.
(111) FIGS. **102A-102B** are graphical representations of feedback control to adjust ultrasonic power applied to an ultrasonic transducer when a sudden drop in temperature of an ultrasonic blade is detected, where
(112) FIG. **102A** is a graphical representation of ultrasonic power as a function of time; and
(113) FIG. **102B** is a plot of ultrasonic blade temperature as a function of time, in accordance with at least one aspect of the present disclosure.
(114) FIG. **103** is a logic flow diagram of a process depicting a control program or a logic configuration to control the temperature of an ultrasonic blade, in accordance with at least one aspect of the present disclosure.
(115) FIG. **104** is a graphical representation of ultrasonic blade temperature as a function of time during a vessel firing, in accordance with at least one aspect of the present disclosure.
(116) FIG. **105** is a logic flow diagram of a process depicting a control program or a logic configuration to control the temperature of an ultrasonic blade between two temperature set points, in accordance with at least one aspect of the present disclosure.
(117) FIG. **106** is a logic flow diagram of a process depicting a control program or a logic configuration to determine the initial temperature of an ultrasonic blade, in accordance with at least one aspect of the present disclosure.
(118) FIG. **107** is a logic flow diagram of a process depicting a control program or a logic configuration to determine when an ultrasonic blade is approaching instability and then adjusting the power to the ultrasonic transducer to prevent instability of the ultrasonic transducer, in accordance with at least one aspect of the present disclosure.
(119) FIG. **108** is a logic flow diagram of a process depicting a control program or a logic configuration to provide ultrasonic sealing with temperature control, in accordance with at least one aspect of the present disclosure.
(120) FIG. **109** are graphical representations of ultrasonic transducer current and ultrasonic blade temperature as a function of time, in accordance with at least one aspect of the present disclosure.
(121) FIG. **110** provides a diagram showing an example system with means for detecting capacitive coupling, in accordance with at least one aspect of the present disclosure.
(122) FIG. **111** is a logic flow diagram depicting a control program or a logic configuration of an example methodology for limiting the effects of capacitive coupling in a surgical system is disclosed, in accordance with at least one aspect of the present disclosure.
(123) FIG. **112** is a logic flow diagram depicting a control program or a logic configuration of an example methodology that may be performed by the surgical system utilizing monopolar energy generation to determine whether to take advantage of parasitic capacitive coupling, in accordance with at least one aspect of the present disclosure.
(124) FIG. **113** is a logic flow diagram depicting a control program or a logic configuration to adjust compression force applied to tissue, based on one or more selected energy modalities, according to a least one aspect of the present disclosure.
(125) FIG. **114** illustrates a mechanical method of adjusting compression force applied by an end effector for different treatment types, in accordance with at least one aspect of the present disclosure.
(126) FIGS. **115A-115B** illustrate a mechanical method of adjusting compression force applied by an end effector for different treatment types, by rotating an ultrasonic blade, in accordance with at least one aspect of the present disclosure.
(127) FIG. **116** shows a diagram illustrating switching between active electrodes of an end effector, in accordance with at least one aspect of the present disclosure.
(128) FIG. **117** depicts a surgical procedure using an electrosurgical system in accordance with at least one aspect of the present disclosure.
(129) FIG. **118** illustrates a block diagram of the electrosurgical system used in FIG. **117** in accordance with at least one aspect of the present disclosure.
(130) FIG. **119** illustrates a return pad of the electrosurgical system of FIG. **117** including a plurality of electrodes in accordance with at least one aspect of the present disclosure.
(131) FIG. **120** illustrates an array of sensing devices in the return pad depicted in FIG. **118** in accordance with at least one aspect of the present disclosure.
(132) FIG. **121** is a graphical representation of a therapeutic RF signal that may be used in an electrosurgical system in accordance with at least one

aspect of the present disclosure.

(133) FIG. 122 is a graphical representation of a nerve stimulation signal that may be incorporated in an electrosurgical system in accordance with at least one aspect of the present disclosure.

(134) FIGS. 123A-123C are graphical representations of signals used by an electrosurgical system that may incorporate features of both the therapeutic RF signal of FIG. 121 and the nerve stimulation signal of FIG. 122 in accordance with at least one aspect of the present disclosure.

(135) FIG. 124 summarizes a method in which such a control for a smart electrosurgical device may be effected in accordance with at least one aspect of the present disclosure.

(136) FIG. 125 illustrates an ultrasonic surgical instrument system, in accordance with at least one aspect of the present disclosure.

(137) FIGS. 126A-126C illustrate a piezoelectric transducer, in accordance with at least one aspect of the present disclosure.

(138) FIG. 127 illustrates a D31 ultrasonic transducer architecture that includes an ultrasonic waveguide and one or more piezoelectric elements fixed to the ultrasonic waveguide, in accordance with at least one aspect of the present disclosure.

(139) FIG. 128 illustrates a cutaway view of an ultrasonic surgical instrument, in accordance with at least one aspect of the present disclosure.

(140) FIG. 129 illustrates an exploded view of the ultrasonic surgical instrument in FIG. 128, in accordance with at least one aspect of the present disclosure.

(141) FIG. 130 illustrates a block diagram of a surgical system, in accordance with at least one aspect of the present disclosure.

(142) FIG. 131 illustrates a perspective view of a surgical instrument including a sensor assembly configured to detect a user-worn magnetic reference, in accordance with at least one aspect of the present disclosure.

(143) FIG. 132A illustrates a sectional view along line 131-131 of a surgical instrument including a sensor assembly configured to detect an integral magnetic reference, in accordance with at least one aspect of the present disclosure.

(144) FIG. 132B illustrates a detail view of the surgical instrument of FIG. 132A in a first position, in accordance with at least one aspect of the present disclosure.

(145) FIG. 132C illustrates a detail view of the surgical instrument of FIG. 132A in a second position, in accordance with at least one aspect of the present disclosure.

(146) FIG. 133A illustrates a perspective view of a surgical instrument including a sensor assembly configured to detect contact thereagainst that is oriented orthogonally, in accordance with at least one aspect of the present disclosure.

(147) FIG. 133B illustrates a perspective view of a surgical instrument including a sensor assembly configured to detect contact thereagainst that is oriented laterally, in accordance with at least one aspect of the present disclosure.

(148) FIG. 134 illustrates a circuit diagram of the surgical instrument of either FIG. 133A or FIG. 133B, in accordance with at least one aspect of the present disclosure.

(149) FIG. 135A illustrates a perspective view of a surgical instrument including a sensor assembly configured to detect closure of the surgical instrument, wherein the surgical instrument is in an open position, in accordance with at least one aspect of the present disclosure.

(150) FIG. 135B illustrates a perspective view of a surgical instrument including a sensor assembly configured to detect closure of the surgical instrument, wherein the surgical instrument is in a first closed position, in accordance with at least one aspect of the present disclosure.

(151) FIG. 135C illustrates a perspective view of a surgical instrument including a sensor assembly configured to detect closure of the surgical instrument, wherein the surgical instrument is in a second closed position, in accordance with at least one aspect of the present disclosure.

(152) FIG. 136A illustrates a perspective view of a surgical instrument including a sensor assembly configured to detect opening of the surgical instrument, in accordance with at least one aspect of the present disclosure.

(153) FIG. 136B illustrates a sectional view along line 135B-135B of the surgical instrument of FIG. 136A, in accordance with at least one aspect of the present disclosure.

(154) FIG. 136C illustrates an exploded perspective view of the surgical instrument of FIG. 136A, in accordance with at least one aspect of the present disclosure.

(155) FIG. 136D illustrates a perspective view of the surgical instrument of FIG. 136A, in accordance with at least one aspect of the present disclosure.

(156) FIG. 136E illustrates a detail view of a portion of FIG. 136D, in accordance with at least one aspect of the present disclosure.

(157) FIG. 136F illustrates a perspective view of the interior face of the arm of the surgical instrument of FIG. 136A, in accordance with at least one aspect of the present disclosure.

(158) FIG. 137 illustrates a perspective view of a surgical instrument including a sensor assembly comprising a pair of sensors for controlling the activation of the surgical instrument, in accordance with at least one aspect of the present disclosure.

(159) FIG. 138 illustrates a perspective view of a surgical instrument comprising a deactivation switch, in accordance with at least one aspect of the present disclosure.

(160) FIG. 139 illustrates a perspective view of a retractor comprising a sensor, in accordance with at least one aspect of the present disclosure.

(161) FIG. 140 illustrates a perspective view of a retractor comprising a display in use at a surgical site, in accordance with at least one aspect of the present disclosure.

(162) FIG. 141 is a timeline depicting situational awareness of a surgical hub, in accordance with at least one aspect of the present disclosure.

DESCRIPTION

(163) Applicant of the present application owns the following U.S. Patent Applications, filed on Dec. 4, 2018, the disclosure of each of which is herein incorporated by reference in its entirety: U.S. patent application Ser. No. 16/209,385, titled METHOD OF HUB COMMUNICATION, PROCESSING, STORAGE AND DISPLAY, now U.S. Patent Application Publication No. 2019/0200844; U.S. patent application Ser. No. 16/209,395, titled METHOD OF HUB COMMUNICATION, now U.S. Patent Application Publication No. 2019/0201136; U.S. patent application Ser. No. 16/209,403, titled METHOD OF CLOUD BASED DATA ANALYTICS FOR USE WITH THE HUB, now U.S. Patent Application Publication No. 2019/0206569; U.S. patent application Ser. No. 16/209,407, titled METHOD OF ROBOTIC HUB COMMUNICATION, DETECTION, AND CONTROL, now U.S. Patent Application Publication No. 2019/0201137; U.S. patent application Ser. No. 16/209,416, titled METHOD OF HUB COMMUNICATION, PROCESSING, DISPLAY, AND CLOUD ANALYTICS, now U.S. Patent Application Publication No. 2019/0206562; U.S. patent application Ser. No. 16/209,423, titled METHOD OF COMPRESSING TISSUE WITHIN A STAPLING DEVICE AND SIMULTANEOUSLY DISPLAYING THE LOCATION OF THE TISSUE WITHIN THE JAWS, now U.S. Patent Application Publication No. 2019/0200981; U.S. patent application Ser. No. 16/209,427, titled METHOD OF USING REINFORCED FLEXIBLE CIRCUITS WITH MULTIPLE SENSORS TO OPTIMIZE PERFORMANCE OF RADIO FREQUENCY DEVICES, now U.S. Patent Application Publication No. 2019/0208641; U.S. patent application Ser. No. 16/209,433, titled METHOD OF SENSING PARTICULATE FROM SMOKE EVACUATED FROM A PATIENT, ADJUSTING THE PUMP SPEED BASED ON THE SENSED INFORMATION, AND COMMUNICATING THE FUNCTIONAL PARAMETERS OF THE SYSTEM TO THE HUB, now U.S. Patent Application Publication No. 2019/0201594; U.S. patent application Ser. No. 16/209,447, titled METHOD FOR SMOKE EVACUATION FOR SURGICAL HUB, now U.S. Patent Application Publication No. 2019/0201045; U.S. patent application Ser. No. 16/209,453, titled METHOD FOR CONTROLLING SMART ENERGY DEVICES, now U.S. Patent Application Publication No. 2019/0201046; U.S. patent application Ser. No. 16/209,465, titled METHOD FOR ADAPTIVE CONTROL SCHEMES FOR SURGICAL NETWORK CONTROL AND INTERACTION, now U.S. Pat. No. 11,304,465; U.S. patent application Ser. No. 16/209,478, titled METHOD FOR SITUATIONAL AWARENESS FOR SURGICAL NETWORK OR SURGICAL NETWORK CONNECTED DEVICE CAPABLE

OF ADJUSTING FUNCTION BASED ON A SENSED SITUATION OR USAGE, now U.S. Patent Application Publication No. 2019/0104919; U.S. patent application Ser. No. 16/209,490, titled METHOD FOR FACILITY DATA COLLECTION AND INTERPRETATION, now U.S. Patent Application Publication No. 2019/0206564; and U.S. patent application Ser. No. 16/209,491, titled METHOD FOR CIRCULAR STAPLER CONTROL ALGORITHM ADJUSTMENT BASED ON SITUATIONAL AWARENESS, now U.S. Pat. No. 11,109,866.

(164) Applicant of the present application owns the following U.S. Patent Applications, filed on Nov. 6, 2018, the disclosure of each of which is herein incorporated by reference in its entirety: U.S. patent application Ser. No. 16/182,224, titled SURGICAL NETWORK, INSTRUMENT, AND CLOUD RESPONSES BASED ON VALIDATION OF RECEIVED DATASET AND AUTHENTICATION OF ITS SOURCE AND INTEGRITY; U.S. patent application Ser. No. 16/182,230, titled SURGICAL SYSTEM FOR PRESENTING INFORMATION INTERPRETED FROM EXTERNAL DATA; U.S. patent application Ser. No. 16/182,233, titled SURGICAL SYSTEMS WITH AUTONOMOUSLY ADJUSTABLE CONTROL PROGRAMS; U.S. patent application Ser. No. 16/182,239, titled ADJUSTMENT OF DEVICE CONTROL PROGRAMS BASED ON STRATIFIED CONTEXTUAL DATA IN ADDITION TO THE DATA; U.S. patent application Ser. No. 16/182,243, titled SURGICAL HUB AND MODULAR DEVICE RESPONSE ADJUSTMENT BASED ON SITUATIONAL AWARENESS; U.S. patent application Ser. No. 16/182,248, titled DETECTION AND ESCALATION OF SECURITY RESPONSES OF SURGICAL INSTRUMENTS TO INCREASING SEVERITY THREATS; U.S. patent application Ser. No. 16/182,251, titled INTERACTIVE SURGICAL SYSTEM; U.S. patent application Ser. No. 16/182,260, titled AUTOMATED DATA SCALING, ALIGNMENT, AND ORGANIZING BASED ON PREDEFINED PARAMETERS WITHIN SURGICAL NETWORKS; U.S. patent application Ser. No. 16/182,267, titled SENSING THE PATIENT POSITION AND CONTACT UTILIZING THE MONO-POLAR RETURN PAD ELECTRODE TO PROVIDE SITUATIONAL AWARENESS TO THE HUB; U.S. patent application Ser. No. 16/182,249, titled POWERED SURGICAL TOOL WITH PREDEFINED ADJUSTABLE CONTROL ALGORITHM FOR CONTROLLING END EFFECTOR PARAMETER; U.S. patent application Ser. No. 16/182,246, titled ADJUSTMENTS BASED ON AIRBORNE PARTICLE PROPERTIES; U.S. patent application Ser. No. 16/182,256, titled ADJUSTMENT OF A SURGICAL DEVICE FUNCTION BASED ON SITUATIONAL AWARENESS; U.S. patent application Ser. No. 16/182,242, titled REAL-TIME ANALYSIS OF COMPREHENSIVE COST OF ALL INSTRUMENTATION USED IN SURGERY UTILIZING DATA FLUIDITY TO TRACK INSTRUMENTS THROUGH STOCKING AND IN-HOUSE PROCESSES; U.S. patent application Ser. No. 16/182,255, titled USAGE AND TECHNIQUE ANALYSIS OF SURGEON/STAFF PERFORMANCE AGAINST A BASELINE TO OPTIMIZE DEVICE UTILIZATION AND PERFORMANCE FOR BOTH CURRENT AND FUTURE PROCEDURES; U.S. patent application Ser. No. 16/182,269, titled IMAGE CAPTURING OF THE AREAS OUTSIDE THE ABDOMEN TO IMPROVE PLACEMENT AND CONTROL OF A SURGICAL DEVICE IN USE; U.S. patent application Ser. No. 16/182,278, titled COMMUNICATION OF DATA WHERE A SURGICAL NETWORK IS USING CONTEXT OF THE DATA AND REQUIREMENTS OF A RECEIVING SYSTEM/USER TO INFLUENCE INCLUSION OR LINKAGE OF DATA AND METADATA TO ESTABLISH CONTINUITY; U.S. patent application Ser. No. 16/182,290, titled SURGICAL NETWORK RECOMMENDATIONS FROM REAL TIME ANALYSIS OF PROCEDURE VARIABLES AGAINST A BASELINE HIGHLIGHTING DIFFERENCES FROM THE OPTIMAL SOLUTION; U.S. patent application Ser. No. 16/182,232, titled CONTROL OF A SURGICAL SYSTEM THROUGH A SURGICAL BARRIER; U.S. patent application Ser. No. 16/182,227, titled SURGICAL NETWORK DETERMINATION OF PRIORITIZATION OF COMMUNICATION, INTERACTION, OR PROCESSING BASED ON SYSTEM OR DEVICE NEEDS; U.S. patent application Ser. No. 16/182,231, titled WIRELESS PAIRING OF A SURGICAL DEVICE WITH ANOTHER DEVICE WITHIN A STERILE SURGICAL FIELD BASED ON THE USAGE AND SITUATIONAL AWARENESS OF DEVICES; U.S. patent application Ser. No. 16/182,229, titled ADJUSTMENT OF STAPLE HEIGHT OF AT LEAST ONE ROW OF STAPLES BASED ON THE SENSED TISSUE THICKNESS OR FORCE IN CLOSING; U.S. patent application Ser. No. 16/182,234, titled STAPLING DEVICE WITH BOTH COMPULSORY AND DISCRETIONARY LOCKOUTS BASED ON SENSED PARAMETERS; U.S. patent application Ser. No. 16/182,240, titled POWERED STAPLING DEVICE CONFIGURED TO ADJUST FORCE, ADVANCEMENT SPEED, AND OVERALL STROKE OF CUTTING MEMBER BASED ON SENSED PARAMETER OF FIRING OR CLAMPING; U.S. patent application Ser. No. 16/182,235, titled VARIATION OF RADIO FREQUENCY AND ULTRASONIC POWER LEVEL IN COOPERATION WITH VARYING CLAMP ARM PRESSURE TO ACHIEVE PREDEFINED HEAT FLUX OR POWER APPLIED TO TISSUE; and U.S. patent application Ser. No. 16/182,238, titled ULTRASONIC ENERGY DEVICE WHICH VARIES PRESSURE APPLIED BY CLAMP ARM TO PROVIDE THRESHOLD CONTROL PRESSURE AT A CUT PROGRESSION LOCATION.

(165) Applicant of the present application owns the following U.S. Patent Applications that were filed on Oct. 26, 2018, the disclosure of each of which is herein incorporated by reference in its entirety: U.S. patent application Ser. No. 16/172,303, titled METHOD FOR OPERATING A POWERED ARTICULATING MULTI-CLIP APPLIER; U.S. patent application Ser. No. 16/172,130, titled CLIP APPLIER COMPRISING INTERCHANGEABLE CLIP RELOADS; U.S. patent application Ser. No. 16/172,066, titled CLIP APPLIER COMPRISING A MOVABLE CLIP MAGAZINE; U.S. patent application Ser. No. 16/172,078, titled CLIP APPLIER COMPRISING A ROTATABLE CLIP MAGAZINE; U.S. patent application Ser. No. 16/172,087, titled CLIP APPLIER COMPRISING CLIP ADVANCING SYSTEMS; U.S. patent application Ser. No. 16/172,094, titled CLIP APPLIER COMPRISING A CLIP CRIMPING SYSTEM; U.S. patent application Ser. No. 16/172,128, titled CLIP APPLIER COMPRISING A RECIPROCATING CLIP ADVANCING MEMBER; U.S. patent application Ser. No. 16/172,168, titled CLIP APPLIER COMPRISING A MOTOR CONTROLLER; U.S. patent application Ser. No. 16/172,164, titled SURGICAL SYSTEM COMPRISING A SURGICAL TOOL AND A SURGICAL HUB; U.S. patent application Ser. No. 16/172,328, titled METHOD OF HUB COMMUNICATION WITH SURGICAL INSTRUMENT SYSTEMS; U.S. patent application Ser. No. 16/172,280, titled METHOD FOR PRODUCING A SURGICAL INSTRUMENT COMPRISING A SMART ELECTRICAL SYSTEM; U.S. patent application Ser. No. 16/172,219, titled METHOD OF HUB COMMUNICATION WITH SURGICAL INSTRUMENT SYSTEMS; U.S. patent application Ser. No. 16/172,248, titled METHOD OF HUB COMMUNICATION WITH SURGICAL INSTRUMENT SYSTEMS; U.S. patent application Ser. No. 16/172,198, titled METHOD OF HUB COMMUNICATION WITH SURGICAL INSTRUMENT SYSTEMS; and U.S. patent application Ser. No. 16/172,155, titled METHOD OF HUB COMMUNICATION WITH SURGICAL INSTRUMENT SYSTEMS.

(166) Applicant of the present application owns the following U.S. Patent Applications, filed on Aug. 28, 2018, the disclosure of each of which is herein incorporated by reference in its entirety: U.S. patent application Ser. No. 16/115,214, titled ESTIMATING STATE OF ULTRASONIC END EFFECTOR AND CONTROL SYSTEM THEREFOR; U.S. patent application Ser. No. 16/115,205, titled TEMPERATURE CONTROL OF ULTRASONIC END EFFECTOR AND CONTROL SYSTEM THEREFOR; U.S. patent application Ser. No. 16/115,233, titled RADIO FREQUENCY ENERGY DEVICE FOR DELIVERING COMBINED ELECTRICAL SIGNALS; U.S. patent application Ser. No. 16/115,208, titled CONTROLLING AN ULTRASONIC SURGICAL INSTRUMENT ACCORDING TO TISSUE LOCATION; U.S. patent application Ser. No. 16/115,220, titled CONTROLLING ACTIVATION OF AN ULTRASONIC SURGICAL INSTRUMENT ACCORDING TO THE PRESENCE OF TISSUE; U.S. patent application Ser. No. 16/115,232, titled DETERMINING TISSUE COMPOSITION VIA AN ULTRASONIC SYSTEM; U.S. patent application Ser. No. 16/115,239, titled DETERMINING THE STATE OF AN ULTRASONIC ELECTROMECHANICAL SYSTEM ACCORDING TO FREQUENCY SHIFT; U.S. patent application Ser. No. 16/115,247, titled DETERMINING THE STATE OF AN ULTRASONIC END EFFECTOR; U.S. patent application Ser. No. 16/115,211, titled SITUATIONAL AWARENESS OF ELECTROSURGICAL SYSTEMS; U.S. patent application Ser. No. 16/115,226, titled MECHANISMS FOR CONTROLLING DIFFERENT ELECTROMECHANICAL SYSTEMS OF AN ELECTROSURGICAL INSTRUMENT; U.S. patent application Ser. No. 16/115,240, titled DETECTION OF END EFFECTOR EMERSION IN LIQUID; U.S. patent application Ser. No. 16/115,249, titled INTERRUPTION OF ENERGY DUE TO INADVERTENT CAPACITIVE COUPLING; U.S. patent application Ser. No. 16/115,256, titled INCREASING RADIO FREQUENCY TO CREATE PADLESS MONOPOLAR LOOP; U.S. patent application Ser. No. 16/115,223, titled BIPOLAR COMBINATION DEVICE THAT AUTOMATICALLY ADJUSTS PRESSURE

BASED ON ENERGY MODALITY; and U.S. patent application Ser. No. 16/115,238, titled ACTIVATION OF SURGICAL DEVICES.

(167) Applicant of the present application owns the following U.S. Patent Applications, filed on Aug. 24, 2018, the disclosure of each of which is herein incorporated by reference in its entirety: U.S. patent application Ser. No. 16/112,129, titled SURGICAL SUTURING INSTRUMENT CONFIGURED TO MANIPULATE TISSUE USING MECHANICAL AND ELECTRICAL POWER; U.S. patent application Ser. No. 16/112,155, titled SURGICAL SUTURING INSTRUMENT COMPRISING A CAPTURE WIDTH WHICH IS LARGER THAN TROCAR DIAMETER; U.S. patent application Ser. No. 16/112,168, titled SURGICAL SUTURING INSTRUMENT COMPRISING A NON-CIRCULAR NEEDLE; U.S. patent application Ser. No. 16/112,180, titled ELECTRICAL POWER OUTPUT CONTROL BASED ON MECHANICAL FORCES; U.S. patent application Ser. No. 16/112,193, titled REACTIVE ALGORITHM FOR SURGICAL SYSTEM; U.S. patent application Ser. No. 16/112,099, titled SURGICAL INSTRUMENT COMPRISING AN ADAPTIVE ELECTRICAL SYSTEM; U.S. patent application Ser. No. 16/112,112, titled CONTROL SYSTEM ARRANGEMENTS FOR A MODULAR SURGICAL INSTRUMENT; U.S. patent application Ser. No. 16/112,119, titled ADAPTIVE CONTROL PROGRAMS FOR A SURGICAL SYSTEM COMPRISING MORE THAN ONE TYPE OF CARTRIDGE; U.S. patent application Ser. No. 16/112,097, titled SURGICAL INSTRUMENT SYSTEMS COMPRISING BATTERY ARRANGEMENTS; U.S. patent application Ser. No. 16/112,109, titled SURGICAL INSTRUMENT SYSTEMS COMPRISING HANDLE ARRANGEMENTS; U.S. patent application Ser. No. 16/112,114, titled SURGICAL INSTRUMENT SYSTEMS COMPRISING FEEDBACK MECHANISMS; U.S. patent application Ser. No. 16/112,117, titled SURGICAL INSTRUMENT SYSTEMS COMPRISING LOCKOUT MECHANISMS; U.S. patent application Ser. No. 16/112,095, titled SURGICAL INSTRUMENTS COMPRISING A LOCKABLE END EFFECTOR SOCKET; U.S. patent application Ser. No. 16/112,121, titled SURGICAL INSTRUMENTS COMPRISING A SHIFTING MECHANISM; U.S. patent application Ser. No. 16/112,151, titled SURGICAL INSTRUMENTS COMPRISING A SYSTEM FOR ARTICULATION AND ROTATION COMPENSATION; U.S. patent application Ser. No. 16/112,154, titled SURGICAL INSTRUMENTS COMPRISING A BIASED SHIFTING MECHANISM; U.S. patent application Ser. No. 16/112,226, titled SURGICAL INSTRUMENTS COMPRISING AN ARTICULATION DRIVE THAT PROVIDES FOR HIGH ARTICULATION ANGLES; U.S. patent application Ser. No. 16/112,062, titled SURGICAL DISSECTORS AND MANUFACTURING TECHNIQUES; U.S. patent application Ser. No. 16/112,098, titled SURGICAL DISSECTORS CONFIGURED TO APPLY MECHANICAL AND ELECTRICAL ENERGY; U.S. patent application Ser. No. 16/112,237, titled SURGICAL CLIP APPLIER CONFIGURED TO STORE CLIPS IN A STORED STATE; U.S. patent application Ser. No. 16/112,245, titled SURGICAL CLIP APPLIER COMPRISING AN EMPTY CLIP CARTRIDGE LOCKOUT; U.S. patent application Ser. No. 16/112,249, titled SURGICAL CLIP APPLIER COMPRISING AN AUTOMATIC CLIP FEEDING SYSTEM; U.S. patent application Ser. No. 16/112,253, titled SURGICAL CLIP APPLIER COMPRISING ADAPTIVE FIRING CONTROL; and U.S. patent application Ser. No. 16/112,257, titled SURGICAL CLIP APPLIER COMPRISING ADAPTIVE CONTROL IN RESPONSE TO A STRAIN GAUGE CIRCUIT.

(168) Applicant of the present application owns the following U.S. Patent Applications, filed on Jun. 29, 2018, the disclosure of each of which is herein incorporated by reference in its entirety: U.S. patent application Ser. No. 16/024,090, titled CAPACITIVE COUPLED RETURN PATH PAD WITH SEPARABLE ARRAY ELEMENTS; U.S. patent application Ser. No. 16/024,057, titled CONTROLLING A SURGICAL INSTRUMENT ACCORDING TO SENSED CLOSURE PARAMETERS; U.S. patent application Ser. No. 16/024,067, titled SYSTEMS FOR ADJUSTING END EFFECTOR PARAMETERS BASED ON PERIOPERATIVE INFORMATION; U.S. patent application Ser. No. 16/024,075, titled SAFETY SYSTEMS FOR SMART POWERED SURGICAL STAPLING; U.S. patent application Ser. No. 16/024,083, titled SAFETY SYSTEMS FOR SMART POWERED SURGICAL STAPLING; U.S. patent application Ser. No. 16/024,094, titled SURGICAL SYSTEMS FOR DETECTING END EFFECTOR TISSUE DISTRIBUTION IRREGULARITIES; U.S. patent application Ser. No. 16/024,138, titled SYSTEMS FOR DETECTING PROXIMITY OF SURGICAL END EFFECTOR TO CANCEROUS TISSUE; U.S. patent application Ser. No. 16/024,150, titled SURGICAL INSTRUMENT CARTRIDGE SENSOR ASSEMBLIES; U.S. patent application Ser. No. 16/024,160, titled VARIABLE OUTPUT CARTRIDGE SENSOR ASSEMBLY; U.S. patent application Ser. No. 16/024,124, titled SURGICAL INSTRUMENT HAVING A FLEXIBLE ELECTRODE; U.S. patent application Ser. No. 16/024,132, titled SURGICAL INSTRUMENT HAVING A FLEXIBLE CIRCUIT; U.S. patent application Ser. No. 16/024,141, titled SURGICAL INSTRUMENT WITH A TISSUE MARKING ASSEMBLY; U.S. patent application Ser. No. 16/024,162, titled SURGICAL SYSTEMS WITH PRIORITIZED DATA TRANSMISSION CAPABILITIES; U.S. patent application Ser. No. 16/024,066, titled SURGICAL EVACUATION SENSING AND MOTOR CONTROL; U.S. patent application Ser. No. 16/024,096, titled SURGICAL EVACUATION SENSOR ARRANGEMENTS; U.S. patent application Ser. No. 16/024,116, titled SURGICAL EVACUATION FLOW PATHS; U.S. patent application Ser. No. 16/024,149, titled SURGICAL EVACUATION SENSING AND GENERATOR CONTROL; U.S. patent application Ser. No. 16/024,180, titled SURGICAL EVACUATION SENSING AND DISPLAY; U.S. patent application Ser. No. 16/024,245, titled COMMUNICATION OF SMOKE EVACUATION SYSTEM PARAMETERS TO HUB OR CLOUD IN SMOKE EVACUATION MODULE FOR INTERACTIVE SURGICAL PLATFORM; U.S. patent application Ser. No. 16/024,258, titled SMOKE EVACUATION SYSTEM INCLUDING A SEGMENTED CONTROL CIRCUIT FOR INTERACTIVE SURGICAL PLATFORM; U.S. patent application Ser. No. 16/024,265, titled SURGICAL EVACUATION SYSTEM WITH A COMMUNICATION CIRCUIT FOR COMMUNICATION BETWEEN A FILTER AND A SMOKE EVACUATION DEVICE; and U.S. patent application Ser. No. 16/024,273, titled DUAL IN-SERIES LARGE AND SMALL DROPLET FILTERS.

(169) Applicant of the present application owns the following U.S. Patent Applications, filed on Mar. 29, 2018, the disclosure of each of which is herein incorporated by reference in its entirety: U.S. patent application Ser. No. 15/940,641, titled INTERACTIVE SURGICAL SYSTEMS WITH ENCRYPTED COMMUNICATION CAPABILITIES; U.S. patent application Ser. No. 15/940,648, titled INTERACTIVE SURGICAL SYSTEMS WITH CONDITION HANDLING OF DEVICES AND DATA CAPABILITIES; U.S. patent application Ser. No. 15/940,656, titled SURGICAL HUB COORDINATION OF CONTROL AND COMMUNICATION OF OPERATING ROOM DEVICES; U.S. patent application Ser. No. 15/940,666, titled SPATIAL AWARENESS OF SURGICAL HUBS IN OPERATING ROOMS; U.S. patent application Ser. No. 15/940,670, titled COOPERATIVE UTILIZATION OF DATA DERIVED FROM SECONDARY SOURCES BY INTELLIGENT SURGICAL HUBS; U.S. patent application Ser. No. 15/940,677, titled SURGICAL HUB CONTROL ARRANGEMENTS; U.S. patent application Ser. No. 15/940,632, titled DATA STRIPPING METHOD TO INTERROGATE PATIENT RECORDS AND CREATE ANONYMIZED RECORD; U.S. patent application Ser. No. 15/940,640, titled COMMUNICATION HUB AND STORAGE DEVICE FOR STORING PARAMETERS AND STATUS OF A SURGICAL DEVICE TO BE SHARED WITH CLOUD BASED ANALYTICS SYSTEMS; U.S. patent application Ser. No. 15/940,645, titled SELF DESCRIBING DATA PACKETS GENERATED AT AN ISSUING INSTRUMENT; U.S. patent application Ser. No. 15/940,649, titled DATA PAIRING TO INTERCONNECT A DEVICE MEASURED PARAMETER WITH AN OUTCOME; U.S. patent application Ser. No. 15/940,654, titled SURGICAL HUB SITUATIONAL AWARENESS; U.S. patent application Ser. No. 15/940,663, titled SURGICAL SYSTEM DISTRIBUTED PROCESSING; U.S. patent application Ser. No. 15/940,668, titled AGGREGATION AND REPORTING OF SURGICAL HUB DATA; U.S. patent application Ser. No. 15/940,671, titled SURGICAL HUB SPATIAL AWARENESS TO DETERMINE DEVICES IN OPERATING THEATER; U.S. patent application Ser. No. 15/940,686, titled DISPLAY OF ALIGNMENT OF STAPLE CARTRIDGE TO PRIOR LINEAR STAPLE LINE; U.S. patent application Ser. No. 15/940,700, titled STERILE FIELD INTERACTIVE CONTROL DISPLAYS; U.S. patent application Ser. No. 15/940,629, titled COMPUTER IMPLEMENTED INTERACTIVE SURGICAL SYSTEMS; U.S. patent application Ser. No. 15/940,704, titled USE OF LASER LIGHT AND RED-GREEN-BLUE COLORATION TO DETERMINE PROPERTIES OF BACK SCATTERED LIGHT; U.S. patent application Ser. No. 15/940,722, titled CHARACTERIZATION OF TISSUE IRREGULARITIES THROUGH THE USE OF MONO-CHROMATIC LIGHT REFRACTIVITY; U.S. patent application Ser. No. 15/940,742, titled DUAL CMOS ARRAY IMAGING; U.S. patent application Ser. No. 15/940,636, titled ADAPTIVE CONTROL PROGRAM UPDATES FOR SURGICAL DEVICES; U.S. patent application Ser. No. 15/940,653, titled ADAPTIVE CONTROL PROGRAM UPDATES FOR SURGICAL HUBS; U.S. patent application Ser. No. 15/940,660, titled CLOUD-BASED

MEDICAL ANALYTICS FOR CUSTOMIZATION AND RECOMMENDATIONS TO A USER; U.S. patent application Ser. No. 15/940,679, titled CLOUD-BASED MEDICAL ANALYTICS FOR LINKING OF LOCAL USAGE TRENDS WITH THE RESOURCE ACQUISITION BEHAVIORS OF LARGER DATA SET; U.S. patent application Ser. No. 15/940,694, titled CLOUD-BASED MEDICAL ANALYTICS FOR MEDICAL FACILITY SEGMENTED INDIVIDUALIZATION OF INSTRUMENT FUNCTION; U.S. patent application Ser. No. 15/940,634, titled CLOUD-BASED MEDICAL ANALYTICS FOR SECURITY AND AUTHENTICATION TRENDS AND REACTIVE MEASURES; U.S. patent application Ser. No. 15/940,706, titled DATA HANDLING AND PRIORITIZATION IN A CLOUD ANALYTICS NETWORK; U.S. patent application Ser. No. 15/940,675, titled CLOUD INTERFACE FOR COUPLED SURGICAL DEVICES; U.S. patent application Ser. No. 15/940,627, titled DRIVE ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; U.S. patent application Ser. No. 15/940,637, titled COMMUNICATION ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; U.S. patent application Ser. No. 15/940,642, titled CONTROLS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; U.S. patent application Ser. No. 15/940,676, titled AUTOMATIC TOOL ADJUSTMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; U.S. patent application Ser. No. 15/940,680, titled CONTROLLERS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; U.S. patent application Ser. No. 15/940,683, titled COOPERATIVE SURGICAL ACTIONS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; U.S. patent application Ser. No. 15/940,690, titled DISPLAY ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; and U.S. patent application Ser. No. 15/940,711, titled SENSING ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS.

(170) Applicant of the present application owns the following U.S. Provisional Patent Applications, filed on Mar. 8, 2018, the disclosure of each of which is herein incorporated by reference in its entirety: U.S. Provisional Patent Application No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR; and U.S. Provisional Patent Application No. 62/640,415, titled ESTIMATING STATE OF ULTRASONIC END EFFECTOR AND CONTROL SYSTEM THEREFOR.

(171) Before explaining various aspects of surgical devices and generators in detail, it should be noted that the illustrative examples are not limited in application or use to the details of construction and arrangement of parts illustrated in the accompanying drawings and description. The illustrative examples may be implemented or incorporated in other aspects, variations and modifications, and may be practiced or carried out in various ways. Further, unless otherwise indicated, the terms and expressions employed herein have been chosen for the purpose of describing the illustrative examples for the convenience of the reader and are not for the purpose of limitation thereof. Also, it will be appreciated that one or more of the following-described aspects, expressions of aspects, and/or examples, can be combined with any one or more of the other following-described aspects, expressions of aspects and/or examples.

(172) Various aspects are directed to improved ultrasonic surgical devices, electrosurgical devices and generators for use therewith. Aspects of the ultrasonic surgical devices can be configured for transecting and/or coagulating tissue during surgical procedures, for example. Aspects of the electrosurgical devices can be configured for transecting, coagulating, scaling, welding and/or desiccating tissue during surgical procedures, for example.

(173) Referring to FIG. 1, a computer-implemented interactive surgical system **100** includes one or more surgical systems **102** and a cloud-based system (e.g., the cloud **104** that may include a remote server **113** coupled to a storage device **105**). Each surgical system **102** includes at least one surgical hub **106** in communication with the cloud **104** that may include a remote server **113**. In one example, as illustrated in FIG. 1, the surgical system **102** includes a visualization system **108**, a robotic system **110**, and a handheld intelligent surgical instrument **112**, which are configured to communicate with one another and/or the hub **106**. In some aspects, a surgical system **102** may include an M number of hubs **106**, an N number of visualization systems **108**, an O number of robotic systems **110**, and a P number of handheld intelligent surgical instruments **112**, where M, N, O, and P are integers greater than or equal to one.

(174) FIG. 3 depicts an example of a surgical system **102** being used to perform a surgical procedure on a patient who is lying down on an operating table **114** in a surgical operating room **116**. A robotic system **110** is used in the surgical procedure as a part of the surgical system **102**. The robotic system **110** includes a surgeon's console **118**, a patient side cart **120** (surgical robot), and a surgical robotic hub **122**. The patient side cart **120** can manipulate at least one removably coupled surgical tool **117** through a minimally invasive incision in the body of the patient while the surgeon views the surgical site through the surgeon's console **118**. An image of the surgical site can be obtained by a medical imaging device **124**, which can be manipulated by the patient side cart **120** to orient the imaging device **124**. The robotic hub **122** can be used to process the images of the surgical site for subsequent display to the surgeon through the surgeon's console **118**.

(175) Other types of robotic systems can be readily adapted for use with the surgical system **102**. Various examples of robotic systems and surgical tools that are suitable for use with the present disclosure are described in U.S. Provisional Patent Application Ser. No. 62/611,339, titled ROBOT ASSISTED SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety.

(176) Various examples of cloud-based analytics that are performed by the cloud **104**, and are suitable for use with the present disclosure, are described in U.S. Provisional Patent Application Ser. No. 62/611,340, titled CLOUD-BASED MEDICAL ANALYTICS, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety.

(177) In various aspects, the imaging device **124** includes at least one image sensor and one or more optical components. Suitable image sensors include, but are not limited to, Charge-Coupled Device (CCD) sensors and Complementary Metal-Oxide Semiconductor (CMOS) sensors.

(178) The optical components of the imaging device **124** may include one or more illumination sources and/or one or more lenses. The one or more illumination sources may be directed to illuminate portions of the surgical field. The one or more image sensors may receive light reflected or refracted from the surgical field, including light reflected or refracted from tissue and/or surgical instruments.

(179) The one or more illumination sources may be configured to radiate electromagnetic energy in the visible spectrum as well as the invisible spectrum. The visible spectrum, sometimes referred to as the optical spectrum or luminous spectrum, is that portion of the electromagnetic spectrum that is visible to (i.e., can be detected by) the human eye and may be referred to as visible light or simply light. A typical human eye will respond to wavelengths in air that are from about 380 nm to about 750 nm.

(180) The invisible spectrum (i.e., the non-luminous spectrum) is that portion of the electromagnetic spectrum that lies below and above the visible spectrum (i.e., wavelengths below about 380 nm and above about 750 nm). The invisible spectrum is not detectable by the human eye. Wavelengths greater than about 750 nm are longer than the red visible spectrum, and they become invisible infrared (IR), microwave, and radio electromagnetic radiation. Wavelengths less than about 380 nm are shorter than the violet spectrum, and they become invisible ultraviolet, x-ray, and gamma ray electromagnetic radiation.

(181) In various aspects, the imaging device **124** is configured for use in a minimally invasive procedure. Examples of imaging devices suitable for use with the present disclosure include, but not limited to, an arthroscope, angioscope, bronchoscope, choledochoscope, colonoscope, cytoscope, duodenoscope, enteroscope, esophagogastro-duodenoscope (gastroscope), endoscope, laryngoscope, nasopharyngo-neproscope, sigmoidoscope, thoracoscope, and ureterscope.

(182) In one aspect, the imaging device employs multi-spectrum monitoring to discriminate topography and underlying structures. A multi-spectral image is one that captures image data within specific wavelength ranges across the electromagnetic spectrum. The wavelengths may be separated by filters or by the use of instruments that are sensitive to particular wavelengths, including light from frequencies beyond the visible light range, e.g., IR and ultraviolet. Spectral imaging can allow extraction of additional information the human eye fails to capture with its receptors for red, green, and blue. The use of multi-spectral imaging is described in greater detail under the heading "Advanced Imaging Acquisition Module" in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety. Multi-spectrum monitoring can be a useful tool in relocating a surgical field after a surgical task is

completed to perform one or more of the previously described tests on the treated tissue.

(183) It is axiomatic that strict sterilization of the operating room and surgical equipment is required during any surgery. The strict hygiene and sterilization conditions required in a “surgical theater,” i.e., an operating or treatment room, necessitate the highest possible sterility of all medical devices and equipment. Part of that sterilization process is the need to sterilize anything that comes in contact with the patient or penetrates the sterile field, including the imaging device **124** and its attachments and components. It will be appreciated that the sterile field may be considered a specified area, such as within a tray or on a sterile towel, that is considered free of microorganisms, or the sterile field may be considered an area, immediately around a patient, who has been prepared for a surgical procedure. The sterile field may include the scrubbed team members, who are properly attired, and all furniture and fixtures in the area.

(184) In various aspects, the visualization system **108** includes one or more imaging sensors, one or more image-processing units, one or more storage arrays, and one or more displays that are strategically arranged with respect to the sterile field, as illustrated in FIG. 2. In one aspect, the visualization system **108** includes an interface for HL7, PACS, and EMR. Various components of the visualization system **108** are described under the heading “Advanced Imaging Acquisition Module” in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety.

(185) As illustrated in FIG. 2, a primary display **119** is positioned in the sterile field to be visible to an operator at the operating table **114**. In addition, a visualization tower **111** is positioned outside the sterile field. The visualization tower **111** includes a first non-sterile display **107** and a second non-sterile display **109**, which face away from each other. The visualization system **108**, guided by the hub **106**, is configured to utilize the displays **107**, **109**, and **119** to coordinate information flow to operators inside and outside the sterile field. For example, the hub **106** may cause the visualization system **108** to display a snapshot of a surgical site, as recorded by an imaging device **124**, on a non-sterile display **107** or **109**, while maintaining a live feed of the surgical site on the primary display **119**. The snapshot on the non-sterile display **107** or **109** can permit a non-sterile operator to perform a diagnostic step relevant to the surgical procedure, for example.

(186) In one aspect, the hub **106** is also configured to route a diagnostic input or feedback entered by a non-sterile operator at the visualization tower **111** to the primary display **119** within the sterile field, where it can be viewed by a sterile operator at the operating table. In one example, the input can be in the form of a modification to the snapshot displayed on the non-sterile display **107** or **109**, which can be routed to the primary display **119** by the hub **106**.

(187) Referring to FIG. 2, a surgical instrument **112** is being used in the surgical procedure as part of the surgical system **102**. The hub **106** is also configured to coordinate information flow to a display of the surgical instrument **112**. For example, in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety. A diagnostic input or feedback entered by a non-sterile operator at the visualization tower **111** can be routed by the hub **106** to the surgical instrument display **115** within the sterile field, where it can be viewed by the operator of the surgical instrument **112**. Example surgical instruments that are suitable for use with the surgical system **102** are described under the heading “Surgical Instrument Hardware” and in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety, for example.

(188) Referring now to FIG. 3, a hub **106** is depicted in communication with a visualization system **108**, a robotic system **110**, and a handheld intelligent surgical instrument **112**. The hub **106** includes a hub display **135**, an imaging module **138**, a generator module **140**, a communication module **130**, a processor module **132**, and a storage array **134**. In certain aspects, as illustrated in FIG. 3, the hub **106** further includes a smoke evacuation module **126** and/or a suction/irrigation module **128**.

(189) During a surgical procedure, energy application to tissue, for sealing and/or cutting, is generally associated with smoke evacuation, suction of excess fluid, and/or irrigation of the tissue. Fluid, power, and/or data lines from different sources are often entangled during the surgical procedure. Valuable time can be lost addressing this issue during a surgical procedure. Detangling the lines may necessitate disconnecting the lines from their respective modules, which may require resetting the modules. The hub modular enclosure **136** offers a unified environment for managing the power, data, and fluid lines, which reduces the frequency of entanglement between such lines.

(190) Aspects of the present disclosure present a surgical hub for use in a surgical procedure that involves energy application to tissue at a surgical site. The surgical hub includes a hub enclosure and a combo generator module slidably receivable in a docking station of the hub enclosure. The docking station includes data and power contacts. The combo generator module includes two or more of an ultrasonic energy generator component, a bipolar RF energy generator component, and a monopolar RF energy generator component that are housed in a single unit. In one aspect, the combo generator module also includes a smoke evacuation component, at least one energy delivery cable for connecting the combo generator module to a surgical instrument, at least one smoke evacuation component configured to evacuate smoke, fluid, and/or particulates generated by the application of therapeutic energy to the tissue, and a fluid line extending from the remote surgical site to the smoke evacuation component.

(191) In one aspect, the fluid line is a first fluid line and a second fluid line extends from the remote surgical site to a suction and irrigation module slidably received in the hub enclosure. In one aspect, the hub enclosure comprises a fluid interface.

(192) Certain surgical procedures may require the application of more than one energy type to the tissue. One energy type may be more beneficial for cutting the tissue, while another different energy type may be more beneficial for sealing the tissue. For example, a bipolar generator can be used to seal the tissue while an ultrasonic generator can be used to cut the sealed tissue. Aspects of the present disclosure present a solution where a hub modular enclosure **136** is configured to accommodate different generators, and facilitate an interactive communication therebetween. One of the advantages of the hub modular enclosure **136** is enabling the quick removal and/or replacement of various modules.

(193) Aspects of the present disclosure present a modular surgical enclosure for use in a surgical procedure that involves energy application to tissue. The modular surgical enclosure includes a first energy-generator module, configured to generate a first energy for application to the tissue, and a first docking station comprising a first docking port that includes first data and power contacts, wherein the first energy-generator module is slidably movable into an electrical engagement with the power and data contacts and wherein the first energy-generator module is slidably movable out of the electrical engagement with the first power and data contacts,

(194) Further to the above, the modular surgical enclosure also includes a second energy-generator module configured to generate a second energy, different than the first energy, for application to the tissue, and a second docking station comprising a second docking port that includes second data and power contacts, wherein the second energy-generator module is slidably movable into an electrical engagement with the power and data contacts, and wherein the second energy-generator module is slidably movable out of the electrical engagement with the second power and data contacts.

(195) In addition, the modular surgical enclosure also includes a communication bus between the first docking port and the second docking port, configured to facilitate communication between the first energy-generator module and the second energy-generator module.

(196) Referring to FIGS. 3-7, aspects of the present disclosure are presented for a hub modular enclosure **136** that allows the modular integration of a generator module **140**, a smoke evacuation module **126**, and a suction/irrigation module **128**. The hub modular enclosure **136** further facilitates interactive communication between the modules **140**, **126**, **128**. As illustrated in FIG. 5, the generator module **140** can be a generator module with integrated monopolar, bipolar, and ultrasonic components supported in a single housing unit **139** slidably insertable into the hub modular enclosure **136**. As illustrated in FIG. 5, the generator module **140** can be configured to connect to a monopolar device **146**, a bipolar device **147**, and an ultrasonic device **148**. Alternatively, the generator module **140** may comprise a series of monopolar, bipolar, and/or ultrasonic generator modules that interact through the hub modular enclosure **136**. The hub modular enclosure **136** can be configured to facilitate the insertion of multiple generators and interactive communication between the generators docked into the hub modular enclosure **136** so that the generators would act as a single

generator.

(197) In one aspect, the hub modular enclosure **136** comprises a modular power and communication backplane **149** with external and wireless communication headers to enable the removable attachment of the modules **140**, **126**, **128** and interactive communication therebetween.

(198) In one aspect, the hub modular enclosure **136** includes docking stations, or drawers, **151**, herein also referred to as drawers, which are configured to slidably receive the modules **140**, **126**, **128**. FIG. 4 illustrates a partial perspective view of a surgical hub enclosure **136**, and a combo generator module **145** slidably receivable in a docking station **151** of the surgical hub enclosure **136**. A docking port **152** with power and data contacts on a rear side of the combo generator module **145** is configured to engage a corresponding docking port **150** with power and data contacts of a corresponding docking station **151** of the hub modular enclosure **136** as the combo generator module **145** is slid into position within the corresponding docking station **151** of the hub module enclosure **136**. In one aspect, the combo generator module **145** includes a bipolar, ultrasonic, and monopolar module and a smoke evacuation module integrated together into a single housing unit **139**, as illustrated in FIG. 5.

(199) In various aspects, the smoke evacuation module **126** includes a fluid line **154** that conveys captured/collected smoke and/or fluid away from a surgical site and to, for example, the smoke evacuation module **126**. Vacuum suction originating from the smoke evacuation module **126** can draw the smoke into an opening of a utility conduit at the surgical site. The utility conduit, coupled to the fluid line, can be in the form of a flexible tube terminating at the smoke evacuation module **126**. The utility conduit and the fluid line define a fluid path extending toward the smoke evacuation module **126** that is received in the hub enclosure **136**.

(200) In various aspects, the suction/irrigation module **128** is coupled to a surgical tool comprising an aspiration fluid line and a suction fluid line. In one example, the aspiration and suction fluid lines are in the form of flexible tubes extending from the surgical site toward the suction/irrigation module **128**. One or more drive systems can be configured to cause irrigation and aspiration of fluids to and from the surgical site.

(201) In one aspect, the surgical tool includes a shaft having an end effector at a distal end thereof and at least one energy treatment associated with the end effector, an aspiration tube, and an irrigation tube. The aspiration tube can have an inlet port at a distal end thereof and the aspiration tube extends through the shaft. Similarly, an irrigation tube can extend through the shaft and can have an inlet port in proximity to the energy deliver implement. The energy deliver implement is configured to deliver ultrasonic and/or RF energy to the surgical site and is coupled to the generator module **140** by a cable extending initially through the shaft.

(202) The irrigation tube can be in fluid communication with a fluid source, and the aspiration tube can be in fluid communication with a vacuum source. The fluid source and/or the vacuum source can be housed in the suction/irrigation module **128**. In one example, the fluid source and/or the vacuum source can be housed in the hub enclosure **136** separately from the suction/irrigation module **128**. In such example, a fluid interface can be configured to connect the suction/irrigation module **128** to the fluid source and/or the vacuum source.

(203) In one aspect, the modules **140**, **126**, **128** and/or their corresponding docking stations on the hub modular enclosure **136** may include alignment features that are configured to align the docking ports of the modules into engagement with their counterparts in the docking stations of the hub modular enclosure **136**. For example, as illustrated in FIG. 4, the combo generator module **145** includes side brackets **155** that are configured to slidably engage with corresponding brackets **156** of the corresponding docking station **151** of the hub modular enclosure **136**. The brackets cooperate to guide the docking port contacts of the combo generator module **145** into an electrical engagement with the docking port contacts of the hub modular enclosure **136**.

(204) In some aspects, the drawers **151** of the hub modular enclosure **136** are the same, or substantially the same size, and the modules are adjusted in size to be received in the drawers **151**. For example, the side brackets **155** and/or **156** can be larger or smaller depending on the size of the module. In other aspects, the drawers **151** are different in size and are each designed to accommodate a particular module.

(205) Furthermore, the contacts of a particular module can be keyed for engagement with the contacts of a particular drawer to avoid inserting a module into a drawer with mismatching contacts.

(206) As illustrated in FIG. 4, the docking port **150** of one drawer **151** can be coupled to the docking port **150** of another drawer **151** through a communications link **157** to facilitate an interactive communication between the modules housed in the hub modular enclosure **136**. The docking ports **150** of the hub modular enclosure **136** may alternatively, or additionally, facilitate a wireless interactive communication between the modules housed in the hub modular enclosure **136**. Any suitable wireless communication can be employed, such as for example Air Titan-Bluetooth.

(207) FIG. 6 illustrates individual power bus attachments for a plurality of lateral docking ports of a lateral modular housing **160** configured to receive a plurality of modules of a surgical hub **206**. The lateral modular housing **160** is configured to laterally receive and interconnect the modules **161**. The modules **161** are slidably inserted into docking stations **162** of lateral modular housing **160**, which includes a backplane for interconnecting the modules **161**. As illustrated in FIG. 6, the modules **161** are arranged laterally in the lateral modular housing **160**. Alternatively, the modules **161** may be arranged vertically in a lateral modular housing.

(208) FIG. 7 illustrates a vertical modular housing **164** configured to receive a plurality of modules **165** of the surgical hub **106**. The modules **165** are slidably inserted into docking stations, or drawers, **167** of vertical modular housing **164**, which includes a backplane for interconnecting the modules **165**. Although the drawers **167** of the vertical modular housing **164** are arranged vertically, in certain instances, a vertical modular housing **164** may include drawers that are arranged laterally. Furthermore, the modules **165** may interact with one another through the docking ports of the vertical modular housing **164**. In the example of FIG. 7, a display **177** is provided for displaying data relevant to the operation of the modules **165**. In addition, the vertical modular housing **164** includes a master module **178** housing a plurality of sub-modules that are slidably received in the master module **178**.

(209) In various aspects, the imaging module **138** comprises an integrated video processor and a modular light source and is adapted for use with various imaging devices. In one aspect, the imaging device is comprised of a modular housing that can be assembled with a light source module and a camera module. The housing can be a disposable housing. In at least one example, the disposable housing is removably coupled to a reusable controller, a light source module, and a camera module. The light source module and/or the camera module can be selectively chosen depending on the type of surgical procedure. In one aspect, the camera module comprises a CCD sensor. In another aspect, the camera module comprises a CMOS sensor. In another aspect, the camera module is configured for scanned beam imaging. Likewise, the light source module can be configured to deliver a white light or a different light, depending on the surgical procedure.

(210) During a surgical procedure, removing a surgical device from the surgical field and replacing it with another surgical device that includes a different camera or a different light source can be inefficient. Temporarily losing sight of the surgical field may lead to undesirable consequences. The module imaging device of the present disclosure is configured to permit the replacement of a light source module or a camera module midstream during a surgical procedure, without having to remove the imaging device from the surgical field.

(211) In one aspect, the imaging device comprises a tubular housing that includes a plurality of channels. A first channel is configured to slidably receive the camera module, which can be configured for a snap-fit engagement with the first channel. A second channel is configured to slidably receive the light source module, which can be configured for a snap-fit engagement with the second channel. In another example, the camera module and/or the light source module can be rotated into a final position within their respective channels. A threaded engagement can be employed in lieu of the snap-fit engagement.

(212) In various examples, multiple imaging devices are placed at different positions in the surgical field to provide multiple views. The imaging module **138** can be configured to switch between the imaging devices to provide an optimal view. In various aspects, the imaging module **138** can be configured to integrate the images from the different imaging device.

(213) Various image processors and imaging devices suitable for use with the present disclosure are described in U.S. Pat. No. 7,995,045, titled COMBINED SBI AND CONVENTIONAL IMAGE PROCESSOR, which issued on Aug. 9, 2011, which is herein incorporated by reference in its

entirety. U.S. Pat. No. 7,982,718, titled *SBI MOTION ARTIFACT REMOVAL APPARATUS AND METHOD*, which issued on Jul. 19, 2011, which is herein incorporated by reference in its entirety, describes various systems for removing motion artifacts from image data. Such systems can be integrated with the imaging module **138**. Furthermore, U.S. Patent Application Publication No. 2011/0306840, titled *CONTROLLABLE MAGNETIC SOURCE TO FIXTURE INTRACORPOREAL APPARATUS*, which published on Dec. 15, 2011, and U.S. Patent Application Publication No. 2014/0243597, titled *SYSTEM FOR PERFORMING A MINIMALLY INVASIVE SURGICAL PROCEDURE*, which published on Aug. 28, 2014, each of which is herein incorporated by reference in its entirety.

(214) FIG. **8** illustrates a surgical data network **201** comprising a modular communication hub **203** configured to connect modular devices located in one or more operating theaters of a healthcare facility, or any room in a healthcare facility specially equipped for surgical operations, to a cloud-based system (e.g., the cloud **204** that may include a remote server **213** coupled to a storage device **205**). In one aspect, the modular communication hub **203** comprises a network hub **207** and/or a network switch **209** in communication with a network router. The modular communication hub **203** also can be coupled to a local computer system **210** to provide local computer processing and data manipulation. The surgical data network **201** may be configured as passive, intelligent, or switching. A passive surgical data network serves as a conduit for the data, enabling it to go from one device (or segment) to another and to the cloud computing resources. An intelligent surgical data network includes additional features to enable the traffic passing through the surgical data network to be monitored and to configure each port in the network hub **207** or network switch **209**. An intelligent surgical data network may be referred to as a manageable hub or switch. A switching hub reads the destination address of each packet and then forwards the packet to the correct port.

(215) Modular devices **1a-1n** located in the operating theater may be coupled to the modular communication hub **203**. The network hub **207** and/or the network switch **209** may be coupled to a network router **211** to connect the devices **1a-1n** to the cloud **204** or the local computer system **210**. Data associated with the devices **1a-1n** may be transferred to cloud-based computers via the router for remote data processing and manipulation. Data associated with the devices **1a-1n** may also be transferred to the local computer system **210** for local data processing and manipulation. Modular devices **2a-2m** located in the same operating theater also may be coupled to a network switch **209**. The network switch **209** may be coupled to the network hub **207** and/or the network router **211** to connect to the devices **2a-2m** to the cloud **204**. Data associated with the devices **2a-2m** may be transferred to the cloud **204** via the network router **211** for data processing and manipulation. Data associated with the devices **2a-2m** may also be transferred to the local computer system **210** for local data processing and manipulation.

(216) It will be appreciated that the surgical data network **201** may be expanded by interconnecting multiple network hubs **207** and/or multiple network switches **209** with multiple network routers **211**. The modular communication hub **203** may be contained in a modular control tower configured to receive multiple devices **1a-1n/2a-2m**. The local computer system **210** also may be contained in a modular control tower. The modular communication hub **203** is connected to a display **212** to display images obtained by some of the devices **1a-1n/2a-2m**, for example during surgical procedures. In various aspects, the devices **1a-1n/2a-2m** may include, for example, various modules such as an imaging module **138** coupled to an endoscope, a generator module **140** coupled to an energy-based surgical device, a smoke evacuation module **126**, a suction/irrigation module **128**, a communication module **130**, a processor module **132**, a storage array **134**, a surgical device coupled to a display, and/or a non-contact sensor module, among other modular devices that may be connected to the modular communication hub **203** of the surgical data network **201**.

(217) In one aspect, the surgical data network **201** may comprise a combination of network hub(s), network switch(es), and network router(s) connecting the devices **1a-1n/2a-2m** to the cloud. Any one of or all of the devices **1a-1n/2a-2m** coupled to the network hub or network switch may collect data in real time and transfer the data to cloud computers for data processing and manipulation. It will be appreciated that cloud computing relies on sharing computing resources rather than having local servers or personal devices to handle software applications. The word “cloud” may be used as a metaphor for “the Internet,” although the term is not limited as such. Accordingly, the term “cloud computing” may be used herein to refer to “a type of Internet-based computing,” where different services—such as servers, storage, and applications—are delivered to the modular communication hub **203** and/or computer system **210** located in the surgical theater (e.g., a fixed, mobile, temporary, or field operating room or space) and to devices connected to the modular communication hub **203** and/or computer system **210** through the Internet. The cloud infrastructure may be maintained by a cloud service provider. In this context, the cloud service provider may be the entity that coordinates the usage and control of the devices **1a-1n/2a-2m** located in one or more operating theaters. The cloud computing services can perform a large number of calculations based on the data gathered by smart surgical instruments, robots, and other computerized devices located in the operating theater. The hub hardware enables multiple devices or connections to be connected to a computer that communicates with the cloud computing resources and storage.

(218) Applying cloud computer data processing techniques on the data collected by the devices **1a-1n/2a-2m**, the surgical data network provides improved surgical outcomes, reduced costs, and improved patient satisfaction. At least some of the devices **1a-1n/2a-2m** may be employed to view tissue states to assess leaks or perfusion of sealed tissue after a tissue sealing and cutting procedure. At least some of the devices **1a-1n/2a-2m** may be employed to identify pathology, such as the effects of diseases, using the cloud-based computing to examine data including images of samples of body tissue for diagnostic purposes. This includes localization and margin confirmation of tissue and phenotypes. At least some of the devices **1a-1n/2a-2m** may be employed to identify anatomical structures of the body using a variety of sensors integrated with imaging devices and techniques such as overlaying images captured by multiple imaging devices. The data gathered by the devices **1a-1n/2a-2m**, including image data, may be transferred to the cloud **204** or the local computer system **210** or both for data processing and manipulation including image processing and manipulation. The data may be analyzed to improve surgical procedure outcomes by determining if further treatment, such as the application of endoscopic intervention, emerging technologies, a targeted radiation, targeted intervention, and precise robotics to tissue-specific sites and conditions, may be pursued. Such data analysis may further employ outcome analytics processing, and using standardized approaches may provide beneficial feedback to either confirm surgical treatments and the behavior of the surgeon or suggest modifications to surgical treatments and the behavior of the surgeon.

(219) In one implementation, the operating theater devices **1a-1n** may be connected to the modular communication hub **203** over a wired channel or a wireless channel depending on the configuration of the devices **1a-1n** to a network hub. The network hub **207** may be implemented, in one aspect, as a local network broadcast device that works on the physical layer of the Open System Interconnection (OSI) model. The network hub provides connectivity to the devices **1a-1n** located in the same operating theater network. The network hub **207** collects data in the form of packets and sends them to the router in half duplex mode. The network hub **207** does not store any media access control/Internet Protocol (MAC/IP) to transfer the device data. Only one of the devices **1a-1n** can send data at a time through the network hub **207**. The network hub **207** has no outing tables or intelligence regarding where to send information and broadcasts all network data across each connection and to a remote server **213** (FIG. **9**) over the cloud **204**. The network hub **207** can detect basic network errors such as collisions, but having all information broadcast to multiple ports can be a security risk and cause bottlenecks.

(220) In another implementation, the operating theater devices **2a-2m** may be connected to a network switch **209** over a wired channel or a wireless channel. The network switch **209** works in the data link layer of the OSI model. The network switch **209** is a multicast device for connecting the devices **2a-2m** located in the same operating theater to the network. The network switch **209** sends data in the form of frames to the network router **211** and works in full duplex mode. Multiple devices **2a-2m** can send data at the same time through the network switch **209**. The network switch **209** stores and uses MAC addresses of the devices **2a-2m** to transfer data.

(221) The network hub **207** and/or the network switch **209** are coupled to the network router **211** for connection to the cloud **204**. The network router **211** works in the network layer of the OSI model. The network router **211** creates a route for transmitting data packets received from the network hub **207** and/or network switch **211** to cloud-based computer resources for further processing and manipulation of the data collected by any one of or all the devices **1a-1n/2a-2m**. The network router **211** may be employed to connect two or more different networks located in different locations, such as,

for example, different operating theaters or theaters of the same healthcare facility or different theaters located in different operating theaters of different healthcare facilities. The network router **211** sends data in the form of packets to the cloud **204** and works in full duplex mode. Multiple devices can send data at the same time. The network router **211** uses IP addresses to transfer data.

(222) In one example, the network hub **207** may be implemented as a USB hub, which allows multiple USB devices to be connected to a host computer. The USB hub may expand a single USB port into several tiers so that there are more ports available to connect devices to the host system computer. The network hub **207** may include wired or wireless capabilities to receive information over a wired channel or a wireless channel. In one aspect, a wireless USB short-range, high-bandwidth wireless radio communication protocol may be employed for communication between the devices **1a-1n** and devices **2a-2m** located in the operating theater.

(223) In other examples, the operating theater devices **1a-1n/2a-2m** may communicate to the modular communication hub **203** via Bluetooth wireless technology standard for exchanging data over short distances (using short-wavelength UHF radio waves in the ISM band from 2.4 to 2.485 GHz) from fixed and mobile devices and building personal area networks (PANs). In other aspects, the operating theater devices **1a-1n/2a-2m** may communicate to the modular communication hub **203** via a number of wireless or wired communication standards or protocols, including but not limited to Wi-Fi (IEEE 802.11 family), WiMAX (IEEE 802.16 family), IEEE 802.20, long-term evolution (LTE), and Ev-DO, HSPA+, HSDPA+, HSUPA+, EDGE, GSM, GPRS, CDMA, TDMA, DECT, and Ethernet derivatives thereof, as well as any other wireless and wired protocols that are designated as 3G, 4G, 5G, and beyond. The computing module may include a plurality of communication modules. For instance, a first communication module may be dedicated to shorter-range wireless communications such as Wi-Fi and Bluetooth, and a second communication module may be dedicated to longer-range wireless communications such as GPS, EDGE, GPRS, CDMA, WiMAX, LTE, Ev-DO, and others.

(224) The modular communication hub **203** may serve as a central connection for one or all of the operating theater devices **1a-1n/2a-2m** and handles a data type known as frames. Frames carry the data generated by the devices **1a-1n/2a-2m**. When a frame is received by the modular communication hub **203**, it is amplified and transmitted to the network router **211**, which transfers the data to the cloud computing resources by using a number of wireless or wired communication standards or protocols, as described herein.

(225) The modular communication hub **203** can be used as a standalone device or be connected to compatible network hubs and network switches to form a larger network. The modular communication hub **203** is generally easy to install, configure, and maintain, making it a good option for networking the operating theater devices **1a-1n/2a-2m**.

(226) FIG. 9 illustrates a computer-implemented interactive surgical system **200**. The computer-implemented interactive surgical system **200** is similar in many respects to the computer-implemented interactive surgical system **100**. For example, the computer-implemented interactive surgical system **200** includes one or more surgical systems **202**, which are similar in many respects to the surgical systems **102**. Each surgical system **202** includes at least one surgical hub **206** in communication with a cloud **204** that may include a remote server **213**. In one aspect, the computer-implemented interactive surgical system **200** comprises a modular control tower **236** connected to multiple operating theater devices such as, for example, intelligent surgical instruments, robots, and other computerized devices located in the operating theater. As shown in FIG. 10, the modular control tower **236** comprises a modular communication hub **203** coupled to a computer system **210**. As illustrated in the example of FIG. 9, the modular control tower **236** is coupled to an imaging module **238** that is coupled to an endoscope **239**, a generator module **240** that is coupled to an energy device **241**, a smoke evacuator module **226**, a suction/irrigation module **228**, a communication module **230**, a processor module **232**, a storage array **234**, a smart device/instrument **235** optionally coupled to a display **237**, and a non-contact sensor module **242**. The operating theater devices are coupled to cloud computing resources and data storage via the modular control tower **236**. A robot hub **222** also may be connected to the modular control tower **236** and to the cloud computing resources. The devices/instruments **235**, visualization systems **208**, among others, may be coupled to the modular control tower **236** via wired or wireless communication standards or protocols, as described herein. The modular control tower **236** may be coupled to a hub display **215** (e.g., monitor, screen) to display and overlay images received from the imaging module, device/instrument display, and/or other visualization systems **208**. The hub display also may display data received from devices connected to the modular control tower in conjunction with images and overlaid images.

(227) FIG. 10 illustrates a surgical hub **206** comprising a plurality of modules coupled to the modular control tower **236**. The modular control tower **236** comprises a modular communication hub **203**, e.g., a network connectivity device, and a computer system **210** to provide local processing, visualization, and imaging, for example. As shown in FIG. 10, the modular communication hub **203** may be connected in a tiered configuration to expand the number of modules (e.g., devices) that may be connected to the modular communication hub **203** and transfer data associated with the modules to the computer system **210**, cloud computing resources, or both. As shown in FIG. 10, each of the network hubs/switches in the modular communication hub **203** includes three downstream ports and one upstream port. The upstream network hub/switch is connected to a processor to provide a communication connection to the cloud computing resources and a local display **217**. Communication to the cloud **204** may be made either through a wired or a wireless communication channel.

(228) The surgical hub **206** employs a non-contact sensor module **242** to measure the dimensions of the operating theater and generate a map of the surgical theater using either ultrasonic or laser-type non-contact measurement devices. An ultrasound-based non-contact sensor module scans the operating theater by transmitting a burst of ultrasound and receiving the echo when it bounces off the perimeter walls of an operating theater as described under the heading "Surgical Hub Spatial Awareness Within an Operating Room" in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, which is herein incorporated by reference in its entirety, in which the sensor module is configured to determine the size of the operating theater and to adjust Bluetooth-pairing distance limits. A laser-based non-contact sensor module scans the operating theater by transmitting laser light pulses, receiving laser light pulses that bounce off the perimeter walls of the operating theater, and comparing the phase of the transmitted pulse to the received pulse to determine the size of the operating theater and to adjust Bluetooth pairing distance limits, for example.

(229) The computer system **210** comprises a processor **244** and a network interface **245**. The processor **244** is coupled to a communication module **247**, storage **248**, memory **249**, non-volatile memory **250**, and input/output interface **251** via a system bus. The system bus can be any of several types of bus structure(s) including the memory bus or memory controller, a peripheral bus or external bus, and/or a local bus using any variety of available bus architectures including, but not limited to, 9-bit bus, Industrial Standard Architecture (ISA), Micro-Charmel Architecture (MSA), Extended ISA (EISA), Intelligent Drive Electronics (IDE), VESA Local Bus (VLB), Peripheral Component Interconnect (PCI), USB, Advanced Graphics Port (AGP), Personal Computer Memory Card International Association bus (PCMCIA), Small Computer Systems Interface (SCSI), or any other proprietary bus.

(230) The processor **244** may be any single-core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In one aspect, the processor may be an LM4F230H5QR ARM Cortex-M4F Processor Core, available from Texas Instruments, for example, comprising an on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle serial random access memory (SRAM), an internal read-only memory (ROM) loaded with StellarisWare® software, a 2 KB electrically erasable programmable read-only memory (EEPROM), and/or one or more pulse width modulation (PWM) modules, one or more quadrature encoder inputs (QEI) analogs, one or more 12-bit analog-to-digital converters (ADCs) with 12 analog input channels, details of which are available for the product datasheet.

(231) In one aspect, the processor **244** may comprise a safety controller comprising two controller-based families such as TMS570 and RM4x, known under the trade name Hercules ARM Cortex R4, also by Texas Instruments. The safety controller may be configured specifically for IEC 61508 and ISO 26262 safety critical applications, among others, to provide advanced integrated safety features while delivering scalable performance, connectivity, and memory options.

(232) The system memory includes volatile memory and non-volatile memory. The basic input/output system (BIOS), containing the basic routines to transfer information between elements within the computer system, such as during start-up, is stored in non-volatile memory. For example, the non-volatile memory can include ROM, programmable ROM (PROM), electrically programmable ROM (EPROM), EEPROM, or flash memory. Volatile memory includes random-access memory (RAM), which acts as external cache memory. Moreover, RAM is available in many forms such as SRAM, dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SLDRAM), and direct Rambus RAM (DRRAM).

(233) The computer system **210** also includes removable/non-removable, volatile/non-volatile computer storage media, such as for example disk storage. The disk storage includes, but is not limited to, devices like a magnetic disk drive, floppy disk drive, tape drive, Jaz drive, Zip drive, LS-60 drive, flash memory card, or memory stick. In addition, the disk storage can include storage media separately or in combination with other storage media including, but not limited to, an optical disc drive such as a compact disc ROM device (CD-ROM), compact disc recordable drive (CD-R Drive), compact disc rewritable drive (CD-RW Drive), or a digital versatile disc ROM drive (DVD-ROM). To facilitate the connection of the disk storage devices to the system bus, a removable or non-removable interface may be employed.

(234) It is to be appreciated that the computer system **210** includes software that acts as an intermediary between users and the basic computer resources described in a suitable operating environment. Such software includes an operating system. The operating system, which can be stored on the disk storage, acts to control and allocate resources of the computer system. System applications take advantage of the management of resources by the operating system through program modules and program data stored either in the system memory or on the disk storage. It is to be appreciated that various components described herein can be implemented with various operating systems or combinations of operating systems.

(235) A user enters commands or information into the computer system **210** through input device(s) coupled to the I/O interface **251**. The input devices include, but are not limited to, a pointing device such as a mouse, trackball, stylus, touch pad, keyboard, microphone, joystick, game pad, satellite dish, scanner, TV tuner card, digital camera, digital video camera, web camera, and the like. These and other input devices connect to the processor through the system bus via interface port(s). The interface port(s) include, for example, a serial port, a parallel port, a game port, and a USB. The output device(s) use some of the same types of ports as input device(s). Thus, for example, a USB port may be used to provide input to the computer system and to output information from the computer system to an output device. An output adapter is provided to illustrate that there are some output devices like monitors, displays, speakers, and printers, among other output devices that require special adapters. The output adapters include, by way of illustration and not limitation, video and sound cards that provide a means of connection between the output device and the system bus. It should be noted that other devices and/or systems of devices, such as remote computer(s), provide both input and output capabilities.

(236) The computer system **210** can operate in a networked environment using logical connections to one or more remote computers, such as cloud computer(s), or local computers. The remote cloud computer(s) can be a personal computer, server, router, network PC, workstation, microprocessor-based appliance, peer device, or other common network node, and the like, and typically includes many or all of the elements described relative to the computer system. For purposes of brevity, only a memory storage device is illustrated with the remote computer(s). The remote computer(s) is logically connected to the computer system through a network interface and then physically connected via a communication connection. The network interface encompasses communication networks such as local area networks (LANs) and wide area networks (WANs). LAN technologies include Fiber Distributed Data Interface (FDDI), Copper Distributed Data Interface (CDDI), Ethernet/IEEE 802.3, Token Ring/IEEE 802.5 and the like. WAN technologies include, but are not limited to, point-to-point links, circuit-switching networks like Integrated Services Digital Networks (ISDN) and variations thereon, packet-switching networks, and Digital Subscriber Lines (DSL).

(237) In various aspects, the computer system **210** of FIG. **10**, the imaging module **238** and/or visualization system **208**, and/or the processor module **232** of FIGS. **9-10**, may comprise an image processor, image-processing engine, media processor, or any specialized digital signal processor (DSP) used for the processing of digital images. The image processor may employ parallel computing with single instruction, multiple data (SIMD) or multiple instruction, multiple data (MIMD) technologies to increase speed and efficiency. The digital image-processing engine can perform a range of tasks. The image processor may be a system on a chip with multicore processor architecture.

(238) The communication connection(s) refers to the hardware/software employed to connect the network interface to the bus. While the communication connection is shown for illustrative clarity inside the computer system, it can also be external to the computer system **210**. The hardware/software necessary for connection to the network interface includes, for illustrative purposes only, internal and external technologies such as modems, including regular telephone-grade modems, cable modems, and DSL modems, ISDN adapters, and Ethernet cards.

(239) FIG. **11** illustrates a functional block diagram of one aspect of a USB network hub **300** device, in accordance with at least one aspect of the present disclosure. In the illustrated aspect, the USB network hub device **300** employs a TUSB2036 integrated circuit hub by Texas Instruments. The USB network hub **300** is a CMOS device that provides an upstream USB transceiver port **302** and up to three downstream USB transceiver ports **304**, **306**, **308** in compliance with the USB 2.0 specification. The upstream USB transceiver port **302** is a differential root data port comprising a differential data minus (DM0) input paired with a differential data plus (DP0) input. The three downstream USB transceiver ports **304**, **306**, **308** are differential data ports where each port includes differential data plus (DP1-DP3) outputs paired with differential data minus (DM1-DM3) outputs.

(240) The USB network hub **300** device is implemented with a digital state machine instead of a microcontroller, and no firmware programming is required. Fully compliant USB transceivers are integrated into the circuit for the upstream USB transceiver port **302** and all downstream USB transceiver ports **304**, **306**, **308**. The downstream USB transceiver ports **304**, **306**, **308** support both full-speed and low-speed devices by automatically setting the slew rate according to the speed of the device attached to the ports. The USB network hub **300** device may be configured either in bus-powered or self-powered mode and includes a hub power logic **312** to manage power.

(241) The USB network hub **300** device includes a serial interface engine **310** (SIE). The SIE **310** is the front end of the USB network hub **300** hardware and handles most of the protocol described in chapter 8 of the USB specification. The SIE **310** typically comprehends signaling up to the transaction level. The functions that it handles could include: packet recognition, transaction sequencing, SOP, EOP, RESET, and RESUME signal detection/generation, clock/data separation, non-return-to-zero invert (NRZI) data encoding/decoding and bit-stuffing, CRC generation and checking (token and data), packet ID (PID) generation and checking/decoding, and/or serial-parallel/parallel-serial conversion. The **310** receives a clock input **314** and is coupled to a suspend/resume logic and frame timer **316** circuit and a hub repeater circuit **318** to control communication between the upstream USB transceiver port **302** and the downstream USB transceiver ports **304**, **306**, **308** through port logic circuits **320**, **322**, **324**. The SIE **310** is coupled to a command decoder **326** via interface logic to control commands from a serial EEPROM via a serial EEPROM interface **330**.

(242) In various aspects, the USB network hub **300** can connect **127** functions configured in up to six logical layers (tiers) to a single computer. Further, the USB network hub **300** can connect to all peripherals using a standardized four-wire cable that provides both communication and power distribution. The power configurations are bus-powered and self-powered modes. The USB network hub **300** may be configured to support four modes of power management: a bus-powered hub, with either individual-port power management or ganged-port power management, and the self-powered hub, with either individual-port power management or ganged-port power management. In one aspect, using a USB cable, the USB network hub **300**, the upstream USB transceiver port **302** is plugged into a USB host controller, and the downstream USB transceiver ports **304**, **306**, **308** are exposed for connecting USB compatible devices, and so forth.

Surgical Instrument Hardware

(243) FIG. **12** illustrates a logic diagram of a control system **470** of a surgical instrument or tool in accordance with one or more aspects of the present disclosure. The system **470** comprises a control circuit. The control circuit includes a microcontroller **461** comprising a processor **462** and a memory **468**. One or more of sensors **472**, **474**, **476**, for example, provide real-time feedback to the processor **462**. A motor **482**, driven by a motor driver **492**, operably couples a longitudinally movable displacement member to drive a clamp arm closure member. A tracking system **480** is

configured to determine the position of the longitudinally movable drive member. The position information is provided to the processor **462**, which can be programmed or configured to determine the position of the longitudinally movable drive member as well as the position of the closure member. Additional motors may be provided at the tool driver interface to control closure tube travel, shaft rotation, articulation, or clamp arm closure, or a combination of the above. A display **473** displays a variety of operating conditions of the instruments and may include touch screen functionality for data input. Information displayed on the display **473** may be overlaid with images acquired via endoscopic imaging modules.

(244) In one aspect, the microcontroller **461** may be any single-core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In one aspect, the main microcontroller **461** may be an LM4F230H5QR ARM Cortex-M4F Processor Core, available from Texas Instruments, for example, comprising an on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle SRAM, and internal ROM loaded with StellarisWare® software, a 2 KB EEPROM, one or more PWM modules, one or more QEI analogs, and/or one or more 12-bit ADCs with 12 analog input channels, details of which are available for the product datasheet.

(245) In one aspect, the microcontroller **461** may comprise a safety controller comprising two controller-based families such as TMS570 and RM4x, known under the trade name Hercules ARM Cortex R4, also by Texas Instruments. The safety controller may be configured specifically for IEC 61508 and ISO 26262 safety critical applications, among others, to provide advanced integrated safety features while delivering scalable performance, connectivity, and memory options.

(246) The microcontroller **461** may be programmed to perform various functions such as precise control over the speed and position of the knife, articulation systems, clamp arm, or a combination of the above. In one aspect, the microcontroller **461** includes a processor **462** and a memory **468**. The electric motor **482** may be a brushed direct current (DC) motor with a gearbox and mechanical links to an articulation or knife system. In one aspect, a motor driver **492** may be an A3941 available from Allegro Microsystems, Inc. Other motor drivers may be readily substituted for use in the tracking system **480** comprising an absolute positioning system. A detailed description of an absolute positioning system is described in U.S. Patent Application Publication No. 2017/0296213, titled SYSTEMS AND METHODS FOR CONTROLLING A SURGICAL STAPLING AND CUTTING INSTRUMENT, which published on Oct. 19, 2017, which is herein incorporated by reference in its entirety.

(247) The microcontroller **461** may be programmed to provide precise control over the speed and position of displacement members and articulation systems. The microcontroller **461** may be configured to compute a response in the software of the microcontroller **461**. The computed response is compared to a measured response of the actual system to obtain an “observed” response, which is used for actual feedback decisions. The observed response is a favorable, tuned value that balances the smooth, continuous nature of the simulated response with the measured response, which can detect outside influences on the system.

(248) In one aspect, the motor **482** may be controlled by the motor driver **492** and can be employed by the firing system of the surgical instrument or tool. In various forms, the motor **482** may be a brushed DC driving motor having a maximum rotational speed of approximately 25,000 RPM. In other arrangements, the motor **482** may include a brushless motor, a cordless motor, a synchronous motor, a stepper motor, or any other suitable electric motor. The motor driver **492** may comprise an H-bridge driver comprising field-effect transistors (FETs), for example. The motor **482** can be powered by a power assembly releasably mounted to the handle assembly or tool housing for supplying control power to the surgical instrument or tool. The power assembly may comprise a battery which may include a number of battery cells connected in series that can be used as the power source to power the surgical instrument or tool. In certain circumstances, the battery cells of the power assembly may be replaceable and/or rechargeable battery cells. In at least one example, the battery cells can be lithium-ion batteries which can be couplable to and separable from the power assembly.

(249) The motor driver **492** may be an A3941 available from Allegro Microsystems, Inc. The A3941 **492** is a full-bridge controller for use with external N-channel power metal-oxide semiconductor field-effect transistors (MOSFETs) specifically designed for inductive loads, such as brush DC motors. The driver **492** comprises a unique charge pump regulator that provides full (>10 V) gate drive for battery voltages down to 7 V and allows the A3941 to operate with a reduced gate drive, down to 5.5 V. A bootstrap capacitor may be employed to provide the above battery supply voltage required for N-channel MOSFETs. An internal charge pump for the high-side drive allows DC (100% duty cycle) operation. The full bridge can be driven in fast or slow decay modes using diode or synchronous rectification. In the slow decay mode, current recirculation can be through the high-side or the low-side FETs. The power FETs are protected from shoot-through by resistor-adjustable dead time. Integrated diagnostics provide indications of undervoltage, overtemperature, and power bridge faults and can be configured to protect the power MOSFETs under most short circuit conditions. Other motor drivers may be readily substituted for use in the tracking system **480** comprising an absolute positioning system.

(250) The tracking system **480** comprises a controlled motor drive circuit arrangement comprising a position sensor **472** according to one aspect of this disclosure. The position sensor **472** for an absolute positioning system provides a unique position signal corresponding to the location of a displacement member. In one aspect, the displacement member represents a longitudinally movable drive member comprising a rack of drive teeth for meshing engagement with a corresponding drive gear of a gear reducer assembly. In other aspects, the displacement member represents the firing member, which could be adapted and configured to include a rack of drive teeth. In yet another aspect, the displacement member represents a longitudinal displacement member to open and close a clamp arm, which can be adapted and configured to include a rack of drive teeth. In other aspects, the displacement member represents a clamp arm closure member configured to close and to open a clamp arm of a stapler, ultrasonic, or electrosurgical device, or combinations of the above. Accordingly, as used herein, the term displacement member is used generically to refer to any movable member of the surgical instrument or tool such as the drive member, the clamp arm, or any element that can be displaced. Accordingly, the absolute positioning system can, in effect, track the displacement of the clamp arm by tracking the linear displacement of the longitudinally movable drive member.

(251) In other aspects, the absolute positioning system can be configured to track the position of a clamp arm in the process of closing or opening. In various other aspects, the displacement member may be coupled to any position sensor **472** suitable for measuring linear displacement. Thus, the longitudinally movable drive member, or clamp arm, or combinations thereof, may be coupled to any suitable linear displacement sensor. Linear displacement sensors may include contact or non-contact displacement sensors. Linear displacement sensors may comprise linear variable differential transformers (LVDT), differential variable reluctance transducers (DVRT), a slide potentiometer, a magnetic sensing system comprising a movable magnet and a series of linearly arranged Hall effect sensors, a magnetic sensing system comprising a fixed magnet and a series of movable, linearly arranged Hall effect sensors, an optical sensing system comprising a movable light source and a series of linearly arranged photo diodes or photo detectors, an optical sensing system comprising a fixed light source and a series of movable linearly, arranged photo diodes or photo detectors, or any combination thereof.

(252) The electric motor **482** can include a rotatable shaft that operably interfaces with a gear assembly that is mounted in meshing engagement with a set, or rack, of drive teeth on the displacement member. A sensor element may be operably coupled to a gear assembly such that a single revolution of the position sensor **472** element corresponds to some linear longitudinal translation of the displacement member. An arrangement of gearing and sensors can be connected to the linear actuator, via a rack and pinion arrangement, or a rotary actuator, via a spur gear or other connection. A power source supplies power to the absolute positioning system and an output indicator may display the output of the absolute positioning system. The displacement member represents the longitudinally movable drive member comprising a rack of drive teeth formed thereon for meshing engagement with a corresponding drive gear of the gear reducer assembly. The displacement member represents the longitudinally movable firing member to open and close a clamp arm.

(253) A single revolution of the sensor element associated with the position sensor **472** is equivalent to a longitudinal linear displacement d.sub.1 of the of the displacement member, where d.sub.1 is the longitudinal linear distance that the displacement member moves from point “a” to point “b”

after a single revolution of the sensor element coupled to a gear reduction member. The sensor arrangement may be connected via a gear reduction that results in the position sensor **472** completing one or more revolutions for the full stroke of the displacement member. The position sensor **472** may complete multiple revolutions for the full stroke of the displacement member.

(254) A series of switches, where n is an integer greater than one, may be employed alone or in combination with a gear reduction to provide a unique position signal for more than one revolution of the position sensor **472**. The state of the switches are fed back to the microcontroller **461** that applies logic to determine a unique position signal corresponding to the longitudinal linear displacement $d_{sub.1} + d_{sub.2} + \dots + d_{sub.n}$ of the displacement member. The output of the position sensor **472** is provided to the microcontroller **461**. The position sensor **472** of the sensor arrangement may comprise a magnetic sensor, an analog rotary sensor like a potentiometer, or an array of analog Hall-effect elements, which output a unique combination of position signals or values.

(255) The position sensor **472** may comprise any number of magnetic sensing elements, such as, for example, magnetic sensors classified according to whether they measure the total magnetic field or the vector components of the magnetic field. The techniques used to produce both types of magnetic sensors encompass many aspects of physics and electronics. The technologies used for magnetic field sensing include search coil, fluxgate, optically pumped, nuclear precession, SQUID, Hall-effect, anisotropic magnetoresistance, giant magnetoresistance, magnetic tunnel junctions, giant magnetoimpedance, magnetostrictive/piezoelectric composites, magnetodiode, magnetotransistor, fiber-optic, magneto-optic, and microelectromechanical systems-based magnetic sensors, among others.

(256) In one aspect, the position sensor **472** for the tracking system **480** comprising an absolute positioning system comprises a magnetic rotary absolute positioning system. The position sensor **472** may be implemented as an AS5055EQFT single-chip magnetic rotary position sensor available from Austria Microsystems, AG. The position sensor **472** is interfaced with the microcontroller **461** to provide an absolute positioning system. The position sensor **472** is a low-voltage and low-power component and includes four Hall-effect elements in an area of the position sensor **472** that is located above a magnet. A high-resolution ADC and a smart power management controller are also provided on the chip. A coordinate rotation digital computer (CORDIC) processor, also known as the digit-by-digit method and Volder's algorithm, is provided to implement a simple and efficient algorithm to calculate hyperbolic and trigonometric functions that require only addition, subtraction, bitshift, and table lookup operations. The angle position, alarm bits, and magnetic field information are transmitted over a standard serial communication interface, such as a serial peripheral interface (SPI) interface, to the microcontroller **461**. The position sensor **472** provides 12 or 14 bits of resolution. The position sensor **472** may be an AS5055 chip provided in a small QFN 16-pin $4 \times 4 \times 0.85$ mm package.

(257) The tracking system **480** comprising an absolute positioning system may comprise and/or be programmed to implement a feedback controller, such as a PID, state feedback, and adaptive controller. A power source converts the signal from the feedback controller into a physical input to the system: in this case the voltage. Other examples include a PWM of the voltage, current, and force. Other sensor(s) may be provided to measure physical parameters of the physical system in addition to the position measured by the position sensor **472**. In some aspects, the other sensor(s) can include sensor arrangements such as those described in U.S. Pat. No. 9,345,481, titled STAPLE CARTRIDGE TISSUE THICKNESS SENSOR SYSTEM, which issued on May 24, 2016, which is herein incorporated by reference in its entirety; U.S. Patent Application Publication No. 2014/0263552, titled STAPLE CARTRIDGE TISSUE THICKNESS SENSOR SYSTEM, which published on Sep. 18, 2014, which is herein incorporated by reference in its entirety; and U.S. patent application Ser. No. 15/628,175, titled TECHNIQUES FOR ADAPTIVE CONTROL OF MOTOR VELOCITY OF A SURGICAL STAPLING AND CUTTING INSTRUMENT, filed Jun. 20, 2017, which is herein incorporated by reference in its entirety. In a digital signal processing system, an absolute positioning system is coupled to a digital data acquisition system where the output of the absolute positioning system will have a finite resolution and sampling frequency. The absolute positioning system may comprise a compare-and-combine circuit to combine a computed response with a measured response using algorithms, such as a weighted average and a theoretical control loop, that drive the computed response towards the measured response. The computed response of the physical system takes into account properties like mass, inertia, viscous friction, inductance resistance, etc., to predict what the states and outputs of the physical system will be by knowing the input.

(258) The absolute positioning system provides an absolute position of the displacement member upon power-up of the instrument, without retracting or advancing the displacement member to a reset (zero or home) position as may be required with conventional rotary encoders that merely count the number of steps forwards or backwards that the motor **482** has taken to infer the position of a device actuator, drive bar, knife, or the like.

(259) A sensor **474**, such as, for example, a strain gauge or a micro-strain gauge, is configured to measure one or more parameters of the end effector, such as, for example, the amplitude of the strain exerted on the anvil during a clamping operation, which can be indicative of the closure forces applied to the anvil. The measured strain is converted to a digital signal and provided to the processor **462**. Alternatively, or in addition to the sensor **474**, a sensor **476**, such as, for example, a load sensor, can measure the closure force applied by the closure drive system to the anvil in a stapler or a clamp arm in an ultrasonic or electrosurgical instrument. The sensor **476**, such as, for example, a load sensor, can measure the firing force applied to a closure member coupled to a clamp arm of the surgical instrument or tool or the force applied by a clamp arm to tissue located in the jaws of an ultrasonic or electrosurgical instrument. Alternatively, a current sensor **478** can be employed to measure the current drawn by the motor **482**. The displacement member also may be configured to engage a clamp arm to open or close the clamp arm. The force sensor may be configured to measure the clamping force on tissue. The force required to advance the displacement member can correspond to the current drawn by the motor **482**, for example. The measured force is converted to a digital signal and provided to the processor **462**.

(260) In one form, the strain gauge sensor **474** can be used to measure the force applied to the tissue by the end effector. A strain gauge can be coupled to the end effector to measure the force on the tissue being treated by the end effector. A system for measuring forces applied to the tissue grasped by the end effector comprises a strain gauge sensor **474**, such as, for example, a micro-strain gauge, that is configured to measure one or more parameters of the end effector, for example. In one aspect, the strain gauge sensor **474** can measure the amplitude or magnitude of the strain exerted on a jaw member of an end effector during a clamping operation, which can be indicative of the tissue compression. The measured strain is converted to a digital signal and provided to a processor **462** of the microcontroller **461**. A load sensor **476** can measure the force used to operate the knife element, for example, to cut the tissue captured between the anvil and the staple cartridge. A load sensor **476** can measure the force used to operate the clamp arm element, for example, to capture tissue between the clamp arm and an ultrasonic blade or to capture tissue between the clamp arm and a jaw of an electrosurgical instrument. A magnetic field sensor can be employed to measure the thickness of the captured tissue. The measurement of the magnetic field sensor also may be converted to a digital signal and provided to the processor **462**.

(261) The measurements of the tissue compression, the tissue thickness, and/or the force required to close the end effector on the tissue, as respectively measured by the sensors **474**, **476**, can be used by the microcontroller **461** to characterize the selected position of the firing member and/or the corresponding value of the speed of the firing member. In one instance, a memory **468** may store a technique, an equation, and/or a lookup table which can be employed by the microcontroller **461** in the assessment.

(262) The control system **470** of the surgical instrument or tool also may comprise wired or wireless communication circuits to communicate with the modular communication hub as shown in FIGS. 8-11.

(263) FIG. 13 illustrates a control circuit **500** configured to control aspects of the surgical instrument or tool according to one aspect of this disclosure. The control circuit **500** can be configured to implement various processes described herein. The control circuit **500** may comprise a microcontroller comprising one or more processors **502** (e.g., microprocessor, microcontroller) coupled to at least one memory circuit **504**. The memory circuit **504** stores machine-executable instructions that, when executed by the processor **502**, cause the processor **502** to execute machine instructions to implement various processes described herein. The processor **502** may be any one of a number of single-core or multicore processors known in the art. The memory circuit **504** may comprise volatile and non-volatile storage media. The processor **502** may include an instruction

processing unit **506** and an arithmetic unit **508**. The instruction processing unit may be configured to receive instructions from the memory circuit **504** of this disclosure.

(264) FIG. **14** illustrates a combinational logic circuit **510** configured to control aspects of the surgical instrument or tool according to one aspect of this disclosure. The combinational logic circuit **510** can be configured to implement various processes described herein. The combinational logic circuit **510** may comprise a finite state machine comprising a combinational logic **512** configured to receive data associated with the surgical instrument or tool at an input **514**, process the data by the combinational logic **512**, and provide an output **516**.

(265) FIG. **15** illustrates a sequential logic circuit **520** configured to control aspects of the surgical instrument or tool according to one aspect of this disclosure. The sequential logic circuit **520** or the combinational logic **522** can be configured to implement various processes described herein. The sequential logic circuit **520** may comprise a finite state machine. The sequential logic circuit **520** may comprise a combinational logic **522**, at least one memory circuit **524**, and a clock **529**, for example. The at least one memory circuit **524** can store a current state of the finite state machine. In certain instances, the sequential logic circuit **520** may be synchronous or asynchronous. The combinational logic **522** is configured to receive data associated with the surgical instrument or tool from an input **526**, process the data by the combinational logic **522**, and provide an output **528**. In other aspects, the circuit may comprise a combination of a processor (e.g., processor **502**, FIG. **13**) and a finite state machine to implement various processes herein. In other aspects, the finite state machine may comprise a combination of a combinational logic circuit (e.g., combinational logic circuit **510**, FIG. **14**) and the sequential logic circuit **520**.

(266) FIG. **16** illustrates a surgical instrument or tool comprising a plurality of motors which can be activated to perform various functions. In certain instances, a first motor can be activated to perform a first function, a second motor can be activated to perform a second function, a third motor can be activated to perform a third function, a fourth motor can be activated to perform a fourth function, and so on. In certain instances, the plurality of motors of robotic surgical instrument **600** can be individually activated to cause firing, closure, and/or articulation motions in the end effector. The firing, closure, and/or articulation motions can be transmitted to the end effector through a shaft assembly, for example.

(267) In certain instances, the surgical instrument system or tool may include a firing motor **602**. The firing motor **602** may be operably coupled to a firing motor drive assembly **604** which can be configured to transmit firing motions, generated by the motor **602** to the end effector, in particular to displace the clamp arm closure member. The closure member may be retracted by reversing the direction of the motor **602**, which also causes the clamp arm to open.

(268) In certain instances, the surgical instrument or tool may include a closure motor **603**. The closure motor **603** may be operably coupled to a closure motor drive assembly **605** which can be configured to transmit closure motions, generated by the motor **603** to the end effector, in particular to displace a closure tube to close the anvil and compress tissue between the anvil and the staple cartridge. The closure motor **603** may be operably coupled to a closure motor drive assembly **605** which can be configured to transmit closure motions, generated by the motor **603** to the end effector, in particular to displace a closure tube to close the clamp arm and compress tissue between the clamp arm and either an ultrasonic blade or jaw member of an electrosurgical device. The closure motions may cause the end effector to transition from an open configuration to an approximated configuration to capture tissue, for example. The end effector may be transitioned to an open position by reversing the direction of the motor **603**.

(269) In certain instances, the surgical instrument or tool may include one or more articulation motors **606a**, **606b**, for example. The motors **606a**, **606b** may be operably coupled to respective articulation motor drive assemblies **608a**, **608b**, which can be configured to transmit articulation motions generated by the motors **606a**, **606b** to the end effector. In certain instances, the articulation motions may cause the end effector to articulate relative to the shaft, for example.

(270) As described above, the surgical instrument or tool may include a plurality of motors which may be configured to perform various independent functions. In certain instances, the plurality of motors of the surgical instrument or tool can be individually or separately activated to perform one or more functions while the other motors remain inactive. For example, the articulation motors **606a**, **606b** can be activated to cause the end effector to be articulated while the firing motor **602** remains inactive. Alternatively, the firing motor **602** can be activated to fire the plurality of staples, and/or to advance the cutting edge, while the articulation motor **606** remains inactive. Furthermore, the closure motor **603** may be activated simultaneously with the firing motor **602** to cause the closure tube or closure member to advance distally as described in more detail hereinbelow.

(271) In certain instances, the surgical instrument or tool may include a common control module **610** which can be employed with a plurality of motors of the surgical instrument or tool. In certain instances, the common control module **610** may accommodate one of the plurality of motors at a time. For example, the common control module **610** can be couplable to and separable from the plurality of motors of the robotic surgical instrument individually. In certain instances, a plurality of the motors of the surgical instrument or tool may share one or more common control modules such as the common control module **610**. In certain instances, a plurality of motors of the surgical instrument or tool can be individually and selectively engaged with the common control module **610**. In certain instances, the common control module **610** can be selectively switched from interfacing with one of a plurality of motors of the surgical instrument or tool to interfacing with another one of the plurality of motors of the surgical instrument or tool.

(272) In at least one example, the common control module **610** can be selectively switched between operable engagement with the articulation motors **606a**, **606b** and operable engagement with either the firing motor **602** or the closure motor **603**. In at least one example, as illustrated in FIG. **16**, a switch **614** can be moved or transitioned between a plurality of positions and/or states. In a first position **616**, the switch **614** may electrically couple the common control module **610** to the firing motor **602**; in a second position **617**, the switch **614** may electrically couple the common control module **610** to the closure motor **603**; in a third position **618a**, the switch **614** may electrically couple the common control module **610** to the first articulation motor **606a**; and in a fourth position **618b**, the switch **614** may electrically couple the common control module **610** to the second articulation motor **606b**, for example. In certain instances, separate common control modules **610** can be electrically coupled to the firing motor **602**, the closure motor **603**, and the articulations motor **606a**, **606b** at the same time. In certain instances, the switch **614** may be a mechanical switch, an electromechanical switch, a solid-state switch, or any suitable switching mechanism.

(273) Each of the motors **602**, **603**, **606a**, **606b** may comprise a torque sensor to measure the output torque on the shaft of the motor. The force on an end effector may be sensed in any conventional manner, such as by force sensors on the outer sides of the jaws or by a torque sensor for the motor actuating the jaws.

(274) In certain instances, as illustrated in FIG. **16**, the common control module **610** may comprise a motor driver **626** which may comprise one or more H-Bridge FETs. The motor driver **626** may modulate the power transmitted from a power source **628** to a motor coupled to the common control module **610** based on input from a microcontroller **620** (the “controller”), for example. In certain instances, the microcontroller **620** can be employed to determine the current drawn by the motor, for example, while the motor is coupled to the common control module **610**, as described above.

(275) In certain instances, the microcontroller **620** may include a microprocessor **622** (the “processor”) and one or more non-transitory computer-readable mediums or memory units **624** (the “memory”). In certain instances, the memory **624** may store various program instructions, which when executed may cause the processor **622** to perform a plurality of functions and/or calculations described herein. In certain instances, one or more of the memory units **624** may be coupled to the processor **622**, for example. In various aspects, the microcontroller **620** may communicate over a wired or wireless channel, or combinations thereof.

(276) In certain instances, the power source **628** can be employed to supply power to the microcontroller **620**, for example. In certain instances, the power source **628** may comprise a battery (or “battery pack” or “power pack”), such as a lithium-ion battery, for example. In certain instances, the battery pack may be configured to be releasably mounted to a handle for supplying power to the surgical instrument **600**. A number of battery cells connected in series may be used as the power source **628**. In certain instances, the power source **628** may be replaceable and/or rechargeable, for example.

(277) In various instances, the processor **622** may control the motor driver **626** to control the position, direction of rotation, and/or velocity of a motor that is coupled to the common control module **610**. In certain instances, the processor **622** can signal the motor driver **626** to stop and/or disable a motor that is coupled to the common control module **610**. It should be understood that the term “processor” as used herein includes any suitable microprocessor, microcontroller, or other basic computing device that incorporates the functions of a computer's central processing unit (CPU) on an integrated circuit or, at most, a few integrated circuits. The processor **622** is a multipurpose, programmable device that accepts digital data as input, processes it according to instructions stored in its memory, and provides results as output. It is an example of sequential digital logic, as it has internal memory. Processors operate on numbers and symbols represented in the binary numeral system.

(278) In one instance, the processor **622** may be any single-core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In certain instances, the microcontroller **620** may be an LM 4F230H5QR, available from Texas Instruments, for example. In at least one example, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising an on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle SRAM, an internal ROM loaded with StellarisWare® software, a 2 KB EEPROM, one or more PWM modules, one or more QEI analogs, one or more 12-bit ADCs with 12 analog input channels, among other features that are readily available for the product datasheet. Other microcontrollers may be readily substituted for use with the module **4410**. Accordingly, the present disclosure should not be limited in this context.

(279) In certain instances, the memory **624** may include program instructions for controlling each of the motors of the surgical instrument **600** that are couplable to the common control module **610**. For example, the memory **624** may include program instructions for controlling the firing motor **602**, the closure motor **603**, and the articulation motors **606a**, **606b**. Such program instructions may cause the processor **622** to control the firing, closure, and articulation functions in accordance with inputs from algorithms or control programs of the surgical instrument or tool.

(280) In certain instances, one or more mechanisms and/or sensors such as, for example, sensors **630** can be employed to alert the processor **622** to the program instructions that should be used in a particular setting. For example, the sensors **630** may alert the processor **622** to use the program instructions associated with firing, closing, and articulating the end effector. In certain instances, the sensors **630** may comprise position sensors which can be employed to sense the position of the switch **614**, for example. Accordingly, the processor **622** may use the program instructions associated with firing the closure member coupled to the clamp arm of the end effector upon detecting, through the sensors **630** for example, that the switch **614** is in the first position **616**; the processor **622** may use the program instructions associated with closing the anvil upon detecting, through the sensors **630** for example, that the switch **614** is in the second position **617**; and the processor **622** may use the program instructions associated with articulating the end effector upon detecting, through the sensors **630** for example, that the switch **614** is in the third or fourth position **618a**, **618b**.

(281) FIG. 17 is a schematic diagram of a robotic surgical instrument **700** configured to operate a surgical tool described herein according to one aspect of this disclosure. The robotic surgical instrument **700** may be programmed or configured to control distal/proximal translation of a displacement member, distal/proximal displacement of a closure tube, shaft rotation, and articulation, either with single or multiple articulation drive links. In one aspect, the surgical instrument **700** may be programmed or configured to individually control a firing member, a closure member, a shaft member, or one or more articulation members, or combinations thereof. The surgical instrument **700** comprises a control circuit **710** configured to control motor-driven firing members, closure members, shaft members, or one or more articulation members, or combinations thereof.

(282) In one aspect, the robotic surgical instrument **700** comprises a control circuit **710** configured to control a clamp arm **716** and a closure member **714** portion of an end effector **702**, an ultrasonic blade **718** coupled to an ultrasonic transducer **719** excited by an ultrasonic generator **721**, a shaft **740**, and one or more articulation members **742a**, **742b** via a plurality of motors **704a-704e**. A position sensor **734** may be configured to provide position feedback of the closure member **714** to the control circuit **710**. Other sensors **738** may be configured to provide feedback to the control circuit **710**. A timer/counter **731** provides timing and counting information to the control circuit **710**. An energy source **712** may be provided to operate the motors **704a-704e**, and a current sensor **736** provides motor current feedback to the control circuit **710**. The motors **704a-704e** can be operated individually by the control circuit **710** in an open-loop or closed-loop feedback control.

(283) In one aspect, the control circuit **710** may comprise one or more microcontrollers, microprocessors, or other suitable processors for executing instructions that cause the processor or processors to perform one or more tasks. In one aspect, a timer/counter **731** provides an output signal, such as the elapsed time or a digital count, to the control circuit **710** to correlate the position of the closure member **714** as determined by the position sensor **734** with the output of the timer/counter **731** such that the control circuit **710** can determine the position of the closure member **714** at a specific time (t) relative to a starting position or the time (t) when the closure member **714** is at a specific position relative to a starting position. The timer/counter **731** may be configured to measure elapsed time, count external events, or time external events.

(284) In one aspect, the control circuit **710** may be programmed to control functions of the end effector **702** based on one or more tissue conditions. The control circuit **710** may be programmed to sense tissue conditions, such as thickness, either directly or indirectly, as described herein. The control circuit **710** may be programmed to select a firing control program or closure control program based on tissue conditions. A firing control program may describe the distal motion of the displacement member. Different firing control programs may be selected to better treat different tissue conditions. For example, when thicker tissue is present, the control circuit **710** may be programmed to translate the displacement member at a lower velocity and/or with lower power. When thinner tissue is present, the control circuit **710** may be programmed to translate the displacement member at a higher velocity and/or with higher power. A closure control program may control the closure force applied to the tissue by the clamp arm **716**. Other control programs control the rotation of the shaft **740** and the articulation members **742a**, **742b**.

(285) In one aspect, the control circuit **710** may generate motor set point signals. The motor set point signals may be provided to various motor controllers **708a-708e**. The motor controllers **708a-708e** may comprise one or more circuits configured to provide motor drive signals to the motors **704a-704e** to drive the motors **704a-704e** as described herein. In some examples, the motors **704a-704e** may be brushed DC electric motors. For example, the velocity of the motors **704a-704e** may be proportional to the respective motor drive signals. In some examples, the motors **704a-704e** may be brushless DC electric motors, and the respective motor drive signals may comprise a PWM signal provided to one or more stator windings of the motors **704a-704e**. Also, in some examples, the motor controllers **708a-708e** may be omitted and the control circuit **710** may generate the motor drive signals directly.

(286) In one aspect, the control circuit **710** may initially operate each of the motors **704a-704e** in an open-loop configuration for a first open-loop portion of a stroke of the displacement member. Based on the response of the robotic surgical instrument **700** during the open-loop portion of the stroke, the control circuit **710** may select a firing control program in a closed-loop configuration. The response of the instrument may include a translation distance of the displacement member during the open-loop portion, a time elapsed during the open-loop portion, the energy provided to one of the motors **704a-704e** during the open-loop portion, a sum of pulse widths of a motor drive signal, etc. After the open-loop portion, the control circuit **710** may implement the selected firing control program for a second portion of the displacement member stroke. For example, during a closed-loop portion of the stroke, the control circuit **710** may modulate one of the motors **704a-704e** based on translation data describing a position of the displacement member in a closed-loop manner to translate the displacement member at a constant velocity.

(287) In one aspect, the motors **704a-704e** may receive power from an energy source **712**. The energy source **712** may be a DC power supply driven by a main alternating current power source, a battery, a super capacitor, or any other suitable energy source. The motors **704a-704e** may be mechanically coupled to individual movable mechanical elements such as the closure member **714**, clamp arm **716**, shaft **740**, articulation **742a**, and articulation **742b** via respective transmissions **706a-706e**. The transmissions **706a-706e** may include one or more gears or other linkage components to couple the motors **704a-704e** to movable mechanical elements. A position sensor **734** may sense a position of the closure member **714**. The position sensor **734** may be or include any type of sensor that is capable of generating position data that indicate a position of the closure member

714. In some examples, the position sensor **734** may include an encoder configured to provide a series of pulses to the control circuit **710** as the closure member **714** translates distally and proximally. The control circuit **710** may track the pulses to determine the position of the closure member **714**. Other suitable position sensors may be used, including, for example, a proximity sensor. Other types of position sensors may provide other signals indicating motion of the closure member **714**. Also, in some examples, the position sensor **734** may be omitted. Where any of the motors **704a-704e** is a stepper motor, the control circuit **710** may track the position of the closure member **714** by aggregating the number and direction of steps that the motor **704** has been instructed to execute. The position sensor **734** may be located in the end effector **702** or at any other portion of the instrument. The outputs of each of the motors **704a-704e** include a torque sensor **744a-744e** to sense force and have an encoder to sense rotation of the drive shaft.

(288) In one aspect, the control circuit **710** is configured to drive a firing member such as the closure member **714** portion of the end effector **702**. The control circuit **710** provides a motor set point to a motor control **708a**, which provides a drive signal to the motor **704a**. The output shaft of the motor **704a** is coupled to a torque sensor **744a**. The torque sensor **744a** is coupled to a transmission **706a** which is coupled to the closure member **714**. The transmission **706a** comprises movable mechanical elements such as rotating elements and a firing member to control the movement of the closure member **714** distally and proximally along a longitudinal axis of the end effector **702**. In one aspect, the motor **704a** may be coupled to the knife gear assembly, which includes a knife gear reduction set that includes a first knife drive gear and a second knife drive gear. A torque sensor **744a** provides a firing force feedback signal to the control circuit **710**. The firing force signal represents the force required to fire or displace the closure member **714**. A position sensor **734** may be configured to provide the position of the closure member **714** along the firing stroke or the position of the firing member as a feedback signal to the control circuit **710**. The end effector **702** may include additional sensors **738** configured to provide feedback signals to the control circuit **710**. When ready to use, the control circuit **710** may provide a firing signal to the motor control **708a**. In response to the firing signal, the motor **704a** may drive the firing member distally along the longitudinal axis of the end effector **702** from a proximal stroke start position to a stroke end position distal to the stroke start position. As the closure member **714** translates distally, the clamp arm **716** closes towards the ultrasonic blade **718**.

(289) In one aspect, the control circuit **710** is configured to drive a closure member such as the clamp arm **716** portion of the end effector **702**. The control circuit **710** provides a motor set point to a motor control **708b**, which provides a drive signal to the motor **704b**. The output shaft of the motor **704b** is coupled to a torque sensor **744b**. The torque sensor **744b** is coupled to a transmission **706b** which is coupled to the clamp arm **716**. The transmission **706b** comprises movable mechanical elements such as rotating elements and a closure member to control the movement of the clamp arm **716** from the open and closed positions. In one aspect, the motor **704b** is coupled to a closure gear assembly, which includes a closure reduction gear set that is supported in meshing engagement with the closure spur gear. The torque sensor **744b** provides a closure force feedback signal to the control circuit **710**. The closure force feedback signal represents the closure force applied to the clamp arm **716**. The position sensor **734** may be configured to provide the position of the closure member as a feedback signal to the control circuit **710**. Additional sensors **738** in the end effector **702** may provide the closure force feedback signal to the control circuit **710**. The pivotable clamp arm **716** is positioned opposite the ultrasonic blade **718**. When ready to use, the control circuit **710** may provide a closure signal to the motor control **708b**. In response to the closure signal, the motor **704b** advances a closure member to grasp tissue between the clamp arm **716** and the ultrasonic blade **718**.

(290) In one aspect, the control circuit **710** is configured to rotate a shaft member such as the shaft **740** to rotate the end effector **702**. The control circuit **710** provides a motor set point to a motor control **708c**, which provides a drive signal to the motor **704c**. The output shaft of the motor **704c** is coupled to a torque sensor **744c**. The torque sensor **744c** is coupled to a transmission **706c** which is coupled to the shaft **740**. The transmission **706c** comprises movable mechanical elements such as rotating elements to control the rotation of the shaft **740** clockwise or counterclockwise up to and over 360°. In one aspect, the motor **704c** is coupled to the rotational transmission assembly, which includes a tube gear segment that is formed on (or attached to) the proximal end of the proximal closure tube for operable engagement by a rotational gear assembly that is operably supported on the tool mounting plate. The torque sensor **744c** provides a rotation force feedback signal to the control circuit **710**. The rotation force feedback signal represents the rotation force applied to the shaft **740**. The position sensor **734** may be configured to provide the position of the closure member as a feedback signal to the control circuit **710**. Additional sensors **738** such as a shaft encoder may provide the rotational position of the shaft **740** to the control circuit **710**.

(291) In one aspect, the control circuit **710** is configured to articulate the end effector **702**. The control circuit **710** provides a motor set point to a motor control **708d**, which provides a drive signal to the motor **704d**. The output shaft of the motor **704d** is coupled to a torque sensor **744d**. The torque sensor **744d** is coupled to a transmission **706d** which is coupled to an articulation member **742a**. The transmission **706d** comprises movable mechanical elements such as articulation elements to control the articulation of the end effector **702**±65°. In one aspect, the motor **704d** is coupled to an articulation nut, which is rotatably journaled on the proximal end portion of the distal spine portion and is rotatably driven thereon by an articulation gear assembly. The torque sensor **744d** provides an articulation force feedback signal to the control circuit **710**. The articulation force feedback signal represents the articulation force applied to the end effector **702**. Sensors **738**, such as an articulation encoder, may provide the articulation position of the end effector **702** to the control circuit **710**.

(292) In another aspect, the articulation function of the robotic surgical system **700** may comprise two articulation members, or links, **742a**, **742b**. These articulation members **742a**, **742b** are driven by separate disks on the robot interface (the rack) which are driven by the two motors **708d**, **708e**. When the separate firing motor **704a** is provided, each of articulation links **742a**, **742b** can be antagonistically driven with respect to the other link in order to provide a resistive holding motion and a load to the head when it is not moving and to provide an articulation motion as the head is articulated. The articulation members **742a**, **742b** attach to the head at a fixed radius as the head is rotated. Accordingly, the mechanical advantage of the push-and-pull link changes as the head is rotated. This change in the mechanical advantage may be more pronounced with other articulation link drive systems.

(293) In one aspect, the one or more motors **704a-704e** may comprise a brushed DC motor with a gearbox and mechanical links to a firing member, closure member, or articulation member. Another example includes electric motors **704a-704e** that operate the movable mechanical elements such as the displacement member, articulation links, closure tube, and shaft. An outside influence is an unmeasured, unpredictable influence of things like tissue, surrounding bodies, and friction on the physical system. Such outside influence can be referred to as drag, which acts in opposition to one of electric motors **704a-704e**. The outside influence, such as drag, may cause the operation of the physical system to deviate from a desired operation of the physical system.

(294) In one aspect, the position sensor **734** may be implemented as an absolute positioning system. In one aspect, the position sensor **734** may comprise a magnetic rotary absolute positioning system implemented as an AS5055EQFT single-chip magnetic rotary position sensor available from Austria Microsystems, AG. The position sensor **734** may interface with the control circuit **710** to provide an absolute positioning system. The position may include multiple Hall-effect elements located above a magnet and coupled to a CORDIC processor, also known as the digit-by-digit method and Volder's algorithm, that is provided to implement a simple and efficient algorithm to calculate hyperbolic and trigonometric functions that require only addition, subtraction, bitshift, and table lookup operations.

(295) In one aspect, the control circuit **710** may be in communication with one or more sensors **738**. The sensors **738** may be positioned on the end effector **702** and adapted to operate with the robotic surgical instrument **700** to measure the various derived parameters such as the gap distance versus time, tissue compression versus time, and anvil strain versus time. The sensors **738** may comprise a magnetic sensor, a magnetic field sensor, a strain gauge, a load cell, a pressure sensor, a force sensor, a torque sensor, an inductive sensor such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor for measuring one or more parameters of the end effector **702**. The sensors **738** may include one or more sensors. The sensors **738** may be located on the clamp arm **716** to determine tissue location using segmented electrodes.

the torque sensors **744a-744e** may be configured to sense force such as firing force, closure force, and/or articulation force, among others. Accordingly, the control circuit **710** can sense (1) the closure load experienced by the distal closure tube and its position, (2) the firing member at the rack and its position, (3) what portion of the ultrasonic blade **718** has tissue on it, and (4) the load and position on both articulation rods.

(296) In one aspect, the one or more sensors **738** may comprise a strain gauge, such as a micro-strain gauge, configured to measure the magnitude of the strain in the clamp arm **716** during a clamped condition. The strain gauge provides an electrical signal whose amplitude varies with the magnitude of the strain. The sensors **738** may comprise a pressure sensor configured to detect a pressure generated by the presence of compressed tissue between the clamp arm **716** and the ultrasonic blade **718**. The sensors **738** may be configured to detect impedance of a tissue section located between the clamp arm **716** and the ultrasonic blade **718** that is indicative of the thickness and/or fullness of tissue located therebetween.

(297) In one aspect, the sensors **738** may be implemented as one or more limit switches, electromechanical devices, solid-state switches, Hall-effect devices, magneto-resistive (MR) devices, giant magneto-resistive (GMR) devices, magnetometers, among others. In other implementations, the sensors **738** may be implemented as solid-state switches that operate under the influence of light, such as optical sensors, IR sensors, ultraviolet sensors, among others. Still, the switches may be solid-state devices such as transistors (e.g., FET, junction FET, MOSFET, bipolar, and the like). In other implementations, the sensors **738** may include electrical conductorless switches, ultrasonic switches, accelerometers, and inertial sensors, among others.

(298) In one aspect, the sensors **738** may be configured to measure forces exerted on the clamp arm **716** by the closure drive system. For example, one or more sensors **738** can be at an interaction point between the closure tube and the clamp arm **716** to detect the closure forces applied by the closure tube to the clamp arm **716**. The forces exerted on the clamp arm **716** can be representative of the tissue compression experienced by the tissue section captured between the clamp arm **716** and the ultrasonic blade **718**. The one or more sensors **738** can be positioned at various interaction points along the closure drive system to detect the closure forces applied to the clamp arm **716** by the closure drive system. The one or more sensors **738** may be sampled in real time during a clamping operation by the processor of the control circuit **710**. The control circuit **710** receives real-time sample measurements to provide and analyze time-based information and assess, in real time, closure forces applied to the clamp arm **716**.

(299) In one aspect, a current sensor **736** can be employed to measure the current drawn by each of the motors **704a-704e**. The force required to advance any of the movable mechanical elements such as the closure member **714** corresponds to the current drawn by one of the motors **704a-704e**. The force is converted to a digital signal and provided to the control circuit **710**. The control circuit **710** can be configured to simulate the response of the actual system of the instrument in the software of the controller. A displacement member can be actuated to move the closure member **714** in the end effector **702** at or near a target velocity. The robotic surgical instrument **700** can include a feedback controller, which can be one of any feedback controllers, including, but not limited to a PID, a state feedback, a linear-quadratic (LQR), and/or an adaptive controller, for example. The robotic surgical instrument **700** can include a power source to convert the signal from the feedback controller into a physical input such as case voltage, PWM voltage, frequency modulated voltage, current, torque, and/or force, for example. Additional details are disclosed in U.S. patent application Ser. No. 15/636,829, titled CLOSED LOOP VELOCITY CONTROL TECHNIQUES FOR ROBOTIC SURGICAL INSTRUMENT, filed Jun. 29, 2017, which is herein incorporated by reference in its entirety.

(300) FIG. **18** illustrates a schematic diagram of a surgical instrument **750** configured to control the distal translation of a displacement member according to one aspect of this disclosure. In one aspect, the surgical instrument **750** is programmed to control the distal translation of a displacement member such as the closure member **764**. The surgical instrument **750** comprises an end effector **752** that may comprise a clamp arm **766**, a closure member **764**, and an ultrasonic blade **768** coupled to an ultrasonic transducer **769** driven by an ultrasonic generator **771**.

(301) The position, movement, displacement, and/or translation of a linear displacement member, such as the closure member **764**, can be measured by an absolute positioning system, sensor arrangement, and position sensor **784**. Because the closure member **764** is coupled to a longitudinally movable drive member, the position of the closure member **764** can be determined by measuring the position of the longitudinally movable drive member employing the position sensor **784**. Accordingly, in the following description, the position, displacement, and/or translation of the closure member **764** can be achieved by the position sensor **784** as described herein. A control circuit **760** may be programmed to control the translation of the displacement member, such as the closure member **764**. The control circuit **760**, in some examples, may comprise one or more microcontrollers, microprocessors, or other suitable processors for executing instructions that cause the processor or processors to control the displacement member, e.g., the closure member **764**, in the manner described. In one aspect, a timer/counter **781** provides an output signal, such as the elapsed time or a digital count, to the control circuit **760** to correlate the position of the closure member **764** as determined by the position sensor **784** with the output of the timer/counter **781** such that the control circuit **760** can determine the position of the closure member **764** at a specific time (t) relative to a starting position. The timer/counter **781** may be configured to measure elapsed time, count external events, or time external events.

(302) The control circuit **760** may generate a motor set point signal **772**. The motor set point signal **772** may be provided to a motor controller **758**. The motor controller **758** may comprise one or more circuits configured to provide a motor drive signal **774** to the motor **754** to drive the motor **754** as described herein. In some examples, the motor **754** may be a brushed DC electric motor. For example, the velocity of the motor **754** may be proportional to the motor drive signal **774**. In some examples, the motor **754** may be a brushless DC electric motor and the motor drive signal **774** may comprise a PWM signal provided to one or more stator windings of the motor **754**. Also, in some examples, the motor controller **758** may be omitted, and the control circuit **760** may generate the motor drive signal **774** directly.

(303) The motor **754** may receive power from an energy source **762**. The energy source **762** may be or include a battery, a super capacitor, or any other suitable energy source. The motor **754** may be mechanically coupled to the closure member **764** via a transmission **756**. The transmission **756** may include one or more gears or other linkage components to couple the motor **754** to the closure member **764**. A position sensor **784** may sense a position of the closure member **764**. The position sensor **784** may be or include any type of sensor that is capable of generating position data that indicate a position of the closure member **764**. In some examples, the position sensor **784** may include an encoder configured to provide a series of pulses to the control circuit **760** as the closure member **764** translates distally and proximally. The control circuit **760** may track the pulses to determine the position of the closure member **764**. Other suitable position sensors may be used, including, for example, a proximity sensor. Other types of position sensors may provide other signals indicating motion of the closure member **764**. Also, in some examples, the position sensor **784** may be omitted. Where the motor **754** is a stepper motor, the control circuit **760** may track the position of the closure member **764** by aggregating the number and direction of steps that the motor **754** has been instructed to execute. The position sensor **784** may be located in the end effector **752** or at any other portion of the instrument.

(304) The control circuit **760** may be in communication with one or more sensors **788**. The sensors **788** may be positioned on the end effector **752** and adapted to operate with the surgical instrument **750** to measure the various derived parameters such as gap distance versus time, tissue compression versus time, and anvil strain versus time. The sensors **788** may comprise a magnetic sensor, a magnetic field sensor, a strain gauge, a pressure sensor, a force sensor, an inductive sensor such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor for measuring one or more parameters of the end effector **752**. The sensors **788** may include one or more sensors.

(305) The one or more sensors **788** may comprise a strain gauge, such as a micro-strain gauge, configured to measure the magnitude of the strain in the clamp arm **766** during a clamped condition. The strain gauge provides an electrical signal whose amplitude varies with the magnitude of the strain. The sensors **788** may comprise a pressure sensor configured to detect a pressure generated by the presence of compressed tissue between the clamp arm **766** and the ultrasonic blade **768**. The sensors **788** may be configured to detect impedance of a tissue section located between the clamp arm **766** and the ultrasonic blade **768** that is indicative of the thickness and/or fullness of tissue located therebetween.

(306) The sensors **788** may be configured to measure forces exerted on the clamp arm **766** by a closure drive system. For example, one or more sensors **788** can be at an interaction point between a closure tube and the clamp arm **766** to detect the closure forces applied by a closure tube to the

clamp arm **766**. The forces exerted on the clamp arm **766** can be representative of the tissue compression experienced by the tissue section captured between the clamp arm **766** and the ultrasonic blade **768**. The one or more sensors **788** can be positioned at various interaction points along the closure drive system to detect the closure forces applied to the clamp arm **766** by the closure drive system. The one or more sensors **788** may be sampled in real time during a clamping operation by a processor of the control circuit **760**. The control circuit **760** receives real-time sample measurements to provide and analyze time-based information and assess, in real time, closure forces applied to the clamp arm **766**.

(307) A current sensor **786** can be employed to measure the current drawn by the motor **754**. The force required to advance the closure member **764** corresponds to the current drawn by the motor **754**. The force is converted to a digital signal and provided to the control circuit **760**.

(308) The control circuit **760** can be configured to simulate the response of the actual system of the instrument in the software of the controller. A displacement member can be actuated to move a closure member **764** in the end effector **752** at or near a target velocity. The surgical instrument **750** can include a feedback controller, which can be one of any feedback controllers, including, but not limited to a PID, a state feedback, LQR, and/or an adaptive controller, for example. The surgical instrument **750** can include a power source to convert the signal from the feedback controller into a physical input such as case voltage, PWM voltage, frequency modulated voltage, current, torque, and/or force, for example.

(309) The actual drive system of the surgical instrument **750** is configured to drive the displacement member, cutting member, or closure member **764**, by a brushed DC motor with gearbox and mechanical links to an articulation and/or knife system. Another example is the electric motor **754** that operates the displacement member and the articulation driver, for example, of an interchangeable shaft assembly. An outside influence is an unmeasured, unpredictable influence of things like tissue, surrounding bodies and friction on the physical system. Such outside influence can be referred to as drag which acts in opposition to the electric motor **754**. The outside influence, such as drag, may cause the operation of the physical system to deviate from a desired operation of the physical system.

(310) Various example aspects are directed to a surgical instrument **750** comprising an end effector **752** with motor-driven surgical sealing and cutting implements. For example, a motor **754** may drive a displacement member distally and proximally along a longitudinal axis of the end effector **752**. The end effector **752** may comprise a pivotable clamp arm **766** and, when configured for use, an ultrasonic blade **768** positioned opposite the clamp arm **766**. A clinician may grasp tissue between the clamp arm **766** and the ultrasonic blade **768**, as described herein. When ready to use the instrument **750**, the clinician may provide a firing signal, for example by depressing a trigger of the instrument **750**. In response to the firing signal, the motor **754** may drive the displacement member distally along the longitudinal axis of the end effector **752** from a proximal stroke begin position to a stroke end position distal of the stroke begin position. As the displacement member translates distally, the closure member **764** with a cutting element positioned at a distal end, may cut the tissue between the ultrasonic blade **768** and the clamp arm **766**.

(311) In various examples, the surgical instrument **750** may comprise a control circuit **760** programmed to control the distal translation of the displacement member, such as the closure member **764**, for example, based on one or more tissue conditions. The control circuit **760** may be programmed to sense tissue conditions, such as thickness, either directly or indirectly, as described herein. The control circuit **760** may be programmed to select a control program based on tissue conditions. A control program may describe the distal motion of the displacement member. Different control programs may be selected to better treat different tissue conditions. For example, when thicker tissue is present, the control circuit **760** may be programmed to translate the displacement member at a lower velocity and/or with lower power. When thinner tissue is present, the control circuit **760** may be programmed to translate the displacement member at a higher velocity and/or with higher power.

(312) In some examples, the control circuit **760** may initially operate the motor **754** in an open loop configuration for a first open loop portion of a stroke of the displacement member. Based on a response of the instrument **750** during the open loop portion of the stroke, the control circuit **760** may select a firing control program. The response of the instrument may include, a translation distance of the displacement member during the open loop portion, a time elapsed during the open loop portion, energy provided to the motor **754** during the open loop portion, a sum of pulse widths of a motor drive signal, etc. After the open loop portion, the control circuit **760** may implement the selected firing control program for a second portion of the displacement member stroke. For example, during the closed loop portion of the stroke, the control circuit **760** may modulate the motor **754** based on translation data describing a position of the displacement member in a closed loop manner to translate the displacement member at a constant velocity. Additional details are disclosed in U.S. patent application Ser. No. 15/720,852, titled SYSTEM AND METHODS FOR CONTROLLING A DISPLAY OF A SURGICAL INSTRUMENT, filed Sep. 29, 2017, which is herein incorporated by reference in its entirety.

(313) FIG. **19** is a schematic diagram of a surgical instrument **790** configured to control various functions according to one aspect of this disclosure. In one aspect, the surgical instrument **790** is programmed to control distal translation of a displacement member such as the closure member **764**. The surgical instrument **790** comprises an end effector **792** that may comprise a clamp arm **766**, a closure member **764**, and an ultrasonic blade **768** which may be interchanged with or work in conjunction with one or more RF electrodes **796** (shown in dashed line). The ultrasonic blade **768** is coupled to an ultrasonic transducer **769** driven by an ultrasonic generator **771**.

(314) In one aspect, sensors **788** may be implemented as a limit switch, electromechanical device, solid-state switches, Hall-effect devices, MR devices, GMR devices, magnetometers, among others. In other implementations, the sensors **638** may be solid-state switches that operate under the influence of light, such as optical sensors, IR sensors, ultraviolet sensors, among others. Still, the switches may be solid-state devices such as transistors (e.g., FET, junction FET, MOSFET, bipolar, and the like). In other implementations, the sensors **788** may include electrical conductorless switches, ultrasonic switches, accelerometers, and inertial sensors, among others.

(315) In one aspect, the position sensor **784** may be implemented as an absolute positioning system comprising a magnetic rotary absolute positioning system implemented as an AS5055EQFT single-chip magnetic rotary position sensor available from Austria Microsystems, AG. The position sensor **784** may interface with the control circuit **760** to provide an absolute positioning system. The position may include multiple Hall-effect elements located above a magnet and coupled to a CORDIC processor, also known as the digit-by-digit method and Volder's algorithm, that is provided to implement a simple and efficient algorithm to calculate hyperbolic and trigonometric functions that require only addition, subtraction, bitshift, and table lookup operations.

(316) In some examples, the position sensor **784** may be omitted. Where the motor **754** is a stepper motor, the control circuit **760** may track the position of the closure member **764** by aggregating the number and direction of steps that the motor has been instructed to execute. The position sensor **784** may be located in the end effector **792** or at any other portion of the instrument.

(317) The control circuit **760** may be in communication with one or more sensors **788**. The sensors **788** may be positioned on the end effector **792** and adapted to operate with the surgical instrument **790** to measure the various derived parameters such as gap distance versus time, tissue compression versus time, and anvil strain versus time. The sensors **788** may comprise a magnetic sensor, a magnetic field sensor, a strain gauge, a pressure sensor, a force sensor, an inductive sensor such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor for measuring one or more parameters of the end effector **792**. The sensors **788** may include one or more sensors.

(318) An RF energy source **794** is coupled to the end effector **792** and is applied to the RF electrode **796** when the RF electrode **796** is provided in the end effector **792** in place of the ultrasonic blade **768** or to work in conjunction with the ultrasonic blade **768**. For example, the ultrasonic blade is made of electrically conductive metal and may be employed as the return path for electrosurgical RF current. The control circuit **760** controls the delivery of the RF energy to the RF electrode **796**.

(319) Additional details are disclosed in U.S. Patent application Ser. No. 15/636,096, titled SURGICAL SYSTEM COUPLABLE WITH STAPLE CARTRIDGE AND RADIO FREQUENCY CARTRIDGE, AND METHOD OF USING SAME, filed Jun. 28, 2017, which is herein incorporated by reference in its entirety.

Adaptive Ultrasonic Blade Control Algorithms

(320) In various aspects smart ultrasonic energy devices may comprise adaptive algorithms to control the operation of the ultrasonic blade. In one

aspect, the ultrasonic blade control algorithms are configured to identify tissue type and adjust device parameters. In one aspect, the ultrasonic blade control algorithms are configured to parameterize tissue type. An algorithm to detect the collagen/elastic ratio of tissue to tune the amplitude of the distal tip of the ultrasonic blade is described in the following section of the present disclosure. Various aspects of smart ultrasonic energy devices are described herein in connection with FIGS. 1-94, for example. Accordingly, the following description of adaptive ultrasonic blade control algorithms should be read in conjunction with FIGS. 1-94 and the description associated therewith.

Tissue Type Identification And Device Parameter Adjustments

(321) In certain surgical procedures it would be desirable to employ adaptive ultrasonic blade control algorithms. In one aspect, adaptive ultrasonic blade control algorithms may be employed to adjust the parameters of the ultrasonic device based on the type of tissue in contact with the ultrasonic blade. In one aspect, the parameters of the ultrasonic device may be adjusted based on the location of the tissue within the jaws of the ultrasonic end effector, for example, the location of the tissue between the clamp arm and the ultrasonic blade. The impedance of the ultrasonic transducer may be employed to differentiate what percentage of the tissue is located in the distal or proximal end of the end effector. The reactions of the ultrasonic device may be based on the tissue type or compressibility of the tissue. In another aspect, the parameters of the ultrasonic device may be adjusted based on the identified tissue type or parameterization. For example, the mechanical displacement amplitude of the distal tip of the ultrasonic blade may be tuned based on the ration of collagen to elastin tissue detected during the tissue identification procedure. The ratio of collagen to elastin tissue may be detected used a variety of techniques including infrared (IR) surface reflectance and emissivity. The force applied to the tissue by the clamp arm and/or the stroke of the clamp arm to produce gap and compression. Electrical continuity across a jaw equipped with electrodes may be employed to determine what percentage of the jaw is covered with tissue.

(322) FIG. 20 is a system 800 configured to execute adaptive ultrasonic blade control algorithms in a surgical data network comprising a modular communication hub, in accordance with at least one aspect of the present disclosure. In one aspect, the generator module 240 is configured to execute the adaptive ultrasonic blade control algorithm(s) 802 as described herein with reference to FIGS. 53-94. In another aspect, the device/instrument 235 is configured to execute the adaptive ultrasonic blade control algorithm(s) 804 as described herein with reference to FIGS. 53-94. In another aspect, both the device/instrument 235 and the device/instrument 235 are configured to execute the adaptive ultrasonic blade control algorithms 802, 804 as described herein with reference to FIGS. 53-94.

(323) The generator module 240 may comprise a patient isolated stage in communication with a non-isolated stage via a power transformer. A secondary winding of the power transformer is contained in the isolated stage and may comprise a tapped configuration (e.g., a center-tapped or a non-center-tapped configuration) to define drive signal outputs for delivering drive signals to different surgical instruments, such as, for example, an ultrasonic surgical instrument, an RF electrosurgical instrument, and a multifunction surgical instrument which includes ultrasonic and RF energy modes that can be delivered alone or simultaneously. In particular, the drive signal outputs may output an ultrasonic drive signal (e.g., a 420V root-mean-square (RMS) drive signal) to an ultrasonic surgical instrument 241, and the drive signal outputs may output an RF electrosurgical drive signal (e.g., a 100V RMS drive signal) to an RF electrosurgical instrument 241. Aspects of the generator module 240 are described herein with reference to FIGS. 21-28B.

(324) The generator module 240 or the device/instrument 235 or both are coupled to the modular control tower 236 connected to multiple operating theater devices such as, for example, intelligent surgical instruments, robots, and other computerized devices located in the operating theater, as described with reference to FIGS. 8-11, for example.

(325) FIG. 21 illustrates an example of a generator 900, which is one form of a generator configured to couple to an ultrasonic instrument and further configured to execute adaptive ultrasonic blade control algorithms in a surgical data network comprising a modular communication hub as shown in FIG. 20. The generator 900 is configured to deliver multiple energy modalities to a surgical instrument. The generator 900 provides RF and ultrasonic signals for delivering energy to a surgical instrument either independently or simultaneously. The RF and ultrasonic signals may be provided alone or in combination and may be provided simultaneously. As noted above, at least one generator output can deliver multiple energy modalities (e.g., ultrasonic, bipolar or monopolar RF, irreversible and/or reversible electroporation, and/or microwave energy, among others) through a single port, and these signals can be delivered separately or simultaneously to the end effector to treat tissue. The generator 900 comprises a processor 902 coupled to a waveform generator 904. The processor 902 and waveform generator 904 are configured to generate a variety of signal waveforms based on information stored in a memory coupled to the processor 902, not shown for clarity of disclosure. The digital information associated with a waveform is provided to the waveform generator 904 which includes one or more DAC circuits to convert the digital input into an analog output. The analog output is fed to an amplifier 1106 for signal conditioning and amplification. The conditioned and amplified output of the amplifier 906 is coupled to a power transformer 908. The signals are coupled across the power transformer 908 to the secondary side, which is in the patient isolation side. A first signal of a first energy modality is provided to the surgical instrument between the terminals labeled ENERGY.sub.1 and RETURN. A second signal of a second energy modality is coupled across a capacitor 910 and is provided to the surgical instrument between the terminals labeled ENERGY.sub.2 and RETURN. It will be appreciated that more than two energy modalities may be output and thus the subscript "n" may be used to designate that up to n ENERGY.sub.n terminals may be provided, where n is a positive integer greater than 1. It also will be appreciated that up to "n" return paths RETURN.sub.n may be provided without departing from the scope of the present disclosure.

(326) A first voltage sensing circuit 912 is coupled across the terminals labeled ENERGY.sub.1 and the RETURN path to measure the output voltage therebetween. A second voltage sensing circuit 924 is coupled across the terminals labeled ENERGY.sub.2 and the RETURN path to measure the output voltage therebetween. A current sensing circuit 914 is disposed in series with the RETURN leg of the secondary side of the power transformer 908 as shown to measure the output current for either energy modality. If different return paths are provided for each energy modality, then a separate current sensing circuit should be provided in each return leg. The outputs of the first and second voltage sensing circuits 912, 924 are provided to respective isolation transformers 916, 922 and the output of the current sensing circuit 914 is provided to another isolation transformer 918. The outputs of the isolation transformers 916, 922, 918 on the primary side of the power transformer 908 (non-patient isolated side) are provided to a one or more ADC circuit 926. The digitized output of the ADC circuit 926 is provided to the processor 902 for further processing and computation. The output voltages and output current feedback information can be employed to adjust the output voltage and current provided to the surgical instrument and to compute output impedance, among other parameters. Input/output communications between the processor 902 and patient isolated circuits is provided through an interface circuit 920. Sensors also may be in electrical communication with the processor 902 by way of the interface circuit 920.

(327) In one aspect, the impedance may be determined by the processor 902 by dividing the output of either the first voltage sensing circuit 912 coupled across the terminals labeled ENERGY.sub.1/RETURN or the second voltage sensing circuit 924 coupled across the terminals labeled ENERGY.sub.2/RETURN by the output of the current sensing circuit 914 disposed in series with the RETURN leg of the secondary side of the power transformer 908. The outputs of the first and second voltage sensing circuits 912, 924 are provided to separate isolation transformers 916, 922 and the output of the current sensing circuit 914 is provided to another isolation transformer 918. The digitized voltage and current sensing measurements from the ADC circuit 926 are provided to the processor 902 for computing impedance. As an example, the first energy modality ENERGY.sub.1 may be ultrasonic energy and the second energy modality ENERGY.sub.2 may be RF energy. Nevertheless, in addition to ultrasonic and bipolar or monopolar RF energy modalities, other energy modalities include irreversible and/or reversible electroporation and/or microwave energy, among others. Also, although the example illustrated in FIG. 21 shows a single return path RETURN may be provided for two or more energy modalities, in other aspects, multiple return paths RETURN.sub.n may be provided for each energy modality ENERGY.sub.n. Thus, as described herein, the ultrasonic transducer impedance may be measured by dividing the output of the first voltage sensing circuit 912 by the current sensing circuit 914 and the tissue impedance may be measured by dividing the output of the second voltage sensing circuit 924 by the current sensing

circuit **914**.

(328) As shown in FIG. **21**, the generator **900** comprising at least one output port can include a power transformer **908** with a single output and with multiple taps to provide power in the form of one or more energy modalities, such as ultrasonic, bipolar or monopolar RF, irreversible and/or reversible electroporation, and/or microwave energy, among others, for example, to the end effector depending on the type of treatment of tissue being performed. For example, the generator **900** can deliver energy with higher voltage and lower current to drive an ultrasonic transducer, with lower voltage and higher current to drive RF electrodes for sealing tissue, or with a coagulation waveform for spot coagulation using either monopolar or bipolar RF electrosurgical electrodes. The output waveform from the generator **900** can be steered, switched, or filtered to provide the frequency to the end effector of the surgical instrument. The connection of an ultrasonic transducer to the generator **900** output would be preferably located between the output labeled ENERGY.sub.1 and RETURN as shown in FIG. **21**. In one example, a connection of RF bipolar electrodes to the generator **900** output would be preferably located between the output labeled ENERGY.sub.2 and RETURN. In the case of monopolar output, the preferred connections would be active electrode (e.g., pencil or other probe) to the ENERGY.sub.2 output and a suitable return pad connected to the RETURN output.

(329) Additional details are disclosed in U.S. Patent Application Publication No. 2017/0086914, titled TECHNIQUES FOR OPERATING GENERATOR FOR DIGITALLY GENERATING ELECTRICAL SIGNAL WAVEFORMS AND SURGICAL INSTRUMENTS, which published on Mar. 30, 2017, which is herein incorporated by reference in its entirety.

(330) As used throughout this description, the term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a non-solid medium. The term does not imply that the associated devices do not contain any wires, although in some aspects they might not. The communication module may implement any of a number of wireless or wired communication standards or protocols, including but not limited to Wi-Fi (IEEE 802.11 family), WiMAX (IEEE 802.16 family), IEEE 802.20, long term evolution (LTE), Ev-DO, HSPA+, HSDPA+, HSUPA+, EDGE, GSM, GPRS, CDMA, TDMA, DECT, Bluetooth, Ethernet derivatives thereof, as well as any other wireless and wired protocols that are designated as 3G, 4G, 5G, and beyond. The computing module may include a plurality of communication modules. For instance, a first communication module may be dedicated to shorter range wireless communications such as Wi-Fi and Bluetooth and a second communication module may be dedicated to longer range wireless communications such as GPS, EDGE, GPRS, CDMA, WiMAX, LTE, Ev-DO, and others.

(331) As used herein a processor or processing unit is an electronic circuit which performs operations on some external data source, usually memory or some other data stream. The term is used herein to refer to the central processor (central processing unit) in a system or computer systems (especially systems on a chip (SoCs)) that combine a number of specialized “processors.”

(332) As used herein, a system on a chip or system on chip (SoC or SOC) is an integrated circuit (also known as an “IC” or “chip”) that integrates all components of a computer or other electronic systems. It may contain digital, analog, mixed-signal, and often radio-frequency functions—all on a single substrate. A SoC integrates a microcontroller (or microprocessor) with advanced peripherals like graphics processing unit (GPU), Wi-Fi module, or coprocessor. A SoC may or may not contain built-in memory.

(333) As used herein, a microcontroller or controller is a system that integrates a microprocessor with peripheral circuits and memory. A microcontroller (or MCU for microcontroller unit) may be implemented as a small computer on a single integrated circuit. It may be similar to a SoC; an SoC may include a microcontroller as one of its components. A microcontroller may contain one or more core processing units (CPUs) along with memory and programmable input/output peripherals. Program memory in the form of Ferroelectric RAM, NOR flash or OTP ROM is also often included on chip, as well as a small amount of RAM. Microcontrollers may be employed for embedded applications, in contrast to the microprocessors used in personal computers or other general purpose applications consisting of various discrete chips.

(334) As used herein, the term controller or microcontroller may be a stand-alone IC or chip device that interfaces with a peripheral device. This may be a link between two parts of a computer or a controller on an external device that manages the operation of (and connection with) that device.

(335) Any of the processors or microcontrollers described herein, may be implemented by any single core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In one aspect, the processor may be an LM4F230H5QR ARM Cortex-M4F Processor Core, available from Texas Instruments, for example, comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle serial random access memory (SRAM), internal read-only memory (ROM) loaded with StellarisWare® software, 2 KB electrically erasable programmable read-only memory (EEPROM), one or more pulse width modulation (PWM) modules, one or more quadrature encoder inputs (QEI) analog, one or more 12-bit Analog-to-Digital Converters (ADC) with 12 analog input channels, details of which are available for the product datasheet.

(336) In one aspect, the processor may comprise a safety controller comprising two controller-based families such as TMS570 and RM4x known under the trade name Hercules ARM Cortex R4, also by Texas Instruments. The safety controller may be configured specifically for IEC 61508 and ISO 26262 safety critical applications, among others, to provide advanced integrated safety features while delivering scalable performance, connectivity, and memory options.

(337) Modular devices include the modules (as described in connection with FIGS. **3** and **9**, for example) that are receivable within a surgical hub and the surgical devices or instruments that can be connected to the various modules in order to connect or pair with the corresponding surgical hub. The modular devices include, for example, intelligent surgical instruments, medical imaging devices, suction/irrigation devices, smoke evacuators, energy generators, ventilators, insufflators, and displays. The modular devices described herein can be controlled by control algorithms. The control algorithms can be executed on the modular device itself, on the surgical hub to which the particular modular device is paired, or on both the modular device and the surgical hub (e.g., via a distributed computing architecture). In some exemplifications, the modular devices' control algorithms control the devices based on data sensed by the modular device itself (i.e., by sensors in, on, or connected to the modular device). This data can be related to the patient being operated on (e.g., tissue properties or insufflation pressure) or the modular device itself (e.g., the rate at which a knife is being advanced, motor current, or energy levels). For example, a control algorithm for a surgical stapling and cutting instrument can control the rate at which the instrument's motor drives its knife through tissue according to resistance encountered by the knife as it advances.

(338) FIG. **22** illustrates one form of a surgical system **1000** comprising a generator **1100** and various surgical instruments **1104**, **1106**, **1108** usable therewith, where the surgical instrument **1104** is an ultrasonic surgical instrument, the surgical instrument **1106** is an RF electrosurgical instrument, and the multifunction surgical instrument **1108** is a combination ultrasonic/RF electrosurgical instrument. The generator **1100** is configurable for use with a variety of surgical instruments. According to various forms, the generator **1100** may be configurable for use with different surgical instruments of different types including, for example, ultrasonic surgical instruments **1104**, RF electrosurgical instruments **1106**, and multifunction surgical instruments **1108** that integrate RF and ultrasonic energies delivered simultaneously from the generator **1100**. Although in the form of FIG. **22** the generator **1100** is shown separate from the surgical instruments **1104**, **1106**, **1108** in one form, the generator **1100** may be formed integrally with any of the surgical instruments **1104**, **1106**, **1108** to form a unitary surgical system. The generator **1100** comprises an input device **1110** located on a front panel of the generator **1100** console. The input device **1110** may comprise any suitable device that generates signals suitable for programming the operation of the generator **1100**. The generator **1100** may be configured for wired or wireless communication.

(339) The generator **1100** is configured to drive multiple surgical instruments **1104**, **1106**, **1108**. The first surgical instrument is an ultrasonic surgical instrument **1104** and comprises a handpiece **1105** (HP), an ultrasonic transducer **1120**, a shaft **1126**, and an end effector **1122**. The end effector **1122** comprises an ultrasonic blade **1128** acoustically coupled to the ultrasonic transducer **1120** and a clamp arm **1140**. The handpiece **1105** comprises a trigger **1143** to operate the clamp arm **1140** and a combination of the toggle buttons **1134a**, **1134b**, **1134c** to energize and drive the ultrasonic blade **1128** or other function. The toggle buttons **1134a**, **1134b**, **1134c** can be configured to energize the ultrasonic transducer **1120** with the generator **1100**.

(340) The generator **1100** also is configured to drive a second surgical instrument **1106**. The second surgical instrument **1106** is an RF electrosurgical instrument and comprises a handpiece **1107** (HP), a shaft **1127**, and an end effector **1124**. The end effector **1124** comprises electrodes in clamp arms **1142a**, **1142b** and return through an electrical conductor portion of the shaft **1127**. The electrodes are coupled to and energized by a bipolar energy source within the generator **1100**. The handpiece **1107** comprises a trigger **1145** to operate the clamp arms **1142a**, **1142b** and an energy button **1135** to actuate an energy switch to energize the electrodes in the end effector **1124**.

(341) The generator **1100** also is configured to drive a multifunction surgical instrument **1108**. The multifunction surgical instrument **1108** comprises a handpiece **1109** (HP), a shaft **1129**, and an end effector **1125**. The end effector **1125** comprises an ultrasonic blade **1149** and a clamp arm **1146**. The ultrasonic blade **1149** is acoustically coupled to the ultrasonic transducer **1120**. The handpiece **1109** comprises a trigger **1147** to operate the clamp arm **1146** and a combination of the toggle buttons **1137a**, **1137b**, **1137c** to energize and drive the ultrasonic blade **1149** or other function. The toggle buttons **1137a**, **1137b**, **1137c** can be configured to energize the ultrasonic transducer **1120** with the generator **1100** and energize the ultrasonic blade **1149** with a bipolar energy source also contained within the generator **1100**.

(342) The generator **1100** is configurable for use with a variety of surgical instruments. According to various forms, the generator **1100** may be configurable for use with different surgical instruments of different types including, for example, the ultrasonic surgical instrument **1104**, the RF electrosurgical instrument **1106**, and the multifunction surgical instrument **1108** that integrates RF and ultrasonic energies delivered simultaneously from the generator **1100**. Although in the form of FIG. **22** the generator **1100** is shown separate from the surgical instruments **1104**, **1106**, **1108**, in another form the generator **1100** may be formed integrally with any one of the surgical instruments **1104**, **1106**, **1108** to form a unitary surgical system. As discussed above, the generator **1100** comprises an input device **1110** located on a front panel of the generator **1100** console. The input device **1110** may comprise any suitable device that generates signals suitable for programming the operation of the generator **1100**. The generator **1100** also may comprise one or more output devices **1112**. Further aspects of generators for digitally generating electrical signal waveforms and surgical instruments are described in US patent publication US-2017-0086914-A1, which is herein incorporated by reference in its entirety.

(343) FIG. **23** is an end effector **1122** of the example ultrasonic device **1104**, in accordance with at least one aspect of the present disclosure. The end effector **1122** may comprise a blade **1128** that may be coupled to the ultrasonic transducer **1120** via a wave guide. When driven by the ultrasonic transducer **1120**, the blade **1128** may vibrate and, when brought into contact with tissue, may cut and/or coagulate the tissue, as described herein. According to various aspects, and as illustrated in FIG. **23**, the end effector **1122** may also comprise a clamp arm **1140** that may be configured for cooperative action with the blade **1128** of the end effector **1122**. With the blade **1128**, the clamp arm **1140** may comprise a set of jaws. The clamp arm **1140** may be pivotally connected at a distal end of a shaft **1126** of the instrument portion **1104**. The clamp arm **1140** may include a clamp arm tissue pad **1163**, which may be formed from TEFLON® or other suitable low-friction material. The pad **1163** may be mounted for cooperation with the blade **1128**, with pivotal movement of the clamp arm **1140** positioning the clamp pad **1163** in substantially parallel relationship to, and in contact with, the blade **1128**. By this construction, a tissue bite to be clamped may be grasped between the tissue pad **1163** and the blade **1128**. The tissue pad **1163** may be provided with a sawtooth-like configuration including a plurality of axially spaced, proximally extending gripping teeth **1161** to enhance the gripping of tissue in cooperation with the blade **1128**. The clamp arm **1140** may transition from the open position shown in FIG. **23** to a closed position (with the clamp arm **1140** in contact with or proximity to the blade **1128**) in any suitable manner. For example, the handpiece **1105** may comprise a jaw closure trigger. When actuated by a clinician, the jaw closure trigger may pivot the clamp arm **1140** in any suitable manner.

(344) The generator **1100** may be activated to provide the drive signal to the ultrasonic transducer **1120** in any suitable manner. For example, the generator **1100** may comprise a foot switch **1430** (FIG. **24**) coupled to the generator **1100** via a footswitch cable **1432**. A clinician may activate the ultrasonic transducer **1120**, and thereby the ultrasonic transducer **1120** and blade **1128**, by depressing the foot switch **1430**. In addition, or instead of the foot switch **1430**, some aspects of the device **1104** may utilize one or more switches positioned on the handpiece **1105** that, when activated, may cause the generator **1100** to activate the ultrasonic transducer **1120**. In one aspect, for example, the one or more switches may comprise a pair of toggle buttons **1134a**, **1134b**, **1134c** (FIG. **22**), for example, to determine an operating mode of the device **1104**. When the toggle button **1134a** is depressed, for example, the ultrasonic generator **1100** may provide a maximum drive signal to the ultrasonic transducer **1120**, causing it to produce maximum ultrasonic energy output. Depressing toggle button **1134b** may cause the ultrasonic generator **1100** to provide a user-selectable drive signal to the ultrasonic transducer **1120**, causing it to produce less than the maximum ultrasonic energy output. The device **1104** additionally or alternatively may comprise a second switch to, for example, indicate a position of a jaw closure trigger for operating the jaws via the clamp arm **1140** of the end effector **1122**. Also, in some aspects, the ultrasonic generator **1100** may be activated based on the position of the jaw closure trigger, (e.g., as the clinician depresses the jaw closure trigger to close the jaws via the clamp arm **1140**, ultrasonic energy may be applied).

(345) Additionally or alternatively, the one or more switches may comprise a toggle button **1134c** that, when depressed, causes the generator **1100** to provide a pulsed output (FIG. **22**). The pulses may be provided at any suitable frequency and grouping, for example. In certain aspects, the power level of the pulses may be the power levels associated with toggle buttons **1134a**, **1134b** (maximum, less than maximum), for example.

(346) It will be appreciated that a device **1104** may comprise any combination of the toggle buttons **1134a**, **1134b**, **1134c** (FIG. **22**). For example, the device **1104** could be configured to have only two toggle buttons: a toggle button **1134a** for producing maximum ultrasonic energy output and a toggle button **1134c** for producing a pulsed output at either the maximum or less than maximum power level per. In this way, the drive signal output configuration of the generator **1100** could be five continuous signals, or any discrete number of individual pulsed signals (**1**, **2**, **3**, **4**, or **5**). In certain aspects, the specific drive signal configuration may be controlled based upon, for example, EEPROM settings in the generator **1100** and/or user power level selection(s).

(347) In certain aspects, a two-position switch may be provided as an alternative to a toggle button **1134c** (FIG. **22**). For example, a device **1104** may include a toggle button **1134a** for producing a continuous output at a maximum power level and a two-position toggle button **1134b**. In a first detented position, toggle button **1134b** may produce a continuous output at a less than maximum power level, and in a second detented position the toggle button **1134b** may produce a pulsed output (e.g., at either a maximum or less than maximum power level, depending upon the EEPROM settings).

(348) In some aspects, the RF electrosurgical end effector **1124**, **1125** (FIG. **22**) may also comprise a pair of electrodes. The electrodes may be in communication with the generator **1100**, for example, via a cable. The electrodes may be used, for example, to measure an impedance of a tissue bite present between the clamp arm **1142a**, **1146** and the blade **1142b**, **1149**. The generator **1100** may provide a signal (e.g., a non-therapeutic signal) to the electrodes. The impedance of the tissue bite may be found, for example, by monitoring the current, voltage, etc. of the signal.

(349) In various aspects, the generator **1100** may comprise several separate functional elements, such as modules and/or blocks, as shown in FIG. **24**, a diagram of the surgical system **1000** of FIG. **22**. Different functional elements or modules may be configured for driving the different kinds of surgical devices **1104**, **1106**, **1108**. For example an ultrasonic generator module may drive an ultrasonic device, such as the ultrasonic device **1104**. An electrosurgery/RF generator module may drive the electrosurgical device **1106**. The modules may generate respective drive signals for driving the surgical devices **1104**, **1106**, **1108**. In various aspects, the ultrasonic generator module and/or the electrosurgery/RF generator module each may be formed integrally with the generator **1100**. Alternatively, one or more of the modules may be provided as a separate circuit module electrically coupled to the generator **1100**. (The modules are shown in phantom to illustrate this option.) Also, in some aspects, the electrosurgery/RF generator module may be formed integrally with the ultrasonic generator module, or vice versa.

(350) In accordance with the described aspects, the ultrasonic generator module may produce a drive signal or signals of particular voltages, currents, and frequencies (e.g. 55,500 cycles per second, or Hz). The drive signal or signals may be provided to the ultrasonic device **1104**, and specifically to the transducer **1120**, which may operate, for example, as described above. In one aspect, the generator **1100** may be configured to produce a drive signal of a particular voltage, current, and/or frequency output signal that can be stepped with high resolution, accuracy, and repeatability.

(351) In accordance with the described aspects, the electrosurgery/RF generator module may generate a drive signal or signals with output power sufficient to perform bipolar electrosurgery using radio frequency (RF) energy. In bipolar electrosurgery applications, the drive signal may be provided, for example, to the electrodes of the electrosurgical device **1106**, for example, as described above. Accordingly, the generator **1100** may be configured for therapeutic purposes by applying electrical energy to the tissue sufficient for treating the tissue (e.g., coagulation, cauterization, tissue welding, etc.).

(352) The generator **1100** may comprise an input device **2150** (FIG. 27B) located, for example, on a front panel of the generator **1100** console. The input device **2150** may comprise any suitable device that generates signals suitable for programming the operation of the generator **1100**. In operation, the user can program or otherwise control operation of the generator **1100** using the input device **2150**. The input device **2150** may comprise any suitable device that generates signals that can be used by the generator (e.g., by one or more processors contained in the generator) to control the operation of the generator **1100** (e.g., operation of the ultrasonic generator module and/or electrosurgery/RF generator module). In various aspects, the input device **2150** includes one or more of: buttons, switches, thumbwheels, keyboard, keypad, touch screen monitor, pointing device, remote connection to a general purpose or dedicated computer. In other aspects, the input device **2150** may comprise a suitable user interface, such as one or more user interface screens displayed on a touch screen monitor, for example. Accordingly, by way of the input device **2150**, the user can set or program various operating parameters of the generator, such as, for example, current (I), voltage (V), frequency (f), and/or period (T) of a drive signal or signals generated by the ultrasonic generator module and/or electrosurgery/RF generator module.

(353) The generator **1100** may also comprise an output device **2140** (FIG. 27B) located, for example, on a front panel of the generator **1100** console. The output device **2140** includes one or more devices for providing a sensory feedback to a user. Such devices may comprise, for example, visual feedback devices (e.g., an LCD display screen, LED indicators), audio feedback devices (e.g., a speaker, a buzzer) or tactile feedback devices (e.g., haptic actuators).

(354) Although certain modules and/or blocks of the generator **1100** may be described by way of example, it can be appreciated that a greater or lesser number of modules and/or blocks may be used and still fall within the scope of the aspects. Further, although various aspects may be described in terms of modules and/or blocks to facilitate description, such modules and/or blocks may be implemented by one or more hardware components, e.g., processors, Digital Signal Processors (DSPs), Programmable Logic Devices (PLDs), Application Specific Integrated Circuits (ASICs), circuits, registers and/or software components, e.g., programs, subroutines, logic and/or combinations of hardware and software components.

(355) In one aspect, the ultrasonic generator drive module and electrosurgery/RF drive module **1110** (FIG. 22) may comprise one or more embedded applications implemented as firmware, software, hardware, or any combination thereof. The modules may comprise various executable modules such as software, programs, data, drivers, application program interfaces (APIs), and so forth. The firmware may be stored in nonvolatile memory (NVM), such as in bit-masked read-only memory (ROM) or flash memory. In various implementations, storing the firmware in ROM may preserve flash memory. The NVM may comprise other types of memory including, for example, programmable ROM (PROM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), or battery backed random-access memory (RAM) such as dynamic RAM (DRAM), Double-Data-Rate DRAM (DDRDRAM), and/or synchronous DRAM (SDRAM).

(356) In one aspect, the modules comprise a hardware component implemented as a processor for executing program instructions for monitoring various measurable characteristics of the devices **1104**, **1106**, **1108** and generating a corresponding output drive signal or signals for operating the devices **1104**, **1106**, **1108**. In aspects in which the generator **1100** is used in conjunction with the device **1104**, the drive signal may drive the ultrasonic transducer **1120** in cutting and/or coagulation operating modes. Electrical characteristics of the device **1104** and/or tissue may be measured and used to control operational aspects of the generator **1100** and/or provided as feedback to the user. In aspects in which the generator **1100** is used in conjunction with the device **1106**, the drive signal may supply electrical energy (e.g., RF energy) to the end effector **1124** in cutting, coagulation and/or desiccation modes. Electrical characteristics of the device **1106** and/or tissue may be measured and used to control operational aspects of the generator **1100** and/or provided as feedback to the user. In various aspects, as previously discussed, the hardware components may be implemented as DSP, PLD, ASIC, circuits, and/or registers. In one aspect, the processor may be configured to store and execute computer software program instructions to generate the step function output signals for driving various components of the devices **1104**, **1106**, **1108**, such as the ultrasonic transducer **1120** and the end effectors **1122**, **1124**, **1125**.

(357) An electromechanical ultrasonic system includes an ultrasonic transducer, a waveguide, and an ultrasonic blade. The electromechanical ultrasonic system has an initial resonant frequency defined by the physical properties of the ultrasonic transducer, the waveguide, and the ultrasonic blade. The ultrasonic transducer is excited by an alternating voltage $V_{\text{sub.g}}(t)$ and current $I_{\text{sub.g}}(t)$ signal equal to the resonant frequency of the electromechanical ultrasonic system. When the electromechanical ultrasonic system is at resonance, the phase difference between the voltage $V_{\text{sub.g}}(t)$ and current $I_{\text{sub.g}}(t)$ signals is zero. Stated another way, at resonance the inductive impedance is equal to the capacitive impedance. As the ultrasonic blade heats up, the compliance of the ultrasonic blade (modeled as an equivalent capacitance) causes the resonant frequency of the electromechanical ultrasonic system to shift. Thus, the inductive impedance is no longer equal to the capacitive impedance causing a mismatch between the drive frequency and the resonant frequency of the electromechanical ultrasonic system. The system is now operating “off-resonance.” The mismatch between the drive frequency and the resonant frequency is manifested as a phase difference between the voltage $V_{\text{sub.g}}(t)$ and current $I_{\text{sub.g}}(t)$ signals applied to the ultrasonic transducer. The generator electronics can easily monitor the phase difference between the voltage $V_{\text{sub.g}}(t)$ and current $I_{\text{sub.g}}(t)$ signals and can continuously adjust the drive frequency until the phase difference is once again zero. At this point, the new drive frequency is equal to the new resonant frequency of the electromechanical ultrasonic system. The change in phase and/or frequency can be used as an indirect measurement of the ultrasonic blade temperature.

(358) As shown in FIG. 25, the electromechanical properties of the ultrasonic transducer may be modeled as an equivalent circuit comprising a first branch having a static capacitance and a second “motional” branch having a serially connected inductance, resistance and capacitance that define the electromechanical properties of a resonator. Known ultrasonic generators may include a tuning inductor for tuning out the static capacitance at a resonant frequency so that substantially all of generator's drive signal current flows into the motional branch. Accordingly, by using a tuning inductor, the generator's drive signal current represents the motional branch current, and the generator is thus able to control its drive signal to maintain the ultrasonic transducer's resonant frequency. The tuning inductor may also transform the phase impedance plot of the ultrasonic transducer to improve the generator's frequency lock capabilities. However, the tuning inductor must be matched with the specific static capacitance of an ultrasonic transducer at the operational resonance frequency. In other words, a different ultrasonic transducer having a different static capacitance requires a different tuning inductor.

(359) FIG. 25 illustrates an equivalent circuit **1500** of a ultrasonic transducer, such as the ultrasonic transducer **1120**, according to one aspect. The circuit **1500** comprises a first “motional” branch having a serially connected inductance $L_{\text{sub.s}}$, resistance $R_{\text{sub.s}}$ and capacitance $C_{\text{sub.s}}$ that define the electromechanical properties of the resonator, and a second capacitive branch having a static capacitance $C_{\text{sub.0}}$. Drive current $I_{\text{sub.g}}(t)$ may be received from a generator at a drive voltage $V_{\text{sub.g}}(t)$, with motional current $I_{\text{sub.m}}(t)$ flowing through the first branch and current $I_{\text{sub.g}}(t) - I_{\text{sub.m}}(t)$ flowing through the capacitive branch. Control of the electromechanical properties of the ultrasonic transducer may be achieved by suitably controlling $I_{\text{sub.g}}(t)$ and $V_{\text{sub.g}}(t)$. As explained above, known generator architectures may include a tuning inductor $L_{\text{sub.t}}$ (shown in phantom in FIG. 25) in a parallel resonance circuit for tuning out the static capacitance $C_{\text{sub.0}}$ at a resonant frequency so that substantially all of the generator's current output $I_{\text{sub.g}}(t)$ flows through the motional branch. In this way, control of the motional branch current $I_{\text{sub.m}}(t)$ is achieved by controlling the generator current output $I_{\text{sub.g}}(t)$. The tuning inductor $L_{\text{sub.t}}$ is specific to the static capacitance $C_{\text{sub.0}}$ of an ultrasonic transducer, however, and a different ultrasonic transducer having a different static capacitance requires a different tuning inductor $L_{\text{sub.t}}$. Moreover, because the tuning inductor $L_{\text{sub.t}}$ is matched to the nominal value of the static capacitance $C_{\text{sub.0}}$ at a single resonant frequency, accurate control of the

motional branch current $I_{\text{sub.m}}(t)$ is assured only at that frequency. As frequency shifts down with transducer frequency, accurate control of the motional branch current is compromised.

(360) Various aspects of the generator **1100** may not rely on a tuning inductor $L_{\text{sub.t}}$ to monitor the motional branch current $I_{\text{sub.m}}(t)$. Instead, the generator **1100** may use the measured value of the static capacitance $C_{\text{sub.0}}$ in between applications of power for a specific ultrasonic surgical device **1104** (along with drive signal voltage and current feedback data) to determine values of the motional branch current $I_{\text{sub.m}}(t)$ on a dynamic and ongoing basis (e.g., in real-time). Such aspects of the generator **1100** are therefore able to provide virtual tuning to simulate a system that is tuned or resonant with any value of static capacitance $C_{\text{sub.0}}$ at any frequency, and not just at a single resonant frequency dictated by a nominal value of the static capacitance $C_{\text{sub.0}}$.

(361) FIG. 26 is a simplified block diagram of one aspect of the generator **1100** for providing inductorless tuning as described above, among other benefits. FIGS. 27A-27C illustrate an architecture of the generator **1100** of FIG. 26 according to one aspect. With reference to FIG. 26, the generator **1100** may comprise a patient isolated stage **1520** in communication with a non-isolated stage **1540** via a power transformer **1560**. A secondary winding **1580** of the power transformer **1560** is contained in the isolated stage **1520** and may comprise a tapped configuration (e.g., a center-tapped or non-center tapped configuration) to define drive signal outputs **1600a**, **1600b**, **1600c** for outputting drive signals to different surgical devices, such as, for example, an ultrasonic surgical device **1104** and an electrosurgical device **1106**. In particular, drive signal outputs **1600a**, **1600b**, **1600c** may output a drive signal (e.g., a 420V RMS drive signal) to an ultrasonic surgical device **1104**, and drive signal outputs **1600a**, **1600b**, **1600c** may output a drive signal (e.g., a 100V RMS drive signal) to an electrosurgical device **1106**, with output **1600b** corresponding to the center tap of the power transformer **1560**. The non-isolated stage **1540** may comprise a power amplifier **1620** having an output connected to a primary winding **1640** of the power transformer **1560**. In certain aspects the power amplifier **1620** may comprise a push-pull amplifier, for example. The non-isolated stage **1540** may further comprise a programmable logic device **1660** for supplying a digital output to a digital-to-analog converter (DAC) **1680**, which in turn supplies a corresponding analog signal to an input of the power amplifier **1620**. In certain aspects the programmable logic device **1660** may comprise a field-programmable gate array (FPGA), for example. The programmable logic device **1660**, by virtue of controlling the power amplifier's **1620** input via the DAC **1680**, may therefore control any of a number of parameters (e.g., frequency, waveform shape, waveform amplitude) of drive signals appearing at the drive signal outputs **1600a**, **1600b**, **1600c**. In certain aspects and as discussed below, the programmable logic device **1660**, in conjunction with a processor (e.g., processor **1740** discussed below), may implement a number of digital signal processing (DSP)-based and/or other control algorithms to control parameters of the drive signals output by the generator **1100**.

(362) Power may be supplied to a power rail of the power amplifier **1620** by a switch-mode regulator **1700**. In certain aspects the switch-mode regulator **1700** may comprise an adjustable buck regulator, for example. As discussed above, the non-isolated stage **1540** may further comprise a processor **1740**, which in one aspect may comprise a DSP processor such as an ADSP-21469 SHARC DSP, available from Analog Devices, Norwood, Mass., for example. In certain aspects the processor **1740** may control operation of the switch-mode power converter **1700** responsive to voltage feedback data received from the power amplifier **1620** by the processor **1740** via an analog-to-digital converter (ADC) **1760**. In one aspect, for example, the processor **1740** may receive as input, via the ADC **1760**, the waveform envelope of a signal (e.g., an RF signal) being amplified by the power amplifier **1620**. The processor **1740** may then control the switch-mode regulator **1700** (e.g., via a pulse-width modulated (PWM) output) such that the rail voltage supplied to the power amplifier **1620** tracks the waveform envelope of the amplified signal. By dynamically modulating the rail voltage of the power amplifier **1620** based on the waveform envelope, the efficiency of the power amplifier **1620** may be significantly improved relative to a fixed rail voltage amplifier scheme. The processor **1740** may be configured for wired or wireless communication.

(363) In certain aspects and as discussed in further detail in connection with FIGS. 28A-28B, the programmable logic device **1660**, in conjunction with the processor **1740**, may implement a direct digital synthesizer (DDS) control scheme to control the waveform shape, frequency and/or amplitude of drive signals output by the generator **1100**. In one aspect, for example, the programmable logic device **1660** may implement a DDS control algorithm **2680** (FIG. 28A) by recalling waveform samples stored in a dynamically-updated look-up table (LUT), such as a RAM LUT which may be embedded in an FPGA. This control algorithm is particularly useful for ultrasonic applications in which an ultrasonic transducer, such as the ultrasonic transducer **1120**, may be driven by a clean sinusoidal current at its resonant frequency. Because other frequencies may excite parasitic resonances, minimizing or reducing the total distortion of the motional branch current may correspondingly minimize or reduce undesirable resonance effects. Because the waveform shape of a drive signal output by the generator **1100** is impacted by various sources of distortion present in the output drive circuit (e.g., the power transformer **1560**, the power amplifier **1620**), voltage and current feedback data based on the drive signal may be input into an algorithm, such as an error control algorithm implemented by the processor **1740**, which compensates for distortion by suitably pre-distorting or modifying the waveform samples stored in the LUT on a dynamic, ongoing basis (e.g., in real-time). In one aspect, the amount or degree of pre-distortion applied to the LUT samples may be based on the error between a computed motional branch current and a desired current waveform shape, with the error being determined on a sample-by-sample basis. In this way, the pre-distorted LUT samples, when processed through the drive circuit, may result in a motional branch drive signal having the desired waveform shape (e.g., sinusoidal) for optimally driving the ultrasonic transducer. In such aspects, the LUT waveform samples will therefore not represent the desired waveform shape of the drive signal, but rather the waveform shape that is required to ultimately produce the desired waveform shape of the motional branch drive signal when distortion effects are taken into account.

(364) The non-isolated stage **1540** may further comprise an ADC **1780** and an ADC **1800** coupled to the output of the power transformer **1560** via respective isolation transformers **1820**, **1840** for respectively sampling the voltage and current of drive signals output by the generator **1100**. In certain aspects, the ADCs **1780**, **1800** may be configured to sample at high speeds (e.g., 80 Msps) to enable oversampling of the drive signals. In one aspect, for example, the sampling speed of the ADCs **1780**, **1800** may enable approximately 200× (depending on drive frequency) oversampling of the drive signals. In certain aspects, the sampling operations of the ADCs **1780**, **1800** may be performed by a single ADC receiving input voltage and current signals via a two-way multiplexer. The use of high-speed sampling in aspects of the generator **1100** may enable, among other things, calculation of the complex current flowing through the motional branch (which may be used in certain aspects to implement DDS-based waveform shape control described above), accurate digital filtering of the sampled signals, and calculation of real power consumption with a high degree of precision. Voltage and current feedback data output by the ADCs **1780**, **1800** may be received and processed (e.g., FIFO buffering, multiplexing) by the programmable logic device **1660** and stored in data memory for subsequent retrieval by, for example, the processor **1740**. As noted above, voltage and current feedback data may be used as input to an algorithm for pre-distorting or modifying LUT waveform samples on a dynamic and ongoing basis. In certain aspects, this may require each stored voltage and current feedback data pair to be indexed based on, or otherwise associated with, a corresponding LUT sample that was output by the programmable logic device **1660** when the voltage and current feedback data pair was acquired. Synchronization of the LUT samples and the voltage and current feedback data in this manner contributes to the correct timing and stability of the pre-distortion algorithm.

(365) In certain aspects, the voltage and current feedback data may be used to control the frequency and/or amplitude (e.g., current amplitude) of the drive signals. In one aspect, voltage and current feedback data may be used to determine impedance phase, e.g., the phase difference between the voltage and current drive signals. The frequency of the drive signal may then be controlled to minimize or reduce the difference between the determined impedance phase and an impedance phase setpoint (e.g., 0°), thereby minimizing or reducing the effects of harmonic distortion and correspondingly enhancing impedance phase measurement accuracy. The determination of phase impedance and a frequency control signal may be implemented in the processor **1740**, for example, with the frequency control signal being supplied as input to a DDS control algorithm implemented by the programmable logic device **1660**.

(366) The impedance phase may be determined through Fourier analysis. In one aspect, the phase difference between the generator voltage $V_{\text{sub.g}}(t)$

and generator current $I_{\text{sub.g}}(t)$ driving signals may be determined using the Fast Fourier Transform (FFT) or the Discrete Fourier Transform (DFT) as follows:

(367)

$$V_g(t) = A_1 \cos(2\pi f_0 t + \phi_1) I_g(t) = A_2 \cos(2\pi f_0 t + \phi_2) V_g(f) = \frac{A_1}{2} (f - f_0) + (f + f_0) \exp(j2\pi f \frac{t}{f_0}) I_g(f) = \frac{A_2}{2} (f - f_0) + (f + f_0) \exp(j2\pi f \frac{t}{f_0})$$

(368) Evaluating the Fourier Transform at the frequency of the sinusoid yields:

$$(369) V_g(f_0) = \frac{A_1}{2} (0) \exp(j\phi_1) \arg V(f_0) = \phi_1 I_g(f_0) = \frac{A_2}{2} (0) \exp(j\phi_2) \arg I(f_0) = \phi_2$$

(370) Other approaches include weighted least-squares estimation, Kalman filtering, and space-vector-based techniques. Virtually all of the processing in an FFT or DFT technique may be performed in the digital domain with the aid of the 2-channel high speed ADC **1780**, **1800**, for example. In one technique, the digital signal samples of the voltage and current signals are Fourier transformed with an FFT or a DFT. The phase angle ϕ at any point in time can be calculated by:

$\phi = 2\pi f t + \phi_{\text{sub.0}}$ where ϕ is the phase angle, f is the frequency, t is time, and $\phi_{\text{sub.0}}$ is the phase at $t=0$.

(371) Another technique for determining the phase difference between the voltage $V_{\text{sub.g}}(t)$ and current $I_{\text{sub.g}}(t)$ signals is the zero-crossing method and produces highly accurate results. For voltage $V_{\text{sub.g}}(t)$ and current $I_{\text{sub.g}}(t)$ signals having the same frequency, each negative to positive zero-crossing of voltage signal $V_{\text{sub.g}}(t)$ triggers the start of a pulse, while each negative to positive zero-crossing of current signal $I_{\text{sub.g}}(t)$ triggers the end of the pulse. The result is a pulse train with a pulse width proportional to the phase angle between the voltage signal and the current signal. In one aspect, the pulse train may be passed through an averaging filter to yield a measure of the phase difference. Furthermore, if the positive to negative zero crossings also are used in a similar manner, and the results averaged, any effects of DC and harmonic components can be reduced. In one implementation, the analog voltage $V_{\text{sub.g}}(t)$ and current $I_{\text{sub.g}}(t)$ signals are converted to digital signals that are high if the analog signal is positive and low if the analog signal is negative. High accuracy phase estimates require sharp transitions between high and low. In one aspect, a Schmitt trigger along with an RC stabilization network may be employed to convert the analog signals into digital signals. In other aspects, an edge triggered RS flip-flop and ancillary circuitry may be employed. In yet another aspect, the zero-crossing technique may employ an exclusive OR (XOR) gate.

(372) Other techniques for determining the phase difference between the voltage and current signals include Lissajous figures and monitoring the image; methods such as the three-voltmeter method, the crossed-coil method, vector voltmeter and vector impedance methods; and using phase standard instruments, phase-locked loops, and other techniques as described in O'Shea, Peter, "Phase Measurement" 2000 CRC Press LLC, which is incorporated by reference herein in its entirety.

(373) In another aspect, for example, the current feedback data may be monitored in order to maintain the current amplitude of the drive signal at a current amplitude setpoint. The current amplitude setpoint may be specified directly or determined indirectly based on specified voltage amplitude and power setpoints. In certain aspects, control of the current amplitude may be implemented by control algorithm, such as, for example, a proportional-integral-derivative (PID) control algorithm, in the processor **1740**. Variables controlled by the control algorithm to suitably control the current amplitude of the drive signal may include, for example, the scaling of the LUT waveform samples stored in the programmable logic device **1660** and/or the full-scale output voltage of the DAC **1680** (which supplies the input to the power amplifier **1620**) via a DAC **1860**.

(374) The non-isolated stage **1540** may further comprise a processor **1900** for providing, among other things, user interface (UI) functionality. In one aspect, the processor **1900** may comprise an Atmel AT91 SAM9263 processor having an ARM 926EJ-S core, available from Atmel Corporation, San Jose, Calif., for example. Examples of UI functionality supported by the processor **1900** may include audible and visual user feedback, communication with peripheral devices (e.g., via a Universal Serial Bus (USB) interface), communication with a foot switch **1430**, communication with an input device **2150** (e.g., a touch screen display) and communication with an output device **2140** (e.g., a speaker). The processor **1900** may communicate with the processor **1740** and the programmable logic device (e.g., via a serial peripheral interface (SPI) bus). Although the processor **1900** may primarily support UI functionality, it may also coordinate with the processor **1740** to implement hazard mitigation in certain aspects. For example, the processor **1900** may be programmed to monitor various aspects of user input and/or other inputs (e.g., touch screen inputs **2150**, foot switch **1430** inputs, temperature sensor inputs **2160**) and may disable the drive output of the generator **1100** when an erroneous condition is detected.

(375) In certain aspects, both the processor **1740** (FIG. 26, 27A) and the processor **1900** (FIG. 26, 27B) may determine and monitor the operating state of the generator **1100**. For processor **1740**, the operating state of the generator **1100** may dictate, for example, which control and/or diagnostic processes are implemented by the processor **1740**. For processor **1900**, the operating state of the generator **1100** may dictate, for example, which elements of a user interface (e.g., display screens, sounds) are presented to a user. The processors **1740**, **1900** may independently maintain the current operating state of the generator **1100** and recognize and evaluate possible transitions out of the current operating state. The processor **1740** may function as the master in this relationship and determine when transitions between operating states are to occur. The processor **1900** may be aware of valid transitions between operating states and may confirm if a particular transition is appropriate. For example, when the processor **1740** instructs the processor **1900** to transition to a specific state, the processor **1900** may verify that the requested transition is valid. In the event that a requested transition between states is determined to be invalid by the processor **1900**, the processor **1900** may cause the generator **1100** to enter a failure mode.

(376) The non-isolated stage **1540** may further comprise a controller **1960** (FIG. 26, 27B) for monitoring input devices **2150** (e.g., a capacitive touch sensor used for turning the generator **1100** on and off, a capacitive touch screen). In certain aspects, the controller **1960** may comprise at least one processor and/or other controller device in communication with the processor **1900**. In one aspect, for example, the controller **1960** may comprise a processor (e.g., a Mega168 8-bit controller available from Atmel) configured to monitor user input provided via one or more capacitive touch sensors. In one aspect, the controller **1960** may comprise a touch screen controller (e.g., a QT5480 touch screen controller available from Atmel) to control and manage the acquisition of touch data from a capacitive touch screen.

(377) In certain aspects, when the generator **1100** is in a "power off" state, the controller **1960** may continue to receive operating power (e.g., via a line from a power supply of the generator **1100**, such as the power supply **2110** (FIG. 26) discussed below). In this way, the controller **1960** may continue to monitor an input device **2150** (e.g., a capacitive touch sensor located on a front panel of the generator **1100**) for turning the generator **1100** on and off. When the generator **1100** is in the "power off" state, the controller **1960** may wake the power supply (e.g., enable operation of one or more DC/DC voltage converters **2130** (FIG. 26) of the power supply **2110**) if activation of the "on/off" input device **2150** by a user is detected. The controller **1960** may therefore initiate a sequence for transitioning the generator **1100** to a "power on" state. Conversely, the controller **1960** may initiate a sequence for transitioning the generator **1100** to the "power off" state if activation of the "on/off" input device **2150** is detected when the generator **1100** is in the "power on" state. In certain aspects, for example, the controller **1960** may report activation of the "on/off" input device **2150** to the processor **1900**, which in turn implements the necessary process sequence for transitioning the generator **1100** to the "power off" state. In such aspects, the controller **1960** may have no independent ability for causing the removal of power from the generator **1100** after its "power on" state has been established.

(378) In certain aspects, the controller **1960** may cause the generator **1100** to provide audible or other sensory feedback for alerting the user that a "power on" or "power off" sequence has been initiated. Such an alert may be provided at the beginning of a "power on" or "power off" sequence and prior to the commencement of other processes associated with the sequence.

(379) In certain aspects, the isolated stage **1520** may comprise an instrument interface circuit **1980** to, for example, provide a communication interface between a control circuit of a surgical device (e.g., a control circuit comprising handpiece switches) and components of the non-isolated stage **1540**, such as, for example, the programmable logic device **1660**, the processor **1740** and/or the processor **1900**. The instrument interface circuit **1980** may exchange information with components of the non-isolated stage **1540** via a communication link that maintains a suitable degree of

electrical isolation between the stages **1520**, **1540**, such as, for example, an infrared (IR)-based communication link. Power may be supplied to the instrument interface circuit **1980** using, for example, a low-dropout voltage regulator powered by an isolation transformer driven from the non-isolated stage **1540**.

(380) In one aspect, the instrument interface circuit **1980** may comprise a programmable logic device **2000** (e.g., an FPGA) in communication with a signal conditioning circuit **2020** (FIG. **26** and FIG. **27C**). The signal conditioning circuit **2020** may be configured to receive a periodic signal from the programmable logic device **2000** (e.g., a 2 kHz square wave) to generate a bipolar interrogation signal having an identical frequency. The interrogation signal may be generated, for example, using a bipolar current source fed by a differential amplifier. The interrogation signal may be communicated to a surgical device control circuit (e.g., by using a conductive pair in a cable that connects the generator **1100** to the surgical device) and monitored to determine a state or configuration of the control circuit. For example, the control circuit may comprise a number of switches, resistors and/or diodes to modify one or more characteristics (e.g., amplitude, rectification) of the interrogation signal such that a state or configuration of the control circuit is uniquely discernible based on the one or more characteristics. In one aspect, for example, the signal conditioning circuit **2020** may comprise an ADC for generating samples of a voltage signal appearing across inputs of the control circuit resulting from passage of interrogation signal therethrough. The programmable logic device **2000** (or a component of the non-isolated stage **1540**) may then determine the state or configuration of the control circuit based on the ADC samples.

(381) In one aspect, the instrument interface circuit **1980** may comprise a first data circuit interface **2040** to enable information exchange between the programmable logic device **2000** (or other element of the instrument interface circuit **1980**) and a first data circuit disposed in or otherwise associated with a surgical device. In certain aspects, for example, a first data circuit **2060** may be disposed in a cable integrally attached to a surgical device handpiece, or in an adaptor for interfacing a specific surgical device type or model with the generator **1100**. In certain aspects, the first data circuit may comprise a non-volatile storage device, such as an electrically erasable programmable read-only memory (EEPROM) device. In certain aspects and referring again to FIG. **26**, the first data circuit interface **2040** may be implemented separately from the programmable logic device **2000** and comprise suitable circuitry (e.g., discrete logic devices, a processor) to enable communication between the programmable logic device **2000** and the first data circuit. In other aspects, the first data circuit interface **2040** may be integral with the programmable logic device **2000**.

(382) In certain aspects, the first data circuit **2060** may store information pertaining to the particular surgical device with which it is associated. Such information may include, for example, a model number, a serial number, a number of operations in which the surgical device has been used, and/or any other type of information. This information may be read by the instrument interface circuit **1980** (e.g., by the programmable logic device **2000**), transferred to a component of the non-isolated stage **1540** (e.g., to programmable logic device **1660**, processor **1740** and/or processor **1900**) for presentation to a user via an output device **2140** and/or for controlling a function or operation of the generator **1100**. Additionally, any type of information may be communicated to first data circuit **2060** for storage therein via the first data circuit interface **2040** (e.g., using the programmable logic device **2000**). Such information may comprise, for example, an updated number of operations in which the surgical device has been used and/or dates and/or times of its usage.

(383) As discussed previously, a surgical instrument may be detachable from a handpiece (e.g., instrument **1106** may be detachable from handpiece **1107**) to promote instrument interchangeability and/or disposability. In such cases, known generators may be limited in their ability to recognize particular instrument configurations being used and to optimize control and diagnostic processes accordingly. The addition of readable data circuits to surgical device instruments to address this issue is problematic from a compatibility standpoint, however. For example, it may be impractical to design a surgical device to maintain backward compatibility with generators that lack the requisite data reading functionality due to, for example, differing signal schemes, design complexity and cost. Other aspects of instruments address these concerns by using data circuits that may be implemented in existing surgical instruments economically and with minimal design changes to preserve compatibility of the surgical devices with current generator platforms.

(384) Additionally, aspects of the generator **1100** may enable communication with instrument-based data circuits. For example, the generator **1100** may be configured to communicate with a second data circuit (e.g., a data circuit) contained in an instrument (e.g., instrument **1104**, **1106** or **1108**) of a surgical device. The instrument interface circuit **1980** may comprise a second data circuit interface **2100** to enable this communication. In one aspect, the second data circuit interface **2100** may comprise a tri-state digital interface, although other interfaces may also be used. In certain aspects, the second data circuit may generally be any circuit for transmitting and/or receiving data. In one aspect, for example, the second data circuit may store information pertaining to the particular surgical instrument with which it is associated. Such information may include, for example, a model number, a serial number, a number of operations in which the surgical instrument has been used, and/or any other type of information. Additionally or alternatively, any type of information may be communicated to the second data circuit for storage therein via the second data circuit interface **2100** (e.g., using the programmable logic device **2000**). Such information may comprise, for example, an updated number of operations in which the instrument has been used and/or dates and/or times of its usage. In certain aspects, the second data circuit may transmit data acquired by one or more sensors (e.g., an instrument-based temperature sensor). In certain aspects, the second data circuit may receive data from the generator **1100** and provide an indication to a user (e.g., an LED indication or other visible indication) based on the received data.

(385) In certain aspects, the second data circuit and the second data circuit interface **2100** may be configured such that communication between the programmable logic device **2000** and the second data circuit can be effected without the need to provide additional conductors for this purpose (e.g., dedicated conductors of a cable connecting a handpiece to the generator **1100**). In one aspect, for example, information may be communicated to and from the second data circuit using a one-wire bus communication scheme implemented on existing cabling, such as one of the conductors used to transmit interrogation signals from the signal conditioning circuit **2020** to a control circuit in a handpiece. In this way, design changes or modifications to the surgical device that might otherwise be necessary are minimized or reduced. Moreover, because different types of communications can be implemented over a common physical channel (either with or without frequency-band separation), the presence of a second data circuit may be “invisible” to generators that do not have the requisite data reading functionality, thus enabling backward compatibility of the surgical device instrument.

(386) In certain aspects, the isolated stage **1520** may comprise at least one blocking capacitor **2960-1** (FIG. **27C**) connected to the drive signal output **1600b** to prevent passage of DC current to a patient. A single blocking capacitor may be required to comply with medical regulations or standards, for example. While failure in single-capacitor designs is relatively uncommon, such failure may nonetheless have negative consequences. In one aspect, a second blocking capacitor **2960-2** may be provided in series with the blocking capacitor **2960-1**, with current leakage from a point between the blocking capacitors **2960-1**, **2960-2** being monitored by, for example, an ADC **2980** for sampling a voltage induced by leakage current. The samples may be received by the programmable logic device **2000**, for example. Based on changes in the leakage current (as indicated by the voltage samples in the aspect of FIG. **26**), the generator **1100** may determine when at least one of the blocking capacitors **2960-1**, **2960-2** has failed.

Accordingly, the aspect of FIG. **26** may provide a benefit over single-capacitor designs having a single point of failure.

(387) In certain aspects, the non-isolated stage **1540** may comprise a power supply **2110** for outputting DC power at a suitable voltage and current. The power supply may comprise, for example, a 400 W power supply for outputting a 48 VDC system voltage. As discussed above, the power supply **2110** may further comprise one or more DC/DC voltage converters **2130** for receiving the output of the power supply to generate DC outputs at the voltages and currents required by the various components of the generator **1100**. As discussed above in connection with the controller **1960**, one or more of the DC/DC voltage converters **2130** may receive an input from the controller **1960** when activation of the “on/off” input device **2150** by a user is detected by the controller **1960** to enable operation of, or wake, the DC/DC voltage converters **2130**.

(388) FIGS. **28A-28B** illustrate certain functional and structural aspects of one aspect of the generator **1100**. Feedback indicating current and voltage output from the secondary winding **1580** of the power transformer **1560** is received by the ADCs **1780**, **1800**, respectively. As shown, the ADCs

1780, 1800 may be implemented as a 2-channel ADC and may sample the feedback signals at a high speed (e.g., 80 Msps) to enable oversampling (e.g., approximately 200× oversampling) of the drive signals. The current and voltage feedback signals may be suitably conditioned in the analog domain (e.g., amplified, filtered) prior to processing by the ADCs **1780, 1800**. Current and voltage feedback samples from the ADCs **1780, 1800** may be individually buffered and subsequently multiplexed or interleaved into a single data stream within block **2120** of the programmable logic device **1660**. In the aspect of FIGS. **28A-28B**, the programmable logic device **1660** comprises an FPGA.

(389) The multiplexed current and voltage feedback samples may be received by a parallel data acquisition port (PDAP) implemented within block **2144** of the processor **1740**. The PDAP may comprise a packing unit for implementing any of a number of methodologies for correlating the multiplexed feedback samples with a memory address. In one aspect, for example, feedback samples corresponding to a particular LUT sample output by the programmable logic device **1660** may be stored at one or more memory addresses that are correlated or indexed with the LUT address of the LUT sample. In another aspect, feedback samples corresponding to a particular LUT sample output by the programmable logic device **1660** may be stored, along with the LUT address of the LUT sample, at a common memory location. In any event, the feedback samples may be stored such that the address of the LUT sample from which a particular set of feedback samples originated may be subsequently ascertained. As discussed above, synchronization of the LUT sample addresses and the feedback samples in this way contributes to the correct timing and stability of the pre-distortion algorithm. A direct memory access (DMA) controller implemented at block **2166** of the processor **1740** may store the feedback samples (and any LUT sample address data, where applicable) at a designated memory location **2180** of the processor **1740** (e.g., internal RAM).

(390) Block **2200** of the processor **1740** may implement a pre-distortion algorithm for pre-distorting or modifying the LUT samples stored in the programmable logic device **1660** on a dynamic, ongoing basis. As discussed above, pre-distortion of the LUT samples may compensate for various sources of distortion present in the output drive circuit of the generator **1100**. The pre-distorted LUT samples, when processed through the drive circuit, will therefore result in a drive signal having the desired waveform shape (e.g., sinusoidal) for optimally driving the ultrasonic transducer.

(391) At block **2220** of the pre-distortion algorithm, the current through the motional branch of the ultrasonic transducer is determined. The motional branch current may be determined using Kirchhoff's Current Law based on, for example, the current and voltage feedback samples stored at memory location **2180** (which, when suitably scaled, may be representative of $I_{sub.g}$ and $V_{sub.g}$ in the model of FIG. **25** discussed above), a value of the ultrasonic transducer static capacitance $C_{sub.0}$ (measured or known a priori) and a known value of the drive frequency. A motional branch current sample for each set of stored current and voltage feedback samples associated with a LUT sample may be determined.

(392) At block **2240** of the pre-distortion algorithm, each motional branch current sample determined at block **2220** is compared to a sample of a desired current waveform shape to determine a difference, or sample amplitude error, between the compared samples. For this determination, the sample of the desired current waveform shape may be supplied, for example, from a waveform shape LUT **2260** containing amplitude samples for one cycle of a desired current waveform shape. The particular sample of the desired current waveform shape from the LUT **2260** used for the comparison may be dictated by the LUT sample address associated with the motional branch current sample used in the comparison. Accordingly, the input of the motional branch current to block **2240** may be synchronized with the input of its associated LUT sample address to block **2240**. The LUT samples stored in the programmable logic device **1660** and the LUT samples stored in the waveform shape LUT **2260** may therefore be equal in number. In certain aspects, the desired current waveform shape represented by the LUT samples stored in the waveform shape LUT **2260** may be a fundamental sine wave. Other waveform shapes may be desirable. For example, it is contemplated that a fundamental sine wave for driving main longitudinal motion of an ultrasonic transducer superimposed with one or more other drive signals at other frequencies, such as a third order harmonic for driving at least two mechanical resonances for beneficial vibrations of transverse or other modes, could be used.

(393) Each value of the sample amplitude error determined at block **2240** may be transmitted to the LUT of the programmable logic device **1660** (shown at block **2280** in FIG. **28A**) along with an indication of its associated LUT address. Based on the value of the sample amplitude error and its associated address (and, optionally, values of sample amplitude error for the same LUT address previously received), the LUT **2280** (or other control block of the programmable logic device **1660**) may pre-distort or modify the value of the LUT sample stored at the LUT address such that the sample amplitude error is reduced or minimized. It will be appreciated that such pre-distortion or modification of each LUT sample in an iterative manner across the entire range of LUT addresses will cause the waveform shape of the generator's output current to match or conform to the desired current waveform shape represented by the samples of the waveform shape LUT **2260**.

(394) Current and voltage amplitude measurements, power measurements and impedance measurements may be determined at block **2300** of the processor **1740** based on the current and voltage feedback samples stored at memory location **2180**. Prior to the determination of these quantities, the feedback samples may be suitably scaled and, in certain aspects, processed through a suitable filter **2320** to remove noise resulting from, for example, the data acquisition process and induced harmonic components. The filtered voltage and current samples may therefore substantially represent the fundamental frequency of the generator's drive output signal. In certain aspects, the filter **2320** may be a finite impulse response (FIR) filter applied in the frequency domain. Such aspects may use the Fast Fourier Transform (FFT) of the output drive signal current and voltage signals. In certain aspects, the resulting frequency spectrum may be used to provide additional generator functionality. In one aspect, for example, the ratio of the second and/or third order harmonic component relative to the fundamental frequency component may be used as a diagnostic indicator.

(395) At block **2340** (FIG. **28B**), a root mean square (RMS) calculation may be applied to a sample size of the current feedback samples representing an integral number of cycles of the drive signal to generate a measurement $I_{sub.rms}$ representing the drive signal output current.

(396) At block **2360**, a root mean square (RMS) calculation may be applied to a sample size of the voltage feedback samples representing an integral number of cycles of the drive signal to determine a measurement $V_{sub.rms}$ representing the drive signal output voltage.

(397) At block **2380**, the current and voltage feedback samples may be multiplied point by point, and a mean calculation is applied to samples representing an integral number of cycles of the drive signal to determine a measurement $P_{sub.r}$ of the generator's real output power.

(398) At block **2400**, measurement $P_{sub.a}$ of the generator's apparent output power may be determined as the product $V_{sub.rms} \cdot I_{sub.rms}$.

(399) At block **2420**, measurement $Z_{sub.m}$ of the load impedance magnitude may be determined as the quotient $V_{sub.rms}/I_{sub.rms}$.

(400) In certain aspects, the quantities $I_{sub.rms}$, $V_{sub.rms}$, $P_{sub.r}$, $P_{sub.a}$ and $Z_{sub.m}$ determined at blocks **2340, 2360, 2380, 2400** and **2420** may be used by the generator **1100** to implement any of a number of control and/or diagnostic processes. In certain aspects, any of these quantities may be communicated to a user via, for example, an output device **2140** integral with the generator **1100** or an output device **2140** connected to the generator **1100** through a suitable communication interface (e.g., a USB interface). Various diagnostic processes may include, without limitation, handpiece integrity, instrument integrity, instrument attachment integrity, instrument overload, approaching instrument overload, frequency lock failure, over-voltage condition, over-current condition, over-power condition, voltage sense failure, current sense failure, audio indication failure, visual indication failure, short circuit condition, power delivery failure, or blocking capacitor failure, for example.

(401) Block **2440** of the processor **1740** may implement a phase control algorithm for determining and controlling the impedance phase of an electrical load (e.g., the ultrasonic transducer) driven by the generator **1100**. As discussed above, by controlling the frequency of the drive signal to minimize or reduce the difference between the determined impedance phase and an impedance phase setpoint (e.g., 0°), the effects of harmonic distortion may be minimized or reduced, and the accuracy of the phase measurement increased.

(402) The phase control algorithm receives as input the current and voltage feedback samples stored in the memory location **2180**. Prior to their use in the phase control algorithm, the feedback samples may be suitably scaled and, in certain aspects, processed through a suitable filter **2460** (which may be identical to filter **2320**) to remove noise resulting from the data acquisition process and induced harmonic components, for example. The filtered voltage and current samples may therefore substantially represent the fundamental frequency of the generator's drive output signal.

(403) At block **2480** of the phase control algorithm, the current through the motional branch of the ultrasonic transducer is determined. This determination may be identical to that described above in connection with block **2220** of the pre-distortion algorithm. The output of block **2480** may

thus be, for each set of stored current and voltage feedback samples associated with a LUT sample, a motional branch current sample.

(404) At block **2500** of the phase control algorithm, impedance phase is determined based on the synchronized input of motional branch current samples determined at block **2480** and corresponding voltage feedback samples. In certain aspects, the impedance phase is determined as the average of the impedance phase measured at the rising edge of the waveforms and the impedance phase measured at the falling edge of the waveforms.

(405) At block **2520** of the phase control algorithm, the value of the impedance phase determined at block **2220** is compared to phase setpoint **2540** to determine a difference, or phase error, between the compared values.

(406) At block **2560** (FIG. **28A**) of the phase control algorithm, based on a value of phase error determined at block **2520** and the impedance magnitude determined at block **2420**, a frequency output for controlling the frequency of the drive signal is determined. The value of the frequency output may be continuously adjusted by the block **2560** and transferred to a DDS control block **2680** (discussed below) in order to maintain the impedance phase determined at block **2500** at the phase setpoint (e.g., zero phase error). In certain aspects, the impedance phase may be regulated to a 0° phase setpoint. In this way, any harmonic distortion will be centered about the crest of the voltage waveform, enhancing the accuracy of phase impedance determination.

(407) Block **2580** of the processor **1740** may implement an algorithm for modulating the current amplitude of the drive signal in order to control the drive signal current, voltage and power in accordance with user specified setpoints, or in accordance with requirements specified by other processes or algorithms implemented by the generator **1100**. Control of these quantities may be realized, for example, by scaling the LUT samples in the LUT **2280** and/or by adjusting the full-scale output voltage of the DAC **1680** (which supplies the input to the power amplifier **1620**) via a DAC **1860**. Block **2600** (which may be implemented as a PID controller in certain aspects) may receive, as input, current feedback samples (which may be suitably scaled and filtered) from the memory location **2180**. The current feedback samples may be compared to a “current demand” $I_{sub.d}$ value dictated by the controlled variable (e.g., current, voltage or power) to determine if the drive signal is supplying the necessary current. In aspects in which drive signal current is the control variable, the current demand $I_{sub.d}$ may be specified directly by a current setpoint **2620A** ($I_{sub.sp}$). For example, an RMS value of the current feedback data (determined as in block **2340**) may be compared to user-specified RMS current setpoint $I_{sub.sp}$ to determine the appropriate controller action. If, for example, the current feedback data indicates an RMS value less than the current setpoint $I_{sub.sp}$, LUT scaling and/or the full-scale output voltage of the DAC **1680** may be adjusted by the block **2600** such that the drive signal current is increased. Conversely, block **2600** may adjust LUT scaling and/or the full-scale output voltage of the DAC **1680** to decrease the drive signal current when the current feedback data indicates an RMS value greater than the current setpoint $I_{sub.sp}$.

(408) In aspects in which the drive signal voltage is the control variable, the current demand $I_{sub.d}$ may be specified indirectly, for example, based on the current required to maintain a desired voltage setpoint **2620B** ($V_{sub.sp}$) given the load impedance magnitude $Z_{sub.m}$ measured at block **2420** (e.g. $I_{sub.d} = V_{sub.sp} / Z_{sub.m}$). Similarly, in aspects in which drive signal power is the control variable, the current demand $I_{sub.d}$ may be specified indirectly, for example, based on the current required to maintain a desired power setpoint **2620C** ($P_{sub.sp}$) given the voltage $V_{sub.rms}$ measured at blocks **2360** (e.g. $I_{sub.d} = P_{sub.sp} / V_{sub.rms}$).

(409) Block **2680** (FIG. **28A**) may implement a DDS control algorithm for controlling the drive signal by recalling LUT samples stored in the LUT **2280**. In certain aspects, the DDS control algorithm may be a numerically-controlled oscillator (NCO) algorithm for generating samples of a waveform at a fixed clock rate using a point (memory location)-skipping technique. The NCO algorithm may implement a phase accumulator, or frequency-to-phase converter, that functions as an address pointer for recalling LUT samples from the LUT **2280**. In one aspect, the phase accumulator may be a D step size, modulo N phase accumulator, where D is a positive integer representing a frequency control value, and N is the number of LUT samples in the LUT **2280**. A frequency control value of $D=1$, for example, may cause the phase accumulator to sequentially point to every address of the LUT **2280**, resulting in a waveform output replicating the waveform stored in the LUT **2280**. When $D>1$, the phase accumulator may skip addresses in the LUT **2280**, resulting in a waveform output having a higher frequency. Accordingly, the frequency of the waveform generated by the DDS control algorithm may therefore be controlled by suitably varying the frequency control value. In certain aspects, the frequency control value may be determined based on the output of the phase control algorithm implemented at block **2440**. The output of block **2680** may supply the input of DAC **1680**, which in turn supplies a corresponding analog signal to an input of the power amplifier **1620**.

(410) Block **2700** of the processor **1740** may implement a switch-mode converter control algorithm for dynamically modulating the rail voltage of the power amplifier **1620** based on the waveform envelope of the signal being amplified, thereby improving the efficiency of the power amplifier **1620**. In certain aspects, characteristics of the waveform envelope may be determined by monitoring one or more signals contained in the power amplifier **1620**. In one aspect, for example, characteristics of the waveform envelope may be determined by monitoring the minima of a drain voltage (e.g., a MOSFET drain voltage) that is modulated in accordance with the envelope of the amplified signal. A minima voltage signal may be generated, for example, by a voltage minima detector coupled to the drain voltage. The minima voltage signal may be sampled by ADC **1760**, with the output minima voltage samples being received at block **2720** of the switch-mode converter control algorithm. Based on the values of the minima voltage samples, block **2740** may control a PWM signal output by a PWM generator **2760**, which, in turn, controls the rail voltage supplied to the power amplifier **1620** by the switch-mode regulator **1700**. In certain aspects, as long as the values of the minima voltage samples are less than a minima target **2780** input into block **2720**, the rail voltage may be modulated in accordance with the waveform envelope as characterized by the minima voltage samples. When the minima voltage samples indicate low envelope power levels, for example, block **2740** may cause a low rail voltage to be supplied to the power amplifier **1620**, with the full rail voltage being supplied only when the minima voltage samples indicate maximum envelope power levels. When the minima voltage samples fall below the minima target **2780**, block **2740** may cause the rail voltage to be maintained at a minimum value suitable for ensuring proper operation of the power amplifier **1620**.

(411) FIG. **29** is a schematic diagram of one aspect of an electrical circuit **2900**, suitable for driving an ultrasonic transducer, such as ultrasonic transducer **1120**, in accordance with at least one aspect of the present disclosure. The electrical circuit **2900** comprises an analog multiplexer **2980**. The analog multiplexer **2980** multiplexes various signals from the upstream channels SCL-A, SDA-A such as ultrasonic, battery, and power control circuit. A current sensor **2982** is coupled in series with the return or ground leg of the power supply circuit to measure the current supplied by the power supply. A field effect transistor (FET) temperature sensor **2984** provides the ambient temperature. A pulse width modulation (PWM) watchdog timer **2988** automatically generates a system reset if the main program neglects to periodically service it. It is provided to automatically reset the electrical circuit **2900** when it hangs or freezes because of a software or hardware fault. It will be appreciated that the electrical circuit **2900** may be configured as an RF driver circuit for driving the ultrasonic transducer or for driving RF electrodes such as the electrical circuit **3600** shown in FIG. **36**, for example. Accordingly, with reference now back to FIG. **29**, the electrical circuit **2900** can be used to drive both ultrasonic transducers and RF electrodes interchangeably. If driven simultaneously, filter circuits may be provided in the corresponding first stage circuits **3404** (FIG. **34**) to select either the ultrasonic waveform or the RF waveform. Such filtering techniques are described in commonly owned U.S. Pat. Pub. No. US-2017-0086910-A1, titled TECHNIQUES FOR CIRCUIT TOPOLOGIES FOR COMBINED GENERATOR, which is herein incorporated by reference in its entirety.

(412) A drive circuit **2986** provides left and right ultrasonic energy outputs. A digital signal that represents the signal waveform is provided to the SCL-A, SDA-A inputs of the analog multiplexer **2980** from a control circuit, such as the control circuit **3200** (FIG. **32**). A digital-to-analog converter **2990** (DAC) converts the digital input to an analog output to drive a PWM circuit **2992** coupled to an oscillator **2994**. The PWM circuit **2992** provides a first signal to a first gate drive circuit **2996a** coupled to a first transistor output stage **2998a** to drive a first Ultrasonic (LEFT) energy output. The PWM circuit **2992** also provides a second signal to a second gate drive circuit **2996b** coupled to a second transistor output stage **2998b** to drive a second Ultrasonic (RIGHT) energy output. A voltage sensor **2999** is coupled between the Ultrasonic LEFT/RIGHT output terminals to measure the output voltage. The drive circuit **2986**, the first and second drive circuits **2996a**, **2996b**, and the first and second transistor output stages

2998a, 2998b define a first stage amplifier circuit. In operation, the control circuit 3200 (FIG. 32) generates a digital waveform 4300 (FIG. 43) employing circuits such as direct digital synthesis (DDS) circuits 4100, 4200 (FIGS. 41 and 42). The DAC 2990 receives the digital waveform 4300 and converts it into an analog waveform, which is received and amplified by the first stage amplifier circuit.

(413) FIG. 30 is a schematic diagram of the transformer 3000 coupled to the electrical circuit 2900 shown in FIG. 29, in accordance with at least one aspect of the present disclosure. The Ultrasonic LEFT/RIGHT input terminals (primary winding) of the transformer 3000 are electrically coupled to the Ultrasonic LEFT/RIGHT output terminals of the electrical circuit 2900. The secondary winding of the transformer 3000 are coupled to the positive and negative electrodes 3074a, 3074b. The positive and negative electrodes 3074a, 3074b of the transformer 3000 are coupled to the positive terminal (Stack 1) and the negative terminal (Stack 2) of an ultrasonic transducer. In one aspect, the transformer 3000 has a turns-ratio of $n_{\text{sub.1}}:n_{\text{sub.2}}$ of 1:50.

(414) FIG. 31 is a schematic diagram of the transformer 3000 shown in FIG. 30 coupled to a test circuit 3165, in accordance with at least one aspect of the present disclosure. The test circuit 3165 is coupled to the positive and negative electrodes 3074a, 3074b. A switch 3167 is placed in series with an inductor/capacitor/resistor (LCR) load that simulates the load of an ultrasonic transducer.

(415) FIG. 32 is a schematic diagram of a control circuit 3200, such as control circuit 3212, in accordance with at least one aspect of the present disclosure. The control circuit 3200 is located within a housing of the battery assembly. The battery assembly is the energy source for a variety of local power supplies 3215. The control circuit comprises a main processor 3214 coupled via an interface master 3218 to various downstream circuits by way of outputs SCL-A and SDA-A, SCL-B and SDA-B, SCL-C and SDA-C, for example. In one aspect, the interface master 3218 is a general purpose serial interface such as an I^{sup}.2C serial interface. The main processor 3214 also is configured to drive switches 3224 through general purposes input/output (GPIO) 3220, a display 3226 (e.g., and LCD display), and various indicators 3228 through GPIO 3222. A watchdog processor 3216 is provided to control the main processor 3214. A switch 3230 is provided in series with a battery 3211 to activate the control circuit 3212 upon insertion of the battery assembly into a handle assembly of a surgical instrument.

(416) In one aspect, the main processor 3214 is coupled to the electrical circuit 2900 (FIG. 29) by way of output terminals SCL-A, SDA-A. The main processor 3214 comprises a memory for storing tables of digitized drive signals or waveforms that are transmitted to the electrical circuit 2900 for driving the ultrasonic transducer 1120, for example. In other aspects, the main processor 3214 may generate a digital waveform and transmit it to the electrical circuit 2900 or may store the digital waveform for later transmission to the electrical circuit 2900. The main processor 3214 also may provide RF drive by way of output terminals SCL-B, SDA-B and various sensors (e.g., Hall-effect sensors, magneto-rheological fluid (MRF) sensors, etc.) by way of output terminals SCL-C, SDA-C. In one aspect, the main processor 3214 is configured to sense the presence of ultrasonic drive circuitry and/or RF drive circuitry to enable appropriate software and user interface functionality.

(417) In one aspect, the main processor 3214 may be an LM 4F230H5QR, available from Texas Instruments, for example. In at least one example, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle serial random access memory (SRAM), internal read-only memory (ROM) loaded with StellarisWare® software, 2 KB electrically erasable programmable read-only memory (EEPROM), one or more pulse width modulation (PWM) modules, one or more quadrature encoder inputs (QED analog, one or more 12-bit Analog-to-Digital Converters (ADC) with 12 analog input channels, among other features that are readily available from the product datasheet. Other processors may be readily substituted and, accordingly, the present disclosure should not be limited in this context.

(418) FIG. 33 shows a simplified block circuit diagram illustrating another electrical circuit 3300 contained within a modular ultrasonic surgical instrument 3334, in accordance with at least one aspect of the present disclosure. The electrical circuit 3300 includes a processor 3302, a clock 3330, a memory 3326, a power supply 3304 (e.g., a battery), a switch 3306, such as a metal-oxide semiconductor field effect transistor (MOSFET) power switch, a drive circuit 3308 (PLL), a transformer 3310, a signal smoothing circuit 3312 (also referred to as a matching circuit and can be, for example, a tank circuit), a sensing circuit 3314, a transducer 1120, and a shaft assembly (e.g. shaft assembly 1126, 1129) comprising an ultrasonic transmission waveguide that terminates at an ultrasonic blade (e.g. ultrasonic blade 1128, 1149) which may be referred to herein simply as the waveguide.

(419) One feature of the present disclosure that severs dependency on high voltage (120 VAC) input power (a characteristic of general ultrasonic cutting devices) is the utilization of low-voltage switching throughout the wave-forming process and the amplification of the driving signal only directly before the transformer stage. For this reason, in one aspect of the present disclosure, power is derived from only a battery, or a group of batteries, small enough to fit either within a handle assembly. State-of-the-art battery technology provides powerful batteries of a few centimeters in height and width and a few millimeters in depth. By combining the features of the present disclosure to provide a self-contained and self-powered ultrasonic device, a reduction in manufacturing cost may be achieved.

(420) The output of the power supply 3304 is fed to and powers the processor 3302. The processor 3302 receives and outputs signals and, as will be described below, functions according to custom logic or in accordance with computer programs that are executed by the processor 3302. As discussed above, the electrical circuit 3300 can also include a memory 3326, preferably, random access memory (RAM), that stores computer-readable instructions and data.

(421) The output of the power supply 3304 also is directed to the switch 3306 having a duty cycle controlled by the processor 3302. By controlling the on-time for the switch 3306, the processor 3302 is able to dictate the total amount of power that is ultimately delivered to the transducer 1120. In one aspect, the switch 3306 is a MOSFET, although other switches and switching configurations are adaptable as well. The output of the switch 3306 is fed to a drive circuit 3308 that contains, for example, a phase detecting phase-locked loop (PLL) and/or a low-pass filter and/or a voltage-controlled oscillator. The output of the switch 3306 is sampled by the processor 3302 to determine the voltage and current of the output signal (V_{sub.IN} and I_{sub.IN}, respectively). These values are used in a feedback architecture to adjust the pulse width modulation of the switch 3306. For instance, the duty cycle of the switch 3306 can vary from about 20% to about 80%, depending on the desired and actual output from the switch 3306.

(422) The drive circuit 3308, which receives the signal from the switch 3306, includes an oscillatory circuit that turns the output of the switch 3306 into an electrical signal having an ultrasonic frequency, e.g., 55 kHz (VCO). As explained above, a smoothed-out version of this ultrasonic waveform is ultimately fed to the ultrasonic transducer 1120 to produce a resonant sine wave along an ultrasonic transmission waveguide.

(423) At the output of the drive circuit 3308 is a transformer 3310 that is able to step up the low voltage signal(s) to a higher voltage. It is noted that upstream switching, prior to the transformer 3310, is performed at low (e.g., battery driven) voltages, something that, to date, has not been possible for ultrasonic cutting and cautery devices. This is at least partially due to the fact that the device advantageously uses low on-resistance MOSFET switching devices. Low on-resistance MOSFET switches are advantageous, as they produce lower switching losses and less heat than a traditional MOSFET device and allow higher current to pass through. Therefore, the switching stage (pre-transformer) can be characterized as low voltage/high current. To ensure the lower on-resistance of the amplifier MOSFET(s), the MOSFET(s) are run, for example, at 10 V. In such a case, a separate 10 VDC power supply can be used to feed the MOSFET gate, which ensures that the MOSFET is fully on and a reasonably low on resistance is achieved. In one aspect of the present disclosure, the transformer 3310 steps up the battery voltage to 120 V root-mean-square (RMS). Transformers are known in the art and are, therefore, not explained here in detail.

(424) In the circuit configurations described, circuit component degradation can negatively impact the circuit performance of the circuit. One factor that directly affects component performance is heat. Known circuits generally monitor switching temperatures (e.g., MOSFET temperatures). However, because of the technological advancements in MOSFET designs, and the corresponding reduction in size, MOSFET temperatures are no longer a valid indicator of circuit loads and heat. For this reason, in accordance with at least one aspect of the present disclosure, the sensing circuit 3314 senses the temperature of the transformer 3310. This temperature sensing is advantageous as the transformer 3310 is run at or very close to its

maximum temperature during use of the device. Additional temperature may cause the core material, e.g., the ferrite, to break down and permanent damage can occur. The present disclosure can respond to a maximum temperature of the transformer **3310** by, for example, reducing the driving power in the transformer **3310**, signaling the user, turning the power off, pulsing the power, or other appropriate responses.

(425) In one aspect of the present disclosure, the processor **3302** is communicatively coupled to the end effector (e.g. **1122**, **1125**), which is used to place material in physical contact with the ultrasonic blade (e.g. **1128**, **1149**). Sensors are provided that measure, at the end effector, a clamping force value (existing within a known range) and, based upon the received clamping force value, the processor **3302** varies the motional voltage V.sub.M. Because high force values combined with a set motional rate can result in high blade temperatures, a temperature sensor **3332** can be communicatively coupled to the processor **3302**, where the processor **3302** is operable to receive and interpret a signal indicating a current temperature of the blade from the temperature sensor **3336** and to determine a target frequency of blade movement based upon the received temperature. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to the trigger (e.g. **1143**, **1147**) to measure the force applied to the trigger by the user. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to a switch button such that displacement intensity corresponds to the force applied by the user to the switch button.

(426) In accordance with at least one aspect of the present disclosure, the PLL portion of the drive circuit **3308**, which is coupled to the processor **3302**, is able to determine a frequency of waveguide movement and communicate that frequency to the processor **3302**. The processor **3302** stores this frequency value in the memory **3326** when the device is turned off. By reading the clock **3330**, the processor **3302** is able to determine an elapsed time after the device is shut off and retrieve the last frequency of waveguide movement if the elapsed time is less than a predetermined value. The device can then start up at the last frequency, which, presumably, is the optimum frequency for the current load.

Modular Battery Powered Handheld Surgical Instrument with Multistage Generator Circuits

(427) In another aspect, the present disclosure provides a modular battery powered handheld surgical instrument with multistage generator circuits. Disclosed is a surgical instrument that includes a battery assembly, a handle assembly, and a shaft assembly where the battery assembly and the shaft assembly are configured to mechanically and electrically connect to the handle assembly. The battery assembly includes a control circuit configured to generate a digital waveform. The handle assembly includes a first stage circuit configured to receive the digital waveform, convert the digital waveform into an analog waveform, and amplify the analog waveform. The shaft assembly includes a second stage circuit coupled to the first stage circuit to receive, amplify, and apply the analog waveform to a load.

(428) In one aspect, the present disclosure provides a surgical instrument, comprising: a battery assembly, comprising a control circuit comprising a battery, a memory coupled to the battery, and a processor coupled to the memory and the battery, wherein the processor is configured to generate a digital waveform; a handle assembly comprising a first stage circuit coupled to the processor, the first stage circuit comprising a digital-to-analog (DAC) converter and a first stage amplifier circuit, wherein the DAC is configured to receive the digital waveform and convert the digital waveform into an analog waveform, wherein the first stage amplifier circuit is configured to receive and amplify the analog waveform; and a shaft assembly comprising a second stage circuit coupled to the first stage amplifier circuit to receive the analog waveform, amplify the analog waveform, and apply the analog waveform to a load; wherein the battery assembly and the shaft assembly are configured to mechanically and electrically connect to the handle assembly.

(429) The load may comprise any one of an ultrasonic transducer, an electrode, or a sensor, or any combinations thereof. The first stage circuit may comprise a first stage ultrasonic drive circuit and a first stage high-frequency current drive circuit. The control circuit may be configured to drive the first stage ultrasonic drive circuit and the first stage high-frequency current drive circuit independently or simultaneously. The first stage ultrasonic drive circuit may be configured to couple to a second stage ultrasonic drive circuit. The second stage ultrasonic drive circuit may be configured to couple to an ultrasonic transducer. The first stage high-frequency current drive circuit may be configured to couple to a second stage high-frequency drive circuit. The second stage high-frequency drive circuit may be configured to couple to an electrode.

(430) The first stage circuit may comprise a first stage sensor drive circuit. The first stage sensor drive circuit may be configured to a second stage sensor drive circuit. The second stage sensor drive circuit may be configured to couple to a sensor.

(431) In another aspect, the present disclosure provides a surgical instrument, comprising: a battery assembly, comprising a control circuit comprising a battery, a memory coupled to the battery, and a processor coupled to the memory and the battery, wherein the processor is configured to generate a digital waveform; a handle assembly comprising a common first stage circuit coupled to the processor, the common first stage circuit comprising a digital-to-analog (DAC) converter and a common first stage amplifier circuit, wherein the DAC is configured to receive the digital waveform and convert the digital waveform into an analog waveform, wherein the common first stage amplifier circuit is configured to receive and amplify the analog waveform; and a shaft assembly comprising a second stage circuit coupled to the common first stage amplifier circuit to receive the analog waveform, amplify the analog waveform, and apply the analog waveform to a load; wherein the battery assembly and the shaft assembly are configured to mechanically and electrically connect to the handle assembly.

(432) The load may comprise any one of an ultrasonic transducer, an electrode, or a sensor, or any combinations thereof. The common first stage circuit may be configured to drive ultrasonic, high-frequency current, or sensor circuits. The common first stage drive circuit may be configured to couple to a second stage ultrasonic drive circuit, a second stage high-frequency drive circuit, or a second stage sensor drive circuit. The second stage ultrasonic drive circuit may be configured to couple to an ultrasonic transducer, the second stage high-frequency drive circuit is configured to couple to an electrode, and the second stage sensor drive circuit is configured to couple to a sensor.

(433) In another aspect, the present disclosure provides a surgical instrument, comprising a control circuit comprising a memory coupled to a processor, wherein the processor is configured to generate a digital waveform; a handle assembly comprising a common first stage circuit coupled to the processor, the common first stage circuit configured to receive the digital waveform, convert the digital waveform into an analog waveform, and amplify the analog waveform; and a shaft assembly comprising a second stage circuit coupled to the common first stage circuit to receive and amplify the analog waveform; wherein the shaft assembly is configured to mechanically and electrically connect to the handle assembly.

(434) The common first stage circuit may be configured to drive ultrasonic, high-frequency current, or sensor circuits. The common first stage drive circuit may be configured to couple to a second stage ultrasonic drive circuit, a second stage high-frequency drive circuit, or a second stage sensor drive circuit. The second stage ultrasonic drive circuit may be configured to couple to an ultrasonic transducer, the second stage high-frequency drive circuit is configured to couple to an electrode, and the second stage sensor drive circuit is configured to couple to a sensor.

(435) FIG. **34** illustrates a generator circuit **3400** partitioned into a first stage circuit **3404** and a second stage circuit **3406**, in accordance with at least one aspect of the present disclosure. In one aspect, the surgical instruments of surgical system **1000** described herein may comprise a generator circuit **3400** partitioned into multiple stages. For example, surgical instruments of surgical system **1000** may comprise the generator circuit **3400** partitioned into at least two circuits: the first stage circuit **3404** and the second stage circuit **3406** of amplification enabling operation of RF energy only, ultrasonic energy only, and/or a combination of RF energy and ultrasonic energy. A combination modular shaft assembly **3414** may be powered by the common first stage circuit **3404** located within a handle assembly **3412** and the modular second stage circuit **3406** integral to the modular shaft assembly **3414**. As previously discussed throughout this description in connection with the surgical instruments of surgical system **1000**, a battery assembly **3410** and the shaft assembly **3414** are configured to mechanically and electrically connect to the handle assembly **3412**. The end effector assembly is configured to mechanically and electrically connect the shaft assembly **3414**.

(436) Turning now to FIG. **34**, the generator circuit **3400** is partitioned into multiple stages located in multiple modular assemblies of a surgical instrument, such as the surgical instruments of surgical system **1000** described herein. In one aspect, a control stage circuit **3402** may be located in the battery assembly **3410** of the surgical instrument. The control stage circuit **3402** is a control circuit **3200** as described in connection with FIG. **32**. The control circuit **3200** comprises a processor **3214**, which includes internal memory **3217** (FIG. **34**) (e.g., volatile and non-volatile memory), and is

electrically coupled to a battery 3211. The battery 3211 supplies power to the first stage circuit 3404, and a third stage circuit 3408, respectively. As previously discussed, the control circuit 3200 generates a digital waveform 4300 (FIG. 43) using circuits and techniques described in connection with FIGS. 41 and 42. Returning to FIG. 34, the digital waveform 4300 may be configured to drive an ultrasonic transducer, high-frequency (e.g., RF) electrodes, or a combination thereof either independently or simultaneously. If driven simultaneously, filter circuits may be provided in the corresponding first stage circuits 3404 to select either the ultrasonic waveform or the RF waveform. Such filtering techniques are described in commonly owned U.S. Pat. Pub. No. US-2017-0086910-A1, titled TECHNIQUES FOR CIRCUIT TOPOLOGIES FOR COMBINED GENERATOR, which is herein incorporated by reference in its entirety.

(437) The first stage circuits 3404 (e.g., the first stage ultrasonic drive circuit 3420, the first stage RF drive circuit 3422, and the first stage sensor drive circuit 3424) are located in a handle assembly 3412 of the surgical instrument. The control circuit 3200 provides the ultrasonic drive signal to the first stage ultrasonic drive circuit 3420 via outputs SCL-A, SDA-A of the control circuit 3200. The first stage ultrasonic drive circuit 3420 is described in detail in connection with FIG. 29. The control circuit 3200 provides the RF drive signal to the first stage RF drive circuit 3422 via outputs SCL-B, SDA-B of the control circuit 3200. The first stage RF drive circuit 3422 is described in detail in connection with FIG. 36. The control circuit 3200 provides the sensor drive signal to the first stage sensor drive circuit 3424 via outputs SCL-C, SDA-C of the control circuit 3200. Generally, each of the first stage circuits 3404 includes a digital-to-analog (DAC) converter and a first stage amplifier section to drive the second stage circuits 3406. The outputs of the first stage circuits 3404 are provided to the inputs of the second stage circuits 3406.

(438) The control circuit 3200 is configured to detect which modules are plugged into the control circuit 3200. For example, the control circuit 3200 is configured to detect whether the first stage ultrasonic drive circuit 3420, the first stage RF drive circuit 3422, or the first stage sensor drive circuit 3424 located in the handle assembly 3412 is connected to the battery assembly 3410. Likewise, each of the first stage circuits 3404 can detect which second stage circuits 3406 are connected thereto and that information is provided back to the control circuit 3200 to determine the type of signal waveform to generate. Similarly, each of the second stage circuits 3406 can detect which third stage circuits 3408 or components are connected thereto and that information is provided back to the control circuit 3200 to determine the type of signal waveform to generate.

(439) In one aspect, the second stage circuits 3406 (e.g., the ultrasonic drive second stage circuit 3430, the RF drive second stage circuit 3432, and the sensor drive second stage circuit 3434) are located in the shaft assembly 3414 of the surgical instrument. The first stage ultrasonic drive circuit 3420 provides a signal to the second stage ultrasonic drive circuit 3430 via outputs US-Left/US-Right. The second stage ultrasonic drive circuit 3430 is described in detail in connection with FIGS. 30 and 31. In addition to a transformer (FIGS. 30 and 31), the second stage ultrasonic drive circuit 3430 also may include filter, amplifier, and signal conditioning circuits. The first stage high-frequency (RF) current drive circuit 3422 provides a signal to the second stage RF drive circuit 3432 via outputs RF-Left/RF-Right. In addition to a transformer and blocking capacitors, the second stage RF drive circuit 3432 also may include filter, amplifier, and signal conditioning circuits. The first stage sensor drive circuit 3424 provides a signal to the second stage sensor drive circuit 3434 via outputs Sensor-1/Sensor-2. The second stage sensor drive circuit 3434 may include filter, amplifier, and signal conditioning circuits depending on the type of sensor. The outputs of the second stage circuits 3406 are provided to the inputs of the third stage circuits 3408.

(440) In one aspect, the third stage circuits 3408 (e.g., the ultrasonic transducer 1120, the RF electrodes 3074a, 3074b, and the sensors 3440) may be located in various assemblies 3416 of the surgical instruments. In one aspect, the second stage ultrasonic drive circuit 3430 provides a drive signal to the ultrasonic transducer 1120 piezoelectric stack. In one aspect, the ultrasonic transducer 1120 is located in the ultrasonic transducer assembly of the surgical instrument. In other aspects, however, the ultrasonic transducer 1120 may be located in the handle assembly 3412, the shaft assembly 3414, or the end effector. In one aspect, the second stage RF drive circuit 3432 provides a drive signal to the RF electrodes 3074a, 3074b, which are generally located in the end effector portion of the surgical instrument. In one aspect, the second stage sensor drive circuit 3434 provides a drive signal to various sensors 3440 located throughout the surgical instrument.

(441) FIG. 35 illustrates a generator circuit 3500 partitioned into multiple stages where a first stage circuit 3504 is common to the second stage circuit 3506, in accordance with at least one aspect of the present disclosure. In one aspect, the surgical instruments of surgical system 1000 described herein may comprise generator circuit 3500 partitioned into multiple stages. For example, the surgical instruments of surgical system 1000 may comprise the generator circuit 3500 partitioned into at least two circuits: the first stage circuit 3504 and the second stage circuit 3506 of amplification enabling operation of high-frequency (RF) energy only, ultrasonic energy only, and/or a combination of RF energy and ultrasonic energy. A combination modular shaft assembly 3514 may be powered by a common first stage circuit 3504 located within the handle assembly 3512 and a modular second stage circuit 3506 integral to the modular shaft assembly 3514. As previously discussed throughout this description in connection with the surgical instruments of surgical system 1000, a battery assembly 3510 and the shaft assembly 3514 are configured to mechanically and electrically connect to the handle assembly 3512. The end effector assembly is configured to mechanically and electrically connect the shaft assembly 3514.

(442) As shown in the example of FIG. 35, the battery assembly 3510 portion of the surgical instrument comprises a first control circuit 3502, which includes the control circuit 3200 previously described. The handle assembly 3512, which connects to the battery assembly 3510, comprises a common first stage drive circuit 3420. As previously discussed, the first stage drive circuit 3420 is configured to drive ultrasonic, high-frequency (RF) current, and sensor loads. The output of the common first stage drive circuit 3420 can drive any one of the second stage circuits 3506 such as the second stage ultrasonic drive circuit 3430, the second stage high-frequency (RF) current drive circuit 3432, and/or the second stage sensor drive circuit 3434. The common first stage drive circuit 3420 detects which second stage circuit 3506 is located in the shaft assembly 3514 when the shaft assembly 3514 is connected to the handle assembly 3512. Upon the shaft assembly 3514 being connected to the handle assembly 3512, the common first stage drive circuit 3420 determines which one of the second stage circuits 3506 (e.g., the second stage ultrasonic drive circuit 3430, the second stage RF drive circuit 3432, and/or the second stage sensor drive circuit 3434) is located in the shaft assembly 3514. The information is provided to the control circuit 3200 located in the handle assembly 3512 in order to supply a suitable digital waveform 4300 (FIG. 43) to the second stage circuit 3506 to drive the appropriate load, e.g., ultrasonic, RF, or sensor. It will be appreciated that identification circuits may be included in various assemblies 3516 in third stage circuit 3508 such as the ultrasonic transducer 1120, the electrodes 3074a, 3074b, or the sensors 3440. Thus, when a third stage circuit 3508 is connected to a second stage circuit 3506, the second stage circuit 3506 knows the type of load that is required based on the identification information.

(443) FIG. 36 is a schematic diagram of one aspect of an electrical circuit 3600 configured to drive a high-frequency current (RF), in accordance with at least one aspect of the present disclosure. The electrical circuit 3600 comprises an analog multiplexer 3680. The analog multiplexer 3680 multiplexes various signals from the upstream channels SCL-A, SDA-A such as RF, battery, and power control circuit. A current sensor 3682 is coupled in series with the return or ground leg of the power supply circuit to measure the current supplied by the power supply. A field effect transistor (FET) temperature sensor 3684 provides the ambient temperature. A pulse width modulation (PWM) watchdog timer 3688 automatically generates a system reset if the main program neglects to periodically service it. It is provided to automatically reset the electrical circuit 3600 when it hangs or freezes because of a software or hardware fault. It will be appreciated that the electrical circuit 3600 may be configured for driving RF electrodes or for driving the ultrasonic transducer 1120 as described in connection with FIG. 29, for example. Accordingly, with reference now back to FIG. 36, the electrical circuit 3600 can be used to drive both ultrasonic and RF electrodes interchangeably.

(444) A drive circuit 3686 provides Left and Right RF energy outputs. A digital signal that represents the signal waveform is provided to the SCL-A, SDA-A inputs of the analog multiplexer 3680 from a control circuit, such as the control circuit 3200 (FIG. 32). A digital-to-analog converter 3690 (DAC) converts the digital input to an analog output to drive a PWM circuit 3692 coupled to an oscillator 3694. The PWM circuit 3692 provides a first signal to a first gate drive circuit 3696a coupled to a first transistor output stage 3698a to drive a first RF+ (Left) energy output. The PWM

circuit **3692** also provides a second signal to a second gate drive circuit **3696b** coupled to a second transistor output stage **3698b** to drive a second RF- (Right) energy output. A voltage sensor **3699** is coupled between the RF Left/RF output terminals to measure the output voltage. The drive circuit **3686**, the first and second drive circuits **3696a**, **3696b**, and the first and second transistor output stages **3698a**, **3698b** define a first stage amplifier circuit. In operation, the control circuit **3200** (FIG. 32) generates a digital waveform **4300** (FIG. 43) employing circuits such as direct digital synthesis (DDS) circuits **4100**, **4200** (FIGS. 41 and 42). The DAC **3690** receives the digital waveform **4300** and converts it into an analog waveform, which is received and amplified by the first stage amplifier circuit.

(445) FIG. 37 is a schematic diagram of the transformer **3700** coupled to the electrical circuit **3600** shown in FIG. 36, in accordance with at least one aspect of the present disclosure. The RF+/RF input terminals (primary winding) of the transformer **3700** are electrically coupled to the RF Left/RF output terminals of the electrical circuit **3600**. One side of the secondary winding is coupled in series with first and second blocking capacitors **3706**, **3708**. The second blocking capacitor is coupled to the second stage RF drive circuit **3774a** positive terminal. The other side of the secondary winding is coupled to the second stage RF drive circuit **3774b** negative terminal. The second stage RF drive circuit **3774a** positive output is coupled to the ultrasonic blade and the second stage RF drive circuit **3774b** negative ground terminal is coupled to an outer tube. In one aspect, a transformer has a turns-ratio of n.sub.1:n.sub.2 of 1:50.

(446) FIG. 38 is a schematic diagram of a circuit **3800** comprising separate power sources for high power energy/drive circuits and low power circuits, in accordance with at least one aspect of the present disclosure. A power supply **3812** includes a primary battery pack comprising first and second primary batteries **3815**, **3817** (e.g., Li-ion batteries) that are connected into the circuit **3800** by a switch **3818** and a secondary battery pack comprising a secondary battery **3820** that is connected into the circuit by a switch **3823** when the power supply **3812** is inserted into the battery assembly. The secondary battery **3820** is a sag preventing battery that has componentry resistant to gamma or other radiation sterilization. For instance, a switch mode power supply **3827** and optional charge circuit within the battery assembly can be incorporated to allow the secondary battery **3820** to reduce the voltage sag of the primary batteries **3815**, **3817**. This guarantees full charged cells at the beginning of a surgery that are easy to introduce into the sterile field. The primary batteries **3815**, **3817** can be used to power motor control circuits **3826** and energy circuits **3832** directly. The motor control circuits **3826** are configured to control a motor, such as motor **3829**. The power supply/battery pack **3812** may comprise a dual type battery assembly including primary Li-ion batteries **3815**, **3817** and secondary NiMH batteries **3820** with dedicated energy cells **3820** to control handle electronics circuits **3830** from dedicated energy cells **3815**, **3817** to run the motor control circuits **3826** and the energy circuits **3832**. In this case the circuit **3810** pulls from the secondary batteries **3820** involved in driving the handle electronics circuits **3830** when the primary batteries **3815**, **3817** involved in driving the energy circuits **3832** and/or motor control circuits **3826** are dropping low. In one various aspect, the circuit **3810** may include a one way diode that would not allow for current to flow in the opposite direction (e.g., from the batteries involved in driving the energy and/or motor control circuits to the batteries involved in driving the electronics circuits).

(447) Additionally, a gamma friendly charge circuit may be provided that includes a switch mode power supply **3827** using diodes and vacuum tube components to minimize voltage sag at a predetermined level. With the inclusion of a minimum sag voltage that is a division of the NiMH voltages (3 NiMH cells) the switch mode power supply **3827** could be eliminated. Additionally a modular system may be provided wherein the radiation hardened components are located in a module, making the module sterilizable by radiation sterilization. Other non-radiation hardened components may be included in other modular components and connections made between the modular components such that the componentry operates together as if the components were located together on the same circuit board. If only two NiMH cells are desired the switch mode power supply **3827** based on diodes and vacuum tubes allows for sterilizable electronics within the disposable primary battery pack.

(448) Turning now to FIG. 39, there is shown a control circuit **3900** for operating a battery **3901** powered RF generator circuit **3902** for use with a surgical instrument, in accordance with at least one aspect of the present disclosure. The surgical instrument is configured to use both ultrasonic vibration and high-frequency current to carry out surgical coagulation/cutting treatments on living tissue, and uses high-frequency current to carry out a surgical coagulation treatment on living tissue.

(449) FIG. 39 illustrates the control circuit **3900** that allows a dual generator system to switch between the RF generator circuit **3902** and the ultrasonic generator circuit **3920** energy modalities for a surgical instrument of the surgical system **1000**. In one aspect, a current threshold in an RF signal is detected. When the impedance of the tissue is low the high-frequency current through tissue is high when RF energy is used as the treatment source for the tissue. According to one aspect, a visual indicator **3912** or light located on the surgical instrument of surgical system **1000** may be configured to be in an on-state during this high current period. When the current falls below a threshold, the visual indicator **3912** is in an off-state. Accordingly, a phototransistor **3914** may be configured to detect the transition from an on-state to an off-state and disengages the RF energy as shown in the control circuit **3900** shown in FIG. 39. Therefore, when the energy button is released and an energy switch **3926** is opened, the control circuit **3900** is reset and both the RF and ultrasonic generator circuits **3902**, **3920** are held off.

(450) With reference to FIG. 39, in one aspect, a method of managing an RF generator circuit **3902** and ultrasound generator circuit **3920** is provided. The RF generator circuit **3902** and/or the ultrasound generator circuit **3920** may be located in the handle assembly **1109**, the ultrasonic transducer/RF generator assembly **1120**, the battery assembly, the shaft assembly **1129**, and/or the nozzle, of the multifunction electrosurgical instrument **1108**, for example. The control circuit **3900** is held in a reset state if the energy switch **3926** is off (e.g., open). Thus, when the energy switch **3926** is opened, the control circuit **3900** is reset and both the RF and ultrasonic generator circuits **3902**, **3920** are turned off. When the energy switch **3926** is squeezed and the energy switch **3926** is engaged (e.g., closed), RF energy is delivered to the tissue and the visual indicator **3912** operated by a current sensing step-up transformer **3904** will be lit while the tissue impedance is low. The light from the visual indicator **3912** provides a logic signal to keep the ultrasonic generator circuit **3920** in the off state. Once the tissue impedance increases above a threshold and the high-frequency current through the tissue decreases below a threshold, the visual indicator **3912** turns off and the light transitions to an off-state. A logic signal generated by this transition turns off a relay **3908**, whereby the RF generator circuit **3902** is turned off and the ultrasonic generator circuit **3920** is turned on, to complete the coagulation and cut cycle.

(451) Still with reference to FIG. 39, in one aspect, the dual generator circuit configuration employs the on-board RF generator circuit **3902**, which is battery **3901** powered, for one modality and a second, on-board ultrasound generator circuit **3920**, which may be on-board in the handle assembly **1109**, battery assembly, shaft assembly **1129**, nozzle, and/or the ultrasonic transducer/RF generator assembly **1120** of the multifunction electrosurgical instrument **1108**, for example. The ultrasonic generator circuit **3920** also is battery **3901** operated. In various aspects, the RF generator circuit **3902** and the ultrasonic generator circuit **3920** may be an integrated or separable component of the handle assembly **1109**. According to various aspects, having the dual RF/ultrasonic generator circuits **3902**, **3920** as part of the handle assembly **1109** may eliminate the need for complicated wiring. The RF/ultrasonic generator circuits **3902**, **3920** may be configured to provide the full capabilities of an existing generator while utilizing the capabilities of a cordless generator system simultaneously.

(452) Either type of system can have separate controls for the modalities that are not communicating with each other. The surgeon activates the RF and Ultrasonic separately and at their discretion. Another approach would be to provide fully integrated communication schemes that share buttons, tissue status, instrument operating parameters (such as jaw closure, forces, etc.) and algorithms to manage tissue treatment. Various combinations of this integration can be implemented to provide the appropriate level of function and performance.

(453) As discussed above, in one aspect, the control circuit **3900** includes the battery **3901** powered RF generator circuit **3902** comprising a battery as an energy source. As shown, RF generator circuit **3902** is coupled to two electrically conductive surfaces referred to herein as electrodes **3906a**, **3906b** (i.e., active electrode **3906a** and return electrode **3906b**) and is configured to drive the electrodes **3906a**, **3906b** with RF energy (e.g., high-frequency current). A first winding **3910a** of the step-up transformer **3904** is connected in series with one pole of the bipolar RF generator circuit **3902** and the return electrode **3906b**. In one aspect, the first winding **3910a** and the return electrode **3906b** are connected to the negative pole of the

bipolar RF generator circuit **3902**. The other pole of the bipolar RF generator circuit **3902** is connected to the active electrode **3906a** through a switch contact **3909** of the relay **3908**, or any suitable electromagnetic switching device comprising an armature which is moved by an electromagnet **3936** to operate the switch contact **3909**. The switch contact **3909** is closed when the electromagnet **3936** is energized and the switch contact **3909** is open when the electromagnet **3936** is de-energized. When the switch contact is closed, RF current flows through conductive tissue (not shown) located between the electrodes **3906a**, **3906b**. It will be appreciated, that in one aspect, the active electrode **3906a** is connected to the positive pole of the bipolar RF generator circuit **3902**.

(454) A visual indicator circuit **3905** comprises the step-up transformer **3904**, a series resistor **R2**, and the visual indicator **3912**. The visual indicator **3912** can be adapted for use with the surgical instrument **1108** and other electrosurgical systems and tools, such as those described herein. The first winding **3910a** of the step-up transformer **3904** is connected in series with the return electrode **3906b** and the second winding **3910b** of the step-up transformer **3904** is connected in series with the resistor **R2** and the visual indicator **3912** comprising a type NE-2 neon bulb, for example.

(455) In operation, when the switch contact **3909** of the relay **3908** is open, the active electrode **3906a** is disconnected from the positive pole of the bipolar RF generator circuit **3902** and no current flows through the tissue, the return electrode **3906b**, and the first winding **3910a** of the step-up transformer **3904**. Accordingly, the visual indicator **3912** is not energized and does not emit light. When the switch contact **3909** of the relay **3908** is closed, the active electrode **3906a** is connected to the positive pole of the bipolar RF generator circuit **3902** enabling current to flow through tissue, the return electrode **3906b**, and the first winding **3910a** of the step-up transformer **3904** to operate on tissue, for example cut and cauterize the tissue.

(456) A first current flows through the first winding **3910a** as a function of the impedance of the tissue located between the active and return electrodes **3906a**, **3906b** providing a first voltage across the first winding **3910a** of the step-up transformer **3904**. A stepped up second voltage is induced across the second winding **3910b** of the step-up transformer **3904**. The secondary voltage appears across the resistor **R2** and energizes the visual indicator **3912** causing the neon bulb to light when the current through the tissue is greater than a predetermined threshold. It will be appreciated that the circuit and component values are illustrative and not limited thereto. When the switch contact **3909** of the relay **3908** is closed, current flows through the tissue and the visual indicator **3912** is turned on.

(457) Turning now to the energy switch **3926** portion of the control circuit **3900**, when the energy switch **3926** is open position, a logic high is applied to the input of a first inverter **3928** and a logic low is applied of one of the two inputs of the AND gate **3932**. Thus, the output of the AND gate **3932** is low and a transistor **3934** is off to prevent current from flowing through the winding of the electromagnet **3936**. With the electromagnet **3936** in the de-energized state, the switch contact **3909** of the relay **3908** remains open and prevents current from flowing through the electrodes **3906a**, **3906b**. The logic low output of the first inverter **3928** also is applied to a second inverter **3930** causing the output to go high and resetting a flip-flop **3918** (e.g., a D-Type flip-flop). At which time, the Q output goes low to turn off the ultrasound generator circuit **3920** circuit and the Q output goes high and is applied to the other input of the AND gate **3932**.

(458) When the user presses the energy switch **3926** on the instrument handle to apply energy to the tissue between the electrodes **3906a**, **3906b**, the energy switch **3926** closes and applies a logic low at the input of the first inverter **3928**, which applies a logic high to other input of the AND gate **3932** causing the output of the AND gate **3932** to go high and turns on the transistor **3934**. In the on state, the transistor **3934** conducts and sinks current through the winding of the electromagnet **3936** to energize the electromagnet **3936** and close the switch contact **3909** of the relay **3908**. As discussed above, when the switch contact **3909** is closed, current can flow through the electrodes **3906a**, **3906b** and the first winding **3910a** of the step-up transformer **3904** when tissue is located between the electrodes **3906a**, **3906b**.

(459) As discussed above, the magnitude of the current flowing through the electrodes **3906a**, **3906b** depends on the impedance of the tissue located between the electrodes **3906a**, **3906b**. Initially, the tissue impedance is low and the magnitude of the current high through the tissue and the first winding **3910a**. Consequently, the voltage impressed on the second winding **3910b** is high enough to turn on the visual indicator **3912**. The light emitted by the visual indicator **3912** turns on the phototransistor **3914**, which pulls the input of an inverter **3916** low and causes the output of the inverter **3916** to go high. A high input applied to the CLK of the flip-flop **3918** has no effect on the Q or the Q outputs of the flip-flop **3918** and Q output remains low and the Q output remains high. Accordingly, while the visual indicator **3912** remains energized, the ultrasound generator circuit **3920** is turned OFF and an ultrasonic transducer **3922** and an ultrasonic blade **3924** of the multifunction electrosurgical instrument are not activated.

(460) As the tissue between the electrodes **3906a**, **3906b** dries up, due to the heat generated by the current flowing through the tissue, the impedance of the tissue increases and the current therethrough decreases. When the current through the first winding **3910a** decreases, the voltage across the second winding **3910b** also decreases and when the voltage drops below a minimum threshold required to operate the visual indicator **3912**, the visual indicator **3912** and the phototransistor **3914** turn off. When the phototransistor **3914** turns off, a logic high is applied to the input of the inverter **3916** and a logic low is applied to the CLK input of the flip-flop **3918** to clock a logic high to the Q output and a logic low to the Q output. The logic high at the Q output turns on the ultrasound generator circuit **3920** to activate the ultrasonic transducer **3922** and the ultrasonic blade **3924** to initiate cutting the tissue located between the electrodes **3906a**, **3906a**. Simultaneously or near simultaneously with the ultrasound generator circuit **3920** turning on, the Q output of the flip-flop **3918** goes low and causes the output of the AND gate **3932** to go low and turn off the transistor **3934**, thereby de-energizing the electromagnet **3936** and opening the switch contact **3909** of the relay **3908** to cut off the flow of current through the electrodes **3906a**, **3906b**.

(461) While the switch contact **3909** of the relay **3908** is open, no current flows through the electrodes **3906a**, **3906b**, tissue, and the first winding **3910a** of the step-up transformer **3904**. Therefore, no voltage is developed across the second winding **3910b** and no current flows through the visual indicator **3912**.

(462) The state of the Q and the Q outputs of the flip-flop **3918** remain the same while the user squeezes the energy switch **3926** on the instrument handle to maintain the energy switch **3926** closed. Thus, the ultrasonic blade **3924** remains activated and continues cutting the tissue between the jaws of the end effector while no current flows through the electrodes **3906a**, **3906b** from the bipolar RF generator circuit **3902**. When the user releases the energy switch **3926** on the instrument handle, the energy switch **3926** opens and the output of the first inverter **3928** goes low and the output of the second inverter **3930** goes high to reset the flip-flop **3918** causing the Q output to go low and turn off the ultrasound generator circuit **3920**. At the same time, the Q output goes high and the circuit is now in an off state and ready for the user to actuate the energy switch **3926** on the instrument handle to close the energy switch **3926**, apply current to the tissue located between the electrodes **3906a**, **3906b**, and repeat the cycle of applying RF energy to the tissue and ultrasonic energy to the tissue as described above.

(463) FIG. 40 illustrates a diagram of a surgical system **4000**, which represents one aspect of the surgical system **1000**, comprising a feedback system for use with any one of the surgical instruments of surgical system **1000**, which may include or implement many of the features described herein. The surgical system **4000** may include a generator **4002** coupled to a surgical instrument that includes an end effector **4006**, which may be activated when a clinician operates a trigger **4010**. In various aspects, the end effector **4006** may include an ultrasonic blade to deliver ultrasonic vibration to carry out surgical coagulation/cutting treatments on living tissue. In other aspects the end effector **4006** may include electrically conductive elements coupled to an electrosurgical high-frequency current energy source to carry out surgical coagulation or cauterization treatments on living tissue and either a mechanical knife with a sharp edge or an ultrasonic blade to carry out cutting treatments on living tissue. When the trigger **4010** is actuated, a force sensor **4012** may generate a signal indicating the amount of force being applied to the trigger **4010**. In addition to, or instead of a force sensor **4012**, the surgical instrument may include a position sensor **4013**, which may generate a signal indicating the position of the trigger **4010** (e.g., how far the trigger has been depressed or otherwise actuated). In one aspect, the position sensor **4013** may be a sensor positioned with an outer tubular sheath or reciprocating tubular actuating member located within the outer tubular sheath of the surgical instrument. In one aspect, the sensor may be a Hall-effect sensor or any suitable transducer that varies its output voltage in response to a magnetic field. The Hall-effect sensor may be used for proximity switching, positioning, speed detection, and current sensing applications. In one aspect, the Hall-effect sensor operates as an analog

transducer, directly returning a voltage. With a known magnetic field, its distance from the Hall plate can be determined.

(464) A control circuit **4008** may receive the signals from the sensors **4012** and/or **4013**. The control circuit **4008** may include any suitable analog or digital circuit components. The control circuit **4008** also may communicate with the generator **4002** and/or a transducer **4004** to modulate the power delivered to the end effector **4006** and/or the generator level or ultrasonic blade amplitude of the end effector **4006** based on the force applied to the trigger **4010** and/or the position of the trigger **4010** and/or the position of the outer tubular sheath described above relative to a reciprocating tubular actuating member located within an outer tubular sheath (e.g., as measured by a Hall-effect sensor and magnet combination). For example, as more force is applied to the trigger **4010**, more power and/or higher ultrasonic blade amplitude may be delivered to the end effector **4006**. According to various aspects, the force sensor **4012** may be replaced by a multi-position switch.

(465) According to various aspects, the end effector **4006** may include a clamp or clamping mechanism. When the trigger **4010** is initially actuated, the clamping mechanism may close, clamping tissue between a clamp arm and the end effector **4006**. As the force applied to the trigger increases (e.g., as sensed by force sensor **4012**) the control circuit **4008** may increase the power delivered to the end effector **4006** by the transducer **4004** and/or the generator level or ultrasonic blade amplitude brought about in the end effector **4006**. In one aspect, trigger position, as sensed by position sensor **4013** or clamp or clamp arm position, as sensed by position sensor **4013** (e.g., with a Hall-effect sensor), may be used by the control circuit **4008** to set the power and/or amplitude of the end effector **4006**. For example, as the trigger is moved further towards a fully actuated position, or the clamp or clamp arm moves further towards the ultrasonic blade (or end effector **4006**), the power and/or amplitude of the end effector **4006** may be increased.

(466) According to various aspects, the surgical instrument of the surgical system **1000** also may include one or more feedback devices for indicating the amount of power delivered to the end effector **4006**. For example, a speaker **4014** may emit a signal indicative of the end effector power. According to various aspects, the speaker **4014** may emit a series of pulse sounds, where the frequency of the sounds indicates power. In addition to, or instead of the speaker **4014**, the surgical instrument may include a visual display **4016**. The visual display **4016** may indicate end effector power according to any suitable method. For example, the visual display **4016** may include a series of LEDs, where end effector power is indicated by the number of illuminated LEDs. The speaker **4014** and/or visual display **4016** may be driven by the control circuit **4008**. According to various aspects, the surgical instrument may include a ratcheting device connected to the trigger **4010**. The ratcheting device may generate an audible sound as more force is applied to the trigger **4010**, providing an indirect indication of end effector power. The surgical instrument may include other features that may enhance safety. For example, the control circuit **4008** may be configured to prevent power from being delivered to the end effector **4006** in excess of a predetermined threshold. Also, the control circuit **4008** may implement a delay between the time when a change in end effector power is indicated (e.g., by speaker **4014** or visual display **4016**), and the time when the change in end effector power is delivered. In this way, a clinician may have ample warning that the level of ultrasonic power that is to be delivered to the end effector **4006** is about to change.

(467) In one aspect, the ultrasonic or high-frequency current generators of the surgical system **1000** may be configured to generate the electrical signal waveform digitally such that the desired using a predetermined number of phase points stored in a lookup table to digitize the wave shape. The phase points may be stored in a table defined in a memory, a field programmable gate array (FPGA), or any suitable non-volatile memory. FIG. **41** illustrates one aspect of a fundamental architecture for a digital synthesis circuit such as a direct digital synthesis (DDS) circuit **4100** configured to generate a plurality of wave shapes for the electrical signal waveform. The generator software and digital controls may command the FPGA to scan the addresses in the lookup table **4104** which in turn provides varying digital input values to a DAC circuit **4108** that feeds a power amplifier. The addresses may be scanned according to a frequency of interest. Using such a lookup table **4104** enables generating various types of wave shapes that can be fed into tissue or into a transducer, an RF electrode, multiple transducers simultaneously, multiple RF electrodes simultaneously, or a combination of RF and ultrasonic instruments. Furthermore, multiple lookup tables **4104** that represent multiple wave shapes can be created, stored, and applied to tissue from a generator.

(468) The waveform signal may be configured to control at least one of an output current, an output voltage, or an output power of an ultrasonic transducer and/or an RF electrode, or multiples thereof (e.g. two or more ultrasonic transducers and/or two or more RF electrodes). Further, where the surgical instrument comprises an ultrasonic components, the waveform signal may be configured to drive at least two vibration modes of an ultrasonic transducer of the at least one surgical instrument. Accordingly, a generator may be configured to provide a waveform signal to at least one surgical instrument wherein the waveform signal corresponds to at least one wave shape of a plurality of wave shapes in a table. Further, the waveform signal provided to the two surgical instruments may comprise two or more wave shapes. The table may comprise information associated with a plurality of wave shapes and the table may be stored within the generator. In one aspect or example, the table may be a direct digital synthesis table, which may be stored in an FPGA of the generator. The table may be addressed by anyway that is convenient for categorizing wave shapes. According to one aspect, the table, which may be a direct digital synthesis table, is addressed according to a frequency of the waveform signal. Additionally, the information associated with the plurality of wave shapes may be stored as digital information in the table.

(469) The analog electrical signal waveform may be configured to control at least one of an output current, an output voltage, or an output power of an ultrasonic transducer and/or an RF electrode, or multiples thereof (e.g., two or more ultrasonic transducers and/or two or more RF electrodes). Further, where the surgical instrument comprises ultrasonic components, the analog electrical signal waveform may be configured to drive at least two vibration modes of an ultrasonic transducer of the at least one surgical instrument. Accordingly, the generator circuit may be configured to provide an analog electrical signal waveform to at least one surgical instrument wherein the analog electrical signal waveform corresponds to at least one wave shape of a plurality of wave shapes stored in a lookup table **4104**. Further, the analog electrical signal waveform provided to the two surgical instruments may comprise two or more wave shapes. The lookup table **4104** may comprise information associated with a plurality of wave shapes and the lookup table **4104** may be stored either within the generator circuit or the surgical instrument. In one aspect or example, the lookup table **4104** may be a direct digital synthesis table, which may be stored in an FPGA of the generator circuit or the surgical instrument. The lookup table **4104** may be addressed by anyway that is convenient for categorizing wave shapes. According to one aspect, the lookup table **4104**, which may be a direct digital synthesis table, is addressed according to a frequency of the desired analog electrical signal waveform. Additionally, the information associated with the plurality of wave shapes may be stored as digital information in the lookup table **4104**.

(470) With the widespread use of digital techniques in instrumentation and communications systems, a digitally-controlled method of generating multiple frequencies from a reference frequency source has evolved and is referred to as direct digital synthesis. The basic architecture is shown in FIG. **41**. In this simplified block diagram, a DDS circuit is coupled to a processor, controller, or a logic device of the generator circuit and to a memory circuit located in the generator circuit of the surgical system **1000**. The DDS circuit **4100** comprises an address counter **4102**, lookup table **4104**, a register **4106**, a DAC circuit **4108**, and a filter **4112**. A stable clock $f_{sub.c}$ is received by the address counter **4102** and the register **4106** drives a programmable-read-only-memory (PROM) which stores one or more integral number of cycles of a sinewave (or other arbitrary waveform) in a lookup table **4104**. As the address counter **4102** steps through memory locations, values stored in the lookup table **4104** are written to the register **4106**, which is coupled to the DAC circuit **4108**. The corresponding digital amplitude of the signal at the memory location of the lookup table **4104** drives the DAC circuit **4108**, which in turn generates an analog output signal **4110**. The spectral purity of the analog output signal **4110** is determined primarily by the DAC circuit **4108**. The phase noise is basically that of the reference clock $f_{sub.out}$. The first analog output signal **4110** output from the DAC circuit **4108** is filtered by the filter **4112** and a second analog output signal **4114** output by the filter **4112** is provided to an amplifier having an output coupled to the output of the generator circuit. The second analog output signal has a frequency $f_{sub.out}$.

(471) Because the DDS circuit **4100** is a sampled data system, issues involved in sampling must be considered: quantization noise, aliasing, filtering, etc. For instance, the higher order harmonics of the DAC circuit **4108** output frequencies fold back into the Nyquist bandwidth, making them unfilterable, whereas, the higher order harmonics of the output of phase-locked-loop (PLL) based synthesizers can be filtered. The lookup table **4104**

contains data for an integral number of cycles. The final output frequency fa can be changed changing the reference clock frequency f.sub.out or by reprogramming the PROM.

(472) The DDS circuit **4100** may comprise multiple lookup tables **4104** where the lookup table **4104** stores a waveform represented by a predetermined number of samples, wherein the samples define a predetermined shape of the waveform. Thus multiple waveforms having a unique shape can be stored in multiple lookup tables **4104** to provide different tissue treatments based on instrument settings or tissue feedback. Examples of waveforms include high crest factor RF electrical signal waveforms for surface tissue coagulation, low crest factor RF electrical signal waveform for deeper tissue penetration, and electrical signal waveforms that promote efficient touch-up coagulation. In one aspect, the DDS circuit **4100** can create multiple wave shape lookup tables **4104** and during a tissue treatment procedure (e.g., “on-the-fly” or in virtual real time based on user or sensor inputs) switch between different wave shapes stored in separate lookup tables **4104** based on the tissue effect desired and/or tissue feedback.

(473) Accordingly, switching between wave shapes can be based on tissue impedance and other factors, for example. In other aspects, the lookup tables **4104** can store electrical signal waveforms shaped to maximize the power delivered into the tissue per cycle (i.e., trapezoidal or square wave). In other aspects, the lookup tables **4104** can store wave shapes synchronized in such way that they make maximizing power delivery by the multifunction surgical instrument of surgical system **1000** while delivering RF and ultrasonic drive signals. In yet other aspects, the lookup tables **4104** can store electrical signal waveforms to drive ultrasonic and RF therapeutic, and/or sub-therapeutic, energy simultaneously while maintaining ultrasonic frequency lock. Custom wave shapes specific to different instruments and their tissue effects can be stored in the non-volatile memory of the generator circuit or in the non-volatile memory (e.g., EEPROM) of the surgical system **1000** and be fetched upon connecting the multifunction surgical instrument to the generator circuit. An example of an exponentially damped sinusoid, as used in many high crest factor “coagulation” waveforms is shown in FIG. 43.

(474) A more flexible and efficient implementation of the DDS circuit **4100** employs a digital circuit called a Numerically Controlled Oscillator (NCO). A block diagram of a more flexible and efficient digital synthesis circuit such as a DDS circuit **4200** is shown in FIG. 42. In this simplified block diagram, a DDS circuit **4200** is coupled to a processor, controller, or a logic device of the generator and to a memory circuit located either in the generator or in any of the surgical instruments of surgical system **1000**. The DDS circuit **4200** comprises a load register **4202**, a parallel delta phase register **4204**, an adder circuit **4216**, a phase register **4208**, a lookup table **4210** (phase-to-amplitude converter), a DAC circuit **4212**, and a filter **4214**. The adder circuit **4216** and the phase register **4208** form part of a phase accumulator **4206**. A clock frequency f.sub.c is applied to the phase register **4208** and a DAC circuit **4212**. The load register **4202** receives a tuning word that specifies output frequency as a fraction of the reference clock frequency signal f.sub.c. The output of the load register **4202** is provided to the parallel delta phase register **4204** with a tuning word M.

(475) The DDS circuit **4200** includes a sample clock that generates the clock frequency f.sub.c, the phase accumulator **4206**, and the lookup table **4210** (e.g., phase to amplitude converter). The content of the phase accumulator **4206** is updated once per clock cycle f.sub.c. When time the phase accumulator **4206** is updated, the digital number, M, stored in the parallel delta phase register **4204** is added to the number in the phase register **4208** by the adder circuit **4216**. Assuming that the number in the parallel delta phase register **4204** is 00 . . . 01 and that the initial contents of the phase accumulator **4206** is 00 . . . 00. The phase accumulator **4206** is updated by 00 . . . 01 per clock cycle. If the phase accumulator **4206** is 32-bits wide, 232 clock cycles (over 4 billion) are required before the phase accumulator **4206** returns to 00 . . . 00, and the cycle repeats.

(476) A truncated output **4218** of the phase accumulator **4206** is provided to a phase-to amplitude converter lookup table **4210** and the output of the lookup table **4210** is coupled to a DAC circuit **4212**. The truncated output **4218** of the phase accumulator **4206** serves as the address to a sine (or cosine) lookup table. An address in the lookup table corresponds to a phase point on the sinewave from 0° to 360°. The lookup table **4210** contains the corresponding digital amplitude information for one complete cycle of a sinewave. The lookup table **4210** therefore maps the phase information from the phase accumulator **4206** into a digital amplitude word, which in turn drives the DAC circuit **4212**. The output of the DAC circuit is a first analog signal **4220** and is filtered by a filter **4214**. The output of the filter **4214** is a second analog signal **4222**, which is provided to a power amplifier coupled to the output of the generator circuit.

(477) In one aspect, the electrical signal waveform may be digitized into 1024 (210) phase points, although the wave shape may be digitized is any suitable number of 2n phase points ranging from 256 (28) to 281, 474, 976, 710, 656 (248), where n is a positive integer, as shown in TABLE 1. The electrical signal waveform may be expressed as A.sub.n(θ.sub.n), where a normalized amplitude A.sub.n at a point n is represented by a phase angle θ.sub.n is referred to as a phase point at point n. The number of discrete phase points n determines the tuning resolution of the DDS circuit **4200** (as well as the DDS circuit **4100** shown in FIG. 41).

(478) TABLE 1 specifies the electrical signal waveform digitized into a number of phase points.

(479) TABLE-US-00001 TABLE 1 N Number of Phase Points 2.sup.n 8 256 10 1,024 12 4,096 14 16,384 16 65,536 18 262,144 20 1,048,576 22 4,194,304 24 16,777,216 26 67,108,864 28 268,435,456 32 4,294,967,296 48 281,474,976,710,656

(480) The generator circuit algorithms and digital control circuits scan the addresses in the lookup table **4210**, which in turn provides varying digital input values to the DAC circuit **4212** that feeds the filter **4214** and the power amplifier. The addresses may be scanned according to a frequency of interest. Using the lookup table enables generating various types of shapes that can be converted into an analog output signal by the DAC circuit **4212**, filtered by the filter **4214**, amplified by the power amplifier coupled to the output of the generator circuit, and fed to the tissue in the form of RF energy or fed to an ultrasonic transducer and applied to the tissue in the form of ultrasonic vibrations which deliver energy to the tissue in the form of heat. The output of the amplifier can be applied to an RF electrode, multiple RF electrodes simultaneously, an ultrasonic transducer, multiple ultrasonic transducers simultaneously, or a combination of RF and ultrasonic transducers, for example. Furthermore, multiple wave shape tables can be created, stored, and applied to tissue from a generator circuit.

(481) With reference back to FIG. 41, for n=32, and M=1, the phase accumulator **4206** steps through 232 possible outputs before it overflows and restarts. The corresponding output wave frequency is equal to the input clock frequency divided by 232. If M=2, then the phase register **1708** “rolls over” twice as fast, and the output frequency is doubled. This can be generalized as follows.

(482) For a phase accumulator **4206** configured to accumulate n-bits (n generally ranges from 24 to 32 in most DDS systems, but as previously discussed n may be selected from a wide range of options), there are 2.sup.n possible phase points. The digital word in the delta phase register, M, represents the amount the phase accumulator is incremented per clock cycle. If f.sub.c is the clock frequency, then the frequency of the output sinewave is equal to:

$$(483) f_0 = \frac{M \cdot \text{Math} \cdot f_c}{2^n}$$

(484) The above equation is known as the DDS “tuning equation.” Note that the frequency resolution of the system is equal to

$$(485) \frac{f_0}{2^n}.$$

For n=32, the resolution is greater than one part in four billion. In one aspect of the DDS circuit **4200**, not all of the bits out of the phase accumulator **4206** are passed on to the lookup table **4210**, but are truncated, leaving only the first 13 to 15 most significant bits (MSBs), for example. This reduces the size of the lookup table **4210** and does not affect the frequency resolution. The phase truncation only adds a small but acceptable amount of phase noise to the final output.

(486) The electrical signal waveform may be characterized by a current, voltage, or power at a predetermined frequency. Further, where any one of the surgical instruments of surgical system **1000** comprises ultrasonic components, the electrical signal waveform may be configured to drive at least two vibration modes of an ultrasonic transducer of the at least one surgical instrument. Accordingly, the generator circuit may be configured to provide an electrical signal waveform to at least one surgical instrument wherein the electrical signal waveform is characterized by a predetermined wave shape stored in the lookup table **4210** (or lookup table **4104** FIG. 41). Further, the electrical signal waveform may be a combination of two or

more wave shapes. The lookup table **4210** may comprise information associated with a plurality of wave shapes. In one aspect or example, the lookup table **4210** may be generated by the DDS circuit **4200** and may be referred to as a direct digital synthesis table. DDS works by first storing a large repetitive waveform in onboard memory. A cycle of a waveform (sine, triangle, square, arbitrary) can be represented by a predetermined number of phase points as shown in TABLE 1 and stored into memory. Once the waveform is stored into memory, it can be generated at very precise frequencies. The direct digital synthesis table may be stored in a non-volatile memory of the generator circuit and/or may be implemented with a FPGA circuit in the generator circuit. The lookup table **4210** may be addressed by any suitable technique that is convenient for categorizing wave shapes. According to one aspect, the lookup table **4210** is addressed according to a frequency of the electrical signal waveform. Additionally, the information associated with the plurality of wave shapes may be stored as digital information in a memory or as part of the lookup table **4210**.

(487) In one aspect, the generator circuit may be configured to provide electrical signal waveforms to at least two surgical instruments simultaneously. The generator circuit also may be configured to provide the electrical signal waveform, which may be characterized two or more wave shapes, via an output channel of the generator circuit to the two surgical instruments simultaneously. For example, in one aspect the electrical signal waveform comprises a first electrical signal to drive an ultrasonic transducer (e.g., ultrasonic drive signal), a second RF drive signal, and/or a combination thereof. In addition, an electrical signal waveform may comprise a plurality of ultrasonic drive signals, a plurality of RF drive signals, and/or a combination of a plurality of ultrasonic and RF drive signals.

(488) In addition, a method of operating the generator circuit according to the present disclosure comprises generating an electrical signal waveform and providing the generated electrical signal waveform to any one of the surgical instruments of surgical system **1000**, where generating the electrical signal waveform comprises receiving information associated with the electrical signal waveform from a memory. The generated electrical signal waveform comprises at least one wave shape. Furthermore, providing the generated electrical signal waveform to the at least one surgical instrument comprises providing the electrical signal waveform to at least two surgical instruments simultaneously.

(489) The generator circuit as described herein may allow for the generation of various types of direct digital synthesis tables. Examples of wave shapes for RF/Electrosurgery signals suitable for treating a variety of tissue generated by the generator circuit include RF signals with a high crest factor (which may be used for surface coagulation in RF mode), a low crest factor RF signals (which may be used for deeper tissue penetration), and waveforms that promote efficient touch-up coagulation. The generator circuit also may generate multiple wave shapes employing a direct digital synthesis lookup table **4210** and, on the fly, can switch between particular wave shapes based on the desired tissue effect. Switching may be based on tissue impedance and/or other factors.

(490) In addition to traditional sine/cosine wave shapes, the generator circuit may be configured to generate wave shape(s) that maximize the power into tissue per cycle (i.e., trapezoidal or square wave). The generator circuit may provide wave shape(s) that are synchronized to maximize the power delivered to the load when driving RF and ultrasonic signals simultaneously and to maintain ultrasonic frequency lock, provided that the generator circuit includes a circuit topology that enables simultaneously driving RF and ultrasonic signals. Further, custom wave shapes specific to instruments and their tissue effects can be stored in a non-volatile memory (NVM) or an instrument EEPROM and can be fetched upon connecting any one of the surgical instruments of surgical system **1000** to the generator circuit.

(491) The DDS circuit **4200** may comprise multiple lookup tables **4104** where the lookup table **4210** stores a waveform represented by a predetermined number of phase points (also may be referred to as samples), wherein the phase points define a predetermined shape of the waveform. Thus multiple waveforms having a unique shape can be stored in multiple lookup tables **4210** to provide different tissue treatments based on instrument settings or tissue feedback. Examples of waveforms include high crest factor RF electrical signal waveforms for surface tissue coagulation, low crest factor RF electrical signal waveform for deeper tissue penetration, and electrical signal waveforms that promote efficient touch-up coagulation. In one aspect, the DDS circuit **4200** can create multiple wave shape lookup tables **4210** and during a tissue treatment procedure (e.g., “on-the-fly” or in virtual real time based on user or sensor inputs) switch between different wave shapes stored in different lookup tables **4210** based on the tissue effect desired and/or tissue feedback.

(492) Accordingly, switching between wave shapes can be based on tissue impedance and other factors, for example. In other aspects, the lookup tables **4210** can store electrical signal waveforms shaped to maximize the power delivered into the tissue per cycle (i.e., trapezoidal or square wave). In other aspects, the lookup tables **4210** can store wave shapes synchronized in such way that they make maximizing power delivery by any one of the surgical instruments of surgical system **1000** when delivering RF and ultrasonic drive signals. In yet other aspects, the lookup tables **4210** can store electrical signal waveforms to drive ultrasonic and RF therapeutic, and/or sub-therapeutic, energy simultaneously while maintaining ultrasonic frequency lock. Generally, the output wave shape may be in the form of a sine wave, cosine wave, pulse wave, square wave, and the like. Nevertheless, the more complex and custom wave shapes specific to different instruments and their tissue effects can be stored in the non-volatile memory of the generator circuit or in the non-volatile memory (e.g., EEPROM) of the surgical instrument and be fetched upon connecting the surgical instrument to the generator circuit. One example of a custom wave shape is an exponentially damped sinusoid as used in many high crest factor “coagulation” waveforms, as shown in FIG. 43.

(493) FIG. 43 illustrates one cycle of a discrete time digital electrical signal waveform **4300**, in accordance with at least one aspect of the present disclosure of an analog waveform **4304** (shown superimposed over the discrete time digital electrical signal waveform **4300** for comparison purposes). The horizontal axis represents Time (t) and the vertical axis represents digital phase points. The digital electrical signal waveform **4300** is a digital discrete time version of the desired analog waveform **4304**, for example. The digital electrical signal waveform **4300** is generated by storing an amplitude phase point **4302** that represents the amplitude per clock cycle T.sub.clk over one cycle or period T.sub.o. The digital electrical signal waveform **4300** is generated over one period T.sub.o by any suitable digital processing circuit. The amplitude phase points are digital words stored in a memory circuit. In the example illustrated in FIGS. 41, 42, the digital word is a six-bit word that is capable of storing the amplitude phase points with a resolution of 26 or 64 bits. It will be appreciated that the examples shown in FIGS. 41, 42 is for illustrative purposes and in actual implementations the resolution can be much higher. The digital amplitude phase points **4302** over one cycle T.sub.o are stored in the memory as a string of string words in a lookup table **4104**, **4210** as described in connection with FIGS. 41, 42, for example. To generate the analog version of the analog waveform **4304**, the amplitude phase points **4302** are read sequentially from the memory from 0 to T.sub.o per clock cycle T.sub.clk and are converted by a DAC circuit **4108**, **4212**, also described in connection with FIGS. 41, 42. Additional cycles can be generated by repeatedly reading the amplitude phase points **4302** of the digital electrical signal waveform **4300** the from 0 to T.sub.o for as many cycles or periods as may be desired. The smooth analog version of the analog waveform **4304** is achieved by filtering the output of the DAC circuit **4108**, **4212** by a filter **4112**, **4214** (FIGS. 41 and 42). The filtered analog output signal **4114**, **4222** (FIGS. 41 and 42) is applied to the input of a power amplifier.

(494) FIG. 44 is a diagram of a control system **12950** configured to provide progressive closure of a closure member (e.g., closure tube) when the displacement member advances distally and couples into a clamp arm (e.g., anvil) to lower the closure force load on the closure member at a desired rate and decrease the firing force load on the firing member according to one aspect of this disclosure. In one aspect, the control system **12950** may be implemented as a nested PID feedback controller. A PID controller is a control loop feedback mechanism (controller) to continuously calculate an error value as the difference between a desired set point and a measured process variable and applies a correction based on proportional, integral, and derivative terms (sometimes denoted P, I, and D respectively). The nested PID controller feedback control system **12950** includes a primary controller **12952**, in a primary (outer) feedback loop **12954** and a secondary controller **12955** in a secondary (inner) feedback loop **12956**. The primary controller **12952** may be a PID controller **12972** as shown in FIG. 45, and the secondary controller **12955** also may be a PID controller **12972** as shown in FIG. 45. The primary controller **12952** controls a primary process **12958** and the secondary controller **12955** controls a secondary process **12960**. The output **12966** of the primary process **12958** is subtracted from a primary set point SP.sub.1 by a first summer **12962**. The first summer **12962** produces a single sum output signal which is applied to the primary controller **12952**. The output of the primary controller **12952** is

the secondary set point SP.sub.2. The output **12968** of the secondary process **12960** is subtracted from the secondary set point SP.sub.2 by a second summer **12964**.

(495) In the context of controlling the displacement of a closure tube, the control system **12950** may be configured such that the primary set point SP.sub.1 is a desired closure force value and the primary controller **12952** is configured to receive the closure force from a torque sensor coupled to the output of a closure motor and determine a set point SP.sub.2 motor velocity for the closure motor. In other aspects, the closure force may be measured with strain gauges, load cells, or other suitable force sensors. The closure motor velocity set point SP.sub.2 is compared to the actual velocity of the closure tube, which is determined by the secondary controller **12955**. The actual velocity of the closure tube may be measured by comparing measuring the displacement of the closure tube with the position sensor and measuring elapsed time with a timer/counter. Other techniques, such as linear or rotary encoders may be employed to measure displacement of the closure tube. The output **12968** of the secondary process **12960** is the actual velocity of the closure tube. This closure tube velocity output **12968** is provided to the primary process **12958** which determines the force acting on the closure tube and is fed back to the adder **12962**, which subtracts the measured closure force from the primary set point SP.sub.1. The primary set point SP.sub.1 may be an upper threshold or a lower threshold. Based on the output of the adder **12962**, the primary controller **12952** controls the velocity and direction of the closure motor. The secondary controller **12955** controls the velocity of the closure motor based on the actual velocity of closure tube measured by the secondary process **12960** and the secondary set point SP.sub.2, which is based on a comparison of the actual firing force and the firing force upper and lower thresholds.

(496) FIG. **45** illustrates a PID feedback control system **12970** according to one aspect of this disclosure. The primary controller **12952** or the secondary controller **12955**, or both, may be implemented as a PID controller **12972**. In one aspect, the PID controller **12972** may comprise a proportional element **12974** (P), an integral element **12976** (I), and a derivative element **12978** (D). The outputs of the P, I, D elements **12974**, **12976**, **12978** are summed by a summer **12986**, which provides the control variable $\mu(t)$ to the process **12980**. The output of the process **12980** is the process variable $y(t)$. A summer **12984** calculates the difference between a desired set point $r(t)$ and a measured process variable $y(t)$. The PID controller **12972** continuously calculates an error value $e(t)$ (e.g., difference between closure force threshold and measured closure force) as the difference between a desired set point $r(t)$ (e.g., closure force threshold) and a measured process variable $y(t)$ (e.g., velocity and direction of closure tube) and applies a correction based on the proportional, integral, and derivative terms calculated by the proportional element **12974** (P), integral element **12976** (I), and derivative element **12978** (D), respectively. The PID controller **12972** attempts to minimize the error $e(t)$ over time by adjustment of the control variable $\mu(t)$ (e.g., velocity and direction of the closure tube).

(497) In accordance with the PID algorithm, the “P” element **12974** accounts for present values of the error. For example, if the error is large and positive, the control output will also be large and positive. In accordance with the present disclosure, the error term $e(t)$ is the different between the desired closure force and the measured closure force of the closure tube. The “I” element **12976** accounts for past values of the error. For example, if the current output is not sufficiently strong, the integral of the error will accumulate over time, and the controller will respond by applying a stronger action. The “D” element **12978** accounts for possible future trends of the error, based on its current rate of change. For example, continuing the P example above, when the large positive control output succeeds in bringing the error closer to zero, it also puts the process on a path to large negative error in the near future. In this case, the derivative turns negative and the D module reduces the strength of the action to prevent this overshoot.

(498) It will be appreciated that other variables and set points may be monitored and controlled in accordance with the feedback control systems **12950**, **12970**. For example, the adaptive closure member velocity control algorithm described herein may measure at least two of the following parameters: firing member stroke location, firing member load, displacement of cutting element, velocity of cutting element, closure tube stroke location, closure tube load, among others.

(499) Ultrasonic surgical devices, such as ultrasonic scalpels, are finding increasingly widespread applications in surgical procedures by virtue of their unique performance characteristics. Depending upon specific device configurations and operational parameters, ultrasonic surgical devices can provide substantially simultaneous transection of tissue and homeostasis by coagulation, desirably minimizing patient trauma. An ultrasonic surgical device may comprise a handpiece containing an ultrasonic transducer, and an instrument coupled to the ultrasonic transducer having a distally-mounted end effector (e.g., a blade tip) to cut and seal tissue. In some cases, the instrument may be permanently affixed to the handpiece. In other cases, the instrument may be detachable from the handpiece, as in the case of a disposable instrument or an interchangeable instrument. The end effector transmits ultrasonic energy to tissue brought into contact with the end effector to realize cutting and sealing action. Ultrasonic surgical devices of this nature can be configured for open surgical use, laparoscopic, or endoscopic surgical procedures including robotic-assisted procedures.

(500) Ultrasonic energy cuts and coagulates tissue using temperatures lower than those used in electrosurgical procedures and can be transmitted to the end effector by an ultrasonic generator in communication with the handpiece. Vibrating at high frequencies (e.g., 55,500 cycles per second), the ultrasonic blade denatures protein in the tissue to form a sticky coagulum. Pressure exerted on tissue by the blade surface collapses blood vessels and allows the coagulum to form a hemostatic seal. A surgeon can control the cutting speed and coagulation by the force applied to the tissue by the end effector, the time over which the force is applied, and the selected excursion level of the end effector.

(501) The ultrasonic transducer may be modeled as an equivalent circuit comprising a first branch having a static capacitance and a second “motional” branch having a serially connected inductance, resistance and capacitance that define the electromechanical properties of a resonator. Known ultrasonic generators may include a tuning inductor for tuning out the static capacitance at a resonant frequency so that substantially all of a generator's drive signal current flows into the motional branch. Accordingly, by using a tuning inductor, the generator's drive signal current represents the motional branch current, and the generator is thus able to control its drive signal to maintain the ultrasonic transducer's resonant frequency. The tuning inductor may also transform the phase impedance plot of the ultrasonic transducer to improve the generator's frequency lock capabilities. However, the tuning inductor must be matched with the specific static capacitance of an ultrasonic transducer at the operational resonant frequency. In other words, a different ultrasonic transducer having a different static capacitance requires a different tuning inductor.

(502) Additionally, in some ultrasonic generator architectures, the generator's drive signal exhibits asymmetrical harmonic distortion that complicates impedance magnitude and phase measurements. For example, the accuracy of impedance phase measurements may be reduced due to harmonic distortion in the current and voltage signals.

(503) Moreover, electromagnetic interference in noisy environments decreases the ability of the generator to maintain lock on the ultrasonic transducer's resonant frequency, increasing the likelihood of invalid control algorithm inputs.

(504) Electrosurgical devices for applying electrical energy to tissue in order to treat and/or destroy the tissue are also finding increasingly widespread applications in surgical procedures. An electrosurgical device may comprise a handpiece and an instrument having a distally-mounted end effector (e.g., one or more electrodes). The end effector can be positioned against the tissue such that electrical current is introduced into the tissue. Electrosurgical devices can be configured for bipolar or monopolar operation. During bipolar operation, current is introduced into and returned from the tissue by active and return electrodes, respectively, of the end effector. During monopolar operation, current is introduced into the tissue by an active electrode of the end effector and returned through a return electrode (e.g., a grounding pad) separately located on a patient's body. Heat generated by the current flowing through the tissue may form hemostatic seals within the tissue and/or between tissues and thus may be particularly useful for sealing blood vessels, for example. The end effector of an electrosurgical device may also comprise a cutting member that is movable relative to the tissue and the electrodes to transect the tissue.

(505) Electrical energy applied by an electrosurgical device can be transmitted to the instrument by a generator in communication with the handpiece. The electrical energy may be in the form of radio frequency (RF) energy. RF energy is a form of electrical energy that may be in the frequency range of 300 kHz to 1 MHz, as described in EN60601-2-2:2009+A11:2011, Definition 201.3.218—HIGH FREQUENCY. For example, the frequencies in monopolar RF applications are typically restricted to less than 5 MHz. However, in bipolar RF applications, the frequency can be

at most any value. Frequencies above 200 kHz are typically used in order to avoid the unwanted stimulation of nerves and muscles which would result from the use of low frequency current. Lower frequencies may be used for bipolar techniques if a risk analysis shows the possibility of neuromuscular stimulation has been mitigated to an acceptable level. Normally, frequencies above 5 MHz are not used in order to minimize the problems associated with high frequency leakage currents. It is generally recognized that 10 mA is the lower threshold of thermal effects on tissue.

(506) During its operation, an electrosurgical device can transmit low frequency RF energy through tissue, which causes ionic agitation, or friction, in effect resistive heating, thereby increasing the temperature of the tissue. Because a sharp boundary may be created between the affected tissue and the surrounding tissue, surgeons can operate with a high level of precision and control, without sacrificing un-targeted adjacent tissue. The low operating temperatures of RF energy may be useful for removing, shrinking, or sculpting soft tissue while simultaneously sealing blood vessels. RF energy may work particularly well on connective tissue, which is primarily comprised of collagen and shrinks when contacted by heat.

(507) Due to their unique drive signal, sensing and feedback needs, ultrasonic and electrosurgical devices have generally required different generators. Additionally, in cases where the instrument is disposable or interchangeable with a handpiece, ultrasonic and electrosurgical generators are limited in their ability to recognize the particular instrument configuration being used and to optimize control and diagnostic processes accordingly. Moreover, capacitive coupling between the non-isolated and patient-isolated circuits of the generator, especially in cases where higher voltages and frequencies are used, may result in exposure of a patient to unacceptable levels of leakage current.

(508) Furthermore, due to their unique drive signal, sensing and feedback needs, ultrasonic and electrosurgical devices have generally required different user interfaces for the different generators. In such conventional ultrasonic and electrosurgical devices, one user interface is configured for use with an ultrasonic instrument whereas a different user interface may be configured for use with an electrosurgical instrument. Such user interfaces include hand and/or foot activated user interfaces such as hand activated switches and/or foot activated switches. As various aspects of combined generators for use with both ultrasonic and electrosurgical instruments are contemplated in the subsequent disclosure, additional user interfaces that are configured to operate with both ultrasonic and/or electrosurgical instrument generators also are contemplated.

(509) Additional user interfaces for providing feedback, whether to the user or other machine, are contemplated within the subsequent disclosure to provide feedback indicating an operating mode or status of either an ultrasonic and/or electrosurgical instrument. Providing user and/or machine feedback for operating a combination ultrasonic and/or electrosurgical instrument will require providing sensory feedback to a user and electrical/mechanical/electro-mechanical feedback to a machine. Feedback devices that incorporate visual feedback devices (e.g., an LCD display screen, LED indicators), audio feedback devices (e.g., a speaker, a buzzer) or tactile feedback devices (e.g., haptic actuators) for use in combined ultrasonic and/or electrosurgical instruments are contemplated in the subsequent disclosure.

(510) Other electrical surgical instruments include, without limitation, irreversible and/or reversible electroporation, and/or microwave technologies, among others. Accordingly, the techniques disclosed herein are applicable to ultrasonic, bipolar or monopolar RF (electrosurgical), irreversible and/or reversible electroporation, and/or microwave based surgical instruments, among others.

(511) Various aspects are directed to improved ultrasonic surgical devices, electrosurgical devices and generators for use therewith. Aspects of the ultrasonic surgical devices can be configured for transecting and/or coagulating tissue during surgical procedures, for example. Aspects of the electrosurgical devices can be configured for transecting, coagulating, scaling, welding and/or desiccating tissue during surgical procedures, for example.

(512) Aspects of the generator utilize high-speed analog-to-digital sampling (e.g., approximately 200× oversampling, depending on frequency) of the generator drive signal current and voltage, along with digital signal processing, to provide a number of advantages and benefits over known generator architectures. In one aspect, for example, based on current and voltage feedback data, a value of the ultrasonic transducer static capacitance, and a value of the drive signal frequency, the generator may determine the motional branch current of an ultrasonic transducer. This provides the benefit of a virtually tuned system, and simulates the presence of a system that is tuned or resonant with any value of the static capacitance (e.g., C.sub.0 in FIG. 25) at any frequency. Accordingly, control of the motional branch current may be realized by tuning out the effects of the static capacitance without the need for a tuning inductor. Additionally, the elimination of the tuning inductor may not degrade the generator's frequency lock capabilities, as frequency lock can be realized by suitably processing the current and voltage feedback data.

(513) High-speed analog-to-digital sampling of the generator drive signal current and voltage, along with digital signal processing, may also enable precise digital filtering of the samples. For example, aspects of the generator may utilize a low-pass digital filter (e.g., a finite impulse response (FIR) filter) that rolls off between a fundamental drive signal frequency and a second-order harmonic to reduce the asymmetrical harmonic distortion and EMI-induced noise in current and voltage feedback samples. The filtered current and voltage feedback samples represent substantially the fundamental drive signal frequency, thus enabling a more accurate impedance phase measurement with respect to the fundamental drive signal frequency and an improvement in the generator's ability to maintain resonant frequency lock. The accuracy of the impedance phase measurement may be further enhanced by averaging falling edge and rising edge phase measurements, and by regulating the measured impedance phase to 0°.

(514) Various aspects of the generator may also utilize the high-speed analog-to-digital sampling of the generator drive signal current and voltage, along with digital signal processing, to determine real power consumption and other quantities with a high degree of precision. This may allow the generator to implement a number of useful algorithms, such as, for example, controlling the amount of power delivered to tissue as the impedance of the tissue changes and controlling the power delivery to maintain a constant rate of tissue impedance increase. Some of these algorithms are used to determine the phase difference between the generator drive signal current and voltage signals. At resonance, the phase difference between the current and voltage signals is zero. The phase changes as the ultrasonic system goes off-resonance. Various algorithms may be employed to detect the phase difference and adjust the drive frequency until the ultrasonic system returns to resonance, i.e., the phase difference between the current and voltage signals goes to zero. The phase information also may be used to infer the conditions of the ultrasonic blade. As discussed with particularity below, the phase changes as a function of the temperature of the ultrasonic blade. Therefore, the phase information may be employed to control the temperature of the ultrasonic blade. This may be done, for example, by reducing the power delivered to the ultrasonic blade when the ultrasonic blade runs too hot and increasing the power delivered to the ultrasonic blade when the ultrasonic blade runs too cold.

(515) Various aspects of the generator may have a wide frequency range and increased output power necessary to drive both ultrasonic surgical devices and electrosurgical devices. The lower voltage, higher current demand of electrosurgical devices may be met by a dedicated tap on a wideband power transformer, thereby eliminating the need for a separate power amplifier and output transformer. Moreover, sensing and feedback circuits of the generator may support a large dynamic range that addresses the needs of both ultrasonic and electrosurgical applications with minimal distortion.

(516) Various aspects may provide a simple, economical means for the generator to read from, and optionally write to, a data circuit (e.g., a single-wire bus device, such as a one-wire protocol EEPROM known under the trade name "1-Wire") disposed in an instrument attached to the handpiece using existing multi-conductor generator/handpiece cables. In this way, the generator is able to retrieve and process instrument-specific data from an instrument attached to the handpiece. This may enable the generator to provide better control and improved diagnostics and error detection. Additionally, the ability of the generator to write data to the instrument makes possible new functionality in terms of, for example, tracking instrument usage and capturing operational data. Moreover, the use of frequency band permits the backward compatibility of instruments containing a bus device with existing generators.

(517) Disclosed aspects of the generator provide active cancellation of leakage current caused by unintended capacitive coupling between non-isolated and patient-isolated circuits of the generator. In addition to reducing patient risk, the reduction of leakage current may also lessen electromagnetic emissions.

(518) These and other benefits of aspects of the present disclosure will be apparent from the description to follow.

(519) It will be appreciated that the terms “proximal” and “distal” are used herein with reference to a clinician gripping a handpiece. Thus, an end effector is distal with respect to the more proximal handpiece. It will be further appreciated that, for convenience and clarity, spatial terms such as “top” and “bottom” may also be used herein with respect to the clinician gripping the handpiece. However, surgical devices are used in many orientations and positions, and these terms are not intended to be limiting and absolute.

(520) FIG. 46 is an elevational exploded view of modular handheld ultrasonic surgical instrument 6480 showing the left shell half removed from a handle assembly 6482 exposing a device identifier communicatively coupled to the multi-lead handle terminal assembly in accordance with one aspect of the present disclosure. In additional aspects of the present disclosure, an intelligent or smart battery is used to power the modular handheld ultrasonic surgical instrument 6480. However, the smart battery is not limited to the modular handheld ultrasonic surgical instrument 6480 and, as will be explained, can be used in a variety of devices, which may or may not have power requirements (e.g., current and voltage) that vary from one another. The smart battery assembly 6486, in accordance with one aspect of the present disclosure, is advantageously able to identify the particular device to which it is electrically coupled. It does this through encrypted or unencrypted identification methods. For instance, a smart battery assembly 6486 can have a connection portion, such as connection portion 6488. The handle assembly 6482 can also be provided with a device identifier communicatively coupled to the multi-lead handle terminal assembly 6491 and operable to communicate at least one piece of information about the handle assembly 6482. This information can pertain to the number of times the handle assembly 6482 has been used, the number of times an ultrasonic transducer/generator assembly 6484 (presently disconnected from the handle assembly 6482) has been used, the number of times a waveguide shaft assembly 6490 (presently connected to the handle assembly 6482) has been used, the type of the waveguide shaft assembly 6490 that is presently connected to the handle assembly 6482, the type or identity of the ultrasonic transducer/generator assembly 6484 that is presently connected to the handle assembly 6482, and/or many other characteristics. When the smart battery assembly 6486 is inserted in the handle assembly 6482, the connection portion 6488 within the smart battery assembly 6486 makes communicating contact with the device identifier of the handle assembly 6482. The handle assembly 6482, through hardware, software, or a combination thereof, is able to transmit information to the smart battery assembly 6486 (whether by self-initiation or in response to a request from the smart battery assembly 6486). This communicated identifier is received by the connection portion 6488 of the smart battery assembly 6486. In one aspect, once the smart battery assembly 6486 receives the information, the communication portion is operable to control the output of the smart battery assembly 6486 to comply with the device's specific power requirements.

(521) In one aspect, the communication portion includes a processor 6493 and a memory 6497, which may be separate or a single component. The processor 6493, in combination with the memory, is able to provide intelligent power management for the modular handheld ultrasonic surgical instrument 6480. This aspect is particularly advantageous because an ultrasonic device, such as the modular handheld ultrasonic surgical instrument 6480, has a power requirement (frequency, current, and voltage) that may be unique to the modular handheld ultrasonic surgical instrument 6480. In fact, the modular handheld ultrasonic surgical instrument 6480 may have a particular power requirement or limitation for one dimension or type of outer tube 6494 and a second different power requirement for a second type of waveguide having a different dimension, shape, and/or configuration.

(522) A smart battery assembly 6486, in accordance with at least one aspect of the present disclosure, therefore, allows a battery assembly to be used amongst several surgical instruments. Because the smart battery assembly 6486 is able to identify to which device it is attached and is able to alter its output accordingly, the operators of various different surgical instruments utilizing the smart battery assembly 6486 no longer need be concerned about which power source they are attempting to install within the electronic device being used. This is particularly advantageous in an operating environment where a battery assembly needs to be replaced or interchanged with another surgical instrument in the middle of a complex surgical procedure.

(523) In a further aspect of the present disclosure, the smart battery assembly 6486 stores in a memory 6497 a record of each time a particular device is used. This record can be useful for assessing the end of a device's useful or permitted life. For instance, once a device is used 20 times, such batteries in the smart battery assembly 6486 connected to the device will refuse to supply power thereto-because the device is defined as a “no longer reliable” surgical instrument. Reliability is determined based on a number of factors. One factor can be wear, which can be estimated in a number of ways including the number of times the device has been used or activated. After a certain number of uses, the parts of the device can become worn and tolerances between parts exceeded. For instance, the smart battery assembly 6486 can sense the number of button pushes received by the handle assembly 6482 and can determine when a maximum number of button pushes has been met or exceeded. The smart battery assembly 6486 can also monitor an impedance of the button mechanism which can change, for instance, if the handle gets contaminated, for example, with saline.

(524) This wear can lead to an unacceptable failure during a procedure. In some aspects, the smart battery assembly 6486 can recognize which parts are combined together in a device and even how many uses a part has experienced. For instance, if the smart battery assembly 6486 is a smart battery according to the present disclosure, it can identify the handle assembly 6482, the waveguide shaft assembly 6490, as well as the ultrasonic transducer/generator assembly 6484, well before the user attempts use of the composite device. The memory 6497 within the smart battery assembly 6486 can, for example, record a time when the ultrasonic transducer/generator assembly 6484 is operated, and how, when, and for how long it is operated. If the ultrasonic transducer/generator assembly 6484 has an individual identifier, the smart battery assembly 6486 can keep track of uses of the ultrasonic transducer/generator assembly 6484 and refuse to supply power to that the ultrasonic transducer/generator assembly 6484 once the handle assembly 6482 or the ultrasonic transducer/generator assembly 6484 exceeds its maximum number of uses. The ultrasonic transducer/generator assembly 6484, the handle assembly 6482, the waveguide shaft assembly 6490, or other components can include a memory chip that records this information as well. In this way, any number of smart batteries in the smart battery assembly 6486 can be used with any number of ultrasonic transducer/generator assemblies 6484, staplers, vessel sealers, etc. and still be able to determine the total number of uses, or the total time of use (through use of the clock), or the total number of actuations, etc. of the ultrasonic transducer/generator assembly 6484, the stapler, the vessel sealer, etc. or charge or discharge cycles. Smart functionality may reside outside the battery assembly 6486 and may reside in the handle assembly 6482, the ultrasonic transducer/generator assembly 6484, and/or the shaft assembly 6490, for example.

(525) When counting uses of the ultrasonic transducer/generator assembly 6484, to intelligently terminate the life of the ultrasonic transducer/generator assembly 6484, the surgical instrument accurately distinguishes between completion of an actual use of the ultrasonic transducer/generator assembly 6484 in a surgical procedure and a momentary lapse in actuation of the ultrasonic transducer/generator assembly 6484 due to, for example, a battery change or a temporary delay in the surgical procedure. Therefore, as an alternative to simply counting the number of activations of the ultrasonic transducer/generator assembly 6484, a real-time clock (RTC) circuit can be implemented to keep track of the amount of time the ultrasonic transducer/generator assembly 6484 actually is shut down. From the length of time measured, it can be determined through appropriate logic if the shutdown was significant enough to be considered the end of one actual use or if the shutdown was too short in time to be considered the end of one use. Thus, in some applications, this method may be a more accurate determination of the useful life of the ultrasonic transducer/generator assembly 6484 than a simple “activations-based” algorithm, which for example, may provide that ten “activations” occur in a surgical procedure and, therefore, ten activations should indicate that the counter is incremented by one. Generally, this type and system of internal clocking will prevent misuse of the device that is designed to deceive a simple “activations-based” algorithm and will prevent incorrect logging of a complete use in instances when there was only a simple de-mating of the ultrasonic transducer/generator assembly 6484 or the smart battery assembly 6486 that was required for legitimate reasons.

(526) Although the ultrasonic transducer/generator assemblies 6484 of the surgical instrument 6480 are reusable, in one aspect a finite number of uses may be set because the surgical instrument 6480 is subjected to harsh conditions during cleaning and sterilization. More specifically, the battery pack is configured to be sterilized. Regardless of the material employed for the outer surfaces, there is a limited expected life for the actual materials

used. This life is determined by various characteristics which could include, for example, the amount of times the pack has actually been sterilized, the time from which the pack was manufactured, and the number of times the pack has been recharged, to name a few. Also, the life of the battery cells themselves is limited. Software of the present disclosure incorporates inventive algorithms that verify the number of uses of the ultrasonic transducer/generator assembly **6484** and smart battery assembly **6486** and disables the device when this number of uses has been reached or exceeded. Analysis of the battery pack exterior in each of the possible sterilizing methods can be performed. Based on the harshest sterilization procedure, a maximum number of permitted sterilizations can be defined and that number can be stored in a memory of the smart battery assembly **6486**. If it is assumed that a charger is non-sterile and that the smart battery assembly **6486** is to be used after it is charged, then the charge count can be defined as being equal to the number of sterilizations encountered by that particular pack.

(527) In one aspect, the hardware in the battery pack may be disabled to minimize or eliminate safety concerns due to continuous drain in from the battery cells after the pack has been disabled by software. A situation can exist where the battery's internal hardware is incapable of disabling the battery under certain low voltage conditions. In such a situation, in an aspect, the charger can be used to "kill" the battery. Due to the fact that the battery microcontroller is OFF while the battery is in its charger, a non-volatile, System Management Bus (SMB) based electrically erasable programmable read only memory (EEPROM) can be used to exchange information between the battery microcontroller and the charger. Thus, a serial EEPROM can be used to store information that can be written and read even when the battery microcontroller is OFF, which is very beneficial when trying to exchange information with the charger or other peripheral devices. This example EEPROM can be configured to contain enough memory registers to store at least (a) a use-count limit at which point the battery should be disabled (Battery Use Count), (b) the number of procedures the battery has undergone (Battery Procedure Count), and/or (c) a number of charges the battery has undergone (Charge Count), to name a few. Some of the information stored in the EEPROM, such as the Use Count Register and Charge Count Register are stored in write-protected sections of the EEPROM to prevent users from altering the information. In an aspect, the use and counters are stored with corresponding bit-inverted minor registers to detect data corruption.

(528) Any residual voltage in the SMBus lines could damage the microcontroller and corrupt the SMBus signal. Therefore, to ensure that the SMBus lines of a battery controller do not carry a voltage while the microcontroller is OFF, relays are provided between the external SMBus lines and the battery microcontroller board.

(529) During charging of the smart battery assembly **6486**, an "end-of-charge" condition of the batteries within the smart battery assembly **6486** is determined when, for example, the current flowing into the battery falls below a given threshold in a tapering manner when employing a constant-current/constant-voltage charging scheme. To accurately detect this "end-of-charge" condition, the battery microcontroller and buck boards are powered down and turned OFF during charging of the battery to reduce any current drain that may be caused by the boards and that may interfere with the tapering current detection. Additionally, the microcontroller and buck boards are powered down during charging to prevent any resulting corruption of the SMBus signal.

(530) With regard to the charger, in one aspect the smart battery assembly **6486** is prevented from being inserted into the charger in any way other than the correct insertion position. Accordingly, the exterior of the smart battery assembly **6486** is provided with charger-holding features. A cup for holding the smart battery assembly **6486** securely in the charger is configured with a contour-matching taper geometry to prevent the accidental insertion of the smart battery assembly **6486** in any way other than the correct (intended) way. It is further contemplated that the presence of the smart battery assembly **6486** may be detectable by the charger itself. For example, the charger may be configured to detect the presence of the SMBus transmission from the battery protection circuit, as well as resistors that are located in the protection board. In such case, the charger would be enabled to control the voltage that is exposed at the charger's pins until the smart battery assembly **6486** is correctly seated or in place at the charger. This is because an exposed voltage at the charger's pins would present a hazard and a risk that an electrical short could occur across the pins and cause the charger to inadvertently begin charging.

(531) In some aspects, the smart battery assembly **6486** can communicate to the user through audio and/or visual feedback. For example, the smart battery assembly **6486** can cause the LEDs to light in a pre-set way. In such a case, even though the microcontroller in the ultrasonic transducer/generator assembly **6484** controls the LEDs, the microcontroller receives instructions to be carried out directly from the smart battery assembly **6486**.

(532) In yet a further aspect of the present disclosure, the microcontroller in the ultrasonic transducer/generator assembly **6484**, when not in use for a predetermined period of time, goes into a sleep mode. Advantageously, when in the sleep mode, the clock speed of the microcontroller is reduced, cutting the current drain significantly. Some current continues to be consumed because the processor continues pinging waiting to sense an input. Advantageously, when the microcontroller is in this power-saving sleep mode, the microcontroller and the battery controller can directly control the LEDs. For example, a decoder circuit could be built into the ultrasonic transducer/generator assembly **6484** and connected to the communication lines such that the LEDs can be controlled independently by the processor **6493** while the ultrasonic transducer/generator assembly **6484** microcontroller is "OFF" or in a "sleep mode." This is a power-saving feature that eliminates the need for waking up the microcontroller in the ultrasonic transducer/generator assembly **6484**. Power is conserved by allowing the generator to be turned off while still being able to actively control the user-interface indicators.

(533) Another aspect slows down one or more of the microcontrollers to conserve power when not in use. For example, the clock frequencies of both microcontrollers can be reduced to save power. To maintain synchronized operation, the microcontrollers coordinate the changing of their respective clock frequencies to occur at about the same time, both the reduction and, then, the subsequent increase in frequency when full speed operation is required. For example, when entering the idle mode, the clock frequencies are decreased and, when exiting the idle mode, the frequencies are increased.

(534) In an additional aspect, the smart battery assembly **6486** is able to determine the amount of usable power left within its cells and is programmed to only operate the surgical instrument to which it is attached if it determines there is enough battery power remaining to predictably operate the device throughout the anticipated procedure. For example, the smart battery assembly **6486** is able to remain in a non-operational state if there is not enough power within the cells to operate the surgical instrument for 20 seconds. According to one aspect, the smart battery assembly **6486** determines the amount of power remaining within the cells at the end of its most recent preceding function, e.g., a surgical cutting. In this aspect, therefore, the smart battery assembly **6486** would not allow a subsequent function to be carried out if, for example, during that procedure, it determines that the cells have insufficient power. Alternatively, if the smart battery assembly **6486** determines that there is sufficient power for a subsequent procedure and goes below that threshold during the procedure, it would not interrupt the ongoing procedure and, instead, will allow it to finish and thereafter prevent additional procedures from occurring.

(535) The following explains an advantage to maximizing use of the device with the smart battery assembly **6486** of the present disclosure. In this example, a set of different devices have different ultrasonic transmission waveguides. By definition, the waveguides could have a respective maximum allowable power limit where exceeding that power limit overstresses the waveguide and eventually causes it to fracture. One waveguide from the set of waveguides will naturally have the smallest maximum power tolerance. Because prior-art batteries lack intelligent battery power management, the output of prior-art batteries must be limited by a value of the smallest maximum allowable power input for the smallest/thinnest/most-frail waveguide in the set that is envisioned to be used with the device/battery. This would be true even though larger, thicker waveguides could later be attached to that handle and, by definition, allow a greater force to be applied. This limitation is also true for maximum battery power. For example, if one battery is designed to be used in multiple devices, its maximum output power will be limited to the lowest maximum power rating of any of the devices in which it is to be used. With such a configuration, one or more devices or device configurations would not be able to maximize use of the battery because the battery does not know the particular device's specific limits.

(536) In one aspect, the smart battery assembly **6486** may be configured to intelligently circumvent the above-mentioned ultrasonic device limitations. The smart battery assembly **6486** can produce one output for one device or a particular device configuration and the same smart battery assembly **6486** can later produce a different output for a second device or device configuration. This universal smart battery surgical system lends itself well to the modern operating room where space and time are at a premium. By having a smart battery pack operate many different devices, the nurses can easily manage the storage, retrieval, and inventory of these packs. Advantageously, in one aspect the smart battery system according to the present disclosure may employ one type of charging station, thus increasing ease and efficiency of use and decreasing cost of surgical room charging equipment.

(537) In addition, other surgical instruments, such as an electric stapler, may have a different power requirement than that of the modular handheld ultrasonic surgical instrument **6480**. In accordance with various aspects of the present disclosure, a smart battery assembly **6486** can be used with any one of a series of surgical instruments and can be made to tailor its own power output to the particular device in which it is installed. In one aspect, this power tailoring is performed by controlling the duty cycle of a switched mode power supply, such as buck, buck-boost, boost, or other configuration, integral with or otherwise coupled to and controlled by the smart battery assembly **6486**. In other aspects, the smart battery assembly **6486** can dynamically change its power output during device operation. For instance, in vessel sealing devices, power management provides improved tissue sealing. In these devices, large constant current values are needed. The total power output needs to be adjusted dynamically because, as the tissue is sealed, its impedance changes. Aspects of the present disclosure provide the smart battery assembly **6486** with a variable maximum current limit. The current limit can vary from one application (or device) to another, based on the requirements of the application or device.

(538) FIG. **47** is a detail view of a trigger **6483** portion and switch of the ultrasonic surgical instrument **6480** shown in FIG. **46**, in accordance with at least one aspect of the present disclosure. The trigger **6483** is operably coupled to the jaw member **6495** of the end effector **6492**. The ultrasonic blade **6496** is energized by the ultrasonic transducer/generator assembly **6484** upon activating the activation switch **6485**. Continuing now with FIG. **46** and also looking to FIG. **47**, the trigger **6483** and the activation switch **6485** are shown as components of the handle assembly **6482**. The trigger **6483** activates the end effector **6492**, which has a cooperative association with the ultrasonic blade **6496** of the waveguide shaft assembly **6490** to enable various kinds of contact between the end effector jaw member **6495** and the ultrasonic blade **6496** with tissue and/or other substances. The jaw member **6495** of the end effector **6492** is usually a pivoting jaw that acts to grasp or clamp onto tissue disposed between the jaw and the ultrasonic blade **6496**. In one aspect, an audible feedback is provided in the trigger that clicks when the trigger is fully depressed. The noise can be generated by a thin metal part that the trigger snaps over while closing. This feature adds an audible component to user feedback that informs the user that the jaw is fully compressed against the waveguide and that sufficient clamping pressure is being applied to accomplish vessel sealing. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to the trigger **6483** to measure the force applied to the trigger **6483** by the user. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to the switch **6485** button such that displacement intensity corresponds to the force applied by the user to the switch **6485** button.

(539) The activation switch **6485**, when depressed, places the modular handheld ultrasonic surgical instrument **6480** into an ultrasonic operating mode, which causes ultrasonic motion at the waveguide shaft assembly **6490**. In one aspect, depression of the activation switch **6485** causes electrical contacts within a switch to close, thereby completing a circuit between the smart battery assembly **6486** and the ultrasonic transducer/generator assembly **6484** so that electrical power is applied to the ultrasonic transducer, as previously described. In another aspect, depression of the activation switch **6485** closes electrical contacts to the smart battery assembly **6486**. Of course, the description of closing electrical contacts in a circuit is, here, merely an example general description of switch operation. There are many alternative aspects that can include opening contacts or processor-controlled power delivery that receives information from the switch and directs a corresponding circuit reaction based on the information.

(540) FIG. **48** is a fragmentary, enlarged perspective view of an end effector **6492**, in accordance with at least one aspect of the present disclosure, from a distal end with a jaw member **6495** in an open position. Referring to FIG. **48**, a perspective partial view of the distal end **6498** of the waveguide shaft assembly **6490** is shown. The waveguide shaft assembly **6490** includes an outer tube **6494** surrounding a portion of the waveguide. The ultrasonic blade **6496** portion of the waveguide **6499** protrudes from the distal end **6498** of the outer tube **6494**. It is the ultrasonic blade **6496** portion that contacts the tissue during a medical procedure and transfers its ultrasonic energy to the tissue. The waveguide shaft assembly **6490** also includes a jaw member **6495** that is coupled to the outer tube **6494** and an inner tube (not visible in this view). The jaw member **6495**, together with the inner and outer tubes and the ultrasonic blade **6496** portion of the waveguide **6499**, can be referred to as an end effector **6492**. As will be explained below, the outer tube **6494** and the non-illustrated inner tube slide longitudinally with respect to each other. As the relative movement between the outer tube **6494** and the non-illustrated inner tube occurs, the jaw member **6495** pivots upon a pivot point, thereby causing the jaw member **6495** to open and close. When closed, the jaw member **6495** imparts a pinching force on tissue located between the jaw member **6495** and the ultrasonic blade **6496**, insuring positive and efficient blade-to-tissue contact.

(541) FIG. **49** is a system diagram **7400** of a segmented circuit **7401** comprising a plurality of independently operated circuit segments **7402**, **7414**, **7416**, **7420**, **7424**, **7428**, **7434**, **7440**, in accordance with at least one aspect of the present disclosure. A circuit segment of the plurality of circuit segments of the segmented circuit **7401** comprises one or more circuits and one or more sets of machine executable instructions stored in one or more memory devices. The one or more circuits of a circuit segment are coupled to for electrical communication through one or more wired or wireless connection media. The plurality of circuit segments are configured to transition between three modes comprising a sleep mode, a standby mode and an operational mode.

(542) In one aspect shown, the plurality of circuit segments **7402**, **7414**, **7416**, **7420**, **7424**, **7428**, **7434**, **7440** start first in the standby mode, transition second to the sleep mode, and transition third to the operational mode. However, in other aspects, the plurality of circuit segments may transition from any one of the three modes to any other one of the three modes. For example, the plurality of circuit segments may transition directly from the standby mode to the operational mode. Individual circuit segments may be placed in a particular state by the voltage control circuit **7408** based on the execution by a processor of machine executable instructions. The states comprise a deenergized state, a low energy state, and an energized state. The deenergized state corresponds to the sleep mode, the low energy state corresponds to the standby mode, and the energized state corresponds to the operational mode. Transition to the low energy state may be achieved by, for example, the use of a potentiometer.

(543) In one aspect, the plurality of circuit segments **7402**, **7414**, **7416**, **7420**, **7424**, **7428**, **7434**, **7440** may transition from the sleep mode or the standby mode to the operational mode in accordance with an energization sequence. The plurality of circuit segments also may transition from the operational mode to the standby mode or the sleep mode in accordance with a deenergization sequence. The energization sequence and the deenergization sequence may be different. In some aspects, the energization sequence comprises energizing only a subset of circuit segments of the plurality of circuit segments. In some aspects, the deenergization sequence comprises deenergizing only a subset of circuit segments of the plurality of circuit segments.

(544) Referring back to the system diagram **7400** in FIG. **49**, the segmented circuit **7401** comprise a plurality of circuit segments comprising a transition circuit segment **7402**, a processor circuit segment **7414**, a handle circuit segment **7416**, a communication circuit segment **7420**, a display circuit segment **7424**, a motor control circuit segment **7428**, an energy treatment circuit segment **7434**, and a shaft circuit segment **7440**. The transition circuit segment comprises a wake up circuit **7404**, a boost current circuit **7406**, a voltage control circuit **7408**, a safety controller **7410** and a POST controller **7412**. The transition circuit segment **7402** is configured to implement a deenergization and an energization sequence, a safety detection protocol, and a POST.

(545) In some aspects, the wake up circuit **7404** comprises an accelerometer button sensor **7405**. In aspects, the transition circuit segment **7402** is configured to be in an energized state while other circuit segments of the plurality of circuit segments of the segmented circuit **7401** are configured to

be in a low energy state or a deenergized state, the accelerometer button sensor **7405** may monitor movement or acceleration of the surgical instrument **6480** described herein. For example, the movement may be a change in orientation or rotation of the surgical instrument. The surgical instrument may be moved in any direction relative to a three dimensional Euclidean space by for example, a user of the surgical instrument. When the accelerometer button sensor **7405** senses movement or acceleration, the accelerometer button sensor **7405** sends a signal to the voltage control circuit **7408** to cause the voltage control circuit **7408** to apply voltage to the processor circuit segment **7414** to transition the processor and a volatile memory to an energized state. In aspects, the processor and the volatile memory are in an energized state before the voltage control circuit **7409** applies voltage to the processor and the volatile memory. In the operational mode, the processor may initiate an energization sequence or a deenergization sequence. In various aspects, the accelerometer button sensor **7405** may also send a signal to the processor to cause the processor to initiate an energization sequence or a deenergization sequence. In some aspects, the processor initiates an energization sequence when the majority of individual circuit segments are in a low energy state or a deenergized state. In other aspects, the processor initiates a deenergization sequence when the majority of individual circuit segments are in an energized state.

(546) Additionally or alternatively, the accelerometer button sensor **7405** may sense external movement within a predetermined vicinity of the surgical instrument. For example, the accelerometer button sensor **7405** may sense a user of the surgical instrument **6480** described herein moving a hand of the user within the predetermined vicinity. When the accelerometer button sensor **7405** senses this external movement, the accelerometer button sensor **7405** may send a signal to the voltage control circuit **7408** and a signal to the processor, as previously described. After receiving the sent signal, the processor may initiate an energization sequence or a deenergization sequence to transition one or more circuit segments between the three modes. In aspects, the signal sent to the voltage control circuit **7408** is sent to verify that the processor is in operational mode. In some aspects, the accelerometer button sensor **7405** may sense when the surgical instrument has been dropped and send a signal to the processor based on the sensed drop. For example, the signal can indicate an error in the operation of an individual circuit segment. One or more sensors may sense damage or malfunctioning of the affected individual circuit segments. Based on the sensed damage or malfunctioning, the POST controller **7412** may perform a POST of the corresponding individual circuit segments.

(547) An energization sequence or a deenergization sequence may be defined based on the accelerometer button sensor **7405**. For example, the accelerometer button sensor **7405** may sense a particular motion or a sequence of motions that indicates the selection of a particular circuit segment of the plurality of circuit segments. Based on the sensed motion or series of sensed motions, the accelerometer button sensor **7405** may transmit a signal comprising an indication of one or more circuit segments of the plurality of circuit segments to the processor when the processor is in an energized state. Based on the signal, the processor determines an energization sequence comprising the selected one or more circuit segments.

Additionally or alternatively, a user of the surgical instruments **6480** described herein may select a number and order of circuit segments to define an energization sequence or a deenergization sequence based on interaction with a graphical user interface (GUI) of the surgical instrument.

(548) In various aspects, the accelerometer button sensor **7405** may send a signal to the voltage control circuit **7408** and a signal to the processor only when the accelerometer button sensor **7405** detects movement of the surgical instrument **6480** described herein or external movement within a predetermined vicinity above a predetermined threshold. For example, a signal may only be sent if movement is sensed for 5 or more seconds or if the surgical instrument is moved 5 or more inches. In other aspects, the accelerometer button sensor **7405** may send a signal to the voltage control circuit **7408** and a signal to the processor only when the accelerometer button sensor **7405** detects oscillating movement of the surgical instrument. A predetermined threshold reduces inadvertent transition of circuit segments of the surgical instrument. As previously described, the transition may comprise a transition to operational mode according to an energization sequence, a transition to low energy mode according to a deenergization sequence, or a transition to sleep mode according to a deenergization sequence. In some aspects, the surgical instrument comprises an actuator that may be actuated by a user of the surgical instrument. The actuation is sensed by the accelerometer button sensor **7405**. The actuator may be a slider, a toggle switch, or a momentary contact switch. Based on the sensed actuation, the accelerometer button sensor **7405** may send a signal to the voltage control circuit **7408** and a signal to the processor.

(549) The boost current circuit **7406** is coupled to a battery. The boost current circuit **7406** is a current amplifier, such as a relay or transistor, and is configured to amplify the magnitude of a current of an individual circuit segment. The initial magnitude of the current corresponds to the source voltage provided by the battery to the segmented circuit **7401**. Suitable relays include solenoids. Suitable transistors include field-effect transistors (FET), MOSFET, and bipolar junction transistors (BJT). The boost current circuit **7406** may amplify the magnitude of the current corresponding to an individual circuit segment or circuit which requires more current draw during operation of the surgical instruments **6480** described herein. For example, an increase in current to the motor control circuit segment **7428** may be provided when a motor of the surgical instrument requires more input power. The increase in current provided to an individual circuit segment may cause a corresponding decrease in current of another circuit segment or circuit segments. Additionally or alternatively, the increase in current may correspond to voltage provided by an additional voltage source operating in conjunction with the battery.

(550) The voltage control circuit **7408** is coupled to the battery. The voltage control circuit **7408** is configured to provide voltage to or remove voltage from the plurality of circuit segments. The voltage control circuit **7408** is also configured to increase or reduce voltage provided to the plurality of circuit segments of the segmented circuit **7401**. In various aspects, the voltage control circuit **7408** comprises a combinational logic circuit such as a multiplexer (MUX) to select inputs, a plurality of electronic switches, and a plurality of voltage converters. An electronic switch of the plurality of electronic switches may be configured to switch between an open and closed configuration to disconnect or connect an individual circuit segment to or from the battery. The plurality of electronic switches may be solid state devices such as transistors or other types of switches such as wireless switches, ultrasonic switches, accelerometers, inertial sensors, among others. The combinational logic circuit is configured to select an individual electronic switch for switching to an open configuration to enable application of voltage to the corresponding circuit segment. The combination logic circuit also is configured to select an individual electronic switch for switching to a closed configuration to enable removal of voltage from the corresponding circuit segment. By selecting a plurality of individual electronic switches, the combination logic circuit may implement a deenergization sequence or an energization sequence. The plurality of voltage converters may provide a stepped-up voltage or a stepped-down voltage to the plurality of circuit segments. The voltage control circuit **7408** may also comprise a microprocessor and memory device.

(551) The safety controller **7410** is configured to perform safety checks for the circuit segments. In some aspects, the safety controller **7410** performs the safety checks when one or more individual circuit segments are in the operational mode. The safety checks may be performed to determine whether there are any errors or defects in the functioning or operation of the circuit segments. The safety controller **7410** may monitor one or more parameters of the plurality of circuit segments. The safety controller **7410** may verify the identity and operation of the plurality of circuit segments by comparing the one or more parameters with predefined parameters. For example, if an RF energy modality is selected, the safety controller **7410** may verify that an articulation parameter of the shaft matches a predefined articulation parameter to verify the operation of the RF energy modality of the surgical instrument **6480** described herein. In some aspects, the safety controller **7410** may monitor, by the sensors, a predetermined relationship between one or more properties of the surgical instrument to detect a fault. A fault may arise when the one or more properties are inconsistent with the predetermined relationship. When the safety controller **7410** determines that a fault exists, an error exists, or that some operation of the plurality of circuit segments was not verified, the safety controller **7410** prevents or disables operation of the particular circuit segment where the fault, error or verification failure originated.

(552) The POST controller **7412** performs a POST to verify proper operation of the plurality of circuit segments. In some aspects, the POST is performed for an individual circuit segment of the plurality of circuit segments prior to the voltage control circuit **7408** applying a voltage to the individual circuit segment to transition the individual circuit segment from standby mode or sleep mode to operational mode. If the individual circuit segment does not pass the POST, the particular circuit segment does not transition from standby mode or sleep mode to operational mode. POST of

the handle circuit segment **7415** may comprise, for example, testing whether the handle control sensors **7418** sense an actuation of a handle control of the surgical instrument **6480** described herein. In some aspects, the POST controller **7412** may transmit a signal to the accelerometer button sensor **7405** to verify the operation of the individual circuit segment as part of the POST. For example, after receiving the signal, the accelerometer button sensor **7405** may prompt a user of the surgical instrument to move the surgical instrument to a plurality of varying locations to confirm operation of the surgical instrument. The accelerometer button sensor **7405** may also monitor an output of a circuit segment or a circuit of a circuit segment as part of the POST. For example, the accelerometer button sensor **7405** can sense an incremental motor pulse generated by the motor **7432** to verify operation. A motor controller of the motor control circuit **7430** may be used to control the motor **7432** to generate the incremental motor pulse.

(553) In various aspects, the surgical instrument **6480** described herein may comprise additional accelerometer button sensors. The POST controller **7412** may also execute a control program stored in the memory device of the voltage control circuit **7408**. The control program may cause the POST controller **7412** to transmit a signal requesting a matching encrypted parameter from a plurality of circuit segments. Failure to receive a matching encrypted parameter from an individual circuit segment indicates to the POST controller **7412** that the corresponding circuit segment is damaged or malfunctioning. In some aspects, if the POST controller **7412** determines based on the POST that the processor is damaged or malfunctioning, the POST controller **7412** may send a signal to one or more secondary processors to cause one or more secondary processors to perform critical functions that the processor is unable to perform. In some aspects, if the POST controller **7412** determines based on the POST that one or more circuit segments do not operate properly, the POST controller **7412** may initiate a reduced performance mode of those circuit segments operating properly while locking out those circuit segments that fail POST or do not operate properly. A locked out circuit segment may function similarly to a circuit segment in standby mode or sleep mode.

(554) The processor circuit segment **7414** comprises the processor and the volatile memory. The processor is configured to initiate an energization or a deenergization sequence. To initiate the energization sequence, the processor transmits an energizing signal to the voltage control circuit **7408** to cause the voltage control circuit **7408** to apply voltage to the plurality or a subset of the plurality of circuit segments in accordance with the energization sequence. To initiate the deenergization sequence, the processor transmits a deenergizing signal to the voltage control circuit **7408** to cause the voltage control circuit **7408** to remove voltage from the plurality or a subset of the plurality of circuit segments in accordance with the deenergization sequence.

(555) The handle circuit segment **7416** comprises handle control sensors **7418**. The handle control sensors **7418** may sense an actuation of one or more handle controls of the surgical instrument **6480** described herein. In various aspects, the one or more handle controls comprise a clamp control, a release button, an articulation switch, an energy activation button, and/or any other suitable handle control. The user may activate the energy activation button to select between an RF energy mode, an ultrasonic energy mode or a combination RF and ultrasonic energy mode. The handle control sensors **7418** may also facilitate attaching a modular handle to the surgical instrument. For example, the handle control sensors **7418** may sense proper attachment of the modular handle to the surgical instrument and indicate the sensed attachment to a user of the surgical instrument. The LCD display **7426** may provide a graphical indication of the sensed attachment. In some aspects, the handle control sensors **7418** senses actuation of the one or more handle controls. Based on the sensed actuation, the processor may initiate either an energization sequence or a deenergization sequence.

(556) The communication circuit segment **7420** comprises a communication circuit **7422**. The communication circuit **7422** comprises a communication interface to facilitate signal communication between the individual circuit segments of the plurality of circuit segments. In some aspects, the communication circuit **7422** provides a path for the modular components of the surgical instrument **6480** described herein to communicate electrically. For example, a modular shaft and a modular transducer, when attached together to the handle of the surgical instrument, can upload control programs to the handle through the communication circuit **7422**.

(557) The display circuit segment **7424** comprises a LCD display **7426**. The LCD display **7426** may comprise a liquid crystal display screen, LED indicators, etc. In some aspects, the LCD display **7426** is an organic light-emitting diode (OLED) screen. A display may be placed on, embedded in, or located remotely from the surgical instrument **6480** described herein. For example, the display can be placed on the handle of the surgical instrument. The display is configured to provide sensory feedback to a user. In various aspects, the LCD display **7426** further comprises a backlight. In some aspects, the surgical instrument may also comprise audio feedback devices such as a speaker or a buzzer and tactile feedback devices such as a haptic actuator.

(558) The motor control circuit segment **7428** comprises a motor control circuit **7430** coupled to a motor **7432**. The motor **7432** is coupled to the processor by a driver and a transistor, such as a FET. In various aspects, the motor control circuit **7430** comprises a motor current sensor in signal communication with the processor to provide a signal indicative of a measurement of the current draw of the motor to the processor. The processor transmits the signal to the display. The display receives the signal and displays the measurement of the current draw of the motor **7432**. The processor may use the signal, for example, to monitor that the current draw of the motor **7432** exists within an acceptable range, to compare the current draw to one or more parameters of the plurality of circuit segments, and to determine one or more parameters of a patient treatment site. In various aspects, the motor control circuit **7430** comprises a motor controller to control the operation of the motor. For example, the motor control circuit **7430** controls various motor parameters, such as by adjusting the velocity, torque and acceleration of the motor **7432**. The adjusting is done based on the current through the motor **7432** measured by the motor current sensor.

(559) In various aspects, the motor control circuit **7430** comprises a force sensor to measure the force and torque generated by the motor **7432**. The motor **7432** is configured to actuate a mechanism of the surgical instruments **6480** described herein. For example, the motor **7432** is configured to control actuation of the shaft of the surgical instrument to realize clamping, rotation and articulation functionality. For example, the motor **7432** may actuate the shaft to realize a clamping motion with jaws of the surgical instrument. The motor controller may determine whether the material clamped by the jaws is tissue or metal. The motor controller may also determine the extent to which the jaws clamp the material. For example, the motor controller may determine how open or closed the jaws are based on the derivative of sensed motor current or motor voltage. In some aspects, the motor **7432** is configured to actuate the transducer to cause the transducer to apply torque to the handle or to control articulation of the surgical instrument. The motor current sensor may interact with the motor controller to set a motor current limit. When the current meets the predefined threshold limit, the motor controller initiates a corresponding change in a motor control operation. For example, exceeding the motor current limit causes the motor controller to reduce the current draw of the motor.

(560) The energy treatment circuit segment **7434** comprises a RF amplifier and safety circuit **7436** and an ultrasonic signal generator circuit **7438** to implement the energy modular functionality of the surgical instrument **6480** described herein. In various aspects, the RF amplifier and safety circuit **7436** is configured to control the RF modality of the surgical instrument by generating an RF signal. The ultrasonic signal generator circuit **7438** is configured to control the ultrasonic energy modality by generating an ultrasonic signal. The RF amplifier and safety circuit **7436** and an ultrasonic signal generator circuit **7438** may operate in conjunction to control the combination RF and ultrasonic energy modality.

(561) The shaft circuit segment **7440** comprises a shaft module controller **7442**, a modular control actuator **7444**, one or more end effector sensors **7446**, and a non volatile memory **7448**. The shaft module controller **7442** is configured to control a plurality of shaft modules comprising the control programs to be executed by the processor. The plurality of shaft modules implements a shaft modality, such as ultrasonic, combination ultrasonic and RF, RF I-blade, and RF-opposable jaw. The shaft module controller **7442** can select shaft modality by selecting the corresponding shaft module for the processor to execute. The modular control actuator **7444** is configured to actuate the shaft according to the selected shaft modality. After actuation is initiated, the shaft articulates the end effector according to the one or more parameters, routines or programs specific to the selected shaft modality and the selected end effector modality. The one or more end effector sensors **7446** located at the end effector may include force sensors, temperature sensors, current sensors or motion sensors. The one or more end effector sensors **7446** transmit data about one or more operations of the end effector,

based on the energy modality implemented by the end effector, the energy modalities include an ultrasonic energy modality, a RF energy modality, or a combination of the ultrasonic energy modality and the RF energy modality. The non volatile memory **7448** stores the shaft control programs. A control program comprises one or more parameters, routines or programs specific to the shaft. In various aspects, the non volatile memory **7448** may be an ROM, EPROM, EEPROM or flash memory. The non volatile memory **7448** stores the shaft modules corresponding to the selected shaft of the surgical instrument **6480** described herein in. The shaft modules may be changed or upgraded in the non volatile memory **7448** by the shaft module controller **7442**, depending on the surgical instrument shaft to be used in operation.

(562) FIG. **50** is a schematic diagram of a circuit **7925** of various components of a surgical instrument with motor control functions, in accordance with at least one aspect of the present disclosure. In various aspects, the surgical instrument **6480** described herein may include a drive mechanism **7930** which is configured to drive shafts and/or gear components in order to perform the various operations associated with the surgical instrument **6480**. In one aspect, the drive mechanism **7930** includes a rotation drivetrain **7932** configured to rotate an end effector, for example, about a longitudinal axis relative to handle housing. The drive mechanism **7930** further includes a closure drivetrain **7934** configured to close a jaw member to grasp tissue with the end effector. In addition, the drive mechanism **7930** includes a firing drive train **7936** configured to open and close a clamp arm portion of the end effector to grasp tissue with the end effector.

(563) The drive mechanism **7930** includes a selector gearbox assembly **7938** that can be located in the handle assembly of the surgical instrument. Proximal to the selector gearbox assembly **7938** is a function selection module which includes a first motor **7942** that functions to selectively move gear elements within the selector gearbox assembly **7938** to selectively position one of the drivetrains **7932**, **7934**, **7936** into engagement with an input drive component of an optional second motor **7944** and motor drive circuit **7946** (shown in dashed line to indicate that the second motor **7944** and motor drive circuit **7946** are optional components).

(564) Still referring to FIG. **50**, the motors **7942**, **7944** are coupled to motor control circuits **7946**, **7948**, respectively, which are configured to control the operation of the motors **7942**, **7944** including the flow of electrical energy from a power source **7950** to the motors **7942**, **7944**. The power source **7950** may be a DC battery (e.g., rechargeable lead-based, nickel-based, lithium-ion based, battery etc.) or any other power source suitable for providing electrical energy to the surgical instrument.

(565) The surgical instrument further includes a microcontroller **7952** ("controller"). In certain instances, the controller **7952** may include a microprocessor **7954** ("processor") and one or more computer readable mediums or memory units **7956** ("memory"). In certain instances, the memory **7956** may store various program instructions, which when executed may cause the processor **7954** to perform a plurality of functions and/or calculations described herein. The power source **7950** can be configured to supply power to the controller **7952**, for example.

(566) The processor **7954** may be in communication with the motor control circuit **7946**. In addition, the memory **7956** may store program instructions, which when executed by the processor **7954** in response to a user input **7958** or feedback elements **7960**, may cause the motor control circuit **7946** to motivate the motor **7942** to generate at least one rotational motion to selectively move gear elements within the selector gearbox assembly **7938** to selectively position one of the drivetrains **7932**, **7934**, **7936** into engagement with the input drive component of the second motor **7944**. Furthermore, the processor **7954** can be in communication with the motor control circuit **7948**. The memory **7956** also may store program instructions, which when executed by the processor **7954** in response to a user input **7958**, may cause the motor control circuit **7948** to motivate the motor **7944** to generate at least one rotational motion to drive the drivetrain engaged with the input drive component of the second motor **7948**, for example.

(567) The controller **7952** and/or other controllers of the present disclosure may be implemented using integrated and/or discrete hardware elements, software elements, and/or a combination of both. Examples of integrated hardware elements may include processors, microprocessors, microcontrollers, integrated circuits, ASICs, PLDs, DSPs, FPGAs, logic gates, registers, semiconductor devices, chips, microchips, chip sets, microcontrollers, system on a chip (SoC), and/or single in-line package (SIP). Examples of discrete hardware elements may include circuits and/or circuit elements such as logic gates, field effect transistors, bipolar transistors, resistors, capacitors, inductors, and/or relays. In certain instances, the controller **7952** may include a hybrid circuit comprising discrete and integrated circuit elements or components on one or more substrates, for example.

(568) In certain instances, the controller **7952** and/or other controllers of the present disclosure may be an LM 4F230H5QR, available from Texas Instruments, for example. In certain instances, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle SRAM, internal ROM loaded with StellarisWare® software, 2 KB EEPROM, one or more PWM modules, one or more QEI analog, one or more 12-bit ADC with 12 analog input channels, among other features that are readily available. Other microcontrollers may be readily substituted for use with the present disclosure. Accordingly, the present disclosure should not be limited in this context.

(569) In various instances, one or more of the various steps described herein can be performed by a finite state machine comprising either a combinational logic circuit or a sequential logic circuit, where either the combinational logic circuit or the sequential logic circuit is coupled to at least one memory circuit. The at least one memory circuit stores a current state of the finite state machine. The combinational or sequential logic circuit is configured to cause the finite state machine to the steps. The sequential logic circuit may be synchronous or asynchronous. In other instances, one or more of the various steps described herein can be performed by a circuit that includes a combination of the processor **7958** and the finite state machine, for example.

(570) In various instances, it can be advantageous to be able to assess the state of the functionality of a surgical instrument to ensure its proper function. It is possible, for example, for the drive mechanism, as explained above, which is configured to include various motors, drivetrains, and/or gear components in order to perform the various operations of the surgical instrument, to wear out over time. This can occur through normal use, and in some instances the drive mechanism can wear out faster due to abuse conditions. In certain instances, a surgical instrument can be configured to perform self-assessments to determine the state, e.g. health, of the drive mechanism and its various components.

(571) For example, the self-assessment can be used to determine when the surgical instrument is capable of performing its function before a re-sterilization or when some of the components should be replaced and/or repaired. Assessment of the drive mechanism and its components, including but not limited to the rotation drivetrain **7932**, the closure drivetrain **7934**, and/or the firing drivetrain **7936**, can be accomplished in a variety of ways. The magnitude of deviation from a predicted performance can be used to determine the likelihood of a sensed failure and the severity of such failure. Several metrics can be used including: Periodic analysis of repeatably predictable events, Peaks or drops that exceed an expected threshold, and width of the failure.

(572) In various instances, a signature waveform of a properly functioning drive mechanism or one or more of its components can be employed to assess the state of the drive mechanism or the one or more of its components. One or more vibration sensors can be arranged with respect to a properly functioning drive mechanism or one or more of its components to record various vibrations that occur during operation of the properly functioning drive mechanism or the one or more of its components. The recorded vibrations can be employed to create the signature waveform. Future waveforms can be compared against the signature waveform to assess the state of the drive mechanism and its components.

(573) Still referring to FIG. **50**, the surgical instrument **7930** includes a drivetrain failure detection module **7962** configured to record and analyze one or more acoustic outputs of one or more of the drivetrains **7932**, **7934**, **7936**. The processor **7954** can be in communication with or otherwise control the module **7962**. As described below in greater detail, the module **7962** can be embodied as various means, such as circuitry, hardware, a computer program product comprising a computer readable medium (for example, the memory **7956**) storing computer readable program instructions that are executable by a processing device (for example, the processor **7954**), or some combination thereof. In some aspects, the processor **36** can include, or otherwise control the module **7962**.

(574) Turning now to FIG. 51, the second effector **8400** comprises RF data sensors **8406**, **8408a**, **8408b** located on the jaw member **8402**. The end effector **8400** comprises a jaw member **8402** and an ultrasonic blade **8404**. The jaw member **8402** is shown clamping tissue **8410** located between the jaw member **8402** and the ultrasonic blade **8404**. A first sensor **8406** is located in a center portion of the jaw member **8402**. Second and third sensors **8408a**, **8408b** are located on lateral portions of the jaw member **8402**. The sensors **8406**, **8408a**, **8408b** are mounted or formed integrally with a flexible circuit **8412** (shown more particularly in FIG. 52) configured to be fixedly mounted to the jaw member **8402**.

(575) The end effector **8400** is an example end effector for a surgical instrument. The sensors **8406**, **8408a**, **8408b** are electrically connected to a control circuit such as the control circuit **7400** (FIG. 63) via interface circuits. The sensors **8406**, **8408a**, **8408b** are battery powered and the signals generated by the sensors **8406**, **8408a**, **8408b** are provided to analog and/or digital processing circuits of the control circuit.

(576) In one aspect, the first sensor **8406** is a force sensor to measure a normal force **F3** applied to the tissue **8410** by the jaw member **8402**. The second and third sensors **8408a**, **8408b** include one or more elements to apply RF energy to the tissue **8410**, measure tissue impedance, down force **F1**, transverse forces **F2**, and temperature, among other parameters. Electrodes **8409a**, **8409b** are electrically coupled to an energy source and apply RF energy to the tissue **8410**. In one aspect, the first sensor **8406** and the second and third sensors **8408a**, **8408b** are strain gauges to measure force or force per unit area. It will be appreciated that the measurements of the down force **F1**, the lateral forces **F2**, and the normal force **F3** may be readily converted to pressure by determining the surface area upon which the force sensors **8406**, **8408a**, **8408b** are acting upon. Additionally, as described with particularity herein, the flexible circuit **8412** may comprise temperature sensors embedded in one or more layers of the flexible circuit **8412**. The one or more temperature sensors may be arranged symmetrically or asymmetrically and provide tissue **8410** temperature feedback to control circuits of an ultrasonic drive circuit and an RF drive circuit.

(577) FIG. 52 illustrates one aspect of the flexible circuit **8412** shown in FIG. 51 in which the sensors **8406**, **8408a**, **8408b** may be mounted to or formed integrally therewith. The flexible circuit **8412** is configured to fixedly attach to the jaw member **8402**. As shown particularly in FIG. 52, asymmetric temperature sensors **8414a**, **8414b** are mounted to the flexible circuit **8412** to enable measuring the temperature of the tissue **8410** (FIG. 51).

(578) FIG. 53 is an alternative system **132000** for controlling the frequency of an ultrasonic electromechanical system **132002** and detecting the impedance thereof, in accordance with at least one aspect of the present disclosure. The system **132000** may be incorporated into a generator. A processor **132004** coupled to a memory **132026** programs a programmable counter **132006** to tune to the output frequency $f_{sub.o}$ of the ultrasonic electromechanical system **132002**. The input frequency is generated by a crystal oscillator **132008** and is input into a fixed counter **132010** to scale the frequency to a suitable value. The outputs of the fixed counter **132010** and the programmable counter **132006** are applied to a phase/frequency detector **132012**. The output of the phase/frequency detector **132012** is applied to an amplifier/active filter circuit **132014** to generate a tuning voltage $V_{sub.t}$ that is applied to a voltage controlled oscillator **132016** (VCO). The VCO **132016** applies the output frequency $f_{sub.o}$ to an ultrasonic transducer portion of the ultrasonic electromechanical system **132002**, shown here modeled as an equivalent electrical circuit. The voltage and current signals applied to the ultrasonic transducer are monitored by a voltage sensor **132018** and a current sensor **132020**.

(579) The outputs of the voltage and current sensors **132018**, **132020** are applied to another phase/frequency detector **132022** to determine the phase angle between the voltage and current as measured by the voltage and current sensors **132018**, **132020**. The output of the phase/frequency detector **132022** is applied to one channel of a high speed analog to digital converter **132024** (ADC) and is provided to the processor **132004** therethrough. Optionally, the outputs of the voltage and current sensors **132018**, **132020** may be applied to respective channels of the two-channel ADC **132024** and provided to the processor **132004** for zero crossing, FFT, or other algorithm described herein for determining the phase angle between the voltage and current signals applied to the ultrasonic electromechanical system **132002**.

(580) Optionally the tuning voltage $V_{sub.t}$, which is proportional to the output frequency $f_{sub.o}$, may be fed back to the processor **132004** via the ADC **132024**. This provides the processor **132004** with a feedback signal proportional to the output frequency $f_{sub.o}$ and can use this feedback to adjust and control the output frequency $f_{sub.o}$.

Estimating the State of the Jaw

Pad Burn Through, Staples, Broken Blade, Bone in Jaw, Tissue in Jaw

(581) A challenge with ultrasonic energy delivery is that ultrasonic acoustics applied on the wrong materials or the wrong tissue can result in device failure, for example, clamp arm pad burn through or ultrasonic blade breakage. It is also desirable to detect what is located in the jaws of an end effector of an ultrasonic device and the state of the jaws without adding additional sensors in the jaws. Locating sensors in the jaws of an ultrasonic end effector poses reliability, cost, and complexity challenges.

(582) Ultrasonic spectroscopy smart blade algorithm techniques may be employed for estimating the state of the jaw (clamp arm pad burn through, staples, broken blade, bone in jaw, tissue in jaw, back-cutting with jaw closed, etc.) based on the impedance

$$(583) Z_g(t) = \frac{V_g(t)}{I_g(t)}$$

of an ultrasonic transducer configured to drive an ultrasonic transducer blade, in accordance with at least one aspect of the present disclosure. The impedance $Z_{sub.g}(t)$, magnitude $|Z|$, and phase ϕ are plotted as a function of frequency f .

(584) Dynamic mechanical analysis (DMA), also known as dynamic mechanical spectroscopy or simply mechanical spectroscopy, is a technique used to study and characterize materials. A sinusoidal stress is applied to a material, and the strain in the material is measured, allowing the determination of the complex modulus of the material. The spectroscopy as applied to ultrasonic devices includes exciting the tip of the ultrasonic blade with a sweep of frequencies (compound signals or traditional frequency sweeps) and measuring the resulting complex impedance at each frequency. The complex impedance measurements of the ultrasonic transducer across a range of frequencies are used in a classifier or model to infer the characteristics of the ultrasonic end effector. In one aspect, the present disclosure provides a technique for determining the state of an ultrasonic end effector (clamp arm, jaw) to drive automation in the ultrasonic device (such as disabling power to protect the device, executing adaptive algorithms, retrieving information, identifying tissue, etc.).

(585) FIG. 54 is a spectra **132030** of an ultrasonic device with a variety of different states and conditions of the end effector where the impedance $Z_{sub.g}(t)$, magnitude $|Z|$, and phase ϕ are plotted as a function of frequency f , in accordance with at least one aspect of the present disclosure. The spectra **132030** is plotted in three-dimensional space where frequency (Hz) is plotted along the x-axis, phase (Rad) is plotted along the y-axis, and magnitude (Ohms) is plotted along the z-axis.

(586) Spectral analysis of different jaw bites and device states produces different complex impedance characteristic patterns (fingerprints) across a range of frequencies for different conditions and states. Each state or condition has a different characteristic pattern in 3D space when plotted. These characteristic patterns can be used to estimate the condition and state of the end effector. FIG. 54 shows the spectra for air **132032**, clamp arm pad **132034**, chamois **132036**, staple **132038**, and broken blade **132040**. The chamois **132036** may be used to characterize different types of tissue.

(587) The spectra **132030** can be evaluated by applying a low-power electrical signal across the ultrasonic transducer to produce a non-therapeutic excitation of the ultrasonic blade. The low-power electrical signal can be applied in the form of a sweep or a compound Fourier series to measure the impedance

$$(588) Z_g(t) = \frac{V_g(t)}{I_g(t)}$$

across the ultrasonic transducer at a range of frequencies in series (sweep) or in parallel (compound signal) using an FFT.

Methods of Classification of New Data

(589) For each characteristic pattern, a parametric line can be fit to the data used for training using a polynomial, a Fourier series, or any other form of parametric equation as may be dictated by convenience. A new data point is then received and is classified by using the Euclidean perpendicular

distance from the new data point to the trajectory that has been fitted to the characteristic pattern training data. The perpendicular distance of the new data point to each of the trajectories (each trajectory representing a different state or condition) is used to assign the point to a state or condition. (590) The probability distribution of distance of each point in the training data to the fitted curve can be used to estimate the probability of a correctly classified new data point. This essentially constructs a two-dimensional probability distribution in a plane perpendicular to the fitted trajectory at each new data point of the fitted trajectory. The new data point can then be included in the training set based on its probability of correct classification to make an adaptive, learning classifier that readily detects high-frequency changes in states but adapts to slow occurring deviations in system performance, such as a device getting dirty or the pad wearing out.

(591) FIG. 55 is a graphical representation of a plot 132042 of a set of 3D training data set (S), where ultrasonic transducer impedance $Z_{\text{sub}}(t)$, magnitude $|Z|$, and phase ϕ are plotted as a function of frequency f , in accordance with at least one aspect of the present disclosure. The 3D training data set (S) plot 132042 is graphically depicted in three-dimensional space where phase (Rad) is plotted along the x-axis, frequency (Hz) is plotted along the y-axis, magnitude (Ohms) is plotted along the z-axis, and a parametric Fourier series is fit to the 3D training data set (S). A methodology for classifying data is based on the 3D training data set (S) is used to generate the plot 132042).

(592) The parametric Fourier series fit to the 3D training data set (S) is given by:

$$(593) \text{ } f_{\text{darw}} = f_{\text{darw}_0} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n}{T} t + b_n \sin \frac{n}{T} t \right)$$

(594) For a new point custom character, the perpendicular distance from custom character to custom character is found by:

$$D = \| \text{custom character} - \text{custom character} \|$$

When:

$$(595) \frac{\partial D}{\partial t} = 0$$

Then:

$$D = D_{\text{sub}} \cdot L$$

(596) A probability distribution of D can be used to estimate the probability of a data point custom character belonging to the group S.

Control

(597) Based on the classification of data measured before, during, or after activation of the ultrasonic transducer/ultrasonic blade, a variety of automated tasks and safety measures can be implemented. Similarly, the state of the tissue located in the end effector and temperature of the ultrasonic blade also can be inferred to some degree, and used to better inform the user of the state of the ultrasonic device or protect critical structures, etc. Temperature control of an ultrasonic blade is described in commonly owned U.S. Provisional Patent Application No. 62/640,417, filed Mar. 8, 2018, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR, which is incorporated herein by reference in its entirety.

(598) Similarly, power delivery can be reduced when there is a high probability that the ultrasonic blade is contacting the clamp arm pad (e.g., without tissue in between) or if there is a probability that the ultrasonic blade has broken or that the ultrasonic blade is touching metal (e.g., a staple). Furthermore, back-cutting can be disallowed if the jaw is closed and no tissue is detected between the ultrasonic blade and the clamp arm pad.

Integration of Other Data to Improve Classification

(599) This system can be used in conjunction with other information provided by sensors, the user, metrics on the patient, environmental factors, etc., by combining the data from this process with the aforementioned data using probability functions and a Kalman filter. The Kalman filter determines the maximum likelihood of a state or condition occurring given a plethora of uncertain measurements of varying confidence. Since this method allows for an assignment of probability to a newly classified data point, this algorithm's information can be implemented with other measures or estimates in a Kalman filter.

(600) FIG. 56 is a logic flow diagram 132044 depicting a control program or a logic configuration to determine jaw conditions based on the complex impedance characteristic pattern (fingerprint), in accordance with at least one aspect of the present disclosure. Prior to determining jaw conditions based on the complex impedance characteristic pattern (fingerprint), a database is populated with reference complex impedance characteristic patterns or a training data sets (S) that characterize various jaw conditions, including, without limitation, air 132032, clamp arm pad 132034, chamois 132036, staple 132038, broken blade 132040, as shown in FIG. 82, and a variety of tissue types and conditions. The chamois dry or wet, full byte or tip, may be used to characterize different types of tissue. The data points used to generate reference complex impedance characteristic patterns or a training data set (S) are obtained by applying a sub-therapeutic drive signal to the ultrasonic transducer, sweeping the driving frequency over a predetermined range of frequencies from below resonance to above resonance, measuring the complex impedance at each of the frequencies, and recording the data points. The data points are then fit to a curve using a variety of numerical methods including polynomial curve fit, Fourier series, and/or parametric equation. A parametric Fourier series fit to the reference complex impedance characteristic patterns or a training data set (S) is described herein.

(601) Once the reference complex impedance characteristic patterns or a training data sets (S) are generated, the ultrasonic instrument measures new data points, classifies the new points, and determines whether the new data points should be added to the reference complex impedance characteristic patterns or a training data sets (S).

(602) Turning now to the logic flow diagram of FIG. 56, in one aspect, the processor or control circuit measures 132046 a complex impedance of an ultrasonic transducer, wherein the complex impedance is defined as

$$(603) Z_g(t) = \frac{V_g(t)}{I_g(t)}$$

The processor or control circuit receives 132048 a complex impedance measurement data point and compares 132050 the complex impedance measurement data point to a data point in a reference complex impedance characteristic pattern. The processor or control circuit classifies 132052 the complex impedance measurement data point based on a result of the comparison analysis and assigns 132054 a state or condition of the end effector based on the result of the comparison analysis.

(604) In one aspect, the processor or control circuit receives the reference complex impedance characteristic pattern from a database or memory coupled to the processor. In one aspect, the processor or control circuit generates the reference complex impedance characteristic pattern as follows. A drive circuit coupled to the processor or control circuit applies a nontherapeutic drive signal to the ultrasonic transducer starting at an initial frequency, ending at a final frequency, and at a plurality of frequencies therebetween. The processor or control circuit measures the impedance of the ultrasonic transducer at each frequency and stores a data point corresponding to each impedance measurement. The processor or control circuit curve fits a plurality of data points to generate a three-dimensional curve of representative of the reference complex impedance characteristic pattern, wherein the magnitude $|Z|$ and phase ϕ are plotted as a function of frequency f . The curve fitting includes a polynomial curve fit, a Fourier series, and/or a parametric equation.

(605) In one aspect, the processor or control circuit receives a new impedance measurement data point and classifies the new impedance measurement data point using a Euclidean perpendicular distance from the new impedance measurement data point to a trajectory that has been fitted to the reference complex impedance characteristic pattern. The processor or control circuit estimates a probability that the new impedance measurement data point is correctly classified. The processor or control circuit adds the new impedance measurement data point to the reference complex impedance characteristic pattern based on the probability of the estimated correct classification of the new impedance measurement data point. In one aspect, the processor or control circuit classifies data based on a training data set (S), where the training data set (S) comprises a plurality of complex impedance measurement data, and curve fits the training data set (S) using a parametric Fourier series, wherein S is defined

herein and wherein the probability distribution is used to estimate the probability of the new impedance measurement data point belonging to the group S.

State of Jaw Classifier Based on Model

(606) There has been an existing interest in classifying matter located within the jaws of an ultrasonic device including tissue types and condition. In various aspects, it can be shown that with high data sampling and sophisticated pattern recognition this classification is possible. The approach is based on impedance as a function of frequency, where magnitude, phase, and frequency are plotted in 3D the patterns look like ribbons as shown in FIGS. 54 and 55 and the logic flow diagram of FIG. 56. This disclosure provides an alternative smart blade algorithm approach that is based on a well-established model for piezoelectric transducers.

(607) By way of example, the equivalent electrical lumped parameter model is known to be an accurate model of the physical piezoelectric transducer. It is based on the Mittag-Leffler expansion of a tangent near a mechanical resonance. When the complex impedance or the complex admittance is plotted as an imaginary component versus a real component, circles are formed. FIG. 57 is a circle plot 132056 of complex impedance plotted as an imaginary component versus real components of a piezoelectric vibrator, in accordance with at least one aspect of the present disclosure. FIG. 58 is a circle plot 132058 of complex admittance plotted as an imaginary component versus real components of a piezoelectric vibrator, in accordance with at least one aspect of the present disclosure. The circles depicted in FIGS. 57 and 58 are taken from the IEEE 177 Standard, which is incorporated herein by reference in its entirety. Tables 1-4 are taken from the IEEE 177 Standard and disclosed herein for completeness.

(608) The circle is created as the frequency is swept from below resonance to above resonance. Rather than stretching the circle out in 3D, a circle is identified and the radius (r) and offsets (a, b) of the circle are estimated. These values are then compared with established values for given conditions. These conditions may be: 1) open nothing in jaws, 2) tip bite 3) full bite and staple in jaws. If the sweep generates multiple resonances, circles of different characteristics will be present for each resonance. Each circle will be drawn out before the next if the resonances are separated. Rather than fitting a 3D curve with a series approximation, the data is fitted with a circle. The radius (r) and offsets (a, b) can be calculated using a processor programmed to execute a variety of mathematical or numerical techniques described below. These values may be estimated by capturing an image of a circle and, using image processing techniques, the radius (r) and offsets (a, b) that define the circle are estimated.

(609) FIG. 59 is a circle plot 132060 of complex admittance for a 55.5 kHz ultrasonic piezoelectric transducer for lumped parameters inputs and outputs specified hereinbelow. Values for a lumped parameter model were used to generate the complex admittance. A moderate load was applied in the model. The obtained admittance circle generated in MathCad is shown in FIG. 59. The circle plot 132060 is formed when the frequency is swept from 54 to 58 kHz.

(610) The lumped parameter input values are:

$$C_o = 3.0 \text{ nF}$$

$$C_s = 8.22 \text{ pF}$$

$$L_s = 1.0 \text{ H}$$

$$R_s = 450 \Omega$$

(611) The outputs of the model based on the inputs are:

$$(612) 0am = \frac{D \cdot \text{Math} \cdot C - B \cdot \text{Math} \cdot C}{A \cdot \text{Math} \cdot C - B^2} = 1.013 \cdot \text{Math} \cdot 10^3 \text{ bm} = \frac{A \cdot \text{Math} \cdot E - B \cdot \text{Math} \cdot D}{A \cdot \text{Math} \cdot C - B^2} = -954.585$$

$$rm = \frac{1}{fpts} \cdot \sqrt{\left(\text{Math} \cdot \left((Zout_{1,i} - am)^2 + (Zout_{2,i} - bm)^2 \right) \right)} = 1.012 \cdot \text{Math} \cdot 10^3$$

(613) The output values are used to plot the circle plot 132060 shown in FIG. 59. The circle plot 132060 has a radius (r) and the center 132062 is offset (a, b) from the origin 132064 as follows:

$$r = 1.012 \cdot 10 \cdot \text{sup.3}$$

$$a = 1.013 \cdot 10 \cdot \text{sup.3}$$

$$b = -954.585$$

(614) The summations A-E specified below are needed to estimate the circle plot 132060 plot for the example given in FIG. 59, in accordance with at least one aspect of the present disclosure. Several algorithms exist to calculate a fit to a circle. A circle is defined by its radius (r) and offsets (a, b) of the center from the origin:

$$r \cdot \text{sup.2} = (x - a) \cdot \text{sup.2} + (y - b) \cdot \text{sup.2}$$

(615) The modified least squares method (Umbach and Jones) is convenient in that there a simple close formed solution for a, b, and r.

$$(616) \hat{a} = \frac{DC - BE}{AC - B^2} \hat{b} = \frac{AE - BD}{AC - B^2} \hat{r} = \frac{1}{n} \cdot \text{Math} \cdot \sqrt{\sum_{i=1}^n (x_i - \hat{a})^2 + (y_i - \hat{b})^2}$$

(617) The caret symbol over the variable "a" indicates an estimate of the true value. A, B, C, D, and E are summations of various products which are calculated from the data. They are included herein for completeness as follows:

$$(618) A := fpts \cdot \text{Math} \cdot \sum_{i=1}^{fpts} (Zout_{1,i})^2 - \left(\sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{1,i}) \right) = 5.463 \cdot \text{Math} \cdot 10^{10}$$

$$B := fpts \cdot \text{Math} \cdot \sum_{i=1}^{fpts} (Zout_{1,i} \cdot \text{Math} \cdot Zout_{2,i}) - \left(\sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{1,i}) \right) \cdot \left(\sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{2,i}) \right) = 5.461 \cdot \text{Math} \cdot 10^7$$

$$C := fpts \cdot \text{Math} \cdot \sum_{i=1}^{fpts} (Zout_{2,i})^2 - \left(\sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{2,i}) \right) = 5.445 \cdot \text{Math} \cdot 10^{10}$$

$$D := 0.5 \cdot \text{Math} \cdot \left(\sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{1,i} \cdot \text{Math} \cdot (Zout_{2,i})^2) - \left(\sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{1,i}) \right) \cdot \left(\sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{2,i})^2 \right) + \sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{1,i}^3) - \left(\sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{1,i}) \right) \cdot \left(\sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{1,i}) \right) \right)$$

$$E := 0.5 \cdot \text{Math} \cdot \left(\sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{2,i} \cdot \text{Math} \cdot (Zout_{1,i})^2) - \left(\sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{2,i}) \right) \cdot \left(\sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{1,i})^2 \right) + \sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{2,i}^3) - \left(\sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{2,i}) \right) \cdot \left(\sum_{i=1}^{fpts} \text{Math} \cdot (Zout_{2,i}) \right) \right)$$

Z1,i is a first vector of the real components referred to as conductance; Z2,i is a second of the imaginary components referred to as susceptance; and Z3,i is a third vector that represents the frequencies at which admittances are calculated.

(619) This disclosure will work for ultrasonic systems and may possibly be applied to electrosurgical systems, even though electrosurgical systems do not rely on a resonance.

(620) FIGS. 60-64 illustrate images taken from an impedance analyzer showing impedance/admittance circle plots for an ultrasonic device with the end effector jaw in various open or closed configurations and loading. The circle plots in solid line depict impedance and the circle plots in broken lines depict admittance, in accordance with at least one aspect of the present disclosure. By way of example, the impedance/admittance circle plots are generated by connecting an ultrasonic device to an impedance analyzer. The display of the impedance analyzer is set to complex impedance and complex admittance, which can be selectable from the front panel of the impedance analyzer. An initial display may be obtained with the jaw of the ultrasonic end effector in an open position and the ultrasonic device in an unloaded state, as described below in connection with FIG. 60, for example. The autoscale display function of the impedance analyzer may be used to generate both the complex impedance and admittance circle plots. The same display is used for subsequent runs of the ultrasonic device with different loading conditions as shown in the subsequent FIGS. 60-64. A LabVIEW application may be employed to upload the data files. In another technique, the display images may be captured with a camera, such as a smartphone camera, like an iPhone or Android. As such, the image of the display may include some "keystone-ing" and in general may not appear to be parallel to the screen. Using this technique, the circle plot traces on the display will appear distorted in the captured image. With this approach, the

material located in the jaws of the ultrasonic end effector can be classified.

(621) The complex impedance and complex admittance are just the reciprocal of one another. No new information should be added by looking at both. Another consideration includes determining how sensitive the estimates are to noise when using complex impedance or complex admittance. (622) In the examples illustrated in FIGS. **60-64**, the impedance analyzer is set up with a range to just capture the main resonance. By scanning over a wider range of frequencies more resonances may be encountered and multiple circle plots may be formed. An equivalent circuit of an ultrasonic transducer may be modeled by a first “motional” branch having a serially connected inductance L_s , resistance R_s and capacitance C_s that define the electromechanical properties of the resonator, and a second capacitive branch having a static capacitance C_0 . In the impedance/admittance plots shown in FIGS. **60-64** that follow, the values of the components of the equivalent circuit are:

$$L_s = L_1 = 1.1068 \text{ H}$$

$$R_s = R_1 = 311.352 \Omega$$

$$C_s = C_1 = 7.43265 \text{ pF}$$

$$C_0 = C_0 = 3.64026 \text{ nF}$$

(623) The oscillator voltage applied to the ultrasonic transducer is 500 mV and the frequency is swept from 55 kHz to 56 kHz. The impedance (Z) scale is 200 Ω/div and the admittance (Y) scale is 500 $\mu\text{S}/\text{div}$. Measurements of values that may characterize the impedance (Z) and admittance (Y) circle plots may be obtained at the locations on the circle plots as indicated by an impedance cursor and an admittance cursor.

State of Jaw: Open with No Loading

(624) FIG. **60** is a graphical display **132066** of an impedance analyzer showing complex impedance (Z)/admittance (Y) circle plots **132068**, **132070** for an ultrasonic device with the jaw open and no loading where a circle plot **132068** in solid line depicts complex impedance and a circle plot **132070** in broken line depicts complex admittance, in accordance with at least one aspect of the present disclosure. The oscillator voltage applied to the ultrasonic transducer is 500 mV and the frequency is swept from 55 kHz to 56 kHz. The impedance (Z) scale is 200 Ω/div and the admittance (Y) scale is 500 $\mu\text{S}/\text{div}$. Measurements of values that may characterize the complex impedance (Z) and admittance (Y) circle plots **132068**, **132070** may be obtained at locations on the circle plots **132068**, **132070** as indicated by the impedance cursor **132072** and the admittance cursor **132074**. Thus, the impedance cursor **132072** is located at a portion of the impedance circle plot **132068** that is equivalent to about 55.55 kHz, and the admittance cursor **132074** is located at a portion of the admittance circle plot **132070** that is equivalent to about 55.29 kHz. As depicted in FIG. **60**, the position of the impedance cursor **132072** corresponds to values of:

$$R = 1.66026 \Omega$$

$X = -697.309 \Omega$ where R is the resistance (real value) and X is the reactance (imaginary value). Similarly, the position of the admittance cursor **132074** corresponds to values of:

$$G = 64.0322 \mu\text{S}$$

$$B = 1.63007 \text{ mS} \text{ where } G \text{ is the conductance (real value) and } B \text{ is susceptance (imaginary value).}$$

State of Jaw: Clamped on Dry Chamois

(625) FIG. **61** is a graphical display **132076** of an impedance analyzer showing complex impedance (Z)/admittance (Y) circle plots **132078**, **132080** for an ultrasonic device with the jaw of the end effector clamped on dry chamois where the impedance circle plot **132078** is shown in solid line and the admittance circle plot **132080** is shown in broken line, in accordance with at least one aspect of the present disclosure. The voltage applied to the ultrasonic transducer is 500 mV and the frequency is swept from 55 kHz to 56 kHz. The impedance (Z) scale is 200 Ω/div and the admittance (Y) scale is 500 $\mu\text{S}/\text{div}$.

(626) Measurements of values that may characterize the complex impedance (Z) and admittance (Y) circle plots **132078**, **132080** may be obtained at locations on the circle plots **132078**, **132080** as indicated by the impedance cursor **132082** and the admittance cursor **132084**. Thus, the impedance cursor **132082** is located at a portion of the impedance circle plot **132078** that is equivalent to about 55.68 kHz, and the admittance cursor **132084** is located at a portion of the admittance circle plot **132080** that is equivalent to about 55.29 kHz. As depicted in FIG. **61**, the position of the impedance cursor **132082** corresponds to values of:

$$R = 434.577 \Omega$$

$X = -758.772 \Omega$ where R is the resistance (real value) and X is the reactance (imaginary value). Similarly, the position of the admittance cursor **132084** corresponds to values of:

$$G = 85.1712 \mu\text{S}$$

$$B = 1.49569 \text{ mS} \text{ where } G \text{ is the conductance (real value) and } B \text{ is susceptance (imaginary value).}$$

State of Jaw: Tip Clamped on Moist Chamois

(627) FIG. **62** is a graphical display **132086** of an impedance analyzer showing complex impedance (Z)/admittance (Y) circle plots **132098**, **132090** for an ultrasonic device with the jaw tip clamped on moist chamois where the impedance circle plot **132088** is shown in solid line and the admittance circle plot **132090** is shown in broken line, in accordance with at least one aspect of the present disclosure. The voltage applied to the ultrasonic transducer is 500 mV and the frequency is swept from 55 kHz to 56 kHz. The impedance (Z) scale is 200 Ω/div and the admittance (Y) scale is 500 $\mu\text{S}/\text{div}$.

(628) Measurements of values that may characterize the complex impedance (Z) and complex admittance (Y) circle plots **132088**, **132090** may be obtained at locations on the circle plots **132088**, **132090** as indicated by the impedance cursor **132092** and the admittance cursor **132094**. Thus, the impedance cursor **132092** is located at a portion of the impedance circle plot **132088** that is equivalent to about 55.68 kHz, and the admittance cursor **132094** is located at a portion of the admittance circle plot **132090** that is equivalent to about 55.29 kHz. As depicted in FIG. **63**, the impedance cursor **132092** corresponds to values of:

$$R = 445.259 \Omega$$

$X = -750.082 \Omega$ where R is the resistance (real value) and X is the reactance (imaginary value). Similarly, the admittance cursor **132094** corresponds to values of:

$$G = 96.2179 \mu\text{S}$$

$$B = 1.50236 \text{ mS} \text{ where } G \text{ is the conductance (real value) and } B \text{ is susceptance (imaginary value).}$$

State of Jaw: Fully Clamped on Moist Chamois

(629) FIG. **63** is a graphical display **132096** of an impedance analyzer showing complex impedance (Z)/admittance (Y) circle plots **132098**, **132100** for an ultrasonic device with the jaw fully clamped on moist chamois where the impedance circle plot **132098** is shown in solid line and the admittance circle plot **132100** is shown in broken line, in accordance with at least one aspect of the present disclosure. The voltage applied to the ultrasonic transducer is 500 mV and the frequency is swept from 55 kHz to 56 kHz. The impedance (Z) scale is 200 Ω/div and the admittance (Y) scale is 500 $\mu\text{S}/\text{div}$.

(630) Measurements of values that may characterize the impedance and admittance circle plots **132098**, **132100** may be obtained at locations on the circle plots **132098**, **132100** as indicated by the impedance cursor **132102** and admittance cursor **132104**. Thus, the impedance cursor **132102** is located at a portion of the impedance circle plot **132098** equivalent to about 55.63 kHz, and the admittance cursor **132104** is located at a portion of the admittance circle plot **132100** equivalent to about 55.29 kHz. As depicted in FIG. **63**, the impedance cursor **132102** corresponds to values of R, the resistance (real value, not shown), and X, the reactance (imaginary value, also not shown).

(631) Similarly, the admittance cursor **132104** corresponds to values of:

$$G = 137.272 \mu\text{S}$$

B=1.48481 mS where G is the conductance (real value) and B is susceptance (imaginary value).

State of Jaw: Open with No Loading

(632) FIG. 64 is a graphical display **132106** of an impedance analyzer showing impedance (Z)/admittance (Y) circle plots where frequency is swept from 48 kHz to 62 kHz to capture multiple resonances of an ultrasonic device with the jaw open and no loading where the area designated by the rectangle **132108** shown in broken line is to help see the impedance circle plots **132110a**, **132110b**, **132110c** shown in solid line and the admittance circle plots **132112a**, **132112b**, **132112c**, in accordance with at least one aspect of the present disclosure. The voltage applied to the ultrasonic transducer is 500 mV and the frequency is swept from 48 kHz to 62 kHz. The impedance (Z) scale is 500 Ω /div and the admittance (Y) scale is 500 μ S/div.

(633) Measurements of values that may characterize the impedance and admittance circle plots **132110a-c**, **132112a-c** may be obtained at locations on the impedance and admittance circle plots **132110a-c**, **132112a-c** as indicated by the impedance cursor **132114** and the admittance cursor **132116**. Thus, the impedance cursor **132114** is located at a portion of the impedance circle plots **132110a-c** equivalent to about 55.52 kHz, and the admittance cursor **132116** is located at a portion of the admittance circle plot **132112a-c** equivalent to about 59.55 kHz. As depicted in FIG. 64, the impedance cursor **132114** corresponds to values of:

$R=1.86163 \text{ k}\Omega$

$X=-536.229 \Omega$ where R is the resistance (real value) and X is the reactance (imaginary value). Similarly, the admittance cursor **132116** corresponds to values of:

$G=649.956 \mu\text{S}$

$B=2.51975 \text{ mS}$ where G is the conductance (real value) and B is susceptance (imaginary value).

(634) Because there are only 400 samples across the sweep range of the impedance analyzer, there are only a few points about a resonance. So, the circle on the right side becomes choppy. But this is only due to the impedance analyzer and the settings used to cover multiple resonances.

(635) When multiple resonances are present, there is more information to improve the classifier. The circle plots **132110a-c**, **132112a-c** fit can be calculated for each as encountered to keep the algorithm running fast. So once there is a cross of the complex admittance, which implies a circle, during the sweep, a fit can be calculated.

(636) Benefits include in-the-jaw classifier based on data and a well-known model for ultrasonic systems. Count and characterizations of circles are well known in vision systems. So data processing is readily available. For example, a closed form solution exists to calculate the radius and axes' offsets for a circle. This technique can be relatively fast.

(637) TABLE 2 is a list of symbols used for lumped parameter model of a piezoelectric transducer (from IEEE 177 Standard).

(638) TABLE-US-00002 TABLE 2 References Symbols Meaning SI Units Equations Tables Figures B.sub.p Equivalent parallel susceptance of mho 2 vibrator C.sub.0 Shunt (parallel) capacitance in the farad 2, 3, 4, 8 5 1, 4 equivalent electric circuit C.sub.1 Motional capacitance in the equivalent farad 2, 3, 4, 6, 8, 9 5 1, 4 electric circuit f Frequency hertz 3 f.sub.a Antiresonance frequency, zero hertz 2, 4 2, 3 susceptance f.sub.m Frequency of maximum admittance hertz 2, 4 2, 3 (minimum impedance) f.sub.n Frequency of minimum admittance hertz 2, 4 2, 3 (maximum impedance) f.sub.p Parallel resonance frequency hertz 2, 3 2, 4 2 (lossless) = $\frac{1}{2 \sqrt{L_1 C_1 C_0}}$ f.sub.r Resonance frequency, zero substance hertz 2, 4 2, 3 f.sub.B

Motional(series)resonancefrequency $\frac{1}{2}$ hertz 2, 3, 6, 7, 9, 11a, 11b, 11c, 2, 4 2, 3, 6, 8 12, G.sub.p Equivalent parallel conductance of 1 vibrator L.sub.1 Motional inductance in the equivalent henry 8, 9 1, 4, 5 electric circuit M Figure of merit of a vibrator = $\frac{Q}{f}$ dimensionless 10, 11a, 11b 3, 4, 5 $M = \frac{1}{s C_0 R_1} Q$ Quality factor $Q = \frac{R_1}{s L_1} = \frac{1}{s C_1 R_1}$ = rM dimensionless 12 3 6, 8 r Capacitance ratio $r = \frac{C_0}{C_1}$ dimensionless 2, 3, 10, 11 2, 3, 4, 5 8 R.sub.a Impedance at zero phase angle near ohm 2, 3 antiresonance R.sub.e Equivalent series resistance of vibrator ohm 1, 2 R.sub.r Impedance at f.sub.r zero phase angle ohm 2, 3 R.sub.1 Motional resistance in the equivalent ohm 4, 8, 10, 11a, 2, 5 1, 3, 4, 6, 7, electric circuit 11b, 11c, 12 8 X.sub.e Equivalent series reactance of vibrator ohm 1, 2 X.sub.0 Reactance of shunt (parallel) ohm 1, 4, 5 5 3, 7 capacitance at series resonance = $\frac{1}{s C_0}$ X.sub.1 Reactance of motional (series) arm of vibrator ohm 2 2 0 $X_1 = L_1 - \frac{1}{s C_1}$ Y Admittance of vibrator mho 1 $Y = G_p + j B_{pz}$ Y.sub.m Maximum admittance of vibrator mho 3 Y.sub.n Minimum admittance of vibrator mho 3 Z Impedance of vibrator ohm 1 $Z = R_{se} + j X_{se}$ Z.sub.m Minimum impedance of vibrator ohm 3 Z.sub.n Maximum impedance of vibrator ohm 3 Absolute value of impedance of ohm 2 2 vibrator $Z = \sqrt{(R_{se})^2 + (X_{se})^2}$ Absolute value of impedance at f.sub.m ohm 2 (minimum impedance) Absolute value of impedance at f.sub.n ohm 2 (maximum impedance) δ Normalized damping factor $\delta = \omega C_{sub.0} R_{sub.1}$ dimensionless 1 2 Ω Normalized frequency factor $\Omega = \text{dimensionless}$ 1 2 $\frac{f - f_s}{f_p - f_s}$ ω Circular (angular) frequency hertz 2 $\omega = 2\pi f$ $\omega_{sub.s}$ Circular frequency at motional hertz resonance $\omega_{sub.s} = 2\pi f_{sub.s}$

(639) TABLE 3 is a list of symbols for the transmission network (from IEEE 177 Standard).

(640) TABLE-US-00003 TABLE 3 References Symbols Meaning SI Units Equations Tables Figures b Normalized compensation factor 1 - dimensionless 4, 10 5 $\frac{1}{4 \sqrt{f_s C_0 L_0}}$ B Normalized admittance factor dimensionless 10 5 C Normalized admittance factor dimensionless 10 5 C.sub.A-B Stray capacitance between the farad terminals A-B (FIG. 4) C.sub.L Load capacitance farad 6 4 C.sub.T Shunt capacitance terminating farad 4, 10 5 4 transmission circuit C.sub.L1 Load capacitance farad 7 C.sub.L2 Load capacitance farad 7 e.sub.2 Output voltage of transmission volt 4 network f.sub.mT Frequency of maximum transmission hertz 10 F.sub.sL1 Motional resonance frequency of hertz 7 combination of vibrator and C.sub.L1 F.sub.sL2 Motional resonance frequency of hertz 7 combination of vibrator and C.sub.L2 i.sub.1 Input current to transmission network ampere 4 L.sub.0 Compensation inductance shunting henry 4 vibrator M.sub.T Figure of merit of transmission dimensionless 4, 10 5 network termination = $\frac{1}{2 \sqrt{f_s C_T R_T}} = \frac{X_T}{R_T}$ R.sub.T Shunt resistance termination of ohm 4, 11a, 11b, 5 4, 6, 7, 8 transmission network 11c, 12 R.sub.sL2 Standard resistor ohm 4, 5 5 7 S Detector sensitivity smallest detectable dimensionless 12 6 current change/current x Normalized frequency factor x = dimensionless 12 $\frac{f}{f_s} - 1 = \tau$ X.sub.A-B Reactance of stray capacitance C.sub.A-B ohm X.sub.T Reactance of C.sub.T at the motional ohm 4 5 resonance frequency $X_T = \frac{1}{2 \sqrt{f_s C_T}}$ x.sub.mT Normalized frequency factor at the dimensionless 5 frequency of maximum transmission $\Delta C_{sub.L}$ $\Delta C_{sub.L} = C_{sub.L2} - C_{sub.L1}$ farad 6, 7 $\Delta f_{sub.1} = f_{sub.sL1} - f_{sub.sL2}$ hertz 6, 7 6 $\Delta f_{sub.1} = f_{sub.sL1} - f_{sub.s}$ hertz 6, 7 $\Delta f_{sub.2} \Delta f_{sub.1} = f_{sub.sL2} - f_{sub.s}$ hertz 6, 7 *Refers to real roots; complex roots to be disregarded.

(641) TABLE 4 is a list of solutions for various characteristic frequencies (from IEEE 177 Standard).

Solutions for the Various Characteristic Frequencies

(642) TABLE-US-00004 TABLE 4 Characteristic Constituent Equation for 57 IEEE Frequencies Meaning Condition Frequency Root 14.S1.sup.1 f.sub.m Frequency of maximum = 0 -2 $\delta_{sup.2}(\Omega + r) - 2\Omega r(1 - \text{lower}^* f_{sub.m} \text{ admittance (minimum } \Omega) - \Omega_{sup.2} = 0$ impedance) f.sub.a Motional (series) resonance X.sub.1 = 0 $\Omega = 0$ f.sub.a frequency f.sub.r Resonance frequency X.sub.e = B.sub.p = 0 $\Omega(1 - \Omega) - \delta_{sup.2} = 0$ lower f.sub.r f.sub.a Antiresonance frequency X.sub.e = B.sub.p = 0 $\Omega(1 - \Omega) - \delta_{sup.2} = 0$ upper f.sub.a f.sub.p Parallel resonance $|\Omega| = 1$ f.sub.p frequency (lossless) = ∞ | R.sub.1 = 0 f.sub.n Frequency of minimum = 0 -2 $\delta_{sup.2}(\Omega + r) - 2\Omega r(1 - \text{upper}^* f_{sub.n} \text{ admittance (maximum } \Omega) - \Omega_{sup.2} = 0$ impedance) *Refers to real roots; complex roots to be disregarded

(643) TABLE 5 is a list of losses of three classes of piezoelectric materials.

(644) TABLE-US-00005 TABLE 5 Type of Piezoelectric Vibrator Q = Mr r Q.sub.r/r min Piezoelectric Ceramics 90-500 2-40 200 Water-Soluble Piezoelectric 200-50,000 3-500 80 Crystals Quartz 10.sub.4-10.sub.7 100-50,000 2000 Minimum Values for the Ratio Q.sub.r/r to be Expected for Various Types of Piezoelectric Vibrators

(645) TABLE 6 illustrates jaw conditions, estimated parameters of a circle based on real time measurements of complex impedance/admittance,

radius (r) and offsets (ae and be) of the reference circle represented by measured variables Re, Ge, Xe, Be, and parameters of a reference circle plots, as described in FIGS. 60-64, based on real time measurements of complex impedance/admittance, radius (rr) and offsets (ar, br) of the reference circle represented by reference variables Rref, Gref, Xref, Bref. These values are then compared with established values for given conditions. These conditions may be: 1) open with nothing in jaws, 2) tip bite 3) full bite and staple in jaws. The equivalent circuit of the ultrasonic transducer was modeled as follows and the frequency was swept from 55 kHz to 56 kHz:

$Ls=L1=1.1068H$

$Rs=R1=311.352\Omega$

$Cs=C1=7.43265\text{ pF}$ and

$C0=C0=3.64026\text{ nF}$.

(646) TABLE-US-00006 TABLE 6 Reference Circle Plot Reference Jaw Conditions R.sub.ref (Ω) G.sub.ref (μS) X.sub.ref (Ω) B.sub.ef (mS) Jaw open and no loading 1.66026 64.0322 -697.309 1.63007 Jaw clamped on dry 434.577 85.1712 -758.772 1.49569 chamois Jaw tip clamped on 445.259 96.2179 -750.082 1.50236 moist chamois Jaw fully clamped on 137.272 1.48481 moist chamois

(647) In use, the ultrasonic generator sweeps the frequency, records the measured variables, and determines estimates Re, Ge, Xe, Be. These estimates are then compared to reference variables Rref, Gref, Xref, Bref stored in memory (e.g., stored in a look-up table) and determines the jaw conditions. The reference jaw conditions shown in TABLE 6 are examples only. Additional or fewer reference jaw conditions may be classified and stored in memory. These variables can be used to estimate the radius and offsets of the impedance/admittance circle.

(648) FIG. 65 is a logic flow diagram 132120 of a process depicting a control program or a logic configuration to determine jaw conditions based on estimates of the radius (r) and offsets (a, b) of an impedance/admittance circle, in accordance with at least one aspect of the present disclosure.

Initially a data base or lookup table is populated with reference values based on reference jaw conditions as described in connection with FIGS. 60-64 and TABLE 6. A reference jaw condition is set and the frequency is swept from a value below resonance to a value above resonance. The reference values Rref, Gref, Xref, Bref that define the corresponding impedance/admittance circle plot are stored in a database or lookup table. During use, under control of a control program or logic configuration a processor or control circuit of the generator or instrument causes the frequency to sweep 132122 from below resonance to above resonance. The processor or control circuit measures and records 132124 (e.g., stores in memory) the variables Re, Ge, Xe, Be that define the corresponding impedance/admittance circle plot and compares 132126 them to the reference values Rref, Gref, Xref, Bref stored in the database or lookup table. The processor or control circuit determines 132128, e.g., estimates, the end effector jaw conditions based on the results of the comparison.

Application of "Smart Blade" Technology

(649) Current ultrasonic and/or combination ultrasonic/RF tissue treatment conditions employ advanced tissue treatment algorithms with a pre-determined current level for each step of the algorithm. Instead of using an advanced hemostasis tissue treatment algorithm with a pre-determined current level for each step of the algorithm, the proposed advanced tissue treatment technique adjusts electrical current delivered to the ultrasonic transducer to drive the ultrasonic blade to a constant temperature using a frequency-temperature control system.

(650) FIGS. 66A-66B are graphical representations of an advanced ultrasonic transducer current controlled hemostasis algorithm. For example, the tissue treatment process may start out by driving the ultrasonic transducer current to generate a high constant temperature for a first predetermined period T1. At the end of the first predetermined period T1, the process drives the ultrasonic transducer current to generate a lower constant temperature of the ultrasonic blade for a second predetermined period T2. The lower temperature of the ultrasonic blade may be suitable to achieve a tissue seal but not a tissue transection. Finally, the process drives the ultrasonic transducer current to increase (ramps back up) the temperature of the ultrasonic blade to a higher constant temperature for a third predetermined period T3. The higher temperature is high enough to complete the transection but is lower than the melting point of the clamp arm pad. For example, the higher ultrasonic blade temperature during the third predetermined period T3 may be selected to be lower than the melting point of TEFLON, for example, which is a material commonly used for the clamp arm pad.

(651) FIG. 66A is a graphical representation 132130 of percent of maximum current delivered into an ultrasonic transducer as a function of time, in accordance with at least one aspect of the present disclosure. The vertical axis represents percentage (%) of maximum current delivered to an ultrasonic transducer and the horizontal axis represents time (sec). The percentage of transducer current is set to a first percentage of maximum current X1% to raise the temperature of the ultrasonic blade during a first period T1. The percentage of transducer current is then lowered to a second percentage of maximum current X2% over a second period T2 to lower the blade temperature to a value that is suitable for sealing tissue but not for transecting tissue. The percentage of transducer current is then increased to a third percentage of maximum current X3% for a third period T3, to raise the blade temperature to a value that is suitable for transecting tissue but is lower than the melting point of the clamp arm pad (e.g., TEFLON). According to the process graphically depicted in FIG. 66A, the same percentage of ultrasonic transducer current profile may be used for all tissue types, loading conditions, etc.

(652) FIG. 66B is a graphical representation 132140 of ultrasonic blade temperature as a function of time and tissue type, in accordance with at least one aspect of the present disclosure. The vertical axis represents temperature ($^{\circ}F$) of the ultrasonic blade and the horizontal axis represents time (sec). This technique may be combined with impedance spectroscopy to detect tissue of various thicknesses. Such as for example, thick tissue versus thin tissue, located in the jaws of the ultrasonic end effector. Once the tissue thickness is detected, the ultrasonic blade temperature may be controlled to accommodate different levels of energy delivery as may be needed across a range of tissue types and adjust the advanced hemostasis algorithm in real-time. Once the tissue type is detected or determined, the ultrasonic blade temperature is set to a nominal temperature Temp.sub.1 by controlling the driving current into the ultrasonic transducer. The temperature of the ultrasonic blade is set to a first temperature Temp.sub.1, which may be raised (+) or lowered (-) based on tissue type over a first period T1. The ultrasonic blade temperature is then lowered to a second temperature Temp.sub.2 over a second period T2 to lower the blade temperature to a value that is suitable for sealing tissue but not for transecting tissue. The second temperature Temp.sub.2, also may be raised (+) or lowered (-) based on the detected tissue type. The ultrasonic blade temperature is then increased to a third temperature Temp.sub.3 over a third period T3 to a value that is suitable for transecting tissue but that is lower than the melting point temperature Tw of the clamp jaw pad material. According to the process depicted in FIG. 66B, the ultrasonic blade temperature can be varied based on tissue types, loading conditions, etc. In addition, the ultrasonic blade temperature versus time profile can be varied by varying the periods T1-T3. Finally, the ultrasonic blade temperature versus time profile can be varied by varying both the temperature of the ultrasonic blade and the time periods T1-T3.

(653) In one example, for audible surgeon feedback, tones can be tied to achieving a certain temperature threshold. This would improve consistency in advanced hemostasis transection times and hemostasis across a range of tissue types.

(654) FIG. 67 is a logic flow diagram 132150 of a process depicting a control program or a logic configuration to control the temperature of an ultrasonic blade based on tissue type, in accordance with at least one aspect of the present disclosure. The tissue type may be determined 132152 using the techniques described in FIGS. 19-21 under the heading ESTIMATING THE STATE OF THE JAW (PAD BURN THROUGH, STAPLES, BROKEN BLADE, BONE IN JAW, TISSUE IN JAW and/or FIGS. 22-30 under the heading STATE OF JAW CLASSIFIER BASED ON MODEL and/or techniques for estimating the temperature of the ultrasonic blade are described in related U.S. Provisional Patent Application No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR, to Nott et al, which is incorporated herein by reference in its entirety. According to the process, a control circuit in the generator or instrument determines the tissue type and sets the initial temperature of the ultrasonic blade to a nominal temperature by controlling the driving current into the ultrasonic transducer. The control circuit raises (+) or lowers (-) the temperature of the ultrasonic blade based on tissue type over a first period T1. The control circuit then lowers the

temperature of the ultrasonic blade over a second period T2 to lower the blade temperature to a value that is suitable for sealing tissue but not for transecting tissue. The control circuit raises (+) or lowers (−) the second temperature based on the detected tissue type. The control circuit increases the temperature of the ultrasonic blade to a third temperature over a third period T3, to a value that is suitable for transecting tissue but is lower than the melting point of the clamp jaw pad material (e.g., TEFLON).

Smart Blade and Power Pulsing

(655) During surgery with an ultrasonic shears device the power delivered to the tissue is set at a predetermined level. That predetermined level is used to transect the tissue throughout the transection procedure. Certain tissues may seal better or cut better/faster if the power delivered varies throughout the transection procedure. A solution is needed to vary the power delivered to the tissue through the blade during the transection process. In various aspects, the tissue type and changes to the tissue during the transection process may be determined using the techniques described in FIGS. 19-21 under the heading ESTIMATING THE STATE OF THE JAW (PAD BURN THROUGH, STAPLES, BROKEN BLADE, BONE IN JAW, TISSUE IN JAW and/or FIGS. 22-30 under the heading STATE OF JAW CLASSIFIER BASED ON MODEL and/or techniques for estimating the temperature of the ultrasonic blade are described in related U.S. Provisional Patent Application No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR, to Nott et al, which is incorporated herein by reference in its entirety.

(656) One solution that provides better ultrasonic transection employs the impedance feedback of the ultrasonic blade. As previously discussed, the impedance of the ultrasonic blade is related to the impedance of the electromechanical ultrasonic system and may be determined by measuring the phase angle between the voltage and current signals applied to the ultrasonic transducer as described herein. This technique may be employed to measure the magnitude and phase of the impedance of the ultrasonic transducer. The impedance of the ultrasonic transducer may be employed to profile factors that may be influencing the ultrasonic blade during use (e.g., force, temperature, vibration, force over time, etc.). This information may be employed to affect the power delivered to the ultrasonic blade during the transection process.

(657) FIG. 68 is a logic flow diagram 132170 of a process depicting a control program or a logic configuration to monitor the impedance of an ultrasonic transducer to profile an ultrasonic blade and deliver power to the ultrasonic blade on the profile according to one aspect of the present disclosure. According to the process, a control circuit determines 132172 (e.g., measures) the impedance (Z) of the ultrasonic transducer during a tissue transection process. The control circuit analyzes and profiles 132174 the ultrasonic blade right after the tissue is fully clamped in jaw of the end effector of the ultrasonic device based on the determined 132172 impedance (Z). The control circuit adjusts 132176 a power output level based on the profile (e.g., high power for dense tissue low power for thin tissue) of the ultrasonic blade. The control circuit controls the generator to momentarily drive the ultrasonic transducer and the ultrasonic blade and then stops. The control circuit again determines 132172 the impedance (Z) of the ultrasonic blade and profiles 132174 the ultrasonic blade based on the determined 132172 impedance (Z). The control circuit controls the generator to adjust the output power level or keep it the same based on the profile of the ultrasonic blade. The control circuit again controls the generator to momentarily drive the ultrasonic transducer and the ultrasonic blade and then stops. The process repeats and determines 132172 the impedance (Z), profiles 132174 the ultrasonic blade, and adjusts 132176 the power level until the impedance profile detected is that of the clam arm pad and then adjusts the power to prevent the clam arm pad from melting.

(658) The process discussed in connection with FIG. 68 allows the ultrasonic transducer power level to be adjusted on the fly as the tissue changes from being heated and cut. Accordingly, if the tissue is initially tough and then weakens or if different layers of tissue are encountered during the transection process, the power level can be optimally adjusted to match the profile of the ultrasonic blade. This method could eliminate the need for the user to set the power level. The ultrasonic device would adapt and choose the right power level based on current tissue conditions and transection process.

(659) This technique provides intelligent control for power level setting based on tissue feedback. This technique may eliminate the need for power settings on the generator and may lead to faster transection times. In one aspect, in an ultrasonic transection medical device including a jaw with an ultrasonic blade, the impedance of the ultrasonically driven blade is used to profile the ultrasonic blade characteristics (force, heat, vibration, etc.) and that profile is used to influence the power output of the transducer during the transection process. Power may be pulsed on and off so that the tissue changes can be read for feedback in between pulses to adjust the power during the transection process.

(660) FIGS. 69A-69D is a series of graphical representations of the impedance of an ultrasonic transducer to profile an ultrasonic blade and deliver power to the ultrasonic blade based on the profile, in accordance with at least one aspect of the present disclosure. FIG. 69A is a graphical representation 132180 of ultrasonic transducer impedance versus time. The generator control circuit reads the initial impedance Z1 which is based on the contents of the jaw and applies a pulsed power P1 to the ultrasonic transducer as shown in FIG. 69B, which is a graphical depiction 132182 of pulsed power versus time. FIG. 69C is a graphical representation 132184 of a new impedance Z2 versus time. The control circuit of the generator reads the new impedance Z2 and applies pulsed power P2 to the ultrasonic transducer to meet the new tissue condition as plotted in FIG. 69D, which is a graphical representation 132186 of pulsed power P2 versus time.

Adjustment of Complex Impedance to Compensate for Lost Power in an Articulating Ultrasonic Device

(661) FIG. 70 is a system 132190 for adjusting complex impedance of the ultrasonic transducer 132192 to compensate for power lost when the ultrasonic blade 132194 is articulated, in accordance with at least one aspect of the present disclosure. The performance of an articulatable ultrasonic blade 132194 is inconsistent throughout the full articulation angle θ from A-B. For example, power is lost when the ultrasonic blade 132194 is articulated. Knowing the articulation angle θ that the ultrasonic blade 132194 is at, the generator 132196 or the surgical instrument 132199 can adjust the complex impedance (Z) to compensate for the power lost when the ultrasonic blade 132194 is articulated. Also, by analyzing the performance of the ultrasonic blade 132194 through its full articulation angle θ , the generator 132196 can execute an algorithm to adjust the complex impedance (Z) to compensate for power loss.

(662) Adjusting complex impedance of the ultrasonic transducer 132192 to compensate for power lost when the ultrasonic blade 132194 may employ the techniques described in FIGS. 19-21 under the heading ESTIMATING THE STATE OF THE JAW (PAD BURN THROUGH, STAPLES, BROKEN BLADE, BONE IN JAW, TISSUE IN JAW and/or FIGS. 22-30 under the heading STATE OF JAW CLASSIFIER BASED ON MODEL and/or techniques for estimating the temperature of the ultrasonic blade are described in related U.S. Provisional Patent Application No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR, to Nott et al, which is incorporated herein by reference in its entirety.

(663) These techniques may be employed to determine the articulation angle θ of the ultrasonic blade 132194 by sweeping through a range of articulation angles θ from A-B at a predetermined angular increment. At each angular increment, activating the ultrasonic transducer 132192 at either a therapeutic or non-therapeutic energy level, measuring the complex impedance (Z) of the ultrasonic transducer 132192, recording a set of complex impedance (Z) measurements, generating reference complex impedance characteristic patterns or a training data sets S as a function of articulation angle θ , and storing the reference complex impedance characteristic patterns or a training data sets S in a memory or database that is accessible by the ultrasonic instrument 132199 during a surgical procedure. During a surgical procedure, the ultrasonic instrument 132199 can determine articulation angle θ by comparing real time complex impedance (Z) measurements of the ultrasonic transducer 132192 with the reference complex impedance characteristic patterns or a training data sets S.

(664) Articulatable ultrasonic waveguides 132198 are described in U.S. Pat. No. 9,095,367 titled Flexible Harmonic Waveguides/Blades For Surgical Instruments, which is incorporated herein by reference. See FIGS. 47-66B and associated description. Measurement of articulation angle is described in U.S. Pat. No. 9,808,244 titled Sensor Arrangements For Absolute Positioning System For Surgical Instruments, which is incorporated herein by reference. See FIGS. 193-196 and associated description.

(665) FIG. 71 is a logic flow diagram **132200** of a process depicting a control program or a logic configuration to compensate output power as a function of articulation angle, in accordance with at least one aspect of the present disclosure. Accordingly, in conjunction with FIG. 70, during use, a control circuit of the generator **132196** or instrument determines **132202** the articulation angle θ of the ultrasonic blade **132194**. The control circuit adjusts **132204** the complex impedance (Z) to compensate for power lost as a function of articulation angle θ . The control circuit applies **132206** the output power of the generator **132196** applied to the ultrasonic transducer **132192** based on the articulation angle θ of the ultrasonic blade **132194**. Using Spectroscopy to Determine Device Use State in Combo Instrument

(666) FIG. 72 is a system **132210** for measuring complex impedance (Z) of an ultrasonic transducer **132212** in real time to determine action being performed by an ultrasonic blade **132214**, in accordance with at least one aspect of the present disclosure. Current surgical instruments include three functions (seal+cut, seal only, and spot coagulation). These functions can be executed by activating two buttons. It would be useful if the surgeon only had to press one button and could receive either the seal only or spot coagulation algorithm based on the desired action to be performed. Ultrasonic spectroscopy can be used to measure the complex impedance (Z) of the ultrasonic blade **132214** in real time. The real time measurements can be compared to predefined data to determine which action is being performed. The different complex impedance (Z) patterns between spot coagulation and seal only enable the generator **132216** to determine which action is being performed and to execute the appropriate algorithm.

(667) Measuring complex impedance (Z) of an ultrasonic transducer **132212** in real time to determine action being performed by an ultrasonic blade **132214** may employ the techniques described in FIGS. 19-21 under the heading ESTIMATING THE STATE OF THE JAW (PAD BURN THROUGH, STAPLES, BROKEN BLADE, BONE IN JAW, TISSUE IN JAW and/or FIGS. 22-30 under the heading STATE OF JAW CLASSIFIER BASED ON MODEL and/or techniques for estimating the temperature of the ultrasonic blade are described in related U.S. Provisional Patent Application No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR, to Nott et al, which is incorporated herein by reference in its entirety.

(668) TABLE 7 is a chart of ultrasonic blade action and corresponding complex impedance. This information is stored in a memory lookup table or database.

(669) TABLE-US-00007 TABLE 7 Ultrasonic Blade Action Impedance (Z) Seal only Z1 Spot coagulation Z2

(670) FIG. 73 is a logic flow diagram **132220** of a process depicting a control program or a logic configuration to determine action being performed by an ultrasonic blade **132214** (FIG. 72) based on the complex impedance pattern, in accordance with at least one aspect of the present disclosure. Prior to implementing the process described in FIG. 73 and in conjunction with FIG. 72, a database or memory lookup table is populated with data of ultrasonic blade **132214** actions and observed complex impedances (Z) associated with the ultrasonic blade **132214** actions. The database or lookup table can be accessed by the ultrasonic instrument **132218** or generator **132216** while executing the ultrasonic blade **132214** action. Accordingly, during a hemostasis procedure, a control circuit of the generator **132216** or instrument **13218** determines **132222** the complex impedance (Z) of the ultrasonic blade **132214**. The control circuit compares **132224** the measured complex impedance (Z) to stored values of complex impedance patterns associated with ultrasonic blade **132214** functions. The control circuit controls the generator **132216** to apply **132226** an output power algorithm to the ultrasonic transducer **132212** based on the comparison.

Vessel Sensing for Adaptive Advanced Hemostasis

(671) In various aspects, the present disclosure provides adaptive vessel sealing modes. In one aspect, the ultrasonic instrument can deliver ultrasonic energy uniquely for veins as opposed to arteries.

(672) In another aspect, the present disclosure provides a technique for identifying the jaw contents of an ultrasonic device. Using this approach, a vessel clamped in the jaw is identified as either a vein or an artery, which can be characterized as differences in vessel wall and pressure. Knowing that a vessel is a vein or an artery can be used to activate a unique advanced hemostasis cycle for each type. A vein requires more time and lower temperature due to the thinner vessel walls, so an advanced hemostasis cycle will include lower current and longer time in the vessel sealing portion of the cycle.

(673) FIG. 74 is a logic flow diagram **132230** depicting a control program or a logic configuration of an adaptive process for identifying a hemostasis vessel, in accordance with at least one aspect of the present disclosure. In accordance with the process, a control circuit of the generator or instrument senses **132232** the vessel located in the jaw of the ultrasonic device using any of the smart blade algorithm techniques for estimating or classifying the state of the jaw of an ultrasonic device described in connection with FIGS. 19-21 under the heading ESTIMATING THE STATE OF THE JAW (PAD BURN THROUGH, STAPLES, BROKEN BLADE, BONE IN JAW, TISSUE IN JAW and/or FIGS. 22-30 under the heading STATE OF JAW CLASSIFIER BASED ON MODEL and/or techniques for estimating the temperature of the ultrasonic blade are described in related U.S. Provisional Patent Application No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR, to Nott et al, which is incorporated herein by reference in its entirety. When a vein is sensed **132234** or an artery is sensed **132236**, the control circuit receives a command for sealing either a vein or an artery and activates **132238** an advanced hemostasis algorithm based on the type of vessel sensed. In one aspect, the command may be originated by a user from a button located on the instrument to activate the appropriate advanced hemostasis algorithm. In other aspects, the command may be originated automatically based on tissue characterization algorithms.

(674) When a vein is sensed **132234**, the control circuit executes **132240** a first algorithm that can seal slower at a lower power level and lower ultrasonic blade temperature. Accordingly, to treat a vein, the control circuit controls the generator to output a lower power P1 and activates the generator for a longer time T1.

(675) When an artery is sensed **132236**, the control circuit executes **132242** a second algorithm that can seal faster at a higher power level and higher ultrasonic blade temperature. Accordingly, to treat an artery, the control circuit controls the generator to output higher power P2 and activates the generator for a shorter time T2.

(676) FIG. 75 is a graphical representation **132250** of ultrasonic transducer current profiles as a function of time for vein and artery vessel types, in accordance with at least one aspect of the present disclosure. The vertical axis is generator output current (I) delivered to the ultrasonic transducer and the horizontal axis is time (sec). With reference also to FIG. 74, the first curve **132252** represents a vein and is treated with lower power (P1 at I1) and longer period (T1) and the second curve **132254** represents an artery and is treated with higher power (P2 at I2) applied for a shorter period (T2) relative to the first curve **132252**.

(677) In another aspect, the present disclosure provides a technique for delivering ultrasonic transducer current (I) in a feedback control loop to achieve a targeted frequency which is associated with a desired ultrasonic blade temperature. When sealing a vein, for example, the feedback control loop will drive to a higher targeted frequency which corresponds to a cooler ultrasonic blade temperature that is suitable (and may be ideal) for sealing the vein. An artery would be driven to a slightly lower frequency target associated with a hotter ultrasonic blade temperature.

(678) FIG. 76 is a logic flow diagram **132260** depicting a control program or a logic configuration of an adaptive process for identifying a hemostasis vessel, in accordance with at least one aspect of the present disclosure. In accordance with the process, a control circuit of the generator or instrument senses **132262** the vessel in the jaw using any of the smart blade algorithm techniques for estimating or classifying the state of the jaw of an ultrasonic device described in connection with FIGS. 19-21 under the heading ESTIMATING THE STATE OF THE JAW (PAD BURN THROUGH, STAPLES, BROKEN BLADE, BONE IN JAW, TISSUE IN JAW and/or FIGS. 22-30 under the heading STATE OF JAW CLASSIFIER BASED ON MODEL and/or techniques for estimating the temperature of the ultrasonic blade are described in related U.S. Provisional Patent Application No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR, to Nott et al, which is incorporated herein by reference in its entirety.

(679) When a vein is sensed **132264**, the control circuit executes a first algorithm to supply **132268** current to the ultrasonic transducer to achieve a targeted seal temperature for a vein. A feedback control loop estimates the temperature of the ultrasonic blade and adjusts the current delivered to the

ultrasonic transducer to control the temperature of the ultrasonic blade. When an artery is sensed **132266**, the control circuit executes a second algorithm to supply **132269** current to the ultrasonic transducer to achieve a targeted seal temperature for an artery. A feedback control loop estimates the temperature of the ultrasonic blade and adjusts the current delivered to the ultrasonic transducer to control the temperature of the ultrasonic blade. (680) FIG. 77 is a graphical representation **132270** of ultrasonic transducer frequency profiles as a function of time for vein and artery vessel types, in accordance with at least one aspect of the present disclosure. The vertical axis represents the frequency (kHz) of the signal applied to the ultrasonic transducer and the horizontal axis represents time (sec). The first curve **132272** represents a vein. A vein requires a cooler ultrasonic blade temperature to effect a seal. The first algorithm controls the temperature of the ultrasonic blade by setting the frequency applied to the ultrasonic transducer to a higher frequency and controls current delivered to the ultrasonic transducer to maintain the set frequency. The second curve **132274** represents an artery. An artery requires a hotter ultrasonic blade temperature to effect a seal. The second algorithm controls the temperature of the ultrasonic blade by setting the frequency applied to the ultrasonic transducer to a lower frequency and controls current delivered to the ultrasonic transducer to maintain the set frequency.

Calcified Vessel Identification

(681) In various aspects, the present disclosure provides various techniques for improving hemostasis when sealing calcified vessels and to address challenges in sealing calcified vessels. In one aspect, the ultrasonic instrument is configured to manage sealing of calcified vessels with intelligence. In one aspect, the jaw contents may be identified using smart blade algorithm techniques for estimating or classifying the state of the jaw of an ultrasonic device described in connection with FIGS. 19-21 under the heading ESTIMATING THE STATE OF THE JAW (PAD BURN THROUGH, STAPLES, BROKEN BLADE, BONE IN JAW, TISSUE IN JAW and/or FIGS. 22-30 under the heading STATE OF JAW CLASSIFIER BASED ON MODEL and/or techniques for estimating the temperature of the ultrasonic blade are described in related U.S. Provisional Patent Application No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR, to Nott et al, which is incorporated herein by reference in its entirety. Accordingly, these techniques may be employed to identify a calcified vessel when clamped in the jaws of the ultrasonic instrument.

(682) Three possible scenarios are disclosed. In one aspect, the user is prompted with a warning from the generator that the jaws are clamping on a calcified vessel and the instrument will not fire. In another aspect, the instruments prompts a user that they the jaws have clamped a calcified vessel and it will not allow the instrument to fire until a minimum amount of compression time (say 10-15 seconds) has elapsed. This additional time allows the calcification/plaque to migrate away from the transection side and improve hemostasis of the seal. In a third aspect, upon grasping a calcified vessel and pressing the activation button, the instrument employs an internal motor to displace the spring stack an additional amount in order to deliver slightly more clamp force and better compression of the calcified vessel.

(683) FIG. 78 is a logic flow diagram **132280** depicting a control program or a logic configuration of a process for identifying a calcified vessel, in accordance with at least one aspect of the present disclosure. According to the process, a control circuit of the generator or instrument identifies a vessel located in the jaw of the ultrasonic device when the jaw clamps **132282** on the vessel. When the control circuit identifies **132284** a calcified vessel, the control circuit sends **132286** a warning message that can be perceived by the user. The message contains information to notify the user that a calcified vessel has been detected. The control circuit then prompts **132288** to maintain compression on the calcified vessel for a predetermined waiting period **T1** (e.g., x-seconds). This will allow the calcification to migrate away from the jaws. At the expiration of the compression waiting period **T1**, the control circuit enables **132290** the activation of the ultrasonic generator. When the control circuit identifies **132292** a “normal (e.g., not calcified) vessel, the control circuit enables **132294** normal activation of the ultrasonic device. Accordingly, the ultrasonic device can execute one or more hemostasis algorithms as described herein.

(684) FIG. 79 is a logic flow diagram **132300** depicting a control program or a logic configuration of a process for identifying a calcified vessel, in accordance with at least one aspect of the present disclosure. According to the process, a control circuit of the generator or instrument identifies a vessel located in the jaw of the ultrasonic device when the jaw clamps **132302** on the vessel. When the control circuit identifies **132304** a calcified vessel, the control circuit sends **132306** a warning message that can be perceived by the user that a calcified vessel was detected. The control circuit disables **132308** or alternatively does not enable activation of the ultrasonic device. When the control circuit identifies **132310** a “normal” vessel (e.g., not calcified), the control circuit enables **132312** normal activation of the ultrasonic device. Accordingly, the ultrasonic device can execute one or more hemostasis algorithms as described herein.

(685) FIG. 80 is a logic flow diagram **132320** depicting a control program or a logic configuration of a process for identifying a calcified vessel, in accordance with at least one aspect of the present disclosure. According to the process, a control circuit of the generator or instrument identifies a vessel located in the jaw of the ultrasonic device when the jaw clamps **132322** on the vessel. When the control circuit identifies **132324** a calcified vessel, the control circuit sends **132326** a warning message that can be perceived by the user that a calcified vessel was detected. The control circuit increases **132328** the jaw clamp force with a motor in order to achieve better compression of the calcified vessel. The control circuit then enables **132330** activation of the ultrasonic energy after a clamp force adjustment. When the control circuit identifies **132332** a “normal” vessel (e.g., not calcified), the control circuit enables **132334** normal activation of the ultrasonic device. Accordingly, the ultrasonic device can execute one or more hemostasis algorithms as described herein.

Detection of Large Vessels During Parenchymal Dissection Using a Smart Blade

(686) During liver resection procedures, surgeons risk cutting large vessels because they are buried inside the parenchyma that is being dissected, and thus cannot be seen. FIGS. 81-86 of this disclosure outlines an application for a “smart blade” (e.g., an ultrasonic blade with feedback to provide jaw content identification) that can detect the difference between parenchymal tissue, and large vessels within the parenchymal tissue by using the magnitude and phase of the impedance measurements over a swept frequency range. During a parenchymal dissection procedure, vessels may be detected using smart blade algorithm techniques for estimating or classifying the state of the jaw of an ultrasonic device described in connection with FIGS. 19-21 under the heading ESTIMATING THE STATE OF THE JAW (PAD BURN THROUGH, STAPLES, BROKEN BLADE, BONE IN JAW, TISSUE IN JAW and/or FIGS. 22-30 under the heading STATE OF JAW CLASSIFIER BASED ON MODEL and/or techniques for estimating the temperature of the ultrasonic blade are described in related U.S. Provisional Patent Application No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR, to Nott et al, which is incorporated herein by reference in its entirety.

(687) During liver resection and dissection procedures of other vascular parenchymal tissues, the surgeon cannot see vessels that are embedded within the parenchyma along the dissection plane. This can cause surgeons to cut large vessels without sealing, resulting in excessive bleeding that causes blood loss to the patient and stress for the surgeon. FIGS. 81-86 describe a solution that offers a method of detecting large vessels embedded in parenchymal tissues without the need to visualize the large vessels using a smart ultrasonic blade application.

(688) The ultrasonic devices described herein may be employed to accomplish following vessel detection prior to initiating a liver resection and dissection procedure. A control circuit of the generator or the ultrasonic device initiates a frequency sweep from below resonance to above resonance of the electromechanical ultrasonic system to enable measurements of the magnitude and phase of the impedance. The results are plotted on a 3D curve as described in connection with FIGS. 19-21. The resulting 3D curve will have a particular form when the ultrasonic blade is in contact with parenchymal tissue and will have other forms when the ultrasonic blade contacts tissue other than parenchymal tissue, as discussed below.

(689) A different 3D curve is generated by the frequency sweep when the ultrasonic blade is contacting a large vessel. When the ultrasonic blade contacts a vessel, the control circuit compares the test frequency sweep of the new (vessel) curve with the frequency sweep of the old (parenchyma) curve and identifies the new (vessel) curve as being different from the old (parenchyma) curve. Based on the comparison results, the control circuit enables an action to be taken by the ultrasonic device to prevent cutting into the large vessel, and to inform the surgeon that a large vessel is located

on or in contact with the ultrasonic blade.

(690) The various actions that can be taken by the ultrasonic device include without limitation, change the therapeutic output of the device to prevent cutting of the vessel or change the tone from the generator to inform the surgeon that a vessel has been detected, or a combination thereof.

(691) Alternatively, various aspects of this technique may be applied to detect blood if a vessel had been cut, allowing the surgeon to quickly seal the vessel, even without seeing the cut vessel.

(692) FIG. 81 is a diagram 132340 of a liver resection 132350 with vessels 132354 (FIG. 82) embedded in parenchymal tissue, in accordance with at least one aspect of the present disclosure. An ultrasonic instrument 132342 including an ultrasonic blade 132344 and clamp arm 132346 is shown cutting into a liver 132348 to create a resection 132350. The ultrasonic instrument 132342 is coupled to a generator 132352 that controls the delivery of energy to the ultrasonic instrument 132342. Either the generator 132252 or the ultrasonic instrument 132342, or both, include a control circuit configured to execute the advanced smart blade algorithms discussed herein.

(693) FIG. 82 is a diagram 132356 of an ultrasonic blade 132344 in the process of cutting through parenchyma without contacting a vessel 132354 embedded in the liver 132348, in accordance with at least one aspect of the present disclosure. During the resection process the control circuit monitors the impedance, magnitude, and phase of the signals driving the ultrasonic transducer to assess the state of the jaw, e.g. the state of the ultrasonic blade 132344, as depicted in FIGS. 85A and 85B. Accordingly, as the ultrasonic blade 132344 resects liver 132348 the ultrasonic transducer produces a first response and when the ultrasonic blade 132344 contacts the embedded vessel 132354 the ultrasonic transducer produces a second response, which is associated with the embedded vessel 132354 type as described herein in connection with FIGS. 54-86.

(694) FIGS. 83A and 83B are graphical representations 132360 of ultrasonic transducer impedance magnitude/phase with the parenchyma curves 132362 shown in bold line, in accordance with at least one aspect of the present disclosure. FIG. 83A is a three-dimensional plot and FIG. 83B is a two-dimensional plot. These curves are generated in accordance with FIGS. 19-21, for example, and associated description under the heading ESTIMATING THE STATE OF THE JAW (PAD BURN THROUGH, STAPLES, BROKEN BLADE, BONE IN JAW, TISSUE IN JAW).

Alternatively, techniques for estimating or classifying the state of the jaw of an ultrasonic device described in connection with FIGS. 22-30 under the heading STATE OF JAW CLASSIFIER BASED ON MODEL and/or techniques for estimating the temperature of the ultrasonic blade are described in related U.S. Provisional Patent Application No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR may be employed.

(695) FIG. 84 is a diagram 132364 of an ultrasonic blade 132344 in the process of cutting through parenchyma and contacting a vessel 132354 embedded in the liver 132348, in accordance with at least one aspect of the present disclosure. As the ultrasonic blade 132344 transects the liver 132348 parenchyma tissue, the ultrasonic blade 132344 contacts the vessel 132354 at a location 132366 and thus shifts the resonant frequency of the ultrasonic transducer as depicted in FIGS. 85A and 85B. The control circuit monitors the impedance, magnitude, and phase of the signals driving the ultrasonic transducer to assess the state of the jaw, e.g. the state of the ultrasonic blade 132344 while contacting the vessel 132354, as depicted in FIGS. 85A and 85B.

(696) FIGS. 85A and 85B are graphical representations 132370 ultrasonic transducer impedance magnitude/phase with large vessel curves 132372 shown in bold line, in accordance with at least one aspect of the present disclosure. FIG. 85A is a three-dimensional plot and FIG. 85B is a two-dimensional plot. These curves are generated in accordance with FIGS. 19-21 and associated description under the heading ESTIMATING THE STATE OF THE JAW (PAD BURN THROUGH, STAPLES, BROKEN BLADE, BONE IN JAW, TISSUE IN JAW. Alternatively, techniques for estimating or classifying the state of the jaw of an ultrasonic device described in connection with FIGS. 22-30 under the heading STATE OF JAW CLASSIFIER BASED ON MODEL and/or techniques for estimating the temperature of the ultrasonic blade are described in related U.S. Provisional Patent Application No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR may be employed.

(697) FIG. 86 is a logic flow diagram 132380 depicting a control program or a logic configuration of a process for treating tissue in parenchyma when a vessel is detected as shown in FIGS. 84-85B, in accordance with at least one aspect of the present disclosure. According to the process, using the techniques for estimating or classifying the state of the jaw of an ultrasonic device described in connection with FIGS. 19-21 under the heading ESTIMATING THE STATE OF THE JAW (PAD BURN THROUGH, STAPLES, BROKEN BLADE, BONE IN JAW, TISSUE IN JAW and/or FIGS. 22-30 under the heading STATE OF JAW CLASSIFIER BASED ON MODEL and/or techniques for estimating the temperature of the ultrasonic blade are described in related U.S. Provisional Patent Application No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR, the control circuit determines if a vessel 132354 is located in contact with the ultrasonic blade 132344. If the control circuit detects a vessel 132382, the control circuit stops 132384 the cutting energy, switches 132386 to a lower power level, and sends 132388 a warning message or alert to the user. For example, the control circuit lowers the excitation voltage/current signal power to a level below what is required for cutting a vessel. The warning message or alert may include emitting a light, emitting a sounding, activating a buzzer, and the like. If a vessel 132354 is not detected, the resection process continues 132390.

Smart Ultrasonic Blade Application for Reusable and Disposable Devices

(698) The smart blade algorithm uses spectroscopy to identify the status of an ultrasonic blade. This capability can be applied to reusable and disposable devices with detachable clamp arms to distinguish if the disposable portion of the device has been installed correctly. The status of the ultrasonic blade may be determined using smart blade algorithm techniques for estimating or classifying the state of the jaw of an ultrasonic device described in connection with FIGS. 19-21 under the heading ESTIMATING THE STATE OF THE JAW (PAD BURN THROUGH, STAPLES, BROKEN BLADE, BONE IN JAW, TISSUE IN JAW and/or FIGS. 22-30 under the heading STATE OF JAW CLASSIFIER BASED ON MODEL and/or techniques for estimating the temperature of the ultrasonic blade are described in related U.S. Provisional Patent Application No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR, to Nott et al, which is incorporated herein by reference in its entirety.

(699) The smart blade algorithm techniques described herein can be employed to identify the status of components of reusable and disposable devices. In one aspect, the status of the ultrasonic blade may be determined to distinguish if disposable portions of reusable and disposable devices have been installed correctly or incorrectly.

(700) FIGS. 87 and 88 is a reusable and disposable ultrasonic device 132400 configured to identify the status of the ultrasonic blade 132402 and determine the clocked status of the clamp arm 132404 to determine whether a portion of the reusable and disposable ultrasonic device 132400 has been installed correctly, in accordance with at least one aspect of the present disclosure. FIG. 88 is an end effector 132406 portion of the reusable and disposable ultrasonic device 132400 shown in FIG. 87. Similarities and differences between the spectroscopy signatures can be used to determine whether the reusable and disposable components of the reusable and disposable ultrasonic device 132400 have been installed correctly or incorrectly.

(701) The reusable and disposable ultrasonic device 132400 shown in FIGS. 87 and 88 includes a reusable handle 132408 and a disposable ultrasonic waveguide/blade 132402. Prior to use, a proximal end 132410 of the disposable ultrasonic waveguide/blade 132402 is inserted 132414 into a distal opening 132412 of the reusable handle 132408 and twisted or rotated clockwise 132416 to lock the disposable ultrasonic waveguide/blade 132402 into the handle 132408 as shown in FIG. 87. If the disposable ultrasonic waveguide/blade 132402 is not fully inserted 132414 and/or fully rotated clockwise 132416, the reusable and disposable ultrasonic device 132400 will not operate properly. For example, improper insertion 132414 and rotation 132416 of the disposable ultrasonic waveguide/blade 132402 will result in poor mechanical coupling of the disposable ultrasonic waveguide/blade 132402 and will produce a different spectroscopy signature. Therefore, the smart blade algorithm techniques described herein can be used to determine if the disposable portion of the reusable and disposable ultrasonic device 132400 has been inserted 132414 and rotated 132416 completely.

(702) In another aspect, the configuration, if the clamp arm **132404** shown in FIG. **88** is clocked (rotated) relative to the ultrasonic blade **132402**, the orientation of the ultrasonic blade **132402** relative to the clamp arm **132404** will be out of alignment. This also will produce a different spectroscopy signature when the reusable and disposable ultrasonic device **132400** is actuated and/or clamped. Therefore, the smart blade algorithm techniques described herein can be used to determine if the disposable portion of the reusable and disposable ultrasonic device **132400** has been properly clocked (rotated) relative to the clamp arm **132404**.

(703) In another aspect, the smart blade algorithm techniques described herein can be used to determine if a disposable portion of the reusable and disposable ultrasonic device **132400** has been pushed in or inserted **132414** all the way into the reusable portion **132408**. This may be applicable to the reusable and disposable ultrasonic device **132400** in FIG. **89** below where a reusable portion such as the handle **132408**, for example, is **132414** inserted into a disposable portion, such as the ultrasonic blade **132402**, for example, prior to operation.

(704) FIG. **89** is a reusable and disposable ultrasonic device **132420** configured to identify the status of the ultrasonic blade **132422** and determine whether the clamp arm **132424** is not completely distal to determine whether a disposable portion **132426** of the reusable and disposable ultrasonic device **132420** has been installed correctly, in accordance with at least one aspect of the present disclosure. If the clamp arm is not installed completely distal there will be a different spectroscopy signature when the device is clamped. In another aspect, if the disposable portion **132426** is not installed completely distal on the reusable component **132428**, the ultrasonic blade **132422** spectroscopy signature will be different when clamped into position. Therefore, the smart blade algorithm techniques described herein can be used to determine if the disposable portion **132426** of the reusable and disposable ultrasonic device **132420** has been fully and properly coupled to the reusable portion **132428**.

(705) FIG. **90** is a logic flow diagram **132430** depicting a control program or a logic configuration to identify the status of components of reusable and disposable ultrasonic devices, in accordance with at least one aspect of the present disclosure. According to the process depicted by the logic flow diagram **132430**, a control circuit of the generator or instrument executes a smart blade algorithm technique and determines **132432** the spectroscopy signature of assembled reusable and disposable ultrasonic devices **132400**, **132420** (FIGS. **88** and **89**) comprising reusable and disposable components. The control circuit compares **132434** the measured spectroscopy signature to a reference spectroscopy signature, where the reference spectroscopy signature is associated with a properly assembled reusable and disposable ultrasonic device **132400**, **132420** and is stored in a database or memory of the generator or instrument. When the control circuit determines **132436** that the measured spectroscopy signature differs from the reference spectroscopy signature, the control circuit disables **132438** the operation of the reusable and disposable ultrasonic device **132400**, **132420** and generates **132440** a warning that can be perceived by the user. The warning may include activating a light source sound source, or vibration source. When the measured spectroscopy signature is the same or substantially similar to the reference spectroscopy signature, the control circuit enables **132442** the normal operation of the reusable and disposable ultrasonic device **132400**, **132420**.

Live Time Tissue Classification Using Electrical Parameters

Sealing without Cutting, RF/Ultrasonic Combination Technology, Tailored Algorithms

(706) In one aspect, the present disclosure provides an algorithm for classifying tissue into groups. The ability to classify tissue in live time will allow for tailoring algorithms to a specific tissue group. The tailored algorithms can optimize seal times and hemostasis across all tissue types. In one aspect, the present disclosure provides a sealing algorithm to provide hemostasis needed for large vessels and quickly seal smaller structures that do not need extended energy activation. The ability to classify these distinct tissue types allows for optimized algorithms for each group in live time.

(707) In this aspect, during the first 0.75 seconds of the activation, 3 RF electrical parameters are used in a plot to classify tissue into distinct groups. These electrical parameters are: Initial RF impedance (taken at 0.15 seconds), Minimum RF impedance in first 0.75 seconds, and the amount of time the RF impedance slope is -0 in milliseconds. A plurality of other times that these data points are taken could be implemented. All of this data is collected in a set amount of time, and then using a Support Vector Machine (SVM) or another classification algorithm the tissue can be classified into a distinct group in live time. Each tissue group would have an algorithm specific to it that would be implemented for the remainder of the activation. Types of SVM's include linear, polynomial, and radial basis function (RBF).

(708) FIG. **91** is a three-dimensional graphical representation **132450** of epidermal growth factor (EGF) radio frequency (RF) tissue impedance classification, in accordance with at least one aspect of the present disclosure. The x-axis represents the minimum RF impedance (Z_{min}) of the tissue, the y-axis represents initial RF impedance (Z_{init}) of the tissue, and the z-axis represents the amount of time that the derivative of the RF impedance (Z) of the tissue is approximately 0. FIG. **91** shows a grouping of large vessels **132452**, e.g., carotids—thick tissue, and small vessels **132454**, e.g., thyros—thin tissue, when using the three RF parameters of Initial RF impedance, Minimum RF Impedance, and the amount of time the derivative (slope) of the RF impedance is approximately zero within the first 0.75 seconds of activation. A distinction of this classification method is that the tissue type can be classified in a set amount of time. The advantage to this method is a tissue specific algorithm can be chosen towards the beginning of the activation, so specialized tissue treatment can begin before the exit out of the RF bathtub. It will be appreciated that in the context of tissue impedance under the influence of RF energy a bathtub region is a curve wherein the tissue impedance drops after the initial application of RF energy and stabilizes until the tissue begins to dry out. Thereafter the tissue impedance increases. Thus, the impedance versus time curve resembles the shape of a “bathtub.”

(709) This data was used to train and test a Support Vector Machine to group thick and thin tissue, and accurately classified 94% of the time.

(710) In one aspect, the present disclosure provides a device comprising one combo RF/Ultrasonic algorithm that is used for all tissue types and it has been identified that seal speeds for thin tissues are longer than necessary, however larger vessels and thicker structures could benefit from an extended activation. This classification scheme will enable the combo RF/ultrasonic device to seal small structures with optimal speeds and burst pressures, and to seal larger structures to ensure maximum hemostasis is achieved.

(711) FIG. **92** is a three-dimensional graphical representation **132460** of epidermal growth factor (EGF) radio frequency (RF) tissue impedance analysis, in accordance with at least one aspect of the present disclosure. The x-axis represents the minimum RF impedance (Z_{min}) of the tissue, the y-axis represents the initial RF impedance (Z_{init}) of the tissue, and the z-axis represents the amount of time that the derivative of the RF impedance (Z) of the tissue is approximately 0. To determine if this classification model of thick tissue **132462** versus thin tissue **132464** was robust to different tissue types, data was added for various benchtop tissue types, and the tissue grouped into two distinct groups. It is possible to separate this data into a plurality of groups if it is deemed beneficial or necessary. The different thick tissue **132462** types include, for example, carotid, jejunum, mesentery, jugular, and liver tissue. The different thin tissue **132464** types include, for example, thyro and thyro vein.

Fine Dissection Mode for Tissue Classification

Tissue Classification to Enable Multiple Modes to Account for Different Surgical Techniques

(712) In one aspect, the present disclosure provides an algorithm for classifying tissue into groups and tailoring an algorithm to classify specific tissue classes in live time. This disclosure builds upon the foundation and details of another potential benefit to classifying tissue as previously discussed herein under the heading LIVE TIME TISSUE CLASSIFICATION USING ELECTRICAL PARAMETERS.

(713) FIG. **93** is a graphical representation **132470** of carotid technique sensitivity where the time that the RF impedance (Z) derivative is approximately 0 is plotted as a function of initial RF impedance, in accordance with at least one aspect of the present disclosure. It is known that different surgical techniques exist in different regions of the world, and vary widely from surgeon to surgeon. For this reason, a technique mode may be provided on the generator to enable more efficient energy delivery based upon the user's specific surgical technique such as, for example, a tip bite of tissue versus a full bite of tissue. A tip bite refers to the end effector of a surgical device grasping tissue at the tip only. A full bite refers to the end effector of a surgical device grasping tissue within the entire end effector. The generator may be configured to detect if a user is consistently operating with a tip bite of tissue or a full bite of tissue. As shown in FIG. **93** initial RF impedance data was measured and plotted for tip bites as Group 1 **132472** and full bites as Group 2 **132474**. As shown, Group 1 **132472** tip bites of tissue register an initial RF impedance $Z_{sub.Init}$ of less

than 250 Ohms and Group 2 **132474** full bites of tissue register an initial impedance $Z_{\text{sub.Init}}$ between 250 Ohms and 500 Ohms, the maximum RF tissue impedance $Z_{\text{sub.Max}}$. Upon detecting whether the user grasps a tip bite of tissue or a full bite of tissue, the algorithm can suggest a predetermined dissection mode. For example, for a tip bite of tissue the algorithm may suggest a fine dissection mode to the user or this option may be selected before a procedure. For example, for a full bite of tissue the algorithm may suggest a course dissection mode to the user or this option may be selected before a procedure. In fine dissection mode, the algorithm may be tailored to optimize energy delivery for this surgical technique by lowering the ultrasonic displacement to protect the clamp arm pad from burning through. Also it is known that tip biting has a greater amount of RF noise, causing longer seal times, and greater variation in seal performance. Fine dissection mode for tip biting could have an algorithm tailored to having a lower RF termination impedance and/or different filtering signal to increase accuracy of energy delivery.

(714) A technique sensitivity analysis was conducted as part of the development work for classification. The testing was conducted by transecting 3-7 mm vessels in a benchtop setting using different surgical techniques such as full bite transection with and without tension, and tip bite transection with and without tension. The initial RF impedance, and the time the slope the RF impedance=0 were all examined as significant factors in classifying the tissue into groups.

(715) It was determined that surgical techniques could be grouped into 3 distinct groups based on the initial RF impedance $Z_{\text{sub.Init}}$. Initial RF impedance $Z_{\text{sub.Init}}$ generally ranging between 0-100 ohms indicates operating in a bloody field. Initial RF impedance $Z_{\text{sub.Init}}$ generally ranging between 100-300 ohms indicates operating under normal conditions, and initial RF impedance $Z_{\text{sub.Init}}$ greater than 300 ohms indicates abuse condition especially where tensions is present.

Temperature Inference

(716) FIGS. **94A-94B** are graphical representations **133000**, **133010** of complex impedance spectra of the same ultrasonic device with a cold (room temperature) and hot ultrasonic blade, in accordance with at least one aspect of the present disclosure. As used herein, a cold ultrasonic blade refers to an ultrasonic blade at room temperature and a hot ultrasonic blade refers to an ultrasonic blade after it is frictionally heated in use. FIG. **94A** is a graphical representation **133000** of impedance phase angle ϕ as a function of resonant frequency $f_{\text{sub.o}}$ of the same ultrasonic device with a cold and hot ultrasonic blade and FIG. **94B** is a graphical representation **133010** of impedance magnitude $|Z|$ as a function of resonant frequency $f_{\text{sub.o}}$ of the same ultrasonic device with a cold and hot ultrasonic blade. The impedance phase angle ϕ and impedance magnitude $|Z|$ are at a minimum at the resonant frequency $f_{\text{sub.o}}$.

(717) The ultrasonic transducer impedance $Z_{\text{sub.g(t)}}$ can be measured as the ratio of the drive signal generator voltage $V_{\text{sub.g(t)}}$ and current $I_{\text{sub.g(t)}}$ drive signals:

$$(718) Z_g(t) = \frac{V_g(t)}{I_g(t)}$$

(719) As shown in FIG. **94A**, when the ultrasonic blade is cold, e.g., at room temperature and not frictionally heated, the electromechanical resonant frequency $f_{\text{sub.o}}$ of the ultrasonic device is approximately 55,500 Hz and the excitation frequency of the ultrasonic transducer is set to 55,500 Hz. Thus, when the ultrasonic transducer is excited at the electromechanical resonant frequency $f_{\text{sub.o}}$ and the ultrasonic blade is cold the phase angle ϕ is at minimum or approximately 0 Rad as indicated by the cold blade plot **133002**. As shown in FIG. **94B**, when the ultrasonic blade is cold and the ultrasonic transducer is excited at the electromechanical resonant frequency $f_{\text{sub.o}}$, the impedance magnitude $|Z|$ is 800 Ω , e.g., the impedance magnitude $|Z|$ is at a minimum impedance, and the drive signal amplitude is at a maximum due to the series resonance equivalent circuit of the ultrasonic electromechanical system as depicted in FIG. **6**.

(720) With reference now back to FIGS. **94A** and **94B**, when the ultrasonic transducer is driven by generator voltage $V_{\text{sub.g(t)}}$ and generator current $I_{\text{sub.g(t)}}$ signals at the electromechanical resonant frequency $f_{\text{sub.o}}$ of 55,500 Hz, the phase angle ϕ between the generator voltage $V_{\text{sub.g(t)}}$ and generator current $I_{\text{sub.g(t)}}$ signals is zero, the impedance magnitude $|Z|$ is at a minimum impedance, e.g., 800 Ω , and the signal amplitude is at a peak or maximum due to the series resonance equivalent circuit of the ultrasonic electromechanical system. As the temperature of the ultrasonic blade increases, due to frictional heat generated in use, the electromechanical resonant frequency $f_{\text{sub.o'}}$ of the ultrasonic device decreases. Since the ultrasonic transducer is still driven by generator voltage $V_{\text{sub.g(t)}}$ and generator current $I_{\text{sub.g(t)}}$ signals at the previous (cold blade) electromechanical resonant frequency $f_{\text{sub.o}}$ of 55,500 Hz, the ultrasonic device operates off-resonance $f_{\text{sub.o'}}$ causing a shift in the phase angle ϕ between the generator voltage $V_{\text{sub.g(t)}}$ and generator current $I_{\text{sub.g(t)}}$ signals. There is also an increase in impedance magnitude $|Z|$ and a drop in peak magnitude of the drive signal relative to the previous (cold blade) electromechanical resonant frequency of 55,500 Hz. Accordingly, the temperature of the ultrasonic blade may be inferred by measuring the phase angle ϕ between the generator voltage $V_{\text{sub.g(t)}}$ and the generator current $I_{\text{sub.g(t)}}$ signals as the electromechanical resonant frequency $f_{\text{sub.o}}$ changes due to the changes in temperature of the ultrasonic blade.

(721) As previously described, an electromechanical ultrasonic system includes an ultrasonic transducer, a waveguide, and an ultrasonic blade. As previously discussed, the ultrasonic transducer may be modeled as an equivalent series resonant circuit (see FIG. **6**) comprising first branch having a static capacitance and a second "motional" branch having a serially connected inductance, resistance and capacitance that define the electromechanical properties of a resonator. The electromechanical ultrasonic system has an initial electromechanical resonant frequency defined by the physical properties of the ultrasonic transducer, the waveguide, and the ultrasonic blade. The ultrasonic transducer is excited by an alternating voltage $V_{\text{sub.g(t)}}$ and current $I_{\text{sub.g(t)}}$ signal at a frequency equal to the electromechanical resonant frequency, e.g., the resonant frequency of the electromechanical ultrasonic system. When the electromechanical ultrasonic system is excited at the resonant frequency, the phase angle ϕ between the voltage $V_{\text{sub.g(t)}}$ and current $I_{\text{sub.g(t)}}$ signals is zero.

(722) Stated in another way, at resonance, the analogous inductive impedance of the electromechanical ultrasonic system is equal to the analogous capacitive impedance of the electromechanical ultrasonic system. As the ultrasonic blade heats up, for example due to frictional engagement with tissue, the compliance of the ultrasonic blade (modeled as an analogous capacitance) causes the resonant frequency of the electromechanical ultrasonic system to shift. In the present example, the resonant frequency of the electromechanical ultrasonic system decreases as the temperature of the ultrasonic blade increases. Thus, the analogous inductive impedance of the electromechanical ultrasonic system is no longer equal to the analogous capacitive impedance of the electromechanical ultrasonic system causing a mismatch between the drive frequency and the new resonant frequency of the electromechanical ultrasonic system. Thus, with a hot ultrasonic blade, the electromechanical ultrasonic system operates "off-resonance." The mismatch between the drive frequency and the resonant frequency is manifested as a phase angle ϕ between the voltage $V_{\text{sub.g(t)}}$ and current $I_{\text{sub.g(t)}}$ signals applied to the ultrasonic transducer.

(723) As previously discussed, the generator electronics can easily monitor the phase angle ϕ between the voltage $V_{\text{sub.g(t)}}$ and current $I_{\text{sub.g(t)}}$ signals applied to the ultrasonic transducer. The phase angle ϕ may be determined through Fourier analysis, weighted least-squares estimation, Kalman filtering, space-vector-based techniques, zero-crossing method, Lissajous figures, three-voltmeter method, crossed-coil method, vector voltmeter and vector impedance methods, phase standard instruments, phase-locked loops, among other techniques previously described. The generator can continuously monitor the phase angle ϕ and adjust the drive frequency until the phase angle ϕ goes to zero. At this point, the new drive frequency is equal to the new resonant frequency of the electromechanical ultrasonic system. The change in phase angle ϕ and/or generator drive frequency can be used as an indirect or inferred measurement of the temperature of the ultrasonic blade.

(724) A variety of techniques are available to estimate temperature from the data in these spectra. Most notably, a time variant, non-linear set of state space equations can be employed to model the dynamic relationship between the temperature of the ultrasonic blade and the measured impedance:

$$(725) Z_g(t) = \frac{V_g(t)}{I_g(t)}$$

across a range of generator drive frequencies, where the range of generator drive frequencies is specific to device model.

Methods of Temperature Estimation

(726) One aspect of estimating or inferring the temperature of an ultrasonic blade may include three steps. First, define a state space model of temperature and frequency that is time and energy dependent. To model temperature as a function of frequency content, a set of non-linear state space equations are used to model the relationship between the electromechanical resonant frequency and the temperature of the ultrasonic blade. Second, apply a Kalman filter to improve the accuracy of the temperature estimator and state space model over time. Third, a state estimator is provided in the feedback loop of the Kalman filter to control the power applied to the ultrasonic transducer, and hence the ultrasonic blade, to regulate the temperature of the ultrasonic blade. The three steps are described hereinbelow.

Step 1

(727) The first step is to define a state space model of temperature and frequency that is time and energy dependent. To model temperature as a function of frequency content, a set of non-linear state space equations are used to model the relationship between the electromechanical resonant frequency and the temperature of the ultrasonic blade. In one aspect, the state space model is defined by:

$$(728) \begin{bmatrix} \dot{F}_n \\ \dot{T} \end{bmatrix} = f(t, T(t), F_n(t), E(t)) \dot{y} = h(t, T(t), F_n(t), E(t))$$

(729) The state space model represents the rate of change of the natural frequency of the electromechanical ultrasonic system $\{\dot{F}\}_{\text{sub.n}}$ and the rate of change of the temperature $\{\dot{T}\}$ of the ultrasonic blade with respect to natural frequency $F_{\text{sub.n}}(t)$, temperature $T(t)$, energy $E(t)$, and time t . $\{\dot{y}\}$ represents the observability of variables that are measurable and observable such as the natural frequency $F_{\text{sub.n}}(t)$ of the electromechanical ultrasonic system, the temperature $T(t)$ of the ultrasonic blade, the energy $E(t)$ applied to the ultrasonic blade, and time t . The temperature $T(t)$ of the ultrasonic blade is observable as an estimate.

Step 2

(730) The second step is to apply a Kalman filter to improve temperature estimator and state space model. FIG. 95 is a diagram of a Kalman filter **133020** to improve the temperature estimator and state space model based on impedance according to the equation:

$$(731) 0Z_g(t) = \frac{V_g(t)}{I_g(t)}$$

which represents the impedance across an ultrasonic transducer measured at a variety of frequencies, in accordance with at least one aspect of the present disclosure.

(732) The Kalman filter **133020** may be employed to improve the performance of the temperature estimate and allows for the augmentation of external sensors, models, or prior information to improve temperature prediction in the midst of noisy data. The Kalman filter **133020** includes a regulator **133022** and a plant **133024**. In control theory a plant **133024** is the combination of process and actuator. A plant **133024** may be referred to with a transfer function which indicates the relation between an input signal and the output signal of a system. The regulator **133022** includes a state estimator **133026** and a controller K **133028**. The state regulator **133026** includes a feedback loop **133030**. The state regulator **133026** receives y , the output of the plant **133024**, as an input and a feedback variable u . The state estimator **133026** is an internal feedback system that converges to the true value of the state of the system. The output of the state estimator **133026** is $\{\text{circumflex over (x)}\}$, the full feedback control variable including $F_{\text{sub.n}}(t)$ of the electromechanical ultrasonic system, the estimate of the temperature $T(t)$ of the ultrasonic blade, the energy $E(t)$ applied to the ultrasonic blade, the phase angle ϕ , and time t . The input into the controller K **133028** is $\{\text{circumflex over (x)}\}$ and the output of the controller K **133028** u is fed back to the state estimator **133026** and t of the plant **133024**.

(733) Kalman filtering, also known as linear quadratic estimation (LQE), is an algorithm that uses a series of measurements observed over time, containing statistical noise and other inaccuracies, and produces estimates of unknown variables that tend to be more accurate than those based on a single measurement alone, by estimating a joint probability distribution over the variables for each timeframe and thus calculating the maximum likelihood estimate of actual measurements. The algorithm works in a two-step process. In a prediction step, the Kalman filter **133020** produces estimates of the current state variables, along with their uncertainties. Once the outcome of the next measurement (necessarily corrupted with some amount of error, including random noise) is observed, these estimates are updated using a weighted average, with more weight being given to estimates with higher certainty. The algorithm is recursive and can run in real time, using only the present input measurements and the previously calculated state and its uncertainty matrix; no additional past information is required.

(734) The Kalman filter **133020** uses a dynamics model of the electromechanical ultrasonic system, known control inputs to that system, and multiple sequential measurements (observations) of the natural frequency and phase angle of the applied signals (e.g., magnitude and phase of the electrical impedance of the ultrasonic transducer) to the ultrasonic transducer to form an estimate of the varying quantities of the electromechanical ultrasonic system (its state) to predict the temperature of the ultrasonic blade portion of the electromechanical ultrasonic system that is better than an estimate obtained using only one measurement alone. As such, the Kalman filter **133020** is an algorithm that includes sensor and data fusion to provide the maximum likelihood estimate of the temperature of the ultrasonic blade.

(735) The Kalman filter **133020** deals effectively with uncertainty due to noisy measurements of the applied signals to the ultrasonic transducer to measure the natural frequency and phase shift data and also deals effectively with uncertainty due to random external factors. The Kalman filter **133020** produces an estimate of the state of the electromechanical ultrasonic system as an average of the predicted state of the system and of the new measurement using a weighted average. Weighted values provide better (i.e., smaller) estimated uncertainty and are more “trustworthy” than unweighted values. The weights may be calculated from the covariance, a measure of the estimated uncertainty of the prediction of the system's state. The result of the weighted average is a new state estimate that lies between the predicted and measured state, and has a better estimated uncertainty than either alone. This process is repeated at every time step, with the new estimate and its covariance informing the prediction used in the following iteration. This recursive nature of the Kalman filter **133020** requires only the last “best guess,” rather than the entire history, of the state of the electromechanical ultrasonic system to calculate a new state.

(736) The relative certainty of the measurements and current state estimate is an important consideration, and it is common to discuss the response of the filter in terms of the gain K of the Kalman filter **133020**. The Kalman gain K is the relative weight given to the measurements and current state estimate, and can be “tuned” to achieve particular performance. With a high gain K, the Kalman filter **133020** places more weight on the most recent measurements, and thus follows them more responsively. With a low gain K, the Kalman filter **133020** follows the model predictions more closely. At the extremes, a high gain close to one will result in a more jumpy estimated trajectory, while low gain close to zero will smooth out noise but decrease the responsiveness.

(737) When performing the actual calculations for the Kalman filter **133020** (as discussed below), the state estimate and covariances are coded into matrices to handle the multiple dimensions involved in a single set of calculations. This allows for a representation of linear relationships between different state variables (such as position, velocity, and acceleration) in any of the transition models or covariances. Using a Kalman filter **133020** does not assume that the errors are Gaussian. However, the Kalman filter **133020** yields the exact conditional probability estimate in the special case that all errors are Gaussian-distributed.

Step 3

(738) The third step uses a state estimator **133026** in the feedback loop **133032** of the Kalman filter **133020** for control of power applied to the ultrasonic transducer, and hence the ultrasonic blade, to regulate the temperature of the ultrasonic blade.

(739) FIG. 96 is a graphical depiction **133040** of three probability distributions employed by the state estimator **133026** of the Kalman filter **133020** shown in FIG. 95 to maximize estimates, in accordance with at least one aspect of the present disclosure. The probability distributions include the prior probability distribution **133042**, the prediction (state) probability distribution **133044**, and the observation probability distribution **133046**. The three probability distributions **133042**, **133044**, **133046** are used in feedback control of power applied to an ultrasonic transducer to regulate

temperature based on impedance across the ultrasonic transducer measured at a variety of frequencies, in accordance with at least one aspect of the present disclosure. The estimator used in feedback control of power applied to an ultrasonic transducer to regulate temperature based on impedance is defined by the expression:

$$(740) Z_g(t) = \frac{V_g(t)}{I_g(t)}$$

which is the impedance across the ultrasonic transducer measured at a variety of frequencies, in accordance with at least one aspect of the present disclosure.

(741) The prior probability distribution **133042** includes a state variance defined by the expression:

$$(\sigma.\text{sub.k.sup.-}).\text{sup.2} = \sigma.\text{sub.k-1.sup.2} + \sigma.\text{sub.P.sub.k.sup.2}$$

The state variance $(\sigma.\text{sub.k.sup.-})$ is used to predict the next state of the system, which is represented as the prediction (state) probability distribution

133044. The observation probability distribution **133046** is the probability distribution of the actual observation of the state of the system where the observation variance $\sigma.\text{sub.m}$ is used to define the gain, which is defined by the following expression:

$$(742) K = \frac{(\hat{k})^2}{(\hat{k})^2 + \frac{\sigma}{m}}$$

Feedback Control

(743) Power input is decreased to ensure that the temperature (as estimated by the state estimator and of the Kalman filter) is controlled.

(744) In one aspect, the initial proof of concept assumed a static, linear relationship between the natural frequency of the electromechanical ultrasonic system and the temperature of the ultrasonic blade. By reducing the power as a function of the natural frequency of the electromechanical ultrasonic system (i.e., regulating temperature with feedback control), the temperature of the ultrasonic blade tip could be controlled directly. In this example, the temperature of the distal tip of the ultrasonic blade can be controlled to not exceed the melting point of the Teflon pad.

(745) FIG. **97A** is a graphical representation **133050** of temperature versus time of an ultrasonic device without temperature feedback control.

Temperature ($^{\circ}\text{C.}$) of the ultrasonic blade is shown along the vertical axis and time (sec) is shown along the horizontal axis. The test was conducted with a chamois located in the jaws of the ultrasonic device. One jaw is the ultrasonic blade and the other jaw is the clamp arm with a TEFLON pad. The ultrasonic blade was excited at the resonant frequency while in frictional engagement with the chamois clamped between the ultrasonic blade and the clamp arm. Over time, the temperature ($^{\circ}\text{C.}$) of the ultrasonic blade increases due to the frictional engagement with the chamois. Over time, the temperature profile **133052** of the ultrasonic blade increases until the chamois sample is cut after about 19.5 seconds at a temperature of 220°C. as indicated at point **133054**. Without temperature feedback control, after cutting the chamois sample, the temperature of the ultrasonic blade increases to a temperature well above the melting point of TEFLON $\sim 380^{\circ}\text{C.}$ up to $\sim 490^{\circ}\text{C.}$ At point **133056** the temperature of the ultrasonic blade reaches a maximum temperature of 490°C. until the TEFLON pad is completely melted. The temperature of the ultrasonic blade drops slightly from the peak temperature at point **133056** after the pad is completely gone.

(746) FIG. **97B** is a plot of temperature versus time of an ultrasonic device with temperature feedback control, in accordance with at least one aspect of the present disclosure. Temperature ($^{\circ}\text{C.}$) of the ultrasonic blade is shown along the vertical axis and the time (sec) is shown along the horizontal axis. The test was conducted with a chamois sample located in the jaws of the ultrasonic device. One jaw is the ultrasonic blade and the other jaw is the clamp arm with a TEFLON pad. The ultrasonic blade was excited at the resonant frequency while in frictional engagement with the chamois clamped between the ultrasonic blade and the clamp arm pad. Over time, the temperature profile **133062** of the ultrasonic blade increases until the chamois sample is cut after about 23 seconds at a temperature of 220°C. as indicated at point **133064**. With temperature feedback control, the temperature of the ultrasonic blade increases up to a maximum temperature of about 380°C. , just below the melting point of TEFLON, as indicated at point **133066** and then is lowered to an average of about 330°C. as indicated generally at region **133068**, thus preventing the TEFLON pad from melting.

Controlled Thermal Management (CTM) for Pad Protection

(747) In one aspect, the present disclosure provides a controlled thermal management (CTM) algorithm to regulate temperature with feedback control. The output of the feedback control can be used to prevent the ultrasonic end effector clamp arm pad from burning through, which is not a desirable effect for ultrasonic surgical instruments. As previously discussed, in general, pad burn through is caused by the continued application of ultrasonic energy to an ultrasonic blade in contact with the pad after tissue grasped in the end effector has been transected.

(748) The CTM algorithm leverages the fact that the resonant frequency of an ultrasonic blade, general made of titanium, varies in proportion to temperature. As the temperature increases, the modulus of elasticity of the ultrasonic blade decreases, and so does the natural frequency of the ultrasonic blade. A factor to consider is that when the distal end of the ultrasonic blade is hot but the waveguide is cold, there is a different frequency difference (Δ) to achieve a predetermined temperature than when the distal end of the ultrasonic blade and the waveguide are both hot.

(749) In one aspect, the CTM algorithm calculates a change in frequency of the ultrasonic transducer drive signal that is required to reach a certain predetermined temperature as a function of the resonant frequency of the ultrasonic electromechanical system at the beginning of activation (at lock). The ultrasonic electromechanical system comprising an ultrasonic transducer coupled to an ultrasonic blade by an ultrasonic waveguide has a predefined resonant frequency that varies with temperature. The resonant frequency of the ultrasonic electromechanical system at lock can be employed to estimate the change in ultrasonic transducer drive frequency that is required to achieve a temperature end point to account for the initial thermal state of the ultrasonic blade. The resonant frequency of the ultrasonic electromechanical system can vary as a function of temperature of the ultrasonic transducer or ultrasonic waveguide or ultrasonic blade or a combination of these components.

(750) FIG. **98** is a graphical representation **133300** of the relationship between initial resonant frequency (frequency at lock) and the change in frequency (Δ frequency) required to achieve a temperature of approximately 340°C. , in accordance with at least one aspect of the present disclosure. The change in frequency required to reach an ultrasonic blade temperature of approximately 340°C. is shown along the vertical axis and the resonant frequency of the electromechanical ultrasonic system at lock is shown along the horizontal axis. Based on measurement data points **133302** shown as scatter plot there is a linear relationship **133304** between the change in frequency required to reach an ultrasonic blade temperature of approximately 340°C. and the resonant frequency at lock.

(751) At resonant frequency lock, the CTM algorithm employs the linear relationship **133304** between the lock frequency and the Δ frequency required to achieve a temperature just below the melting point of a TEFLON pad (approximately 340°C.). Once the frequency is within a certain buffer distance from a lower bound on frequency, as shown in FIG. **99**, a feedback control system **133310** comprising an ultrasonic generator **133312** regulates the electrical current (i) set point applied to the ultrasonic transducer of the ultrasonic electromechanical system **133314** to prevent the frequency (f) of the ultrasonic transducer from decreasing lower than a predetermined threshold, in accordance with at least one aspect of the present disclosure. Decreasing the electrical current set point decreases the displacement of the ultrasonic blade, which in turn decreases the temperature of the ultrasonic blade and increases the natural frequency of the ultrasonic blade. This relationship allows a change in the electrical current applied to the ultrasonic transducer to regulate the natural frequency of the ultrasonic blade and indirectly control the temperature of the ultrasonic blade or the ultrasonic electromechanical system **133314**. In one aspect, the generator **133312** may be implemented as the ultrasonic generator described with reference to FIGS. **2**, **7**, **8A-8C**, and **9A-9B**, for example. The feedback control system **133310** may be implemented as the PID controller described with reference to FIGS. **16-17**, for example.

(752) FIG. **100** is a flow diagram **133320** of a process or logic configuration of a controlled thermal management (CTM) algorithm to protect the clamp arm pad in an ultrasonic end effector, in accordance with at least one aspect of the present disclosure. The process or logic configuration illustrated by way of the flow diagram **133320** may be executed by the ultrasonic generator **133312** as described herein or by control circuits located in the ultrasonic instrument or a combination thereof. As previously discussed, the generator **133312** may be implemented as the generator described

with reference to FIGS. 2, 7, 8A-8C, and 9A-9B, for example.

(753) In one aspect, initially the control circuit in the generator **133312** activates the ultrasonic instrument by applying an electrical current to the ultrasonic transducer. The resonant frequency of the ultrasonic electromechanical system is initially locked at initial conditions where the ultrasonic blade temperature is cold or close to room temperature. As the temperature of the ultrasonic blade increases due to frictional contact with tissue, for example, the control circuit monitors the change or delta in the resonant frequency of the ultrasonic electromechanical system and determines **133324** whether the delta frequency threshold for a predetermined blade temperature has been reached. If the delta frequency is below the threshold, the process continues along the NO branch and the control circuit continues to seek **133325** the new resonant frequency and monitor the delta frequency. When the delta frequency meets or exceeds the delta frequency threshold, the process continues along the YES branch and calculates **133326** a new lower frequency limit (threshold), which corresponds to the melting point of the clamp arm pad. In one non-limiting example, the clamp arm pad is made of TEFLON and the melting point is approximately 340° C.

(754) Once a new frequency lower limit is calculated **133326**, the control circuit determines **133328** if the resonant frequency is near the newly calculated lower frequency limit. For example, in the case of a TEFLON clamp arm pad, the control circuit determines **133328** if the ultrasonic blade temperature is approaching 350° C., for example, based on the current resonant frequency. If the current resonant frequency is above the lower frequency limit, the process continues along the NO branch and applies **133330** a normal level of electrical current to the ultrasonic transducer suitable for tissue transection. Alternatively, if the current resonant frequency is at or below the lower frequency limit, the process continues along the YES branch and regulates **133332** the resonant frequency by modifying the electrical current applied to the ultrasonic transducer. In one aspect, the control circuit employs a PID controller as described with reference to FIGS. 16-17, for example. The control circuit regulates **133332** the frequency in a loop to determine **133328** when the frequency is near the lower limit until the “seal and cut” surgical procedure is terminated and the ultrasonic transducer is deactivated. Since the CTM algorithm depicted by the logic flow diagram **133320** only has an effect at or near the melting point of the clamp arm pad, the CTM algorithm is activated after the tissue is transected.

(755) Burst pressure testing conducted on samples indicates that there is no impact on the burst pressure of the seal when the CTM process or logic configuration depicted by the logic flow diagram **133320** is employed to seal and cut vessels or other tissue. Furthermore, based on test samples, transection times were affected. Moreover, temperature measurements confirm that the ultrasonic blade temperature is bounded by the CTM algorithm compared to devices without CTM feedback algorithm control and devices that underwent 10 firings at maximum power for ten seconds against the pad with 5 seconds rest between firings showed significantly reduced pad wear whereas no device without CTM algorithm feedback control lasted more than 2 firings in this abuse test.

(756) FIG. **101** is a graphical representation **133340** of temperature versus time comparing the desired temperature of an ultrasonic blade with a smart ultrasonic blade and a conventional ultrasonic blade, in accordance with at least one aspect of the present disclosure. Temperature (deg. C.) is shown along the vertical axis and Time (sec) is shown along the horizontal axis. In the plot, the dash-dot line is a temperature threshold **133342** that represents the desired temperature of the ultrasonic blade. The solid line is a temperature versus time curve **133344** of a smart ultrasonic blade under the control of the CTM algorithm described with reference to FIGS. **99** and **100**. The dotted line is a temperature versus time curve **133346** of a regular ultrasonic blade that is not under the control of the CTM algorithm described with reference to FIGS. **99** and **100**. As shown. Once the temperature of the smart ultrasonic blade under the control of the CTM algorithm exceeds the desired temperature threshold (~340° C.), the CTM algorithm takes control and regulates the temperature of the smart ultrasonic blade to match the threshold as closely as possible until the transection procedure is completed and the power to the ultrasonic transducer is deactivated or cut off.

(757) In another aspect, the present disclosure provides a CTM algorithm for a “seal only” tissue effect by an ultrasonic device, such as ultrasonic shears, for example. Generally speaking, ultrasonic surgical instruments typically seal and cut tissue simultaneously. Creating an ultrasonic device configured to seal only without cutting has not been difficult to achieve using ultrasonic technology alone due to the uncertainty of knowing when the seal was completed before initiating the cutting. In one aspect, the CTM algorithm may be configured to protect the end effector clamp arm pad by allowing the temperature of the ultrasonic blade to exceed the temperature required for cutting (transecting) the tissue but not to exceed the melting point of the clamp arm pad. In another aspect, the CTM seal only algorithm may be tuned to exceed the sealing temperature of tissue (approximately 115° C. to approximately 180° C. based on experimentation) but not to exceed the cutting (transecting) temperature of tissue (approximately 180° C. to approximately 350° C.). In the latter configuration, the CTM seal only algorithm provides a “seal only” tissue effect that has been successfully demonstrated. In a linear fit that calculates the change in frequency with respect to the initial lock frequency, as shown in FIG. **98**, for example, changing the intercept of the fit regulates the final steady state temperature of the ultrasonic blade. By adjusting the intercept parameter, the ultrasonic blade can be set to never exceed approximately 180° C. resulting in the tissue sealing but not cutting. In one aspect, increasing the clamp force may improve the sealing process without impacting clamp arm pad burn through because the temperature of the blade is controlled by the CTM seal only algorithm. As previously described, the CTM seal only algorithm may be implemented by the generator and PID controller described with reference to FIGS. 2, 7, 8A-8C, 9A-9B, and 16-17, for example. Accordingly, the flow diagram **133320** shown in FIG. **100** may be modified such that the control circuit calculates **133326** a new lower frequency limit (threshold) to correspond with a “seal only” temperature such as, for example, approximately 180° C., determine **133328** when the frequency is near the lower limit, and regulate **133332** the temperature until the “seal only” surgical procedure is terminated and the ultrasonic transducer is deactivated.

(758) In another aspect, the present disclosure provides a cool thermal monitoring (CTMo) algorithm configured to detect when atraumatic grasping is feasible. Acoustic ultrasonic energy results in an ultrasonic blade temperature of approximately 230° C. to approximately 300° C. to achieve the desired effect of cutting or transecting tissue. Because heat is retained in the metal body of the ultrasonic blade for a period of time after deactivation of the ultrasonic transducer, the residual heat stored in the ultrasonic blade can cause tissue damage if the ultrasonic end effector is used to grasp tissue before the ultrasonic blade has had an opportunity to cool down.

(759) In one aspect, the CTMo algorithm calculates a change in the natural frequency of the ultrasonic electromechanical system from the natural frequency at a hot state to a natural frequency at a temperature where atraumatic grasping is possible without damaging the tissue grasped by the end effector. Directly or a predetermined period of time after activating the ultrasonic transducer, a non-therapeutic signal (approximately 5 mA) is applied to the ultrasonic transducer containing a bandwidth of frequencies, approximately 48,000 Hz to 52,000 Hz, for example, at which the natural frequency is expected to be found. A FFT algorithm, or other mathematically efficient algorithm of detecting the natural frequency of the ultrasonic electromechanical system, of the impedance of the ultrasonic transducer measured during the stimulation of the ultrasonic transducer with the non-therapeutic signal will indicate the natural frequency of the ultrasonic blade as being the frequency at which the impedance magnitude is at a minimum. Continually stimulating the ultrasonic transducer in this manner provides continual feedback of the natural frequency of the ultrasonic blade within a frequency resolution of the FFT or other algorithm for estimating or measuring the natural frequency. When a change in natural frequency is detected that corresponds to a temperature that is feasible for atraumatic grasping, a tone, or a LED, or an on screen display or other form of notification, or a combination thereof, is provided to indicate that the device is capable of atraumatic grasping.

(760) In another aspect, the present disclosure provides a CTM algorithm configured to tone for seal and end of cut or transection. Providing “tissue sealed” and “end of cut” notifications is a challenge for conventional ultrasonic devices because temperature measurement cannot easily be directly mounted to the ultrasonic blade and the clamp arm pad is not explicitly detected by the blade using sensors. A CTM algorithm can indicate temperature state of the ultrasonic blade and can be employed to indicate the “end of cut” or “tissue sealed”, or both, states because these are temperature-based events.

(761) In one aspect, a CTM algorithm according to the present disclosure detects the “end of cut” state and activates a notification. Tissue typically cuts at approximately 210° C. to approximately 320° C. with high probability. A CTM algorithm can activate a tone at 320° C. (or similar) to indicate

that further activation of the tissue is not productive so that the tissue is probably cut and the ultrasonic blade is now running against the clamp arm pad, which is acceptable when the CTM algorithm is active because it controls the temperature of the ultrasonic blade. In one aspect, the CTM algorithm is programmed to control or regulate power to the ultrasonic transducer to maintain the temperature of the ultrasonic blade to approximately 320° C. when the temperature of the ultrasonic blade is estimated to have reached 320° C. Initiating a tone at this point provides an indication that the tissue has been cut. The CTM algorithm is based on a variation in frequency with temperature. After determining an initial state temperature (based on initial frequency), the CTM algorithm can calculate a frequency change that corresponds to a temperature that implies when the tissue is cut. For example, if the starting frequency is 51,000 Hz, the CTM algorithm will calculate the change in frequency required to achieve 320° C. which might be -112 Hz. It will then initiate control to maintain that frequency set point (e.g., 50,888 Hz) thereby regulating the temperature of the ultrasonic blade. Similarly, a frequency change can be calculated based on an initial frequency that indicates when the ultrasonic blade is at a temperature which indicates that the tissue is probably cut. At this point, the CTM algorithm does not have to control power, but simply initiate a tone to indicate the state of the tissue or the CTM algorithm can control frequency at this point to maintain that temperature if desired. Either way, the “end of cut” is indicated.

(762) In one aspect, a CTM algorithm according to the present disclosure detects the “tissue sealed” state and activates a notification. Similar to the end of cut detection, tissue seals between approximately 105° C. and approximately 200° C. The change in frequency from an initial frequency required to indicate that a temperature of the ultrasonic blade has reached 200° C., which indicates a seal only state, can be calculated at the onset of activation of the ultrasonic transducer. The CTM algorithm can activate a tone at this point and if the surgeon wishes to obtain a seal only state, the surgeon could stop activation or to achieve a seal only state the surgeon could stop activation of the ultrasonic transducer and automatically initiate a specific seal only algorithm from this point on or the surgeon could continue activation of the ultrasonic transducer to achieve a tissue cut state.

Application of Smart Ultrasonic Blade Technology

(763) When an ultrasonic blade is immersed in a fluid-filled surgical field, the ultrasonic blade cools down during activation rendering less effective for sealing and cutting tissue in contact therewith. The cooling down of the ultrasonic blade may lead to longer activation times and/or hemostasis issues because adequate heat is not delivered to the tissue. In order to overcome the cooling of the ultrasonic blade, more energy delivery may be required to shorten the transection times and achieve suitable hemostasis under these fluid immersion conditions. Using a frequency-temperature feedback control system, if the ultrasonic blade temperature is detected to, either start out below, or remain below a certain temperature for a certain period of time, the output power of the generator can be increased to compensate for cooling due to blood/saline/other fluid present in the surgical field.

(764) Accordingly, the frequency-temperature feedback control system described herein can improve the performance of an ultrasonic device especially when the ultrasonic blade is located or immersed, partially or wholly, in a fluid-filled surgical field. The frequency-temperature feedback control system described herein minimizes long activation times and/or potential issues with ultrasonic device performance in fluid-filled surgical field.

(765) As previously described, the temperature of the ultrasonic blade may be inferred by detecting the impedance of the ultrasonic transducer given by the following expression:

$$(766) Z_g(t) = \frac{V_g(t)}{I_g(t)}$$

or equivalently, detecting the phase angle ϕ between voltage $V_{\text{sub.g}}(t)$ current $I_{\text{sub.g}}(t)$ signals applied to the ultrasonic transducer. The phase angle ϕ information also may be used to infer the conditions of the ultrasonic blade. As discussed with particularity herein, the phase angle ϕ changes as a function of the temperature of the ultrasonic blade. Therefore, the phase angle ϕ information may be employed to control the temperature of the ultrasonic blade. This may be done, for example, by reducing the power delivered to the ultrasonic blade when the ultrasonic blade runs too hot and increasing the power delivered to the ultrasonic blade when the ultrasonic blade runs too cold. FIGS. 102A-102B are graphical representations of temperature feedback control for adjusting ultrasonic power applied to an ultrasonic transducer when a sudden drop in temperature of an ultrasonic blade is detected.

(767) FIG. 102A is a graphical representation of ultrasonic power output 133070 as a function of time, in accordance with at least one aspect of the present disclosure. Power output of the ultrasonic generator is shown along the vertical axis and time (sec) is shown along the horizontal axis. FIG. 102B is a graphical representation of ultrasonic blade temperature 133080 as a function of time, in accordance with at least one aspect of the present disclosure. Ultrasonic blade temperature is shown along the vertical axis and time (sec) is shown along the horizontal axis. The temperature of the ultrasonic blade increases with the application of constant power 133072 as shown in FIG. 102A. During use, the temperature of the ultrasonic blade suddenly drops. This may result from a variety of conditions, however, during use, it may be inferred that the temperature of the ultrasonic blade drops when it is immersed in a fluid-filled surgical field (e.g., blood, saline, water, etc.). At time t_0 , the temperature of the ultrasonic blade drops below the desired minimum temperature 133082 and the frequency-temperature feedback control algorithm detects the drop in temperature and begins to increase or “ramp up” the power as shown by the power ramp 133074 delivered to the ultrasonic blade to start raising the temperature of the ultrasonic blade above the desired minimum temperature 133082.

(768) With reference to FIGS. 102A and 102B, the ultrasonic generator is outputs substantially constant power 133072 as long the temperature of the ultrasonic blade remains above the desired minimum temperature 133082. At $t_{\text{sub.0}}$, processor or control circuit in the generator or instrument or both detects the drop in temperature of the ultrasonic blade below the desired minimum temperature 133072 and initiates a frequency-temperature feedback control algorithm to raise the temperature of the ultrasonic blade above the minimum desired temperature 133082. Accordingly, the generator power begins to ramp 133074 at $t_{\text{sub.1}}$ corresponding to the detection of a sudden drop in the temperature of the ultrasonic blade at $t_{\text{sub.0}}$. Under the frequency-temperature feedback control algorithm, the power continues to ramp 133074 until the temperature of the ultrasonic blade is above the desired minimum temperature 133082.

(769) FIG. 103 is a logic flow diagram 133090 of a process depicting a control program or a logic configuration to control the temperature of an ultrasonic blade, in accordance with at least one aspect of the present disclosure. According to the process, the processor or control circuit of the generator or instrument or both executes one aspect of a frequency-temperature feedback control algorithm discussed in connection with FIGS. 102A and 102B to apply 133092 a power level to the ultrasonic transducer to achieve a desired temperature at the ultrasonic blade. The generator monitors 133094 the phase angle ϕ between the voltage $V_{\text{sub.g}}(t)$ and current $I_{\text{sub.g}}(t)$ signals applied to drive the ultrasonic transducer. Based on the phase angle ϕ , the generator infers 133096 the temperature of the ultrasonic blade using the techniques described herein in connection with FIGS. 94A-96. The generator determines 133098 whether the temperature of the ultrasonic blade is below a desired minimum temperature by comparing the inferred temperature of the ultrasonic blade to a predetermined desired temperature. The generator then adjusts the power level applied to the ultrasonic transducer based on the comparison. For example, the process continues along NO branch when the temperature of the ultrasonic blade is at or above the desired minimum temperature and continues along YES branch when the temperature of the ultrasonic blade is below the desired minimum temperature. When the temperature of the ultrasonic blade is below the desired minimum temperature, the generator increases 133100 the power level to the ultrasonic transducer, e.g., by increasing the voltage $V_{\text{sub.g}}(t)$ and/or current $I_{\text{sub.g}}(t)$ signals, to raise the temperature of the ultrasonic blade and continues increasing the power level applied to the ultrasonic transducer until the temperature of the ultrasonic blade increases above the minimum desired temperature.

Adaptive Advanced Tissue Treatment Pad Saver Mode

(770) FIG. 104 is a graphical representation 133110 of ultrasonic blade temperature as a function of time during a vessel firing, in accordance with at least one aspect of the present disclosure. A plot 133112 of ultrasonic blade temperature is graphed along the vertical axis as a function of time along

the horizontal axis. The frequency-temperature feedback control algorithm combines the temperature of the ultrasonic blade feedback control with the jaw sensing ability. The frequency-temperature feedback control algorithm provides optimal hemostasis balanced with device durability and can deliver energy intelligently for best sealing while protecting the clamp arm pad.

(771) As shown in FIG. **104**, the optimum temperature **133114** for vessel sealing is marked as a first target temperature T.sub.1 and the optimum temperature **133116** for “infinite” clamp arm pad life is marked as a second target temperature T.sub.2. The frequency-temperature feedback control algorithm infers the temperature of the ultrasonic blade and maintains the temperature of the ultrasonic blade between the first and second target temperature thresholds T.sub.1 and T.sub.2. The generator power output is thus driven to achieve optimal ultrasonic blade temperatures for sealing vessels and prolonging the life of the clamp arm pad.

(772) Initially, the temperature of the ultrasonic blade increases as the blade heats up and eventually exceeds the first target temperature threshold T.sub.1. The frequency-temperature feedback control algorithm takes over to control the temperature of the blade to T.sub.1 until the vessel transection is completed **133118** at to and the ultrasonic blade temperature drops below the second target temperature threshold T.sub.2. A processor or control circuit of the generator or instrument or both detects when the ultrasonic blade contacts the clamp arm pad. Once the vessel transection is completed at to and detected, the frequency-temperature feedback control algorithm switches to controlling the temperature of the ultrasonic blade to the second target threshold T.sub.2 to prolong the life of the clam arm pad. The optimal clamp arm pad life temperature for a TEFLON clamp arm pad is approximately 325° C. In one aspect, the advanced tissue treatment can be announced to the user at a second activation tone.

(773) FIG. **105** is a logic flow diagram **133120** of a process depicting a control program or a logic configuration to control the temperature of an ultrasonic blade between two temperature set points as depicted in FIG. **104**, in accordance with at least one aspect of the present disclosure. According to the process, the generator executes one aspect of the frequency-temperature feedback control algorithm to apply **133122** a first power level to the ultrasonic transducer, e.g., by adjusting the voltage V.sub.g(t) and/or the current I.sub.g(t) signals applied to the ultrasonic transducer, to set the ultrasonic blade temperature to a first target T.sub.1 optimized for vessel sealing. As previously described, the generator monitors **133124** the phase angle ϕ between the voltage V.sub.g(t) and current I.sub.g(t) signals applied to the ultrasonic transducer and based on the phase angle ϕ , the generator infers **133126** the temperature of the ultrasonic blade using the techniques described herein in connection with FIGS. **94A-96**. According to the frequency-temperature feedback control algorithm, a processor or control circuit of the generator or instrument or both maintains the ultrasonic blade temperature at the first target temperature T.sub.1 until the transection is completed. The frequency-temperature feedback control algorithm may be employed to detect the completion of the vessel transection process. The processor or control circuit of the generator or instrument or both determines **133128** when the vessel transection is complete. The process continues along NO branch when the vessel transection is not complete and continues along YES branch when the vessel transection is complete.

(774) When the vessel transection is not complete, the processor or control circuit of the generator or instrument or both determines **133130** if the temperature of the ultrasonic blade is set at temperature T.sub.1 optimized for vessel sealing and transecting. If the ultrasonic blade temperature is set at T.sub.1, the process continues along the YES branch and the processor or control circuit of the generator or instrument or both continues to monitor **133124** the phase angle ϕ between the voltage V.sub.g(t) and current I.sub.g(t) signals applied to the ultrasonic transducer and based on the phase angle ϕ . If the ultrasonic blade temperature is not set at T.sub.1, the process continues along NO branch and the processor or control circuit of the generator or instrument or both continues to apply **133122** a first power level to the ultrasonic transducer.

(775) When the vessel transection is complete, the processor or control circuit of the generator or instrument or both applies **133132** a second power level to the ultrasonic transducer to set the ultrasonic blade to a second target temperature T.sub.2 optimized for preserving or extending the life of the clamp arm pad. The processor or control circuit of the generator or instrument or both determines **133134** if the temperature of the ultrasonic blade is at set temperature T.sub.2. If the temperature of the ultrasonic blade is set at T2, the process completes **133136** the vessel transection procedure.

Start Temperature of Blade

(776) Knowing the temperature of the ultrasonic blade at the beginning of a transection can enable the generator to deliver the proper quantity of power to heat up the blade for a quick cut or if the blade is already hot add only as much power as would be needed. This technique can achieve more consistent transection times and extend the life of the clam arm pad (e.g., a TEFLON clamp arm pad). Knowing the temperature of the ultrasonic blade at the beginning of the transection can enable the generator to deliver the right amount of power to the ultrasonic transducer to generate a desired amount of displacement of the ultrasonic blade.

(777) FIG. **106** is a logic flow diagram **133140** of a process depicting a control program or a logic configuration to determine the initial temperature of an ultrasonic blade, in accordance with at least one aspect of the present disclosure. To determine the initial temperature of an ultrasonic blade, at the manufacturing plant, the resonant frequencies of ultrasonic blades are measured at room temperature or at a predetermined ambient temperature. The baseline frequency values are recorded and stored in a lookup table of the generator or instrument or both. The baseline values are used to generate a transfer function. At the start of an ultrasonic transducer activation cycle, the generator measures **133142** the resonant frequency of the ultrasonic blade and compares **133144** the measured resonant frequency to the baseline resonant frequency value and determines the difference in frequency (Δf). The Δf is compared to the lookup table or transfer function for corrected ultrasonic blade temperature. The resonant frequency of the ultrasonic blade may be determined by sweeping the frequency of the voltage V.sub.g(t) and current I.sub.g(t) signals applied to the ultrasonic transducer. The resonant frequency is that frequency at which the phase angle ϕ voltage V.sub.g(t) and current I.sub.g(t) signals is zero as described herein.

(778) Once the resonant frequency of the ultrasonic blade is determined, the processor or control circuit of the generator or instrument or both determines **133146** the initial temperature of the ultrasonic blade based on the difference between the measured resonant frequency and the baseline resonant frequency. The generator sets the power level delivered the ultrasonic transducer, e.g., by adjusting the voltage V.sub.g(t) or current I.sub.g(t) drive signals or both, to one of the following values prior to activating the ultrasonic transducer.

(779) The processor or control circuit of the generator or instrument or both determines **133148** if the initial temperature of the ultrasonic blade is low. If the initial temperature of the ultrasonic blade is low, the process continues along YES branch and the processor or control circuit of the generator or instrument or both applies **133152** a high power level to the ultrasonic transducer to increase the temperature of the ultrasonic blade and completes **133156** the vessel transection procedure.

(780) If the initial temperature of the ultrasonic blade is not low, the process continues along NO branch and the processor or control circuit of the generator or instrument or both determines **133150** if the initial temperature of the ultrasonic blade is high. If the initial temperature of the ultrasonic blade is high, the process proceeds along YES branch and the processor or control circuit of the generator or instrument or both applies **133154** a low power level to the ultrasonic transducer to decrease the temperature of the ultrasonic blade and completes **133156** the vessel transection procedure. If the initial temperature of the ultrasonic blade is not high, the process continues along NO branch and the processor or control circuit of the generator or instrument or both completes **133156** the vessel transection.

Smart Blade Technology to Control Blade Instability

(781) The temperature of an ultrasonic blade and the contents within the jaws of an ultrasonic end effector can be determined using the frequency-temperature feedback control algorithms described herein. The frequency/temperature relationship of the ultrasonic blade is employed to control ultrasonic blade instability with temperature.

(782) As described herein, there is a well-known relationship between the frequency and temperature in ultrasonic blades. Some ultrasonic blades exhibit displacement instability or modal instability in the presence of increasing temperature. This known relationship may be employed to interpret when an ultrasonic blade is approaching instability and then adjusting the power level driving the ultrasonic transducer (e.g., by adjusting the driving

voltage $V_{sub.g(t)}$ or current $I_{sub.g(t)}$ signals, or both, applied to the ultrasonic transducer) to modulate the temperature of the ultrasonic blade to prevent instability of the ultrasonic blade.

(783) FIG. **107** is a logic flow diagram **133160** of a process depicting a control program or a logic configuration to determine when an ultrasonic blade is approaching instability and then adjusting the power to the ultrasonic transducer to prevent instability of the ultrasonic transducer, in accordance with at least one aspect of the present disclosure. The frequency/temperature relationship of an ultrasonic blade that exhibits a displacement or modal instability is mapped by sweeping the frequency of the drive voltage $V_{sub.g(t)}$ or current $I_{sub.g(t)}$ signals, or both, over the temperature of the ultrasonic blade and recording the results. A function or relationship is developed that can be used/interpreted by a control algorithm executed by the generator. Trigger points can be established using the relationship to notify the generator that an ultrasonic blade is approaching the known blade instability. The generator executes a frequency-temperature feedback control algorithm processing function and closed loop response such that the driving power level is reduced (e.g., by lowering the driving voltage $V_{sub.g(t)}$ or current $I_{sub.g(t)}$, or both, applied to the ultrasonic transducer) to modulate the temperature of the ultrasonic blade at or below the trigger point to prevent a given blade from reaching instability.

(784) Advantages include simplification of ultrasonic blade configurations such that the instability characteristics of the ultrasonic blade do not need to be designed out and can be compensated using the present instability control technique. The present instability control technique also enables new ultrasonic blade geometries and can improve stress profile in heated ultrasonic blades. Additionally, ultrasonic blades can be configured to diminish performance of the ultrasonic blade if used with generators that do not employ this technique.

(785) In accordance with the process depicted by the logic flow diagram **133160**, the processor or control circuit of the generator or instrument or both monitors **133162** the phase angle ϕ between the voltage $V_{sub.g(t)}$ and current $I_{sub.g(t)}$ signals applied to the ultrasonic transducer. The processor or control circuit of the generator or instrument or both infers **133164** the temperature of the ultrasonic blade based on the phase angle ϕ between the voltage $V_{sub.g(t)}$ and current $I_{sub.g(t)}$ signals applied to the ultrasonic transducer. The processor or control circuit of the generator or instrument or both compares **133166** the inferred temperature of the ultrasonic blade to an ultrasonic blade instability trigger point threshold. The processor or control circuit of the generator or instrument or both determines **133168** whether the ultrasonic blade is approaching instability. If not, the process proceed along the NO branch and monitors **133162** the phase angle ϕ , infers **133164** the temperature of the ultrasonic blade, and compares **133166** the inferred temperature of the ultrasonic blade to an ultrasonic blade instability trigger point threshold until the ultrasonic blade approaches instability. The process then proceeds along the YES branch and the processor or control circuit of the generator or instrument or both adjusts **133170** the power level applied to the ultrasonic transducer to modulate the temperature of the ultrasonic blade.

Ultrasonic Sealing Algorithm with Temperature Control

(786) Ultrasonic sealing algorithms for ultrasonic blade temperature control can be employed to improve hemostasis utilizing a frequency-temperature feedback control algorithm described herein to exploit the frequency/temperature relationship of ultrasonic blades.

(787) In one aspect, a frequency-temperature feedback control algorithm may be employed to alter the power level applied to the ultrasonic transducer based on measured resonant frequency (using spectroscopy) which relates to temperature, as described in various aspects of the present disclosure. In one aspect, the frequency-temperature feedback control algorithm may be activated by an energy button on the ultrasonic instrument.

(788) It is known that optimal tissue effects may be obtained by increasing the power level driving the ultrasonic transducer (e.g., by increasing the driving voltage $V_{sub.g(t)}$ or current $I_{sub.g(t)}$, or both, applied to the ultrasonic transducer) early on in the sealing cycle to rapidly heat and desiccate the tissue, then lowering the power level driving the ultrasonic transducer (e.g., by lowering the driving voltage $V_{sub.g(t)}$ or current $I_{sub.g(t)}$, or both, applied to the ultrasonic transducer) to slowly allow the final seal to form. In one aspect, a frequency-temperature feedback control algorithm according to the present disclosure sets a limit on the temperature threshold that the tissue can reach as the tissue heats up during the higher power level stage and then reduces the power level to control the temperature of the ultrasonic blade based on the melting point of the clamp jaw pad (e.g., TEFLON) to complete the seal. The control algorithm can be implemented by activating an energy button on the instrument for a more responsive/adaptive sealing to reduce more the complexity of the hemostasis algorithm.

(789) FIG. **108** is a logic flow diagram **133180** of a process depicting a control program or a logic configuration to provide ultrasonic sealing with temperature control, in accordance with at least one aspect of the present disclosure. According to the control algorithm, the processor or control circuit of the generator or instrument or both activates **133182** ultrasonic blade sensing using spectroscopy (e.g., smart blade) and measures **133184** the resonant frequency of the ultrasonic blade (e.g., the resonant frequency of the ultrasonic electromechanical system) to determine the temperature of the ultrasonic blade using a frequency-temperature feedback control algorithm (spectroscopy) as described herein. As previously described, the resonant frequency of the ultrasonic electromechanical system is mapped to obtain the temperature of the ultrasonic blade as a function of resonant frequency of the electromechanical ultrasonic system.

(790) A first desired resonant frequency f of the ultrasonic electromechanical system corresponds to a first desired temperature Z° of the ultrasonic blade. In one aspect, the first desired ultrasonic blade temperature Z° is an optimal temperature (e.g., 450°C.) for tissue coagulation. A second desired frequency $f_{sub.r}$ of the ultrasonic electromechanical system corresponds to a second desired temperature ZZ° of the ultrasonic blade. In one aspect, the second desired ultrasonic blade temperature ZZ° is a temperature of 330°C. , which is below the melting point of the clamp arm pad, which is approximately 380°C. for TEFLON.

(791) The processor or control circuit of the generator or instrument or both compares **133186** the measured resonant frequency of the ultrasonic electromechanical system to the first desired frequency f . In other words, the process determines whether the temperature of the ultrasonic blade is less than the temperature for optimal tissue coagulation. If the measured resonant frequency of the ultrasonic electromechanical system is less than the first desired frequency $f_{sub.x}$, the process continues along the NO branch and the processor or control circuit of the generator or instrument or both increases **133188** the power level applied to the ultrasonic transducer to increase the temperature of the ultrasonic blade until the measured resonant frequency of the ultrasonic electromechanical system exceeds the first desired frequency $f_{sub.x}$. In which case, the tissue coagulation process is completed and the process controls the temperature of the ultrasonic blade to the second desired temperature corresponding to the second desired frequency $f_{sub.y}$.

(792) The process continues along the YES branch and the processor or control circuit of the generator or instrument or both decreases **133190** the power level applied to the ultrasonic transducer to decrease the temperature of the ultrasonic blade. The processor or control circuit of the generator or instrument or both measures **133192** the resonant frequency of the ultrasonic electromechanical system and compares the measured resonant frequency to the second desired frequency $f_{sub.Y}$. If the measured resonant frequency is not less than the second desired frequency $f_{sub.Y}$, the processor or control circuit of the generator or instrument or both decreases **133190** the ultrasonic power level until the measured resonant frequency is less than the second desired frequency $f_{sub.Y}$. The frequency-temperature feedback control algorithm to maintain the measured resonant frequency of the ultrasonic electromechanical system below the second desired frequency $f_{sub.y}$, e.g., the temperature of the ultrasonic blade is less than the temperature of the melting point of the clamp arm pad then, the generator executes the increases the power level applied to the ultrasonic transducer to increase the temperature of the ultrasonic blade until the tissue transection process is complete **133196**.

(793) FIG. **109** is a graphical representation **133200** of ultrasonic transducer current and ultrasonic blade temperature as a function of time, in accordance with at least one aspect of the present disclosure. FIG. **109** illustrates the results of the application of the frequency-temperature feedback control algorithm described in FIG. **109**. The graphical representation **133200** depicts a first plot **133202** of ultrasonic blade temperature as a function of time with respect to a second plot **133204** of ultrasonic transducer current $I_{sub.g(t)}$ as a function of time. As shown, the transducer $I_{sub.g(t)}$ is maintained constant until the ultrasonic blade temperature reaches 450° , which is an optimal coagulation temperature. Once the ultrasonic blade temperature reaches 450° , the frequency-temperature feedback control algorithm decrease the transducer current $I_{sub.g(t)}$ until the temperature of

the ultrasonic blade drops to below 330°, which is below the melting point of a TEFLON pad, for example.

Limiting Capacitive Coupling and its Effects

(794) Aspects of the present disclosure are presented for a surgical instrument with improved device capabilities for reducing undesired operational side effects. In particular, the surgical instrument may include means for limiting capacitive coupling to improve monopolar isolation for use independently or in cooperation with another advanced energy modality. Capacitive coupling occurs generally when there is a transfer of energy between nodes, induced by an electric field. During surgery, capacitive coupling may occur when two or more electrical surgical instruments are being used in or around a patient. While in some cases capacitive coupling may be desirable, as additional devices may be powered inductively by capacitive coupling, having capacitive coupling occur accidentally during surgery or around a patient generally can have extremely deleterious consequences. Parasitic or accidental capacitive coupling may occur in unknown or unpredictable locations, causing energy to be applied to unintended areas. When the patient is under anesthesia and unable to provide any response, parasitic capacitive coupling can burn a patient while the surgeon would not know it is even occurring. It is therefore desirable to limit the parasitic or accidental capacitive coupling in surgical instruments and during surgery generally.

(795) In some aspects, a system including a surgical instrument and a generator may be configured to interrupt the transmission of energy from the generator to the surgical instrument when capacitive coupling has been detected. One or more safety fuses, sensors, controls, and/or algorithms may be in place to automatically trigger an interruption of the generator in these scenarios. Alerts, including audio signals, vibrations, and visual messages may issue to inform the surgery team that the generator was interrupted due to the detection of capacitive coupling.

(796) In some aspects, the system includes means for detecting that a capacitive coupling event has occurred. For example, an algorithm that includes inputs from one or more sensors for monitoring events around the system may apply situational awareness and other programmatic means to conclude that capacitive coupling is occurring somewhere within the system and react accordingly. A system having situational awareness means that the system may be configured to anticipate scenarios that may arise based on present environmental and system data and determining that the present conditions follow a pattern that gives rise to predictable next steps. As an example, the system may apply situational awareness in the context of handling capacitive coupling events by recalling instances in similarly situated surgeries where various sensor data is detected. The sensor data may indicate an increase in current at two particular locations along a closed loop electrosurgical system, that based on previous data of similarly situated surgeries, indicates a high likelihood that a capacitive coupling event is imminent.

(797) In some aspects, the surgical instruments may be modified in structure to limit the occurrence of capacitive coupling, or in other cases reduce the collateral damage caused by capacitive coupling. For example, additional insulation placed strategically in or around the surgical instrument may help limit the incidence of capacitive coupling. In other cases, the end effector of the surgical instrument may include modified structures that reduce the incidence of current displacement, such as rounding the tips of the end effector or specifically shaping the blade of the end effector to behave more like a monopolar blade while still acting as a bipolar device.

(798) In some aspects, the system may include passive means for mitigating or limiting the effects of the capacitive coupling. For example, the system may include leads that can shunt the energy to a neutral node through conductive passive components. In general, any and all of these aspects may be combined or included in a single system to address the challenges posed by multiple electrical components liable to cause capacitive coupling during patient surgery.

(799) FIG. 110 provides a diagram showing an example system **134000** with means for detecting capacitive coupling, in accordance with at least one aspect of the present disclosure. The system **134000** includes a monopolar ESU generator **134002** that is electrically coupled to a surgical instrument **134008**. The surgical instrument **134008** is used to perform surgery on a patient, where patient tissue **134016** is shown to represent the surgical site of the patient where surgery is being performed. The surgical instrument **134008** may include means for applying electrosurgical or ultrasonic energy to an end effector, and in some cases may include a blade and/or a pair of jaws to grasp or clamp onto tissue. The energy powered by the ESU generator **134002** may touch the patient through the end effector, via any of the possible various components of the end effector. At least a portion of the patient may rest on a return path pad **134014**, such as a Smart Megasoft Pad™, for example, that is configured to divert excess energy away from the patient when the surgical instrument **134008** touches the patient and applies electrosurgical energy.

(800) Because of the multiple electrical sources near the patient, parasitic capacitive coupling is ever present and always a danger to harm the patient during surgery. Because the patient is not expected to express any reaction during surgery, if unknown or unpredicted capacitive coupling occurs, the patient may experience burns in unintended places as a result. In general, energy anomalies like capacitive coupling should be minimized or otherwise corrected in order to improve patient safety. To limit the occurrence of capacitive coupling or other types of energy anomalies, multiple smart sensors or monitors, such as CT1 (**134006**), CT2 (**134010**), and CT3 (**134012**) smart sensors may be integrated into the electrosurgical system as indicators to determine whether excess or inductive energy is radiating outside the one or more of the electrical sources. As shown in FIG. 110, the smart sensors CT1 (**134006**), CT2 (**134010**) and CT3 (**134012**) are placed at likely locations where energy may inductively radiate. The sensors or monitors may be configured to detect capacitance, and if placed at strategic locations within the system, a reading of capacitance may imply that capacitive leakage is occurring near the sensor or monitor. Coupled with knowledge of other sensors nearby or throughout the system not indicating a reading of capacitance, one may conclude that capacitive leakage is occurring in close proximity to the sensor or monitor that is providing a positive indication. Other sensors may be used, such as capacitive leakage monitors or detectors. These sensors may be configured to provide an alert, such as lighting up or delivering a noise or transmitting a signal ultimately to a display monitor. In addition, the monopolar ESU **134002** may be configured to automatically trigger an interruption in energy generation to stop any further capacitive coupling from occurring.

(801) In some aspects, a neutral electrode **134004** may be included in the monopolar ESU **134002** and may be electrically coupled to the return path pad **134014**, such as a Smart Megasoft Pad®, for example, as another solution to reduce capacitive coupling. Energy can reach the neutral node **134004** conductively as the electrosurgical instrument **134008** touches the patient, the patient is touching the return path pad **134014**, and the pad is conductively connected to the neutral electrode **134004**. Thus, energy can be diverted to the neutral node **134004** from the monopolar ESU **134002** or the surgical instrument **134008** and thereby reduce the incidence of capacitive coupling.

(802) In some aspects, a cloud analytics system communicatively coupled to the monopolar ESU, such as through a medical hub, may be configured to employ situational awareness that can help anticipate when capacitive coupling may occur during surgery. The cloud analytics system and/or the medical hub may utilize a capacitive coupling algorithm to monitor the incidence of energy flowing through the surgical system, and based on previous data about the state of energy in the system for a similar situated procedure, may conclude there is a likelihood that capacitive coupling may occur if no additional action is taken. For example, during a surgery involving prescribed methods for how to the surgical instrument and how much power should be employed during particular steps in the surgery, the cloud analytics module may draw from previous surgeries of the same and note that capacitive coupling has a stronger likelihood to occur after a particular step in the surgery. While monitoring the steps in the surgery, when the same or very similar energy profiles occur during or just before the expected step that tends to induce capacitive coupling, the cloud analytics system may deliver an alert that indicates this is likely to cause capacitive coupling. The surgeon may be given the option to reduce peak voltage in the surgical instrument **134008** or interrupt the power generation by the monopolar ESU **134002**, or the cloud analytics module may automatically cause the medical hub to take these measures. This may lead to eliminating the possibility of capacitive coupling before it has a chance to occur, or at least may limit any unintended effects caused by a momentary occurrence of capacitive coupling.

(803) In some aspects, the surgical instrument as shown in FIG. 110 may include structural means for reducing or preventing capacitive coupling. For example, insulation in the shaft of the surgical instrument **134008** may reduce the incidence of inductance. In other cases, the monopolar wire connecting the monopolar ESU **134002** to the surgical instrument **134008** may be shielded. As another example, interrupting plastic elements within the shaft may be intermittently present to prevent capacitive coupling from transmitting long distance within the shaft. Other insulator-type elements

may be used to achieve similar effects. In some aspects, the monopolar wire electrically connecting the surgical instrument **134008** to the generator **134002** may be shielded to also reduce the incidence of capacitive coupling.

(804) In some aspects, the structure of the end effector may be modified to reduce the effects of capacitive coupling as the end effector makes contact with the patient. As one example, the jaws of the end effector may be designed to have only one side of the each of the jaws directed to deliver energy, thereby causing the end effector to act like a monopolar blade while still actually functionally structured as a bipolar device. In one example of this, the ends or tips of the end effector may be shaped like a duck bill, with rounded ends to reduce any voltage peaks that might arise out of pointed ends. The direction of energy in the end effector may still be directed to an area or a point along the duck billed ends, but the dispersion of any excess energy may be blunted by the duck billed end. As another example, the blade may be structured to be slightly thicker on one side, such as having a triangular cross-sectional area, and having a thin standing upper blade element on the opposite. This may allow any energy being delivered to the blade to be focused to a point, which may help the surgical instrument act like a monopolar blade while still being a bipolar device. In this way, energy will not be dispersed that would make the surgical instrument more prone to causing capacitive coupling. As a final example, the jaws of the surgical instrument may have electrodes placed on the inside of the end effector, allowing the outer portions of the end effector to act like a shield to ward against capacitive coupling. The electrodes may still be placed sufficiently to contact the tissue of the patient during a surgery, while having one or more edges of the end effector shield the energy from dispersing beyond the focused surgical area.

(805) FIG. **111** is a logic flow diagram **134100** depicting a control program or a logic configuration of an example methodology for limiting the effects of capacitive coupling in a surgical system is disclosed, according to some aspects. The example methodology may be consistent with the descriptions above regarding several enumerated means for limiting capacitive coupling or mitigating its effects during surgery using one or more surgical instruments.

(806) As shown and consistent with the examples discussed above, the methodology **134100** may start with the surgical system being configured to monitor **134102** energy generation. For example, multiple sensors may be placed strategically at potential vulnerable points more liable to leak energy that can cause capacitive coupling. These sensors may be configured to deliver an alert when an energy anomaly occurs.

(807) Continuing on, the sensors or other detecting means may detect **134104** a voltage anomaly, such as a voltage peak or voltage spike, at one or more locations along the surgical system that would not normally be expected to produce such energy production. The system may be configured to conclude these scenarios may give rise to parasitic capacitive coupling, potentially burning the patient unbeknownst to the surgery team in the absence of any alerts. As a result, an alert or message may be delivered indicating the energy anomaly and the danger of capacitive coupling occurring.

(808) In some aspects, situational awareness also may be used to anticipate **134106** when capacitive coupling is more likely to occur during the usual course of a surgery. Situational awareness may be used to refer back to past surgical operations of similar type or circumstances to identify what variables may be present when capacitive coupling was determined to have occurred. If there are certain steps in the procedure that are more likely to cause capacitive coupling, the system may anticipate these situations by particularly monitoring the sensors at these times, and/or taking preemptive measures to reduce the incidences of capacitive coupling.

(809) If capacitive coupling is detected or believed to be imminent, based on the above the methodology **134100** executed by the surgical system, measures taken to reduce, eliminate, or mitigate the effects of capacitive coupling can include to automatically interrupt **134108** energy generation at the monopolar energy generator, according to some aspects. It is noted that some loss in surgical operation may occur momentarily at the time this interruption is enabled, but preventing unintended damage to the patient would be paramount in any case. The surgery can continue as planned after a brief moment of interruption.

(810) Measuring the energy out relative to energy in, taking advantage of parasitic leakage to improve pad contact, or turn power off, generator knows how much current it's generating, and you're measuring the energy that is put out.

Increasing Frequency in the Presence of Capacitive Coupling

(811) In some aspects, the presence of parasitic capacitive coupling can be harnessed to perform energy coagulation or energy cautery. In certain instances, it may be desirable to increase the energy generation of the electrosurgical instrument in order to drive the monopolar circuit to ground, through the body of the patient. While there may be a number of instances where the monopolar circuit is completed through the conductivity drawn by a conductive return pad **134014**, such as the Smart Megasoft Pad® (see FIG. **110**), in some cases the pad may be defective or worn, such that the conductivity of the return pad **134014** is not sufficient to draw the current of the electrosurgical instrument (e.g., **134008**) through the body of the patient. In such cases, the current may lack a sufficient ground for the energy to travel to, effectively making the body of the patient act like a short circuit. This may render the electrosurgery ineffective, as the energy delivered by the surgical instrument **134008** will not pass through the tissue of the patient and therefore not heat the tissue as intended. A similar situation may occur when there is no return pad at all. That is, without a conductive return pad **134014**, such as the Smart Megasoft Pad®, to provide a wide conductive return path, there may be no ground available that is connected to the patient. This also may lead to the patient acting as a short circuit if energy from the surgical instrument were applied to the patient.

(812) To adjust for these situations, in some aspects, the monopolar energy generation may be increased to a very high frequency, such as 500 KHz to 3-4 Mhz, to take advantage of parasitic patient leakage to do padless electrosurgery (or electrosurgery with insufficient conductivity in the pad). By increasing the alternating current frequency, the parasitic leakage current will increase. The stronger leakage current can then more effectively radiatively traverse through the body of the patient. After reaching through the body of the patient, the leakage current of the capacitive coupling may be more effectively radiatively coupled to a ground state as a result, which may effectively drive the current radiatively into another object that acts as ground. For example, if the AC frequency is high enough, the current leakage may reach the monopolar generator grounding terminal. This will help to remove the short circuit effect of the patient, thereby allowing for the energy coagulation to take place. Therefore, in the situations where there is a non-padded system, or a system with poor conductivity in a pad, it may be desirable to increase current leakage in order to take advantage of higher leakage return that can be used to complete the monopolar circuit. That is, in some cases, the return path may be formed by the radiative current leakage caused by capacitive coupling. To help ensure that the radiative return path reaches a ground plane, the energy of the surgical instrument may be increased to a very high frequency.

(813) In some cases, the poorly conducting return pad may be connected purposely to an earth ground, or table, or to a closest support surface, while the return connector on the generator may be connected to earth ground as well. This will divert the circuit to flow through the radiative return path, rather than have any energy attempt to travel through the poorly conducting return pad and back to the generator, which may cause burns on the patient.

(814) It is noted that when there is a padded system, and the pad provides sufficient conductivity under the patient, the typical monopolar circuit that drives the current through the body and into the return pad may be a preferred method. In these cases, it may be useful to build isolation barriers to the externally connected power source, such as the energy generator **134002** (see FIG. **110**). Alternatively, battery powered instruments may be the more ideal system for reducing the leakage current that will help isolate the energy path through the conductive return pad.

(815) In some aspects, the surgical system may include a detection circuit configured to determine the capacity of the return path pad. The detection circuit may then provide information as to whether it would be better to utilize the radiative current leakage to complete the circuit, rather than try to rely on a poorly conducting return path pad, or simply no pad at all. The detection circuit may measure an amount of conductivity in the return path pad. If the measure of conductivity satisfies a predetermined threshold, the system may determine that the return path pad may be used to perform the surgery and provide a return path for the monopolar energy. If the conductivity is below the threshold, then the detection circuit may be configured to send a signal to the system, such as at a processor in the surgical hub or the monopolar generator, that the frequency of the monopolar energy should be increased drastically and the return path pad should be eliminated or at least isolated from consideration. Increasing the frequency will then

complete the monopolar circuit by creating a radiative return path.

(816) In some aspects, the monopolar generator may include one or more control circuits coupled to one or more sensors that are configured to determine if the current leakage has reached the grounding terminal of the monopolar generator. The sensor, combined with the detection circuit and a control circuit of the monopolar generator may be used to create a closed feedback loop system that may automatically adjust the frequency to create a sufficient return path based on high leakage current. For example, the detection circuit may determine if there is sufficient conductivity in the return path pad. If not, the control circuit of the monopolar generator may cause the energy generation to increase the AC frequency. The sensor at the monopolar generator may continuously monitor if any radiative current leakage has reached the ground terminal of the monopolar generator, based on the increased frequency. The control circuit may gradually increase the frequency until it is detected that the radiative current leakage has reached the ground terminal. Therefore, the surgical system may rely on a predetermined frequency threshold if it is determined there is no return path pad or an insufficient conductivity in the pad, or a closed feedback system may be used to find a sufficiently high frequency that can create a return path through radiative coupling.

(817) FIG. 112 is a logic flow diagram 134200 depicting a control program or a logic configuration of an example methodology that may be performed by the surgical system utilizing monopolar energy generation to determine whether to take advantage of parasitic capacitive coupling. Consistent with the descriptions above, a detection circuit as part of the surgical system may be configured to measure 134202 a level of conductivity in the return path of a monopolar electrosurgical setup. The return path may originally be identified to go through a conductive pad, such as a Soft Megasoft Pad® or other return path conductive pad. In some cases, the conductivity of the pad may offer poor conductivity. In other cases, no pad may exist as part of the surgery setup. This may cause the patient body to act as a short circuit of the monopolar circuit, which would reduce or eliminate the effectiveness of trying to apply monopolar energy to a surgical site at the patient.

(818) The detection circuit may determine 134204 that the measure of conductivity falls below a predetermined threshold, indicating that the level of conductivity in the return path is sufficiently poor, which prevents completion of the monopolar circuit. As a result, the surgical system may cause the generator to increase 134206 the current leakage by increasing the frequency of the alternating current in the monopolar generator. The surgical system may instead utilize the radiative current leakage to create a return path. When the frequency is increased, the current leakage will also increase, which thereby increases the reach of the radiative current leakage to reach a ground plane and complete the circuit. Thus, by increasing the frequency, the poor conductivity of the return path pad 134014—or even lack of any pad at all—may be subverted. In some cases, the increase in leakage may be determined based on a closed feedback sensor system that adjusts the frequency until it is determined that the radiative current leakage has reached the ground terminal at the monopolar generator.

(819) In some aspects, the surgical system also may provide an instruction to isolate 134208 any return path pads and to attach the return connector of the monopolar generator to an earth ground. These measures may be taken to eliminate other alternative return paths that may inadvertently cause burns at undesirable locations in the patient.

Adjustment of Compression Force Applied to Tissue Based on Proportion of Energy Modalities

(820) Aspects of the present disclosure are presented for a surgical instrument that is situationally aware. The surgical instrument may be any suitable surgical instrument described in the present disclosure. For the sake of clarity, surgical instrument 112 is referenced. In particular, the surgical instrument 112 may be a bipolar combination surgical instrument 112 which may automatically adjust a compression force applied by the end effector of the surgical instrument 112 based on a selected energy modality. In one aspect, the bipolar combination surgical instrument 112 may be configured to deliver energy according to a bipolar radio frequency (RF) and an ultrasonic energy modality. More specifically, the automatic adjustment may be based on a proportion of two different selected energy modalities. This automatic adjustment of the compression force is an example of a situational awareness characteristic of the surgical instrument 112 that may improve the efficacy and quality of a surgical procedure performed with the surgical instrument 112. The clamp pressure applied by the end effector can be indicative of the compression force applied to tissue being treated. As discussed above, selectable energy modalities include ultrasonic, bipolar or monopolar radio frequency (electrosurgical), irreversible or reversible electroporation, and microwave energy modality implemented by a generator of the surgical instrument 112. In one aspect, the two selected energy modalities are bipolar RF energy and ultrasonic energy. Additionally or alternatively to adjusting the compression force applied to tissue based on the selected energy modality, the compression force may also be adjusted based on a power parameter. The power parameter may refer to the relative proportion of total energy applied during a surgical procedure that is allocated between bipolar RF and ultrasonic energy, respectively, for example.

(821) In general, the compression force adjustment may be actively performed during performance of a surgical procedure. Such active adjustment can mean that the clinician operating the surgical instrument 112 does not need to manually adjust the clamp arm, waveguide or ultrasonic blade, or end effector to modify the compression force applied to tissue being treated in the jaws of the end effector. As described below in further detail, a control circuit or generator of the situationally aware surgical instrument 112 may automatically adjust tissue compression force by executing an algorithm. The algorithm is executable to determine an appropriate compression force by considering the particular proportion or blend of the selected energy modalities. For example, the relative proportion of time spent applying each energy modality can be considered in determining suitable tissue compression forces to be applied during the course of the performed surgical procedure. The relative amplitudes of each energy modality could be considered as well. Also, each type of energy modality can correspond to a certain pressure or range of pressures, which could also change depending on other parameters such as the power and timing of the delivered energy modality. This energy-pressure relationship can also be understood as energy delivered and pressure existing on a spectrum together. That is, the more compression pressure that is applied, the more effective the application of energy to treat tissue will be. Consequently, less energy may be required when more compression pressure is applied. Energy is delivered according to the selected energy modality.

(822) The selected energy modalities could be applied simultaneously to tissue. Alternatively, the generator may switch between providing a drive output signal according to a first energy modality such as bipolar RF and providing the drive output signal according to a second energy modality such as ultrasonic. That is, the delivery of RF can follow ultrasonic and vice versa. The extent of switching may be used in the algorithm to determine a suitable automatic adjustment to tissue compression force. For example, when the generator switches from delivering ultrasonic energy to bipolar RF energy, the applied tissue compression force may be increased. When multiple energy modalities are applied simultaneously, the relative proportion of the energy modalities may be used to determine the compression force adjustment. For example, the control circuit could the proportion of ultrasonic drive signals to RF drive signals provided by the generator. As described further below, electrical or mechanical methods of adjusting tissue compression may be used and such methods may or may not involve control by the processor, control circuit, or generator as appropriate. In some aspects, the tissue compression adjustment algorithm may be stored in memory and may be updated by an algorithm program update transmitted from the corresponding surgical hub (e.g., surgical hub 106, 206) to the surgical instrument 112. In turn, the hub may receive this algorithm program update from a cloud computing system (e.g., cloud 104, 204). The hub may also store the update locally in a memory device of the hub. Additionally or alternatively, the control circuit or generator of the surgical instrument 112 can modify the algorithm as suitable, such as by a clinician changing the parameters of the algorithm through a user interface of the surgical instrument 112.

(823) Mechanical methods of adjusting tissue compression force include adjusting the waveguide or ultrasonic blade and adjusting the clamp arm linkage mechanism (e.g., linkage components of the transmissions 706a-706e) of the surgical instrument 112. The ultrasonic blade can be for example, an offset oval ultrasonic blade where the end effector clamp arm and offset ultrasonic blade are rotatable relative to each other to define a tissue gap distance between the end effector jaws. By adjusting the tissue gap distance, various tissue compression forces are possible. In this way, the ultrasonic blade is adjustable to generate a smaller tissue gap (and thus relatively higher compression force) when the RF energy modality is selected and to generate a larger tissue gap (and thus relatively lower compression force) when the ultrasonic energy modality is selected. The end

effector jaw members can also be adjusted to a desired tissue gap size independent of the wave guide. For example, in one aspect, a clamp arm linkage mechanism is provided to change the clamp arm actuator rod stroke according to the selected energy modality. The clamp arm actuator rod (e.g., articulation actuator such as via output shaft of the motor **704a**, **704b** coupled to moveable mechanical elements of transmissions **706a**, **706b**) is coupled to a linkage pivot, which in turn is operationally coupled to a mechanical selector of a surgical instrument **112**.

(824) The mechanical selector may be a mechanically actuated switch such as a momentary-action manual switch, mechanical-bail switch, capacitive touch switch, membrane switch, or other suitable mechanical switch. The mechanical switch may be controlled by the control circuit or generator to change the linkage pivot or linkage mechanism coupled to the end effector clamp arm such that the clamp arm exerts different compression force according to the selected energy modality. As such, in one position of the mechanical switch, the clamp arm may be linked so that when the actuator rod is actuated, relatively high compressive forces are applied. In a second position of the mechanical switch, the clamp arm may be linked so that when the actuator rod is actuated, relatively low compressive forces are applied. In one aspect, the first position corresponds to the ultrasonic energy modality, while the second position corresponds to the bipolar RF energy modality.

(825) Electrical methods of adjusting tissue compression force are also possible. For example, compression force can be automatically adjusted by the situationally aware surgical instrument **112** with the use of an electroactive polymer (EAP). The EAP can be, for example an electric EAP (e.g., ferroelectric polymer), ionic EAP (e.g., ionomeric polymer-metal composite), non-ionic EAP, conductive polymer or other suitable EAP. The EAP can be arranged in parallel with an RF energy circuit of the surgical instrument **112**, where the RF energy circuit is configured to implement the delivery of RF energy. Accordingly, the selection of an RF energy modality would cause current to flow through the RF energy circuit as RF energy is being delivered, which would also cause the EAP to expand. Upon expansion of the EAP, the tissue gap size decreases and results in the application of greater compression force.

(826) Additionally or alternatively, multiple sets of electrodes (e.g. electrodes **796**, **3074a**, **3074b**) may be provided and activated according to a particular sequence. This type of surgical treatment can be called hybrid activation. In hybrid activation, multiple different electrodes in the end effector are provided to perform different surgical functions. For example, a first set of electrodes may be used for the sealing surgical stage and a second set of electrodes may be used for the cutting surgical stage. To this end, a switch, filter, or other suitable wiring is provided to route the drive signal delivered by the generator to one or more appropriate electrodes in the end effector. For example, the situationally aware surgical instrument **112** may determine that an end effector electrode is configured for sealing rather than cutting. Consequently, an RF drive signal of relatively low voltage and high current may be driven through the output port of the generator to the pre-defined sealing electrodes for sealing. Similarly, an RF drive signal of greater power than the one used for sealing may be driven to different pre-defined cutting electrodes. The surgical instrument **112** may determine when the shift from the sealing electrodes to the cutting electrodes should occur based on a measured impedance threshold. When the impedance threshold is reached, the surgical instrument **112** may determine that a sufficiently secure seal has been created and that the cutting stage of the surgical procedure may begin. In general, the drive signal from the generator output port is routed appropriately to the corresponding electrode. Thus, drive signals can be routed to the appropriate treatment electrodes based on the appropriate stage of the surgical operation being performed. Also, the tissue compression force can be adjusted to a suitable level based on the power, time, or proportion of the drive signal or signals that are delivered to the tissue.

(827) In some aspects, the automatic adjustment of jaw clamp pressure may be based on an algorithm implemented by a control circuit of the surgical instrument **112**. As described above, the control circuit is configured to set and change various control parameters of the surgical instrument **112**, including clamp pressure, power delivered to treat tissue, and amplitude and frequency of waveform/drive signal output by the generator. Each energy modality may generally correspond to a compression force. For example, the RF (whether bipolar or monopolar) energy modality generally requires a higher extent of tissue pressure compared to the ultrasonic energy modality. The high tissue compression forces used for low operating temperature RF energy may be advantageous for treating soft and connective tissue. Bipolar RF energy may require even higher tissue compression forces as opposed to those used in monopolar RF energy applications. In one aspect, the compression force adjustment algorithm may be adjusted based on a proportion of two or more different energy modalities. That is, two energy modalities could be applied during a surgical procedure and the proportion of time spent on each modality could be used in an algorithm to calculate suitable tissue compression forces to be applied during the surgical procedure. Energy according to the blended multiple energy modalities could be delivered simultaneously or substantially simultaneously. The situationally aware compression force adjustment algorithm may also be dynamically modified or updated via the surgical instrument **112** receiving an update from a corresponding surgical hub or the cloud.

(828) FIG. **113** is a logic flow diagram **135000** depicting a control program or a logic configuration to adjust compression force applied to tissue, based on one or more selected energy modalities, according to at least one aspect of the present disclosure. The compression force is adjustable for a surgical procedure performed with a surgical instrument **112**. A control circuit or processor of the surgical instrument **112** (FIGS. **10-17**) or hub **106** (e.g., processor **244** FIG. **8**) determines **135002** a type of tissue (which includes any tissue described herein, but is referred to as tissue **8410** for the sake of clarity) being treated by the surgical instrument **112**. Tissue types include, for example, connective tissue (e.g., blood vessels), muscular tissue, and bronchus tissue. Tissue types could be detected or determined in a number of ways, such as by using spectral analysis of tissue bites. As discussed above, spectral analysis of different jaw bites and device states produces different complex impedance characteristic patterns (fingerprint) across a range of frequencies for different conditions and states. Spectroscopy may be applied to surgical instrument **112** by exciting the tip of the ultrasonic blade of the surgical instrument **112** with a sweep of frequencies, for example. The complex impedance characteristic patterns across a range of frequencies could be used in a model or classifier to infer tissue types.

(829) Aside from inferring tissue type, tissue characteristics such as tissue thickness and stiffness may be determined, for example. The determination or inference of tissue type and tissue characteristics may be performed by the control circuit, including control circuit **500**, **710**, **760**, **3200**, **3300**, **3402**, **3502**, **3686**, **3900**. For the sake of clarity, control circuit **3900** is referenced in this portion of the present disclosure. Control circuit **3900** may comprise the processors described above, as appropriate, including processors **822**, **1740**, **1900**, **3214**, **3302**. In one aspect, the control circuit **3900** may cause the generator to apply a non-therapeutic signal to the end effector over a range of frequencies. Subsequently, the control circuit **3900** can determine the tissue impedance based on the impedance characteristic pattern derived from spectral analysis of the non-therapeutic signal, as discussed above. For the sake of clarity, generator **1100** is referred to in this portion although the generator may be any generator described here, including generator **800**, **900**, **1100**, **4002**. Also for the sake of clarity, end effector **8400** is referenced in this portion although the end effector may be any end effector described above, including end effector **702**, **752**, **792**, **1122**. In some aspects, the generator **1100** comprises the control circuit **3900**; that is, the control circuit **3900** can be a component of the generator **1100**.

(830) Based on the tissue type and characteristic determination the situationally aware surgical instrument **112** might infer **135002** a type of surgical procedure or tissue treatment. Alternatively, a suitable surgical procedure could be manually determined or input by the clinician using the surgical instrument **112**. Based on the surgical procedure to be performed, a first energy modality is selected **135004** by the control circuit **3900** so that energy may be delivered by the generator **1100** to tissue **8410** grasped by the end effector **8400**. Delivering energy may comprise outputting drive signals according to the selected energy modality (e.g., ultrasonic drive signals) or electrical signal waveforms (e.g., current waveform determined based on LUT **2260** and used to drive the ultrasonic transducer **1120** or digital waveform **4300**). As described above, the drive signal may have the waveform shape of the waveform generated by the generator **1100** or control circuit **3900**. Based on the surgical procedure, a second energy modality is selected **135006** by the control circuit **3900**. More than two energy modalities could also be selected by the control circuit **3900**. The control circuit **3900** or generator **1100** may then generate signal waveforms according to or based on the selected energy modalities or a tissue treatment algorithm, which could be received from the corresponding hub **106**, **206** or cloud **104**, **204**. As discussed above, energy modalities include ultrasonic, bipolar or monopolar RF, irreversible and/or reversible electroporation, and/or microwave energy, among others. Energy modalities can be selected depending

on the type of treatment being performed. For example, the first and second energy modalities could be the ultrasonic energy modality and bipolar RF energy modality, respectively.

(831) A tissue treatment algorithm is determined **135008**. As discussed above, the surgical instrument **112** may receive **112** may receive the tissue treatment algorithm from an external source. Additionally or alternatively, the control circuit **3900** may determine the tissue treatment algorithm or an adjustment to the received algorithm based on the determined tissue type and selected energy modalities from steps **135002**, **135004**, **135006**. In particular, the tissue treatment algorithm may define parameters of the surgical treatment such as the power and timing of the drive signal and the proportion of the energy modalities. Such parameters also may be dynamically adjusted by control circuit **3900** or generator **1100** during performance of the surgical procedure. In general, the tissue treatment algorithm could be inferred by the surgical instrument **112**, received by the surgical instrument **112** through the corresponding surgical hub or cloud, or manually set by the clinician. The generator **1100** may infer what energy modality to apply based on feedback conditions or other sensed data received by the generator. For example, undeformed tissue thickness could be detected via a contact sensor **738** and could be used as one example characteristic in the inference of tissue type. As discussed above, different energy modalities may be advantageous for different tissue types and treatments. For example, ultrasonic energy may be well suited for treating smaller tissue **8410** (e.g., a small blood vessel). In particular, treatment with an ultrasonic blade may be appropriate for ultrasonic coagulation of a small vessel.

(832) In contrast, RF energy may be more suitable for cauterization of larger tissue. To this end, as discussed above, the generator **1100** may deliver energy with higher voltage and lower current to drive an ultrasonic transducer, with lower voltage and higher current to drive RF electrodes for sealing tissue **8410**, or with a coagulation waveform for spot coagulation using either monopolar or bipolar RF electrosurgical electrodes. In one aspect, the treatment tissue algorithm may specify the times during a surgical procedure that a particular energy modality is applied (e.g., RF versus ultrasonic). For example, the algorithm may define that a mixture of RF and ultrasonic energy are delivered during the sealing stage of the surgical procedure and only ultrasonic energy is delivered during the cutting stage. To this end, the generator **1100** may deliver ultrasonic and electrosurgical RF energy simultaneously from its output port to provide the desired output drive signal. The mixture of RF and ultrasonic could be applied simultaneously or substantially simultaneously as a blended energy modality or the generator **1100** could be configured to switch between delivering energy according to the RF and ultrasonic energy modalities (e.g., by switching between RF generator circuit **3902** and the ultrasonic generator circuit **3920** energy modalities). The treatment tissue algorithm may also specify the power at which the energy modalities are applied. For example, as discussed above, the waveform generator **904** and processor **902** of the generator **1100** are configured to generate signal waveforms of various amplitudes (the power parameter could be set by controlling the input of the power amplifier **1620** to set a particular waveform amplitude). The treatment tissue algorithm can also define how the amplitude, frequency and shape of the waveforms output by the generator **1100** changes over the course of the surgical procedure being performed.

(833) The generator **1100** delivers **135010** energy to the tissue **8410** according to the desired tissue treatment algorithm. It may be possible to change the tissue treatment algorithm during performance of the surgical procedure, such as via manual input by the clinician or data transmitted by the corresponding hub **106**, **206** or cloud **104**, **204**. Also, as discussed above, tissue compression force and energy delivered to the tissue **8410** can vary inversely. As such, in some situations when the amplitude of the waveform output by the generator **1100** increases, the compression force may be decreased. Similarly, when the amplitude decreases, the compression force may be increased. In other situations, the compression force may stay the same even as the amplitude changes. Similarly, the compression force may change even as the amplitude stays constant. The situationally aware surgical instrument **112** may determine the proportion of the first energy modality versus the second energy modality. For example, a clock generator (e.g., clock **3330**) may produce a clock signal used to track the duration that RF waveforms are applied and the duration that ultrasonic waveforms are applied during a surgical operation. Accordingly, the control circuit **3900** calculates **135012** the proportion of the first energy modality to the second energy modality. The control circuit **3900** may calculate **135012** the proportion based on a time or duration of the respective waveforms/drive signals of the selected energy modalities are applied as well as a frequency or amplitude of the respective waveforms. The proportion of additional energy modalities (e.g., a third energy modality) may also be calculated **135012** if such additional modalities are used.

(834) In one aspect, the compression force corresponding to one energy modality is relatively higher. That is, a range of compression force at which RF energy is applied may be greater than the range of compression force at which ultrasonic energy is applied. The proportion of energy modalities may be used to determine an appropriate level to set the pressure exerted by the end effector **8400**. In other words, the proportion may be used to determine an appropriate tissue gap size of the end effector **8400** jaws. For some surgical instruments **112**, the jaws of the end effector **8400** may be considered the clamp arm and the ultrasonic blade or waveguide. For the sake of clarity, clamp arm **716** is referenced in this portion although the clamp arm may be any clamp arm described in the present disclosure, including clamp arm **716**, **766**, **1140**, **1142a**, **1142b**. Similarly, ultrasonic blade **718** is referenced in this portion of the present disclosure although the ultrasonic blade may be any ultrasonic blade described here, including ultrasonic blade **718**, **768**, **1128**. When trigger such as trigger **4010** is actuated, the end effector **8400** closes such as tissue **8410** is clamped between the clamp arm **716** and ultrasonic blade **718**. In one aspect, the first energy modality such as the ultrasonic energy modality corresponds to a higher level of compression force while the second energy modality such as the RF energy modality corresponds to a lower level of compression force. The processor or control circuit may adjust **135014** tissue compression based on the calculated **135012** proportion. For example, a surgical procedure in which more ultrasonic energy and less RF energy are delivered may result in an overall algorithmic adjustment to greater tissue compression force. Similarly, a procedure in which less ultrasonic energy and more RF energy are delivered may result in an overall algorithmic adjustment to lesser tissue compression force.

(835) When ultrasonic energy and RF energy are delivered simultaneously or substantially simultaneously, the overall proportion of ultrasonic to RF energy may be used to determine the appropriate compression force. More generally, the proportion of selected energy modalities is useable by the control circuit **3900** to determine the adjustment. In one example, a lower proportion of ultrasonic to RF energy would correspond to a higher compression force compared to the compression force corresponding to a higher proportion. Also, the timing and power parameter of the delivered energy modalities during the surgical procedure may be considered for the adjustment **135014** by the control circuit **3900**. For example, if relatively high power RF energy is delivered at the sealing stage, then this may result in a smaller increase or adjustment to compression force than compared to a situation in which relatively high power RF energy is delivered at the cutting stage. Moreover, the tissue compression force adjustment may be made for only a discrete portion of the surgical procedure being performed. The timing and duration of the compression force adjustment can be determined based on the calculated proportion, selected tissue treatment algorithm, and tissue type and characteristics, among other possible considerations. As discussed in further detail below, both mechanical and electrical methods are provided to adjust the tissue gap size and consequently the applied pressure/compression force. That is, the control circuit **3900** may use the disclosed mechanical or electrical methods to adjust the size of the gap defined between the clamp arm **716** and ultrasonic blade **718**. Audible feedback could be provided in the trigger **4010**, for example, to indicate that the applied compression force was adjusted. Also, the amount of the adjustment could be displayed in a display of the surgical instrument **112**, based on the control circuit **3900** assessing the compression forces applied to the clamp arm **716**. The logic flow diagram **135000** of FIG. **113** could also be performed by a control circuit or processor the generator **1100**.

(836) FIG. **114** illustrates a mechanical method of adjusting compression force applied by an end effector **135100** for different treatment types, according to one aspect of the present disclosure. End effector **135100** of the surgical instrument **112** is the same as or similar to other end effectors described above, such as end effector **702**, **752**, **792**, **1122**, **8400**. Accordingly, end effector **135100** may comprise two jaw members. In various aspects, the jaw members are a clamp arm **135102** and a waveguide or blade **135104**. The clamp arm **135102**, which is the same or similar to clamp arm **716**, **766**, **1140**, **1142a**, **1142b**, and the blade **135104**, which is the same or similar to ultrasonic blade **718**, **768**, **1128**, can be configured to rotate. For example, as shown in FIG. **114-115B**, the ultrasonic blade **135104** is rotatable three hundred and sixty degrees. Additionally or alternatively, the

clamp arm **135101** is also rotatable three hundred and sixty degrees. In this way, as illustrated in FIG. **114**, the ultrasonic blade **135104** can be transitioned between a horizontal or landscape orientation to a vertical or portrait orientation, including intermediate positions, which may be in between the horizontal and vertical orientations or may exceed ninety degrees.

(837) Stated differently, the ultrasonic blade **135104** has a zero degree orientation corresponding to a horizontal orientation and a ninety degree orientation corresponding to a vertical orientation. With the clamp arm **135102** held in a constant or substantially constant position (shown in FIG. **114**), the ultrasonic blade **135104** is rotatable to define a spectrum of clamp pressure. An opposite configuration is also possible such that the ultrasonic blade **135104** is the constant jaw member rather than the clamp arm **135102**. In one aspect, as the ultrasonic blade **135104** rotates from zero degrees to ninety degrees, the tissue compression force increases from a low force to a high force. The tissue gap resulting from the zero degree orientation may be called low clamp **135106** while the tissue gap resulting from ninety degrees orientation may be called high clamp **135108**. Various orientations of ultrasonic blade **135104** correspond to the same level of compression pressure. For example, zero degrees and one hundred and eighty degrees both correspond to low clamp **135106**. FIG. **114** depicts the low clamp **135106**, high clamp **135108**, and a third orientation **135110**. The third orientation **135110** depicted in FIG. **114** is slightly greater than ninety degrees, such as a one hundred and twenty degree orientation. Consequently, this third orientation **135110** defines a tissue gap size that is slightly smaller than the tissue gap corresponding to high clamp **135108**.

(838) In the horizontal orientation (low clamp) **135106**, the ultrasonic blade **135104** may be configured for tissue sealing (e.g., coagulation or cauterization). In the vertical orientation (high clamp) **135108**, the ultrasonic blade **135104** may be configured for tissue cutting or dissection. As discussed above, the RF energy modality may generally correspond to a greater tissue compression force. Accordingly, in one aspect, the horizontal orientation **135106** corresponds to the ultrasonic energy modality while the vertical orientation **135108** corresponds to the RF energy modality. Other intermediate positions, including third orientation **135110**, may be used as appropriate during the surgical operation. For example, an RF waveform of relatively high power and an intermediate orientation such as 60 degrees may be used when surgical treatment initially commences. Furthermore, the ultrasonic blade **135104** orientation may change throughout a performed surgical procedure according to the selected **135008** tissue treatment algorithm, for example. In another aspect, the ultrasonic blade **135104** may be an oval shape and offset relative to the clamp arm **135102**.

(839) Other mechanical methods of adjusting compression force are disclosed. For example, the clamp arm **135102** or ultrasonic blade **135104** may be moveable such that the end effector **135100** is configurable between a closed configuration, an open configuration, and intermediate positions in between to define various clamp pressures. In one aspect, a mechanical switch such as a momentary-action manual switch, mechanical-bail switch, capacitive touch switch, membrane switch, or other suitable mechanical switch of the surgical instrument **112** can transition between two positions. The control circuit **3900** may control operation of the mechanical switch. The first and second positions may correspond to a first and second compression force level, which in turn may correspond to a first and second energy modality, respectively. Thus, for example, in the first position, the mechanical switch could cause an adjustment to the stroke or longitudinal movement of an actuator rod. The actuator rod may refer to a suitable end effector **135100** actuation mechanism, such as the linkage components of transmissions **706a-706e** to couple motors **704a-704e**.

(840) In particular, the actuator rod may comprise linkage elements similar to or the same as transmissions **706a-706e** used to transmit mechanical energy from the motors **704a-704b** to actuate or close closure member **714** and clamp arm **716**, respectfully. The actuation stroke of the actuator rod could be adjusted by the control circuit **3900** to achieve the different compression forces of the end effector **1351000**. With the adjustment caused by the first position of the mechanical switch, actuation of the actuator rod may result in a clamp arm **135102** orientation corresponding to a relatively large tissue gap. Conversely, in the second position, the mechanical switch could cause a different adjustment such that actuation of the actuator rod may result in a clamp arm **135102** orientation corresponding to a relatively small tissue gap. The first position could correspond to the delivery of energy according to the ultrasonic modality while the second position could correspond to the delivery of energy according to the RF modality. In one aspect, by shifting between the first and second position, the mechanical switch shifts the pivotal linkage or coupling between the actuator rod and the clamp arm **135102**. In this way, the control circuit **3900** can control the mechanical switch to implement adjustments to actuator stroke which in turn results in various clamp arm **135102** configurations (between and including open and closed configurations) that correspond to different tissue compression forces. Other suitable mechanical means of adjusting the actuator stroke or clamp arm configuration are also possible.

(841) FIGS. **115A-115B** illustrate a mechanical method of adjusting compression force applied by an end effector **135200** for different treatment types, by rotating an ultrasonic blade **135204**, according to aspects of the present disclosure. End effector **135200** and ultrasonic blade **135204** are the same as or similar to end effector **135100** and ultrasonic blade **135104**, respectively. The ultrasonic blade **135204** is rotatable between a vertical and a horizontal configuration, as shown in FIGS. **115A-115B**. In one aspect, the end effector **135200** comprises a jaw member **135202** (e.g., clamp arm **135102**, **716**), flexible circuit **135206** and the ultrasonic blade **135204**. Additionally or alternatively, the jaw member **135202** may be rotatable as well. FIG. **115A** depicts tissue **135208** located between the jaw member **135202** and the ultrasonic blade **135204**. In the horizontal orientation shown, the ultrasonic blade **135204** is at or substantially at a zero degree orientation. Accordingly, relatively low compression force is applied to the tissue **135208**. In one aspect, the ultrasonic blade **135204** is configured for tissue sealing (e.g., cauterization) in the horizontal orientation. The ultrasonic blade **135204** also comprises side lobe sections **135210a**, **135210b** to enhance tissue dissection and uniform sections **135212a**, **135212b** to enhance tissue sealing. As discussed above, the control circuit **3900** may control rotation of the jaw member **135202** or ultrasonic blade **135204**.

(842) FIG. **115B** depicts the vertical orientation in which the ultrasonic blade **135204** is at or substantially at a ninety degree orientation. In another aspect, the ultrasonic blade **135204** is configured for tissue dissection in the vertical orientation. The flexible circuit **135206** may include electrodes such that when a RF energy modality is selected, the electrodes are configured to deliver high-frequency RF current to the tissue **135208**. The electrodes may be the same or similar to electrodes described in the present disclosure, such as electrodes **796**, **3074a**, **3074b**, **3906a**, **3906b**. Lower-frequency RF current is also possible. When the RF energy modality is selected, the ultrasonic blade **135204** may act as the electrical ground for the RF waveform output from the generator **1100**. That is, the RF electrodes (e.g., RF electrodes **796**) could be coupled to the positive pole while the ultrasonic blade **135204** is coupled to the negative or return pole. In some configurations, the polarity may be reversed such that the RF electrodes **796** are coupled to the negative pole and the ultrasonic blade **135204** is coupled to the positive pole. In one aspect, when both the RF and ultrasonic energy modalities are selected, the RF current conducted by the electrodes **796** is used to seal the tissue **135208** and the ultrasonic blade **135204** is used to dissect tissue based on ultrasonic vibrations propagating through ultrasonic blade **135204**. As discussed above with respect to FIG. **115A-115B**, other intermediate orientations of the rotatable ultrasonic blade **135204** may also be possible, so that additional levels of different compression forces may be adjusted and applied to tissue **135208** as appropriate.

(843) Electrical methods of adjusting compression force between treatment types are also disclosed. For example, a suitable EAP may be used as an electrostatic actuator for changing tissue gap size. EAPs are voltage activated elastomers which can be electronic EAPs such as electrostrictive elastomers and dielectric electroactive polymers (DEAPs) or ionic EAPs such as ionic polymer metal composites (IPMCs). EAPs have an electromechanical thickness or other strain that is induced by electrostatic forces. Stated differently, when a voltage is applied to an EAP, the EAP bends, contracts, or expands. An EAP actuator may be used in the surgical instrument **112** to change the tissue gap size for changing the applied compression forces based on application of a voltage potential to the EAP. The EAP actuator may be controlled by the control circuit **3900**. For example, the EAP may be configured to expand, which causes the application of more pressure on the blade **135204** or electrodes **796** positioned in the end effector **135200**. To this end, the EAP may be positioned in the end effector **135200** between the power source (e.g., generator) and RF electrodes **796**. For example, the EAP could be part of the flexible circuit **135206**. In this way, as an RF waveform is output from the generator **1100**, voltage is applied to the EAP, which causes the EAP to expand and apply a force on the RF electrodes such that the tissue gap size decreases. In general, the EAP may expand as the generator **1100** delivers energy according to the selected energy modalities.

(844) Additionally or alternatively, when the EAP expands, this causes a force to be applied to the blade **135200** such that the tissue gap size decreases. Conversely, the EAP may be positioned and configured such that electrostatic actuation causes the EAP to contract, which reduces the

compression force applied to the tissue. The change in the size of the EAP may be proportional to the voltage used for EAP actuation and the adjustment to compression force that results. In one aspect, the EAP may also be positioned in the shaft of the surgical instrument **112**, for example, such that electrostatic actuation causes the EAP to exert further force on the clamp arm **135102** or the end effector **135200**. Similarly, the EAP could be configured to remove or reduce force applied by the clamp arm **135102**. In this way, the EAP actuator may be employed to change the end effector configuration, which spans between the open and closed configurations. In general, the EAP could be provided such that the control circuit **3900** can increase compression force as a greater proportion of energy is delivered according to the RF energy modality. Similarly, the EAP could be used to adjust pressure as another energy modality such as the ultrasonic energy modality is selected. The EAP could be part of the flexible circuit **135206**, for example.

(845) In general, the end effector **135200** of the surgical instrument **112** may comprise an ultrasonic blade **135200** and a clamp arm **135102**, which may function as the first and second jaws of the end effector **135200**. The end effector **135200** is configured to clamp tissue therebetween the jaws, fire fasteners through the clamped tissue **135208**, sever the clamped tissue **135208**, and grasp tissue **135208** for application of energy according to the selected energy modality. Moreover, the force applied to the tissue **135208** by the end effector **135200** may be measured by the strain gauge sensor **474**, such as by measuring the amplitude or magnitude of the strain exerted on a jaw member of the end effector **135200** during a clamping operation. As discussed above, energy may be delivered according to multiple energy modalities such as RF and ultrasonic energy, in conjunction to achieve surgical sealing, cutting, and coagulation functions. Although energy from the generator **1100** could be delivered simultaneously such as through simultaneously delivering or outputting waveforms from the generator **1100** to the RF electrodes **796** and the ultrasonic transducer **1120** in conjunction, such energy delivery can also switch between different energy modalities.

(846) FIG. **116** shows a diagram **135300** illustrating switching between active electrodes of an end effector **135200** according to one aspect of the present disclosure. Again, the electrodes may be the same or similar to electrodes **796**, **3074a**, **3074b**, **3906a**, **3906b**. As discussed above, the situationally aware surgical instrument **112** may infer or determine an appropriate tissue treatment algorithm for the surgical procedure being performed. The tissue compression force adjustment is made based on this algorithm and the proportion of selected energy modalities. In addition, the tissue compression force adjustment may be made based on switching from a passive electrode to an active electrode for treatment, as shown in FIG. **116**. For example, the situationally aware surgical instrument **112** may detect a selected function of the surgical instrument **112** such as sealing or cutting. Based on the selected function, a switch, filter, or other suitable wiring such as a relay or transistor may be provided to control routing the waveform (e.g., waveform **4300**) output by the generator **1100** to an appropriate electrode. FIG. **116** depicts the end effector **135200** with a first set of two treatment electrodes “A” **135302** and a second set of two treatment electrodes “B” **135304**, both of which may alternate between active or passive states. The two sets of treatment electrodes **135302**, **135304** are provided on both jaws of the end effector **135200**. Based on which treatment electrode is active or passive, the applied compression force may be adjusted.

(847) In one aspect, the “A” electrodes **135302** are configured for the sealing stage and the “B” electrodes **135304** are configured for the cutting stage. These configurations could be used by the surgical instrument **112** to determine if and what compression force adjustment is required when an energy modality is selected. For example, when the “A” electrodes **135302** become passive while the “B” electrodes **135304** become active, the surgical instrument **112** may adjust the compression force. This could be an adjustment to decrease the compression force, such as because the additional power delivered during the cutting stage might not require as much corresponding compression force. The extent of the adjustment can depend on the proportion of one energy modality (e.g., RF) to another (e.g., ultrasonic) as calculated throughout the duration of the performed surgical operation. In one aspect, the switch is configured to route the output energy by the generator **1100** to the sealing electrodes “A” **135302** while the surgical instrument **112** is used for coagulation. Multiple impedance thresholds **135306**, **135308** may be provided, which are indicated on the impedance graph (Z) **135312** in FIG. **116** as dotted lines. FIG. **116** shows that when threshold **135306** is crossed at point **135310**, the crossing may indicate when optimal tissue coagulation is complete. That is, when the measured tissue impedance reaches point **135310**, it may be determined that a sufficiently secure tissue seal has formed.

(848) As discussed above, impedance may be measured by dividing the output of the voltage sensing circuit and the current sensing circuit or by using spectral analysis, for example. When coagulation is complete, the switch may transition to a second position to route a different waveform from the generator **1100**, in which the amplitude of the different waveform is raised relative to the waveform used for coagulation. The different waveform may be applied for surgical cutting rather than for coagulation. Accordingly, the switch may route this different waveform to the “B” electrodes **135304**. As can be seen in the power graph **135314** of FIG. **116**, the amplitude of the waveform is greater than the amplitude of the coagulation waveform. Although the power levels shown in power graph **135314** are constant, dynamic power levels may be used as well. As discussed above, the increase in power from “A” electrodes **135302** to “B” electrodes **135304** may trigger an adjustment to tissue compression, which may be determined based on the proportion of one selected energy modality to another. In another aspect, the surgical cutting achieved via the “B” electrodes **135304** is a knifeless cutting. Although the energy modality selected for the tissue treatment illustrated in FIG. **116** may be RF, other energy modalities may be used for such treatment and over the course of a performed surgical operation.

Advanced RF Energy Device Including Nerve Stimulation Signal with Therapeutic Waveforms

(849) As disclosed above, in some surgical procedures, a medical professional may employ an electrosurgical device to seal or cut tissues such as blood vessels. Such devices effect a medical therapy by passing electrical energy, for example current at radiofrequencies (RF), through the tissue to be treated. Some electrosurgical devices are termed bipolar devices in that both an electrode to source the electrical energy (the active electrode) and a return electrode are housed in the same surgical probe. It will be appreciated that a surgical probe may comprise a handpiece or a robotically controlled instrument or a combination thereof.

(850) Alternative devices may be termed monopolar devices. In such devices, only the active electrode is housed in the surgical probe. The electrical current entering the patient's tissue may return to the electrical energy generator via an electrical path through the gurney on which the patient reposes or through a specific return electrode pad. In some aspects, the patient may repose on the electrode pad, or the electrode pad may be placed on the patient at a location close to the surgical site where the surgical probe is deployed. It may be recognized that the current path through a patient undergoing a procedure using a monopolar device may be less well characterized than the current path through a patient undergoing a procedure using a bipolar device. Consequently, some non-target tissue may be inadvertently cauterized, cut, or otherwise damaged by a monopolar electrosurgical device. Such unintended injury to excitable tissue may result in the patient experiencing muscle weakness, pain, numbness, paralysis and/or other undesired outcomes.

(851) It is therefore desirable that a monopolar electrosurgical device incorporate features to determine if the device is close enough to excitable tissue to cause inadvertent injury. Such features may be used by one or more subsystems of the electrosurgical device as a basis for notifying the medical professional of the proximity of such tissue to the monopolar electrode. Additionally, such features may be used by one or more subsystems of an intelligent electrosurgical device to reduce or eliminate the amount of therapeutic energy delivered to tissue deemed to be close to non-target excitable tissue. In some intelligent medical devices that combine electrosurgical (RF) with ultrasonic therapeutic modes, features to determine if the device is close enough to excitable tissue to cause inadvertent injury when the device is operating in the electrosurgical (RF) mode may result in the device switching to the ultrasonic mode.

(852) Electrosurgical devices for applying electrical energy to tissue in order to treat and/or destroy the tissue are also finding increasingly widespread applications in surgical procedures. An electrosurgical device typically includes a surgical probe, an instrument having a distally-mounted end effector (e.g., one or more electrodes). The end effector can be positioned against the tissue such that electrical current is introduced into the tissue. Electrosurgical devices can be configured for bipolar or monopolar operation. During bipolar operation, current is introduced into and returned from the tissue by active and return electrodes, respectively, of the end effector. During monopolar operation, current is introduced into the

tissue by an active electrode located at a distal end of the surgical probe and returned through a return electrode (e.g., a grounding pad) located on a patient's body. Heat generated by the current flowing through the tissue may form hemostatic seals within the tissue and/or between tissues and thus may be particularly useful for sealing blood vessels, for example. The end effector of an electrosurgical device also may include a cutting member that is movable relative to the tissue and the electrodes to transect the tissue.

(853) FIG. 117 depicts a typical monopolar electrosurgical system **136000**. The electrosurgical system **136000** can include a controller **136010**, a generator **136012**, an electrosurgical instrument **136015**, and a return pad **136020** which includes one or more return electrodes. Typically, the generator **136012** may source an electrical signal to the electrosurgical instrument **136015** along a first conducting electrical path **136017** and may receive a return signal from the one or more return electrodes along a second conducting electrical path **136023**. FIG. 117 depicts an example of a health care professional **136025** treating a patient **136027** using an electrosurgical instrument **136015** such as an active monopolar electrode.

(854) FIG. 118 is a schematic block diagram of the patient and electrical components depicted in FIG. 117. The generator **136012** may be a separate component from the controller **136010** or the controller **136010** may include the electrical generator **136012**. The controller **136010** may control the operation of the generator **136012**, including controlling an electrical output thereof. As disclosed below, the controller **136010** may control one or more output waveforms of the electrical generator **136012** including the control of a variety of characteristics including amplitude characteristics, frequency characteristics, and phase characteristics of the output signal of the electrical generator **136012**. The controller **136010** may further receive signals from any number of additional components including, without limitation, manual control actuators (switches, push buttons, slides, and similar), sensors, or data signals transmitted by any number of communication devices, computers, smart surgical devices, and imaging systems. The controller **136010** may be composed of any type or types of computer processor devices, one or more memory components (static and/or dynamic memory components), and communication components configured to transmit and/or receive data signals (analog or digital) as may be required for the functioning of the controller. The memory components of the controller **136010** may contain one or more instructions that, when read by the one or more computer processor devices, may direct the operation of the controller. Examples of such instructions and their intended results are disclosed below.

(855) Electrical energy may be sourced by the electrical generator **136012** and received by a surgical instrument **136015** such as an active monopolar electrode. In some aspects, the active electrode may be in electrical communication with an electrical source terminal of the electrical generator **136012** to receive the electrical energy. In some aspects, the surgical instrument **136015** may receive an electrical signal over a first conducting electrical path **136017** such as a wire or other cabling.

(856) During the procedure, the patient **136027** may lie supine on a return pad **136020**. The return pad **136020** may be in electrical communication with the electrical generator **136012** via an electrical return terminal, and the electrical energy sourced into the patient **136027** by the electrosurgical instrument **136015**, such as an active electrode, may be returned to the electrical generator **136012** through the return pad **136020**. In some aspects, the return pad **136020** may be in electrical communication with the electrical return terminal over a second conducting electrical path **136023**, such as a wire or other cabling.

(857) In some aspects, the generator **136012** may supply alternating current at radiofrequency levels to the electrosurgical instrument **136015**. In some alternative aspects, the electrosurgical instrument **136015** may also incorporate features for ultrasonic therapeutic modes, and the generator **136012** may also be configured to generate power to drive one or more ultrasonic therapeutic components. The electrosurgical instrument **136015**, which typically includes an electrode tip (i.e., an active electrode) which can be positioned at a target tissue of a patient **136027**, receives the alternating current from the generator **136012** and delivers the alternating current to the target tissue via the electrode tip. The alternating current received by the electrode tip may be from the generator **136012** via a first conducting electrical path **136017**. The alternating current is received at the target tissue, and the resistance from the tissue creates heat which provides the desired effect (e.g., sealing and/or cutting) at the surgical site. The alternating current received at the target tissue is conducted through the patient's body and ultimately is received by the one or more return electrodes of the return pad **136020**. The alternating current received by the return pad **136020** may be conducted back to the generator via a second conducting electrical path **136023** to complete the closed path followed by the alternating current. The one or more return electrodes are configured to carry the amount of current introduced by the electrode tip. The return pad **136020** may be attached to the patient's body or may be separated a small distance from the patient's body (i.e., capacitive coupling). The alternating current received by the one or more return electrodes is passed back to the generator **136012** to complete the closed path followed by the alternating current.

(858) For an electrosurgical system **136000** which utilizes capacitive coupling to complete the current path between the patient's body and the return electrode, the patient's body effectively acts as a first capacitive plate of a capacitor and the return electrode pad effectively acts as a second capacitive plate of a capacitor.

(859) In some aspects, the return pad **136020** may include a single return electrode which incorporates an array of multiple sensing devices. In some alternative aspects, the return pad **136020** may include an array of return electrodes, where an array of sensing devices may be incorporated into the array of return electrodes. In one non-limiting example, the return pad **136020** may include multiple return electrodes in which each of the return electrodes includes a sensing device.

(860) By incorporating an array of sensing devices into the return electrode pad **136020**, the sensing devices may be used to detect either a nerve control signal applied to the patient or a movement of an anatomical feature of the patient resulting from an application of the nerve control signal. The sensing devices may include, without limitation, one or more pressure sensors, one or more accelerometers, or combinations thereof. In some non-limiting aspects, a sensing device may be configured to output a signal indicative of the detected nerve control signal and/or the detected movement of an anatomical feature of the patient. Using Coulomb's law and the respective locations of the active electrode, the patient's body and the sensing devices, the detected nerve control signal and/or movement of an anatomical feature of the patient can be analyzed to determine the location of a nerve within the patient's body.

(861) In some aspects, for example as depicted in FIG. 120, a return pad **136120** may include a plurality of electrodes **136125** which can be capacitively coupled to the patient's body and collectively are configured to carry the amount of current introduced into the patient's body by the electrosurgical instrument. For this capacitive coupling, the patient's body effectively acts as one plate of a capacitor and collectively the plurality of electrodes **136125** of the return pad **136120** effectively act together as the other plate of the capacitor. A more detailed description of capacitive coupling can be found, for example, in U.S. Pat. No. 6,214,000, titled CAPACITIVE REUSABLE ELECTROSURGICAL RETURN ELECTRODE, issued Apr. 10, 2001 and in U.S. Pat. No. 6,582,424, titled CAPACITIVE REUSABLE ELECTROSURGICAL RETURN ELECTRODE, issued Jun. 24, 2003, the entire contents of which are each incorporated herein by reference and in their respective entireties.

(862) FIG. 118 illustrates a plurality of electrodes **136125a-d** of the return pad of FIG. 117, in accordance with at least one aspect of the present disclosure. Although four electrodes **136125a-d** are shown in FIG. 118, it will be appreciated that the return pad **136120** may include any number of electrodes **136125**. For example, according to various aspects, the return pad **136120** may include two electrodes, eight electrodes, sixteen electrodes, or any number of electrodes that may be fabricated in the return pad **136120**. It should be recognized that the number of electrodes may be an even integer or an odd integer. Also, although the individual electrodes **136125a-d** are shown in FIG. 118 as being substantially rectangular, it will be appreciated that the individual electrodes can be of any suitable shape.

(863) The electrodes **136125a-d** of the return pad **136120** may serve as the return electrodes of the electrosurgical system of FIGS. 117 and 118, and can also be considered to be segmented electrodes as the electrodes **136125a-d** can be selectively decoupled from the patient's body and/or the generator. In some aspects, the electrodes **136125a-d** of the return pad **136120** can be coupled together to effectively act as one large electrode. For example, according to various aspects, each of the electrodes **136125a-d** of the return pad **136120** can be connected by respective conductive members **136130a-d** to inputs of a switching device **136135** as shown in FIG. 119. When the switching device **136135** is in an open position, as

shown in FIG. 119, the respective electrodes **136125a-d** of the return pad **136120** are decoupled from one another as well as from the patient's body and/or the generator. In contrast, when the switching device **136135** is in a closed position, the respective electrodes **136125a-d** of the return pad **136120** are coupled together to effectively act as a single large electrode. It may be recognized that differing combinations of electrodes **136125a-d** may be coupled together by the switching device **136135** to form any group or groups of electrodes. For example, if a patient is disposed in a supine position on the return pad **136120** with the patient's head proximate to the switching device **136135**, electrodes **136125a** and **136125c** may be coupled together and electrodes **136125b** and **136125d** may be coupled together thereby sensing electrical currents flowing through the lower torso and upper torso, respectively. Alternatively, if a patient is disposed in a supine position on the return pad **136120** with the patient's head proximate to the switching device **136135**, electrodes **136125a** and **136125b** may be coupled together and electrodes **136125c** and **136125d** may be coupled together thereby sensing electrical currents flowing through the right torso and left torso, respectively.

(864) The switching device **136135** can be controlled by a processing circuit (e.g., a processing circuit of the generator of the electrosurgical system, of a hub of an electrosurgical system, etc.). For purposes of simplicity, the processing circuit is not shown in FIG. 119. According to various aspects, the switching device **136135** can be incorporated into the return pad **136120**. According to other aspects, the switching device **136135** can be incorporated into the second conducting electrical path of the electrosurgical system of FIGS. 117 and 118. The return pad **136120** can also include a plurality of sensing devices.

(865) FIG. 120 illustrates an array of sensing devices **136140a-d** of the return pad in accordance with at least one aspect of the present disclosure. According to various aspects, the number of sensing devices **136140a-d** may correspond to the number of electrodes **136125a-d** such that there is one sensing device for each electrode (for example, sensing device **136140a** with electrode **136125a**, sensing device **136140b** with electrode **136125b**, sensing device **136140c** with electrode **136125c**, and sensing device **136140d** with electrode **136125d**). Each sensing device **136140a-d** may be mounted to or integrated with a corresponding electrode **136125a-d**, respectively. However, although the number of sensing devices **136140a-d** associated with the corresponding electrodes **136125a-d** may correspond to the number of electrodes, it will be appreciated that the return pad may include any number of sensing devices. For example, for aspects of the return pad which includes sixteen electrodes, the return pad may only include four or eight sensing devices. Although the sensing devices **136140a-d** are shown in FIG. 120 as being centered on the corresponding electrodes **136125a-d**, respectively, it will be appreciated that the sensing devices **136140a-d** can be positioned on any portion of the corresponding electrodes **136125a-d**. It may be further understood that the position of a particular sensing device on a particular electrode is independent of a position of any other sensing device on its respective electrode.

(866) The sensing devices **136140a-d** are configured to detect a monopolar nerve control signal applied to the patient and/or a movement of an anatomical feature of the patient (e.g., a muscle twitch) resulting from application of the nerve control signal. The monopolar nerve control signal may be applied by the surgical instrument of the electrosurgical system of FIGS. 117 and 118, or may be applied by a different surgical instrument which is coupled to a different generator. Each sensing device **136140a-d** may include, for example, a pressure sensor, an accelerometer, or combinations thereof, and is configured to output a signal indicative of the detected nerve control signal and/or the detected movement of an anatomical feature of the patient. In some non-limiting examples, a sensing device composed of a pressure sensor may include for example, a piezoresistive strain gauge, a capacitive pressure sensor, an electromagnetic pressure sensor, and/or a piezoelectric pressure sensor either alone or in combination. In some non-limiting examples, a sensing device composed of an accelerometer may include, for example, a mechanical accelerometer, a capacitive accelerometer, a piezoelectric accelerometer, an electromagnetic accelerometer, and/or a microelectromechanical system (MEMS) accelerometer either alone or in combination. The respective output signals of the respective sensing devices **136140a-d** may be in the form of analog signals and/or digital signals.

(867) Using Coulomb's law and the respective locations of the active electrode of the surgical instrument, the patient's body and the respective sensing devices, the respective output signals of the respected sensing devices **136140a-d**, which are indicative of a detected nerve control signal and/or movement of an anatomical feature of the patient, can be analyzed to determine the location of a nerve within the patient's body. Coulomb's law states that $E = K(Q/r \cdot \sup{2})$, where E is the threshold current required at a nerve to stimulate the nerve, K is a constant, Q is the minimal current from the nerve stimulation electrode and r is the distance from the nerve. The further the nerve stimulation electrode is from the nerve (r increases), the current required to stimulate the nerve is proportionately greater. Thus, the amount of stimulation of an excitable tissue as measured by a sensing device **136150a-d** may be related to the distance of the nerve stimulation electrode to the excitable tissue at constant current stimulation. In some aspects, an output signal of a sensing device **136140a-d** may also be dependent on the distance of the excitable tissue to the sensing device **136140a-d**. It may be recognized that multiple sensing devices **136140a-d** may be used to triangulate the position of an electrically stimulated excitable tissue based on the geometry and position of the multiple sensing devices **136140a-d**. A constant current stimulus can thus be utilized to estimate the distance from the nerve stimulation electrode to the nerve. Alternatively, current stimulus composed of varying amounts of current may be used to improve the determination of the position of the excitable tissue through the triangulation method associated with multiple sensing devices **136140a-d**. In general, the respective strengths of the output signals of the respective sensing devices are indicative of how close or far the respective sensing devices are from the stimulated nerve of the patient.

(868) According to various aspects, the analysis of the respective output signals of the respective sensing devices can be performed by a processing circuit of the generator of the electrosurgical system of FIGS. 117 and 118, by a processing circuit of a nerve monitoring system which is separate from the generator of the electrosurgical system thereof, by a processing circuit of a hub of an electrosurgical system, etc. The analysis can be performed in real time or in near-real time. According to various aspects, the respective output signals serve as inputs to a monopolar nerve stimulation algorithm which is executed by the processing circuit.

(869) As shown in FIG. 120, according to various aspects, the output signals of the respective sensing devices **136140a-d** can be input into a multiple input—single output switching device **136137** (e.g., a multiplexer) via respective conductive members **136142a-d**, respectively. By controlling the selection signals **S0, S1** to the multiple input—single output switching device **136137**, the multiple input—single output switching device **136137** can be controlled to output only one of the output signals of the respective sensing devices **136140a-d** at a time for the above-described analysis. As one non-limiting example, with reference to FIG. 120, by setting the selection signals **S0, S1** to 0,0, the output signal from the sensing device **136140c** can be output by the multiple input—single output switching device **136137** for analysis by the applicable processing circuit. In another non-limiting example, setting the selection signals **S0, S1** to 0,1, the output signal from the sensing device **136140a** can be output by the multiple input—single output switching device **136137** for analysis by the applicable processing circuit. Similarly, by setting the selection signals **S0, S1** to 1,0, the output signal from the sensing device **136140d** can be output by the multiple input—single output switching device **136137** for analysis by the applicable processing circuit. And, by extension, by setting the selection signals **S0, S1** to 1,1, the output signal from the sensing device **136140b** can be output by the multiple input—single output switching device **136137** for analysis by the applicable processing circuit.

(870) The selection signals **S0, S1** can be provided to the multiple input—single output switching device **136137** by a processing circuit such as, as non-limiting examples, a processing circuit of the generator of the electrosurgical system of FIGS. 117 and 118, a processing circuit of a nerve monitoring system which is separate from the generator of the electrosurgical system, by a processing circuit of a hub of an electrosurgical system, and similar. For purposes of simplicity, the processing circuit is not shown in FIG. 120. By providing the various selection signals at a fast enough rate, the output signals of the respective sensing devices **136125a-d** can effectively be scanned at a rate which allows for the timely analysis of all of the output signals of the respective sensing devices **136125a-d** to determine the position of the stimulated nerve.

(871) According to various aspects, the multiple input—single output switching device **136137** can be incorporated into the return pad. According to other aspects, the multiple input—single output switching device **136137** can be incorporated into the second conducting electrical path **136023** of the electrosurgical system **136000** of FIG. 117.

(872) The control of the multiple input—single output switching device **136137** as disclosed in FIG. **120** may be in the context of a four input—four output switching device, corresponding to the four sensing devices **136140a-d** depicted in FIG. **120**. It will be appreciated that for aspects in which there are more than four sensing devices (e.g., sixteen sensing devices), the output signals of the more than four sensing devices may serve as inputs to a multiple input—single output switching device having more than two selection signals (e.g., **S0**, **S1**, **S2** and **S3**).

(873) For aspects where the output signals of the sensing devices (for example **136140a-d** are analog signals, the output of the multiple input—single output switching device **136137** can be converted into a corresponding digital signal by an analog-to-digital converter **136145** prior to the performance of the analysis of the output signals by the applicable processing circuit.

(874) Returning to FIG. **117**, according to various aspects, the detection of the nerve control signal and/or the movement of an anatomical feature of the patient by the sensing devices can be performed while the electrodes **136125a-d** of the return pad **136120** are coupled to one another or while the electrodes **136125a-d** are uncoupled from one another. For example, with regard to performing the detection when the respective electrodes **136125a-d** of the return pad **136120** are uncoupled from one another, after positioning the patient on the operating table but before starting a surgical procedure, the return pad **136120** can be placed in a “sensing mode” by controlling the switching device **136135** to uncouple the respective electrodes **136125a-d** of the return pad **136120** from one another. While the respective electrodes **136125a-d** are uncoupled from one another, a nerve and/or a nerve bundle can be stimulated with an electrosurgical instrument as described above, and the respective output signals of the sensing devices of the return pad **136120** can be analyzed as described above to identify where the nerve, nerve bundle and/or nerve nexuses associated therewith are located. The locations of the nerve, nerve bundle and/or nerve nexuses may be input into a monopolar nerve stimulation algorithm profile. Once the locations of the nerve, nerve bundle and/or nerve nexuses are input into the monopolar nerve stimulation algorithm profile, the locations of the nerve, nerve bundle and/or nerve nexuses may be effectively isolated from the capacitive operation of the electrodes of the return pad **136120**. The locations of the nerve, nerve bundle and/or nerve nexuses may be used as sensing nodes of the monopolar nerve stimulation algorithm profile to inform the surgeon as the surgeon approaches a nerve and/or a nerve bundle while performing a tissue cutting procedure. According to various aspects, the surgeon may be informed of the nearby location of the nerve and/or nerve bundle via an audible warning, a visual warning, a tactile (such as vibratory) warning, etc.

(875) Returning to FIG. **118**, with regard to performing the detection when the respective electrodes of the return pad **136015** are coupled with one another, according to various aspects, the generator **136012** of the electrosurgical system can generate a high frequency waveform (the alternating current at radio frequency) which may be modulated on a carrier wave having a sufficiently low frequency to stimulate a nerve of the patient. This modulation may allow for the sensing of the nerve control signal and/or the movement of an anatomical feature concurrently with the capacitive coupling of the respective electrodes of the return pad **136020** with the patient's body **136027**. By applying a specific waveform to the patient **136027** and sensing a specific response, there is a high level of confidence that the movement of the anatomical feature may be correlated with the applied waveform and not due to random patient motion. The modulation can be adjusted over time to stimulate different nerve sizes. According to various aspects, the modulation can be varied in amplitude over time in order to allow the applicable processing circuit to determine the distance the nerve and/or nerve bundle is from the signal without having to constantly stimulate the nerve and/or nerve bundle.

(876) The electrical energy applied by a surgical probe of an electrosurgical device to the tissue may be in the form of radio frequency (RF) energy that may be in a frequency range described in EN 60601-2-2:2009+A11:2011, Definition 201.3.218—HIGH FREQUENCY. For example, the frequencies in monopolar RF applications are typically restricted to less than 5 MHz. Frequencies above 200 kHz can be typically used for MONOPOLAR applications in order to avoid the unwanted stimulation of nerves and muscles which would result from the use of low frequency current. Lower frequencies may be used for BIPOLAR techniques if the RISK ANALYSIS shows the possibility of neuromuscular stimulation has been mitigated to an acceptable level. Normally, frequencies above 5 MHz are not used in order to minimize the problems associated with HIGH FREQUENCY LEAKAGE CURRENTS. However, higher frequencies may be used in the case of BIPOLAR techniques. It is generally recognized that 10 mA is the lower threshold of thermal effects on tissue.

(877) It may be recognized that an electrosurgical device may take advantage of the response of excitable tissue to electrical frequencies below 200 kHz in order to determine if such excitable tissue is sufficiently proximate to the end effector of the electrosurgical device to be potentially damaged thereby. FIG. **121** illustrates an RF signal **136210** that may be used in an electrosurgical device to cut or cauterize tissue. Such an RF signal **136210** may be termed a therapeutic signal because it has a frequency that may effect a therapeutic result such as cauterizing or cutting tissue. For purely illustrative purposes, the x-axis may represent time wherein each division represents 10 μsecs, and the y-axis (amplitude) has an arbitrary value. The RF signal **136210** depicted in FIG. **121** may therefore have a frequency of about 1 MHz. It may be understood that an RF therapeutic signal may have any frequency, amplitude, and/or phase characteristics sufficient to effect a therapeutic application such as sealing, cauterizing, ablating, or cutting a tissue.

(878) FIG. **122** depicts a signal **136220** that may be used to stimulate excitable tissue such as nerves or muscle. Again, solely for illustrative purposes, the signal **136220** depicted in FIG. **122** may extend over about 20 μsecs, and, if repeated, would constitute a waveform having a frequency of about 50 kHz. Such an electrical signal **136220** may be termed a stimulating signal because it has a frequency that may simulate excitable tissues such as nerve or muscle tissue. It may be understood that a waveform of a stimulating signal may differ from the signal **136220** presented in FIG. **122** in any aspect such as duration, frequency, or amplitude. In general, a stimulating signal **136220** may have any appropriate waveform or amplitude while having a frequency within a range that is capable of stimulating such excitable tissue. As indicated, such waveforms as depicted in FIGS. **121** and **122** are illustrative only. In one alternative example, a therapeutic RF signal may have a frequency of about 330 kHz and a waveform to stimulate excitable tissue may have a frequency of about 2 kHz.

(879) It may be understood that an intelligent electrosurgical device may be configured to emit either a therapeutic signal or a stimulating signal or a combination thereof. FIGS. **123A-123C** present examples of combinations of therapeutic signals and stimulating signals. The electrical generator may source an output current composed of any number or combination of characteristics of the therapeutic signal and characteristics of a tissue stimulating signal. Non-limiting examples of characteristics of a therapeutic signal may include a therapeutic signal frequency, a therapeutic signal amplitude, and a therapeutic signal phase. Non-limiting examples of characteristics of a tissue stimulating signal may include a stimulating signal frequency, a stimulating signal amplitude, and a stimulating signal phase. It may be recognized that a therapeutic signal may be characterized by any number of frequencies, phases, and amplitudes. Additionally, it may be recognized that a tissue stimulating signal may be characterized by any number of frequencies, phases, and amplitudes. In some aspects, the controller may be configured to control an electrical generator to provide an electrical output composed of a combination or combinations of characteristics of a therapeutic signal and characteristics of a tissue stimulating signal.

(880) FIG. **123A** depicts a non-limiting example of a first combination signal **136230** composed of a first therapeutic signal **136212a**, a stimulating signal **136222**, and a second therapeutic signal **136212b**. As depicted, one or more stimulating signals (such as signal **136220**, FIG. **122**) may alternate with one or more therapeutic signals (such as signal **136210**, FIG. **121**). It may be understood that the length of time for the application of the one or more therapeutic signals (such as **136212a,b**) may be arbitrary and may depend on the length of time that a medical professional may wish to apply it. It may also be understood that the stimulating signal **136222** may be transmitted at any time during the application of a therapeutic signal. It may be further understood that one or more zero-amplitude signals may be interspersed between one or more therapeutic signals and one or more stimulating signals. Multiple stimulating signals may be transmitted in succession before a subsequent therapeutic signal is transmitted.

(881) FIG. **123B** presents a non-limiting example of a second combination signal **136240** of a therapeutic signal and a stimulating signal. In FIG. **123B**, the stimulating signal (**136220** depicted in FIG. **122**) may be used to modulate the amplitude of the therapeutic signal (**136210** depicted in FIG. **121**). In some aspects, the stimulating signal **136220** may be applied directly to an amplitude modulation circuit to modulate the amplitude of a

therapeutic signal **136210**. In alternative aspects, the stimulating signal **136220** may be offset and scaled before being used to modulate the amplitude of the therapeutic signal **136210**. As an example, the stimulating signal **136220** in FIG. 122 may be offset by +4.5 V and the resulting signal may be scaled by 4.5 V so that the amplitude of the therapeutic signal **136210** is modulated by a positive-valued modulation signal that may range in value from about 0.1V to about 2V. One may readily recognize that any simple transformation of a stimulating signal **136220** may be used to modulate the amplitude of a therapeutic signal **136210**. It may be recognized that the amplitude of the therapeutic signal **136210** may be modulated by the stimulating signal **136220** at any time or for any number of times during the application of the therapeutic signal. The amplitude of the therapeutic signal **136210** may be modulated in the same manner over the course of multiple periods of modulation. Alternative, each amplitude modulation may differ according to the offset and/or scaling transformation of the stimulating signal **136220**.

(882) FIG. 123C presents a non-limiting example of a third combination signal **136250** of a therapeutic signal and a stimulating signal. In FIG. 123C the stimulating signal (**136220** depicted in FIG. 122) may be used as a DC offset to the therapeutic signal (**136210** depicted in FIG. 121). It may be recognized that the stimulating signal **136220** may also be altered according to any offset or scaling transformation before being applied as a DC offset to the therapeutic signal **136210**. It may be recognized that a DC offset based on the stimulating signal **136220** may be applied at any time to the therapeutic signal **136210** and may be applied multiple times over the course of the application of the therapeutic signal **136210**. The DC offset applied to the therapeutic signal **136210** may be the same over the course of multiple periods of offset application. Alternative, each DC offset to the therapeutic signal **136210** may differ according to the offset and/or scaling transformation of the stimulating signal.

(883) It may be understood that the combination of a stimulating signal with a therapeutic signal is not limited to the examples disclosed above and depicted in FIGS. 123A-123C. A stimulating signal may be combined with a therapeutic signal in the same manner throughout an electrosurgical procedure. Alternatively, a stimulating signal may be combined with a therapeutic signal in any of a number of different ways throughout the electrosurgical procedure. In some aspects, a stimulating signal may be combined with a therapeutic signal based on a choice made by a health care professional during the electrosurgical procedure. For example, the surgical probe may include one or more controls to permit the operator of the electrosurgical device to choose a mode of combination of the stimulating signal with the therapeutic signal. The surgical probe may also include one or more controls to permit the operator of the electrosurgical device to choose when the stimulating signal may be applied. In some alternative aspects, the surgical probe may include controls to permit a user to vary one or more characteristics of the therapeutic signal and/or the stimulating signal. Non-limiting examples of such signal characteristics may include one or more frequencies, one or more phases, and one or more amplitudes. In some alternative aspects, the control or controls of the stimulating signal and the therapeutic signal, their respective characteristics, or their combination may be located on the control unit of the electrosurgical device, or may be incorporated in a foot-operated controller.

(884) In some aspects, a smart electrosurgical device may include a processor, memory components, and instructions resident in the memory components for adjusting a therapeutic signal output based on a distance of the active electrode from excitable tissues. In some aspects, such processor, memory components, and instructions may form components of the controller. In some aspects, such processor, memory components, and instructions may form components of the electrical generator. In some aspects, such processor, memory components, and instructions may form components of a computer system separate from the smart electrosurgical device.

(885) FIG. 124 summarizes a one non-limiting method **136300** in which such a control may be effected. A controller may configure a generator to combine **136310** a stimulating signal with a therapeutic signal to form an electrode emitted signal. The controller may then cause an electrode to transmit **136320** the electrode emitted signal from an active electrode into a patient tissue. The controller may then receive **136330** a signal from a return signal pad in electrical communication with at least a portion of the patient. The signal returned by the return signal pad may include a signal generated by any one or more sensing devices disposed within the return pad. The controller may analyze **136340** the return signal from the return signal pad. It may be recognized that the analysis **136340** may include any one or more pre-processing methods including, without limitation, noise filtering, signal extraction, baseline adjustment, or any other method that may permit the controller to identify the return signal from the patient. Based on the return signal or any suitable manipulation of the return signal, the controller may determine **136350** that an excitable tissue has been stimulated by the emitted electrode signal. When the controller has determined **136350** that an excitable tissue has been stimulated by the emitted electrode signal, the controller may determine **136360** a distance of the excitable tissue from the active electrode. The controller may then adjust **136370** an amplitude of the therapeutic signal when the distance of the excitable tissue from the active electrode is less than a threshold value. In some aspects, the threshold value may be determined by a user of the electrosurgical system. In some other aspects, the threshold value may be based on a plurality of data acquired by the electrosurgical system or a HUB system of which the electrosurgical system is a part. In some aspects, the threshold value may be based on one or more mathematical models, physiological models (such as animal models), or on data acquired during an electrosurgical procedure on the patient.

(886) In some further aspects, a smart electrosurgical device may include processor readable instructions within a memory component that, when executed by a processor, may cause the processor associated with a control unit to combine a stimulating signal with a therapeutic signal. Such instructions may include, without limitation: determining the type of stimulating signal (for example, amplitude, duration, and waveform); determining the type of signal combination (for example alternating, amplitude modulation, DC offset, or other type of combination); determining the timing of the signal combination (that is, when, during a therapeutic activity, the therapeutic signal and the stimulating signals are combined, for example periodically, randomly, or at a single time); or determining types of signal transformations of the stimulating signal before being combined with the therapeutic signal.

(887) In some aspects, the smart electrosurgical device may include processor readable instructions stored within a memory component that, when executed by a processor, may cause the processor within the control unit to cause an active monopolar electrode to emit a therapeutic signal, a combined therapeutic signal and stimulating signal, or a stimulating signal upon contact with a patient's tissue. In some aspects, the smart electrosurgical device may include processor readable instructions within a memory component that, when executed by a processor, may cause the processor within the control unit to combine a therapeutic signal and a stimulating signal, to form an electrode emitted signal and to transmit the emitted signal from the active electrode into a patient tissue. In some aspects, the smart electrosurgical device may include processor readable instructions within a memory component that, when executed by a processor, may cause the processor within the control unit to receive one or more return signals from the patient, the return signals comprising electrical current returned from the current emitted by the active monopolar electrode and received by a return signal pad. In some aspects, the smart electrosurgical device may include processor readable instructions within a memory component that, when executed by a processor, may cause the processor within the control unit to receive one or more output signals of the one or more sensing devices associated with a return pad in contact with the patient. In some aspects, the smart electrosurgical device may include processor readable instructions within a memory component that, when executed by a processor, may cause the processor within the control unit to analyze the one or more output signals received from the one or more sensing devices associated with a return pad in contact with the patient.

(888) In some aspects, the smart electrosurgical device may include processor readable instructions within a memory component that, when executed by a processor, may cause the processor within the control unit to determine that an excitable tissue had been stimulated by the stimulating signal. In some examples, the one or more sensing devices may include an accelerometer associated with the return pad. In one non-limiting example, an output of an accelerometer may reflect to motion of a muscle in contact therewith which is activated by the stimulating signal. The amount of muscle motion may result at least in part from the amount of stimulating current received by either the muscle tissue or a nerve innervating the muscle. Because tissue may act as a resistive element to the propagation of the stimulating signal, the amount of muscle activation may indicate a distance of the active electrode from either the muscle or the innervating nerves.

(889) In some aspects, the patient may rest in a supine position on the return pad, and the sensor outputs of the return pad, such as one or more accelerometers, may indicate an amount of muscle motion of a patient's back muscles in contact with the return pad. In an alternative aspect, a return

pad may be placed on a muscle or muscle group proximal to the position of the surgical site wherein the electrostimulation device may be operated. In some examples, the return pad may be placed on a portion of superficial abdominal muscles (such as the rectus abdominis muscles) for an abdominal surgery. In some examples, the return pad may be placed on a side portion of the abdomen to monitor stimulation of the external oblique or the anterior serratus muscles.

(890) In some aspects, the smart electrosurgical device may include processor readable instructions within a memory component that, when executed by a processor, may cause the processor within the control unit to calculate or determine a distance of an excitable tissue from a distal end of the active electrode based at least in part on a return signal or one or more output signals from the one or more sensing devices associated with a return pad in contact with the patient. In some aspects, the smart electrosurgical device may include processor readable instructions within a memory component that, when executed by a processor, may cause the processor within the control unit to adjust one or more of an amplitude, a frequency, and a phase of a therapeutic signal based at least in part on a distance of an excitable tissue from the distal end of the active electrode. In some aspects, the amplitude, frequency, and/or phase of a therapeutic signal may be adjusted when the distance of an excitable tissue from the active electrode is less than a first pre-determined value. In some aspects, adjusting the amplitude, frequency, or phase of a therapeutic signal may result in the electrosurgical systems emitting no therapeutic signal when the distance of an excitable tissue from the active electrode is less than a second pre-determined value.

(891) In some additional aspects, the active electrode of a surgical probe of the electrosurgical device may be applied to a tissue solely to determine a distance of excitable tissue from the active electrode. In such a use, the medical professional using the device may operate it solely in a stimulating mode, without applying therapeutic signals to the active electrode. In a stimulating mode, the user of the device may operate one or more controls configured to ramp a characteristic of the stimulating signal to determine under which conditions an excitable tissues is stimulated thereby. For example, a user may operate controls configured to ramp a voltage or current amplitude of the stimulating signal from a low value to a high value. When a signal is received from a sensor (for example, an accelerometer sensing muscle movement), the electrosurgical device may then calculate an approximate distance from the active electrode to the excitable tissue based at least in part on the amplitude of the stimulating signal. In another example, a user may operate controls configured to ramp a frequency of the stimulating signal from a low value to a high value. When a signal is received from a sensor (for example, an accelerometer sensing muscle movement), the electrosurgical device may then calculate an approximate distance from the active electrode to the excitable tissue based at least in part on the frequency of the stimulating signal.

(892) In some aspects, an electrosurgical device or a smart electrosurgical device may be incorporated into a surgical HUB system. The HUB system may incorporate a number of hand-held medical devices, robotic medical devices, image acquisition devices, image display devices, communication devices, processing devices, networking devices, and other electronic devices that may operate in a concerted and coordinated fashion. In some aspects, the HUB may include such devices located within a single surgical suite, located within a plurality of surgical suites, or located within any number of computer server locations. The computer memory modules, instructions, and processors disclosed above in the context of the control of a smart, stand-alone electrosurgical device may be distributed among any of the components of the surgical HUB system as may be appropriate.

(893) In some aspects, additional information that may be acquired by the components of the surgical HUB system may be used to improve the operation of a smart electrosurgical device. For example, cameras and imaging systems directed at a surgical site may provide imaging information that can be used to determine the location of the distal end of the active electrode with respect to tissue in the surgical site. The image-based location of the distal end of the active electrode may be used with the return pad sensor output to refine the distance between the active electrode and any excitable tissue in the patient. In some alternative examples, the HUB system may include data comprising anatomical models related to the location of nerve and muscle tissue. Such model information may also be used along with the image-based localization of the active electrode and the return pad sensor output to better determine the proximity of the active electrode to known excitable tissue.

(894) Although the functions and devices disclosed above may be related solely to an electrosurgical device, it may be recognized that such functions and devices may also be incorporated into multi-mode surgical devices that include functions associated with an electrosurgical device. For example, a multi-mode surgical device may incorporate features associated with an electrosurgical device along with features associated with an ultrasonic surgical device. In addition to the functions disclosed above regarding altering the properties of an electrosurgical therapeutic signal, a multi-mode device may include other functions. For example, a surgical device may use either RF energy or ultrasound for a therapeutic effect, for example cutting a tissue. In such a multi-mode device, RF energy may be initially applied to a tissue for purposes of cutting material, but the multi-mode device may be configured to switch to an ultrasound mode if the end effector of the multi-mode device is determined to be too close to excitable tissue.

Ultrasonic Surgical Instrument Architecture

(895) FIG. 125 illustrates one aspect of an ultrasonic system **137010**. One aspect of the ultrasonic system **137010** comprises an ultrasonic signal generator **137012** coupled to an ultrasonic transducer **137014**, a hand piece assembly **137060** comprising a hand piece housing **137016**, and an ultrasonic blade **137050**. The ultrasonic transducer **137014**, which is known as a “Langevin stack,” generally includes a transduction portion **137018**, a first resonator or end-bell **137020**, and a second resonator or fore-bell **137022**, and ancillary components. In various aspects, the ultrasonic transducer **137014** is preferably an integral number of one-half system wavelengths ($n\lambda/2$) in length as will be described in more detail below. An acoustic assembly **137024** can include the ultrasonic transducer **137014**, a mount **137026**, a velocity transformer **137028**, and a surface **137030**.

(896) It will be appreciated that the terms “proximal” and “distal” are used herein with reference to a clinician gripping the hand piece assembly **137060**. Thus, the ultrasonic blade **137050** is distal with respect to the more proximal hand piece assembly **137060**. It will be further appreciated that, for convenience and clarity, spatial terms such as “top” and “bottom” also are used herein with respect to the clinician gripping the hand piece assembly **137060**. However, surgical instruments are used in many orientations and positions, and these terms are not intended to be limiting and absolute.

(897) The distal end of the end-bell **137020** is connected to the proximal end of the transduction portion **137018**, and the proximal end of the fore-bell **137022** is connected to the distal end of the transduction portion **137018**. The fore-bell **137022** and the end-bell **137020** have a length determined by a number of variables, including the thickness of the transduction portion **137018**, the density and modulus of elasticity of the material used to manufacture the end-bell **137020** and the fore-bell **137022**, and the resonant frequency of the ultrasonic transducer **137014**. The fore-bell **137022** may be tapered inwardly from its proximal end to its distal end to amplify the ultrasonic vibration amplitude of the velocity transformer **137028**, or, alternately, fore-bell **137022** may have no amplification.

(898) Referring again to FIG. 125, end-bell **137020** can include a threaded member extending therefrom which can be configured to be threadably engaged with a threaded aperture in fore-bell **137022**. In various aspects, piezoelectric elements, such as piezoelectric elements **137032**, for example, can be compressed between end-bell **137020** and fore-bell **137022** when end-bell **137020** and fore-bell **137022** are assembled together. Piezoelectric elements **137032** may be fabricated from any suitable material, such as, for example, lead zirconate-titanate, lead meta-niobate, lead titanate, and/or any suitable piezoelectric crystal material, for example.

(899) In various aspects, as discussed in greater detail below, transducer **137014** can further comprise electrodes, such as positive electrodes **137034** and negative electrodes **137036**, for example, which can be configured to create a voltage potential across one or more piezoelectric elements **137032**. Each of the positive electrodes **137034**, negative electrodes **137036**, and the piezoelectric elements **137032** can comprise a bore extending through the center which can be configured to receive the threaded member of end-bell **137020**. In various aspects, the positive and negative electrodes **137034** and **137036** are electrically coupled to wires **137038** and **137040**, respectively, wherein the wires **137038** and **137040** can be encased within a cable **137042** and electrically connectable to the ultrasonic signal generator **137012** of the ultrasonic system **137010**.

(900) In various aspects, the ultrasonic transducer **137014** of the acoustic assembly **137024** converts the electrical signal from the ultrasonic signal

generator **137012** into mechanical energy that results in primarily longitudinal vibratory motion of the ultrasonic transducer **137014** and the ultrasonic blade **137050** at ultrasonic frequencies. An ultrasonic surgical generator **137012** can include, for example, the generator **1100** (FIG. **18**) or the generator **137012** (FIG. **125**). When the acoustic assembly **137024** is energized, a vibratory motion standing wave is generated through the acoustic assembly **137024**. A suitable vibrational frequency range may be about 20 Hz to 120 kHz and a well-suited vibrational frequency range may be about 30-70 kHz and one example operational vibrational frequency may be approximately 55.5 k Hz.

(901) The amplitude of the vibratory motion at any point along the acoustic assembly **137024** may depend upon the location along the acoustic assembly **137024** at which the vibratory motion is measured. A minimum or zero crossing in the vibratory motion standing wave is generally referred to as a node (i.e., where motion is usually minimal), and an absolute value maximum or peak in the standing wave is generally referred to as an anti-node (i.e., where motion is usually maximal). The distance between an anti-node and its nearest node is one-quarter wavelength ($\lambda/4$).

(902) As outlined above, the wires **137038**, **137040** transmit an electrical signal from the ultrasonic signal generator **137012** to the positive electrodes **137034** and the negative electrodes **137036**. The piezoelectric elements **137032** are energized by the electrical signal supplied from the ultrasonic signal generator **137012** in response to a foot switch **137044**, for example, to produce an acoustic standing wave in the acoustic assembly **137024**. The electrical signal causes disturbances in the piezoelectric elements **137032** in the form of repeated small displacements resulting in large compression forces within the material. The repeated small displacements cause the piezoelectric elements **137032** to expand and contract in a continuous manner along the axis of the voltage gradient, producing longitudinal waves of ultrasonic energy.

(903) In various aspects, the ultrasonic energy produced by transducer **137014** can be transmitted through the acoustic assembly **137024** to the ultrasonic blade **137050** via an ultrasonic transmission waveguide **137046**. In order for the acoustic assembly **137024** to deliver energy to the ultrasonic blade **137050**, the components of the acoustic assembly **137024** are acoustically coupled to the ultrasonic blade **137050**. For example, the distal end of the ultrasonic transducer **137014** may be acoustically coupled at the surface **137030** to the proximal end of the ultrasonic transmission waveguide **137046** by a threaded connection such as a stud **137048**.

(904) The components of the acoustic assembly **137024** can be acoustically tuned such that the length of any assembly is an integral number of one-half wavelengths ($n\lambda/2$), where the wavelength λ is the wavelength of a pre-selected or operating longitudinal vibration drive frequency f_d of the acoustic assembly **137024**, and where n is any positive integer. It is also contemplated that the acoustic assembly **137024** may incorporate any suitable arrangement of acoustic elements.

(905) The ultrasonic blade **137050** may have a length substantially equal to an integral multiple of one-half system wavelengths ($\lambda/2$). A distal end **137052** of the ultrasonic blade **137050** may be disposed at, or at least near, an antinode in order to provide the maximum, or at least nearly maximum, longitudinal excursion of the distal end. When the transducer assembly is energized, in various aspects, the distal end **137052** of the ultrasonic blade **137050** may be configured to move in the range of, for example, approximately 10 to 500 microns peak-to-peak and preferably in the range of approximately 30 to 150 microns at a predetermined vibrational frequency.

(906) As outlined above, the ultrasonic blade **137050** may be coupled to the ultrasonic transmission waveguide **137046**. In various aspects, the ultrasonic blade **137050** and the ultrasonic transmission guide **137046** as illustrated are formed as a single unit construction from a material suitable for transmission of ultrasonic energy such as, for example, Ti6Al4V (an alloy of titanium including aluminum and vanadium), aluminum, stainless steel, and/or any other suitable material. Alternately, the ultrasonic blade **137050** may be separable (and of differing composition) from the ultrasonic transmission waveguide **137046**, and coupled by, for example, a stud, weld, glue, quick connect, or other suitable known methods. The ultrasonic transmission waveguide **137046** may have a length substantially equal to an integral number of one-half system wavelengths ($\lambda/2$), for example. The ultrasonic transmission waveguide **137046** may be preferably fabricated from a solid core shaft constructed out of material that propagates ultrasonic energy efficiently, such as titanium alloy (i.e., Ti6Al4V) or an aluminum alloy, for example.

(907) In the aspect illustrated in FIG. **125**, the ultrasonic transmission waveguide **137046** comprises a plurality of stabilizing silicone rings or compliant supports **137056** positioned at, or at least near, a plurality of nodes. The silicone rings **137056** can dampen undesirable vibration and isolate the ultrasonic energy from a sheath **137058** at least partially surrounding waveguide **137046**, thereby assuring the flow of ultrasonic energy in a longitudinal direction to the distal end **137052** of the ultrasonic blade **137050** with maximum efficiency.

(908) As shown in FIG. **125**, the sheath **137058** can be coupled to the distal end of the handpiece assembly **137060**. The sheath **137058** generally includes an adapter or nose cone **137062** and an elongated tubular member **137064**. The tubular member **137064** is attached to and/or extends from the adapter **137062** and has an opening extending longitudinally therethrough. In various aspects, the sheath **137058** may be threaded or snapped onto the distal end of the housing **137016**. In at least one aspect, the ultrasonic transmission waveguide **137046** extends through the opening of the tubular member **137064** and the silicone rings **137056** can contact the sidewalls of the opening and isolate the ultrasonic transmission waveguide **137046** therein. In various aspects, the adapter **137062** of the sheath **137058** is preferably constructed from Ultem®, for example, and the tubular member **137064** is fabricated from stainless steel, for example. In at least one aspect, the ultrasonic transmission waveguide **137046** may have polymeric material, for example, surrounding it in order to isolate it from outside contact.

(909) As described above, a voltage, or power source can be operably coupled with one or more of the piezoelectric elements of a transducer, wherein a voltage potential applied to each of the piezoelectric elements can cause the piezoelectric elements to expand and contract, or vibrate, in a longitudinal direction. As also described above, the voltage potential can be cyclical and, in various aspects, the voltage potential can be cycled at a frequency which is the same as, or nearly the same as, the resonant frequency of the system of components comprising transducer **137014**, waveguide **137046**, and ultrasonic blade **137050**, for example. In various aspects, however, certain of the piezoelectric elements within the transducer may contribute more to the standing wave of longitudinal vibrations than other piezoelectric elements within the transducer. More particularly, a longitudinal strain profile may develop within a transducer wherein the strain profile may control, or limit, the longitudinal displacements that some of the piezoelectric elements can contribute to the standing wave of vibrations, especially when the system is being vibrated at or near its resonant frequency.

(910) The piezoelectric elements **137032** are configured into a “Langevin stack,” in which the piezoelectric elements **137032** and their activating electrodes **137034** and **137036** (together, transducer **137014**) are interleaved. The mechanical vibrations of the activated piezoelectric elements **137032** propagate along the longitudinal axis of the transducer **137014**, and are coupled via the acoustic assembly **137024** to the end of the waveguide **137046**. Such a mode of operation of a piezoelectric element is frequently described as the D33 mode of the element, especially for ceramic piezoelectric elements comprising, for example, lead zirconate-titanate, lead meta-niobate, or lead titanate. The D33 mode of a ceramic piezoelectric element is illustrated in FIGS. **126A-126C**.

(911) FIG. **126A** depicts a piezoelectric element **137200** fabricated from a ceramic piezoelectric material. A piezoelectric ceramic material is a polycrystalline material comprising a plurality of individual microcrystalline domains. Each microcrystalline domain possesses a polarization axis along which the domain may expand or contract in response to an imposed electric field. However, in a native ceramic, the polarization axes of the microcrystalline domains are arranged randomly, so there is no net piezoelectric effect in the bulk ceramic. A net re-orientation of the polarization axes may be induced by subjecting the ceramic to a temperature above the Curie temperature of the material and placing the material in a strong electrical field. Once the temperature of the sample is dropped below the Curie temperature, a majority of the individual polarization axes will be re-oriented and fixed in a bulk polarization direction. FIG. **126A** illustrates such a piezoelectric element **137200** after being polarized along the inducing electric field axis P. While the un-polarized piezoelectric element **137200** lacks any net piezoelectric axis, the polarized element **137200** can be described as possessing a polarization axis, d3, parallel to the inducing field axis P direction. For completeness, an axis orthogonal to the d3 axis may be termed a d1 axis. The dimensions of the piezoelectric element **137200** are labeled as length (L), width (W), and thickness (T).

(912) FIGS. **126B** and **126C** illustrate the mechanical deformations of a piezoelectric element **137200** that may be induced by subjecting the

piezoelectric element **137200** to an actuating electrical field **E** oriented along the d3 (or P) axis. FIG. **126B** illustrates the effect of an electric field **E** having the same direction as the polarization field **P** along the d3 axis on a piezoelectric element **137205**. As illustrated in FIG. **126B**, the piezoelectric element **137205** may deform by expanding along the d3 axis while compressing along the d1 axis. FIG. **126C** illustrates the effect of an electric field **E** having the opposing direction to the polarization field **P** along the d3 axis on a piezoelectric element **137210**. As illustrated in FIG. **126C**, the piezoelectric element **137210** may deform by compressing along the d3 axis, while expanding along the d1 axis. Vibrational coupling along the d3 axis during the application of an electric field along the d3 axis may be termed D33 coupling or activation using a D33 mode of a piezoelectric element. The transducer **137014** illustrated in FIG. **125** can use the D33 mode of the piezoelectric elements **137032** for transmitting mechanical vibrations along the waveguide **46** to the ultrasonic blade **137050**. Because the piezoelectric element also deforms along the d1 axis, vibrational coupling along the d1 axis during the application of an electric field along the d3 axis may also be an effective source of mechanical vibrations. Such coupling may be termed D31 coupling or activation using a D31 mode of a piezoelectric element.

(913) As illustrated by FIGS. **126A-126C**, during operation in the D31 mode, transverse expansion of piezoelectric elements **137200**, **137205**, **137210** may be mathematically modeled by the following equation:

$$(914) \frac{L}{T} = \frac{W}{T} = \frac{V_{d31}}{T}$$

(915) In the equation, **L**, **W**, and **T** refer to the length, width and thickness dimensions of a piezoelectric element, respectively. **V.sub.d31** denotes the voltage applied to a piezoelectric element operating in the D31 mode. The quantity of transverse expansion resulting from the D31 coupling described above is represented by ΔL (i.e., expansion of the piezoelectric element along the length dimension) and ΔW (i.e., expansion of the piezoelectric element along the width dimension). Additionally, the transverse expansion equation models the relationship between ΔL and ΔW and the applied voltage **V.sub.d31**. Disclosed below are aspects of ultrasonic surgical instruments based on D31 activation by a piezoelectric element.

(916) In various aspects, as described below, an ultrasonic surgical instrument can comprise a transducer configured to produce longitudinal vibrations, and a surgical instrument having a transducer base plate (e.g., a transducer mounting portion) operably coupled to the transducer, an end effector, and waveguide therebetween. In certain aspects, as also described below, the transducer can produce vibrations which can be transmitted to the end effector, wherein the vibrations can drive the transducer base plate, the waveguide, the end effector, and/or the other various components of the ultrasonic surgical instrument at, or near, a resonant frequency. In resonance, a longitudinal strain pattern, or longitudinal stress pattern, can develop within the transducer, the waveguide, and/or the end effector, for example. In various aspects, such a longitudinal strain pattern, or longitudinal stress pattern, can cause the longitudinal strain, or longitudinal stress, to vary along the length of the transducer base plate, waveguide, and/or end effector, in a sinusoidal, or at least substantially sinusoidal, manner. In at least one aspect, for example, the longitudinal strain pattern can have maximum peaks and zero points, wherein the strain values can vary in a non-linear manner between such peaks and zero points.

(917) FIG. **127** illustrates an ultrasonic surgical instrument **137250** that includes an ultrasonic waveguide **137252** attached to an ultrasonic transducer **137264** by a bonding material, where the ultrasonic surgical instrument **137250** is configured to operate in a D31 mode, according to one aspect of this disclosure. The ultrasonic transducer **137264** includes first and second piezoelectric elements **137254a**, **137254b** attached to the ultrasonic waveguide **137252** by a bonding material. The piezoelectric elements **137254a**, **137254b** include electrically conductive plates **137256a**, **137256b** to electrically couple one pole of a voltage source suitable to drive the piezoelectric elements **137254a**, **137254b** (e.g., usually a high voltage). The opposite pole of the voltage source is electrically coupled to the ultrasonic waveguide **137252** by electrically conductive joints **137258a**, **137258b**. In one aspect, the electrically conductive plates **137256a**, **137256b** are coupled to a positive pole of the voltage source and the electrically conductive joints **137258a**, **137258b** are electrically coupled to ground potential through the metal ultrasonic waveguide **137252**. In one aspect, the ultrasonic waveguide **137252** is made of titanium or titanium alloy (i.e., Ti6Al4V) and the piezoelectric elements **137254a**, **137254b** are made of PZT. The poling axis (**P**) of the piezoelectric elements **137254a**, **137254b** is indicated by the direction arrow **137260**. The motion axis of the ultrasonic waveguide **137252** in response to excitation of the piezoelectric elements **137254a**, **137245b** is shown by a motion arrow **137262** at the distal end of the ultrasonic waveguide **137252** generally referred to as the ultrasonic blade portion of the ultrasonic waveguide **137252**. The motion axis **137262** is orthogonal to the poling axis (**P**) **137260**.

(918) In conventional D33 ultrasonic transducer architectures as shown in FIG. **125**, the bolted piezoelectric elements **137032** utilize electrodes **137034**, **137036** to create electrical contact to both sides of each piezoelectric element **137033**. The D31 architecture **137250** according to one aspect of this disclosure, however, employs a different technique to create electrical contact to both sides of each piezoelectric element **137254a**, **137254b**. Various techniques for providing electrical contact to the piezoelectric elements **137254a**, **137254b** include bonding electrical conductive elements (e.g., wires) to the free surface of each piezoelectric element **137254a**, **137254b** for the high potential connection and bonding each piezoelectric element **137254a**, **137254b** the to the ultrasonic waveguide **137252** for the ground connection using solder, conductive epoxy, or other techniques described herein. Compression can be used to maintain electrical contact to the acoustic train without making a permanent connection. This can cause an increase in device thickness and should be controlled to avoid damaging the piezoelectric elements **137254a**, **137254b**. Low compression can damage the piezoelectric element **137254a**, **137254b** by a spark gap and high compression can damage the piezoelectric elements **137254a**, **137254b** by local mechanical wear. In other techniques, metallic spring contacts may be employed to create electrical contact with the piezoelectric elements **137254a**, **137254b**. Other techniques may include foil-over-foam gaskets, conductive foam, and solder. In some aspects, there is an electrical connection to both sides of the piezoelectric elements **137254a**, **137254b** in the D31 acoustic train configuration. The electrical ground connection can be made to the metal ultrasonic waveguide **137252**, which is electrically conductive, if there is electrical contact between the piezoelectric elements **137254a**, **137254b** and the ultrasonic waveguide **137252**.

(919) In conventional D33 ultrasonic transducer architectures as shown in FIG. **125**, a bolt provides compression that acoustically couples the piezoelectric elements rings to the ultrasonic waveguide. The D31 architecture **137250** according to one aspect of this disclosure employs a variety of different techniques to acoustically couple the piezoelectric elements **137254a**, **137254b** to the ultrasonic waveguide **137252**. Some illustrative techniques are disclosed in U.S. patent application Ser. No. 15/679,940, titled ULTRASONIC TRANSDUCER TECHNIQUES FOR ULTRASONIC SURGICAL INSTRUMENT, filed Aug. 17, 2017, which is hereby incorporated by reference in its entirety.

(920) FIGS. **128** and **129** illustrate various views of an ultrasonic surgical instrument **137400**. In various aspects, the surgical instrument **137400** can be embodied generally as a pair of ultrasonic shears, as shown. In aspects where the ultrasonic surgical instrument **137400** is embodied as a pair of ultrasonic shears, the surgical instrument **137400** can include a first arm **137412a** pivotally connected to a second arm **137412b** at a pivot point **137413** (e.g., by a fastener). The first arm **137412a** includes a clamp arm **137416** positioned at its distal end that includes a cooperating surface (e.g., a pad) that is configured to cooperate with an ultrasonic blade **137415** extending distally from the second arm **137412b**. The clamp arm **137416** and the ultrasonic blade **137415** can collectively define an end effector **137410**. Actuating the first arm **137412a** in a first direction causes the clamp arm **137416** to pivot towards the ultrasonic blade **137415** and actuating the first arm **137412a** in a second direction causes the clamp arm **137416** to pivot away from the ultrasonic blade **137415**. In some aspects, the clamp arm **13716** further includes a pad constructed from a polymeric or other compliant material and engages the ultrasonic blade **137415**. The surgical instrument **137400** further includes a transducer assembly, such as is described above with respect to FIGS. **125-127**. The transducer assembly can be arranged in, e.g., a D31 or D33 architecture. The surgical instrument **137400** further comprises a housing **137414** enclosing various components of an ultrasonic system **137010** (FIG. **125**), including first and second piezoelectric elements **137419a**, **137419b** of an ultrasonic transducer **137418** arranged in a D31 architecture, a transducer base plate **137428** (e.g., a transducer mounting portion) comprising flat faces on opposite sides to receive the piezoelectric elements **137419a**, **137419b**, and a waveguide **137417** that longitudinally translates vibrations from the ultrasonic transducer **137418** to the ultrasonic blade **137415**. Further, the surgical instrument **137400** is connectable to an ultrasonic signal generator for driving the ultrasonic transducer **137418**, as described above. The waveguide **137417** can comprise a plurality of stabilizing silicone rings or compliant supports **137411** positioned at, or at least near, a plurality of nodes (i.e., points located

at a minimum or zero crossing (standing wave). The compliant supports are configured to dampen undesirable lateral vibration in order to ensure that ultrasonic energy is transmitted longitudinally to the ultrasonic blade **137415**. The waveguide **137417** extends through the housing **137414** and the second arm **137412b** and terminates at the ultrasonic blade **137415**, externally to the housing **137414**. The ultrasonic blade **137415** and the clamp arm **137416** are cooperating elements that are configured to grasp tissue, allowing the end effector **137410** to clamp and cut/coagulate tissue. Moving the clamp arm **137416** towards the ultrasonic blade **137415** causes tissue situated therebetween to contact the ultrasonic blade **137415**, allowing the ultrasonic blade **137415** to operate against the grasped tissue. As the ultrasonic blade **137415** ultrasonically vibrates against the grasped tissue, the ultrasonic blade **137415** generates frictional forces that cause the tissue to coagulate and eventually sever along the cutting length of the ultrasonic blade **137415**.

(921) The cutting length of the surgical instrument **137400** corresponds to the lengths of the ultrasonic blade **137415** and the cooperating surface of the clamp arm **137416**. Tissue that is held between the ultrasonic blade **137415** and the cooperating surface of the clamp arm **137416** for a sufficient period of time is cut by the ultrasonic blade **137415**, as described above. The ultrasonic blade **137415** and the corresponding portion of the clamp arm **137416** can have a variety of shapes. In various aspects, the ultrasonic blade **137415** and/or clamp arm **137416** can be substantially linear in shape or have a curvature. In some aspects, the portion of the clamp arm **137416** configured to bring tissue into contact with the ultrasonic blade **137415** can correspond to the shape of the ultrasonic blade **137415** so that the clamp arm **137416** is aligned therewith.

(922) Various additional details regarding ultrasonic transducer assemblies and ultrasonic shears can be found in U.S. patent application Ser. No. 15/679,940, titled ULTRASONIC TRANSDUCER TECHNIQUES FOR ULTRASONIC SURGICAL INSTRUMENT, filed Aug. 17, 2017, which is hereby incorporated by reference in its entirety.

Advanced Energy Device Activation Options

(923) FIG. **130** illustrates a block diagram of a surgical system **137500**, in accordance with at least one aspect of the present disclosure. The surgical system **137500** can include, for example, the surgical system **1000** depicted in FIG. **22** and/or the ultrasonic surgical instrument system **137010** depicted in FIG. **125**. The surgical system **137500** can include a surgical instrument **137400**, such as the ultrasonic surgical instrument **1104** (FIG. **22**), that is electrically connectable to an electrosurgical generator **137504**, such as generator **1100** (FIGS. **22**, **24**) or the generator **137012** (FIG. **125**), capable of producing ultrasonic energy, monopolar or bipolar radiofrequency (RF) energy, other types of energy, and/or combinations thereof for driving the surgical instrument **137400**.

(924) In the aspect depicted in FIG. **130**, the surgical instrument **137400** includes a transducer assembly **137510** that comprises at least two piezoelectric elements. The transducer assembly **137510** is operably coupled to the ultrasonic blade **137512** such that the transducer assembly **137510** can ultrasonically oscillate the ultrasonic blade **137512** when the transducer assembly **137510** is activated, as is described in connection with FIGS. **125-127**. The transducer assembly **137510** is in turn electrically coupled to the generator **137504** to receive energy therefrom. Accordingly, when energized by the generator **137504**, the transducer assembly **137510** is configured to ultrasonically oscillate the ultrasonic blade **137512** in order to sever and/or coagulate tissue captured by the surgical instrument **137400**.

(925) In another aspect, the surgical instrument **137400** includes one or more electrodes **796** (FIG. **19**) or other conducting elements located at the end effector **792** (FIG. **19**). The electrodes **796** are in turn electrically coupled to the generator **137504** to receive energy therefrom. When energized by the generator **137504**, the electrodes **796** are configured to apply RF energy in order to sever and/or coagulate tissue captured by the surgical instrument **137400**, as is described in connection with FIG. **17**.

(926) The surgical instrument **137400** further includes a control circuit **137506** that is communicably coupled to a sensor **137508** and communicably coupleable to the generator **137504**. The control circuit **137506** can include, for example, a processor coupled to primary and/or secondary computer memory for executing instructions stored on the memory, a microcontroller, an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), and other such devices. The sensor **137508** is configured to sense a property of the environment and/or the surgical instrument **137400** and provide an output corresponding to the presence or magnitude of the sensed property. The control circuit **137506** is in turn configured to selectively control the activation of the transducer assembly **137510** and/or electrodes **796** according to whether the sensed property is above, below, or at a threshold value. In other words, the control circuit **137506** is configured to control the activation of the transducer assembly **137510** and/or electrodes **796** according to the sensor output relative to a threshold. In one aspect, the threshold can be stored in a memory of the surgical instrument **137400** and retrieved by the control circuit **137506** to compare the output signal from the sensor **137508** thereagainst.

(927) In various other exemplifications, the control circuit **137506** and/or sensor **137508** may be external to the surgical instrument **137400**. In these exemplifications, the control circuit **137506** and/or sensor **137508** can be communicably coupled to each other and/or the generator **137504** via any wired communication protocol (e.g., I.sup.2C) or wireless communication protocol (e.g., Bluetooth) and include the appropriate hardware and/or software to effectuate the particular communication protocol. In still other exemplifications, the generator **137504** can be integral, internal, or otherwise incorporated with the surgical instrument **137400**, rather than being external thereto, as depicted in FIG. **18** and FIG. **125**.

(928) FIGS. **131-132C** illustrate various views of a surgical instrument **137400** including a sensor assembly **137508** configured to detect a magnetic reference **137404**, in accordance with at least one aspect of the present disclosure. In the following description of FIGS. **131-132C**, reference should also be made to FIG. **130**. In one aspect, the sensor assembly **137508** includes a sensor **137402** that is configured to detect the position or state (e.g., opened or closed) of the surgical instrument **137400** by detecting the corresponding position or location of a magnetic reference **137404**. The sensor **137402** can include, for example, a Hall effect sensor that is configured to detect the location of the magnetic reference **137404** relative thereto. Accordingly, the magnetic reference **137404** is configured such that its position corresponds to the position and/or state of the surgical instrument **137400**. The Hall effect sensor can include, for example, a Hall element configured to detect the relative distance between the magnetic reference **137404** and the sensor **137402** or an assembly of multiple Hall elements configured to detect the multidimensional position or orientation of the magnetic reference **137404** relative to the sensor **137402** (e.g., a TLV493D-A1B6 3D magnetic sensor from Infineon Technologies). Further, the Hall effect sensor can include linear Hall effect sensors (i.e., Hall effect sensors where the output varies linearly with the magnetic flux density) or threshold Hall effect sensors (i.e., Hall effect sensors where the output drops sharply according to decreasing magnetic flux density).

(929) In the aspect depicted in FIG. **131**, the magnetic reference **137404** includes a wearable magnet **137406**. Accordingly, the sensor **137402** is configured to detect the relative position of the wearable magnet **137406** as worn, for example, on the hand of a surgeon. In various aspects, the relative position of the wearable magnet **137406** with respect to the sensor **137402** can include, for example, the relative distance between the wearable magnet **137406** and the sensor **137402** and/or the relative orientation of the wearable magnet **137406** with respect to the sensor **137402**. In one example, the wearable magnet **137406** can be incorporated with or positioned in or on a ring that is worn on a finger of the surgeon (e.g., over a surgical glove). In another example, the wearable magnet **137406** can be attached or integral to a surgical glove wearable by the surgeon. In these aspects, as the wearable magnet **137406** is located on the surgeon's hand and the surgeon's hand grips the arm **137412** of the surgical instrument **137400** during use thereof, the position of the wearable magnet **137406** as detected by the sensor **137402** corresponds to the relative position of the arm **137412** of the surgical instrument **137400**. By sensing the relative position of the arm **137412** of the surgical instrument **137400**, the control circuit **137506** can thereby determine whether the surgical instrument **137400** is opened, closed, or in an intermediate position therebetween.

(930) In another exemplification, the positions of the wearable magnet **137406** and the sensor **137402** can be reversed from the aspect described above. In other words, the magnet can be positioned on or in the surgical instrument **137400** and the sensor **137402** can be positioned on or worn by the surgeon (e.g., incorporated into a ring or a surgical glove, as described above). Otherwise, this exemplification functions in a similar manner to the exemplification that is described above.

(931) In the aspect depicted in FIGS. **132A-C**, the magnetic reference **137404** includes an integral magnet **137408** positioned in or on a movable component of the surgical instrument **137400**, such as the arm **137412** thereof. Accordingly, the sensor **137402** is configured to detect the relative

position of the integral magnet **137408** within the arm **137412** of the surgical instrument **137400**. The integral magnet **137408** and the sensor **137402** can each be positioned such that opening and closing the surgical instrument **137400** causes the integral magnet **137408** to move relative to the sensor **137402**. In the depicted aspect, the integral magnet **137408** can be positioned on or in the movable arm **137412** of the surgical instrument **137400** and the sensor **137402** can be positioned on or in the housing **137414** of the surgical instrument **137400**. In these aspects, the position of the integral magnet **137408** detected by the sensor **137402** corresponds to the relative position of the arm **137412** of the surgical instrument **137400**. By sensing the relative position of the arm **137412** of the surgical instrument **137400**, the control circuit **137506** can thereby determine whether the surgical instrument **137400** is opened, closed, or in an intermediate position therebetween.

(932) In another exemplification, the positions of the integral magnet **137408** and the sensor **137402** can be reversed from the aspect described above. In other words, the integral magnet **137408** can be positioned on or in the housing **137414** of the surgical instrument **137400** and the sensor **137402** can be positioned on or in the corresponding movable component (e.g., the arm **137412**) of the surgical instrument **137400** that is being tracked. Otherwise, this exemplification functions in a similar manner to the exemplification that is described above.

(933) The sensor **137402** is configured to produce an output that corresponds to the position of the magnetic reference **137404** relative thereto (e.g., the distance between the magnetic reference **137404** and the sensor **137402** and/or the orientation of the magnetic reference **137404** with respect to the sensor **137402**). Thus, as the magnetic reference **137404** and/or the sensor **137402** move with respect to each other as the surgical instrument **137400** is closed, opened, or otherwise manipulated by a surgeon, the sensor **137400** is able to detect the relative position of the magnetic reference **137404** according to the sensed magnetic field of the magnetic reference **137404**. The sensor **137402** can then produce an output corresponding to the sensed magnetic field of the magnetic reference **137404**. In one aspect where the sensor **137402** includes a Hall effect sensor, the sensor output can be a voltage, wherein the magnitude of the output voltage corresponds to the strength of the magnetic field from the magnetic reference **137404** sensed by the sensor **137402**.

(934) In one aspect, the control circuit **137506** is configured to receive the output from the sensor **137402** and then compare the output of the sensor **137402** to a threshold. The control circuit **137506** can further activate or deactivate the surgical instrument **137400** according to the comparison between the output of the sensor **137402** and the threshold. The threshold can be, e.g., predetermined or set by a user of the surgical instrument **137400**. The output of the sensor **137402** can correspond to the position of the arm of the surgical instrument **137400** (either directly, as in the aspect depicted in FIGS. **132A-C**, or indirectly, as in the aspect depicted in FIG. **131**), which in turn controls the position of the clamp arm **137416** (FIG. **128**) relative to the ultrasonic blade **137512**. Therefore, the output of the sensor **137402** corresponds to the position of the clamp arm **137416** of the surgical instrument **137402** between, e.g., an open position and a closed position. Further in these exemplifications, the threshold can correspond to a threshold distance between the magnetic reference **137404** and the sensor **137402**. FIG. **132B**, for example, can represent an open position for the surgical instrument **137400** (i.e., the integral magnet **137408** is not within a threshold distance to the sensor **137402**) and FIG. **132C**, for example, can represent a closed position of the surgical instrument **137400** (i.e., the integral magnet **137408** is within a threshold distance to the sensor **137402**).

(935) In one example, the control circuit **137506** can determine whether the magnetic reference **137404** is positioned less than or equal to a threshold distance from the sensor **137402**. In this example, if the control circuit **137506** determines that the sensor output exceeds the threshold, then the control circuit **137506** can activate the surgical instrument **137400**. In another example, the control circuit **137506** can determine whether the magnetic reference **137404** is positioned greater than or equal to a threshold distance from the sensor **137402**. In this example, if the control circuit **137506** determines that the voltage output of the sensor **137402** is less than or equal to the threshold, then the control circuit **137506** can activate the surgical instrument **137400**. The control circuit **137506** can activate the surgical **137400** by transmitting a signal to the generator **137504** that cause the generator **137504** to energize the transducer assembly **137510** and/or RF electrodes **796** to cut and/or coagulate tissue captured by the surgical instrument **137400**. In sum, in some aspects the control circuit **137506** can be configured to determine whether the surgical instrument **137400** is sufficiently closed and, if it is, then activate the surgical instrument **137400**.

(936) In other aspects, the control circuit **137506** can be configured take other actions if it determines that the surgical instrument **137400** is sufficiently closed, such as providing a prompt to the user or transmitting data to a surgical hub **106**, as described in connection with FIGS. **1-11**. In still other aspects, the control circuit **137506** can be configured to determine whether the surgical instrument **137400** is sufficiently opened or at some particular position (or range of positions) between the opened and closed positions. If the surgical instrument **137400** is at or within the defined position(s), the control circuit **137506** can accordingly activate the surgical instrument **137400**, deactivate the surgical instrument **137400**, or take a variety of other actions.

(937) In some aspects, the control circuit **137506** can be configured to detect tapping, rubbing, and other types of motions based upon the amplitude, frequency, and/or direction of the motion of the magnetic reference **137404** detected via the sensor **137402**. Such motions can be detected because the change in the strength of the magnetic field over time detected by the sensor **137402** can be characterized (empirically or otherwise) and defined for different types of motions. For example, a tapping motion could be detectable according to the frequency in the change of the magnetic field detected by the sensor **137402** in a direction substantially perpendicular to the longitudinal axis of the surgical instrument **137400**. As another example, a rubbing motion could be detectable according to the frequency in the change of the magnetic field detected by the sensor **137402** in a direction substantially parallel to the longitudinal axis of the surgical instrument **137400**. In some aspects, the control circuit **137506** can be configured to change the state, mode, and/or properties of the surgical instrument **137400** according to the detected motions. For example, the control circuit **137506** could be configured to activate the surgical instrument **137400** upon detecting a tapping motion via the sensor **137402**.

(938) FIGS. **133A-B** illustrates perspective views of a surgical instrument **137400** including a sensor assembly **137508** configured to detect contact thereagainst and FIG. **134** illustrates a corresponding circuit diagram, in accordance with at least one aspect of the present disclosure. In the following description of FIGS. **133A-134**, reference should also be made to FIG. **130**. In one aspect, the sensor assembly **137508** can include a touch sensor **137420** that is configured to detect force, contact, and/or pressure thereagainst. The touch sensor **137420** can comprise, e.g., a force-sensitive resistor (FSR) **137421**. In one exemplification depicted in FIG. **133A**, the touch sensor **137420** is oriented transverse to the longitudinal axis of the surgical instrument **137400**. In this exemplification, the touch sensor **137420** defines a surface extending orthogonally from the housing **137414** relative to the longitudinal axis of the surgical instrument **137400**. In another exemplification depicted in FIG. **133B**, the touch sensor **137420** extends along the lateral surface(s) of the housing **137414**. In this exemplification, the touch sensor **137420** can be integral to or positioned in or on the housing **137414** of the surgical instrument **137400**. In either of these exemplifications, the touch sensor **137420** can be utilized by a surgeon to, e.g., activate the transducer assembly **137510** of the surgical instrument **137400** or otherwise provide input to the surgical instrument **137400** (e.g., in order to control one or more functions of the surgical instrument **137400**).

(939) In one aspect where the touch sensor **137420** includes a FSR **137421**, as depicted in FIG. **134**, the surgical instrument **137400** can include a circuit to control the activation of the electrosurgical generator **137426** electrically connectable to the surgical instrument **137400**. In this exemplification, the FSR **137421** is electrically coupled to an analog-to-digital converter (ADC) **137422** and a control circuit **137424** (e.g., a microcontroller or an ASIC). As a force *F* is applied to the FSR **137421**, the voltage output of the FSR **137721** varies accordingly. The ADC **137422** then converts the analog signal from the FSR **137421** to a digital signal, which is then supplied to the control circuit **137424**. In one exemplification, the control circuit **137424** can then compare the received signal (which is indicative of the output voltage of the FSR **137421**, which in turn is indicative of the force *F* or pressure experienced by the FSR **137421**) to a threshold to determine whether to activate the electrosurgical generator **137426**. In one exemplification, if the received signal does exceed the threshold, the control circuit **137424** can transmit a signal to the electrosurgical generator **137426** to activate it and energize the transducer assembly **137510** and/or RF electrodes **796** to cut and/or coagulate tissue captured by the surgical instrument **137400**. In another exemplification, the control circuit **137424** can transmit the output of the FSR **137421** or a

signal indicative thereof to a control circuit of the electrosurgical generator **137426**, which then compares the received signal with a threshold to determine whether to activate electrosurgical generator **137426**. If the force **F** or pressure experienced by the FSR **137421** to a threshold to determine whether to activate electrosurgical generator **137426**. If the received signal does exceed the threshold, the control circuit of the electrosurgical generator **137426** can cause the electrosurgical generator **137426** to begin supplying energy (via, e.g., a drive signal) to the transducer assembly **137510** of the surgical instrument **137400** that is electrically connected thereto. In sum, in some aspects the control circuit **137506** can determine whether a sufficient amount of force is being applied to the touch sensor **137420** and then activate the transducer assembly **137510** accordingly.

(940) FIGS. **135A-C** illustrate perspective views of a surgical instrument **137400** including a sensor assembly **137429** configured to detect closure of the surgical instrument **137400**, in accordance with at least one aspect of the present disclosure. In the following description of FIGS. **135A-C**, reference should also be made to FIG. **130**. In one aspect, the closure sensor assembly **137429** can include a closure sensor **137430** configured to detect when the arm **137412** of the surgical instrument **137400** is in a closed position and, in some aspects, whether additional force is being applied to the arm **137412** after the surgical instrument **137400** is in the closed position. In one exemplification, the closure sensor **137430** comprises a two-stage tactile switch that is configured to detect, at a first stage, when the arm of the surgical instrument is in a closed position and, further, is configured to detect, at a second stage, when additional force or pressure is being applied after the arm **137412** of the surgical instrument **137400** is in the closed position. Such a closure sensor **137430** can be utilized to, for example, allow the surgical instrument **137400** to be closed without necessarily automatically activating the transducer assembly **137510** and/or RF electrodes **796**.

(941) In one aspect depicted in FIGS. **135A-C**, the closure sensor **137430** is positioned on the housing **137414** such that the arm **137412** engages the closure sensor **137430** when the arm **137412** is rotated in a first direction **R.sub.1** into a closed position, as shown in FIG. **135B**, from an open position, as shown in FIG. **135A**. When the arm **137412** is in the closed position, the arm **137412** can bottom out against the housing **137414** (shroud) and/or the closure sensor **137430**. When the surgical instrument **137400** is opened (or otherwise not closed) or when the arm **137412** of the surgical instrument **137400** is closed, but no additional force is being applied thereto, the closure sensor **137430** can be in the first position or the first state, as depicted in FIG. **135B**. When the arm **137412** of the surgical instrument **137400** is closed and an additional force **F.sub.1** is applied to the arm **137412**, the closure sensor **137430** can be in the second position or the second state, as depicted in FIG. **135C**. In some aspects, when the arm **137412** is in the initial closed position, the arm **137412** can be at a first angle $\theta_{sub.1}$ from the housing **137414**, and when a force **F.sub.1** is applied to the arm **137412** in the initial closed position, the force **F.sub.1** can cause the closure sensor **137430** to depress, such that the arm **137412** is a second angle $\theta_{sub.2}$ from the housing **137414**.

(942) In one aspect, the output of the closure sensor **137430** can vary according to the position and/or state that the closure sensor **137430** is in. In other words, when the closure sensor **137430** is in the first state, it can provide a first output to the control circuit **137506** of the surgical instrument **137400**, and when the closure sensor **137430** is in the second state, it can provide a second output to the control circuit **137506** of the surgical instrument **137400**. Thus, the closure sensor **137430** can be configured to detect whether (i) the surgical instrument **137400** is closed and (ii) when the surgical instrument **137400** is closed, whether additional force is being applied. In one aspect, the transducer assembly **137510** and/or RF electrodes **796** can be activated and/or supplied energy only when the closure sensor **137430** is in the second state/position. This aspect would allow surgeons to activate the surgical instrument **137400** solely through manipulation of the arm **137412**, but without losing the ability to grasp and manipulate tissue absent activation of the activation of the transducer assembly **137510** and/or RF electrodes **796**.

(943) In one aspect, the control circuit **137506** is configured to receive the output from the closure sensor **137430** and then compare the output of the closure sensor **137430** to a threshold to determine whether the closure sensor **137430** is in the second position/state. The threshold can be, e.g., predetermined or set by a user of the surgical instrument **137400**. In the exemplifications described above where the closure sensor **137430** detects whether the arm **137412** of the surgical instrument **137400** is being closed and, further, whether additional force is being applied to the arm **137412** when the arm **137412** is closed, the output of the closure sensor **137430** thus varies accordingly. Further in these exemplifications, the threshold can correspond to a threshold force being applied to the arm **137412** (and thus the closure sensor **137430**) after the arm **137412** is closed. For example, if the control circuit **137506** determines that the closure sensor **137430** output exceeds the threshold, then the control circuit **137506** can activate the transducer assembly **137510** and/or RF electrodes **796** by sending a signal to the generator **137504** that cause the generator **137504** to begin supplying energy to the transducer assembly. In sum, in some aspects the control circuit **137506** can determine whether a sufficient amount of force is being applied to the closed arm **137412** of the surgical instrument **137400** and, if it is, then activate the transducer assembly **137510** and/or RF electrodes **796**.

(944) FIGS. **136A-F** illustrates various views of a surgical instrument **137400** including a sensor assembly **137439** configured to detect opening of the surgical instrument **137400**, in accordance with at least one aspect of the present disclosure. In the following description of FIGS. **136A-F**, reference should also be made to FIG. **130**. In one aspect, the opening sensor assembly **137439** includes an opening sensor **137440** that is configured to detect when the arm **137412** of the surgical instrument **137400** is rotated in a second direction **R.sub.2** into an open position. In one exemplification, the opening sensor **137440** comprises a tactile switch (e.g., a one-stage tactile switch) that is configured to detect when the arm **137412** of the surgical instrument **137400** is in a sufficiently open position. The opening sensor **137440** can be utilized to, for example, allow the surgical instrument **137400** to be energized when it is in a fully or sufficiently open position for performing back (anterior) scoring and other such surgical techniques that utilize the application of electrosurgical or ultrasonic energy to unclamped tissue, without causing the surgical instrument **137400** to be energized any time the surgical instrument **137400** is opened to any degree.

(945) In various aspects, the opening sensor **137440** can be positioned at or adjacently to the pivot point **137413** of the surgical instrument **137400**. In one aspect depicted in FIGS. **136A-F**, the opening sensor **137440** is positioned within a recess **137443** on a first lateral portion of the housing **137414**. A corresponding tab **137442** is positioned on a second lateral portion of the housing **137414** and is configured to move through the recess **137443** and contact the opening sensor **137440**, applying a force **F.sub.2** thereto, when the clamp arm **137416** of the surgical instrument **137400** is sufficiently open (i.e., opened to at least a particular angle). When the opening sensor **137440** is uncontacted by the tab **137442**, the opening sensor **137440** can be in the first position or the first state. When the arm **137412** of the surgical instrument **137400** is opened to a sufficient angle such that the tab **137422** contacts the opening sensor **137440**, the opening sensor **137440** can be in the second position or the second state, as depicted in FIG. **136D**. In one aspect, the output of the opening sensor **137440** can vary according to the position and/or state that the opening sensor **137440** is in. In other words, when the opening sensor **137440** is in the first state, it can provide a first output to the control circuit **137506** of the surgical instrument **137400**, and when the opening sensor **137440** is in the second state, it can provide a second output to the control circuit **137506** of the surgical instrument **137400**. Thus, the opening sensor **137440** is able to detect whether the surgical instrument **137400** is opened at least to the angle that causes the opening sensor **137440** to be triggered or activated (e.g., by a force **F.sub.2** being applied thereto). In one aspect, the transducer assembly **137510** and/or RF electrodes **796** can be activated and/or energized, as described above, only when the sensor is in the second state/position.

(946) In one aspect, the control circuit **137506** is configured to receive the output from the opening sensor **137440** and then compare the output of the opening sensor **137440** to a threshold, where the threshold corresponds to the opening sensor **137440** being in the second position/state. The threshold can be, e.g., predetermined or set by a user of the surgical instrument **137400**. In the exemplifications described above where the opening sensor **137440** detects whether the arm **137412** of the surgical instrument **137400** is open to a particular angle, the output of the opening sensor **137440** thus varies accordingly. Further in these exemplifications, the threshold can correspond to a threshold angle at which the arm **137412** of the surgical instrument **137400** is positioned. In one aspect, if the control circuit **137506** determines that the output of the opening sensor **137440** exceeds the threshold, then the control circuit **137506** can activate the transducer assembly **137510** and/or RF electrodes **796** by sending a signal to the generator **137504** that cause the generator **137504** to begin supplying energy to the transducer assembly **137510** and/or RF electrodes **796**. In sum, in some aspects the control circuit **137506** can determine whether the arm **137412** of the surgical instrument **137400** is open to a sufficient

angle and, if it is, then activate the transducer assembly **137510** and/or RF electrodes **796**.
(947) In certain aspects, the sensor assemblies for activating a surgical instrument **137400** described above in connection with FIGS. **131-136F** can be implemented in various combinations with each other. For example, FIG. **137** illustrates an exemplification of a surgical instrument **137400** where the sensor assembly **137508** includes both the closure sensor assembly **137429** described in connection with FIGS. **135A-C** and the opening sensor assembly **137439** described in connection with FIGS. **136A-F**. The various aspects of sensor assemblies **137508** described herein can be combined together in a surgical instrument **137400** in order to provide supplementary and/or alternative methods for activating and/or providing input to the surgical instrument **137400**. It should be noted that the exemplification depicted in FIG. **137** is intended to be merely illustrative and other exemplifications of surgical instruments **137400** can include any other combination of the aforementioned sensor assemblies **137508**.
(948) FIG. **138** illustrates a perspective view of a surgical instrument comprising a deactivation control **137450**, in accordance with at least one aspect of the present disclosure. In various aspects, the surgical instrument **137400** can include a deactivation control **137450** for controlling whether one or more of the various sensors of the surgical instrument **137400**, such as various sensor assemblies **137508** described above with respect to FIGS. **131-136F**, are active. The deactivation control **137450** can include, for example, a physical toggle or switch disposed on the housing **137414** of the surgical instrument **137400** or a touchscreen display. The deactivation control **137450** can be communicably coupled to the control circuit **137506** of the surgical instrument **137400** and, depending upon the input from the deactivation control **137450**, the control circuit **137506** can, for example, deactivate the sensor assembly **137508** controlled by the deactivation control **137450** or otherwise ignore the output of or not take any actions in response to the output from the sensor assembly **137508** controlled by the deactivation control **137450**.
(949) In reference to FIGS. **128-138**, the surgical instrument **137400** can further include an indicator, such as an LED, a display, and other such output devices. The indicator can be coupled to the control circuit **137506** and controlled thereby. In some aspects, the control circuit **137506** can be configured to activate the indicator in response to an input received from the sensor assembly **137508**. For example, the control circuit **137506** can be configured to activate the indicator when the control circuit **137506** determines that the surgical instrument **137400** is in a closed position (e.g., as sensed via a sensor assembly **137508**).

Smart Retractor

(950) FIG. **139** illustrates a perspective view of a retractor **137600** comprising a sensor **137602**, in accordance with at least one aspect of the present disclosure. In various aspects, a retractor **137600** for securing a surgical site opening **137650** can include a sensor **137602** that is removably or integrally affixed thereto. In one aspect, the sensor **137602** is removably affixable to the retractor **137600** via a magnet. The sensor **137602** can be configured to detect when the retractor **137600** is tapped, jostled, moved, or otherwise manipulated by a user (e.g., a surgeon). In one exemplification, the sensor **137602** can include a vibration sensor (e.g., an ADIS16223 digital tri-axial vibration sensor) that is configured to detect vibration or movement by the retractor **137600** to which the sensor **137602** is affixed. In one aspect, the sensor **137602** can be reusable, i.e., the sensor **137602** can maintain its effectiveness through sterilization processes (because the sensor **137602** is affixed to a retractor **137600**, which is in the surgical field, it would be sterilized after being used in a surgical procedure if it was to be reused). The sensor **137602** can be configured to detect different types of motions or actions (e.g., tapping) by a user according to the amplitude, frequency, and/or direction of the detected motion or vibration of the retractor **137600**.

(951) The sensor **137602** can be configured to transmit a signal indicative of the detected vibration or movement of the retractor **137600**. In one aspect, the sensor **137602** can be communicably coupled to a surgical instrument **137606** (e.g., a surgical instrument or an electrosurgical instrument) and/or another device (e.g., a generator) via, for example, a wired connection **137604**. Based upon the motion or movement detected by the sensor **137602**, the sensor **137602** can change the state of the surgical instrument(s) **137606** and/or other device(s) that are communicably coupled to the sensor **137602**. The state of the surgical instrument(s) **137606** and/or other device(s) can correspond to, for example, a mode that the instrument(s) **137606** and/or device(s) are in or a property of the instrument(s) **137606** and/or device(s). For example, when the sensor **137602** detects that the retractor **137600** is being tapped, the sensor **137602** can transmit a signal to a surgical instrument **137606** that is communicably coupled with it that causes the surgical instrument to **137606** to change from an inactive state to an activate state (or vice versa). As another example, when the sensor **137602** detects that the retractor **137600** is being touched, the sensor **137602** can transmit a signal to a surgical generator that is communicably coupled with it that causes the generator to change from an inactive mode to an activate mode (or vice versa). In some aspects, the retractor sensor **137602** can be configured to transmit data and/or signals to a surgical hub **106**, as described in connection with FIGS. **1-11**, which can then in turn take various actions, such as controlling the surgical instrument(s) **137606** and/or other device(s), as described above.

(952) FIG. **140** illustrates a perspective view of a retractor **137902** comprising a display in use at a surgical site **137900**, in accordance with at least one aspect of the present disclosure. A surgical retractor **137902** helps the surgeon and operating room professionals hold an incision or wound open during surgical procedures. The surgical retractor **137902** aids in holding back underlying organs or tissues, allowing doctors/nurses better visibility and access to the exposed area. A retractor **137902** can include a display **137904** or other control device that is configured to display alerts and/or information associated with the surgical procedure being performed, provide a means of controlling the instruments or devices being utilized during the course of the surgical procedure or the environment in which the surgical procedure is being performed (e.g., the operating room), and perform other such functions. In the depicted aspect, the control device is integral to the retractor **137902**. In another aspect, the control device can include, for example, a portable electronic device including a touchscreen display (e.g., a tablet computer) that is removably affixable to the retractor **137902**. In yet another aspect, the control device includes a flexible sticker display that is attachable to the body/skin of the patient or another surface.
(953) In one aspect, the control device includes an input device (e.g., a keypad, a capacitive touchscreen, or a combination thereof) for receiving input from a user, an output device (e.g., a display) for providing alerts, information, or other output to a user, an energy source (e.g., a coin cell, a battery, a photovoltaic cell, or a combination thereof); and a network interface controller for a communication protocol (e.g., Wi-Fi, Bluetooth) for communicably connecting the control device to surgical instruments, devices within the operating room (e.g., a surgical hub **106** as described in FIGS. **1-11**), and/or other equipment (surgical or otherwise). The control device can be configured to provide a graphical user interface (GUI) for displaying information to the user (e.g., a surgeon) and receiving input or commands from the user. In one aspect, the control device further includes a light source **137906** (e.g., an array of LEDs) that is configured to illuminate the surgical field of view **137908** that the retractor **137902** is being utilized to secure.

(954) In one aspect, the control device is removably affixable to the surgical retractor **137902**. In another aspect, the control device is integral to the retractor **137902**, defining a “smart” surgical retractor **137902**. The smart surgical retractor **137902** may comprise an input display operated by the smart surgical retractor **137902**. The smart surgical retractor **137902** may comprise a wireless communication device to communicate with a device connected to a generator module coupled to the surgical hub. Using the input display of the smart surgical retractor **137902**, the surgeon can adjust power level or mode of the generator module to cut and/or coagulate tissue. If using automatic on/off for energy delivery on closure of an end effector on the tissue, the status of automatic on/off may be indicated by a light, screen, or other device located on the smart retractor housing. Power being used may be changed and displayed.

(955) In various aspects, the control device can be configured to control various functions of the surgical instruments that are communicably connected to the control device, such as the power parameters (e.g., for an electrosurgical instrument and/or an ultrasonic instrument) or operating modes (e.g., “cut” and “coagulation” modes for an electrosurgical instrument, or automatic) of the surgical instruments. In various aspects, the control device can be configured to display information related to the surgical procedure being performed and/or information related to the equipment being used during the course of the surgical procedure, such as the temperature of an ultrasonic blade (end effector), alerts or alarms that are generated during the course of the surgical procedure, or the location of nerves within the surgical field. The alerts or alarms can be generated by, for example, the surgical instruments and/or a surgical hub **106** to which the surgical instruments (or other modular surgical devices) are

communicably connected. In various aspects, the control device can be configured to control functions of the environment in which the surgical procedure is being performed (e.g., an operating room), such as the intensity and/or position of the field lights within an operating room. (956) In various aspects, the control device can be configured to sense what surgical instruments or other equipment are within the vicinity of the control device and then cause any surgical instruments or other equipment that connected to the control device to pass their operational controls to the control device. In one aspect, the smart surgical retractor **137902** can sense or know what device/instrument the surgeon is using, either through the surgical hub **106** or RFID or other device placed on the device/instrument or the smart surgical retractor **137902**, and provide an appropriate display. Alarms and alerts may be activated when conditions require. Other features include displaying the temperature of the ultrasonic blade, nerve monitoring, light source or fluorescence. The light source **137906** may be employed to illuminate the surgical field of view **137908** and to charge photocells on single use sticker display that stick onto the smart retractor **137902**. In another aspect, the smart surgical retractor **137902** may include an augmented reality projected on the patient's anatomy (e.g., a vein viewer).

(957) In other aspects, the control device can comprise a smart flexible sticker display attachable to the body/skin of a patient. The smart flexible sticker display can be applied to, for example, the body/skin of a patient between the area exposed by the surgical retractors. In one aspect, the smart flexible sticker display may be powered by light, an on board battery, or a ground pad. The flexible sticker display may communicate via short range wireless (e.g., Bluetooth) to a device, may provide readouts, lock power, or change power. The smart flexible sticker display also comprises photocells to power the smart flexible sticker display using ambient light energy. The flexible sticker display includes a display **137904** of a control panel user interface to enable the surgeon to control devices or other modules coupled to the surgical hub.

(958) Various additional details regarding smart retractors can be found in U.S. patent application Ser. No. 15/940,686, titled DISPLAY OF ALIGNMENT OF STAPLE CARTRIDGE TO PRIOR LINEAR STAPLE LINE, filed Mar. 29, 2018, which is hereby incorporated by reference in its entirety.

Situational Awareness of Electrosurgical Systems

(959) Electrosurgical instruments are utilized to treat various tissue types by application of energy to the tissue. As described in connection with FIGS. **22-24**, an electrosurgical instrument (e.g. surgical instruments **1104**, **1106**, **1108**) may be connected to a generator **1100**, and include an end effector (e.g. end effectors **1122**, **1124**, **1125**) configured to grasp and transmit therapeutic energy to tissue.

(960) In various aspects, the end effector can be used to seal, weld, or coagulate tissue such as, for example, a blood vessel by application of energy to the blood vessel while grasped by the end effector. Since blood vessels are generally surrounded by protective tissue, the tissue has to be separated to expose the blood vessels for an effective seal to be achieved. But tissue separation requires lower energy than tissue sealing or coagulation. Also, the amount of tissue to be separated varies depending on, for example, the anatomical location of the blood vessel, the state of the tissue, and the type of surgery being performed.

(961) One technique of treating a tissue that includes a blood vessel involves separating and moving the inner muscle layer of the blood vessel away from the adventitia layer prior to sealing and/or transecting the blood vessel. In order to more effectively separate the tissue layers of the blood vessel, a low level energy sufficient to separate but not coagulate or seal the tissue can be generated and transmitted to the tissue. Subsequently, a high level energy is employed to seal or coagulate the tissue.

(962) A successful energy treatment of tissue grasped by an end effector depends on selecting a suitable energy mode of operation for each closure stage of the end effector. Furthermore, closure stages of an end effector may be determined by various situational parameters such as, for example, tissue type, anatomical location, and/or composition. Various suitable situational parameters are described under the heading "SITUATIONAL AWARENESS" in connection with FIG. **141**.

(963) Aspects of the present disclosure present various processes for selecting different energy modes for different closure stages of an end effector of an electrosurgical instrument. The selection can be based, at least in part, on one or more situational parameters.

Situational Awareness

(964) Situational awareness is the ability of some aspects of a surgical system to determine or infer information related to a surgical procedure from data received from databases and/or instruments. The information can include the type of procedure being undertaken, the type of tissue being operated on, or the body cavity that is the subject of the procedure. With the contextual information related to the surgical procedure, the surgical system can, for example, improve the manner in which it controls the modular devices (e.g. a robotic arm and/or robotic surgical tool) that are connected to it and provide contextualized information or suggestions to the surgeon during the course of the surgical procedure.

(965) Referring now to FIG. **141**, a timeline **5200** depicting situational awareness of a hub, such as the surgical hub **106** or **206**, for example, is depicted. The timeline **5200** is an illustrative surgical procedure and the contextual information that the surgical hub **106**, **206** can derive from the data received from the data sources at each step in the surgical procedure. The timeline **5200** depicts the typical steps that would be taken by the nurses, surgeons, and other medical personnel during the course of a lung segmentectomy procedure, beginning with setting up the operating theater and ending with transferring the patient to a post-operative recovery room.

(966) The situationally aware surgical hub **106**, **206** receives data from the data sources throughout the course of the surgical procedure, including data generated each time medical personnel utilize a modular device that is paired with the surgical hub **106**, **206**. The surgical hub **106**, **206** can receive this data from the paired modular devices and other data sources and continually derive inferences (i.e., contextual information) about the ongoing procedure as new data is received, such as which step of the procedure is being performed at any given time. The situational awareness system of the surgical hub **106**, **206** is able to, for example, record data pertaining to the procedure for generating reports, verify the steps being taken by the medical personnel, provide data or prompts (e.g., via a display screen) that may be pertinent for the particular procedural step, adjust modular devices based on the context (e.g., activate monitors, adjust the field of view (FOV) of the medical imaging device, or change the energy level of an ultrasonic surgical instrument or RF electrosurgical instrument), and take any other such action described above.

(967) As the first step **5202** in this illustrative procedure, the hospital staff members retrieve the patient's EMR from the hospital's EMR database. Based on select patient data in the EMR, the surgical hub **106**, **206** determines that the procedure to be performed is a thoracic procedure.

(968) Second step **5204**, the staff members scan the incoming medical supplies for the procedure. The surgical hub **106**, **206** cross-references the scanned supplies with a list of supplies that are utilized in various types of procedures and confirms that the mix of supplies corresponds to a thoracic procedure. Further, the surgical hub **106**, **206** is also able to determine that the procedure is not a wedge procedure (because the incoming supplies either lack certain supplies that are necessary for a thoracic wedge procedure or do not otherwise correspond to a thoracic wedge procedure).

(969) Third step **5206**, the medical personnel scan the patient band via a scanner that is communicably connected to the surgical hub **106**, **206**. The surgical hub **106**, **206** can then confirm the patient's identity based on the scanned data.

(970) Fourth step **5208**, the medical staff turns on the auxiliary equipment. The auxiliary equipment being utilized can vary according to the type of surgical procedure and the techniques to be used by the surgeon, but in this illustrative case they include a smoke evacuator, insufflator, and medical imaging device. When activated, the auxiliary equipment that are modular devices can automatically pair with the surgical hub **106**, **206** that is located within a particular vicinity of the modular devices as part of their initialization process. The surgical hub **106**, **206** can then derive contextual information about the surgical procedure by detecting the types of modular devices that pair with it during this pre-operative or initialization phase.

In this particular example, the surgical hub **106**, **206** determines that the surgical procedure is a VATS procedure based on this particular combination of paired modular devices. Based on the combination of the data from the patient's EMR, the list of medical supplies to be used in the procedure, and the type of modular devices that connect to the hub, the surgical hub **106**, **206** can generally infer the specific procedure that the surgical team will be performing. Once the surgical hub **106**, **206** knows what specific procedure is being performed, the surgical hub **106**, **206** can then retrieve the steps of that procedure from a memory or from the cloud and then cross-reference the data it subsequently receives from the connected data sources (e.g.,

modular devices and patient monitoring devices) to infer what step of the surgical procedure the surgical team is performing.

(971) Fifth step **5210**, the staff members attach the EKG electrodes and other patient monitoring devices to the patient. The EKG electrodes and other patient monitoring devices are able to pair with the surgical hub **106, 206**. As the surgical hub **106, 206** begins receiving data from the patient monitoring devices, the surgical hub **106, 206** thus confirms that the patient is in the operating theater.

(972) Sixth step **5212**, the medical personnel induce anesthesia in the patient. The surgical hub **106, 206** can infer that the patient is under anesthesia based on data from the modular devices and/or patient monitoring devices, including EKG data, blood pressure data, ventilator data, or combinations thereof, for example. Upon completion of the sixth step **5212**, the pre-operative portion of the lung segmentectomy procedure is completed and the operative portion begins.

(973) Seventh step **5214**, the patient's lung that is being operated on is collapsed (while ventilation is switched to the contralateral lung). The surgical hub **106, 206** can infer from the ventilator data that the patient's lung has been collapsed, for example. The surgical hub **106, 206** can infer that the operative portion of the procedure has commenced as it can compare the detection of the patient's lung collapsing to the expected steps of the procedure (which can be accessed or retrieved previously) and thereby determine that collapsing the lung is the first operative step in this particular procedure.

(974) Eighth step **5216**, the medical imaging device (e.g., a scope) is inserted and video from the medical imaging device is initiated. The surgical hub **106, 206** receives the medical imaging device data (i.e., video or image data) through its connection to the medical imaging device. Upon receipt of the medical imaging device data, the surgical hub **106, 206** can determine that the laparoscopic portion of the surgical procedure has commenced. Further, the surgical hub **106, 206** can determine that the particular procedure being performed is a segmentectomy, as opposed to a lobectomy (note that a wedge procedure has already been discounted by the surgical hub **106, 206** based on data received at the second step **5204** of the procedure). The data from the medical imaging device **124** (FIG. 2) can be utilized to determine contextual information regarding the type of procedure being performed in a number of different ways, including by determining the angle at which the medical imaging device is oriented with respect to the visualization of the patient's anatomy, monitoring the number or medical imaging devices being utilized (i.e., that are activated and paired with the surgical hub **106, 206**), and monitoring the types of visualization devices utilized. For example, one technique for performing a VATS lobectomy places the camera in the lower anterior corner of the patient's chest cavity above the diaphragm, whereas one technique for performing a VATS segmentectomy places the camera in an anterior intercostal position relative to the segmental fissure. Using pattern recognition or machine learning techniques, for example, the situational awareness system can be trained to recognize the positioning of the medical imaging device according to the visualization of the patient's anatomy. As another example, one technique for performing a VATS lobectomy utilizes a single medical imaging device, whereas another technique for performing a VATS segmentectomy utilizes multiple cameras. As yet another example, one technique for performing a VATS segmentectomy utilizes an infrared light source (which can be communicably coupled to the surgical hub as part of the visualization system) to visualize the segmental fissure, which is not utilized in a VATS lobectomy. By tracking any or all of this data from the medical imaging device, the surgical hub **106, 206** can thereby determine the specific type of surgical procedure being performed and/or the technique being used for a particular type of surgical procedure.

(975) Ninth step **5218**, the surgical team begins the dissection step of the procedure. The surgical hub **106, 206** can infer that the surgeon is in the process of dissecting to mobilize the patient's lung because it receives data from the RF or ultrasonic generator indicating that an energy instrument is being fired. The surgical hub **106, 206** can cross-reference the received data with the retrieved steps of the surgical procedure to determine that an energy instrument being fired at this point in the process (i.e., after the completion of the previously discussed steps of the procedure) corresponds to the dissection step. In certain instances, the energy instrument can be an energy tool mounted to a robotic arm of a robotic surgical system.

(976) Tenth step **5220**, the surgical team proceeds to the ligation step of the procedure. The surgical hub **106, 206** can infer that the surgeon is ligating arteries and veins because it receives data from the surgical stapling and cutting instrument indicating that the instrument is being fired. Similarly to the prior step, the surgical hub **106, 206** can derive this inference by cross-referencing the receipt of data from the surgical stapling and cutting instrument with the retrieved steps in the process. In certain instances, the surgical instrument can be a surgical tool mounted to a robotic arm of a robotic surgical system.

(977) Eleventh step **5222**, the segmentectomy portion of the procedure is performed. The surgical hub **106, 206** can infer that the surgeon is transecting the parenchyma based on data from the surgical stapling and cutting instrument, including data from its cartridge. The cartridge data can correspond to the size or type of staple being fired by the instrument, for example. As different types of staples are utilized for different types of tissues, the cartridge data can thus indicate the type of tissue being stapled and/or transected. In this case, the type of staple being fired is utilized for parenchyma (or other similar tissue types), which allows the surgical hub **106, 206** to infer that the segmentectomy portion of the procedure is being performed.

(978) Twelfth step **5224**, the node dissection step is then performed. The surgical hub **106, 206** can infer that the surgical team is dissecting the node and performing a leak test based on data received from the generator indicating that an RF or ultrasonic instrument is being fired. For this particular procedure, an RF or ultrasonic instrument being utilized after parenchyma was transected corresponds to the node dissection step, which allows the surgical hub **106, 206** to make this inference. It should be noted that surgeons regularly switch back and forth between surgical stapling/cutting instruments and surgical energy (i.e., RF or ultrasonic) instruments depending upon the particular step in the procedure because different instruments are better adapted for particular tasks. Therefore, the particular sequence in which the stapling/cutting instruments and surgical energy instruments are used can indicate what step of the procedure the surgeon is performing. Moreover, in certain instances, robotic tools can be utilized for one or more steps in a surgical procedure and/or handheld surgical instruments can be utilized for one or more steps in the surgical procedure. The surgeon(s) can alternate between robotic tools and handheld surgical instruments and/or can use the devices concurrently, for example. Upon completion of the twelfth step **5224**, the incisions are closed up and the post-operative portion of the procedure begins.

(979) Thirteenth step **5226**, the patient's anesthesia is reversed. The surgical hub **106, 206** can infer that the patient is emerging from the anesthesia based on the ventilator data (i.e., the patient's breathing rate begins increasing), for example.

(980) Lastly, the fourteenth step **5228** is that the medical personnel remove the various patient monitoring devices from the patient. The surgical hub **106, 206** can thus infer that the patient is being transferred to a recovery room when the hub loses EKG, BP, and other data from the patient monitoring devices. As can be seen from the description of this illustrative procedure, the surgical hub **106, 206** can determine or infer when each step of a given surgical procedure is taking place according to data received from the various data sources that are communicably coupled to the surgical hub **106, 206**.

(981) Situational awareness is further described in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, which is herein incorporated by reference in its entirety. In certain instances, operation of a robotic surgical system, including the various robotic surgical systems disclosed herein, for example, can be controlled by the hub **106, 206** based on its situational awareness and/or feedback from the components thereof and/or based on information from the cloud **102**.

EXAMPLES

(982) Various aspects of the subject matter described herein are set out in the following numbered examples.

(983) Example 1: A method for characterizing a state of an end effector of an ultrasonic device, the ultrasonic device comprising an electromechanical ultrasonic system defined by a predetermined resonant frequency, the electromechanical ultrasonic system further comprising an ultrasonic transducer coupled to an ultrasonic blade, the method comprising: applying, by an energy source, a power level to the ultrasonic transducer; measuring, by a control circuit coupled to a memory, an impedance value of the ultrasonic transducer; comparing, by the control circuit, the impedance value to a reference impedance value stored in the memory; classifying, by the control circuit, the impedance value based on the

comparison; characterizing, by the control circuit, the state of the electromechanical ultrasonic system based on the classification of the impedance value; and adjusting, by the control circuit, the power level applied to the ultrasonic transducer based on the characterization of the state of the end effector.

(984) Example 2: The method of Example 1, wherein measuring, by a control circuit coupled to a memory, an impedance value of the ultrasonic transducer comprises measuring a complex impedance value defined as a ratio of a voltage signal $V_g(t)$ applied by the energy source to the ultrasonic transducer to a current signal $I_g(t)$ applied by the energy source to the ultrasonic transducer.

(985) Example 3: The method of Example 2, wherein comparing, by the control circuit, the impedance value to a reference impedance value stored in the memory comprises comparing, by the control circuit, the impedance value to a data point in a reference complex impedance characteristic pattern.

(986) Example 4: The method of any one or more of Examples 1-3, wherein measuring, by a control circuit coupled to a memory, an impedance value of the ultrasonic transducer comprises: applying, by the energy source, a drive signal to the ultrasonic transducer, wherein the drive signal is a periodic signal defined by a magnitude and a frequency; sweeping, by the control circuit, the frequency of the drive signal from below a first resonance to above the first resonance of the electromechanical ultrasonic system; and measuring and recording, by the control circuit, impedance/admittance circle variables Re , Ge , Xe , and Be .

(987) Example 5: The method of Example 4, wherein comparing, by the control circuit, the impedance value to reference impedance value stored in the memory comprises comparing, by the control circuit, the measured impedance/admittance circle variables Re , Ge , Xe , Be to the reference impedance/admittance circle variables $Rref$, $Gref$, $Xref$, and $Bref$.

(988) Example 6: The method of any one or more of Examples 1-5, wherein characterizing, by the control circuit, the state of the electromechanical ultrasonic system based on the classification of the impedance value comprises determining the proper installation of two or more components of the ultrasonic device.

(989) Example 7: The method of any one or more of Examples 1-6, wherein characterizing, by the control circuit, the state of the electromechanical ultrasonic system based on the classification of the impedance value comprises determining an amount of power delivered to the ultrasonic device by the energy source to compensate for an articulation angle of an articulatable ultrasonic blade coupled to the ultrasonic transducer.

(990) Example 8: A method for characterizing a function of an end effector of an ultrasonic device, the ultrasonic device comprising an electromechanical ultrasonic system defined by a predetermined resonant frequency, the electromechanical ultrasonic system further comprising an ultrasonic transducer coupled to an ultrasonic blade, the method comprising: applying, by an energy source, a power level to the ultrasonic transducer, measuring, by a control circuit coupled to a memory, an impedance value of the ultrasonic transducer, comparing, by the control circuit, the impedance value to a reference impedance value stored in the memory; classifying, by the control circuit, the impedance value based on the comparison; characterizing, by the control circuit, the function of the electromechanical ultrasonic system based on the classification of the impedance value; and adjusting, by the control circuit, the power level applied to the ultrasonic transducer based on the characterization of the function of the end effector.

(991) Example 9: The method of Example 8, wherein measuring, by a control circuit coupled to a memory, an impedance value of the ultrasonic transducer comprises measuring a complex impedance value defined as a ratio of a voltage signal $V_g(t)$ applied by the energy source to the ultrasonic transducer to a current signal $I_g(t)$ applied by the energy source to the ultrasonic transducer.

(992) Example 10: The method of Example 9, wherein comparing, by the control circuit, the impedance value to a reference impedance value stored in the memory comprises comparing, by the control circuit, the impedance value to a data point in a reference complex impedance characteristic pattern.

(993) Example 11: The method of any one or more of Examples 8-10, wherein adjusting, by the control circuit, the power level applied to the ultrasonic transducer based on the characterization of the function of the end effector comprises adjusting, by the control circuit, the power level applied to the ultrasonic transducer based on a determination that a tissue transection process is complete.

(994) Example 12: The method of any one or more of Examples 8-11, wherein characterizing, by the control circuit, the function of the electromechanical ultrasonic system based on the classification of the impedance value comprises determining, by the control circuit, that the ultrasonic blade is contacting a vessel.

(995) Example 13: The method of Example 12, wherein measuring, by a control circuit coupled to a memory, an impedance value of the ultrasonic transducer comprises measuring, by the control circuit, a complex impedance of the ultrasonic transducer, wherein the complex impedance is defined as

$$(996) Z_g(t) = \frac{V_g(t)}{I_g(t)}.$$

(997) Example 14: The method of Example 13, further comprising: receiving, by the control circuit, a complex impedance measurement data point; comparing, by the control circuit, the complex impedance measurement data point to a data point in a reference complex impedance characteristic pattern; classifying, by the control circuit, the complex impedance measurement data point based on a result of the comparison analysis; and determining, by the control circuit, that the ultrasonic blade is contacting the vessel based on the result of the comparison analysis.

(998) Example 15: The method of any one or more of Examples 12-14, further comprising generating, by the control circuit, a warning that the ultrasonic blade is contacting the vessel.

(999) Example 16: A method for characterizing a tissue in contact with an end effector of an ultrasonic device, the ultrasonic device comprising an electromechanical ultrasonic system defined by a predetermined resonant frequency, the electromechanical ultrasonic system further comprising an ultrasonic transducer coupled to an ultrasonic blade, the method comprising: applying, by an energy source, a power level to the ultrasonic transducer, measuring, by a control circuit coupled to a memory, an impedance value of the ultrasonic transducer, comparing, by the control circuit, the impedance value to a reference impedance value stored in the memory; classifying, by the control circuit, the impedance value based on the comparison; characterizing, by the control circuit, the tissue in contact with the end effector based on the classification of the impedance value; and adjusting, by the control circuit, the power level applied to the ultrasonic transducer based on the characterization of the tissue in contact with the end effector.

(1000) Example 17: The method of Example 16, wherein measuring, by a control circuit coupled to a memory, an impedance value of the ultrasonic transducer comprises measuring a complex impedance value defined as a ratio of a voltage signal $V_g(t)$ applied by the energy source to the ultrasonic transducer to a current signal $I_g(t)$ applied by the energy source to the ultrasonic transducer.

(1001) Example 18: The method of Example 17, further comprising: pulsing, by the control circuit, the power level delivered to the ultrasonic transducer by the energy source; determining, by the control circuit, changes in tissue characteristics of tissue located in the end effector, wherein the changes in tissue characteristics is determined between pulses; and adjusting, by the processor or control circuit, power delivered to the ultrasonic transducer based on the tissue changes.

(1002) Example 19: The method of any one or more of Examples 16-18, wherein measuring, by a control circuit coupled to a memory, an impedance value of the ultrasonic transducer comprises: applying, by the energy source, a drive signal to the ultrasonic transducer, wherein the drive signal is a periodic signal defined by a magnitude and a frequency; sweeping, by the control circuit, the frequency of the drive signal from below a first resonance to above the first resonance of the electromechanical ultrasonic system; and measuring and recording, by the control circuit, impedance/admittance circle variables Re , Ge , Xe , and Be .

(1003) Example 20: The method of claim any one or more of Examples 16-19, wherein characterizing, by the control circuit, the tissue in contact

with the end effector based on the classification value comprises classifying the tissue into a distinct group in live time.

(1004) Various additional aspects of the subject matter described herein are set out in the following numbered examples.

(1005) Example 1: A surgical system comprising: an energy generator, a surgical instrument electrically coupled to the energy generator and configured to transmit electrosurgical energy to tissue of a patient at a surgical site; at least one sensor configured to detect energy anomaly; and at least one processor communicatively coupled to the at least one sensor, and configured to: receive data from the at least one sensor that energy is being emitted at an unintended location within the surgical system; transmit an alert indicating that parasitic capacitive coupling is occurring; and transmit an interrupt to the energy generator to temporarily interrupt energy generation.

(1006) Example 2: The surgical system of Example 1, further comprising an energy dissipating pad electrically coupled to a neutral electrode in the energy generator, wherein the energy dissipating pad is configured to conductively connect to the patient and dissipate energy away from the patient when energy is applied to the patient tissue by the surgical instrument.

(1007) Example 3: The surgical system of any one of Examples 1 or 2, wherein the neutral electrode is configured to passively shunt energy away from the patient during a capacitive coupling event.

(1008) Example 4: The surgical system of any one of Examples 1-3, wherein the surgical instrument comprises: an end effector comprising a pair of jaws; and a shaft electrically coupled to the end effector and configured to deliver energy to the end effector from the energy generator.

(1009) Example 5: The surgical system Example 4, wherein the jaws comprise rounded tips configured to reduce peak voltage spikes as the jaws come into contact with tissue of the patient.

(1010) Example 6: The surgical system of Example 4, wherein the shaft comprises interrupting insulator elements configured to prevent capacitive coupling from transmitting long distance within the shaft.

(1011) Example 7: The surgical system of Example 4, wherein the surgical instrument comprises a triangular one-sided blade with a thin standing upper blade element configured to reduce inductive energy transmission beyond the upper blade element.

(1012) Example 8: The surgical system of Example 4, wherein the end effector comprises one or more electrodes located on the inside portion of the jaws and configured to channel excess energy away from tissue of the patient.

(1013) Example 9: A method of a surgical system for detecting parasitic capacitive coupling, the surgical system comprising an energy generator, a surgical instrument, and at least one sensor for detecting an energy anomaly, the method comprising: receiving data from the at least one sensor that energy is being emitted at an unintended location within the surgical system; transmitting an alert indicating that parasitic capacitive coupling is occurring; and transmitting an interrupt to the energy generator to temporarily interrupt energy generation.

(1014) Example 10: A surgical instrument configured to mitigate parasitic capacitive coupling during surgery, comprising: an electrical input configured to be electrically coupled to an energy generator, a shaft; an end effector at a distal end of the shaft; at least one sensor configured to detect energy anomaly; and at least one processor communicatively coupled to the at least one sensor, and configured to: receive data from the at least one sensor that energy is being emitted at an unintended location within the surgical system; transmit an alert indicating that parasitic capacitive coupling is occurring; and transmit an interrupt to the energy generator to temporarily interrupt energy generation.

(1015) Example 11: The surgical instrument of Example 10, wherein the end effector comprises a pair of jaws, and the jaws comprise rounded tips configured to reduce peak voltage spikes as the jaws come into contact with tissue of the patient.

(1016) Example 12: The surgical instrument of Example 10 or 11, wherein the shaft comprises interrupting insulator elements configured to prevent capacitive coupling from transmitting long distance within the shaft.

(1017) Example 13: The surgical instrument of any one of Examples 10-12, further comprising a triangular one-sided blade with a thin standing upper blade element configured to reduce inductive energy transmission beyond the upper blade element.

(1018) Example 14: The surgical instrument of any one of Examples 10-13, wherein the end effector comprises one or more electrodes located on an inside portion of the jaws and configured to channel excess energy away from tissue of the patient.

(1019) Various additional aspects of the subject matter described herein are set out in the following numbered examples.

(1020) Example 1: A surgical system comprising: a monopolar energy generator, a surgical instrument electrically coupled to the monopolar energy generator comprising an electrode and configured to transmit electrosurgical energy through the electrode to tissue of a patient at a surgical site; at least one detection circuit configured to: measure an amount of conductivity in a return path of the electrosurgical energy; determine that the amount conductivity in the return path falls below a predetermined threshold; and transmit a signal to cause the monopolar generator to increase current leakage in the surgical system by increasing alternating current frequency in the electrosurgical energy generation; wherein the monopolar energy generator comprises a sensor configured to determine that a monopolar energy circuit is completed by detecting that the current leakage has reached a ground terminal in the monopolar energy generator.

(1021) Example 2: The surgical system of Example 1, wherein increasing the current leakage allows for monopolar electrosurgery of the patient to be performed using the surgical instrument.

(1022) Example 3: The surgical system of Example 1 or 2, wherein the monopolar energy generator further comprises a control circuit configured to: receive an indication from the sensor that the current leakage has not yet reached the ground terminal in the monopolar energy generator, and in response to the indication, further increase the alternating current frequency.

(1023) Example 4: The surgical system of Example 3, wherein the control circuit is further configured to: receive a second indication from the sensor that, in response to further increasing the alternating current frequency, the current leakage has reached the ground terminal in the monopolar energy generator, and in response to the second indication, cease increasing the alternating current frequency.

(1024) Example 5: The surgical system of any one of Examples 1-4, wherein the surgical system is further configured to provide an instruction to isolate any return path pads away from the surgical system to minimize conductivity flowing through any of the return path pads.

(1025) Example 6: The surgical system of any one of Examples 1-5, wherein increasing the frequency comprises increasing the frequency to a range of 500 KHz to 4 MHz.

(1026) Example 7: A monopolar energy generator of a surgical system coupled to a surgical instrument configured to transmit electrosurgical energy to tissue of a patient at a surgical site, the energy generator comprising: a power supply configured to generator monopolar electrosurgical energy; a completion circuit sensor, a control circuit; and a ground terminal; wherein the control circuit is configured to: receive a signal from a detection circuit that an amount of conductivity in a return path of the monopolar electrosurgical energy falls below a predetermined threshold; and in response to the signal; cause the power supply to increase current leakage by increasing alternating current frequency; wherein the completion circuit sensor is configured to determine that a monopolar energy circuit is completed by detecting that the current leakage has reached the ground terminal.

(1027) Example 8: The monopolar energy generator of Example 7, wherein increasing the current leakage allows for monopolar electrosurgery of the patient to be performed using the surgical instrument.

(1028) Example 9: The monopolar energy generator of Example 7 or 8, wherein the control circuit is further configured to: receive an indication from the completion circuit sensor that the current leakage has not yet reached the ground terminal; and in response to the indication, further increase the alternating current frequency.

(1029) Example 10: The monopolar energy generator of Example 9, wherein the control circuit is further configured to: receive a second indication from the sensor that, in response to further increasing the alternating current frequency, the current leakage has reached the ground terminal in the monopolar energy generator, and in response to the second indication, cease increasing the alternating current frequency.

(1030) Example 11: The monopolar energy generator of any one of Examples 7-10, further configured to provide an instruction to isolate any return path pads away from the surgical system to minimize conductivity flowing through any of the return path pads.

(1031) Example 12: A surgical system comprising a monopolar energy generator of any one of Examples 7-10, wherein increasing the frequency comprises increasing the frequency to a range of 500 KHz to 4 MHz.

(1032) Example 13: A closed loop method of a surgical system, the surgical system comprising a monopolar energy generator, a surgical instrument coupled to the energy generator, and a detection circuit communicatively coupled to the energy generator, the method comprising: generating, by the energy generator, electrosurgical energy to the surgical instrument; transmitting, by the surgical instrument, electrosurgical energy through a electrode to tissue of a patient at a surgical site; measuring, by the detection circuit, an amount of conductivity in a return path of the electrosurgical energy; determining, by the detection circuit, that the amount conductivity in the return path falls below a predetermined threshold; transmitting, by the detection circuit, a signal to the monopolar energy generator to cause the energy generator to increase current leakage in the surgical system by increasing alternating current frequency in the electrosurgical energy generation; and determining, by a sensor in the monopolar energy generator, that a monopolar energy circuit is completed by detecting that the current leakage has reached a ground terminal in the monopolar energy generator.

(1033) Example 14: The method of Example 13, wherein increasing the current leakage allows for monopolar electrosurgery of the patient to be performed using the surgical instrument.

(1034) Example 15: The method of Example 13 or 14, further comprising: receiving an indication from the sensor that the current leakage has not yet reached the ground terminal in the monopolar energy generator, and in response to the indication, further increasing the alternating current frequency.

(1035) Example 16: The method of Example 15, further comprising: receiving a second indication from the sensor that, in response to further increasing the alternating current frequency, the current leakage has reached the ground terminal in the monopolar energy generator, and in response to the second indication, ceasing increasing the alternating current frequency.

(1036) Example 17: The method of anyone of Examples 13-16, further comprising providing an instruction to isolate any return path pads away from the surgical system to minimize conductivity flowing through any of the return path pads.

(1037) Example 18: The method of any one of Examples 13-17, wherein increasing the frequency comprises increasing the frequency to a range of 500 KHz to 4 MHz.

(1038) Various additional aspects of the subject matter described herein are set out in the following examples.

(1039) Example 1: A method of adjusting a compression force applied by a surgical instrument, wherein the surgical instrument comprises an end effector and a clamp arm configured to receive energy modalities from a generator configured to deliver a plurality of energy modalities to the surgical instrument. The method comprises determining, by a control circuit, tissue impedance of tissue in contact with an end effector of the surgical instrument; determining, by the control circuit, a tissue type based on the tissue impedance; selecting, by the control circuit, a first energy modality of the plurality of energy modalities to deliver to the surgical instrument; generating, by the control circuit, a first signal waveform based on the first energy modality; selecting, by the control circuit, a second energy modality of the plurality of energy modalities to deliver to the surgical instrument; generating, by the control circuit, a second signal waveform based on the second energy modality; outputting, by the generator, the first and second signal waveform to deliver energy to the end effector, and adjusting, by the control circuit, a compression force applied by the end effector by changing a size of a gap between the tissue and the clamp arm based on a proportion of the first signal waveform to the second signal waveform.

(1040) Example 2: The method of Example 1, wherein the first energy modality is a radio frequency (RF) energy modality and the second energy modality is an ultrasonic energy modality.

(1041) Example 3: The method of Example 1 or 2, wherein determining the tissue impedance comprises: applying, by the generator, a non-therapeutic electrical signal to the end effector over a range of frequencies; and determining, by the control circuit, an impedance characteristic pattern based on spectral analysis of the non-therapeutic electrical signal.

(1042) Example 4: The method of any one of Examples 1-3, wherein the proportion is determined by the control circuit based on a time that each of the first and second signal waveform is applied during a surgical treatment cycle or amplitude of each of the first and second signal waveform or a combination thereof.

(1043) Example 5: The method of any one of Examples 1-4, wherein adjusting the compression force comprises actuating a mechanical switch coupled to the clamp arm, wherein a first position of the mechanical switch corresponds to a first actuation of the clamp arm resulting in high compression force, and wherein a second position of the mechanical switch corresponds to a second actuation of the clamp arm resulting in low compression force.

(1044) Example 6: The method of any one of Examples 1-5, wherein adjusting the compression force comprises expanding an electroactive polymer coupled to the clamp arm, and wherein expanding the electroactive polymer based on applying the first and second signal waveform to the end effector.

(1045) Example 7: A surgical instrument comprises a control circuit. The control circuit is configured to communicatively couple to a generator configured to deliver a plurality of energy modalities to an end effector of the surgical instrument, wherein the control circuit is further configured to: determine tissue impedance of tissue in contact with an end effector of the surgical instrument; determine a tissue type of based on the tissue impedance; select a first energy modality of the plurality of energy modalities; generate a first signal waveform based on the first energy modality; select a second energy modality of the plurality of energy modalities; generate a second signal waveform based on the second energy modality; and adjust a compression force applied by an end effector to tissue by changing a gap between tissue and an end effector based on a proportion of the first signal waveform to the second signal waveform.

(1046) Example 8: The surgical instrument of Example 7, further comprising an end effector coupled to the control circuit, wherein the end effector comprises a clamp arm and an ultrasonic blade.

(1047) Example 9: The surgical instrument of Example 7 or 8, further comprising a generator coupled to the control circuit.

(1048) Example 10: The surgical instrument of any one of Examples 7-10, wherein the control circuit determines proportion based on a time that each of the first and second signal waveform are applied during a surgical treatment cycle or amplitude of each of the first and second signal waveform or a combination thereof.

(1049) Example 11: The surgical instrument of any one of Examples 7-10, wherein the control circuit adjusts the compression force based on actuating a mechanical switch coupled to the clamp arm, wherein a first position of the mechanical switch corresponds to a first actuation of the clamp arm resulting in high compression force, and wherein a second position of the mechanical switch corresponds to a second actuation of the clamp arm resulting in low compression force.

(1050) Example 12: The surgical instrument of any one of Examples 7-11, wherein the control circuit adjusts the compression force based on expansion of an electroactive polymer coupled to the clamp arm, and wherein the electroactive polymer expands based on applying the first and second signal waveform to the end effector.

(1051) Example 13: A surgical system comprises a surgical hub configured to receive a tissue treatment algorithm transmitted from a cloud computing system, wherein the surgical hub is communicatively coupled to the cloud computing system; and a surgical instrument communicatively coupled to the surgical hub, wherein the surgical instrument comprises: an end effector comprising: a clamp arm; and an ultrasonic blade; a generator configured to deliver a plurality of energy modalities to the end effector, a control circuit communicatively coupled to the end effector and the generator, wherein the control circuit is configured to treat tissue, and wherein the control circuit is configured to: determine tissue impedance of tissue in contact with the end effector, determine tissue type based on the tissue impedance; select a first energy modality of the plurality of energy modalities; generate a first signal waveform based on the first energy modality; select a second energy modality of the plurality of energy modalities; generate a second signal waveform based on the second energy modality; apply the first and second signal waveform to the end effector, and adjust a compression force applied by the end effector by changing a size of a gap between the tissue and the waveguide based on a proportion of the first

signal waveform to the second signal waveform.

(1052) Example 14: The surgical instrument of Example 13, wherein the first energy modality is a radio frequency (RF) energy modality and the second energy modality is an ultrasonic energy modality.

(1053) Example 15: The surgical instrument of Example 13 or 14, wherein to determine the tissue impedance, the control circuit is configured: apply a non-therapeutic electrical signal to the end effector over a range of frequencies; and determine an impedance characteristic pattern based on spectral analysis of the non-therapeutic electrical signal.

(1054) Example 16: The surgical instrument of any one of Examples 13-15, wherein the control circuit determines the proportion based on a time that each of the first and second signal waveform are applied during a surgical treatment cycle or an amplitude of each of the first and second signal waveform or a combination thereof.

(1055) Example 17: The surgical instrument of any one of Examples 13-16, wherein to adjust the compression force, the control circuit is configured to actuate a mechanical switch coupled to the clamp arm, wherein a first position of the mechanical switch corresponds to a first actuation of the clamp arm resulting in high compression force, and wherein a second position of the mechanical switch corresponds to a second actuation of the clamp arm resulting in low compression force.

(1056) Example 18: The surgical instrument of any one of Examples 13-17, wherein to adjust the compression force, the control circuit is configured to expand an electroactive polymer coupled to the clamp arm, and to expand the electroactive polymer based on the first and second signal waveforms applied to the end effector.

(1057) Example 19: The surgical instrument of any one of Examples 14-18, wherein the RF energy modality corresponds to a first range of compression force and the ultrasonic energy modality to a second range of compression force, and wherein the first range of compression force is greater than the second range of compression force.

(1058) Example 20: The surgical instrument of any one of Examples 13-19, wherein the surgical instrument comprises a passive electrode and an active electrode.

(1059) Various additional aspects of the subject matter described herein are set out in the following numbered examples.

(1060) Example 1: An electrosurgical device, comprising: a controller comprising an electrical generator, a surgical probe comprising a distal active electrode, wherein the active electrode is in electrical communication with an electrical source terminal of the electrical generator, and a return pad in electrical communication with an electrical return terminal of the electrical generator, wherein the electrical generator is configured to source an electrical current from the electrical source terminal, and wherein the electrical current sourced by the electrical generator combines characteristics of a therapeutic electrical signal and characteristics of an excitable tissue stimulating signal.

(1061) Example 2: The electrosurgical device of Example 1, wherein the therapeutic electrical signal is a radiofrequency signal having a frequency greater than 200 kHz and less than 5 MHz.

(1062) Example 3: The electrosurgical device of any one or more of Examples 1 through 2, wherein the excitable tissue stimulating signal is an AC signal having a frequency less than 200 kHz.

(1063) Example 4: The electrosurgical device of any one or more of Examples 1 through 3, wherein the electrical current sourced by the electrical generator comprises at least one alternating therapeutic electrical signal and at least one alternating excitable tissue stimulating signal.

(1064) Example 5: The electrosurgical device of any one or more of Examples 1 through 4, wherein the electrical current sourced by the electrical generator comprises a therapeutic electrical signal amplitude modulated by the excitable tissue stimulating signal.

(1065) Example 6: The electrosurgical device of any one or more of Examples 1 through 5, wherein the electrical current sourced by the electrical generator comprises a therapeutic electrical signal DC offset by the excitable tissue stimulating signal.

(1066) Example 7: The electrosurgical device of any one or more of Examples 1 through 6, wherein the return pad further comprises at least one sensing device having a sensing device output, and the sensing device is configured to determine a stimulation of an excitable tissue by the excitable tissue stimulating signal.

(1067) Example 8: The electrosurgical device of Example 7, wherein the controller is configured to receive the sensing device output.

(1068) Example 9: The electrosurgical device of Example 8, wherein the controller comprises a processor and at least one memory component in data communication with the processor, and wherein the at least one memory component stores one or more instructions that, when executed by the processor, cause the processor to determine a distance of the active electrode from an excitable tissue based at least in part on the sensor output received by the controller.

(1069) Example 10: The electrosurgical device of Example 9, wherein the at least one memory component stores one or more instructions that, when executed by the processor, cause the processor to alter a value of at least one characteristic of the therapeutic electrical signal when the distance of the active electrode from an excitable tissue is less than a predetermined value.

(1070) Example 11: An electrosurgical system comprising: a processor, and a memory coupled to the processor, the memory configured to store instructions executable by the processor to: cause an electrical generator to combine one or more characteristics of a therapeutic signal with one or more characteristics of an excitable tissue stimulating signal to form a combination signal; cause the electrical generator to transmit the combination signal into a tissue of a patient through an active electrode in physical contact with the patient; and receive a sensing device output signal from a sensing device disposed within a return pad in physical contact with the patient.

(1071) Example 12: The electrosurgical system of Example 11, wherein the memory is configured to further store instructions executable by the processor to: determine, based at least in part on the sensing device output signal, a distance from the active electrode to an excitable tissue.

(1072) Example 13: The electrosurgical system of Example 12, wherein the memory is configured to further store instructions executable by the processor to: cause the controller to alter one or more characteristics of the therapeutic signal when the distance from the active electrode to the excitable tissue is less than a predetermined value.

(1073) Example 14: The electrosurgical system of any one or more of Examples 11-13, wherein the instructions executable by the processor to cause an electrical generator to combine one or more characteristics of a therapeutic signal with one or more characteristics of an excitable tissue stimulating signal to form a combination signal comprises instructions executable by the processor to cause the electrical generator to alternate the therapeutic signal and the excitable tissue stimulating signal.

(1074) Example 15: The electrosurgical system of any one or more of Examples 11-14, wherein the instructions executable by the processor to cause an electrical generator to combine one or more characteristics of a therapeutic signal with one or more characteristics of an excitable tissue stimulating signal to form a combination signal comprises instructions executable by the processor to cause the electrical generator to modulate an amplitude of the therapeutic signal by an amplitude of the excitable tissue stimulating signal.

(1075) Example 16: The electrosurgical system of any one or more of Examples 11-15, wherein the instructions executable by the processor to cause an electrical generator to combine one or more characteristics of a therapeutic signal with one or more characteristics of an excitable tissue stimulating signal to form a combination signal comprises instructions executable by the processor to cause the electrical generator to offset a DC value of the therapeutic signal by an amplitude of the excitable tissue stimulating signal.

(1076) Example 17: An electrosurgical system comprising: a control circuit configured to: control an electrical output of an electrical generator, in which the electrical output comprises one or more characteristics of a therapeutic signal and one or more characteristics of an excitable tissue stimulating signal; receive a sensing device signal from at least one sensing device configured to measure an activity of an excitable tissue of a patient; determine a distance between a location of an active electrode configured to transmit the electrical output of the electrical generator into a patient tissue and a location of the at least one sensing device; and alter the electrical output of the electrical generator in at least one characteristic of

the therapeutic signal when the distance between the location of the active electrode configured to transmit the electrical output of the electrical generator into the patient tissue and the location of the at least one sensing device is less than a pre-determined value.

(1077) Example 18: The electrosurgical system of Example 17, wherein the control circuit configured to alter the electrical output of the electrical generator in at least one characteristic of the therapeutic signal when the distance between the location of the active electrode configured to transmit the electrical output of the electrical generator into the patient tissue and the location of the at least one sensing device is less than a pre-determined value comprises a control circuit configured to minimize the at least one characteristic of the therapeutic signal.

(1078) Example 19: A non-transitory computer readable medium storing computer readable instructions which, when executed, causes a machine to: control an electrical output of an electrical generator, in which the electrical output comprises one or more characteristics of a therapeutic signal and one or more characteristics of an excitable tissue stimulating signal; receive a sensing device signal from at least one sensing device configured to measure an activity of an excitable tissue of a patient; determine a distance between a location of an active electrode configured to transmit the electrical output of the electrical generator into a patient tissue and a location of the at least one sensing device; and alter the electrical output of the electrical generator in at least one characteristic of the therapeutic signal when the distance between the location of the active electrode configured to transmit the electrical output of the electrical generator into the patient tissue and the location of the at least one sensing device is less than a pre-determined value.

(1079) Various additional aspects of the subject matter described herein are set out in the following numbered examples:

(1080) Example 1: A surgical instrument comprising: an ultrasonic blade, an arm pivotable relative to the ultrasonic blade between an open position and a closed position, a transducer assembly coupled to the ultrasonic blade, a sensor configured to sense a position of the arm between the open position and the closed position, and a control circuit coupled to the transducer assembly and the sensor. The transducer assembly comprises at least two piezoelectric elements configured to ultrasonically oscillate the ultrasonic blade. The control circuit is configured to activate the transducer assembly according to a position of the arm detected by the sensor relative to a threshold position.

(1081) Example 2: The surgical instrument of Example 1, wherein the sensor comprises a Hall effect sensor.

(1082) Example 3: The surgical instrument of Example 2, wherein the arm comprises a magnet detectable by the Hall effect sensor.

(1083) Example 4: The surgical instrument of Example 2, wherein the Hall effect sensor is configured to detect a magnet disposed on a user.

(1084) Example 5: The surgical instrument of any one of Examples 1-4, wherein the threshold position corresponds to the open position.

(1085) Example 6: The surgical instrument of any one of Examples 1-5, wherein the threshold position corresponds to the closed position.

(1086) Example 7: A surgical instrument comprising: an ultrasonic blade, an arm pivotable relative to the ultrasonic blade between an open position and a closed position, a transducer assembly coupled to the ultrasonic blade, a first sensor configured to sense a first force as the arm transitions to the closed position, a second sensor configured to sense a second force as the arm transitions to the open position, and a control circuit coupled to the transducer assembly, the first sensor, and the second sensor. The transducer assembly comprises at least two piezoelectric elements configured to ultrasonically oscillate the ultrasonic blade. The control circuit is configured to activate the transducer assembly according to the first force sensed by the first sensor relative to a first threshold and the second force sensed by the second sensor relative to a second threshold.

(1087) Example 8: The surgical instrument of Example 7, wherein the first sensor comprise a tactile switch.

(1088) Example 9: The surgical instrument of Example 8, wherein the tactile switch comprises a two-stage tactile switch.

(1089) Example 10: The surgical instrument of Example 9, wherein the first threshold correspond to a second stage of the two-stage tactile switch.

(1090) Example 11: The surgical instrument of any one of Examples 7-10, wherein the first sensor is disposed on a housing of the surgical instrument such that the arm bears thereagainst as the arm transitions to the closed position.

(1091) Example 12: The surgical instrument of any one of Examples 7-11, wherein the second sensor comprise a tactile switch.

(1092) Example 13: The surgical instrument of Example 12, wherein the tactile switch comprises a one-stage tactile switch.

(1093) Example 14: The surgical instrument of any one of Examples 7-13, wherein the second threshold correspond to a non-zero force.

(1094) Example 15: The surgical instrument of any one of Examples 7-14, wherein the second sensor is disposed adjacent to a rotation point between the arm and the ultrasonic blade such that the arm bears against the second sensor as the arm transitions to the open position.

(1095) Example 16: A surgical instrument comprising: an ultrasonic blade, a transducer assembly coupled to the ultrasonic blade, a sensor configured to sense a force thereagainst, and a control circuit coupled to the transducer assembly and the sensor. The transducer assembly comprises at least two piezoelectric elements configured to ultrasonically oscillate the ultrasonic blade. The control circuit is configured to activate the transducer assembly according to the force sensed by the sensor relative to a threshold force.

(1096) Example 17: The surgical instrument of Example 16, wherein the sensor comprises a force sensitive resistor.

(1097) Example 18: The surgical instrument of Example 16 or 17, wherein the control circuit is configured to activate the transducer assembly when the force sensed by the sensor exceeds the threshold force.

(1098) Example 19: The surgical instrument of any one of Examples 16-18, wherein the sensor is disposed on an exterior surface of the surgical instrument.

(1099) Example 20: The surgical instrument of any one of Examples 16-19, wherein an output of the sensor varies according to a degree of force thereagainst and the control circuit is configured to activate the transducer assembly according to the output of the sensor relative to a threshold representative of the threshold force.

(1100) Various additional aspects of the subject matter described herein are set out in the following numbered examples

(1101) Example 1: A method of estimating a state of an end effector of an ultrasonic device, the ultrasonic device including an electromechanical ultrasonic system defined by a predetermined resonant frequency, the electromechanical ultrasonic system including an ultrasonic transducer coupled to an ultrasonic blade, the method comprising: measuring, by a control circuit, a complex impedance of an ultrasonic transducer, wherein the complex impedance is defined as

$$(1102) Z_g(t) = \frac{V_g(t)}{I_g(t)};$$

receiving, by the control circuit, a complex impedance measurement data point; comparing, by the control circuit, the complex impedance measurement data point to a data point in a reference complex impedance characteristic pattern; classifying, by the control circuit, the complex impedance measurement data point based on a result of the comparison analysis; and assigning, by the control circuit, a state or condition of the end effector based on the result of the comparison analysis.

(1103) Example 2: The method of Example 1, comprising: receiving, by the control circuit, the reference complex impedance characteristic pattern from a database or memory coupled to the control circuit; and generating, by the control circuit, the reference complex impedance characteristic pattern as follows: applying, by a drive circuit coupled to the control circuit, a nontherapeutic drive signal to the ultrasonic transducer starting at an initial frequency, ending at a final frequency, and at a plurality of frequencies therebetween; measuring, by the control circuit, the impedance of the ultrasonic transducer at each frequency; storing, by the control circuit, a data point corresponding to each impedance measurement; and curve fitting, by the control circuit, a plurality of data points to generate a three-dimensional curve of representative of the reference complex impedance characteristic pattern, wherein the magnitude $|Z|$ and phase ϕ are plotted as a function of frequency f .

(1104) Example 3: The method of Example 2, where the curve fitting includes a polynomial curve fit, a Fourier series, and/or a parametric equation.

(1105) Example 4: The method of any one of Examples 1-3, comprising: receiving, by the control circuit, a new impedance measurement data point; and classifying, by the control circuit, the new impedance measurement data point using a Euclidean perpendicular distance from the new impedance measurement data point to a trajectory which has been fitted to the reference complex impedance characteristic pattern.

(1106) Example 5: The method of Example 4, comprising estimating, by the control circuit, a probability that the new impedance measurement data point is correctly classified.

(1107) Example 6: The method of Example 5, comprising adding, by the control circuit, the new impedance measurement data point to the reference complex impedance characteristic pattern based on the probability of the estimated correct classification of the new impedance measurement data point.

(1108) Example 7: The method of Example 4, comprising: classifying by the control circuit, data based on a set of training data S, where the set of training data S comprises a plurality of complex impedance measurement data; curve fitting, by the control circuit, the set of training data S using a parametric Fourier series; wherein S is defined by:

(1109)
$$\overset{\text{Math.}}{p} = \overset{\text{Math.}}{a}_0 + \overset{\text{Math.}}{\sum_{n=1}^{\infty}} (\overset{\text{Math.}}{a}_n \cos \overset{\text{Math.}}{\omega_L t} + \overset{\text{Math.}}{b}_n \sin \overset{\text{Math.}}{\omega_L t})$$
 wherein, for a new impedance measurement data point , a perpendicular distance from  to  is found by:

$D = \| \text{custom character} - \text{custom character} \|$

when:

(1110) $\frac{\partial D}{\partial t} = 0$

then:

$D = D.\text{sub.} \perp$

(1111) wherein the probability distribution of D is used to estimate the probability of the new impedance measurement data point  belonging to the group S.

(1112) Example 8: The method of claim 1, wherein the control circuit is located at a surgical hub in communication with the ultrasonic electromechanical system.

(1113) Example 9: A generator for estimating a state of an end effector of an ultrasonic device, the ultrasonic device including an electromechanical ultrasonic system defined by a predetermined resonant frequency, the electromechanical ultrasonic system including an ultrasonic transducer coupled to an ultrasonic blade, the generator comprising: a control circuit coupled to a memory, the control circuit configured to: measure a complex impedance of an ultrasonic transducer, wherein the complex impedance is defined as

(1114) $Z_g(t) = \frac{V_g(t)}{I_g(t)}$;

receive a complex impedance measurement data point; compare the complex impedance measurement data point to a data point in a reference complex impedance characteristic pattern; classify the complex impedance measurement data point based on a result of the comparison analysis; and assign a state or condition of the end effector based on the result of the comparison analysis.

(1115) Example 10: The generator of Example 9, further comprising: a drive circuit coupled to the control circuit, the drive circuit configured to apply a nontherapeutic drive signal to the ultrasonic transducer starting at an initial frequency, ending at a final frequency, and at a plurality of frequencies therebetween; wherein the control circuit is further configured to generate the reference complex impedance characteristic pattern; wherein the control circuit is configured to receive the reference complex impedance characteristic pattern from a database or the memory coupled to the control circuit; measure the impedance of the ultrasonic transducer at each frequency; store in the memory a data point corresponding to each impedance measurement; and curve fit a plurality of data points to generate a three-dimensional curve of representative of the reference complex impedance characteristic pattern, wherein the magnitude |Z| and phase ϕ are plotted as a function of frequency f.

(1116) Example 11: The generator of any one of Example 10, wherein the curve fit includes a polynomial curve fit, a Fourier series, and/or a parametric equation.

(1117) Example 12: The generator of any one of Examples 9-11, wherein the control circuit is further configured to: receive a new impedance measurement data point; and classify the new impedance measurement data point using a Euclidean perpendicular distance from the new impedance measurement data point to a trajectory which has been fitted to the reference complex impedance characteristic pattern.

(1118) Example 13: The generator of Example 11, wherein the control circuit is further configured to estimate a probability that the new impedance measurement data point is correctly classified.

(1119) Example 14: The generator of Example 13, wherein the control circuit is further configured to add the new impedance measurement data point to the reference complex impedance characteristic pattern based on the probability of the estimated correct classification of the new impedance measurement data point.

(1120) Example 15: The generator of Example 13, wherein the control circuit is further configured to: classify data based on a set of training data S, where the set of training data S comprises a plurality of complex impedance measurement data; curve fit the set of training data S using a parametric Fourier series; wherein S is defined by:

(1121)
$$0 \overset{\text{Math.}}{p} = \overset{\text{Math.}}{a}_0 + \overset{\text{Math.}}{\sum_{n=1}^{\infty}} (\overset{\text{Math.}}{a}_n \cos \overset{\text{Math.}}{\omega_L t} + \overset{\text{Math.}}{b}_n \sin \overset{\text{Math.}}{\omega_L t})$$
 wherein, for a new impedance measurement data point , a perpendicular distance from  to  is found by:

$D = \| \text{custom character} - \text{custom character} \|$

when:

(1122) $\frac{\partial D}{\partial t} = 0$

then:

$D = D.\text{sub.} \perp$

wherein the probability distribution of D is used to estimate the probability of the new impedance measurement data point  to the group S.

(1123) Example 16: The generator of claim 9, wherein the control circuit and the memory are located at a surgical hub in communication with the ultrasonic electromechanical system.

(1124) Example 17: An ultrasonic device for estimating a state of an end effector thereof, the ultrasonic device comprising: an electromechanical ultrasonic system defined by a predetermined resonant frequency, the electromechanical ultrasonic system comprising an ultrasonic transducer coupled to an ultrasonic blade; a control circuit coupled to a memory, the control circuit configured to: measure a complex impedance of the ultrasonic transducer, wherein the complex impedance is defined as

(1125) $Z_g(t) = \frac{V_g(t)}{I_g(t)}$;

receive a complex impedance measurement data point; compare the complex impedance measurement data point to a data point in a reference complex impedance characteristic pattern; classify the complex impedance measurement data point based on a result of the comparison analysis; and assign a state or condition of the end effector based on the result of the comparison analysis.

(1126) Example 18: The ultrasonic device of Example 17, further comprising: a drive circuit coupled to the control circuit, the drive circuit configured to apply a nontherapeutic drive signal to the ultrasonic transducer starting at an initial frequency, ending at a final frequency, and at a plurality of frequencies therebetween; wherein the control circuit is further configured to generate the reference complex impedance characteristic pattern; wherein the control circuit is configured to receive the reference complex impedance characteristic pattern from a database or the memory coupled to the control circuit; measure the impedance of the ultrasonic transducer at each frequency; store in the memory a data point corresponding to each impedance measurement; and curve fit a plurality of data points to generate a three-dimensional curve of representative of the reference

complex impedance characteristic pattern, wherein the magnitude $|Z|$ and phase ϕ are plotted as a function of frequency f .

(1127) Example 19: The ultrasonic device of Example 18, wherein the curve fit includes a polynomial curve fit, a Fourier series, and/or a parametric equation.

(1128) Example 20: The ultrasonic device of any one of Examples 17-19, wherein the control circuit is further configured to: receive a new impedance measurement data point; and classify the new impedance measurement data point using a Euclidean perpendicular distance from the new impedance measurement data point to a trajectory which has been fitted to the reference complex impedance characteristic pattern.

(1129) Example 21: The ultrasonic device of Example 20, wherein the control circuit is further configured to estimate a probability that the new impedance measurement data point is correctly classified.

(1130) Example 22: The ultrasonic device of Example 21, wherein the control circuit is further configured to add the new impedance measurement data point to the reference complex impedance characteristic pattern based on the probability of the estimated correct classification of the new impedance measurement data point.

(1131) Example 23: The ultrasonic device of Example 21, wherein the control circuit is further configured to: classify data based on a set of training data S , where the set of training data S comprises a plurality of complex impedance measurement data; curve fit the set of training data S using a parametric Fourier series; wherein S is defined by:

(1132)
$$p^{\text{Math.}} = a_0^{\text{Math.}} + \sum_{n=1}^{\infty} \left(a_n^{\text{Math.}} \cos \frac{n}{L} t + b_n^{\text{Math.}} \sin \frac{n}{L} t \right)$$
 wherein, for a new impedance measurement data point  custom character, a perpendicular distance from  custom character to  custom character is found by:

$$D = \left\| \text{custom character} - \text{custom character} \right\|$$

when:

(1133)
$$\frac{\partial D}{\partial t} = 0$$

then:

$$D = D_{\text{sub.} \perp}$$

wherein the probability distribution of D is used to estimate the probability of the new impedance measurement data point  custom character belonging to the group S .

(1134) Example 24: The ultrasonic device of any one of Examples 17-23, wherein the control circuit and the memory are located at a surgical hub in communication with the ultrasonic electromechanical system.

(1135) Example 25: A method of estimating a state of an end effector of an ultrasonic device, the ultrasonic device including an electromechanical ultrasonic system defined by a predetermined resonant frequency, the electromechanical ultrasonic system including an ultrasonic transducer coupled to an ultrasonic blade, the method comprising: applying, by a drive circuit, a drive signal to an ultrasonic transducer, wherein the drive signal is a periodic signal defined by a magnitude and frequency; sweeping, by a processor or control circuit, the frequency of the drive signal from below resonance to above resonance of the electromagnetic ultrasonic system; measuring and recording, by the processor or control circuit, impedance/admittance circle variables $R_{\text{sub.e}}$, $G_{\text{sub.e}}$, $X_{\text{sub.e}}$, $B_{\text{sub.e}}$; comparing, by the processor or control circuit, measured impedance/admittance circle variables $R_{\text{sub.e}}$, $G_{\text{sub.e}}$, $X_{\text{sub.e}}$, $B_{\text{sub.e}}$ to reference impedance/admittance circle variables $R_{\text{sub.ref}}$, $G_{\text{sub.ref}}$, $X_{\text{sub.ref}}$, $B_{\text{sub.ref}}$; and determining, by the processor or control circuit, a state or condition of the end effector based on the result of the comparison analysis.

(1136) Various aspects of the subject matter described herein are set out in the following numbered examples:

(1137) Example 1: A method of determining a temperature of an ultrasonic blade, the method comprising: determining, by a control circuit coupled to a memory, an actual resonant frequency of an ultrasonic electromechanical system comprising an ultrasonic transducer coupled to an ultrasonic blade by an ultrasonic waveguide, wherein the actual resonant frequency is correlated to an actual temperature of the ultrasonic blade; retrieving, from the memory by the control circuit, a reference resonant frequency of the ultrasonic electromechanical system, wherein the reference resonant frequency is correlated to a reference temperature of the ultrasonic blade; and inferring, by the control circuit, the temperature of the ultrasonic blade based on the difference between the actual resonant frequency and the reference resonant frequency.

(1138) Example 2: The method of Example 1, wherein determining, by the control circuit, the actual resonant frequency of the ultrasonic electromechanical system comprises: determining, by the control circuit, a phase angle ϕ between a voltage $V_{\text{sub.g}}(t)$ and a current $I_{\text{sub.g}}(t)$ signal applied to the ultrasonic transducer.

(1139) Example 3: The method of Example 2, further comprising generating, by the control circuit, a temperature estimator and state space model of the inferred temperature of the ultrasonic blade as a function of the resonant frequency of the ultrasonic electromechanical system based on a set of non-linear state space equations.

(1140) Example 4: The method of Example 3, wherein the state space model is defined by:

(1141)
$$\begin{bmatrix} \dot{F}_n \\ \dot{T} \end{bmatrix} = f(t, T(t), F_n(t), E(t)) \dot{y} = h(t, T(t), F_n(t), E(t)).$$

(1142) Example 5: The method of Example 4, further comprising applying, by the control circuit, a Kalman filter to improve the temperature estimator and state space model.

(1143) Example 6: The method of Example 5, further comprising: applying, by the control circuit, a state estimator in a feedback loop of the Kalman filter, controlling, by the control circuit, power applied to the ultrasonic transducer, and regulating, by the control circuit, the temperature of the ultrasonic blade.

(1144) Example 7: The method of Example 6, wherein a state variance of the state estimator of the Kalman filter is defined by:

$(\sigma_{\text{sub.k.sup.-}})_{\text{sup.2}} = \sigma_{\text{sub.k-1.sup.2}} + \sigma_{\text{sub.P.sub.k.sup.2}}$ and a gain K of the Kalman filter is defined by:

(1145)
$$K = \frac{\left(\frac{k}{k} \right)^2}{\left(\frac{k}{k} \right)^2 + \frac{2}{m}}.$$

(1146) Example 8: The method of Example 1, wherein the control circuit and memory are located at a surgical hub in communication with the ultrasonic electromechanical system.

(1147) Example 9: A generator for determining a temperature of an ultrasonic blade, the generator comprising: a control circuit coupled to a memory, the control circuit configured to: determine an actual resonant frequency of an ultrasonic electromechanical system comprising an ultrasonic transducer coupled to an ultrasonic blade by an ultrasonic waveguide, wherein the actual resonant frequency is correlated to an actual temperature of the ultrasonic blade; retrieve from the memory a reference resonant frequency of the ultrasonic electromechanical system, wherein the reference resonant frequency is correlated to a reference temperature of the ultrasonic blade; and infer the temperature of the ultrasonic blade based on the difference between the actual resonant frequency and the reference resonant frequency.

(1148) Example 10: The generator of Example 9, wherein to determine the actual resonant frequency of the ultrasonic electromechanical system, the control circuit is further configured to: determine a phase angle ϕ between a voltage $V_{\text{sub.g}}(t)$ and a current $I_{\text{sub.g}}(t)$ signal applied to the ultrasonic transducer.

(1149) Example 11: The generator of Example 10, wherein the control circuit is further configured to generate a temperature estimator and state space model of the inferred temperature of the ultrasonic blade as a function of the resonant frequency of the ultrasonic electromechanical system based on a set of non-linear state space equations.

(1150) Example 12: The generator of Example 11, wherein the state space model is defined by:

$$(1151) \left[\begin{matrix} \dot{F}_n \\ \dot{T} \end{matrix} \right] = f(t, T(t), F_n(t), E(t)) \dot{y} = h(t, T(t), F_n(t), E(t)).$$

(1152) Example 13: The generator of Example 12, wherein the control circuit is further configured to apply a Kalman filter to improve the temperature estimator and state space model.

(1153) Example 14: The generator of Example 13, wherein the control circuit is further configured to: apply a state estimator in a feedback loop of the Kalman filter, control power applied to the ultrasonic transducer, and regulate the temperature of the ultrasonic blade.

(1154) Example 15: The generator of Example 14, wherein a state variance of the state estimator of the Kalman filter is defined by: $(\sigma_{\text{sub.k.sup.-}})_{\text{sup.2}} = \sigma_{\text{sub.k-1.sup.2}} + \sigma_{\text{sub.P.sub.k.sup.2}}$ and a gain K of the Kalman filter is defined by:

$$(1155) K = \frac{\begin{pmatrix} \dot{k} \\ \dot{k} \end{pmatrix}}{\begin{pmatrix} \dot{k} \\ \dot{k} \end{pmatrix}^2 + \frac{1}{m}}.$$

(1156) Example 16: The generator of Example 9, wherein the control circuit and memory are located at a surgical hub in communication with the generator.

(1157) Example 17: An ultrasonic device for determining a temperature of an ultrasonic blade, the ultrasonic device comprising: a control circuit coupled to a memory, the control circuit configured to: determine an actual resonant frequency of an ultrasonic electromechanical system comprising an ultrasonic transducer coupled to an ultrasonic blade by an ultrasonic waveguide, wherein the actual resonant frequency is correlated to an actual temperature of the ultrasonic blade; retrieve from the memory a reference resonant frequency of the ultrasonic electromechanical system, wherein the reference resonant frequency is correlated to a reference temperature of the ultrasonic blade; and infer the temperature of the ultrasonic blade based on the difference between the actual resonant frequency and the reference resonant frequency.

(1158) Example 18: The ultrasonic device of Example 17, wherein to determine the actual resonant frequency of the ultrasonic electromechanical system, the control circuit is further configured to: determine a phase angle ϕ between a voltage $V_{\text{sub.g(t)}}$ and a current $I_{\text{sub.g(t)}}$ signal applied to the ultrasonic transducer.

(1159) Example 19: The ultrasonic device of Example 18, wherein the control circuit is further configured to generate a temperature estimator and state space model of the inferred temperature of the ultrasonic blade as a function of the resonant frequency of the ultrasonic electromechanical system based on a set of non-linear state space equations.

(1160) Example 20: The ultrasonic device of Example 19, wherein the state space model is defined by:

$$(1161) \left[\begin{matrix} \dot{F}_n \\ \dot{T} \end{matrix} \right] = f(t, T(t), F_n(t), E(t)) \dot{y} = h(t, T(t), F_n(t), E(t)).$$

(1162) Example 21: The ultrasonic device of Example 20, wherein the control circuit is further configured to apply a Kalman filter to improve the temperature estimator and state space model.

(1163) Example 22: The ultrasonic device of Example 21, wherein the control circuit is further configured to: apply a state estimator in a feedback loop of the Kalman filter, control power applied to the ultrasonic transducer, and regulate the temperature of the ultrasonic blade.

(1164) Example 23: The ultrasonic device of Example 22, wherein a state variance of the state estimator of the Kalman filter is defined by: $(\sigma_{\text{sub.k.sup.-}})_{\text{sup.2}} = \sigma_{\text{sub.k-1.sup.2}} + \sigma_{\text{sub.P.sub.k.sup.2}}$ and a gain K of the Kalman filter is defined by:

$$(1165) 0K = \frac{\begin{pmatrix} \dot{k} \\ \dot{k} \end{pmatrix}}{\begin{pmatrix} \dot{k} \\ \dot{k} \end{pmatrix}^2 + \frac{1}{m}}.$$

(1166) Example 24: The ultrasonic instrument of Example 17, wherein the control circuit and memory are located at a surgical hub in communication with the ultrasonic instrument.

(1167) While several forms have been illustrated and described, it is not the intention of the applicant to restrict or limit the scope of the appended claims to such detail. Numerous modifications, variations, changes, substitutions, combinations, and equivalents to those forms may be implemented and will occur to those skilled in the art without departing from the scope of the present disclosure. Moreover, the structure of each element associated with the described forms can be alternatively described as a means for providing the function performed by the element. Also, where materials are disclosed for certain components, other materials may be used. It is therefore to be understood that the foregoing description and the appended claims are intended to cover all such modifications, combinations, and variations as falling within the scope of the disclosed forms. The appended claims are intended to cover all such modifications, variations, changes, substitutions, modifications, and equivalents.

(1168) The foregoing detailed description has set forth various forms of the devices and/or processes via the use of block diagrams, flowcharts, and/or examples. Insofar as such block diagrams, flowcharts, and/or examples contain one or more functions and/or operations, it will be understood by those within the art that each function and/or operation within such block diagrams, flowcharts, and/or examples can be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or virtually any combination thereof. Those skilled in the art will recognize that some aspects of the forms disclosed herein, in whole or in part, can be equivalently implemented in integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computer systems), as one or more programs running on one or more processors (e.g., as one or more programs running on one or more microprocessors), as firmware, or as virtually any combination thereof, and that designing the circuitry and/or writing the code for the software and or firmware would be well within the skill of one of skill in the art in light of this disclosure. In addition, those skilled in the art will appreciate that the mechanisms of the subject matter described herein are capable of being distributed as one or more program products in a variety of forms, and that an illustrative form of the subject matter described herein applies regardless of the particular type of signal bearing medium used to actually carry out the distribution.

(1169) Instructions used to program logic to perform various disclosed aspects can be stored within a memory in the system, such as dynamic random access memory (DRAM), cache, flash memory, or other storage. Furthermore, the instructions can be distributed via a network or by way of other computer readable media. Thus a machine-readable medium may include any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computer), but is not limited to, floppy diskettes, optical disks, compact disc, read-only memory (CD-ROMs), and magneto-optical disks, read-only memory (ROMs), random access memory (RAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), magnetic or optical cards, flash memory, or a tangible, machine-readable storage used in the transmission of information over the Internet via electrical, optical, acoustical or other forms of propagated signals (e.g., carrier waves, infrared signals, digital signals, etc.). Accordingly, the non-transitory computer-readable medium includes any type of tangible machine-readable medium suitable for storing or transmitting electronic instructions or information in a form readable by a machine (e.g., a computer).

(1170) As used in any aspect herein, the term “control circuit” may refer to, for example, hardwired circuitry, programmable circuitry (e.g., a computer processor comprising one or more individual instruction processing cores, processing unit, processor, microcontroller, microcontroller unit, controller, digital signal processor (DSP), programmable logic device (PLD), programmable logic array (PLA), or field programmable gate array (FPGA)), state machine circuitry, firmware that stores instructions executed by programmable circuitry, and any combination thereof. The control circuit may, collectively or individually, be embodied as circuitry that forms part of a larger system, for example, an integrated circuit (IC), an application-specific integrated circuit (ASIC), a system on-chip (SoC), desktop computers, laptop computers, tablet computers, servers, smart phones, etc. Accordingly, as used herein, “control circuit” includes, but is not limited to, electrical circuitry having at least one discrete electrical circuit, electrical circuitry having at least one integrated circuit, electrical circuitry having at least one application specific integrated circuit, electrical circuitry forming a general purpose computing device configured by a computer program (e.g., a general purpose computer configured by a computer program which at least partially carries out processes and/or devices described herein, or a microprocessor configured by a computer

program which at least partially carries out processes and/or devices described herein), electrical circuitry forming a memory device (e.g., forms of random access memory), and/or electrical circuitry forming a communications device (e.g., a modem, communications switch, or optical-electrical equipment). Those having skill in the art will recognize that the subject matter described herein may be implemented in an analog or digital fashion or some combination thereof.

(1171) As used in any aspect herein, the term “logic” may refer to an app, software, firmware and/or circuitry configured to perform any of the aforementioned operations. Software may be embodied as a software package, code, instructions, instruction sets and/or data recorded on non-transitory computer readable storage medium. Firmware may be embodied as code, instructions or instruction sets and/or data that are hard-coded (e.g., nonvolatile) in memory devices.

(1172) As used in any aspect herein, the terms “component,” “system,” “module” and the like can refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution.

(1173) As used in any aspect herein, an “algorithm” refers to a self-consistent sequence of steps leading to a desired result, where a “step” refers to a manipulation of physical quantities and/or logic states which may, though need not necessarily, take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It is common usage to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like. These and similar terms may be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities and/or states.

(1174) A network may include a packet switched network. The communication devices may be capable of communicating with each other using a selected packet switched network communications protocol. One example communications protocol may include an Ethernet communications protocol which may be capable permitting communication using a Transmission Control Protocol/Internet Protocol (TCP/IP). The Ethernet protocol may comply or be compatible with the Ethernet standard published by the Institute of Electrical and Electronics Engineers (IEEE) titled “IEEE 802.3 Standard”, published in December, 2008 and/or later versions of this standard. Alternatively or additionally, the communication devices may be capable of communicating with each other using an X.25 communications protocol. The X.25 communications protocol may comply or be compatible with a standard promulgated by the International Telecommunication Union-Telecommunication Standardization Sector (ITU-T). Alternatively or additionally, the communication devices may be capable of communicating with each other using a frame relay communications protocol. The frame relay communications protocol may comply or be compatible with a standard promulgated by Consultative Committee for International Telegraph and Telephone (CCITT) and/or the American National Standards Institute (ANSI). Alternatively or additionally, the transceivers may be capable of communicating with each other using an Asynchronous Transfer Mode (ATM) communications protocol. The ATM communications protocol may comply or be compatible with an ATM standard published by the ATM Forum titled “ATM-MPLS Network Interworking 2.0” published August 2001, and/or later versions of this standard. Of course, different and/or after-developed connection-oriented network communication protocols are equally contemplated herein.

(1175) Unless specifically stated otherwise as apparent from the foregoing disclosure, it is appreciated that, throughout the foregoing disclosure, discussions using terms such as “processing,” “computing,” “calculating,” “determining,” “displaying,” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

(1176) One or more components may be referred to herein as “configured to,” “configurable to,” “operable/operative to,” “adapted/adaptable,” “able to,” “conformable/conformed to,” etc. Those skilled in the art will recognize that “configured to” can generally encompass active-state components and/or inactive-state components and/or standby-state components, unless context requires otherwise.

(1177) The terms “proximal” and “distal” are used herein with reference to a clinician manipulating the handle portion of the surgical instrument. The term “proximal” refers to the portion closest to the clinician and the term “distal” refers to the portion located away from the clinician. It will be further appreciated that, for convenience and clarity, spatial terms such as “vertical,” “horizontal,” “up”, and “down” may be used herein with respect to the drawings. However, surgical instruments are used in many orientations and positions, and these terms are not intended to be limiting and/or absolute.

(1178) Those skilled in the art will recognize that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to claims containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations.

(1179) In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that typically a disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms unless context dictates otherwise. For example, the phrase “A or B” will be typically understood to include the possibilities of “A” or “B” or “A and B.”

(1180) With respect to the appended claims, those skilled in the art will appreciate that recited operations therein may generally be performed in any order. Also, although various operational flow diagrams are presented in a sequence(s), it should be understood that the various operations may be performed in other orders than those which are illustrated, or may be performed concurrently. Examples of such alternate orderings may include overlapping, interleaved, interrupted, reordered, incremental, preparatory, supplemental, simultaneous, reverse, or other variant orderings, unless context dictates otherwise. Furthermore, terms like “responsive to,” “related to,” or other past-tense adjectives are generally not intended to exclude such variants, unless context dictates otherwise.

(1181) It is worthy to note that any reference to “one aspect,” “an aspect,” “an exemplification,” “one exemplification,” and the like means that a particular feature, structure, or characteristic described in connection with the aspect is included in at least one aspect. Thus, appearances of the phrases “in one aspect,” “in an aspect,” “in an exemplification,” and “in one exemplification” in various places throughout the specification are not necessarily all referring to the same aspect. Furthermore, the particular features, structures or characteristics may be combined in any suitable manner in one or more aspects.

(1182) Any patent application, patent, non-patent publication, or other disclosure material referred to in this specification and/or listed in any Application Data Sheet is incorporated by reference herein, to the extent that the incorporated materials is not inconsistent herewith. As such, and to the extent necessary, the disclosure as explicitly set forth herein supersedes any conflicting material incorporated herein by reference. Any material, or portion thereof, that is said to be incorporated by reference herein, but which conflicts with existing definitions, statements, or other disclosure material set forth herein will only be incorporated to the extent that no conflict arises between that incorporated material and the existing disclosure material.

(1183) In summary, numerous benefits have been described which result from employing the concepts described herein. The foregoing description of the one or more forms has been presented for purposes of illustration and description. It is not intended to be exhaustive or limiting to the precise form disclosed. Modifications or variations are possible in light of the above teachings. The one or more forms were chosen and described in order to illustrate principles and practical application to thereby enable one of ordinary skill in the art to utilize the various forms and with various modifications as are suited to the particular use contemplated. It is intended that the claims submitted herewith define the overall scope.

Claims

1. A surgical system comprising: an electrosurgical device; a control tower comprising a generator module configured to supply energy to the electrosurgical device; and a robotic control wherein the robotic control is configured to generate a device control signal and an energy control signal; wherein the robotic control is in signal communication with the control tower, such that a first communication path between the robotic control and the control tower is configured to convey the device control signal and the energy control signal, wherein the control tower is in signal communication with the electrosurgical device, such that a second communication path between the control tower and the electrosurgical device is configured to convey the device control signal and the energy control signal to the electrosurgical device; wherein the second communication path is further configured to convey a return control signal from the electrosurgical device to the control tower; wherein the electrosurgical device is in signal communication with the robotic control, such that a third communication path between the electrosurgical device and the robotic control is configured to convey the return control signal from the electrosurgical device to the robotic control; and wherein the return control signal comprises an electrical parameter, the electrical parameter comprising impedance of a tissue interacting with the electrosurgical device.
 2. The surgical system of claim 1, wherein the energy control signal comprises a power level of the energy from the generator module supplied to the electrosurgical device.
 3. The surgical system of claim 1, wherein the return control signal further comprises a mechanical parameter.
 4. The surgical system of claim 1, wherein the device control signal comprises an actuation parameter.
 5. The surgical system of claim 4, wherein the actuation parameter comprises a force to fire the electrosurgical device.
 6. The surgical system of claim 4, wherein the actuation parameter comprises a force to close the electrosurgical device.
 7. The surgical system of claim 4, wherein the actuation parameter comprises a closure status of the electrosurgical device.
 8. The surgical system of claim 1, wherein the device control signal comprises a position of the electrosurgical device.
 9. The surgical system of claim 1, wherein the device control signal and the energy control signal are determined based on a surgical procedure to be performed.
 10. The surgical system of claim 1, wherein the control tower further comprises a wireless communication module, and the first communication path are wireless.
 11. The surgical system of claim 1, wherein the robotic control is in signal communication with a remote server, wherein the energy control signal and device control signal are generated based on an initial signal from the remote server.
 12. A surgical system comprising: an electrosurgical device; a control tower comprising a generator module configured to supply energy to the electrosurgical device; and a robotic control wherein the robotic control is configured to generate a device control signal and an energy control signal; wherein the robotic control is in signal communication with the control tower, such that a first communication path between the robotic control and the control tower is configured to convey the device control signal and the energy control signal, wherein the control tower is in signal communication with the electrosurgical device, such that a second communication path between the control tower and the electrosurgical device is configured to convey the device control signal and the energy control signal to the electrosurgical device; wherein the second communication path is further configured to convey a return control signal from the electrosurgical device to the control tower; wherein the device control signal and the energy control signal are determined based on a surgical procedure to be performed; and wherein the robotic control comprises a control circuit comprising a processor and a memory, wherein the memory is configured to store instructions executable by the processor to: determine, based on the return control signal, an actual parameter of the electrosurgical device; determine, based on the surgical procedure to be performed, an expected parameter of the electrosurgical device; compare the actual parameter of the electrosurgical device to the expected parameter of the electrosurgical device; communicate the comparison to the control tower; and cause the control tower to adjust at least one of the energy control signal and the device control signal based on the comparison.
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